

Mathematics for Economists with Applications

经济学家数学及其应用

James Bergin

詹姆斯·伯金

目录

第一章 Introduction / 绪论	13
1.1 Introduction / 绪论	13
1.2 A Brief Summary by Chapter / 各章内容简介	13
第二章 Matrices and Systems of Equations / 矩阵与方程组	15
2.1 Introduction / 引言	15
2.1.1 Some Motivating Examples / 若干引例	15
2.2 Vectors / 向量	16
2.3 Matrices / 矩阵	16
2.3.1 Matrix Addition and Subtraction / 矩阵加法与减法	16
2.3.2 Matrix Multiplication / 矩阵乘法	17
2.3.3 Matrix Transpose / 矩阵转置	17
2.3.4 Matrix Inversion / 矩阵求逆	17
2.3.5 Additional Remarks on Determinants / 关于行列式的补充说明	17
2.3.6 Cramer's Rule / 克莱姆法则	18
2.3.7 Gaussian Elimination / 高斯消元法	18
2.4 Linear Dependence and Independence / 线性相关与线性无关	19
2.4.1 Matrix Rank and Linear Independence / 矩阵的秩与线性无关性	19
2.5 Solutions of Systems of Equations / 方程组的解	19

2.5.1	Solving Systems of Equations: A Geometric View / 方程组求解：几何视角	20
2.5.2	Solutions of Equation Systems: Rank and Linear Independence / 方程组的解：秩与线性无关性	20
2.6	Special Matrices / 特殊矩阵	20
第三章	Linear Algebra: Applications / 线性代数：应用	23
3.1	Introduction / 引言	23
3.2	The Market Model / 市场模型	23
3.3	Adjacency Matrices / 邻接矩阵	24
3.4	Input–Output Analysis / 投入—产出分析	24
3.5	Inflation and Unemployment Dynamics / 通货膨胀与失业动态	24
3.6	Stationary (Invariant) Distributions / 平稳（不变）分布	25
3.6.1	Convergence to a Stationary Distribution / 收敛到平稳分布	25
3.6.2	Computation of the Invariant Distribution / 不变分布的计算	25
3.7	Econometrics / 计量经济学	26
3.7.1	Derivation of the Least Squares Estimator / 最小二乘估计量的推导	26
第四章	Linear Programming / 线性规划	29
4.1	Introduction / 引论	29
4.1.1	Formulation / 问题的表述	31
4.1.2	The feasible region / 可行区域	35
4.1.3	Finding an optimal solution / 寻找最优解	37
4.1.4	Dual prices and slack variables / 对偶价格与松弛变量	40
4.1.5	Some examples / 若干例子	44
4.1.6	Duality reconsidered / 对偶性再探讨	48
4.1.7	Basic solutions / 基本解	51
4.1.8	Duality principles / 对偶原理	58

4.2	Exercises for Chapter 4 / 第 4 章习题	60
第五章	Functions of One Variable / 一元函数	63
5.1	Introduction / 引论	63
5.2	Some examples / 若干例子	64
5.2.1	Demand functions / 需求函数	64
5.2.2	Present value and the interest rate / 现值与利率	65
5.2.3	Taxation / 税收	66
5.2.4	Options / 期权	67
5.3	Continuity and differentiability / 连续性与可微性	68
5.3.1	Preliminaries / 预备知识	68
5.3.2	Inequalities and absolute value / 不等式与绝对值	69
5.3.3	Continuity / 连续性	69
5.3.4	Differentiability / 可微性	70
5.4	Types of function / 函数的类型	71
5.4.1	The log and exponential functions / 对数函数与指数函数	72
5.5	Growth rates / 增长率	73
5.5.1	Additivity of continuous growth rates / 连续增长率的可加性	74
5.5.2	The derivatives of e^x and $\ln x$ / e^x 与 $\ln x$ 的导数	75
5.6	Some rules of calculus / 微积分的一些法则	75
5.6.1	l'Hôpital's rule / 洛必达法则	76
5.6.2	Higher derivatives / 高阶导数	77
5.7	Exercises for Chapter 5 / 第 5 章习题	77
第六章	Functions of One Variable: Applications / 一元函数的应用	79
6.1	Introduction / 引论	79
6.2	Optimization / 最优化	80

6.2.1	The first-order condition: issues / 一阶条件：若干问题	81
6.2.2	The second-order condition / 二阶条件	81
6.2.3	Applications: profit maximization and related problems / 应用：利润 最大化及相关问题	82
6.3	Concavity and optimization / 凹性与最优化	83
6.3.1	Summary / 小结	84
6.4	Externalities / 外部性	85
6.4.1	The free-rider problem / 搭便车问题	85
6.5	Comparative statics / 比较静态分析	86
6.6	Elasticity of demand / 需求弹性	87
6.6.1	Elasticity, revenue and profit maximization / 弹性、收益与利润最大化	87
6.7	Taxation and monopoly / 税收与垄断	88
6.8	Taxation incidence in supply and demand / 供需中的税收归宿	89
6.8.1	A unit tax / 单位税	89
6.8.2	An ad-valorem tax / 从价税	89
6.8.3	Elasticities and taxation / 弹性与税收	90
6.9	Ramsey pricing / 拉姆齐定价	91
6.10	Exercises for Chapter 6 / 第 6 章习题	92
6.11	Systems of Equations, Differentials and Derivatives / 方程组、微分与导数 .	92
6.11.1	Introduction / 引言	92
6.11.2	Partial Derivatives / 偏导数	98
6.11.3	Level Contours / 水平等高线	102
6.11.4	The Elasticity of Substitution / 替代弹性	105
6.12	Taylor Series / 泰勒级数	119
6.12.1	Introduction / 引言	119
6.12.2	Taylor Series Approximations: An Example / 泰勒级数近似：一个例子	121
6.12.3	Taylor Series Expansions / 泰勒级数展开	124

6.12.4	Approximations and Accuracy / 近似与精度	126
6.12.5	Applications of Taylor Series Expansions / 泰勒级数展开的应用 . . .	127
6.12.6	Expected Utility Theory and Behavior Towards Risk / 期望效用理论 与风险态度	132
6.12.7	Diversification and Portfolio Selection / 分散化与投资组合选择 . . .	136
6.12.8	The Multivariate Version of Taylor's Theorem / 泰勒定理的多元版本	139
6.12.9	Proof of Taylor's Theorem / 泰勒定理的证明	140
6.13	Vectors / 向量	144
6.13.1	Introduction / 引言	144
6.13.2	Vectors: Length and Distance / 向量: 长度与距离	145
6.13.3	Vectors: Direction and Angles / 向量: 方向与角度	147
6.13.4	Hyperplanes and Direction of Increase / 超平面与增加方向	148
6.13.5	Farkas' Lemma / 法卡斯引理	150
6.14	Quadratic Forms / 二次型	153
6.14.1	10.1 Introduction / 引言	153
6.14.2	10.2 Quadratic Forms / 二次型	154
6.14.3	10.3 Positive and Negative Definite Matrices / 正定和负定矩阵 . . .	160
6.14.4	10.4 Definiteness with Equality Constraints / 带等式约束的定性 . .	168
6.15	Multivariate Optimization / 多变量最优化	173
6.15.1	11.1 Introduction / 引言	173
6.15.2	11.2 The Two-Variable Case / 两变量情形	174
6.15.3	11.3 Optimization: n Variables / n 变量最优化	190
6.15.4	11.4 Concavity, Convexity and the Hessian Matrix / 凹性、凸性与 Hessian 矩阵	196
6.16	Equality-Constrained Optimization / 等式约束最优化	203
6.16.1	12.1 Introduction / 引言	203
6.16.2	12.2 The Two-Variable Case / 两变量情形	206

6.16.3	12.3 Motivation for First- and Second-Order Conditions / 一阶与二阶条件的动机	209
6.16.4	12.4 The n -Variable Case / n 变量情形	211
6.16.5	12.5 Interpretation of the Lagrange Multiplier / 拉格朗日乘数的解释	214
6.16.6	12.6 Optimization: Concavity and Convexity / 最优化: 凹性与凸性	215
6.16.7	12.7 Optimization with Many Constraints / 多约束最优化	218
第十三章 Inequality-Constrained Optimization / 不等式约束最优化		223
13.1	Introduction / 引言	223
13.1.1	A motivating example / 一个引入性例子	224
13.1.2	Overview / 概述	227
13.2	Optimization / 最优化	227
13.2.1	The constraint qualification / 约束资格条件	230
13.2.2	Complementary slackness / 互补松弛	231
13.2.3	Minimization and maximization / 最小化与最大化	232
13.2.4	Global and local optima / 全局最优与局部最优	233
13.2.5	Non-negativity constraints / 非负性约束	233
13.2.6	Necessary conditions and non-convexities / 必要条件与非凸性	235
13.2.7	The Lagrange multiplier / 拉格朗日乘子	235
13.2.8	Equality and inequality constraints / 等式与不等式约束	236
第十四章 Integration / 积分		239
14.1	Introduction / 引言	239
14.2	The integral / 积分	240
14.3	Common integrals and rules of integration / 常见积分与积分法则	241
14.4	Measuring welfare: surplus / 度量福利: 剩余	241
14.4.1	Consumer surplus / 消费者剩余	241
14.4.2	Producer surplus / 生产者剩余	242

14.4.3	Consumer surplus and welfare / 消费者剩余与福利	243
14.5	Cost, supply and profit / 成本、供给与利润	243
14.6	Taxation and consumer surplus / 税收与消费者剩余	244
14.7	Present value / 现值	244
14.8	Leibnitz's rule / 莱布尼茨法则	245
14.9	Inequality measures / 不平等度量	246
14.10	Ramsey pricing / 拉姆齐定价	247
14.11	Welfare measures / 福利度量	247
14.11.1	Consumer surplus / 消费者剩余	247
14.11.2	Compensating variation / 补偿变化	248
14.11.3	Equivalent variation / 等价变化	249
14.11.4	Comparison of welfare measures / 福利度量的比较	249
第十五章	Eigenvalues and Eigenvectors / 特征值与特征向量	253
15.1	Introduction / 引言	253
15.2	Definitions and basic properties / 定义与基本性质	254
15.3	An application: dynamic models / 应用: 动态模型	256
15.3.1	Supply and demand dynamics / 供需动态	257
15.3.2	A model of population growth / 人口增长模型	257
15.4	The Perron–Frobenius theorem / Perron–Frobenius 定理	258
15.5	Some applications / 一些应用	259
15.5.1	Webpage ranking / 网页排名	259
15.5.2	Leslie matrices / Leslie 矩阵	260
15.6	Eigenvalues and eigenvectors of symmetric matrices / 对称矩阵的特征值与特征向量	261
15.7	Real and complex eigenvalues and eigenvectors / 实与复特征值和特征向量	262
15.8	Diagonalization / 对角化	262

15.9 Properties of eigenvalues and eigenvectors / 特征值与特征向量的性质	263
15.9.1 Largest eigenvalues and eigenvalue multiplicity / 最大特征值与特征 值重数	264
第一部分 Dynamic Models, Probability & Distributions / 动态 模型、概率与分布	267
第十六章 Differential Equations / 微分方程	269
16.1 Introduction / 引言	269
16.1.1 Preliminary discussion / 初步讨论	270
16.2 First-order linear differential equations / 一阶线性微分方程	272
16.2.1 Constant coefficients / 常系数	273
16.2.2 Variable coefficients / 变系数	277
16.3 Some non-linear first-order differential equations / 一些非线性一阶微分方程	282
16.3.1 The logistic model / 逻辑斯蒂模型	283
16.3.2 The Bernoulli equation / 伯努利方程	284
16.4 First-order differential equations: existence of solutions / 一阶微分方程: 解 的存在性	287
16.5 Second- and higher-order differential equations / 二阶及高阶微分方程 . . .	288
16.5.1 Constant coefficients / 常系数	289
16.5.2 Derivation of solution / 解的推导	290
16.6 Systems of differential equations / 微分方程组	292
16.7 Stability / 稳定性	294
第十七章 Linear Difference Equations / 线性差分方程	299
17.1 Introduction / 引言	299
17.2 Motivating examples / 启发性例子	300
17.3 First-order linear difference equations / 一阶线性差分方程	302

17.3.1	Solving first-order linear difference equations / 求解一阶线性差分方程	303
17.3.2	The constant forcing function / 常值强制函数	303
17.3.3	The cobweb model / 蛛网模型	305
17.3.4	Equations with variable forcing function / 变系数强制函数方程	306
17.4	Second-order linear difference equations / 二阶线性差分方程	307
17.4.1	The inhomogeneous equation / 非齐次方程	307
17.4.2	The homogeneous equation / 齐次方程	308
17.4.3	The general solution and stability / 通解与稳定性	309
17.4.4	The Samuelson multiplier-accelerator model / 萨缪尔森乘数-加速数模型	309
17.5	Vector difference equations / 向量差分方程	310
17.5.1	The particular solution / 特解	310
17.5.2	The complementary solution / 互补解	311
17.6	The n -variable case / n 变量情形	311
17.7	Miscellaneous calculations / 补充计算	312
第十八章	Probability and Distributions / 概率与分布	315
18.1	Introduction / 引言	315
18.2	Random variables / 随机变量	316
18.3	The distribution function and density / 分布函数与密度	317
18.3.1	The conditional distribution / 条件分布	319
18.3.2	Joint distributions / 联合分布	319
18.4	Discrete distributions / 离散分布	321
18.4.1	The Bernoulli distribution / 伯努利分布	321
18.4.2	The binomial distribution / 二项分布	321
18.4.3	The Poisson distribution / 泊松分布	322
18.4.4	The Poisson process / 泊松过程	322

18.5 Continuous distributions / 连续分布	323
18.5.1 The uniform distribution / 均匀分布	323
18.5.2 The exponential distribution / 指数分布	324
18.5.3 The normal distribution / 正态分布	324
18.5.4 Moment-generating functions / 矩母函数	325
18.5.5 The chi-square distribution / 卡方分布	326
18.5.6 The t distribution / t 分布	327
18.5.7 The F distribution / F 分布	327

第一章 Introduction / 绪论

1.1 Introduction / 绪论

Original (English)

This introductory chapter provides an overview of the mathematical tools that will be developed in the text. The focus is on those areas of mathematics most relevant to economic analysis: linear algebra, multivariate calculus, optimization, and dynamics.

翻译 (Chinese)

本章为绪论，概述本书将要展开的数学工具。重点放在与经济分析最为相关的数学领域：线性代数、多元微积分、最优化以及动态分析。

1.2 A Brief Summary by Chapter / 各章内容简介

Original (English)

A brief summary by chapter is provided to guide the reader through the structure of the book. Each chapter builds on material developed earlier, so the ordering is important for those reading the book sequentially.

翻译 (Chinese)

本节提供各章内容简介，帮助读者把握全书的整体结构。各章内容以前面章节为基础逐步展开，因此对于按顺序阅读的读者而言，章节的编排顺序很重要。

第二章 Matrices and Systems of Equations / 矩阵与方程组

2.1 Introduction / 引言

Original (English)

This chapter introduces matrices and their use in representing and solving systems of linear equations. Linear equations and matrix algebra are fundamental tools in economic modelling, appearing in everything from simple supply-and-demand models to sophisticated econometric estimation.

翻译 (Chinese)

本章介绍矩阵及其在表示与求解线性方程组中的应用。线性方程与矩阵代数是经济建模的基本工具，从简单的供需模型到复杂的计量经济学估计，到处都有它们的身影。

2.1.1 Some Motivating Examples / 若干引例

Original (English)

We begin with several simple economic examples that illustrate why matrices and systems of linear equations arise naturally. These examples include a simple market equilibrium problem, a national-income model, and an input-output production model.

翻译 (Chinese)

我们从几个简单的经济学例子入手，说明矩阵与线性方程组何以自然地出现。这些例子包括一个简单的市场均衡问题、一个国民收入模型以及一个投入—产出生产模型。

2.2 Vectors / 向量

Original (English)

Vectors are ordered lists of numbers. Geometrically, a vector can be viewed as a point or a directed line segment in n -dimensional space. Algebraically, vectors are the building blocks of matrix operations.

翻译 (Chinese)

向量是有序数组。从几何上看，向量可视为 n 维空间中的一个点或一条有向线段。从代数上看，向量是矩阵运算的基本构件。

2.3 Matrices / 矩阵

Original (English)

A matrix is a rectangular array of numbers. The size or order of a matrix is given by the number of rows and columns it contains. An $m \times n$ matrix has m rows and n columns.

翻译 (Chinese)

矩阵是一个矩形的数表。矩阵的大小或阶由其行数和列数决定。一个 $m \times n$ 矩阵有 m 行和 n 列。

2.3.1 Matrix Addition and Subtraction / 矩阵加法与减法

Original (English)

Two matrices of the same dimension can be added (or subtracted) by adding (or subtracting) corresponding entries. If A and B are both $m \times n$, then $C = A + B$ is also $m \times n$, with $c_{ij} = a_{ij} + b_{ij}$.

翻译 (Chinese)

两个维数相同的矩阵可以通过对应元素相加（或相减）来进行加法（或减法）运算。若 A 与 B 均为 $m \times n$ 矩阵，则 $C = A + B$ 也是 $m \times n$ 矩阵，且 $c_{ij} = a_{ij} + b_{ij}$ 。

2.3.2 Matrix Multiplication / 矩阵乘法

Original (English)

The product $C = AB$ is defined when the number of columns of A equals the number of rows of B . If A is $m \times r$ and B is $r \times n$, then C is $m \times n$ with entries $c_{ij} = \sum_{k=1}^r a_{ik}b_{kj}$.

翻译 (Chinese)

当 A 的列数等于 B 的行数时, 乘积 $C = AB$ 有定义。若 A 为 $m \times r$ 矩阵, B 为 $r \times n$ 矩阵, 则 C 为 $m \times n$ 矩阵, 其元素为 $c_{ij} = \sum_{k=1}^r a_{ik}b_{kj}$ 。

2.3.3 Matrix Transpose / 矩阵转置

Original (English)

The transpose of an $m \times n$ matrix A , denoted A' or A^T , is the $n \times m$ matrix obtained by interchanging rows and columns: $(A')_{ij} = A_{ji}$.

翻译 (Chinese)

$m \times n$ 矩阵 A 的转置, 记作 A' 或 A^T , 是一个 $n \times m$ 矩阵, 通过互换行与列得到: $(A')_{ij} = A_{ji}$ 。

2.3.4 Matrix Inversion / 矩阵求逆

Original (English)

A square matrix A is invertible (or nonsingular) if there exists a matrix A^{-1} such that $AA^{-1} = A^{-1}A = I$, where I is the identity matrix. Not every square matrix has an inverse; those that do not are called singular.

翻译 (Chinese)

若存在矩阵 A^{-1} 使得 $AA^{-1} = A^{-1}A = I$ (其中 I 为单位矩阵), 则方阵 A 可逆 (或称非奇异)。并非每个方阵都有逆矩阵; 没有逆矩阵的方阵称为奇异矩阵。

2.3.5 Additional Remarks on Determinants / 关于行列式的补充说明

Original (English)

The determinant of a square matrix, denoted $\det(A)$ or $|A|$, is a scalar that provides information about the matrix. A square matrix is invertible if and only if its determinant is nonzero. The determinant also has a geometric interpretation as the volume scaling factor of the linear transformation represented by the matrix.

翻译 (Chinese)

方阵的行列式，记作 $\det(A)$ 或 $|A|$ ，是一个能够提供矩阵信息的标量。方阵可逆当且仅当其行列式非零。行列式还具有几何解释，即矩阵所表示的线性变换的体积缩放因子。

2.3.6 Cramer's Rule / 克莱姆法则

Original (English)

Cramer's rule gives an explicit formula for the solution of a system of n linear equations in n unknowns, provided the coefficient matrix has nonzero determinant. The i -th variable is $x_i = |A_i|/|A|$, where A_i is the matrix obtained by replacing the i -th column of A with the right-hand-side vector b .

翻译 (Chinese)

克莱姆法则给出了 n 个方程 n 个未知数的线性方程组解的显式公式，前提是系数矩阵的行列式非零。第 i 个变量为 $x_i = |A_i|/|A|$ ，其中 A_i 是将 A 的第 i 列替换为右端向量 b 后得到的矩阵。

2.3.7 Gaussian Elimination / 高斯消元法

Original (English)

Gaussian elimination is a systematic procedure for solving systems of linear equations by reducing the augmented matrix to row-echelon form (or reduced row-echelon form) through a sequence of elementary row operations.

翻译 (Chinese)

高斯消元法是一种求解线性方程组的系统化方法，通过一系列初等行变换将增广矩阵化为行阶梯形（或简化行阶梯形）。

2.4 Linear Dependence and Independence / 线性相关与线性无关

Original (English)

A set of vectors $\{v_1, v_2, \dots, v_k\}$ is linearly dependent if there exist scalars $c_1, c_2, \dots, c_k$, not all zero, such that $c_1v_1 + c_2v_2 + \dots + c_kv_k = 0$. Otherwise the vectors are linearly independent.

翻译 (Chinese)

向量组 $\{v_1, v_2, \dots, v_k\}$ 称为线性相关的，若存在不全为零的标量 $c_1, c_2, \dots, c_k$ 使得 $c_1v_1 + c_2v_2 + \dots + c_kv_k = 0$ 。否则，这些向量是线性无关的。

2.4.1 Matrix Rank and Linear Independence / 矩阵的秩与线性无关性

Original (English)

The rank of a matrix is the maximum number of linearly independent rows (or columns) in the matrix. Row rank and column rank are always equal. The rank provides information about the dimension of the solution space of the associated homogeneous system.

翻译 (Chinese)

矩阵的秩是矩阵中线性无关行（或列）的最大数目。行秩与列秩始终相等。秩提供了关于相应齐次方程组解空间维数的信息。

2.5 Solutions of Systems of Equations / 方程组的解

Original (English)

A system of linear equations may have no solution, exactly one solution, or infinitely many solutions. The existence and uniqueness of solutions depend on the relationship between the rank of the coefficient matrix and the rank of the augmented matrix.

翻译 (Chinese)

一个线性方程组可能无解，恰有一解，或有无穷多解。解的存在性与唯一性取决于系数矩阵的秩与增广矩阵的秩之间的关系。

2.5.1 Solving Systems of Equations: A Geometric View / 方程组求解：几何视角

Original (English)

Geometrically, a system of linear equations can be viewed as the intersection of hyperplanes. Each equation defines a hyperplane, and the solution set is the intersection of all these hyperplanes. This geometric perspective helps in understanding the possible solution configurations.

翻译 (Chinese)

从几何上看，线性方程组可视为超平面的交。每个方程定义一个超平面，解集就是所有这些超平面的交集。这一几何视角有助于理解可能的解的各种情形。

2.5.2 Solutions of Equation Systems: Rank and Linear Independence / 方程组的解：秩与线性无关性

Original (English)

The solution structure of $Ax = b$ is determined by the rank of A and the rank of the augmented matrix $[A \mid b]$. When these ranks are equal, solutions exist; the difference between the number of unknowns and the rank gives the dimension of the solution space for the homogeneous system.

翻译 (Chinese)

$Ax = b$ 的解的结构由 A 的秩和增广矩阵 $[A \mid b]$ 的秩决定。当这两个秩相等时，解存在；未知数个数与秩的差给出了齐次方程组解空间的维数。

2.6 Special Matrices / 特殊矩阵

Original (English)

Several classes of matrices have special properties that are useful in economic applications: symmetric matrices ($A' = A$), idempotent matrices ($A^2 = A$), orthogonal matrices ($A'A = I$), diagonal matrices, triangular matrices, and positive definite matrices.

翻译 (Chinese)

有几类矩阵因其特殊性质在经济应用中非常有用：对称矩阵 ($A' = A$)、幂等矩阵 ($A^2 = A$)、正交矩阵 ($A'A = I$)、对角矩阵、三角矩阵以及正定矩阵。

Exercises for Chapter 2 / 第 2 章习题

Original (English)

Exercises for Chapter 2 cover matrix operations, determinants, solving systems of equations using Gaussian elimination and Cramer's rule, linear dependence, rank, and special matrices.

翻译 (Chinese)

第 2 章习题涵盖矩阵运算、行列式、运用高斯消元法和克莱姆法则求解方程组、线性相关性、秩以及特殊矩阵等内容。

第三章 Linear Algebra: Applications / 线性代数：应用

3.1 Introduction / 引言

Original (English)

This chapter applies the linear-algebra techniques developed in Chapter 2 to several economic problems. The applications range from market models and production analysis to dynamic models of inflation and econometric estimation.

翻译 (Chinese)

本章将第 2 章展开的线性代数工具应用于若干经济问题。应用范围涵盖市场模型和生产分析，以及通货膨胀动态模型与计量经济学估计。

3.2 The Market Model / 市场模型

Original (English)

A simple linear market model expresses supply and demand as linear functions of price. The equilibrium price vector solves a system of linear equations. Matrix algebra provides a compact and powerful framework for analysing multi-market equilibrium.

翻译 (Chinese)

简单的线性市场模型将供给和需求表示为价格的线性函数。均衡价格向量是一组线性方程组的解。矩阵代数为分析多市场均衡提供了一个紧凑而强大的框架。

3.3 Adjacency Matrices / 邻接矩阵

Original (English)

An adjacency matrix represents the connections in a network or graph. In economics, adjacency matrices appear in the analysis of social networks, trade linkages, and industrial organisation. Powers of the adjacency matrix give information about indirect connections.

翻译 (Chinese)

邻接矩阵表示网络或图中的连接关系。在经济学中，邻接矩阵出现在社会网络、贸易关联和产业组织分析中。邻接矩阵的幂提供了关于间接连接的信息。

3.4 Input–Output Analysis / 投入—产出分析

Original (English)

Input–output analysis, developed by Wassily Leontief, describes the interdependence among the different sectors of an economy. The input–output matrix records the flows of goods between sectors. The Leontief inverse $(I - A)^{-1}$ is central to computing the total (direct and indirect) requirements needed to meet a given final demand.

翻译 (Chinese)

投入—产出分析由 Wassily Leontief 提出，描述了经济中不同部门之间的相互依赖关系。投入—产出矩阵记录了部门间商品的流动。Leontief 逆矩阵 $(I - A)^{-1}$ 对于计算满足给定最终需求所需的总（直接和间接）需求是核心的。

3.5 Inflation and Unemployment Dynamics / 通货膨胀与失业动态

Original (English)

Dynamic models of inflation and unemployment can be written as systems of linear difference equations. The long-run behaviour of the system depends on the eigenvalues of the coefficient matrix. This section shows how matrix methods illuminate the dynamics of macroeconomic variables.

翻译 (Chinese)

通货膨胀与失业的动态模型可以写成线性差分方程组的形式。系统的长期行为取决于系数矩阵的特征值。本节展示矩阵方法如何揭示宏观经济变量的动态特征。

3.6 Stationary (Invariant) Distributions / 平稳（不变）分布

Original (English)

A stationary (or invariant) distribution of a Markov chain is a probability distribution π satisfying $\pi P = \pi$, where P is the transition matrix. Such distributions describe the long-run behaviour of the chain and are important in many economic applications, including income mobility and firm-size dynamics.

翻译 (Chinese)

马尔可夫链的平稳（或不变）分布是一个概率分布 π ，满足 $\pi P = \pi$ ，其中 P 为转移矩阵。这类分布描述了链的长期行为，在许多经济应用中都很重要，包括收入流动性和企业规模动态。

3.6.1 Convergence to a Stationary Distribution / 收敛到平稳分布

Original (English)

Under appropriate conditions (irreducibility and aperiodicity), a Markov chain converges to its unique stationary distribution regardless of the initial distribution. The speed of convergence is governed by the second-largest eigenvalue of the transition matrix.

翻译 (Chinese)

在适当的条件（不可约性和非周期性）下，马尔可夫链无论从何种初始分布出发，都会收敛到其唯一的平稳分布。收敛速度由转移矩阵的第二大特征值决定。

3.6.2 Computation of the Invariant Distribution / 不变分布的计算

Original (English)

The invariant distribution can be computed by solving the linear system $\pi(I - P) = 0$ subject to the normalisation constraint $\sum_i \pi_i = 1$. Standard matrix methods, including Gaussian elimination, can be used to find π .

翻译 (Chinese)

不变分布可以通过求解线性方程组 $\pi(I - P) = 0$ 并附加归一化约束 $\sum_i \pi_i = 1$ 来计算。标准矩阵方法，包括高斯消元法，可用于求出 π 。

3.7 Econometrics / 计量经济学

Original (English)

Linear algebra is the mathematical foundation of econometrics. The linear regression model $y = X\beta + \varepsilon$ is expressed in matrix notation, and the least-squares estimator is derived using matrix differentiation and projection.

翻译 (Chinese)

线性代数是计量经济学的数学基础。线性回归模型 $y = X\beta + \varepsilon$ 用矩阵记号表示，最小二乘估计量则通过矩阵微分与投影推导得出。

3.7.1 Derivation of the Least Squares Estimator / 最小二乘估计量的推导

Original (English)

The ordinary least squares (OLS) estimator minimises the sum of squared residuals $e'e = (y - X\beta)'(y - X\beta)$. The first-order condition yields the normal equations $X'X\hat{\beta} = X'y$, and, provided $X'X$ is invertible, the OLS estimator is $\hat{\beta} = (X'X)^{-1}X'y$.

翻译 (Chinese)

普通最小二乘 (OLS) 估计量极小化残差平方和 $e'e = (y - X\beta)'(y - X\beta)$ 。一阶条件给出正规方程 $X'X\hat{\beta} = X'y$ ，且只要 $X'X$ 可逆，OLS 估计量即为 $\hat{\beta} = (X'X)^{-1}X'y$ 。

Exercises for Chapter 3 / 第 3 章习题

Original (English)

Exercises for Chapter 3 cover the market model, adjacency matrices, input-output analysis, inflation and unemployment dynamics, stationary distributions, and the least-squares estimator.

翻译 (Chinese)

第 3 章习题涵盖市场模型、邻接矩阵、投入—产出分析、通货膨胀与失业动态、平稳分布以及最小二乘估计量等内容。

第四章 Linear Programming / 线性规划

4.1 Introduction / 引论

Original (English)

Linear programming is an optimization technique for finding optimal solutions to problems where the objective and constraints are linear. In what follows, the general linear programming problem and methods of solution are described, along with various features of this class of problem. A linear program is characterized by a linear objective, linear constraints and restrictions on the signs of choice variables. Linearity of the constraint set leads to a particular geometric structure on the set possible choice variables. For low-dimensional programs this permits a graphical treatment of the problem and, more generally, the geometric structure leads naturally to an algorithmic method for finding solutions. Section 4.1 develops the basic features of a linear program. This consists of formulation of the programming problem, description of key issues in solving these programs and the valuation of resource constraints through the dual prices. A broader understanding of linear programming requires knowledge of basic solutions, which are discussed in Section 4.2. Here the geometric view described in Section 4.1 is reinterpreted in terms of equation systems. Section 4.3 describes classic results in linear programming where the original program (the primal) is related to a second program (the dual). Before developing the discussion, a simple example will illustrate how these programs arise.

翻译 (Chinese)

线性规划是一种最优化技术，用于求解目标函数和约束条件均为线性的问题的最优解。下面将介绍一般线性规划问题及其求解方法，以及此类问题的各种特征。线性规划的特点是：线性目标函数、线性约束条件以及对选择变量符号的限制。约束集的线性性质导致可行选择变量集具有特定的几何结构。对于低维规划，这允许对问题进行图形化处理；更一般地，

这种几何结构自然地导出了一种算法求解方法。第 4.1 节阐述线性规划的基本特征，包括规划问题的表述、求解这些规划的关键问题以及通过对偶价格对资源约束进行估值。更深入地理解线性规划需要了解基本解的概念，这些将在第 4.2 节讨论。在此，第 4.1 节描述的几何观点将用方程组的语言重新解释。第 4.3 节描述线性规划的经典结果，其中原始规划（原始问题）与第二个规划（对偶问题）相关联。在展开讨论之前，先通过一个简单的例子来说明这些规划是如何产生的。

EXAMPLE 4.1:

Original (English)

A firm produces two goods, 1 and 2. Production requires the application of two techniques: processing and finishing. A unit of good 1 requires two hours of processing time and four hours of finishing time; a unit of good 2 requires three hours of processing and three hours of finishing time. The total amount of time that the processing machine can run for each day is 13 hours, while the finishing machine can run for 17 hours. Each unit of good 1 sells for \$3 and each unit of good 2 sells for \$4. The problem is to decide how much of each good should be produced to maximize revenue — a problem that may be formulated as a linear program.

Let x denote the number of units of good 1 produced, and let y denote the number of units of good 2 produced. Then the revenue is equal to $3x + 4y$, since each unit of 1 generates \$3 of revenue and each unit of 2 generates \$4 of revenue. This defines the linear objective. Next, x units of good 1 require two hours of processing time, while y units of good 2 require three hours of processing time: altogether, producing x units of 1 and y units of 2 requires $2x + 3y$ hours of processing time. Similarly, producing x units of 1 and y units of 2 requires $4x + 3y$ hours of finishing time. The total amount of processing time is 13 hours, so the production plan must satisfy $2x + 3y \leq 13$, the first linear constraint, and likewise $4x + 3y \leq 17$ defines the finishing constraint. Finally, negative amounts cannot be produced: $x, y \geq 0$. Putting all these conditions together, the full problem may be summarized:

翻译 (Chinese)

一家企业生产两种商品，商品 1 和商品 2。生产需要两种工艺：加工和精加工。一单位商品 1 需要两小时加工时间和四小时精加工时间；一单位商品 2 需要三小时加工时间和三小时精加工时间。加工机器每天可运行的总时间为 13 小时，而精加工机器可运行 17 小时。每单位商品 1 售价为 \$3，每单位商品 2 售价为 \$4。问题是要决定每种商品各应生产多少以最大化收入——该问题可以表述为一个线性规划。

设 x 表示商品 1 的产量， y 表示商品 2 的产量。则收入等于 $3x + 4y$ ，因为每单位商品 1

产生 \$3 收入，每单位商品 2 产生 \$4 收入。这定义了线性目标。接下来， x 单位商品 1 需要两小时加工时间，而 y 单位商品 2 需要三小时加工时间：总共，生产 x 单位商品 1 和 y 单位商品 2 需要 $2x + 3y$ 小时加工时间。类似地，生产 x 单位商品 1 和 y 单位商品 2 需要 $4x + 3y$ 小时精加工时间。加工总时间为 13 小时，因此生产计划必须满足 $2x + 3y \leq 13$ ，即第一个线性约束；同样， $4x + 3y \leq 17$ 定义了精加工约束。最后，不能生产负数量： $x, y \geq 0$ 。将所有这些条件放在一起，完整的问题可以概括如下：

$$\begin{aligned} \max \quad & 3x + 4y \\ \text{subject to:} \quad & 2x + 3y \leq 13 \quad (\text{processing}) \\ & 4x + 3y \leq 17 \quad (\text{finishing}) \\ & x, y \geq 0. \end{aligned}$$

4.1.1 Formulation / 问题的表述

Original (English)

In a linear-programming problem a collection of variables $(x_1, x_2, \dots, x_n)$ must be chosen to maximize (or minimize) a linear function,

$$f(x_1, x_2, \dots, x_n) = c_1x_1 + c_2x_2 + \dots + c_nx_n,$$

subject to a finite number of linear constraints of the form:

$$a_1x_1 + a_2x_2 + \dots + a_nx_n \leq b, \quad a_1x_1 + a_2x_2 + \dots + a_nx_n = b, \quad \text{or} \quad a_1x_1 + a_2x_2 + \dots + a_nx_n \geq b.$$

In addition, the x_i variables are required to be chosen non-negative. A typical linear program may be written:

翻译 (Chinese)

在线性规划问题中，需要选择一组变量 $(x_1, x_2, \dots, x_n)$ 以最大化（或最小化）一个线性函数，

$$f(x_1, x_2, \dots, x_n) = c_1x_1 + c_2x_2 + \dots + c_nx_n,$$

并满足有限个如下形式的线性约束：

$$a_1x_1 + a_2x_2 + \dots + a_nx_n \leq b, \quad a_1x_1 + a_2x_2 + \dots + a_nx_n = b, \quad \text{或} \quad a_1x_1 + a_2x_2 + \dots + a_nx_n \geq b.$$

此外，变量 x_i 要求为非负。一个典型的线性规划可以写为：

$$\begin{aligned}
 \max \quad & c_1x_1 + c_2x_2 + \cdots + c_nx_n \\
 \text{subject to:} \quad & a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n \leq b_1 \\
 & a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n \leq b_2 \\
 & \quad \quad \quad \cdots \quad \quad \quad \cdots \quad \quad \quad \cdots \\
 & a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n \leq b_m \\
 & x_1, x_2, \dots, x_n \geq 0.
 \end{aligned}$$

Original (English)

This is a linear program with n variables and m constraints. If we write $c = (c_1, c_2, \dots, c_n)$,

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}, \quad b = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix}, \quad x = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix},$$

then the program may be written in matrix form:

$$\max \quad c'x \quad \text{subject to:} \quad Ax \leq b, \quad x \geq 0. \quad (4.1)$$

翻译 (Chinese)

这是一个具有 n 个变量和 m 个约束的线性规划。若记 $c = (c_1, c_2, \dots, c_n)$,

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}, \quad b = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix}, \quad x = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix},$$

则该规划可以用矩阵形式表示为:

$$\max \quad c'x \quad \text{满足:} \quad Ax \leq b, \quad x \geq 0. \quad (4.1)$$

Original (English)

In parallel with the maximization problem, there is minimization formulation:

$$\min \quad c'x \quad \text{subject to:} \quad Ax \geq b, \quad x \geq 0. \quad (4.2)$$

翻译 (Chinese)

与最大化问题相对应, 还有最小化的表述:

$$\min \quad c'x \quad \text{满足:} \quad Ax \geq b, \quad x \geq 0. \quad (4.2)$$

Mixed constraints and unrestricted variable signs / 混合约束与无限制变量符号

Original (English)

The formulations for maximization and minimization given in Equations 4.1 and 4.2 have a standardized form with non-negativity constraints. In general, maximization and minimization problems can be cast in this way. The following discussion illustrates. Consider the program:

$$\begin{aligned} \max \quad & c_1x_1 + c_2x_2 \\ \text{subject to:} \quad & a_{11}x_1 + a_{12}x_2 \leq b_1 \\ & a_{21}x_1 + a_{22}x_2 \geq b_2 \\ & x_1 \geq 0. \end{aligned}$$

In this program, both types of inequality appear, and only x_1 is required to be non-negative. Replacing x_2 with $y - z$ and requiring y and z to be non-negative overcomes the second issue — picking $y = 0$ and z any desired positive number gives x_2 equal to the negative of that number.

$$\begin{aligned} \max \quad & c_1x_1 + c_2y - c_2z \\ \text{subject to:} \quad & a_{11}x_1 + a_{12}y - a_{12}z \leq b_1 \\ & a_{21}x_1 + a_{22}y - a_{22}z \geq b_2 \\ & x_1, y, z \geq 0. \end{aligned}$$

The first issue is solved by multiplying the relevant constraint by -1 to reverse the inequality: for example $5 \geq 4$ is equivalent to $-5 \leq -4$ and $ax \geq b$ is equivalent to $-ax \leq -b$. In this example, multiplying $a_{21}x_1 + a_{22}y - a_{22}z \geq b_2$ by -1 gives:

$$\begin{aligned} \max \quad & c_1x_1 + c_2y - c_2z \\ \text{subject to:} \quad & a_{11}x_1 + a_{12}y - a_{12}z \leq b_1 \\ & -a_{21}x_1 - a_{22}y + a_{22}z \leq b_2 \\ & x_1, y, z \geq 0. \end{aligned}$$

Similarly, given an equality constraint $a_ix_i = b_i$, this is equivalent to two constraints, $a_ix_i \leq b_i$ and $a_ix_i \geq b_i$. Note that the convention is to formulate the minimization program with the inequalities having the “ $\geq$ ” form: in matrix terms, $Ax \geq b$. In the linear program, constraints of the form $Ax \leq b$ or $Ax \geq b$ are called feasibility constraints; the constraints that $x_i \geq 0$ for all i are called non-negativity constraints. Taking both types of constraint together, refer to them as the program constraints or constraints.

翻译 (Chinese)

方程 4.1 和 4.2 给出的最大化和最小化表述具有带非负约束的标准化形式。一般来说，最大化和最小化问题都可以转化为这种形式。下面的讨论将加以说明。考虑如下规划：

$$\begin{aligned} \max \quad & c_1x_1 + c_2x_2 \\ \text{满足:} \quad & a_{11}x_1 + a_{12}x_2 \leq b_1 \\ & a_{21}x_1 + a_{22}x_2 \geq b_2 \\ & x_1 \geq 0. \end{aligned}$$

在这个规划中，两种类型的不等式都出现了，且只有 x_1 要求非负。用 $y - z$ 替换 x_2 并要求 y 和 z 非负，可以解决第二个问题——取 $y = 0$ 和 z 为任意所需正数，即可使 x_2 等于该数的相反数。

$$\begin{aligned} \max \quad & c_1x_1 + c_2y - c_2z \\ \text{满足:} \quad & a_{11}x_1 + a_{12}y - a_{12}z \leq b_1 \\ & a_{21}x_1 + a_{22}y - a_{22}z \geq b_2 \\ & x_1, y, z \geq 0. \end{aligned}$$

第一个问题通过将相关约束乘以 -1 以反转不等式来解决：例如 $5 \geq 4$ 等价于 $-5 \leq -4$ ， $ax \geq b$ 等价于 $-ax \leq -b$ 。在此例中，将 $a_{21}x_1 + a_{22}y - a_{22}z \geq b_2$ 乘以 -1 得：

$$\begin{aligned} \max \quad & c_1x_1 + c_2y - c_2z \\ \text{满足:} \quad & a_{11}x_1 + a_{12}y - a_{12}z \leq b_1 \\ & -a_{21}x_1 - a_{22}y + a_{22}z \leq b_2 \\ & x_1, y, z \geq 0. \end{aligned}$$

类似地，给定等式约束 $a_ix_i = b_i$ ，这等价于两个约束： $a_ix_i \leq b_i$ 和 $a_ix_i \geq b_i$ 。注意，惯例是将最小化规划表述为具有“ $\geq$ ”形式的不等式：矩阵形式为 $Ax \geq b$ 。在线性规划中，形如 $Ax \leq b$ 或 $Ax \geq b$ 的约束称为可行性约束；对所有 i 要求 $x_i \geq 0$ 的约束称为非负约束。将两类约束合在一起，统称为规划约束或约束条件。

Original (English)

The next sections consider a number of important concepts in linear programming. Section 4.1.2 discusses the feasible region — those choices that satisfy all the constraints of the problem. Following this, Section 4.1.3 considers the problem of finding an optimal solution. In Section 4.1.4, dual prices and slack variables are introduced. Dual prices attach a price or valuation to each constraint, while slack variables indicate whether constraints are binding or not. Finally, the dual program associated with a linear program (called “the primal”) is introduced (Section 4.1.6). The dual

program is derived from the primal program, is closely related and provides additional perspective on the primal program.

翻译 (Chinese)

接下来的几节将考虑线性规划中的若干重要概念。第 4.1.2 节讨论可行区域——即满足问题所有约束的那些选择。随后，第 4.1.3 节考虑寻找最优解的问题。在第 4.1.4 节中，引入对偶价格和松弛变量。对偶价格为每个约束赋予一个价格或估值，而松弛变量则指示约束是否紧。最后，引入与线性规划（称为“原始问题”）相关联的对偶规划（第 4.1.6 节）。对偶规划由原始规划推导而来，与之密切相关，并提供了对原始规划的额外视角。

4.1.2 The feasible region / 可行区域

Original (English)

A vector $x = (x_1, \dots, x_n)$ is called feasible if it satisfies the linear constraint inequalities and the non-negativity constraints.

Definition 4.1: The set of x vectors that satisfy the constraints $Ax \leq b$ and $x \geq 0$ is called the feasible set or feasible region.

Geometrically, it has a special shape — it is a convex polyhedron. The following examples give the feasible regions defined by inequality constraints (on the left side), and the corresponding graphs show the feasible regions as the shaded areas.

翻译 (Chinese)

如果一个向量 $x = (x_1, \dots, x_n)$ 满足线性约束不等式和非负约束，则称其为可行的。

定义 4.1: 满足约束 $Ax \leq b$ 和 $x \geq 0$ 的 x 向量的集合称为可行集或可行区域。

从几何上看，它具有特殊的形状——它是一个凸多面体。以下例子给出了由不等式约束定义的可行区域（左侧），相应图形以阴影区域表示可行区域。

EXAMPLE 4.2:

Original (English)

In this example, the feasible region is defined by two inequality constraints and the non-negativity

constraints.

$$2x + y \leq 6$$

$$3x + 4y \leq 12$$

$$x, y \geq 0$$

翻译 (Chinese)

在本例中，可行区域由两个不等式约束和非负约束定义。

$$2x + y \leq 6$$

$$3x + 4y \leq 12$$

$$x, y \geq 0$$

EXAMPLE 4.3:

Original (English)

The next example has three constraints. The additional constraint adds corners to the feasible region, and indicates the impact of additional constraints.

$$2x + 2y \leq 10$$

$$3x + 6y \leq 18$$

$$2x + 12y \leq 24$$

$$x, y \geq 0$$

翻译 (Chinese)

下一个例子有三个约束。额外的约束为可行区域增加了角点，并表明了额外约束的影响。

$$2x + 2y \leq 10$$

$$3x + 6y \leq 18$$

$$2x + 12y \leq 24$$

$$x, y \geq 0$$

EXAMPLE 4.4:

Original (English)

The feasible region is defined by linear equations, but this region may be quite complex. The

next example illustrates increasing complexity. In this example, the second constraint is written as $2x + 6y \geq 12$, which is equivalent to $-2x - 6y \leq 12$.

$$3x + 4y \leq 12$$

$$2x + 6y \geq 12$$

$$2x + 8y \leq 20$$

$$x, y \geq 0$$

翻译 (Chinese)

可行区域由线性方程定义，但该区域可能相当复杂。下一个例子说明了日益增加的复杂性。在此例中，第二个约束写为 $2x + 6y \geq 12$ ，等价于 $-2x - 6y \leq 12$ 。

$$3x + 4y \leq 12$$

$$2x + 6y \geq 12$$

$$2x + 8y \leq 20$$

$$x, y \geq 0$$

4.1.3 Finding an optimal solution / 寻找最优解

Original (English)

Once the feasible region is identified, one can begin the search for an optimal solution: a point in the feasible region such that there is no other point in the feasible region that improves the objective value.

Definition 4.2: Consider the linear program: $\max c'x$ subject to $Ax \leq b$ and $x \geq 0$. If $\bar{x}$ is feasible and $c'\bar{x} \geq c'x$ for all feasible x , then $\bar{x}$ is called an optimal solution. Similarly, in the program: $\min c'x$ subject to $Ax \geq b$ and $x \geq 0$. If $\bar{x}$ is feasible and $c'\bar{x} \leq c'x$ for all feasible x , then $\bar{x}$ is called an optimal solution.

When it is possible to graph the feasible region (with two choice variables), and there are not too many constraints, the problem can be solved by inspection. To illustrate, consider Example 4.1 again.

翻译 (Chinese)

一旦确定了可行区域，就可以开始寻找最优解：可行区域中使得没有其他任何可行点能改进目标函数值的点。

定义 4.2: 考虑线性规划: $\max c'x$ 满足 $Ax \leq b$ 和 $x \geq 0$ 。若 $\bar{x}$ 是可行的, 且对所有可行 x 有 $c'\bar{x} \geq c'x$, 则称 $\bar{x}$ 为最优解。类似地, 在规划 $\min c'x$ 满足 $Ax \geq b$ 和 $x \geq 0$ 中, 若 $\bar{x}$ 是可行的, 且对所有可行 x 有 $c'\bar{x} \leq c'x$, 则称 $\bar{x}$ 为最优解。

当可以用图形表示可行区域 (有两个选择变量) 且约束不太多时, 可以通过观察来求解问题。为说明这一点, 再次考虑例 4.1。

EXAMPLE 4.5:

Original (English)

Recall the program from Example 4.1:

$$\begin{aligned} \max \quad & 3x + 4y \\ \text{subject to:} \quad & 2x + 3y \leq 13 \\ & 4x + 3y \leq 17 \\ & x, y \geq 0. \end{aligned}$$

Consider the objective function, $\pi(x, y) = 3x + 4y$. An equation such as $\pi(x, y) = 2$ defines a line giving those combinations of (x, y) such that $\pi(x, y) = 2$. Referring to π as profit, the set $\{(x, y) \mid \pi(x, y) = k\}$ is called an “iso-profit” line with profit k . Iso-profit lines for the objective function, along with the feasible region, may easily be graphed. In the figure, the iso-profit lines are drawn for $k = 8, 12$ and 18 .

From the figure, the highest value for the objective, 18 , is achieved at $(2, 3)$, the corner of the feasible region determined by equations $2x + 3y = 13$ and $4x + 3y = 17$. Using Cramer’s rule:

$$\begin{vmatrix} 2 & 3 \\ 4 & 3 \end{vmatrix} = 6 - 12 = -6, \quad \begin{vmatrix} 13 & 3 \\ 17 & 3 \end{vmatrix} = 39 - 51 = -12, \quad \begin{vmatrix} 2 & 13 \\ 4 & 17 \end{vmatrix} = 34 - 52 = -18.$$

Thus, the solution is $x = (x, y) = \left(\frac{-12}{-6}, \frac{-18}{-6}\right) = (2, 3)$. Note that the feasible region is defined by its corners. Determining the objective value at each of the corners gives the table:

Corner: (x, y)	Objective value
$(0, 0)$	$0 = 3(0) + 4(0)$
$(0, \frac{13}{3})$	$17\frac{1}{3} = 3(0) + 4(\frac{13}{3})$
$(2, 3)$	$18 = 3(2) + 4(3)$
$(\frac{17}{4}, 0)$	$12\frac{3}{4} = 3(\frac{17}{4}) + 4(0)$

By comparing the objective value at each corner, observe that the highest value occurs at corner $(2, 3)$. This is not accidental. An optimal solution can always be found at a corner.

翻译 (Chinese)

回顾例 4.1 中的规划：

$$\begin{aligned} \max \quad & 3x + 4y \\ \text{满足:} \quad & 2x + 3y \leq 13 \\ & 4x + 3y \leq 17 \\ & x, y \geq 0. \end{aligned}$$

考虑目标函数 $\pi(x, y) = 3x + 4y$ 。像 $\pi(x, y) = 2$ 这样的方程定义了一条直线，给出了使得 $\pi(x, y) = 2$ 的 (x, y) 组合。将 π 视为利润，集合 $\{(x, y) \mid \pi(x, y) = k\}$ 称为利润为 k 的“等利润”线。目标函数的等利润线以及可行区域可以很容易地绘制出来。图中画出了 $k = 8, 12$ 和 18 的等利润线。

从图中可以看出，目标函数的最大值 18 在 $(2, 3)$ 处取得，这是由方程 $2x + 3y = 13$ 和 $4x + 3y = 17$ 确定的可行区域的角点。使用克莱姆法则：

$$\begin{vmatrix} 2 & 3 \\ 4 & 3 \end{vmatrix} = 6 - 12 = -6, \quad \begin{vmatrix} 13 & 3 \\ 17 & 3 \end{vmatrix} = 39 - 51 = -12, \quad \begin{vmatrix} 2 & 13 \\ 4 & 17 \end{vmatrix} = 34 - 52 = -18.$$

因此，解为 $x = (x, y) = \left(\frac{-12}{-6}, \frac{-18}{-6}\right) = (2, 3)$ 。注意可行区域由其角点定义。确定每个角点的目标函数值如下表：

角点: (x, y)	目标函数值
$(0, 0)$	$0 = 3(0) + 4(0)$
$(0, \frac{13}{3})$	$17\frac{1}{3} = 3(0) + 4(\frac{13}{3})$
$(2, 3)$	$18 = 3(2) + 4(3)$
$(\frac{17}{4}, 0)$	$12\frac{3}{4} = 3(\frac{17}{4}) + 4(0)$

通过比较每个角点的目标函数值，可以观察到最大值出现在角点 $(2, 3)$ 处。这并非偶然。最优解总是可以在某个角点处找到。

Original (English)

An important theorem in linear programming (see Theorem 4.2, later) says that if a linear program has an optimal solution, then it has an optimal solution at a corner. This does not say that every optimal solution is at a corner — just that there is an optimal solution at a corner (assuming the problem has an optimal solution). The key benefit of the theorem is that in searching for an optimal solution we can restrict attention to the corners of the feasible region. When there are many variables and a graphical solution is not possible, being able to restrict attention to corners

is very useful. One method for solving a linear program, the simplex method, compares corners, moving from one to another to improve the objective and stopping when no further improvement is possible. (New methods move inside the feasible set during the search.)

In more than two dimensions, the meaning of “corner” may be less clear. In higher dimensions, the appropriate concept is vertex or extreme point of the feasible region. Note in the figures that no corner is a (strict) convex combination of other points in the feasible region. We may define a corner of the feasible region to be a point, x , in the feasible region with the property that we cannot find two other points in the feasible region, say y, z (different from x), such that x lies on a straight line connecting these two points. The only points in the figures satisfying this criterion are the corners. This criterion is independent of the dimension of the feasible region (the number of elements in the x vector) and defines a vertex or extreme point of the feasible set in general. This is discussed further in Section 4.2.

翻译 (Chinese)

线性规划中的一个重要定理（见后文定理 4.2）指出，如果线性规划有最优解，那么它在某个角点处有最优解。这并非说每个最优解都在角点处——只是说存在一个角点上的最优解（假设问题有最优解）。该定理的关键好处在于，在寻找最优解时，我们可以将注意力限制在可行区域的角点上。当变量很多且无法进行图形求解时，能够将注意力限制在角点上是非常有用的。求解线性规划的一种方法——单纯形法——通过比较角点，从一个角点移动到另一个以改进目标函数值，直至无法进一步改进为止。（较新的方法在搜索过程中在可行集内部移动。）

在二维以上，“角点”的含义可能不太明确。在更高维度中，适当的概念是可行区域的顶点或极点。注意图中没有角点是可行区域中其他点的（严格）凸组合。我们可以将可行区域的角点定义为可行区域中具有如下性质的点 x ：我们无法在可行区域中找到另外两个点，比如 y, z （不同于 x ），使得 x 位于连接这两点的直线上。图中满足这一准则的唯一一点是各个角点。该准则与可行区域的维数（ x 向量中元素的数量）无关，并一般地定义了可行集的顶点或极点。这将在第 4.2 节进一步讨论。

4.1.4 Dual prices and slack variables / 对偶价格与松弛变量

Original (English)

In a linear program, the inequality constraints identify resource constraints: usage of a particular resource may not exceed a particular level. Subject to these constraints, the objective is optimized. Therefore, when a resource constraint is varied (some right-hand side parameter b_i is altered), this

generally has an impact on the solution and corresponding objective value. This change in the objective value is measured by the dual price.

翻译 (Chinese)

在线性规划中，不等式约束确定了资源约束：特定资源的使用量不得超过特定水平。在这些约束下，目标函数被最优化。因此，当某个资源约束发生变化（某个右端参数 b_i 被改变）时，这通常会对解及相应的目标函数值产生影响。目标函数值的这种变化通过对偶价格来衡量。

Dual prices / 对偶价格

Original (English)

The dual or shadow price of the i th constraint is defined as the change in the value of the optimized objective due to a unit change in the right-hand side of the i th constraint. For example, suppose that the program has been solved and the solution is at a corner $\bar{x}$ with associated objective value $c'\bar{x}$. Then, $\bar{x}$ is located at the intersection of some set of constraints. Changing the right-hand side (RHS) of one of those constraints slightly will change the intersection point slightly, to say $\tilde{x}$. Suppose that the RHS of the i th constraint is changed from b_i to $\tilde{b}_i$ and suppose that the corner at which the solution is located is defined by the same set of constraints (the solution occurs at the $\tilde{x}$).

(This is the typical situation. Formally, the solution basis does not change. See Example 4.6, Remark 4.1 and the discussion of basic solutions in Section 4.2.) Let the new solution be $\tilde{x}$. Then the dual price of the i th constraint is defined:

$$DP_i = \frac{\text{Change in objective}}{\text{Change in RHS of constraint } i} = \frac{c'\tilde{x} - c'\bar{x}}{\tilde{b}_i - b_i}.$$

翻译 (Chinese)

第 i 个约束的对偶价格或影子价格定义为：第 i 个约束右端项每单位变化所导致的最优化目标函数值的变化。例如，假设规划已解出，解位于角点 $\bar{x}$ 处，相应的目标函数值为 $c'\bar{x}$ 。那么， $\bar{x}$ 位于某组约束的交点处。略微改变其中一个约束的右端项（RHS）将使交点略微移动，移至例如 $\tilde{x}$ 处。假设第 i 个约束的 RHS 从 b_i 变为 $\tilde{b}_i$ ，且解所在的角点由同一组约束定义（解出现在 $\tilde{x}$ 处）。

（这是典型情况。形式上，解的基不变。参见例 4.6、注记 4.1 以及第 4.2 节关于基本解的

讨论。) 设新解为 $\tilde{x}$ 。则第 i 个约束的对偶价格定义为:

$$DP_i = \frac{\text{目标函数的变化}}{\text{约束 } i \text{ 的 RHS 的变化}} = \frac{c'\tilde{x} - c'\bar{x}}{\tilde{b}_i - b_i}.$$

EXAMPLE 4.6:

Original (English)

To illustrate, continuing with the previous example, consider the impact of changing the right-hand side (RHS1) of the first constraint from 13 to 14. In the notation above, $b_1 = 13$, $\tilde{b}_1 = 14$. This defines a new program:

$$\begin{aligned} \max \quad & 3x + 4y \\ \text{subject to:} \quad & 2x + 3y \leq 14 \quad (1) \\ & 4x + 3y \leq 17 \quad (2) \\ & x, y \geq 0. \end{aligned}$$

When the RHS of constraint 1 changes from 13 to 14, the new solution using Cramer's rule is:

$$\begin{vmatrix} 2 & 3 \\ 4 & 3 \end{vmatrix} = 6 - 12 = -6, \quad \begin{vmatrix} 14 & 3 \\ 17 & 3 \end{vmatrix} = 42 - 51 = -9, \quad \begin{vmatrix} 2 & 14 \\ 4 & 17 \end{vmatrix} = 34 - 56 = -22.$$

Then the new solution is at $\tilde{x} = (x, y) = \left(\frac{9}{6}, \frac{22}{6}\right)$.

The objective value at the new solution is:

$$c'\tilde{x} = (3, 4) \cdot \left(\frac{9}{6}, \frac{22}{6}\right) = \frac{1}{6}[3(9) + 4(22)] = \frac{1}{6} \cdot 115 = 19.1667 = 19\frac{1}{6}.$$

Therefore,

$$DP_1 = \frac{\text{objective}}{\text{RHS1}} = \frac{c'\tilde{x} - c'\bar{x}}{\tilde{b}_1 - b_1} = \frac{19.1667 - 18}{14 - 13} = 1.1667 = 1\frac{1}{6}.$$

The same reasoning applies to constraint 2. Consider raising the RHS (RHS2) of constraint 2 from 17 to 18. The new solution is at $\tilde{x} = (x, y) = \left(\frac{15}{6}, \frac{16}{6}\right)$ with objective value:

$$c'\tilde{x} = (3, 4) \cdot \left(\frac{15}{6}, \frac{16}{6}\right) = \frac{1}{6}[3(15) + 4(16)] = \frac{1}{6} \cdot 109 = 18.1667.$$

Therefore,

$$DP_2 = \frac{\text{objective}}{\text{RHS2}} = \frac{c'\tilde{x} - c'\bar{x}}{\tilde{b}_2 - b_2} = \frac{18.1667 - 18}{18 - 17} = 0.1667 = \frac{1}{6}.$$

翻译 (Chinese)

为了说明，继续前面的例子，考虑将第一个约束的右端项（RHS1）从 13 变为 14 的影响。采用上述记号， $b_1 = 13$ ， $\tilde{b}_1 = 14$ 。这定义了一个新规划：

$$\begin{aligned} \max \quad & 3x + 4y \\ \text{满足:} \quad & 2x + 3y \leq 14 \quad (1) \\ & 4x + 3y \leq 17 \quad (2) \\ & x, y \geq 0. \end{aligned}$$

当约束 1 的 RHS 从 13 变为 14 时，使用克莱姆法则得到的新解为：

$$\begin{vmatrix} 2 & 3 \\ 4 & 3 \end{vmatrix} = 6 - 12 = -6, \quad \begin{vmatrix} 14 & 3 \\ 17 & 3 \end{vmatrix} = 42 - 51 = -9, \quad \begin{vmatrix} 2 & 14 \\ 4 & 17 \end{vmatrix} = 34 - 56 = -22.$$

则新解为 $\tilde{x} = (x, y) = \left(\frac{9}{6}, \frac{22}{6}\right)$ 。

新解处的目标函数值为：

$$c'\tilde{x} = (3, 4) \cdot \left(\frac{9}{6}, \frac{22}{6}\right) = \frac{1}{6}[3(9) + 4(22)] = \frac{1}{6} \cdot 115 = 19.1667 = 19\frac{1}{6}.$$

因此，

$$DP_1 = \frac{\text{目标函数}}{\text{RHS1}} = \frac{c'\tilde{x} - c'\bar{x}}{\tilde{b}_1 - b_1} = \frac{19.1667 - 18}{14 - 13} = 1.1667 = 1\frac{1}{6}.$$

同样的推理适用于约束 2。考虑将约束 2 的 RHS（RHS2）从 17 提高到 18。新解为 $\tilde{x} = (x, y) = \left(\frac{15}{6}, \frac{16}{6}\right)$ ，目标函数值为：

$$c'\tilde{x} = (3, 4) \cdot \left(\frac{15}{6}, \frac{16}{6}\right) = \frac{1}{6}[3(15) + 4(16)] = \frac{1}{6} \cdot 109 = 18.1667.$$

因此，

$$DP_2 = \frac{\text{目标函数}}{\text{RHS2}} = \frac{c'\tilde{x} - c'\bar{x}}{\tilde{b}_2 - b_2} = \frac{18.1667 - 18}{18 - 17} = 0.1667 = \frac{1}{6}.$$

Slack variables / 松弛变量

Original (English)

Given the constraints $Ax \leq b$ and $x \geq 0$, suppose that the optimal solution has been found and call it $\bar{x}$. The slack variables are defined as the vector s that satisfies $A\bar{x} + s = b$ or $s = b - A\bar{x}$. (In the case where the constraints have the form $Ax \geq b$ and $x \geq 0$, the slack variables are defined: $A\bar{x} - s = b$ or $s = A\bar{x} - b$. This convention ensures that the slack variables are non-negative.)

Continuing with the example, $s = (s_1, s_2)$ is defined by the equations:

$$2x + 3y + s_1 = 13 \quad \text{or} \quad 2(2) + 3(3) + s_1 = 13 \quad \text{or} \quad 13 + s_1 = 13$$

$$4x + 3y + s_2 = 17 \quad \text{or} \quad 4(2) + 3(3) + s_2 = 17 \quad \text{or} \quad 17 + s_2 = 17$$

so that $s = (s_1, s_2) = (0, 0)$. When a slack variable is 0, the corresponding constraint is binding. Here both slack variables are 0, and the constraints bind. In general, changing the RHS of a binding constraint changes the solution and the value of the objective. Therefore, in general, if the slack variable associated with a constraint is 0, the corresponding dual price is non-zero. Conversely, if the slack variable associated with a constraint is positive, then the constraint is not binding: changing it has no effect on the solution and so the dual price is zero.

翻译 (Chinese)

给定约束 $Ax \leq b$ 和 $x \geq 0$, 假设已找到最优解并记为 $\bar{x}$ 。松弛变量定义为满足 $A\bar{x} + s = b$ 或 $s = b - A\bar{x}$ 的向量 s 。(在约束形式为 $Ax \geq b$ 和 $x \geq 0$ 的情况下, 松弛变量定义为: $A\bar{x} - s = b$ 或 $s = A\bar{x} - b$ 。这一约定确保了松弛变量非负。)

继续该例子, $s = (s_1, s_2)$ 由以下方程定义:

$$2x + 3y + s_1 = 13 \quad \text{或} \quad 2(2) + 3(3) + s_1 = 13 \quad \text{或} \quad 13 + s_1 = 13$$

$$4x + 3y + s_2 = 17 \quad \text{或} \quad 4(2) + 3(3) + s_2 = 17 \quad \text{或} \quad 17 + s_2 = 17$$

因此 $s = (s_1, s_2) = (0, 0)$ 。当松弛变量为 0 时, 相应的约束是紧的。这里两个松弛变量均为 0, 约束是紧的。一般来说, 改变紧约束的 RHS 会改变解和目标函数值。因此, 一般来说, 如果与某个约束相关联的松弛变量为 0, 则相应的对偶价格非零。反之, 如果与某个约束相关联的松弛变量为正, 则该约束不是紧的: 改变它不会对解产生影响, 因此对偶价格为零。

4.1.5 Some examples / 若干例子

Original (English)

In this section, two examples are considered in detail to summarize the discussion so far.

翻译 (Chinese)

本节详细讨论两个例子以总结到目前为时的讨论。

EXAMPLE 4.7:

Original (English)

Consider the following linear program:

$$\begin{aligned} \max \quad & 4x + 7y \\ \text{subject to:} \quad & 2x + 6y \leq 28 \quad (1) \\ & 3x + 4y \leq 22 \quad (2) \\ & 5x + 4y \leq 34 \quad (3) \\ & x, y \geq 0. \end{aligned}$$

Solution: Constraints 1 and 2 intersect at $(x, y) = (2, 4)$; constraints 1 and 3 at $(x, y) = (\frac{92}{22}, \frac{72}{22})$; and constraints 2 and 3 at $(x, y) = (6, 1)$. These are found by solving each pair of equations in turn. For example, $2x + 6y = 28$ and $3x + 4y = 22$ solve to give $(x, y) = (2, 4)$, which turns out to be the optimal solution:

$$\begin{pmatrix} 2 & 6 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 28 \\ 22 \end{pmatrix}.$$

Thus

$$\begin{vmatrix} 2 & 6 \\ 3 & 4 \end{vmatrix} = -10, \quad \begin{vmatrix} 28 & 6 \\ 22 & 4 \end{vmatrix} = -20, \quad \begin{vmatrix} 2 & 28 \\ 3 & 22 \end{vmatrix} = -40.$$

This gives $(x, y) = \left(\frac{-20}{-10}, \frac{-40}{-10} \right) = (2, 4)$. To see that it is optimal, recall from earlier discussion that comparing the objective value at each feasible corner is one way to solve the program. Another way is to compare the slopes of the constraints and the objective.

Constraint	Slope
Constraint 1	$-\frac{1}{3}$
Constraint 2	$-\frac{3}{4}$
Constraint 3	$-\frac{5}{4}$
Objective	$-\frac{4}{7}$

The slope of the objective is between the slope of constraints 1 and 2, so the optimal solution will be where these constraints intersect — at $(2, 4)$. The value of the objective at the solution is: $4(2) + 7(4) = 36$.

Parameter changes: If, for example, the objective were changed to $6x + 7y$ with a slope of $\frac{6}{7}$, this would be steeper than constraint 2 and the solution would be located at $(x, y) = (6, 1)$. If the objective were altered to be parallel to the second constraint (for example, $6x + 8y$) there would be an infinite number of solutions ($(x, y) = (2, 4)$, $(x, y) = (6, 1)$ and all points on the line connecting them).

Slack variables: Since both constraints 1 and 2 are satisfied with equality, $s_1 = s_2 = 0$, while $s_3 = 34 - 5(2) - 4(4) = 34 - 26 = 8$.

Dual prices:

1. To calculate the dual price in constraint 1, replace 28 by 29. Then the new solution is $(x, y) = (\frac{16}{10}, \frac{43}{10})$. The value of the objective is $4(\frac{16}{10}) + 7(\frac{43}{10}) = \frac{365}{10} = 36\frac{1}{2}$. Consequently, $DP_1 = \frac{1}{2}$.
2. To calculate the dual price in constraint 2, replace 22 by 23. Then the new solution is $(x, y) = (\frac{26}{10}, \frac{38}{10})$. The value of the objective is $4(\frac{26}{10}) + 7(\frac{38}{10}) = \frac{370}{10} = 37$. Consequently, $DP_2 = 1$.
3. Since changing the right-hand side of constraint 3 does not affect the solution, the associated dual price is 0.

翻译 (Chinese)

考虑如下线性规划:

$$\max \quad 4x + 7y$$

$$\text{满足: } 2x + 6y \leq 28 \quad (1)$$

$$3x + 4y \leq 22 \quad (2)$$

$$5x + 4y \leq 34 \quad (3)$$

$$x, y \geq 0.$$

解: 约束 1 和 2 交于 $(x, y) = (2, 4)$; 约束 1 和 3 交于 $(x, y) = (\frac{92}{22}, \frac{72}{22})$; 约束 2 和 3 交于 $(x, y) = (6, 1)$ 。这些是通过依次求解每对方程得到的。例如, $2x + 6y = 28$ 和 $3x + 4y = 22$ 求解得 $(x, y) = (2, 4)$, 这恰好是最优解。为验证其最优性, 比较各可行角点的目标函数值是一种求解方法。另一种方法是比较约束和目标函数的斜率。目标函数的斜率介于约束 1 和 2 的斜率之间, 因此最优解将位于这些约束的交点处——即 $(2, 4)$ 。解处的目标函数值为: $4(2) + 7(4) = 36$ 。

参数变化: 若目标函数改为 $6x + 7y$, 斜率为 $\frac{6}{7}$, 这将比约束 2 更陡, 解将位于 $(x, y) = (6, 1)$ 。若目标函数被改变为与第二个约束平行 (例如 $6x + 8y$), 则将有无穷多个解。

松弛变量: 由于约束 1 和 2 均以等式满足, $s_1 = s_2 = 0$, 而 $s_3 = 34 - 5(2) - 4(4) = 34 - 26 = 8$ 。

对偶价格:

1. 为计算约束 1 的对偶价格, 将 28 替换为 29。新解为 $(x, y) = (\frac{16}{10}, \frac{43}{10})$, 目标函数值为 $36\frac{1}{2}$ 。因此 $DP_1 = \frac{1}{2}$ 。

2. 为计算约束 2 的对偶价格, 将 22 替换为 23。新解为 $(x, y) = (\frac{26}{10}, \frac{38}{10})$, 目标函数值为 37。因此 $DP_2 = 1$ 。
3. 由于改变约束 3 的右端项不影响解, 相应的对偶价格为 0。

EXAMPLE 4.8:**Original (English)**

Consider the following linear program:

$$\begin{aligned}
 \min \quad & 4x + 2y \\
 \text{subject to:} \quad & 4x + 8y \geq 40 \quad (1) \\
 & 3x + 2y \geq 24 \quad (2) \\
 & 6x + 2y \geq 30 \quad (3) \\
 & x, y \geq 0.
 \end{aligned}$$

Solution: Comparing the slopes of the objective and constraints gives the solution at the intersection of constraints 2 and 3. Using Cramer's rule: $(x, y) = (2, 9)$. The value of the objective at the solution is: $4(2) + 2(9) = 26$.

Slack variables: The values of the slack variables at the optimal solution, (x, y) are defined from the equations:

$$\begin{aligned}
 4x + 8y - s_1 &= 40 \\
 3x + 2y - s_2 &= 24 \\
 6x + 2y - s_3 &= 30,
 \end{aligned}$$

giving $s_1 = 4(2) + 8(9) - 40 = 80 - 40 = 40$, and $s_2 = 0$, $s_3 = 0$.

Dual prices:

1. Because constraint 1 is not binding, its shadow price is 0: $DP_1 = 0$.
2. To determine the dual price of the second constraint, increase RHS2 by 1 from 24 to 25. The new solution is $(x, y) = (\frac{10}{6}, 10)$ and the objective value is $4(\frac{10}{6}) + 2(10) = 26.6667$. Thus $DP_2 = 0.6667$.
3. Finally, to determine DP_3 , raise the RHS of constraint 3 by 1 unit and find the new intersection with the original constraint 2. The new solution is $(x, y) = (\frac{14}{6}, \frac{51}{6})$ and the objective value is 26.3337. Therefore $DP_3 = 0.3337$.

翻译 (Chinese)

考虑如下线性规划：

$$\begin{aligned} \min \quad & 4x + 2y \\ \text{满足:} \quad & 4x + 8y \geq 40 \quad (1) \\ & 3x + 2y \geq 24 \quad (2) \\ & 6x + 2y \geq 30 \quad (3) \\ & x, y \geq 0. \end{aligned}$$

解：比较目标函数和约束的斜率可知，解位于约束 2 和 3 的交点处。使用克莱姆法则： $(x, y) = (2, 9)$ 。解处的目标函数值为： $4(2) + 2(9) = 26$ 。

松弛变量：在最优解 (x, y) 处，松弛变量的值由以下方程定义：

$$\begin{aligned} 4x + 8y - s_1 &= 40 \\ 3x + 2y - s_2 &= 24 \\ 6x + 2y - s_3 &= 30, \end{aligned}$$

得 $s_1 = 4(2) + 8(9) - 40 = 80 - 40 = 40$, $s_2 = 0$, $s_3 = 0$ 。

对偶价格：

1. 由于约束 1 不是紧的，其影子价格为 0： $DP_1 = 0$ 。
2. 为确定第二个约束的对偶价格，将 RHS2 从 24 增加到 25。新解为 $(x, y) = (\frac{10}{6}, 10)$ ，目标函数值为 26.6667。因此 $DP_2 = 0.6667$ 。
3. 最后，为确定 DP_3 ，将约束 3 的 RHS 增加 1 单位，并找到其与原始约束 2 的新交点。新解为 $(x, y) = (\frac{14}{6}, \frac{51}{6})$ ，目标函数值为 26.3337。因此 $DP_3 = 0.3337$ 。

4.1.6 Duality reconsidered / 对偶性再探讨

Original (English)

The following discussion provides a different perspective on dual prices and their computation. An example will motivate the discussion. The next example takes an earlier example and rearranges the parameters and objective to give a program called the dual program. Every program has a dual program, giving the terminology primal and dual to refer to the initial and transformed programs. It turns out that the optimal solution payoff values are the same in both programs, the dual prices in the primal are the solution values in the dual, and the solution values in the primal are the dual prices in the dual program.

翻译 (Chinese)

下面的讨论为对偶价格及其计算提供了一个不同的视角。一个例子将引出这一讨论。下一个例子取自之前的例子，并重新排列参数和目标函数以得到一个称为对偶规划的规划。每个规划都有一个对偶规划，因此用术语原始问题和对偶问题来分别指代初始规划和变换后的规划。结果表明，两个规划的最优解支付值是相同的，原始问题中的对偶价格是对偶问题中的解值，而原始问题中的解值是对偶规划中的对偶价格。

EXAMPLE 4.9:**Original (English)**

Consider the following linear program:

Program A

$$\begin{aligned} \min \quad & 13w + 17z \\ \text{subject to:} \quad & 2w + 4z \geq 3 \quad (1) \\ & 3w + 3z \geq 4 \quad (2) \\ & w, z \geq 0. \end{aligned}$$

Observe that this program bears a close relation to one discussed earlier (see Examples 4.5 and 4.6):

Program B

$$\begin{aligned} \max \quad & 3x + 4y \\ \text{subject to:} \quad & 2x + 3y \leq 13 \quad (1) \\ & 4x + 3y \leq 17 \quad (2) \\ & x, y \geq 0. \end{aligned}$$

Program A is called the dual of program B. Notice that A is obtained from B by exchanging the objective coefficients and RHS constraint parameters, and taking the transpose of the coefficient matrix. The remarkable fact relating these programs is the following. Let $(\bar{x}, \bar{y})$ solve program B, with objective value $\bar{B}$ and dual prices (DP_1^B, DP_2^B) . Similarly, let $(\bar{w}, \bar{z})$ solve program A, with objective value $\bar{A}$ and dual prices (DP_1^A, DP_2^A) . Then

$$(1) \bar{B} = \bar{A}, \quad (2) (\bar{x}, \bar{y}) = (DP_1^A, DP_2^A), \quad (3) (\bar{w}, \bar{z}) = (DP_1^B, DP_2^B).$$

Recall from Examples 4.5 and 4.6, that the solution to program B has: $\bar{B} = 18$, $(\bar{x}, \bar{y}) = (2, 3)$ and $(DP_1, DP_2) = (1\frac{1}{6}, \frac{1}{6})$. The solution to program A is at the intersection of the two constraints; using Cramer's rule gives solution $(\bar{w}, \bar{z}) = (\frac{7}{6}, \frac{1}{6})$. The objective value is $\bar{A} = 13(\frac{7}{6}) + 17(\frac{1}{6}) = \frac{108}{6} = 18$.

To calculate the dual prices in program A, consider first an increase of $\frac{1}{4}$ in the RHS of constraint 1. The new solution is $(\bar{w}, \bar{z}) = (\frac{25}{24}, \frac{7}{24})$. The new objective value is then $13(\frac{25}{24}) + 17(\frac{7}{24}) = \frac{444}{24} = 18\frac{1}{2}$. Thus, $DP_1^A = \frac{\frac{1}{4}}{\frac{1}{4}} = 2$. For constraint 2, changing the RHS to $4\frac{1}{4}$ gives new solution $(\bar{w}, \bar{z}) = (\frac{8}{6}, \frac{1}{12})$. The new objective value is then $13(\frac{16}{12}) + 17(\frac{1}{12}) = \frac{225}{12} = 18\frac{3}{4}$. Thus, $DP_2^A = \frac{\frac{3}{4}}{\frac{1}{4}} = 3$.

Summarizing: $DP_1^A = 2$, $DP_2^A = 3$. Thus, $(\bar{x}, \bar{y}) = (DP_1^A, DP_2^A) = (2, 3)$. Recall that the dual prices for program B were $DP_1^B = 1.1667 = \frac{7}{6}$ and $DP_2^B = 0.1667 = \frac{1}{6}$ and $\bar{B} = 18$. Thus, $(\bar{w}, \bar{z}) = (DP_1^B, DP_2^B) = (\frac{7}{6}, \frac{1}{6})$ and $\bar{A} = \bar{B} = 18$.

翻译 (Chinese)

考虑如下线性规划:

规划 A

$$\begin{aligned} \min \quad & 13w + 17z \\ \text{满足:} \quad & 2w + 4z \geq 3 \quad (1) \\ & 3w + 3z \geq 4 \quad (2) \\ & w, z \geq 0. \end{aligned}$$

注意到该规划与之前讨论的一个规划 (见例 4.5 和 4.6) 密切相关:

规划 B

$$\begin{aligned} \max \quad & 3x + 4y \\ \text{满足:} \quad & 2x + 3y \leq 13 \quad (1) \\ & 4x + 3y \leq 17 \quad (2) \\ & x, y \geq 0. \end{aligned}$$

规划 A 称为规划 B 的对偶。注意, A 是通过交换目标系数和 RHS 约束参数, 并取系数矩阵的转置而从 B 得到的。联系这些规划的一个显著事实如下。设 $(\bar{x}, \bar{y})$ 为规划 B 的解, 目标函数值为 $\bar{B}$, 对偶价格为 (DP_1^B, DP_2^B) 。类似地, 设 $(\bar{w}, \bar{z})$ 为规划 A 的解, 目标函数值为 $\bar{A}$, 对偶价格为 (DP_1^A, DP_2^A) 。则

$$(1) \bar{B} = \bar{A}, \quad (2) (\bar{x}, \bar{y}) = (DP_1^A, DP_2^A), \quad (3) (\bar{w}, \bar{z}) = (DP_1^B, DP_2^B).$$

回顾例 4.5 和 4.6, 规划 B 的解为: $\bar{B} = 18$, $(\bar{x}, \bar{y}) = (2, 3)$, 以及对偶价格 $(DP_1, DP_2) = (1\frac{1}{6}, \frac{1}{6})$ 。规划 A 的解位于两个约束的交点处; 使用克莱姆法则得解 $(\bar{w}, \bar{z}) = (\frac{7}{6}, \frac{1}{6})$ 。目标函数值为 $\bar{A} = 13(\frac{7}{6}) + 17(\frac{1}{6}) = \frac{108}{6} = 18$ 。

计算规划 A 中的对偶价格: 首先考虑约束 1 的 RHS 增加 $\frac{1}{4}$, 得 $DP_1^A = 2$ 。对于约束 2, 将 RHS 改为 $4\frac{1}{4}$, 得 $DP_2^A = 3$ 。

总结: $DP_1^A = 2$, $DP_2^A = 3$ 。因此, $(\bar{x}, \bar{y}) = (DP_1^A, DP_2^A) = (2, 3)$ 。回顾规划 B 的对偶价格为 $DP_1^B = 1.1667 = \frac{7}{6}$ 和 $DP_2^B = 0.1667 = \frac{1}{6}$, $\bar{B} = 18$ 。因此, $(\bar{w}, \bar{z}) = (DP_1^B, DP_2^B) = (\frac{7}{6}, \frac{1}{6})$ 且 $\bar{A} = \bar{B} = 18$ 。

4.1.7 Basic solutions / 基本解

Original (English)

The previous discussion dealt with programs having constraints of the form $Ax \leq b$ or $Ax \geq b$. In what follows, it is shown that such inequality-constrained problems may be reformulated to equality-constrained programs. The equality-constrained formulation is then used to discuss basic solutions.

翻译 (Chinese)

前面的讨论处理了具有 $Ax \leq b$ 或 $Ax \geq b$ 形式约束的规划。下面将说明此类不等式约束问题可以重新表述为等式约束规划。然后利用等式约束的表述来讨论基本解。

Equality-constrained programs / 等式约束规划

Original (English)

Consider the linear program: $\max c'x$ subject to $Ax \leq b$ and $x \geq 0$. Here $c \sim n \times 1$, $x \sim n \times 1$, $A \sim m \times n$, $s \sim m \times 1$ and $I \sim m \times m$, the identity matrix. The program may be written in a variety of equivalent ways:

I	II
$\max \quad c'x$	$\max \quad c'x + 0's$
s.t. $Ax \leq b$	s.t. $Ax + Is = b$
$x \geq 0$	$x, s \geq 0$
III	IV
$\max \quad (\bar{c}, 0)' \cdot (x, s)$	$\max \quad \bar{c}' \cdot z$
s.t. $(\bar{A} : I)(x, s) = b$	s.t. $\bar{A}z = b$
$(x, s) \geq 0$	$z \geq 0$

Formulation II can be expressed succinctly as follows. Define new matrices and vectors: $\bar{c}' = (c, 0)$, $\bar{A} = (A : I)$, $z = (x, s)$. In the initial problem, the dimensions of the matrices and vectors were

$c \sim n \times 1$, $x \sim n \times 1$, $A \sim m \times n$. Since A is an $m \times n$ matrix, this means that there are m constraints (leaving aside the non-negativity constraints on the x). The equality-constrained model is obtained by adding “slack” variables s — one for each constraint. So there are m slack variables added to give formulation II. Thus $\bar{c}$ is a lengthened vector with zeros added to c ; likewise, z involves lengthening the x vector with s , and $\bar{A}$ is a matrix formed by placing A and I , the $m \times m$ identity matrix, side by side. Therefore, $z \sim (n+m) \times 1$, $\bar{c} \sim (n+m) \times 1$ and $\bar{A} \sim m \times (n+m)$. Programs II, III and IV are equivalent problems — just involving relabeling. Furthermore, I and II are equivalent problems, so that the inequality-constrained program in I may be replaced by the equality-constrained program, IV.

翻译 (Chinese)

考虑线性规划: $\max c'x$ 满足 $Ax \leq b$ 和 $x \geq 0$ 。这里 $c \sim n \times 1$, $x \sim n \times 1$, $A \sim m \times n$, $s \sim m \times 1$ 且 $I \sim m \times m$ 为单位矩阵。该规划可以用多种等价方式表示:

(表格见上)

表述 II 可以简洁地表达如下。定义新的矩阵和向量: $\bar{c} = (c, 0)$, $\bar{A} = (A : I)$, $z = (x, s)$ 。在初始问题中, 矩阵和向量的维度为 $c \sim n \times 1$, $x \sim n \times 1$, $A \sim m \times n$ 。因为 A 是 $m \times n$ 矩阵, 这意味着有 m 个约束 (暂且不考虑 x 的非负约束)。等式约束模型通过添加 “松弛” 变量 s ——每个约束一个——而得到。因此添加了 m 个松弛变量以得到表述 II。因而 $\bar{c}$ 是一个通过在 c 后添加零而延长的向量; 同样地, z 涉及用 s 延长 x 向量, 而 $\bar{A}$ 是通过将 A 和 I ($m \times m$ 单位矩阵) 并排放置而形成的矩阵。因此, $z \sim (n+m) \times 1$, $\bar{c} \sim (n+m) \times 1$ 且 $\bar{A} \sim m \times (n+m)$ 。规划 II、III 和 IV 是等价问题——仅涉及重新标记。此外, I 和 II 是等价问题, 因此 I 中的不等式约束规划可以用等式约束规划 IV 代替。

Original (English)

Theorem 4.1: The linear programs I and II have the same solutions: if $\bar{x}$ solves I then for some $s \geq 0$, $(\bar{x}, s)$ solves II, and if $(\tilde{x}, \tilde{s})$ solves II then $\tilde{x}$ solves I.

翻译 (Chinese)

定理 4.1: 线性规划 I 和 II 具有相同的解: 若 $\bar{x}$ 是 I 的解, 则对某个 $s \geq 0$, $(\bar{x}, s)$ 是 II 的解; 若 $(\tilde{x}, \tilde{s})$ 是 II 的解, 则 $\tilde{x}$ 是 I 的解。

Original (English)

Remark 4.2: The minimization problem is analogous. This can be expressed:

$$\min \bar{c}'z \quad \text{subject to:} \quad \bar{A}z = b, \quad z \geq 0,$$

where $\bar{A} = (A : -I)$, $\bar{c} = (c, 0)$, and $z = (x, s)$. Again, solving I and II yields the same solution.

翻译 (Chinese)

注记 4.2: 最小化问题是类似的。这可以表示为:

$$\min \bar{c}'z \quad \text{满足: } \bar{A}z = b, z \geq 0,$$

其中 $\bar{A} = (A : -I)$, $\bar{c} = (c, 0)$, $z = (x, s)$ 。同样地, 求解 I 和 II 得到相同的解。

Definition and identification of basic solutions / 基本解的定义与识别

Original (English)

In this section, basic solutions are introduced and related to the vertices of the feasible region. In view of Theorem 4.1, we can focus on the equality-constrained formulation for the following discussion (and discard the “bar” notation). Thus, the linear program will be written: $\max c'x$, subject to: $Ax = b$ and $x \geq 0$.

Recall that a feasible solution in the inequality-constrained program is any vector x satisfying the constraints: $Ax \leq b$ and $x \geq 0$. A feasible solution $\hat{x}$ is an optimal feasible solution if there is no other feasible $\tilde{x}$, such that $c'\tilde{x} > c'\hat{x}$ in the case of a maximization problem and $c'\tilde{x} < c'\hat{x}$ in the case of a minimization problem. With the equality-constrained program, a vector x satisfying $Ax = b$ and $x \geq 0$ is said to be feasible. Let m be the number of rows in A and recall that A has more columns than rows.

We assume that A has full rank and since A has more columns than rows this means that the rank of A equals the number of rows: m . This is called the full rank assumption: the m rows of A are linearly independent. This assumption involves no loss of generality. If A does not have full rank then there are two possibilities. Either some of the constraints in $Ax = b$ are redundant, or the constraints are inconsistent — there is no x such that $Ax = b$.

Suppose we select m linearly independent columns from the matrix A : say these columns together form a matrix B . Such a set of columns is called a basis. It is possible that not all sets of m columns of A are linearly independent, but since A has full rank there is at least one set of m linearly independent columns.

Since B is non-singular, the equation system $Bx_B = b$ has a unique solution. For example, suppose that the first m columns of A are linearly independent and let B be the matrix formed by these first m columns. Let the remaining columns of A form some matrix C , so that $A = (B : C)$. Then if x_B satisfies $Bx_B = b$, let $x = (x_B, 0)$ and observe that $Ax = Bx_B + C \cdot 0 = Bx_B = b$. Thus

$x = (x_B : 0)$ satisfies $Ax = b$. However, this x may not satisfy $x \geq 0$ — some elements of the vector x may be negative.

Given $Ax = b$, let B be any non-singular $m \times m$ submatrix of columns of A . If all elements of x not associated with columns of B are set to 0, and the resulting set of equations solved, the unique x determined in this way is called a basic solution of the system $Ax = b$ with respect to the basis B . The components of x associated with columns of B are called basic variables.

If one or more of the basic variables in a basic solution has a value of 0, that solution is said to be a degenerate basic solution. Therefore, in a non-degenerate basic solution there are m non-zero components (and the remaining components are 0). In a degenerate basic solution, there are fewer than m components non-zero.

Recall that if x satisfies both $Ax = b$ and $x \geq 0$, x is said to be feasible (for these constraints). If x is a basic solution satisfying $x \geq 0$ it is called a basic feasible solution. If this solution is also degenerate, it is called a degenerate basic feasible solution. The following result, called the fundamental theorem of linear programming, is central in linear programming.

翻译 (Chinese)

在本节中，引入基本解并将其与可行区域的顶点相关联。鉴于定理 4.1，我们可以在接下来的讨论中专注于等式约束表述（并丢弃“bar”记号）。因此，线性规划将写为： $\max c'x$ ，满足： $Ax = b$ 和 $x \geq 0$ 。

回顾在不等式约束规划中，可行解是满足约束 $Ax \leq b$ 和 $x \geq 0$ 的任何向量 x 。可行解 $\hat{x}$ 是最优可行解，如果在最大化问题中不存在其他可行解 $\tilde{x}$ 使得 $c'\tilde{x} > c'\hat{x}$ ，在最小化问题中不存在其他可行解 $\tilde{x}$ 使得 $c'\tilde{x} < c'\hat{x}$ 。对于等式约束规划，满足 $Ax = b$ 和 $x \geq 0$ 的向量 x 称为可行的。设 m 为 A 的行数，并回顾 A 的列数多于行数。

我们假设 A 具有满秩，且由于 A 的列数多于行数，这意味着 A 的秩等于行数： m 。这称为满秩假设： A 的 m 行线性无关。该假设不丧失一般性。如果 A 不具有满秩，则有两种可能：要么 $Ax = b$ 中的某些约束是冗余的，要么约束是不一致的——不存在 x 使得 $Ax = b$ 。

假设我们从矩阵 A 中选择 m 个线性无关的列：称这些列共同构成矩阵 B 。这样一组列称为一个基。 A 中并非所有 m 列的集合都是线性无关的，但由于 A 具有满秩，至少存在一组 m 个线性无关的列。

由于 B 是非奇异的，方程组 $Bx_B = b$ 有唯一解。例如，假设 A 的前 m 列线性无关，令 B 为由这前 m 列构成的矩阵。令 A 的其余列构成某个矩阵 C ，使得 $A = (B : C)$ 。那么若 x_B 满足 $Bx_B = b$ ，令 $x = (x_B, 0)$ ，则 $Ax = Bx_B + C \cdot 0 = Bx_B = b$ 。因此 $x = (x_B : 0)$ 满足 $Ax = b$ 。然而，这个 x 可能不满足 $x \geq 0$ ——向量 x 的某些元素可能为负。

给定 $Ax = b$, 令 B 为 A 的列的任意非奇异 $m \times m$ 子矩阵。如果将与 B 的列无关的所有 x 的元素设为 0, 并求解所得的方程组, 以这种方式确定的唯一 x 称为方程组 $Ax = b$ 关于基 B 的基本解。与 B 的列相关联的 x 的分量称为基本变量。

如果基本解中的一个或多个基本变量的值为 0, 则该解称为退化的基本解。因此, 在非退化的基本解中, 有 m 个非零分量 (其余分量为 0)。在退化的基本解中, 少于 m 个分量非零。

回顾如果 x 同时满足 $Ax = b$ 和 $x \geq 0$, 则称 x (对这些约束) 是可行的。如果 x 是满足 $x \geq 0$ 的基本解, 则称其为基本可行解。如果该解也是退化的, 则称为退化的基本可行解。以下结果——称为线性规划的基本定理——在线性规划中居于核心地位。

Original (English)

Theorem 4.2: Given a linear program of the form I or II,

$$\text{I} \quad \max c'x \quad \text{s.t.} \quad Ax \leq b, x \geq 0$$

$$\text{II} \quad \max c'x \quad \text{s.t.} \quad Ax = b, x \geq 0.$$

1. If there is a feasible solution there is a basic feasible solution.
2. If there is an optimal feasible solution, there is an optimal basic feasible solution.

The importance of this theorem (from the computational point of view) is that the search for an optimal solution can be restricted to the basic solutions. Since there are only a finite number of these, there are only a finite number of x to be considered in the search for an optimal solution.

翻译 (Chinese)

定理 4.2: 给定形式 I 或 II 的线性规划,

$$\text{I} \quad \max c'x \quad \text{s.t.} \quad Ax \leq b, x \geq 0$$

$$\text{II} \quad \max c'x \quad \text{s.t.} \quad Ax = b, x \geq 0.$$

1. 若存在可行解, 则存在基本可行解。
2. 若存在最优可行解, 则存在最优基本可行解。

该定理的重要性 (从计算的角度看) 在于, 对最优解的搜索可以限制在基本解上。由于基本解只有有限个, 在搜索最优解时只需考虑有限个 x 。

EXAMPLE 4.10:

Original (English)

$$\begin{aligned}
 & \max \quad 2x + 5y \\
 & \text{subject to:} \quad 4x + y \leq 24 \\
 & \quad \quad \quad x + 3y \leq 21 \\
 & \quad \quad \quad x + y \leq 9 \\
 & \quad \quad \quad x, y \geq 0.
 \end{aligned}$$

This is rearranged in equality constraint form as:

$$\begin{aligned}
 & \max \quad 2x + 5y + 0s_1 + 0s_2 + 0s_3 \\
 & \text{subject to:} \quad 4x + y + s_1 = 24 \\
 & \quad \quad \quad x + 3y + s_2 = 21 \\
 & \quad \quad \quad x + y + s_3 = 9 \\
 & \quad \quad \quad x, y, s_1, s_2, s_3 \geq 0.
 \end{aligned}$$

In matrix form, let:

$$\bar{A} = \begin{pmatrix} 4 & 1 & 1 & 0 & 0 \\ 1 & 3 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 \end{pmatrix}, \quad b = \begin{pmatrix} 24 \\ 21 \\ 9 \end{pmatrix}, \quad \bar{c} = (2, 5, 0, 0, 0), \quad z = (x, y, s_1, s_2, s_3).$$

The problem may then be written: $\max \bar{c}'z$ subject to: $\bar{A}z = b, z \geq 0$.

In general, given a matrix with m rows and p columns ($p > m$), the number of $m \times m$ submatrices that can be formed is $\frac{p!}{m!(p-m)!}$. In the present case, the coefficient matrix has $m = 3$ and $p = 5$, and so the maximum number of distinct 3×3 matrices is $\frac{5!}{3!(5-3)!} = 10$.

The ten basic solutions are in the form $z = (x, y, s_1, s_2, s_3)$:

$$\begin{aligned}
 z_1 &= (3, 6, 6, 0, 0) \\
 z_2 &= (5, 4, 0, 4, 0) \\
 z_3 &= (4.636, 5.545, 0, 0, -1.09) \\
 z_4 &= (9, 0, -12, 12, 0) \\
 z_5 &= (21, 0, -60, 0, -12) \\
 z_6 &= (6, 0, 0, 15, 3) \\
 z_7 &= (0, 9, 15, -6, 0) \\
 z_8 &= (0, 7, 17, 0, 2) \\
 z_9 &= (0, 24, 0, -51, -15) \\
 z_{10} &= (0, 0, 24, 21, 9).
 \end{aligned}$$

This list of z is the list of basic solutions. Note now that there are some basic solutions that have negative entries. Once we eliminate those with negative entries, we are left with the basic feasible solutions. The non-feasible basic solutions are: z_3, z_4, z_5, z_7 and z_9 . The basic feasible solutions are $z_1, z_2, z_6, z_8, z_{10}$. Denoting the objective value at z by $\pi(z)$:

$$\pi(z_1) = 2(3) + 5(6) = 36, \pi(z_2) = 2(5) + 5(4) = 30, \pi(z_6) = 2(6) + 5(0) = 12,$$

$$\pi(z_8) = 2(0) + 5(7) = 35, \pi(z_{10}) = 2(0) + 5(0) = 0.$$

So, the solution is at z_1 , with $x = 3$ and $y = 6$.

翻译 (Chinese)

$$\begin{aligned} \max \quad & 2x + 5y \\ \text{满足:} \quad & 4x + y \leq 24 \\ & x + 3y \leq 21 \\ & x + y \leq 9 \\ & x, y \geq 0. \end{aligned}$$

将其重新排列为等式约束形式:

$$\begin{aligned} \max \quad & 2x + 5y + 0s_1 + 0s_2 + 0s_3 \\ \text{满足:} \quad & 4x + y + s_1 = 24 \\ & x + 3y + s_2 = 21 \\ & x + y + s_3 = 9 \\ & x, y, s_1, s_2, s_3 \geq 0. \end{aligned}$$

矩阵形式为: $\max \bar{c}'z$ 满足 $\bar{A}z = b, z \geq 0$ 。

一般来说, 给定一个具有 m 行和 p 列 ($p > m$) 的矩阵, 可以构成的 $m \times m$ 子矩阵的个数为 $\frac{p!}{m!(p-m)!}$ 。在本例中, 系数矩阵有 $m = 3$ 和 $p = 5$, 因此不同的 3×3 矩阵最多有 $\frac{5!}{3!(5-3)!} = 10$ 个。

十个基本解的形式为 $z = (x, y, s_1, s_2, s_3)$: (见上)

该 z 列表即为基本解列表。注意有些基本解包含负值。一旦排除那些有负值的解, 剩下的就是基本可行解。不可行的基本解是: z_3, z_4, z_5, z_7 和 z_9 。基本可行解是 $z_1, z_2, z_6, z_8, z_{10}$ 。因此, 解在 z_1 处, $x = 3$ 且 $y = 6$ 。

4.1.8 Duality principles / 对偶原理

Original (English)

There is a general principle behind the close connection between the primal and dual programs. It turns out that both have the same objective values. The following discussion formalizes these observations. Consider the primal (P) and dual (D):

$$P \quad \max c'x \quad \text{s.t.} \quad Ax \leq b, x \geq 0$$

$$D \quad \min b'y \quad \text{s.t.} \quad A'y \geq c, y \geq 0.$$

The dual is formed from the primal by exchanging objective coefficients with the RHS vector; reversing the inequalities in the linear constraints; and transposing the constraint matrix. Taking the dual program associated with the dual gives the primal.

翻译 (Chinese)

原始问题与对偶问题之间的紧密联系背后有一个一般原理。结果表明，两者具有相同的目标函数值。下面的讨论将这些观察形式化。考虑原始问题 (P) 和对偶问题 (D)：

$$P \quad \max c'x \quad \text{满足} \quad Ax \leq b, x \geq 0$$

$$D \quad \min b'y \quad \text{满足} \quad A'y \geq c, y \geq 0.$$

对偶问题由原始问题通过交换目标系数与 RHS 向量、反转线性约束中的不等式以及转置约束矩阵而构成。取与对偶问题相关联的对偶规划即得到原始问题。

Original (English)

Theorem 4.3: For any feasible x and any feasible y , $b'y \geq c'x$.

Theorem 4.4: If $\bar{x}$ and $\bar{y}$ are feasible and satisfy $c'\bar{x} = b'\bar{y}$, then $\bar{x}$ is optimal in the primal and $\bar{y}$ is optimal in the dual.

Theorem 4.5: If $\bar{x}$ and $\bar{y}$ are solutions to the primal and dual respectively, then $c'\bar{x} = b'\bar{y}$.

Theorem 4.6: Suppose that $F_P = \emptyset$, so that there is no x feasible for the primal. Then there is no $\hat{y} \in F_D$ satisfying $b'\hat{y} \leq b'y$ for all $y \in F_D$.

Thus, if either program has an optimal solution, then so does the other, and the objective values are equal: if $\bar{x}$ solves the primal and $\bar{y}$ solves the dual, then $c'\bar{x} = b'\bar{y}$. If one program is not feasible, the other has no optimal solution.

翻译 (Chinese)

定理 4.3: 对任何可行的 x 和任何可行的 y , 有 $b'y \geq c'x$ 。

定理 4.4: 若 $\bar{x}$ 和 $\bar{y}$ 是可行的且满足 $c'\bar{x} = b'\bar{y}$, 则 $\bar{x}$ 是原始问题的最优解, $\bar{y}$ 是对偶问题的最优解。

定理 4.5: 若 $\bar{x}$ 和 $\bar{y}$ 分别是原始问题和对偶问题的解, 则 $c'\bar{x} = b'\bar{y}$ 。

定理 4.6: 假设 $F_P = \emptyset$, 即不存在原始问题的可行解 x 。则不存在 $\hat{y} \in F_D$ 满足对所有 $y \in F_D$ 有 $b'\hat{y} \leq b'y$ 。

因此, 如果任一规划有最优解, 那么另一个也有, 且目标函数值相等: 若 $\bar{x}$ 是原始问题的解, $\bar{y}$ 是对偶问题的解, 则 $c'\bar{x} = b'\bar{y}$ 。如果一个规划不可行, 则另一个没有最优解。

Duality and dual prices / 对偶性与对偶价格

Original (English)

Theorem 4.4 may be used to provide insight on dual prices. More generally, if the parameters b in the dual objective do not define objective iso-cost lines that are parallel to any of the constraints, then small changes in b lead to no change in the optimal solution of the dual. So, given b with solution $\bar{y}$ to the dual program, if some b_i is changed to $b_i + \Delta b_i$, then the solution to the dual remains unchanged at $\bar{y}$ and the value of the dual objective at the optimal solution to the new program is $\sum_{j \neq i} b_j \bar{y}_j + (b_i + \Delta b_i) \bar{y}_i$. Turning to the primal, with initial RHS, b , the optimal solution is $\bar{x}$. Changing the RHS of the i th equation to $b_i + \Delta b_i$ determines a new solution with each $\bar{x}_i$ shifted by a small amount — so that $\bar{x}_i$ moves to $\bar{x}_i + \Delta x_i$, say. From Theorem 4.4,

$$c_1 \bar{x}_1 + c_2 \bar{x}_2 + \cdots + c_n \bar{x}_n = b_1 \bar{y}_1 + b_2 \bar{y}_2 + \cdots + b_m \bar{y}_m$$

and after the RHS change:

$$c_1 (\bar{x}_1 + \Delta x_1) + \cdots + c_n (\bar{x}_n + \Delta x_n) = b_1 \bar{y}_1 + \cdots + (b_i + \Delta b_i) \bar{y}_i + \cdots + b_m \bar{y}_m.$$

Using the pre-change equality and dividing gives:

$$\frac{c_1 \Delta x_1 + c_2 \Delta x_2 + \cdots + c_n \Delta x_n}{\Delta b_j} = \bar{y}_j.$$

Writing the left side as $\frac{\Delta \pi}{\Delta b_j}$, this gives $\frac{\Delta \pi}{\Delta b_j} = \bar{y}_j$. So, for the “regular” case, the change in the objective resulting from a change in the RHS is given by the corresponding dual price.

翻译 (Chinese)

定理 4.4 可用于提供对对偶价格的洞见。更一般地, 如果对偶目标函数中的参数 b 不定义

与任何约束平行的等成本线，那么 b 的微小变化不会导致对偶问题最优解的变化。因此，在原始问题中，初始 RHS b 对应的最优解为 $\bar{x}$ 。将第 i 个方程的 RHS 变为 $b_i + \Delta b_i$ 确定了一个新解，其中每个 $\bar{x}_i$ 移动了很小的量——即 $\bar{x}_i$ 移动到 $\bar{x}_i + \Delta x_i$ 。根据定理 4.4，利用变化前的等式并除以 Δb_j 得：

$$\frac{c_1 \Delta x_1 + c_2 \Delta x_2 + \cdots + c_n \Delta x_n}{\Delta b_j} = \bar{y}_j.$$

将左边写为 $\frac{\Delta \pi}{\Delta b_j}$ ，得 $\frac{\Delta \pi}{\Delta b_j} = \bar{y}_j$ 。因此，对于“正则”情形，RHS 变化所导致的目标函数变化由相应的对偶价格给出。

4.2 Exercises for Chapter 4 / 第 4 章习题

Original (English)

Exercise 4.1 You have a choice of two foods to eat per day. Food A contains 3 grams of vitamin A per ounce, 4 grams of vitamin C and 1 gram of vitamin E. Food B contains 1 gram of vitamin A, 5 grams of vitamin C and 3 grams of vitamin E. The minimum daily requirements of vitamins A, C and E are 22, 66 and 20 grams respectively. You are also on a diet, so you want to minimize your calorie intake. An ounce of food A has 30 calories and an ounce of food B has 20. (a) Formulate the linear programming problem. (b) What are the vertices of the convex set bounded by the constraints? (c) At what point do you minimize your calorie intake subject to the nutritional requirements? (d) Assume that each calorie you consume costs you \$2. What would you pay for a vitamin pill that would reduce the amount of vitamin A you need from food to 20 grams?

翻译 (Chinese)

习题 4.1 你每天可以吃两种食物。食物 A 每盎司含 3 克维生素 A、4 克维生素 C 和 1 克维生素 E。食物 B 每盎司含 1 克维生素 A、5 克维生素 C 和 3 克维生素 E。维生素 A、C 和 E 的每日最低需求量分别为 22、66 和 20 克。你还在节食，因此希望最小化卡路里摄入量。一盎司食物 A 含 30 卡路里，一盎司食物 B 含 20 卡路里。(a) 表述该线性规划问题。(b) 由约束所围成的凸集的顶点是什么？(c) 在满足营养需求的前提下，你在哪一点最小化卡路里摄入量？(d) 假设你摄入的每卡路里花费你 \$2。你愿意为一片能将你从食物中获取的维生素 A 需求量降低至 20 克的维生素片支付多少钱？

Original (English)

Exercise 4.2 For the following systems of constraints, determine and plot the feasible regions.

(a) $4x + 3y \leq 12, x - y \geq 0$

(b) $x \geq 0, y \geq 0, x + y \leq 12$

(c) $2x + y \leq 50, x + 2y \leq 40, x \geq 0, y \geq 0$

(d) $x + 2y \leq 12, 6x + 4y \leq 48, x \geq 0, y \geq 0$

(e) $3x + y \leq 12, 6x + 5y \leq 30, x + 2y \leq 14, x \geq 0, y \geq 0$

(f) $20x + 15y \leq 60,000, 4x + 9y \leq 25,200, 4x + 6y \leq 18,000, 30x + 20y \leq 87,000, x \geq 0, y \geq 0$

翻译 (Chinese)

习题 4.2 对以下约束组，确定并绘制可行区域。

Original (English)

Exercise 4.3 Solve the following linear programming problems:

Max: $5x + 4y$

s.t. $x + 2y \leq 12, 6x + 4y \leq 48, -x + 2y \leq 8, x, y \geq 0.$

Min: $x + 3y$

s.t. $3x + y \leq 12, 6x + 5y \leq 30, x + 2y \leq 14, x, y \geq 0.$

Part I. In the maximization problem, calculate the change in the value of the objective as a result of: (a) Changing the RHS from 12 to 10, and from this deduce the dual price associated with the first constraint. (b) Changing the RHS from 48 to 40, and from this deduce the dual price associated with the second constraint. (c) Changing the RHS from 8 to 7, and from this deduce the dual price associated with the third constraint.

Part II. In the minimization problem, calculate the change in the value of the objective as a result of: (d) Changing the RHS from 12 to 11, and from this deduce the dual price associated with the first constraint. (e) Changing the RHS from 30 to 32, and from this deduce the dual price associated with the second constraint. (f) Changing the RHS from 14 to 15, and from this deduce the dual price associated with the third constraint.

Part III. For both problems, introduce slack variables and write the programs as equality-constrained problems. In both cases: (g) Identify all the basic solutions. (h) Identify all the basic feasible solutions.

翻译 (Chinese)

习题 4.3 求解以下线性规划问题：（见上）

第一部分：在最大化问题中，计算因以下变化而导致的目标函数值的变化：(a) (c) 分别为改变各约束的 RHS 并推导相应的对偶价格。

第二部分：在最小化问题中，做类似计算。

第三部分：对两个问题，引入松弛变量并将规划写为等式约束问题，识别所有基本解和基本可行解。

Original (English)

Exercise 4.4–4.16 Additional exercises on linear programming: solving programs, identifying basic solutions and basic feasible solutions, calculating dual prices, plotting constraints, and analyzing parameter changes. (See textbook for full exercise statements.)

翻译 (Chinese)

习题 4.4–4.16 关于线性规划的附加习题：求解规划、识别基本解和基本可行解、计算对偶价格、绘制约束以及分析参数变化。（完整习题陈述见教材。）

第五章 Functions of One Variable / 一元函数

5.1 Introduction / 引论

Original (English)

Functions arise naturally in the study of economic problems. For example, a demand function relates quantity demanded to price. Properties of such functions are routinely studied — the slope and elasticity measure responsiveness of demand to price variations. The techniques presented here introduce the tools required for such calculations. Section 5.2 describes a variety of functions arising in different areas (taxation, finance and present value calculations). Section 5.3 introduces the basic notions of continuity and differentiability of a function. Certain families of functions, such as logarithmic, polynomial and exponential functions, appear frequently in economic applications; some of these are introduced in Section 5.4 and their shapes described. Growth rates and the algebra of growth calculations are developed in Section 5.5. Finally, Section 5.6 provides rules for the differentiation where the function is itself formed from other functions (such as the ratio or product of functions).

翻译 (Chinese)

函数自然地出现在经济问题的研究中。例如，需求函数将需求量与价格联系起来。此类函数的性质被常规地研究——斜率和弹性衡量需求对价格变化的响应程度。这里介绍的技术提供了进行此类计算所需的工具。第 5.2 节描述了在不同领域（税收、金融和现值计算）中出现的各种函数。第 5.3 节介绍函数的连续性和可微性的基本概念。某些函数族，如对数函数、多项式函数和指数函数，在经济应用中频繁出现；其中一些在第 5.4 节中介绍并描述了它们的形状。增长率和增长计算的代数在第 5.5 节中展开。最后，第 5.6 节提供了当函数本身由其他函数构成（如函数的比率或乘积）时的微分法则。

Original (English)

A function, f , from a set X to a set Y is a rule associating with each element of a set X some element in the set Y . This is written $f : X \rightarrow Y$. In words, to each point in X , f associates a point, $y = f(x)$ in Y . The set X is called the domain and the set Y is called the range. The inverse demand function, relating price to quantity demanded, $p(Q)$, is a familiar example. The domain is the set of non-negative numbers, $\mathbb{R}_+$ (quantity must be positive), and the range is also the set of non-negative numbers (price is non-negative). Thus, $p : \mathbb{R}_+ \rightarrow \mathbb{R}_+$.

翻译 (Chinese)

从集合 X 到集合 Y 的函数 f 是一个规则，它将集合 X 的每个元素与集合 Y 中的某个元素相关联。记作 $f : X \rightarrow Y$ 。换言之，对于 X 中的每个点， f 关联着 Y 中的一个点 $y = f(x)$ 。集合 X 称为定义域，集合 Y 称为值域。将价格与需求量联系起来的反需求函数 $p(Q)$ 是一个熟悉的例子。定义域是非负实数集 $\mathbb{R}_+$ （数量必须为正），值域也是非负实数集（价格非负）。因此， $p : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ 。

5.2 Some examples / 若干例子**5.2.1 Demand functions / 需求函数****Original (English)**

Consumer demand theory describes an individual in terms of utility derived from consumption of a vector of goods $(x_1, x_2, \dots, x_n)$. With a fixed income y and price p_i for good i , utility maximization leads to choices for the goods that depend on prices and income: $x_i(p_1, p_2, \dots, p_n, y)$. In the two-good case, utility maximization leads to the selection of (x_1, x_2) . The values of x_1 and x_2 depend on the prices (p_1, p_2) and income, y . These are called the demand functions and are written $x_1(p_1, p_2, y)$ and $x_2(p_1, p_2, y)$.

With J individuals, write the demand for good i by individual j as $x_i^j(p_1, p_2, \dots, p_n, y^j)$, where y^j is income of individual j . Then, the aggregate demand for good i is:

$$X_i(p_1, p_2, \dots, p_n, y^1, y^2, \dots, y^J) = \sum_{j=1}^J x_i^j(p_1, p_2, \dots, p_n, y^j).$$

If we consider the case where all prices other than that of good j , $p_j = \bar{p}_j$, $j \neq i$, and all incomes, $y^j = \bar{y}^j$, are fixed then the aggregate demand is:

$$D_i(p_i) = X_i(\bar{p}_1, \bar{p}_2, \dots, \bar{p}_{i-1}, p_i, \bar{p}_{i+1}, \dots, \bar{p}_n, \bar{y}^1, \bar{y}^2, \dots, \bar{y}^J).$$

For simplicity, ignore the subscripts and write, $D(p)$, for the aggregate demand as a function of own price. Writing Q for the quantity demanded, $Q = D(p)$. This gives aggregate demand for the good as a function of the good's price. With this, the responsiveness of demand to price variation may be considered. The slope of the demand function is $\frac{Q'-Q}{p'-p} = \frac{D(p')-D(p)}{p'-p}$. Similarly, the percentage change in quantity resulting from a percentage change in price is given by $\frac{Q'-Q}{Q} / \frac{p'-p}{p}$. The negative of this term gives the elasticity of demand:

$$\varepsilon(p) = -\frac{p}{D(p)} \frac{D(p') - D(p)}{p' - p}.$$

翻译 (Chinese)

消费者需求理论用从商品向量 $(x_1, x_2, \dots, x_n)$ 的消费中获得的效用来描述个体。在固定收入 y 和商品 i 的价格 p_i 下, 效用最大化导致商品的选择依赖于价格和收入: $x_i(p_1, p_2, \dots, p_n, y)$ 。在两商品情形中, 效用最大化导致选择 (x_1, x_2) 。 x_1 和 x_2 的值取决于价格 (p_1, p_2) 和收入 y 。这些称为需求函数, 记作 $x_1(p_1, p_2, y)$ 和 $x_2(p_1, p_2, y)$ 。

在有 J 个个体的情况下, 将个体 j 对商品 i 的需求记作 $x_i^j(p_1, p_2, \dots, p_n, y^j)$ 。则商品 i 的总需求为上述求和形式。

如果考虑除商品 j 外的所有价格 $p_j = \bar{p}_j$, $j \neq i$, 以及所有收入 $y^j = \bar{y}^j$ 都固定的情况, 则总需求为 $D_i(p_i)$ 。为简便起见, 忽略下标, 将总需求作为自身价格的函数记作 $D(p)$ 。记 Q 为需求量, $Q = D(p)$ 。需求的斜率由 $\frac{Q'-Q}{p'-p}$ 给出。类似地, 价格百分比变化导致的数量百分比变化给出了需求弹性。

5.2.2 Present value and the interest rate / 现值与利率

Original (English)

The present value of a cash flow may be viewed as a function of the interest rate. A bond is characterized by a face value P , an interest rate κ and a maturity date T , and it pays the holder a fixed payment, $f = \kappa P$, each time period, and the face value P at the end of the T th period. Thus, the cash flow generated by the bond is: $(f, f, f, \dots, f, P)$, where f is paid at the end of periods 1, 2, up to $T - 1$. If the market interest rate is r , x dollars invested today for one period is worth $y = x(1 + r)$ at the end of the period. Put differently, to have y dollars at the end of the period, $x = \frac{y}{(1+r)}$ dollars should be invested now.

The current value of a payoff y one period from now is $\frac{y}{1+r}$. The same reasoning shows that the current value of a payoff y t periods from now is $\frac{y}{(1+r)^t}$. Therefore, the flow $(f, f, f, \dots, f, P)$ has

current market value of:

$$V = \frac{f}{1+r} + \frac{f}{(1+r)^2} + \cdots + \frac{f}{(1+r)^t} + \cdots + \frac{f}{(1+r)^{T-1}} + \frac{P}{(1+r)^T}.$$

One simple case occurs when the payments continue indefinitely (and there is no end period at which P would be paid). In this case,

$$V = f \frac{1}{1+r} \left[1 + \frac{1}{1+r} + \frac{1}{(1+r)^2} + \cdots \right].$$

Using the fact that for $0 < \gamma < 1$, $1 + \gamma + \gamma^2 + \cdots = \frac{1}{1-\gamma}$, when $\gamma = \frac{1}{1+r}$, the infinite sum becomes $\frac{1}{1-\frac{1}{1+r}} = \frac{1+r}{r}$. Therefore, $V = f \frac{1}{1+r} \cdot \frac{1+r}{r} = \frac{f}{r}$.

Write $V(r)$ to denote this dependence, the valuing of the bond as a function of the current market interest rate: $V(r) = \frac{f}{r}$.

翻译 (Chinese)

现金流的现值可以看作是利率的函数。债券的特征是面值 P 、利率 κ 和到期日 T ，它每期向持有者支付固定金额 $f = \kappa P$ ，并在第 T 期末支付面值 P 。因此，债券产生的现金流为： $(f, f, f, \dots, f, P)$ ，其中 f 在第 1, 2 直至 $T-1$ 期末支付。如果市场利率为 r ，今天投资 x 美元一期后价值为 $y = x(1+r)$ 。换句话说，要在一期末获得 y 美元，现在应投资 $x = \frac{y}{(1+r)}$ 美元。

一期后支付 y 的当前价值为 $\frac{y}{1+r}$ 。同样的推理表明， t 期后支付 y 的当前价值为 $\frac{y}{(1+r)^t}$ 。因此，现金流 $(f, f, f, \dots, f, P)$ 的当前市场价值如上所示。一种简单情况是支付无限期持续（没有支付 P 的结束期）。此时 $V = \frac{f}{r}$ 。将这种依赖性记为 $V(r)$ ： $V(r) = \frac{f}{r}$ 。

5.2.3 Taxation / 税收

Original (English)

In a market with supply and demand curves $S(p)$ and $D(p)$, the introduction of a sales tax creates a wedge between the price paid and the price received. With $D(p)$ and supply $S(p)$, the market clearing price is p_0 , determined as the solution to: $D(p_0) = S(p_0)$. Suppose now that an ad-valorem tax is introduced, so that if the seller receives p_t , the buyer pays $(1+t)p_t$. Let $t_0 = 0$ be the tax initially (no tax) and t the tax imposed. Market clearing requires that the amount supplied at price p_t equals the amount demanded at price $(1+t)p_t$: $D((1+t)p_t) = S(p_t)$. With $Q(0) = D(p_0) = S(p_0)$ and $Q(t) = D((1+t)p_t) = S(p_t)$, the variation on quantity traded as a result of the tax increment

is $Q(t) - Q(0) < 0$, and the rate of change of quantity traded relative to the tax variation is given by $\frac{Q(t)-Q(0)}{t}$.

翻译 (Chinese)

在具有供给曲线 $S(p)$ 和需求曲线 $D(p)$ 的市场中, 引入销售税会在支付价格和收到价格之间造成一个楔子。由 $D(p_0) = S(p_0)$ 确定市场出清价格 p_0 。现在假设引入从价税, 使得如果卖方收到 p_t , 买方支付 $(1+t)p_t$ 。市场出清要求价格为 p_t 时的供给量等于价格为 $(1+t)p_t$ 时的需求量: $D((1+t)p_t) = S(p_t)$ 。由于税收增加导致的交易量变化为 $Q(t) - Q(0) < 0$, 交易量相对于税收变化的变化率由 $\frac{Q(t)-Q(0)}{t}$ 给出。

5.2.4 Options / 期权

Original (English)

At any point in time, a stock trades at some price p , which varies over time. Derivative assets can be created as a function of the stock price p , $f(p)$. Such derivative assets are called “options”. In finance, an option gives the owner the right to buy or sell an asset. An option giving the right to purchase an asset at some specified price, $\bar{p}$, called the exercise or strike price, at (or before) a specified date, T , is a call option. An option giving the right to sell is called a put option. When the right can be exercised only at the date T , the option is a European option. When the right can be exercised on or before the date T , the option is an American option.

Suppose that the strike price of a European call option is $\bar{p}$ and that $\tilde{p}$ is the (as yet unknown) stock price at period T . Then, if the price $\tilde{p}$ turns out to be above $\bar{p}$, the owner can exercise the right to purchase at price $\bar{p}$ and sell on the market at price $\tilde{p}$, so the net gain is $\tilde{p} - \bar{p}$. If $\tilde{p} < \bar{p}$, the owner will choose not to exercise, and the net benefit is 0. So, the return per call option owned is $\pi(\tilde{p}) = \max\{\tilde{p} - \bar{p}, 0\}$.

In contrast, a put option gives the right to sell a unit of the asset at a strike price $\hat{p}$. This has a return of $\pi(\tilde{p}) = \max\{\hat{p} - \tilde{p}, 0\}$. Such options may be combined to provide more complex return streams.

翻译 (Chinese)

在任何时点, 股票以某个价格 p 交易, 该价格随时间变化。衍生资产可以作为股票价格 p 的函数 $f(p)$ 来创建。此类衍生资产称为“期权”。在金融学中, 期权赋予持有者买入或卖出一项资产的权利。赋予在指定日期 T (或之前) 以指定价格 $\bar{p}$ (称为执行价或行权价) 购买

资产的权利的期权是看涨期权。赋予出售权利的期权称为看跌期权。当该权利只能在日期 T 行使时，该期权是欧式期权。当该权利可以在日期 T 或之前行使时，该期权是美式期权。

假设欧式看涨期权的行权价为 $\bar{p}$ ， $\tilde{p}$ 为 T 时刻的股票价格。如果 $\tilde{p} > \bar{p}$ ，持有者可以行权以价格 $\bar{p}$ 购买并以市场价格 $\tilde{p}$ 出售，净收益为 $\tilde{p} - \bar{p}$ 。如果 $\tilde{p} < \bar{p}$ ，持有者将选择不行权，净收益为 0。因此，每份看涨期权的回报为 $\pi(\tilde{p}) = \max\{\tilde{p} - \bar{p}, 0\}$ 。相比之下，看跌期权赋予以行权价 $\hat{p}$ 出售资产的权利，其回报为 $\pi(\tilde{p}) = \max\{\hat{p} - \tilde{p}, 0\}$ 。

5.3 Continuity and differentiability / 连续性与可微性

5.3.1 Preliminaries / 预备知识

Original (English)

Definition 5.1: Functions may be categorized in terms of surjectivity, injectivity and bijectivity.

- If $f(X) = Y$, where $f(X) = \{y \mid \text{there is an } x \in X, f(x) = y\}$, we say f is onto Y or surjective. In words, for any point $y \in Y$, there is some $x \in X$ with $f(x) = y$.
- The function f is called 1-1 (one-to-one) or injective if $f(x_1) = f(x_2)$ implies that $x_1 = x_2$; a 1-1 function cannot have the same value at two different points in the domain.
- If f is both 1-1 and onto, it is called bijective. If f is a bijective mapping, there is a function $g : Y \rightarrow X$, called the inverse of f with $g(f(x)) = x$ and $f(g(y)) = y$. The function g is written f^{-1} .

翻译 (Chinese)

定义 5.1: 函数可以按照满射性、单射性和双射性进行分类。

- 若 $f(X) = Y$ ，其中 $f(X) = \{y \mid \text{存在 } x \in X, f(x) = y\}$ ，则称 f 是到 Y 上的或满射的。换言之，对 Y 中的任意点 y ，存在某个 $x \in X$ 使得 $f(x) = y$ 。
- 若 $f(x_1) = f(x_2)$ 蕴含 $x_1 = x_2$ ，则称函数 f 为 1-1（一一对应）或单射的；一一函数不能在定义域的两个不同点取相同的值。
- 若 f 既是 1-1 又是满射，则称其为双射。若 f 是双射映射，则存在函数 $g : Y \rightarrow X$ ，称为 f 的逆，满足 $g(f(x)) = x$ 和 $f(g(y)) = y$ 。函数 g 记作 f^{-1} 。

5.3.2 Inequalities and absolute value / 不等式与绝对值

Original (English)

For a real number a , the expression $|a|$ is called the absolute value of a , and is defined: $|a| = a$, if $a \geq 0$ and $|a| = -a$, if $a < 0$:

$$|a| = \begin{cases} -a & \text{if } a < 0, \\ a & \text{if } a \geq 0. \end{cases}$$

Proposition 5.1: The absolute value operation satisfies a number of simple rules. For real numbers, a , b and c :

1. If $a \geq 0$, then $|x| \leq a$ if and only if $-a \leq x \leq a$;
2. $|a + b| = |b + a|$;
3. $|a \cdot b| = |a||b|$;
4. $|a + b| \leq |a| + |b|$;
5. $|a + b| \geq |a| - |b|$;
6. $|\frac{a}{b}| = \frac{|a|}{|b|}$, if $b \neq 0$.

翻译 (Chinese)

对于实数 a , 表达式 $|a|$ 称为 a 的绝对值, 定义为: 若 $a \geq 0$ 则 $|a| = a$, 若 $a < 0$ 则 $|a| = -a$ 。

命题 5.1: 绝对值运算满足若干简单规则。对实数 a , b 和 c :

1. 若 $a \geq 0$, 则 $|x| \leq a$ 当且仅当 $-a \leq x \leq a$;
2. $|a + b| = |b + a|$;
3. $|a \cdot b| = |a||b|$;
4. $|a + b| \leq |a| + |b|$;
5. $|a + b| \geq |a| - |b|$;
6. $|\frac{a}{b}| = \frac{|a|}{|b|}$, 若 $b \neq 0$ 。

5.3.3 Continuity / 连续性

Original (English)

Definition 5.2: Let $x_1, x_2, \dots, x_n = \{x_n\}_{n=1}^{\infty}$ be a sequence of real numbers. We say that x_n converges to a real number c if given any positive number, $\varepsilon > 0$, however small, there is an integer n^* , such that $|x_n - c| < \varepsilon$, whenever $n \geq n^*$. This is written $|x_n - c| \rightarrow 0$ as $n \rightarrow \infty$, or $x_n \rightarrow c$, or $\lim_n x_n = c$.

Definition 5.3: A function f is continuous at c if given $\varepsilon > 0$, $\exists \delta > 0$ such that

$$|f(x) - f(c)| < \varepsilon \quad \text{whenever} \quad |x - c| < \delta.$$

In this case, write $\lim_{x \rightarrow c} f(x) = f(c)$. Call f continuous, if f is continuous at every $c \in X$.

翻译 (Chinese)

定义 5.2: 设 $x_1, x_2, \dots, x_n = \{x_n\}_{n=1}^{\infty}$ 为一实数序列。若给定任意正数 $\varepsilon > 0$, 无论多小, 都存在整数 n^* , 使得当 $n \geq n^*$ 时 $|x_n - c| < \varepsilon$, 则称 x_n 收敛于实数 c 。记作 $|x_n - c| \rightarrow 0$ 当 $n \rightarrow \infty$, 或 $x_n \rightarrow c$, 或 $\lim_n x_n = c$ 。

定义 5.3: 若给定 $\varepsilon > 0$, $\exists \delta > 0$ 使得当 $|x - c| < \delta$ 时 $|f(x) - f(c)| < \varepsilon$, 则函数 f 在 c 处连续。此时记 $\lim_{x \rightarrow c} f(x) = f(c)$ 。若 f 在每一 $c \in X$ 处均连续, 则称 f 连续。

Original (English)

Some simple rules follow from the definitions. If $\lim_{x \rightarrow c} f(x) = a$ and $\lim_{x \rightarrow c} g(x) = b$, then:

1. For any constant λ , $\lim_{x \rightarrow c} \lambda f(x) = \lambda a$.
2. $\lim_{x \rightarrow c} [f(x) + g(x)] = a + b$,
3. $\lim_{x \rightarrow c} f(x) \cdot g(x) = ab$,
4. $\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \frac{a}{b}$, if $b \neq 0$.
5. If $f : X \rightarrow Y$, and $g : Y \rightarrow Z$ are continuous functions, then $h : X \rightarrow Z$, defined by $h(x) = g(f(x))$ is a continuous function: $\lim_{x \rightarrow c} h(x) = h(c)$.

翻译 (Chinese)

由定义可得一些简单规则。若 $\lim_{x \rightarrow c} f(x) = a$ 且 $\lim_{x \rightarrow c} g(x) = b$, 则:

1. 对任意常数 λ , $\lim_{x \rightarrow c} \lambda f(x) = \lambda a$ 。
2. $\lim_{x \rightarrow c} [f(x) + g(x)] = a + b$,
3. $\lim_{x \rightarrow c} f(x) \cdot g(x) = ab$,
4. $\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \frac{a}{b}$, 若 $b \neq 0$ 。
5. 若 $f : X \rightarrow Y$ 和 $g : Y \rightarrow Z$ 是连续函数, 则由 $h(x) = g(f(x))$ 定义的 $h : X \rightarrow Z$ 是连续函数: $\lim_{x \rightarrow c} h(x) = h(c)$ 。

5.3.4 Differentiability / 可微性

Original (English)

Definition 5.4: The function f is differentiable at $\bar{x}$ if, as x converges to $\bar{x}$, $\frac{f(x)-f(\bar{x})}{(x-\bar{x})}$ converges to a number. Formally, f is differentiable at $\bar{x}$ if the following limit exists:

$$\lim_{x \rightarrow \bar{x}} \frac{f(x) - f(\bar{x})}{x - \bar{x}} = \lim_{\Delta x \rightarrow 0} \frac{f(\bar{x} + \Delta x) - f(\bar{x})}{\Delta x} = \lim_{\varepsilon \rightarrow 0} \frac{f(\bar{x} + \varepsilon) - f(\bar{x})}{\varepsilon}.$$

If the limit exists it is denoted $\frac{df(\bar{x})}{dx}$ or $f'(\bar{x})$ and is called the derivative of f .

翻译 (Chinese)

定义 5.4: 若当 x 收敛于 $\bar{x}$ 时, $\frac{f(x)-f(\bar{x})}{(x-\bar{x})}$ 收敛于一个数, 则函数 f 在 $\bar{x}$ 处可微。形式上, 若下列极限存在, 则 f 在 $\bar{x}$ 处可微:

$$\lim_{x \rightarrow \bar{x}} \frac{f(x) - f(\bar{x})}{x - \bar{x}} = \lim_{\Delta x \rightarrow 0} \frac{f(\bar{x} + \Delta x) - f(\bar{x})}{\Delta x} = \lim_{\varepsilon \rightarrow 0} \frac{f(\bar{x} + \varepsilon) - f(\bar{x})}{\varepsilon}.$$

若该极限存在, 则记作 $\frac{df(\bar{x})}{dx}$ 或 $f'(\bar{x})$, 并称为 f 的导数。

Original (English)

Remark 5.1: It is always the case that differentiability of a function implies that the function is continuous. The converse is not true. For example, $|x|$ is continuous but not differentiable at $x = 0$.

Remark 5.2: Differentiability plays an important role in locating the maximum or minimum of a function. Consider $f(x) = x(1 - x) = x - x^2$, so $f'(x) = 1 - 2x$. Thus $f'(x)$ is positive when $x < \frac{1}{2}$, negative when $x > \frac{1}{2}$ and equal to 0 at $x = \frac{1}{2}$. This suggests that the use of derivatives will be helpful in finding maxima (and minima) of functions.

翻译 (Chinese)

注记 5.1: 函数的可微性总是蕴含该函数连续。反之则不成立。例如, $|x|$ 是连续的但在 $x = 0$ 处不可微。

注记 5.2: 可微性在定位函数的最大值或最小值中起着重要作用。考虑 $f(x) = x(1 - x) = x - x^2$, 则 $f'(x) = 1 - 2x$ 。因此当 $x < \frac{1}{2}$ 时 $f'(x)$ 为正, 当 $x > \frac{1}{2}$ 时为负, 在 $x = \frac{1}{2}$ 处等于 0。这表明导数的使用将有助于寻找函数的极大值 (和极小值)。

5.4 Types of function / 函数的类型

Original (English)

Exponential functions: The exponential function is defined $f(x) = e^x$, where $e \approx 2.718$. More generally, exponential functions have the functional form $f(x) = a^x$ or $f(x) = ba^{cx}$.

Polynomial functions: A polynomial function is a function of the form: $f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$.

Trigonometric functions: The three primary trigonometric functions are the sine, cosine and tangent functions.

Logarithmic functions: The logarithm of x to the base b is that number y such that $x = b^y$. One special case is where $b = e$, Euler's number. Routinely, $\log_e x$ is written $\ln x$.

翻译 (Chinese)

指数函数: 指数函数定义为 $f(x) = e^x$, 其中 $e \approx 2.718$ 。更一般地, 指数函数具有形式 $f(x) = a^x$ 或 $f(x) = ba^{cx}$ 。

多项式函数: 多项式函数是形如 $f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$ 的函数。

三角函数: 三个基本三角函数是正弦、余弦和正切函数。

对数函数: 以 b 为底的 x 的对数是满足 $x = b^y$ 的数 y 。一种特殊情况是 $b = e$ (欧拉数)。通常将 $\log_e x$ 写作 $\ln x$ 。

Plots of various functions / 各类函数的图像

Original (English)

The following table gives some well known functions and their derivatives.

$f(x)$	x^a	a^x	$\log_e x$	e^{ax}	$\sin(x)$	$\cos(x)$	$\tan(x)$
$f'(x)$	ax^{a-1}	$a^x \cdot \ln(a)$	$\frac{1}{x}$	ae^{ax}	$\cos(x)$	$-\sin(x)$	$\frac{1}{\cos^2(x)}$

翻译 (Chinese)

下表给出了一些众所周知的函数及其导数。

5.4.1 The log and exponential functions / 对数函数与指数函数

Original (English)

Definition 5.5: The log of a number x to the base b is the power to which b must be raised to equal that number. Thus, $a = \log_b x$ means $x = b^a$. So, $\log_b x$ is the number that satisfies $x = b^{\log_b x}$.

Proposition 5.2: The log function satisfies:

1. The log of x^a to base b is equal to a times the log of x to the base b : $\log_b x^a = a \log_b x$.
2. The log of a product is equal to the sum of the logs: $\log_b xy = \log_b x + \log_b y$.
3. The log of a ratio is equal to the difference of the logs: $\log_b \frac{x}{y} = \log_b x - \log_b y$.
4. Change of base: $\log_c x = \log_c b \cdot \log_b x$, or $\frac{\log_c x}{\log_c b} = \log_b x$.

翻译 (Chinese)

定义 5.5: 以 b 为底的数 x 的对数是使得 b 的该次幂等于该数的幂。因此, $a = \log_b x$ 意味着 $x = b^a$ 。故 $\log_b x$ 是满足 $x = b^{\log_b x}$ 的数。

命题 5.2: 对数函数满足:

1. x^a 以 b 为底的对数等于 a 乘以 x 以 b 为底的对数: $\log_b x^a = a \log_b x$ 。
2. 乘积的对数等于对数之和: $\log_b xy = \log_b x + \log_b y$ 。
3. 比率的对数等于对数之差: $\log_b \frac{x}{y} = \log_b x - \log_b y$ 。
4. 换底公式: $\log_c x = \log_c b \cdot \log_b x$ 。

Original (English)

Remark 5.3: The mathematical constant $e \approx 2.71828$, Euler's number, may be defined in a number of ways:

$$e = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots = \sum_{j=0}^{\infty} \frac{x^j}{j!} \quad (\text{evaluated at } x = 1).$$

Yet another equivalent definition gives e as the limit of a sequence of numbers:

$$e = \lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x.$$

翻译 (Chinese)

注记 5.3: 数学常数 $e \approx 2.71828$, 即欧拉数, 可以用多种方式定义。另一种等价定义将 e 作为数列的极限:

$$e = \lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x.$$

5.5 Growth rates / 增长率

Original (English)

Suppose an investor invests one dollar for a year at a 100% interest rate. If the interest is calculated once at the end of the year, the loan is worth $1 \times (1 + r) = 2$. With n rounds of interest calculation, the value of the loan at the end of the year is $(1 + \frac{1}{n})^n$. As n becomes large, this approaches e .

Suppose now that the interest rate is r instead of 1. With n rounds of interest calculation, the final value is $(1 + \frac{r}{n})^{nt}$. From the previous calculations:

$$\left(1 + \frac{r}{n}\right)^{nt} \rightarrow (e^r)^t = e^{rt}.$$

Thus, the value of P dollars invested now with continuous interest accumulation is $V = Pe^{rt}$. Conversely, V dollars in t periods with continuous discounting is worth $Ve^{-rt}(=P)$ now.

翻译 (Chinese)

假设一位投资者以 100% 的利率投资一美元一年。如果利息在年底计算一次，贷款价值为 $1 \times (1 + r) = 2$ 。经过 n 轮利息计算，年底贷款价值为 $(1 + \frac{1}{n})^n$ 。当 n 变得很大时，这趋近于 e 。

现在假设利率为 r 而非 1。经过 n 轮利息计算，最终价值为 $(1 + \frac{r}{n})^{nt}$ 。根据之前的计算，这趋近于 e^{rt} 。因此，现在投资 P 美元在连续计息下的价值为 $V = Pe^{rt}$ 。反之， t 期后的 V 美元按连续贴现的当前价值为 $Ve^{-rt}(=P)$ 。

5.5.1 Additivity of continuous growth rates / 连续增长率的可加性

Original (English)

One advantage of using continuous compound rates is that growth rates are additive. Consider three time periods, $s < t < u$, with some variable observed at those three dates: y_s, y_t, y_u . Using the continuous growth formula:

$$\frac{y_t}{y_s} = e^{\gamma_{st}}, \quad \frac{y_u}{y_t} = e^{\gamma_{tu}}, \quad \frac{y_u}{y_s} = e^{\gamma_{su}},$$

where γ_{ij} is the continuous growth rate from time i to time j . Then, multiplying,

$$e^{\gamma_{su}} = \frac{y_u}{y_s} = \frac{y_t}{y_s} \cdot \frac{y_u}{y_t} = e^{\gamma_{st}} e^{\gamma_{tu}} = e^{\gamma_{st} + \gamma_{tu}},$$

so that $\gamma_{su} = \gamma_{st} + \gamma_{tu}$. Thus, the growth rate over the full period is equal to the sum of the growth rates in the two subperiods.

翻译 (Chinese)

使用连续复利的一个优点是增长率具有可加性。考虑三个时点 $s < t < u$ ，在这些日期观测

到的某个变量为 y_s, y_t, y_u 。使用连续增长公式: $\frac{y_t}{y_s} = e^{\gamma_{st}}$, 等等。其中 γ_{ij} 是从时刻 i 到 j 的连续增长率。然后相乘得 $e^{\gamma_{su}} = e^{\gamma_{st} + \gamma_{tu}}$, 故 $\gamma_{su} = \gamma_{st} + \gamma_{tu}$ 。因此, 整个时期的增长率等于两个子时期增长率之和。

5.5.2 The derivatives of e^x and $\ln x$ / e^x 与 $\ln x$ 的导数

Original (English)

Given a constant k , the derivative of e^{kx} is given by

$$\frac{d}{dx}\{e^{kx}\} = ke^{kx}.$$

The derivative of $\ln x$ is given by

$$\frac{d}{dx}\{\ln x\} = \frac{1}{x}.$$

翻译 (Chinese)

给定常数 k , e^{kx} 的导数为 $\frac{d}{dx}\{e^{kx}\} = ke^{kx}$ 。 $\ln x$ 的导数为 $\frac{d}{dx}\{\ln x\} = \frac{1}{x}$ 。

5.6 Some rules of calculus / 微积分的一些法则

Original (English)

Some basic rules of calculus apply for the differentiation of the product, ratio and composition of continuous functions.

1. The derivative of a product of functions:

$$\frac{d}{dx}\{f(x)g(x)\} = f'(x)g(x) + g'(x)f(x).$$

2. The derivative of a ratio of functions ($g(x) \neq 0$):

$$\frac{d}{dx}\left\{\frac{f(x)}{g(x)}\right\} = \frac{g(x)f'(x) - f(x)g'(x)}{[g(x)]^2}.$$

3. The chain rule (differentiating the composition of functions):

$$\frac{d}{dx}\{f(g(x))\} = f'(g(x))g'(x).$$

In the context of the exponential and log functions, the chain rule gives:

$$\frac{d}{dx} e^{f(x)} = f'(x)e^{f(x)}, \quad \frac{d}{dx} \ln f(x) = \frac{1}{f(x)} f'(x) \quad \text{given } f(x) > 0.$$

翻译 (Chinese)

微积分的一些基本法则适用于连续函数的乘积、比率和复合的微分。

1. 函数乘积的导数: $\frac{d}{dx} \{f(x)g(x)\} = f'(x)g(x) + g'(x)f(x)$ 。
2. 函数比率的导数 ($g(x) \neq 0$): $\frac{d}{dx} \left\{ \frac{f(x)}{g(x)} \right\} = \frac{g(x)f'(x) - f(x)g'(x)}{[g(x)]^2}$ 。
3. 链式法则 (复合函数的微分): $\frac{d}{dx} \{f(g(x))\} = f'(g(x))g'(x)$ 。

在指数函数和对数函数的背景下, 链式法则给出: $\frac{d}{dx} e^{f(x)} = f'(x)e^{f(x)}$, $\frac{d}{dx} \ln f(x) = \frac{1}{f(x)} f'(x)$ (假设 $f(x) > 0$)。

5.6.1 l'Hôpital's rule / 洛必达法则

Original (English)

Theorem 5.1: Suppose that (i) f and g are differentiable on $(a - \varepsilon, a + \varepsilon) \setminus \{a\}$ for some $\varepsilon > 0$, (ii) $\lim_{x \rightarrow a} f(x) = 0$ and $\lim_{x \rightarrow a} g(x) = 0$, (iii) $g'(x) \neq 0$ on $(a - \varepsilon, a + \varepsilon) \setminus \{a\}$. Then,

$$\lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} = k \quad \text{implies} \quad \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = k.$$

Similarly, suppose that (i) f and g are differentiable on $(a - \varepsilon, a + \varepsilon) \setminus \{a\}$ for some $\varepsilon > 0$, (ii) $\lim_{x \rightarrow a} f(x) = \infty$ and $\lim_{x \rightarrow a} g(x) = \infty$, (iii) $g'(x) \neq 0$ on $(a - \varepsilon, a + \varepsilon) \setminus \{a\}$. Then,

$$\lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} = k \quad \text{implies} \quad \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = k.$$

Thus, under general conditions, the limit of the ratio is equal to the limit of the ratio of the derivatives.

翻译 (Chinese)

定理 5.1: 假设 (i) f 和 g 在 $(a - \varepsilon, a + \varepsilon) \setminus \{a\}$ 上对某 $\varepsilon > 0$ 可微, (ii) $\lim_{x \rightarrow a} f(x) = 0$ 且 $\lim_{x \rightarrow a} g(x) = 0$, (iii) 在 $(a - \varepsilon, a + \varepsilon) \setminus \{a\}$ 上 $g'(x) \neq 0$ 。则 $\lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} = k$ 蕴含 $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = k$ 。类似地, 可以对 $f(x)$ 和 $g(x)$ 均趋于无穷的情形陈述类似结果。因此, 在一般条件下, 比率的极限等于导数比率的极限。

5.6.2 Higher derivatives / 高阶导数

Original (English)

For notational convenience, the second derivative is denoted $f''(x)$. Technically, the second derivative is defined in exactly the same way as the first derivative:

$$f''(x) = \lim_{\varepsilon \rightarrow 0} \frac{f'(x + \varepsilon) - f'(x)}{\varepsilon}.$$

Thus, the second derivative is the derivative of the first derivative. Likewise, higher-order derivatives may be determined step by step. Properties of the second derivative are central in the discussion of optimization. If the second derivative is positive, the slope of the function is increasing at that point; if the second derivative is negative, the slope of the function is decreasing.

翻译 (Chinese)

为记号方便, 二阶导数记作 $f''(x)$ 。技术上, 二阶导数的定义与一阶导数完全相同: $f''(x) = \lim_{\varepsilon \rightarrow 0} \frac{f'(x+\varepsilon) - f'(x)}{\varepsilon}$ 。因此, 二阶导数是一阶导数的导数。类似地, 高阶导数可以逐步确定。二阶导数的性质在最优化讨论中居于核心地位。若二阶导数为正, 函数在该点的斜率递增; 若二阶导数为负, 函数的斜率递减。

5.7 Exercises for Chapter 5 / 第 5 章习题

Original (English)

Exercise 5.1–5.14 Exercises on continuity, differentiability, l'Hôpital's rule, derivatives, growth rates, elasticity, and applications. (See textbook for full exercise statements.)

翻译 (Chinese)

习题 5.1–5.14 关于连续性、可微性、洛必达法则、导数、增长率、弹性及其应用的习题。(完整习题陈述见教材。)

第六章 Functions of One Variable: Applications / 一元函数的应用

6.1 Introduction / 引论

Original (English)

In this chapter, calculus techniques are applied to the study of a variety of problems, beginning with optimization in Section 6.2. In an optimization problem, given a function f of one variable x , the problem is to find the value of x , $\bar{x}$ that maximizes f . Such an $\bar{x}$ is characterized by the condition that $f(\bar{x}) \geq f(x)$ for all x . If the choice of x is restricted to some set X , then the requirement is that $f(\bar{x}) \geq f(x)$ for all $x \in X$. Calculus techniques are used to find $\bar{x}$ by considering how f varies with x . These techniques are local in nature and so locate “local” maxima (or minima). In contrast, $\bar{x}$ is a global maximizer if $f(\bar{x})$ is as large as $f(x)$ for any possible x . Identifying whether a value $\bar{x}$ is a local or global maximizer is considered in Section 6.3. This discussion focuses on the shape of the function f , and the use of calculus techniques to identify the shape.

Following this discussion of optimization, a number of applications are examined. The discussion of externalities in Section 6.4 illustrates the use of these techniques to study the welfare implications of traffic congestion. Comparative statics, introduced in Section 6.5, concerns the change in a variable resulting from an underlying parameter change. A key measurement in economics is the elasticity, measuring the response of a dependent variable to a change in another variable — this is introduced in Section 6.6. Sections 6.7 and 6.8 consider comparative static calculations in a taxation context. Finally, Section 6.9 considers Ramsey pricing, using variational arguments to study efficient taxation.

翻译 (Chinese)

在本章中，微积分技术被应用于研究各种问题，首先从第 6.2 节的最优化开始。在最优化问题中，给定一个单变量 x 的函数 f ，问题是找到使 f 最大化的 x 值 $\bar{x}$ 。这样的 $\bar{x}$ 由条

件 $f(\bar{x}) \geq f(x)$ 对所有 x 成立来刻画。如果 x 的选择被限制在某个集合 X 中，则要求是 $f(\bar{x}) \geq f(x)$ 对所有 $x \in X$ 成立。微积分技术通过考虑 f 如何随 x 变化来寻找 $\bar{x}$ 。这些技术在本质上是局部的，因此定位“局部”极大值（或极小值）。相比之下，如果 $f(\bar{x})$ 与任何可能的 x 对应的 $f(x)$ 一样大，则 $\bar{x}$ 是全局最大化者。识别一个值 $\bar{x}$ 是局部还是全局最大化者在第 6.3 节考虑。该讨论侧重于函数 f 的形状，以及使用微积分技术识别该形状。

在此最优化讨论之后，考察若干应用。第 6.4 节关于外部性的讨论说明了使用这些技术来研究交通拥堵的福利含义。第 6.5 节引入的比较静态分析关注由底层参数变化导致的变量变化。经济学中的一个关键度量是弹性，衡量因变量对另一个变量变化的响应——这在第 6.6 节中介绍。第 6.7 和 6.8 节在税收背景下考虑比较静态计算。最后，第 6.9 节考虑拉姆齐定价，使用变分论证来研究有效税收。

6.2 Optimization / 最优化

Original (English)

Definition 6.1: Say that x^* is a global maximum if $f(x^*) \geq f(x)$ for all $x \in X$, and a global minimum if $f(x^*) \leq f(x)$ for all $x \in X$. If $f(x^*) \geq f(x)$ for all $x \in X$, say that x^* maximizes f . If $f(x^*) \leq f(x)$ for all $x \in X$, say that x^* minimizes f .

The discussion here focuses on the problem of maximizing f (to minimize f , maximize $-f$). Furthermore, throughout the discussion only unconstrained optimization is considered: at any point x , one can move in any direction, to alternative choices. That fact allows one to use derivatives to characterize optimal choices.

A necessary condition for a point x^* to be an optimum is that $f'(x^*) = 0$, so that the function is flat at x^* : if $f'(x^*) > 0$ increasing x raises the value of the function, and if $f'(x^*) < 0$ reducing x raises the value of the function.

Theorem 6.1: Let f be a differentiable function. Then if x^* maximizes $f(x)$, $f'(x^*) = 0$.

翻译 (Chinese)

定义 6.1: 若对所有 $x \in X$ 有 $f(x^*) \geq f(x)$ ，则称 x^* 为全局最大值点；若对所有 $x \in X$ 有 $f(x^*) \leq f(x)$ ，则称 x^* 为全局最小值点。若对所有 $x \in X$ 有 $f(x^*) \geq f(x)$ ，则称 x^* 最大化 f 。若对所有 $x \in X$ 有 $f(x^*) \leq f(x)$ ，则称 x^* 最小化 f 。

这里的讨论聚焦于最大化 f 的问题（要最小化 f ，最大化 $-f$ ）。此外，整个讨论中仅考虑无约束最优化：在任何点 x ，可以向任意方向移动到替代选择。这一事实允许我们使用导数来刻画最优选择。

点 x^* 成为最优解的一个必要条件是 $f'(x^*) = 0$ ，即函数在 x^* 处是平坦的：若 $f'(x^*) > 0$ ，增加 x 会提高函数值；若 $f'(x^*) < 0$ ，减少 x 会提高函数值。

定理 6.1: 设 f 为可微函数。则若 x^* 最大化 $f(x)$ ，有 $f'(x^*) = 0$ 。

6.2.1 The first-order condition: issues / 一阶条件：若干问题

Original (English)

The first-order condition is a necessary condition for a point x^* to be either a maximum or a minimum, for if $f'(x^*) \neq 0$ it means that the function is either increasing or decreasing at x^* . However, this is only a necessary condition. Consider the function $f(x) = x^3$: $f'(0) = 0$, but 0 is neither a maximum nor a minimum. Thus, while the condition $f'(x) = 0$ locates an x such that the function f is flat at that point, additional conditions must be checked to determine if x is optimal.

翻译 (Chinese)

一阶条件是点 x^* 成为最大值或最小值的必要条件，因为如果 $f'(x^*) \neq 0$ ，意味着函数在 x^* 处要么递增要么递减。然而，这仅仅是一个必要条件。考虑函数 $f(x) = x^3$ ： $f'(0) = 0$ ，但 0 既非最大值也非最小值。因此，虽然条件 $f'(x) = 0$ 定位了一个使得函数 f 在该点平坦的 x ，但必须检查附加条件以确定 x 是否最优。

6.2.2 The second-order condition / 二阶条件

Original (English)

Theorem 6.2: Suppose that f is a continuous function with continuous first and second derivatives, $f'(x)$ and $f''(x)$. Then

$$f(x^* + \Delta x) = f(x^*) + f'(x^*)\Delta x + \frac{1}{2}f''(p)(\Delta x)^2 \quad (6.1)$$

for some p between x^* and $x^* + \Delta x$.

Theorem 6.3: Let f be a continuous function with continuous first and second derivatives. Suppose that $f'(x^*) = 0$. Then:

1. $f''(x^*) < 0$ implies that x^* is a local maximum.
2. $f''(x^*) > 0$ implies that x^* is a local minimum.

The condition $f''(x) < 0$ is called a second-order condition.

翻译 (Chinese)

定理 6.2: 假设 f 是具有连续一阶和二阶导数 $f'(x)$ 和 $f''(x)$ 的连续函数。则对某个介于 x^* 和 $x^* + \Delta x$ 之间的 p , 有 $f(x^* + \Delta x) = f(x^*) + f'(x^*)\Delta x + \frac{1}{2}f''(p)(\Delta x)^2$ 。

定理 6.3: 设 f 为具有连续一阶和二阶导数的连续函数。假设 $f'(x^*) = 0$ 。则:

1. $f''(x^*) < 0$ 蕴含 x^* 是局部最大值点。
2. $f''(x^*) > 0$ 蕴含 x^* 是局部最小值点。

条件 $f''(x) < 0$ 称为二阶条件。

6.2.3 Applications: profit maximization and related problems / 应用: 利润最大化及相关问题

Original (English)

A simple application of these results is the monopoly revenue optimization problem, where inverse demand is given by $p = p(Q)$ and the cost by $c(Q)$. Thus, total revenue is $R(Q) = Qp(Q)$ and profit is $\pi(Q) = R(Q) - c(Q)$.

The first-order condition for profit maximization is:

$$\pi'(Q) = R'(Q) - c'(Q) = 0.$$

This yields the profit-maximizing rule: set marginal revenue equal to marginal cost. With $\pi(Q) = R(Q) - c(Q) = Qp(Q) - c(Q)$, differentiating gives

$$\pi'(Q) = [p(Q) + Qp'(Q)] - c'(Q)$$

and rearranging:

$$\pi'(Q) = p(Q) \left[1 + \frac{Q}{p} \frac{dp}{dQ} \right] - c'(Q).$$

Example 6.3: In the linear case $p(Q) = a - bQ$ and $c(Q) = cQ$, where c is the constant marginal cost. Then $R(Q) = aQ - bQ^2$ and $\pi(Q) = (a - c)Q - bQ^2$. Also, $\pi'(Q) = (a - c) - 2bQ$. Setting $\pi'(Q) = 0$, the solution for Q is $Q_m = \frac{a-c}{2b}$. The second-order condition is $\pi''(Q) \leq 0$ and in the linear case $\pi''(Q) = -2b < 0$. This may be compared with the competitive model, where quantity Q_c is such that price equals marginal cost: $p(Q) = c(Q)$. With linearity $a - bQ = c$ or $Q_c = \frac{a-c}{b}$ for the competitive output.

翻译 (Chinese)

这些结果的一个简单应用是垄断收入最优化问题，其中反需求由 $p = p(Q)$ 给出，成本由 $c(Q)$ 给出。因此，总收入为 $R(Q) = Qp(Q)$ ，利润为 $\pi(Q) = R(Q) - c(Q)$ 。

利润最大化的一阶条件为： $\pi'(Q) = R'(Q) - c'(Q) = 0$ 。这产生了利润最大化法则：令边际收益等于边际成本。

例 6.3: 在线性情形中 $p(Q) = a - bQ$ 且 $c(Q) = cQ$ ，其中 c 为常数边际成本。则 $R(Q) = aQ - bQ^2$ 且 $\pi(Q) = (a - c)Q - bQ^2$ 。令 $\pi'(Q) = 0$ ，解出 $Q_m = \frac{a-c}{2b}$ 。二阶条件 $\pi''(Q) = -2b < 0$ 满足。这可以与竞争模型相比较，其中产量 Q_c 满足价格等于边际成本： $Q_c = \frac{a-c}{b}$ 。

6.3 Concavity and optimization / 凹性与最优化

Original (English)

Definition 6.2: A function f is concave if for any x, y and any $\lambda \in [0, 1]$

$$f(\lambda x + (1 - \lambda)y) \geq \lambda f(x) + (1 - \lambda)f(y).$$

Definition 6.3: A function g is convex if for any x, y and any $\lambda \in [0, 1]$

$$g(\lambda x + (1 - \lambda)y) \leq \lambda g(x) + (1 - \lambda)g(y).$$

Theorem 6.4: If f is concave and $f'(x^*) = 0$, then x^* is a global maximum. If f is convex and $f'(x^*) = 0$, then x^* is a global minimum.

Definition 6.4: A function f is strictly concave if for $x \neq y$

$$f(\lambda x + (1 - \lambda)y) > \lambda f(x) + (1 - \lambda)f(y), \quad \lambda \in (0, 1).$$

Strict concavity guarantees a unique optimum.

Theorem 6.5: If f is a strictly concave function then f has a unique maximum.

Theorem 6.6: Given the function f , suppose that $f''(x) < 0$ everywhere, then the function is strictly concave. If $f''(x) \leq 0$ for all x , the function is concave.

Theorem 6.7:

1. A twice differentiable function f is concave if and only if $f''(x) \leq 0, \forall x$.
2. The function f is convex if and only if $f''(x) \geq 0, \forall x$.
3. Strict concavity of f implies that $f''(x) \leq 0$ for all x , and $f''(x) < 0$ for all x implies strict concavity.

4. Strict convexity of f implies that $f''(x) \geq 0$ for all x , and $f''(x) > 0$ for all x implies strict convexity.

翻译 (Chinese)

定义 6.2: 若对任意 x, y 和任意 $\lambda \in [0, 1]$ 有 $f(\lambda x + (1 - \lambda)y) \geq \lambda f(x) + (1 - \lambda)f(y)$, 则函数 f 是凹的。

定义 6.3: 若对任意 x, y 和任意 $\lambda \in [0, 1]$ 有 $g(\lambda x + (1 - \lambda)y) \leq \lambda g(x) + (1 - \lambda)g(y)$, 则函数 g 是凸的。

定理 6.4: 若 f 是凹的且 $f'(x^*) = 0$, 则 x^* 是全局最大值点。若 f 是凸的且 $f'(x^*) = 0$, 则 x^* 是全局最小值点。

定义 6.4: 若对 $x \neq y$ 有 $f(\lambda x + (1 - \lambda)y) > \lambda f(x) + (1 - \lambda)f(y)$, $\lambda \in (0, 1)$, 则函数 f 是严格凹的。严格凹性保证了唯一的最优解。

定理 6.5: 若 f 是严格凹函数, 则 f 有唯一的最大值点。

定理 6.6: 给定函数 f , 假设处处有 $f''(x) < 0$, 则该函数是严格凹的。若对所有 x 有 $f''(x) \leq 0$, 则该函数是凹的。

定理 6.7: (1) 二阶可微函数 f 是凹的当且仅当 $f''(x) \leq 0, \forall x$ 。(2) 函数 f 是凸的当且仅当 $f''(x) \geq 0, \forall x$ 。(3) f 的严格凹性蕴含对所有 x 有 $f''(x) \leq 0$, 且对所有 x 有 $f''(x) < 0$ 蕴含严格凹性。

6.3.1 Summary / 小结

Original (English)

The following set of points summarizes the previous discussion on optimization.

- $f'(x^*) = 0$ is necessary for a maximum or minimum.
- $f''(x^*) < 0$ and $f'(x^*) = 0$ is sufficient for x^* to be a unique local maximum.
- $f''(x) \leq 0$ for a neighborhood of x^* and $f'(x^*) = 0$ is sufficient for x^* to be a local maximum.
- $f''(x) \leq 0$ for all x and $f'(x^*) = 0$ implies that x^* is a global maximum.
- $f(x)$ concave implies that $f''(x) \leq 0$ for all x .
- $f''(x) < 0$ for all x implies f is strictly concave, which implies that f has a unique global maximum.

翻译 (Chinese)

以下几点总结了之前关于最优化的讨论。

- $f'(x^*) = 0$ 是最大值或最小值的必要条件。
- $f''(x^*) < 0$ 且 $f'(x^*) = 0$ 是 x^* 为唯一局部最大值点的充分条件。
- 在 x^* 的邻域内 $f''(x) \leq 0$ 且 $f'(x^*) = 0$ 是 x^* 为局部最大值点的充分条件。
- 对所有 x 有 $f''(x) \leq 0$ 且 $f'(x^*) = 0$ 蕴含 x^* 是全局最大值点。
- $f(x)$ 凹蕴含对所有 x 有 $f''(x) \leq 0$ 。
- 对所有 x 有 $f''(x) < 0$ 蕴含 f 是严格凹的, 这蕴含 f 有唯一的全局最大值点。

6.4 Externalities / 外部性

6.4.1 The free-rider problem / 搭便车问题

Original (English)

Consider a public bridge that drivers can use for free. For each individual driver, the benefit from using the bridge is b . With n drivers using the bridge, the cost (in terms of delay) to each driver is $c(n)$, so when there are n drivers in total crossing the bridge the net benefit or utility to each of those drivers is $u = b - c(n)$. Assume that $c(0) = 0$, $c'(n) > 0$ and $c''(n) > 0$ for all $n \geq 0$.

Equilibrium behavior / 均衡行为

The social cost from bridge usage is the cost per user times the number of users, $s(c) = nc(n)$, and the total benefit is nb . Therefore, the social welfare is:

$$w(n) = nb - s(c) = nb - nc(n).$$

Maximizing this:

$$w'(n) = b - [nc'(n) + c(n)].$$

Socially efficient usage of the bridge occurs when $w'(n) = 0$. Let the solution be $\bar{n}$:

$$b = c(\bar{n}) + \bar{n}c'(\bar{n}). \quad (6.8)$$

But drivers independently decide whether to use the bridge or not, and usage increases as long as $b > c(n)$, so the equilibrium usage level ($\hat{n}$) is determined as the solution $b - c(n) = 0$. Comparing these, the market equilibrium level of usage $\hat{n}$ is larger than the socially optimal level, $\bar{n}$.

A policy solution / 政策解决方案

Suppose the government could charge a toll for bridge usage so a driver must pay t to cross the bridge. The manager then charges a toll of $t = \bar{n}c'(\bar{n})$. So, if $\bar{n}$ people use the bridge, each individual compares b with $c(\bar{n}) + \bar{n}c'(\bar{n})$, the travel cost plus the toll charge. With the tax, equilibrium bridge usage is determined by the value of n that satisfies $b = c(n) + nc'(n)$, which is $\bar{n}$.

翻译 (Chinese)

考虑一座司机可以免费使用的公共桥梁。对每个个体司机，使用桥梁的收益为 b 。当有 n 个司机使用桥梁时，每个司机的成本（以延误计）为 $c(n)$ ，因此当总共有 n 个司机过桥时，每个司机的净收益或效用为 $u = b - c(n)$ 。假设 $c(0) = 0$ ， $c'(n) > 0$ 且 $c''(n) > 0$ 对所有 $n \geq 0$ 成立。

桥梁使用的社会成本为每个使用者的成本乘以使用人数： $s(c) = nc(n)$ ，总收益为 nb 。因此社会福利为： $w(n) = nb - nc(n)$ 。最大化此式： $w'(n) = b - [nc'(n) + c(n)]$ 。

当 $w'(n) = 0$ 时桥梁使用达到社会效率水平。设解为 $\bar{n}$ ： $b = c(\bar{n}) + \bar{n}c'(\bar{n})$ 。

但司机独立决定是否使用桥梁，只要 $b > c(n)$ 使用就会增加，因此均衡使用水平 ($\hat{n}$) 由 $b - c(n) = 0$ 确定。比较两者，市场均衡使用水平 $\hat{n}$ 大于社会最优水平 $\bar{n}$ 。

假设政府可以对桥梁使用收费，司机必须支付 t 才能过桥。管理者收取 $t = \bar{n}c'(\bar{n})$ 的通行费。因此，如果有 $\bar{n}$ 人使用桥梁，每个个体将 b 与 $c(\bar{n}) + \bar{n}c'(\bar{n})$ （旅行成本加通行费）进行比较。有了税收后，均衡桥梁使用量由满足 $b = c(n) + nc'(n)$ 的 n 值决定，即 $\bar{n}$ 。

6.5 Comparative statics / 比较静态分析**Original (English)**

Comparative statics studies the impact on a solution variable of changes in the underlying parameters. Consider an optimization problem where the function to be maximized depends explicitly on some parameter or parameters. For example, consider the maximization of the function

$$f(x; a, c) = a \cdot \ln(x + 1) - cx.$$

Maximizing gives $\frac{a}{x+1} - c = 0$ and from the first-order condition, the solution is $x(a, c) = \frac{a}{c} - 1$. So, for example, taking c to be a constant, $\frac{dx}{da} = \frac{1}{c}$, giving the impact on the solution of a change in the parameter a .

翻译 (Chinese)

比较静态分析研究底层参数变化对解变量的影响。考虑一个最优化问题，其中被最大化的函数显式依赖于某个或多个参数。例如，考虑函数 $f(x; a, c) = a \cdot \ln(x+1) - cx$ 的最大化。最大化得 $\frac{a}{x+1} - c = 0$ ，由一阶条件得解为 $x(a, c) = \frac{a}{c} - 1$ 。因此，例如，将 c 视为常数， $\frac{dx}{da} = \frac{1}{c}$ 给出了参数 a 变化对解的影响。

6.6 Elasticity of demand / 需求弹性**Original (English)**

The elasticity of demand measures the responsiveness of demand to price changes and is defined as the percentage change in quantity demanded due to a percentage change in price and denoted ε :

$$\varepsilon = \frac{\% \text{ change in } Q}{\% \text{ change in } P} = \frac{\frac{\Delta Q}{Q}}{\frac{\Delta P}{P}} = \frac{P}{Q} \frac{\Delta Q}{\Delta P} \rightarrow \frac{P}{Q} \frac{dQ}{dP}.$$

Often, because demand is downward sloping, the elasticity is defined as $-\frac{P}{Q} \frac{dQ}{dP}$, so that the elasticity is expressed as a positive number for downward sloping demand.

6.6.1 Elasticity, revenue and profit maximization / 弹性、收益与利润最大化

Recall that the profit function of a monopoly was written:

$$\pi(Q) = P(Q)Q - c(Q) = R(Q) - c(Q).$$

From this expression:

$$\pi'(Q) = R'(Q) - c'(Q) = P(Q) \left[1 + \frac{Q}{P} \frac{dP}{dQ} \right] - c'(Q) = P(Q) \left[1 + \frac{1}{\varepsilon} \right] - c'(Q).$$

Since a profit-maximizing monopolist sets $\pi'(Q) = 0$ and solves for Q , this implies that the solution satisfies:

$$P(Q) \left[1 + \frac{1}{\varepsilon(Q)} \right] = c'(Q),$$

and since $\varepsilon(Q) < 0$, this implies that $P(Q) > c'(Q)$ at the solution so that the profit-maximizing monopolist sets a price greater than marginal cost. Rearranging,

$$\frac{P(Q) - c'(Q)}{P(Q)} = -\frac{1}{\varepsilon(Q)},$$

so that the markup of price over marginal cost is equal to the reciprocal of the elasticity.

翻译 (Chinese)

需求弹性衡量需求对价格变化的响应程度，定义为需求量百分比变化除以价格百分比变化，记为 ε ：

$$\varepsilon = \frac{\% Q \text{ 的变化}}{\% P \text{ 的变化}} = \frac{P}{Q} \frac{dQ}{dP}.$$

通常，由于需求是向下倾斜的，弹性定义为 $-\frac{P}{Q} \frac{dQ}{dP}$ ，使得对于向下倾斜的需求弹性表示为正数。

回顾垄断企业的利润函数写为： $\pi(Q) = P(Q)Q - c(Q) = R(Q) - c(Q)$ 。由此表达式：

$$\pi'(Q) = P(Q) \left[1 + \frac{1}{\varepsilon} \right] - c'(Q).$$

由于利润最大化垄断者令 $\pi'(Q) = 0$ 并求解 Q ，这意味着解满足 $P(Q) \left[1 + \frac{1}{\varepsilon(Q)} \right] = c'(Q)$ 。由于 $\varepsilon(Q) < 0$ ，这意味着在解处 $P(Q) > c'(Q)$ ，因此利润最大化垄断者设定的价格高于边际成本。重新排列得：

$$\frac{P(Q) - c'(Q)}{P(Q)} = -\frac{1}{\varepsilon(Q)},$$

即价格超过边际成本的加成等于弹性的倒数。

6.7 Taxation and monopoly / 税收与垄断

Original (English)

Suppose that the government wishes to induce a monopolist to produce the quantity at which price equals marginal cost. To achieve this outcome, suppose that the government gives a per-unit subsidy of s and imposes a lump sum tax of T . The profit of the firm is then:

$$\pi_g(Q) = Q(P(Q) + s) - cQ - T.$$

The first-order condition is $\pi'_g(Q) = 0 = P(Q) + QP'(Q) + s - c = 0$, which is equivalent to: $P(Q) \left[1 + \frac{1}{\varepsilon(Q)} \right] = c - s$.

If the subsidy is chosen equal to the price-elasticity ratio at the competitive output, the monopolist's optimal choice is the competitive output. In the case of linear demand $\frac{dQ}{dP} = -\frac{1}{b}$, so that s should be chosen to satisfy $Q_c = \frac{s}{b}$ or $s = bQ_c$.

翻译 (Chinese)

假设政府希望诱导垄断者生产价格等于边际成本时的产量。为达到这一结果，假设政府给予每单位 s 的补贴并征收总额税 T 。企业利润则为： $\pi_g(Q) = Q(P(Q) + s) - cQ - T$ 。一阶条件等价于： $P(Q) \left[1 + \frac{1}{\varepsilon(Q)}\right] = c - s$ 。如果补贴选择等于竞争产量处的价格-弹性比，垄断者的最优选择就是竞争产量。在线性需求情形中，应选择 $s = bQ_c$ 。

6.8 Taxation incidence in supply and demand / 供需中的税收归宿

Original (English)

When a tax is imposed on a good, this generally impacts both the buyer and seller. The price paid by the buyer will typically rise, and the price received by the seller will typically fall — with the gap equal to the tax. Suppose that in some market the government charges a tax t per unit sold. Thus the price paid by the purchaser exceeds the price received by the seller by the amount t : $P_d(Q) = P_s(Q) + t$.

6.8.1 A unit tax / 单位税

Given an initial value for t , the corresponding quantity, $Q(t)$, satisfies $P_d(Q(t)) = P_s(Q(t)) + t$, ($Q(0) = Q_e$). Differentiating gives:

$$P'_d(Q(t)) \frac{dQ(t)}{dt} - 1 = P'_s(Q(t)) \frac{dQ(t)}{dt}.$$

Let the terms ε_s and ε_d denote the elasticities of supply and demand respectively. Since $P'_d(Q) < 0$ and $P'_s(Q) > 0$, $\varepsilon_d < 0$ and $\varepsilon_s > 0$. Solving for $\frac{dQ}{dt}$ (evaluated at $t = 0$):

$$\left. \frac{dQ}{dt} \right|_{t=0} = \frac{-1}{\frac{P_e}{Q_e} \left(\frac{1}{\varepsilon_s} - \frac{1}{\varepsilon_d} \right)}.$$

6.8.2 An ad-valorem tax / 从价税

With an ad-valorem tax, $(1 + t)P_s(Q(t)) = P_d(Q(t))$. Differentiating and evaluating at $t = 0$ gives:

$$\left. \frac{dQ}{dt} \right|_{t=0} = -\frac{Q_e}{\frac{1}{\varepsilon_s} - \frac{1}{\varepsilon_d}}.$$

翻译 (Chinese)

当对一种商品征税时，这通常会影响到买卖双方。买方支付的价格通常会上升，卖方收到的价格通常会下降——差额等于税额。假设在某个市场中政府对每单位销售征收 t 的税。因此购买者支付的价格超过销售者收到的价格恰好 t ： $P_d(Q) = P_s(Q) + t$ 。

给定 t 的初始值，相应的数量 $Q(t)$ 满足 $P_d(Q(t)) = P_s(Q(t)) + t$ ， $(Q(0) = Q_e)$ 。微分并求解 $\frac{dQ}{dt}$ （在 $t = 0$ 处求值）得上述公式。由于 $\varepsilon_s > 0$ 且 $\varepsilon_d < 0$ ，有 $\frac{dQ}{dt} < 0$ 。注意当弹性很小时， t 变化对 Q 的影响很大。

对于从价税， $(1+t)P_s(Q(t)) = P_d(Q(t))$ 。微分并在 $t = 0$ 处求值得出类似公式。

6.8.3 Elasticities and taxation / 弹性与税收**Original (English)**

Considering the formula associated with the unit tax,

$$\frac{dQ}{dt} = \frac{-1}{\frac{P}{Q} \left(\frac{1}{\varepsilon_s} - \frac{1}{\varepsilon_d} \right)},$$

the following observations highlight some aspects of elasticities and their relationship to the responsiveness of demand and price to changes in taxation. If ε_s is very large, the quantity supplied is very responsive to price changes, so the supply curve is flat. When ε_d is also very large, then the demand curve is also flat. In this case, both $\frac{1}{\varepsilon_s}$ and $\frac{1}{\varepsilon_d}$ are very small so that $\frac{dQ}{dt}$ is a large negative number. Therefore, large elasticities lead to a large change in quantity traded as a result of the tax. When both ε_s and ε_d are small, this corresponds to the situation where neither supply nor demand is responsive to price changes — the curves are both steep. In this case, $\frac{1}{\varepsilon_s}$ and $\frac{1}{\varepsilon_d}$ are large so that $\frac{dQ}{dt}$ is a small negative number.

翻译 (Chinese)

考虑单位税相关的公式： $\frac{dQ}{dt} = \frac{-1}{\frac{P}{Q} \left(\frac{1}{\varepsilon_s} - \frac{1}{\varepsilon_d} \right)}$ ，以下观察突出了弹性及其与需求和价格对税收变化响应程度之间关系的某些方面。如果 ε_s 非常大，供给量对价格变化非常敏感，因此供给曲线是平坦的。当 ε_d 也非常大时，需求曲线也是平坦的。在这种情况下， $\frac{1}{\varepsilon_s}$ 和 $\frac{1}{\varepsilon_d}$ 都非常小，因此 $\frac{dQ}{dt}$ 是一个很大的负数。因此，较大的弹性导致税收引起的交易量变化较大。当 ε_s 和 ε_d 都很小时，对应于供给和需求对价格变化都不敏感的情况——两条曲线都很陡峭。在这种情况下， $\frac{1}{\varepsilon_s}$ 和 $\frac{1}{\varepsilon_d}$ 很大，因此 $\frac{dQ}{dt}$ 是一个小的负数。

6.9 Ramsey pricing / 拉姆齐定价

Original (English)

Consider the efficiency problem that arises when there is a fixed cost in production, so that pricing at marginal cost generates a loss. In the absence of a subsidy to production, price must be set above marginal cost. Consider the task of recovering fixed cost by pricing above marginal cost in different markets where fixed cost is jointly shared in production, while minimizing loss of consumer and producer surplus.

Consider two markets with demands $p_x(x) = a - bx$ and $p_y(y) = \alpha - \beta y$, where both items are produced by a firm with cost function $c(x, y) = F + c[x + y]$, so that there is common constant marginal cost c and a fixed cost F . Pricing at marginal cost is efficient but generates a loss for the firm. If prices are set above marginal cost so that the fixed cost F is recouped, what is the optimal pricing scheme (minimizing loss of surplus, subject to raising the fixed cost F)?

Consider reducing the quantity in each market by the same amount. The revenue generated per unit of dead-weight loss in each market: ρ_x and ρ_y :

$$\rho_x = \frac{R_x}{DWL_x} = 2 \frac{q}{q - \tilde{q}}, \quad \rho_y = \frac{R_y}{DWL_y} = 2 \frac{q}{q - \tilde{q}}.$$

Let $\rho = \rho_x = \rho_y$. Viewed as a function of $(q, \tilde{q})$, $\rho(q, \tilde{q})$ falls as $\tilde{q}$ falls, so the revenue gain per unit of dead-weight loss is greatest when $\tilde{q}$ is close to q . And the optimal quantity reduction is the same in both markets. In the case where marginal costs are different, setting the ratios equal gives $\frac{\tilde{q}_x}{q_x - \tilde{q}_x} = \frac{\tilde{q}_y}{q_y - \tilde{q}_y}$, so that $\frac{\tilde{q}_x}{q_x} = \frac{\tilde{q}_y}{q_y}$. Therefore, the quantities should be reduced in the same proportion.

翻译 (Chinese)

考虑当生产中存在固定成本时出现的效率问题，此时按边际成本定价会产生亏损。在没有生产补贴的情况下，价格必须设定在边际成本之上。考虑在固定成本于生产中共同分担的不同市场中，通过按高于边际成本定价来回收固定成本，同时最小化消费者和生产者剩余的损失。

考虑两个市场，其需求分别为 $p_x(x) = a - bx$ 和 $p_y(y) = \alpha - \beta y$ ，两种产品均由一家企业以成本函数 $c(x, y) = F + c[x + y]$ 生产，因此存在共同的常数边际成本 c 和固定成本 F 。按边际成本定价是有效的，但会给企业带来亏损。如果定价高于边际成本以回收固定成本 F ，什么是最优定价方案？

考虑在每个市场减少相同数量的产量。每个市场中每单位无谓损失的收益为 ρ_x 和 ρ_y 。令 $\rho = \rho_x = \rho_y$ 。将 ρ 视为 $(q, \tilde{q})$ 的函数， $\rho(q, \tilde{q})$ 随 $\tilde{q}$ 下降而下降，因此当 $\tilde{q}$ 接近 q 时单位无

谓损失的收益增益最大。最优的数量减少在两个市场中是相同的。在边际成本不同的情况下，令比率相等得 $\frac{\bar{q}_x}{q_x} = \frac{\bar{q}_y}{q_y}$ 。因此，数量应按相同比例减少。

6.10 Exercises for Chapter 6 / 第 6 章习题

Original (English)

Exercise 6.1 Let $C(q)$ denote the total cost of producing q units of a good. Show that the average cost is minimized at the point where average and marginal costs coincide.

Exercise 6.2 The price-cost margin (markup), m , for a monopolist is defined as $m = \frac{\text{price} - \text{marginal cost}}{\text{price}}$. Find the relationship between ε , the elasticity of demand and m .

Exercise 6.3–6.16 Additional exercises on monopoly, taxation, comparative statics, elasticity, and related applications. (See textbook for full exercise statements.)

翻译 (Chinese)

习题 6.1 令 $C(q)$ 表示生产 q 单位商品的总成本。证明平均成本在平均成本与边际成本相等处达到最小。

习题 6.2 垄断者的价格-成本边际（加成） m 定义为 $m = \frac{\text{价格} - \text{边际成本}}{\text{价格}}$ 。找出需求弹性 ε 与 m 之间的关系。

习题 6.3–6.16 关于垄断、税收、比较静态分析、弹性及相关应用的附加习题。（完整习题陈述见教材。）

6.11 Systems of Equations, Differentials and Derivatives / 方程组、微分与导数

6.11.1 Introduction / 引言

Original (English)

In many economic problems, functions of more than one variable arise naturally. Preferences are represented by a utility function that depends on goods consumed, $u(x_1, x_2, \dots, x_n)$, output in production is represented as a function of inputs, $f(k_1, k_2, \dots, k_n)$, and so on. As with the single-variable case, differential techniques play an important role in a variety of applications, such as

comparative statics and optimization. For example, the impact of variation in one variable on the value of a function routinely arises in considering its marginal impact. In the production case, the marginal productivity of input k_1 is given by $[f(k_1 + \Delta, k_2, \dots, k_n) - f(k_1, k_2, \dots, k_n)]/\Delta$, and measures the impact on output of a small change in input k_1 . Such a calculation is called a partial derivative, since only variation in one of a number of variables is considered. Developing these techniques leads to a number of applications and provides the necessary tools to develop some important results — in particular the implicit function theorem.

翻译 (Chinese)

在许多经济问题中,多元函数自然地出现。偏好由依赖于所消费商品的效用函数表示, $u(x_1, x_2, \dots, x_n)$; 生产中的产出表示为投入的函数, $f(k_1, k_2, \dots, k_n)$, 等等。与单变量情形一样, 微分技术在诸如比较静态分析和最优化等多种应用中发挥着重要作用。例如, 一个变量的变化对函数值的影响在考虑其边际效应时常常出现。在生产情形中, 投入 k_1 的边际生产率由 $[f(k_1 + \Delta, k_2, \dots, k_n) - f(k_1, k_2, \dots, k_n)]/\Delta$ 给出, 它衡量投入 k_1 的微小变化对产出的影响。这种计算称为偏导数, 因为只考虑多个变量中的一个的变化。发展这些技术会带来许多应用, 并提供了推导某些重要结果——特别是隐函数定理——所需的基本工具。

Original (English)

Following some introductory discussion, partial derivatives are defined in Section 7.2. This is the natural extension of the derivative of a function of one variable. After this, level contours are introduced (Section 7.3). A level contour identifies the set of x that yields a specific value for the function: in microeconomic theory, indifference curves and isoquants are level contours of utility and production functions respectively. In two dimensions particularly, level contours provide a useful way to visualize the shape of a function. Concepts such as the marginal rate of substitution between goods in consumption or production are explained naturally in terms of level contours and are defined using partial derivatives.

翻译 (Chinese)

在介绍性讨论之后, 第 7.2 节定义了偏导数。这是单变量函数导数的自然推广。此后, 第 7.3 节介绍了水平等高线。水平等高线确定了使得函数取特定值的 x 的集合: 在微观经济理论中, 无差异曲线和等产量线分别是效用函数和生产函数的水平等高线。特别是在二维情形中, 水平等高线提供了一种可视化函数形状的有用方法。消费或生产中商品之间的边际替代率等概念可以自然地用水平等高线来解释, 并用偏导数来定义。

Original (English)

One important use of partial derivatives is in locating maxima and minima of a function: values of $(x_1, \dots, x_n)$ that make the function as large or as small as possible. This technique is illustrated in Section 7.5. The remaining sections consider (non-linear) systems of equations. Section 7.7 outlines some of the basic issues relating to non-linear systems, such as the existence and uniqueness of solutions. Following this, the implicit function theorem is introduced in Section 7.8. This theorem studies how variation of one or more parameters in a system of equations affects each variable in the system.

翻译 (Chinese)

偏导数的一个重要用途是确定函数的最大值和最小值：即使函数尽可能大或尽可能小的 $(x_1, \dots, x_n)$ 的取值。这一技术在第 7.5 节中加以说明。其余各节考虑（非线性）方程组。第 7.7 节概述了与非线性系统相关的一些基本问题，如解的存在性和唯一性。此后，第 7.8 节引入了隐函数定理。该定理研究方程组中一个或多个参数的变化如何影响系统中的每个变量。

Original (English)

For a linear function of many variables, the impact of parameter variation is easy to determine. The function $f(x_1, x_2) = ax_1 + bx_2$ is a linear function of two variables and $f(x_1, x_2, \dots, x_n) = a_1x_1 + a_2x_2 + \dots + a_nx_n$ is a linear function of n variables, $(x_1, x_2, \dots, x_n)$. The impact of a change in x_1 on the value of $f(x_1, x_2) = ax_1 + bx_2$ may be computed directly:

$$f(x_1 + \Delta x_1, x_2) - f(x_1, x_2) = (a[x_1 + \Delta x_1] + bx_2) - (ax_1 + bx_2) = a\Delta x_1.$$

From this, the change in f per unit change in x_1 may be calculated as:

$$\frac{f(x_1 + \Delta x_1, x_2) - f(x_1, x_2)}{\Delta x_1} = a.$$

Similar calculations may be applied for non-linear functions to compute the impact of variation in some variable on the function.

翻译 (Chinese)

对于多元线性函数，参数变化的影响很容易确定。函数 $f(x_1, x_2) = ax_1 + bx_2$ 是二元线性函数， $f(x_1, x_2, \dots, x_n) = a_1x_1 + a_2x_2 + \dots + a_nx_n$ 是 n 元 $(x_1, x_2, \dots, x_n)$ 的线性函数。 x_1 的变化对 $f(x_1, x_2) = ax_1 + bx_2$ 的值的影响可以直接计算：

$$f(x_1 + \Delta x_1, x_2) - f(x_1, x_2) = (a[x_1 + \Delta x_1] + bx_2) - (ax_1 + bx_2) = a\Delta x_1.$$

由此, f 的单位 x_1 变化的变化量可以计算为:

$$\frac{f(x_1 + \Delta x_1, x_2) - f(x_1, x_2)}{\Delta x_1} = a.$$

类似的计算也可以应用于非线性函数, 以计算某个变量的变化对函数的影响。

Original (English)

Example 7.1: Consider

$$f(x_1, x_2) = (x_1 x_2)^2 + \cos(x_1) \sin(x_2),$$

suppose that x_1 is increased to $x_1 + \Delta x_1$:

$$\begin{aligned} f(x_1 + \Delta x_1, x_2) &= ([x_1 + \Delta x_1] \cdot x_2)^2 + \cos(x_1 + \Delta x_1) \sin(x_2) \\ &= [x_1^2 + 2x_1 \Delta x_1 + (\Delta x_1)^2] \cdot x_2^2 + \cos(x_1 + \Delta x_1) \sin(x_2) \\ &= x_1^2 \cdot x_2^2 + [2x_1 \Delta x_1 + (\Delta x_1)^2] \cdot x_2^2 + \cos(x_1 + \Delta x_1) \sin(x_2). \end{aligned}$$

Subtracting $f(x_1, x_2)$, then

$$f(x_1 + \Delta x_1, x_2) - f(x_1, x_2) = [2x_1 \Delta x_1 + (\Delta x_1)^2] \cdot x_2^2 + [\cos(x_1 + \Delta x_1) - \cos(x_1)] \sin(x_2).$$

Dividing by Δx_1 ,

$$\frac{f(x_1 + \Delta x_1, x_2) - f(x_1, x_2)}{\Delta x_1} = 2x_1 x_2^2 + \frac{[\cos(x_1 + \Delta x_1) - \cos(x_1)]}{\Delta x_1} \sin(x_2) + (\Delta x_1) \cdot x_2^2.$$

As Δx_1 goes to 0, the term $(\Delta x_1) \cdot x_2^2$ goes to 0 and $\frac{[\cos(x_1 + \Delta x_1) - \cos(x_1)]}{\Delta x_1}$ approaches the derivative of $\cos(x_1)$, which is $-\sin(x_1)$. Therefore, as Δx_1 approaches 0,

$$\frac{f(x_1 + \Delta x_1, x_2) - f(x_1, x_2)}{\Delta x_1} \rightarrow 2x_1 x_2^2 - \sin(x_1) \sin(x_2).$$

This example shows that calculating the impact of parameter or variable changes on a function of many variables or parameters is not significantly different from similar computations in the one-variable case.

翻译 (Chinese)

例 7.1: 考虑

$$f(x_1, x_2) = (x_1 x_2)^2 + \cos(x_1) \sin(x_2),$$

假设 x_1 增加到 $x_1 + \Delta x_1$:

$$\begin{aligned} f(x_1 + \Delta x_1, x_2) &= ([x_1 + \Delta x_1] \cdot x_2)^2 + \cos(x_1 + \Delta x_1) \sin(x_2) \\ &= [x_1^2 + 2x_1 \Delta x_1 + (\Delta x_1)^2] \cdot x_2^2 + \cos(x_1 + \Delta x_1) \sin(x_2) \\ &= x_1^2 \cdot x_2^2 + [2x_1 \Delta x_1 + (\Delta x_1)^2] \cdot x_2^2 + \cos(x_1 + \Delta x_1) \sin(x_2). \end{aligned}$$

减去 $f(x_1, x_2)$, 则

$$f(x_1 + \Delta x_1, x_2) - f(x_1, x_2) = [2x_1\Delta x_1 + (\Delta x_1)^2] \cdot x_2^2 + [\cos(x_1 + \Delta x_1) - \cos(x_1)] \sin(x_2).$$

除以 Δx_1 ,

$$\frac{f(x_1 + \Delta x_1, x_2) - f(x_1, x_2)}{\Delta x_1} = 2x_1x_2^2 + \frac{[\cos(x_1 + \Delta x_1) - \cos(x_1)]}{\Delta x_1} \sin(x_2) + (\Delta x_1) \cdot x_2^2.$$

当 Δx_1 趋于 0 时, 项 $(\Delta x_1) \cdot x_2^2$ 趋于 0, 且 $\frac{[\cos(x_1 + \Delta x_1) - \cos(x_1)]}{\Delta x_1}$ 趋近于 $\cos(x_1)$ 的导数, 即 $-\sin(x_1)$ 。因此, 当 Δx_1 趋近于 0 时,

$$\frac{f(x_1 + \Delta x_1, x_2) - f(x_1, x_2)}{\Delta x_1} \rightarrow 2x_1x_2^2 - \sin(x_1) \sin(x_2).$$

这个例子表明, 计算参数或变量变化对多元函数的影响与单变量情形中的类似计算并没有显著不同。

Original (English)

The need for such calculations comes up routinely in economics. Consider maximizing utility $u(x_1, x_2)$ subject to a budget constraint, $p_1x_1 + p_2x_2 = y$, where p_i is the price of good i and y is income. Rearranging the budget constraint: $x_2 = \frac{y}{p_2} - \frac{p_1}{p_2}x_1$. Substituting this into the utility function gives

$$\hat{u}(x_1, y, p_1, p_2) = u\left(x_1, \frac{y}{p_2} - \frac{p_1}{p_2}x_1\right).$$

Taking (y, p_1, p_2) as fixed (beyond the control of the consumer), maximizing utility amounts to choosing x_1 to maximize $\hat{u}(x_1, y, p_1, p_2)$. The impact of variations in x_1 on $\hat{u}$ requires calculation of the direct effect via x_1 and the indirect effect via $x_2 = \frac{y}{p_2} - \frac{p_1}{p_2}x_1$. When x_1 is chosen optimally, the maximized value of $\hat{u}$ gives the utility of the individual from choosing optimally — let $\bar{u}(y, p_1, p_2) = \max_{x_1} \hat{u}(x_1, y, p_1, p_2)$. The function $\bar{u}$ gives the utility of the individual as a function of (y, p_1, p_2) , so one may consider the impact of income and price variations on the individual's welfare. The impact on utility of an increase in income from y to $y + \Delta y$ is given by $\bar{u}(y + \Delta y, p_1, p_2) - \bar{u}(y, p_1, p_2)$, and the per-unit change is given by

$$\frac{\bar{u}(y + \Delta y, p_1, p_2) - \bar{u}(y, p_1, p_2)}{\Delta y}.$$

This expression is a partial derivative — calculating the impact on $\bar{u}$ of a small change in y , with the other parameters affecting $\bar{u}$ held fixed. The partial derivative is the analogous concept to the derivative of a function of one variable, for functions of more than one variable. In this example, partial derivatives appear both in the choice of x_1 and the subsequent calculation of the marginal utility of income.

翻译 (Chinese)

对此类计算的需求在经济学中经常出现。考虑在预算约束 $p_1x_1 + p_2x_2 = y$ 下最大化效用 $u(x_1, x_2)$ ，其中 p_i 是商品 i 的价格， y 是收入。重新整理预算约束： $x_2 = \frac{y}{p_2} - \frac{p_1}{p_2}x_1$ 。将其代入效用函数得

$$\hat{u}(x_1, y, p_1, p_2) = u\left(x_1, \frac{y}{p_2} - \frac{p_1}{p_2}x_1\right).$$

将 (y, p_1, p_2) 视为固定的（消费者无法控制），最大化效用等价于选择 x_1 来最大化 $\hat{u}(x_1, y, p_1, p_2)$ 。 x_1 的变化对 $\hat{u}$ 的影响需要计算通过 x_1 的直接效应和通过 $x_2 = \frac{y}{p_2} - \frac{p_1}{p_2}x_1$ 的间接效应。当 x_1 被最优选择时， $\hat{u}$ 的最大化值给出了个体从最优选择中获得的效用——令 $\bar{u}(y, p_1, p_2) = \max_{x_1} \hat{u}(x_1, y, p_1, p_2)$ 。函数 $\bar{u}$ 给出了个体效用作为 (y, p_1, p_2) 的函数，因此可以考虑收入和价格变化对个体福利的影响。收入从 y 增加到 $y + \Delta y$ 对效用的影响由 $\bar{u}(y + \Delta y, p_1, p_2) - \bar{u}(y, p_1, p_2)$ 给出，单位变化由

$$\frac{\bar{u}(y + \Delta y, p_1, p_2) - \bar{u}(y, p_1, p_2)}{\Delta y}.$$

给出。这个表达式是一个偏导数——计算 y 的微小变化对 $\bar{u}$ 的影响，同时保持影响 $\bar{u}$ 的其他参数不变。偏导数是对多元函数而言与单变量函数导数相类似的概念。在这个例子中，偏导数既出现在 x_1 的选择中，也出现在随后的收入边际效用的计算中。

Original (English)

In production theory, a production function $f(k, l)$ associates a level of output $Q = f(k, l)$ to inputs of capital (k) and labor (l). If capital is increased from k to $k + \Delta$, then output increases to $f(k + \Delta, l)$ and the change in output is $f(k + \Delta, l) - f(k, l)$. Dividing this by Δ gives $[f(k + \Delta, l) - f(k, l)]/\Delta$, the change in output per unit change in the capital input. As Δ goes to 0, assuming f is differentiable with respect to k , the limiting value is called the marginal product of capital.

翻译 (Chinese)

在生产理论中，生产函数 $f(k, l)$ 将产出水平 $Q = f(k, l)$ 与资本 (k) 和劳动 (l) 的投入联系起来。如果资本从 k 增加到 $k + \Delta$ ，则产出增加到 $f(k + \Delta, l)$ ，产出的变化为 $f(k + \Delta, l) - f(k, l)$ 。将其除以 Δ 得到 $[f(k + \Delta, l) - f(k, l)]/\Delta$ ，即每单位资本投入变化的产出变化。当 Δ 趋于 0 时，假设 f 关于 k 可微，其极限值称为资本的边际产品。

Original (English)

In demand theory, when a profile of goods $(x_1, \dots, x_n)$ is chosen to maximize utility subject to a budget constraint $p_1x_1 + \dots + p_nx_n = y$, the resulting demands depend on the profile of prices and income. Thus, the demand for good i is $x_i(p_1, \dots, p_n, y)$. When prices other than p_i and income y

are constant, changes in p_i lead to changes in x_i , and this connection between x_i and p_i gives the demand function for good i . As p_i varies, the resulting change in x_i is

$$\frac{[x_i(p_1, \dots, p_{i-1}, p_i + \Delta p_i, p_{i+1}, p_n, y) - x_i(p_1, \dots, p_{i-1}, p_i, p_{i+1}, p_n, y)]}{\Delta p_i}.$$

The partial derivative with respect to one variable depends on the values of the other variables entering the function. For example, the impact of a price change on demand depends on the prices of other goods and income. In Figure 7.1, the partial derivative with respect to p_i at $(p_1, \dots, p_i, \dots, p_n, y)$ is different from the partial derivative at $(\tilde{p}_1, \dots, p_i, \dots, \tilde{p}_n, \tilde{y})$, since $|\Delta \tilde{x}_i| > |\Delta x_i|$.

翻译 (Chinese)

在需求理论中, 当选择商品组合 $(x_1, \dots, x_n)$ 在预算约束 $p_1 x_1 + \dots + p_n x_n = y$ 下最大化效用, 由此产生的需求取决于价格组合和收入。因此, 对商品 i 的需求为 $x_i(p_1, \dots, p_n, y)$ 。当除 p_i 以外的其他价格和收入 y 不变时, p_i 的变化导致 x_i 的变化, x_i 与 p_i 之间的这种联系给出了商品 i 的需求函数。随着 p_i 的变化, 由此产生的 x_i 的变化为

$$\frac{[x_i(p_1, \dots, p_{i-1}, p_i + \Delta p_i, p_{i+1}, p_n, y) - x_i(p_1, \dots, p_{i-1}, p_i, p_{i+1}, p_n, y)]}{\Delta p_i}.$$

关于一个变量的偏导数取决于进入函数的其他变量的值。例如, 价格变化对需求的影响取决于其他商品的价格和收入。在图 7.1 中, 在 $(p_1, \dots, p_i, \dots, p_n, y)$ 处关于 p_i 的偏导数与在 $(\tilde{p}_1, \dots, p_i, \dots, \tilde{p}_n, \tilde{y})$ 处不同, 因为 $|\Delta \tilde{x}_i| > |\Delta x_i|$ 。

Original (English)

The next section defines the partial derivative of a function — a natural extension of the derivative of a function of a single variable, to functions of many variables. Following this, level contours are introduced and then, using partial derivatives, the equilibrium output in a duopoly is determined.

翻译 (Chinese)

下一节定义了函数的偏导数——单变量函数导数向多元函数的自然推广。此后, 引入了水平等高线, 然后利用偏导数确定了双寡头市场中的均衡产出。

6.11.2 Partial Derivatives / 偏导数

Original (English)

Let $f : X \rightarrow \mathbb{R}$, where $X \subseteq \mathbb{R}^n$, so $f(x) = f(x_1, x_2, \dots, x_n)$. If all the x 's except one are held fixed,

we can view f as a function of one variable, and then consider the derivative of f with respect to this variable. This is called the partial derivative. The partial derivative of f with respect to x_i is denoted $\frac{\partial f}{\partial x_i}$ and defined as:

$$\frac{\partial f}{\partial x_i} = \lim_{\Delta x_i \rightarrow 0} \frac{f(x_1, x_2, \dots, x_{i-1}, x_i + \Delta x_i, x_{i+1}, \dots, x_n) - f(x_1, x_2, \dots, x_n)}{\Delta x_i}.$$

Thus, $\frac{\partial f}{\partial x_i}$ is defined provided this limit exists for all $\Delta x_i \rightarrow 0$. The partial derivative of f with respect to x_i measures the change in f due to a small change in x_i , when other variables are held constant.

翻译 (Chinese)

设 $f: X \rightarrow \mathbb{R}$, 其中 $X \subseteq \mathbb{R}^n$, 则 $f(x) = f(x_1, x_2, \dots, x_n)$ 。如果除一个变量外所有 x 保持固定, 我们可以将 f 视为单变量函数, 然后考虑 f 关于该变量的导数。这称为偏导数。 f 关于 x_i 的偏导数记为 $\frac{\partial f}{\partial x_i}$, 定义为:

$$\frac{\partial f}{\partial x_i} = \lim_{\Delta x_i \rightarrow 0} \frac{f(x_1, x_2, \dots, x_{i-1}, x_i + \Delta x_i, x_{i+1}, \dots, x_n) - f(x_1, x_2, \dots, x_n)}{\Delta x_i}.$$

因此, $\frac{\partial f}{\partial x_i}$ 在该极限对所有 $\Delta x_i \rightarrow 0$ 存在时即有定义。 f 关于 x_i 的偏导数度量了当其他变量保持不变时, x_i 的微小变化所引起的 f 的变化。

Original (English)

Example 7.2: With $f(x_1, x_2) = (x_1 x_2)^2 + \cos(x_1) \sin(x_2)$,

$$\frac{\partial f}{\partial x_1} = 2(x_1 x_2)x_2 - \sin(x_1) \sin(x_2) = 2x_1 x_2^2 - \sin(x_1) \sin(x_2),$$

$$\frac{\partial f}{\partial x_2} = 2(x_1 x_2)x_1 + \cos(x_1) \cos(x_2) = 2x_1^2 x_2 + \cos(x_1) \cos(x_2).$$

翻译 (Chinese)

例 7.2: 对于 $f(x_1, x_2) = (x_1 x_2)^2 + \cos(x_1) \sin(x_2)$,

$$\frac{\partial f}{\partial x_1} = 2(x_1 x_2)x_2 - \sin(x_1) \sin(x_2) = 2x_1 x_2^2 - \sin(x_1) \sin(x_2),$$

$$\frac{\partial f}{\partial x_2} = 2(x_1 x_2)x_1 + \cos(x_1) \cos(x_2) = 2x_1^2 x_2 + \cos(x_1) \cos(x_2).$$

Original (English)

Remark 7.1: Often, for notational convenience, the partial derivative of f with respect to x_i

is denoted f_{x_i} or f_i when no confusion can arise. Also, where there are just two variables in consideration, these may be labeled x and y with partial derivatives f_x and f_y . With more than two variables, it is often more convenient to label by index, x_i , $i = 1, \dots, n$.

翻译 (Chinese)

注 7.1: 通常, 为了记号的方便, 在不会引起混淆的情况下, f 关于 x_i 的偏导数记为 f_{x_i} 或 f_i 。此外, 当只考虑两个变量时, 可以标记为 x 和 y , 偏导数为 f_x 和 f_y 。对于多于两个变量的情形, 按指标 x_i , $i = 1, \dots, n$ 标记通常更为方便。

Original (English)

Note that for $f(x_1, x_2, \dots, x_n)$, $\frac{\partial f}{\partial x_i}$ is a function of all the x_i . One can take the partial derivative of this function — the partial derivative of the partial derivative. Thus, the partial derivative of $\frac{\partial f}{\partial x_i}$ with respect to x_i is $\frac{\partial}{\partial x_i} \frac{\partial f}{\partial x_i}$ and the partial derivative of $\frac{\partial f}{\partial x_i}$ with respect to x_j is $\frac{\partial}{\partial x_j} \frac{\partial f}{\partial x_i}$. For ease of notation, these are sometimes written as $\frac{\partial^2 f}{\partial x_i^2}$ and $\frac{\partial^2 f}{\partial x_j \partial x_i}$ or as $f_{x_i x_i}$ and $f_{x_j x_i}$. In the two-variable case with $f(x, y)$, the partial derivatives are $f_x = \frac{\partial f}{\partial x}$ and $f_y = \frac{\partial f}{\partial y}$ and the second partial derivatives are written as $f_{xx} = \frac{\partial}{\partial x} \frac{\partial f}{\partial x}$, $f_{yy} = \frac{\partial}{\partial y} \frac{\partial f}{\partial y}$, $f_{xy} = \frac{\partial}{\partial x} \frac{\partial f}{\partial y}$, and $f_{yx} = \frac{\partial}{\partial y} \frac{\partial f}{\partial x}$.

翻译 (Chinese)

注意, 对于 $f(x_1, x_2, \dots, x_n)$, $\frac{\partial f}{\partial x_i}$ 是所有 x_i 的函数。我们可以对该函数求偏导数——即偏导数的偏导数。因此, $\frac{\partial f}{\partial x_i}$ 关于 x_i 的偏导数是 $\frac{\partial}{\partial x_i} \frac{\partial f}{\partial x_i}$, $\frac{\partial f}{\partial x_i}$ 关于 x_j 的偏导数是 $\frac{\partial}{\partial x_j} \frac{\partial f}{\partial x_i}$ 。为了记号的简便, 这些有时写作 $\frac{\partial^2 f}{\partial x_i^2}$ 和 $\frac{\partial^2 f}{\partial x_j \partial x_i}$, 或者写作 $f_{x_i x_i}$ 和 $f_{x_j x_i}$ 。在二元函数 $f(x, y)$ 的情形中, 偏导数为 $f_x = \frac{\partial f}{\partial x}$ 和 $f_y = \frac{\partial f}{\partial y}$, 二阶偏导数写作 $f_{xx} = \frac{\partial}{\partial x} \frac{\partial f}{\partial x}$, $f_{yy} = \frac{\partial}{\partial y} \frac{\partial f}{\partial y}$, $f_{xy} = \frac{\partial}{\partial x} \frac{\partial f}{\partial y}$ 和 $f_{yx} = \frac{\partial}{\partial y} \frac{\partial f}{\partial x}$ 。

Original (English)

Example 7.3: To illustrate, consider the two-variable case with $f(x, y) = x^\alpha y^\beta$. Then:

$$f_x(x, y) = \frac{\partial f}{\partial x} = \alpha x^{\alpha-1} y^\beta = \frac{\alpha}{x} f(x, y),$$

$$f_y(x, y) = \frac{\partial f}{\partial y} = \beta x^\alpha y^{\beta-1} = \frac{\beta}{y} f(x, y).$$

Notice that $f_x(x, y) = \frac{\partial f}{\partial x}(x, y) = \frac{\partial \{f(x, y)\}}{\partial x} = \alpha x^{\alpha-1} y^\beta$ depends on x and y — it is a function of (x, y) . The function $f_x(x, y)$ has a partial derivative with respect to x , which is $\frac{\partial}{\partial x} \frac{\partial f}{\partial x} = \alpha(\alpha-1)x^{\alpha-2} y^\beta$. Similarly, $f_x(x, y)$ has a partial derivative with respect to y given by $\frac{\partial}{\partial y} \frac{\partial f}{\partial x} = \alpha \beta x^{\alpha-1} y^{\beta-1}$.

翻译 (Chinese)

例 7.3: 举例来说, 考虑二元情形 $f(x, y) = x^\alpha y^\beta$ 。则:

$$f_x(x, y) = \frac{\partial f}{\partial x} = \alpha x^{\alpha-1} y^\beta = \frac{\alpha}{x} f(x, y),$$

$$f_y(x, y) = \frac{\partial f}{\partial y} = \beta x^\alpha y^{\beta-1} = \frac{\beta}{y} f(x, y).$$

注意 $f_x(x, y) = \frac{\partial f}{\partial x}(x, y) = \frac{\partial \{f(x, y)\}}{\partial x} = \alpha x^{\alpha-1} y^\beta$ 依赖于 x 和 y ——它是 (x, y) 的函数。函数 $f_x(x, y)$ 有一个关于 x 的偏导数, 即 $\frac{\partial}{\partial x} \frac{\partial f}{\partial x} = \alpha(\alpha-1)x^{\alpha-2} y^\beta$ 。类似地, $f_x(x, y)$ 有一个关于 y 的偏导数, 由 $\frac{\partial}{\partial y} \frac{\partial f}{\partial x} = \alpha \beta x^{\alpha-1} y^{\beta-1}$ 给出。

Original (English)

In Example 7.2, the second partial derivatives are given by:

$$f_{x_1 x_1} = \frac{\partial}{\partial x_1} \frac{\partial f}{\partial x_1} = 2x_2^2 - \cos(x_1) \sin(x_2),$$

$$f_{x_2 x_1} = \frac{\partial}{\partial x_2} \frac{\partial f}{\partial x_1} = 4x_1 x_2 - \sin(x_1) \cos(x_2),$$

$$f_{x_2 x_2} = \frac{\partial}{\partial x_1} \frac{\partial f}{\partial x_2} = 2x_1^2 - \cos(x_1) \sin(x_2),$$

$$f_{x_1 x_2} = \frac{\partial}{\partial x_1} \frac{\partial f}{\partial x_2} = 4x_1 x_2 - \sin(x_1) \cos(x_2).$$

Note that in Example 7.3, $f_{x_2 x_1} = f_{x_1 x_2}$. This is not accidental. When the partial derivatives exist and are continuous, then the second partial derivatives satisfy a special relation known as Young's theorem.

翻译 (Chinese)

在例 7.2 中, 二阶偏导数由下列式子给出:

$$f_{x_1 x_1} = \frac{\partial}{\partial x_1} \frac{\partial f}{\partial x_1} = 2x_2^2 - \cos(x_1) \sin(x_2),$$

$$f_{x_2 x_1} = \frac{\partial}{\partial x_2} \frac{\partial f}{\partial x_1} = 4x_1 x_2 - \sin(x_1) \cos(x_2),$$

$$f_{x_2 x_2} = \frac{\partial}{\partial x_1} \frac{\partial f}{\partial x_2} = 2x_1^2 - \cos(x_1) \sin(x_2),$$

$$f_{x_1 x_2} = \frac{\partial}{\partial x_1} \frac{\partial f}{\partial x_2} = 4x_1 x_2 - \sin(x_1) \cos(x_2).$$

注意在例 7.3 中, $f_{x_2 x_1} = f_{x_1 x_2}$ 。这并非偶然。当偏导数存在且连续时, 二阶偏导数满足一个称为杨氏定理的特殊关系。

Original (English)

Theorem 7.1: Suppose the function $f(x_1, \dots, x_n)$ has continuous partial derivatives, $f_{x_i}(x_1, \dots, x_n)$, in a neighborhood of $(x_1^*, \dots, x_n^*)$, and suppose they are differentiable, $\frac{\partial^2 f}{\partial x_i \partial x_j}$ exists for each i, j ; then the second partial derivatives are equal:

$$f_{x_i x_j} = \frac{\partial^2 f}{\partial x_i \partial x_j} = \frac{\partial}{\partial x_i} \left(\frac{\partial f}{\partial x_j} \right) = \frac{\partial}{\partial x_j} \left(\frac{\partial f}{\partial x_i} \right) = \frac{\partial^2 f}{\partial x_j \partial x_i} = f_{x_j x_i}.$$

翻译 (Chinese)

定理 7.1: 假设函数 $f(x_1, \dots, x_n)$ 在 $(x_1^*, \dots, x_n^*)$ 的邻域内有连续的偏导数 $f_{x_i}(x_1, \dots, x_n)$, 并且假设它们是可微的, $\frac{\partial^2 f}{\partial x_i \partial x_j}$ 对每个 i, j 都存在; 则二阶偏导数相等:

$$f_{x_i x_j} = \frac{\partial^2 f}{\partial x_i \partial x_j} = \frac{\partial}{\partial x_i} \left(\frac{\partial f}{\partial x_j} \right) = \frac{\partial}{\partial x_j} \left(\frac{\partial f}{\partial x_i} \right) = \frac{\partial^2 f}{\partial x_j \partial x_i} = f_{x_j x_i}.$$

Original (English)

Example 7.4: Consider $f(x, y) = \ln(xy) \cdot \sin y$, so that $f_x = \frac{y}{xy} \sin y = \frac{1}{x} \sin y$ and $f_{yx} = \frac{1}{x} \cos y$. Differentiating first with respect to y gives: $f_y = \frac{x}{xy} \sin y + \ln(xy) \cos y = \frac{1}{y} \sin y + \ln(xy) \cos y$. And then differentiating with respect to x : $f_{xy} = \frac{y}{xy} \cos y = \frac{1}{x} \cos y$. So, $f_{xy} = f_{yx}$.

翻译 (Chinese)

例 7.4: 考虑 $f(x, y) = \ln(xy) \cdot \sin y$, 则 $f_x = \frac{y}{xy} \sin y = \frac{1}{x} \sin y$ 且 $f_{yx} = \frac{1}{x} \cos y$ 。先对 y 求导得: $f_y = \frac{x}{xy} \sin y + \ln(xy) \cos y = \frac{1}{y} \sin y + \ln(xy) \cos y$ 。然后再对 x 求导: $f_{xy} = \frac{y}{xy} \cos y = \frac{1}{x} \cos y$ 。因此, $f_{xy} = f_{yx}$ 。

6.11.3 Level Contours / 水平等高线**Original (English)**

Level contours arise frequently in economics, for example, as isoquants in production theory or as indifference curves in consumer theory. Given a function of n variables, $f(x_1, \dots, x_n)$, the level contour of f at level or height $\bar{f}$ is the set of values of $(x_1, \dots, x_n)$ such that $f(x_1, \dots, x_n) = \bar{f}$. If this set is denoted $L_{\bar{f}}$, then

$$L_{\bar{f}} = \{(x_1, \dots, x_n) \mid f(x_1, \dots, x_n) = \bar{f}\}.$$

Different values of $\bar{f}$ determine different level contours. The level contour is a way of visualizing a function of more than one variable.

翻译 (Chinese)

水平等高线在经济学中经常出现，例如，作为生产理论中的等产量线或消费者理论中的无差异曲线。给定一个 n 元函数 $f(x_1, \dots, x_n)$ ， f 在水平或高度 $\bar{f}$ 处的水平等高线是使得 $f(x_1, \dots, x_n) = \bar{f}$ 的 $(x_1, \dots, x_n)$ 的取值的集合。如果该集合记为 $L_{\bar{f}}$ ，则

$$L_{\bar{f}} = \{(x_1, \dots, x_n) \mid f(x_1, \dots, x_n) = \bar{f}\}.$$

不同的 $\bar{f}$ 值确定了不同的水平等高线。水平等高线是可视化多元函数的一种方法。

Original (English)

In Figure 7.2, the function $f(x, y) = x^{\frac{1}{2}}y^{\frac{1}{2}}$ is plotted, and three level contours graphed, giving the (x, y) pairs associated with the level contours $\{(x, y) \mid f(x, y) = 1\}$, $\{(x, y) \mid f(x, y) = 2\}$ and $\{(x, y) \mid f(x, y) = 3\}$.

翻译 (Chinese)

在图 7.2 中，绘制了函数 $f(x, y) = x^{\frac{1}{2}}y^{\frac{1}{2}}$ ，并画出了三条水平等高线，给出了与水平等高线 $\{(x, y) \mid f(x, y) = 1\}$ 、 $\{(x, y) \mid f(x, y) = 2\}$ 和 $\{(x, y) \mid f(x, y) = 3\}$ 相关的 (x, y) 对。

Original (English)

Given a utility function $u(x, y)$, the indifference curve corresponding to utility level $\bar{u}$ is defined as the set of (x, y) combinations that provide that level of utility, $\bar{u}$: $I_{\bar{u}} = \{(x, y) \mid u(x, y) = \bar{u}\}$. The isoquant corresponding to a production function f is the set of input combinations (x, y) that yield a given level of output Q : $I_Q = \{(x, y) \mid f(x, y) = Q\}$. Considering the production function f , since a small change in x has impact f_x on the value of output and a small change in y has impact f_y , the total change in output from small changes (dx, dy) in (x, y) is $df = f_x(x, y)dx + f_y(x, y)dy$. This is called the total derivative of f at (x, y) . Note that df depends on (x, y) and (dx, dy) . An isoquant is a level contour of the function f . Starting from a point (x, y) on the contour, small variations in x and y that leave the value of f unchanged satisfy $df = 0$. This determines the rate of change in y required to offset a change in x so as to leave the output level unchanged (see Figure 7.3):

$$dy = -\frac{f_x(x, y)}{f_y(x, y)}dx.$$

翻译 (Chinese)

给定一个效用函数 $u(x, y)$ ，对应于效用水平 $\bar{u}$ 的无差异曲线定义为提供该效用水平 $\bar{u}$ 的 (x, y) 组合的集合： $I_{\bar{u}} = \{(x, y) \mid u(x, y) = \bar{u}\}$ 。对应于生产函数 f 的等产量线是产生给定产出水平 Q 的投入组合 (x, y) 的集合： $I_Q = \{(x, y) \mid f(x, y) = Q\}$ 。考虑生产函数 f ，由于

x 的微小变化对产出值有影响 f_x , y 的微小变化有影响 f_y , (x, y) 的微小变化 (dx, dy) 引起的产出总变化为 $df = f_x(x, y)dx + f_y(x, y)dy$ 。这称为 f 在 (x, y) 处的全导数。注意 df 依赖于 (x, y) 和 (dx, dy) 。等产量线是函数 f 的水平等高线。从等高线上的一点 (x, y) 出发, 使得 f 的值保持不变的 x 和 y 的微小变动满足 $df = 0$ 。这确定了为使产出水平保持不变、需要抵消 x 的变化的 y 的变化率 (见图 7.3):

$$dy = -\frac{f_x(x, y)}{f_y(x, y)}dx.$$

Original (English)

Example 7.5: This type of calculation may be used in other contexts. Consider a government that imposes an ad-valorem tax on all goods at the rate τ . Let s_i be the total sales of good i and let p_i be the price of good i , both constant over time. Total tax revenue is $T = \sum_{i=1}^n (1 + \tau)s_i p_i$. Suppose now that prices increase at the rate $\pi(t)$ so that at time t , the price of good i is $p_i(1 + \pi(t))$. Suppose also, that the government adjusts the tax rate τ over time to keep the total tax revenue constant: $T = \sum_{i=1}^n (1 + \tau(t))(1 + \pi(t))s_i p_i = (1 + \tau(t))(1 + \pi(t)) \sum_{i=1}^n s_i p_i$. Then, differentiating with respect to t ,

$$0 = \frac{d\tau(t)}{dt}(1 + \pi(t)) + (1 + \tau(t))\frac{d\pi(t)}{dt} \sum_{i=1}^n s_i p_i.$$

Therefore,

$$0 = \frac{d\tau(t)}{dt}(1 + \pi(t)) + (1 + \tau(t))\frac{d\pi(t)}{dt}.$$

Thus, taxes should be adjusted over time according to the formula:

$$\frac{d\tau(t)}{dt} = -\frac{(1 + \tau(t))}{(1 + \pi(t))} \frac{d\pi(t)}{dt}.$$

翻译 (Chinese)

例 7.5: 这类计算可用于其他情境。考虑一个政府对所有商品按税率 τ 征收从价税。设 s_i 为商品 i 的总销售额, p_i 为商品 i 的价格, 二者均随时间保持不变。总税收收入为 $T = \sum_{i=1}^n (1 + \tau)s_i p_i$ 。现在假设价格以速率 $\pi(t)$ 上涨, 因此在时刻 t , 商品 i 的价格为 $p_i(1 + \pi(t))$ 。再假设政府随时间调整税率 τ 以保持总税收收入不变: $T = \sum_{i=1}^n (1 + \tau(t))(1 + \pi(t))s_i p_i = (1 + \tau(t))(1 + \pi(t)) \sum_{i=1}^n s_i p_i$ 。然后, 对 t 求导,

$$0 = \frac{d\tau(t)}{dt}(1 + \pi(t)) + (1 + \tau(t))\frac{d\pi(t)}{dt} \sum_{i=1}^n s_i p_i.$$

因此,

$$0 = \frac{d\tau(t)}{dt}(1 + \pi(t)) + (1 + \tau(t))\frac{d\pi(t)}{dt}.$$

于是，税率应按照以下公式随时间调整：

$$\frac{d\tau(t)}{dt} = -\frac{(1 + \tau(t))}{(1 + \pi(t))} \frac{d\pi(t)}{dt}.$$

6.11.4 The Elasticity of Substitution / 替代弹性

Original (English)

The elasticity of demand measures the slope $\frac{dQ}{dP}$ of a curve (the demand function) relative to the ratio of the variables at that point on the curve, $\frac{Q}{P}$. This is the slope of the curve at a point, divided by the slope of the ray from the origin to that point. The elasticity of substitution measures how that ray varies as the slope varies (all in percentage terms). In principle, this measure can be applied to the level surface of any function, but two particularly important functions in economics are the utility and production functions. Consider Figure 7.4, where the indifference curve u is given and a change in slope from a to b depicted. The marginal rates of substitution at a and b are denoted $MRS(a)$ and $MRS(b)$ and denote the slope of the indifference curve at the respective points. The ray slope from the origin to a is the ratio of consumption of x_2 relative to x_1 , $\frac{x_2^a}{x_1^a}$, when the marginal rate of substitution is $MRS(a) = \frac{u_{x_1}(a)}{u_{x_2}(a)}$. When the marginal rate of substitution changes to $MRS(b)$, the slope of the ray becomes $\frac{x_2^b}{x_1^b}$. Therefore, the percentage change in the ratio of consumption resulting from the percentage change in the MRS is given by the following expression, called the elasticity of substitution:

$$\sigma = \frac{\frac{x_2^b}{x_1^b} - \frac{x_2^a}{x_1^a}}{\frac{x_2^a}{x_1^a}} \bigg/ \frac{MRS(b) - MRS(a)}{MRS(a)} = \frac{\frac{x_2^b}{x_1^b} - \frac{x_2^a}{x_1^a}}{\frac{x_2^a}{x_1^a}} \cdot \frac{MRS(a)}{MRS(b) - MRS(a)}. \quad (7.1)$$

翻译 (Chinese)

需求弹性衡量一条曲线（需求函数）的斜率 $\frac{dQ}{dP}$ 相对于曲线上该点的变量比值 $\frac{Q}{P}$ 。这是曲线在某点的斜率除以从原点到该点的射线的斜率。替代弹性衡量该射线如何随斜率变化而变化（均以百分比形式）。原则上，这一度量可以应用于任何函数的水平曲面，但经济学中两个特别重要的函数是效用函数和生产函数。考虑图 7.4，其中给出了无差异曲线 u ，并描绘了从 a 到 b 的斜率变化。在 a 和 b 处的边际替代率记为 $MRS(a)$ 和 $MRS(b)$ ，表示无差异曲线在各相应点的斜率。从原点到 a 的射线斜率为 x_2 相对于 x_1 的消费比率 $\frac{x_2^a}{x_1^a}$ ，此时的边际替代率为 $MRS(a) = \frac{u_{x_1}(a)}{u_{x_2}(a)}$ 。当边际替代率变为 $MRS(b)$ 时，射线的斜率变为 $\frac{x_2^b}{x_1^b}$ 。因此，消费比率的百分比变化除以 MRS 的百分比变化由以下表达式给出，称为替代弹性：

$$\sigma = \frac{\frac{x_2^b}{x_1^b} - \frac{x_2^a}{x_1^a}}{\frac{x_2^a}{x_1^a}} \bigg/ \frac{MRS(b) - MRS(a)}{MRS(a)} = \frac{\frac{x_2^b}{x_1^b} - \frac{x_2^a}{x_1^a}}{\frac{x_2^a}{x_1^a}} \cdot \frac{MRS(a)}{MRS(b) - MRS(a)}. \quad (7.1)$$

Original (English)

In Equation 7.1, with u given, the ratio $\frac{x_2}{x_1}$ is viewed as a function of MRS, with the elasticity tracking the change in the ratio resulting from the change in MRS. Considering this measure at a point corresponds to the difference between $MRS(a)$ and $MRS(b)$ becoming small (approaching 0 in the limit). Here, this leads to an expression in terms of the $\ln(x)$ function. Recall that the derivative of $\ln(x)$ is $\frac{1}{x}$ so that $\frac{1}{x} = \frac{d \ln(x)}{dx} \approx \frac{\ln(x+\Delta) - \ln(x)}{\Delta}$. Thus, $\ln(x + \Delta) - \ln(x) \approx \frac{\Delta}{x}$. From this perspective, the numerator in Equation 7.1 is $d \ln\left(\frac{x_2}{x_1}\right)$ and the denominator is $d \ln(MRS)$. In differential terms:

$$\sigma = \frac{d \ln\left(\frac{x_2}{x_1}\right)}{d \ln(MRS)} = \frac{d \ln\left(\frac{x_2}{x_1}\right)}{d \ln\left(\frac{u_{x_1}}{u_{x_2}}\right)}, \quad (7.2)$$

recalling the fact that MRS is obtained from $du = 0 = u_{x_1}dx_1 + u_{x_2}dx_2$, giving $\frac{dx_2}{dx_1} = -\frac{u_{x_1}}{u_{x_2}}$, giving $MRS = \frac{u_{x_1}}{u_{x_2}}$ (measuring the slope in absolute terms).

翻译 (Chinese)

在方程 (7.1) 中, 给定 u , 比值 $\frac{x_2}{x_1}$ 被视为 MRS 的函数, 弹性跟踪由 MRS 变化引起的该比值的变化。在一点上考虑这一度量对应于 $MRS(a)$ 和 $MRS(b)$ 之间的差异变得很小 (在极限中趋近于 0)。这里, 这导致了一个用 $\ln(x)$ 函数表示的表达式。回想 $\ln(x)$ 的导数是 $\frac{1}{x}$, 因此 $\frac{1}{x} = \frac{d \ln(x)}{dx} \approx \frac{\ln(x+\Delta) - \ln(x)}{\Delta}$ 。于是, $\ln(x + \Delta) - \ln(x) \approx \frac{\Delta}{x}$ 。从这个角度看, 方程 (7.1) 中的分子是 $d \ln\left(\frac{x_2}{x_1}\right)$, 分母是 $d \ln(MRS)$ 。用微分形式表示:

$$\sigma = \frac{d \ln\left(\frac{x_2}{x_1}\right)}{d \ln(MRS)} = \frac{d \ln\left(\frac{x_2}{x_1}\right)}{d \ln\left(\frac{u_{x_1}}{u_{x_2}}\right)}, \quad (7.2)$$

回想 MRS 是从 $du = 0 = u_{x_1}dx_1 + u_{x_2}dx_2$ 得到的, 给出 $\frac{dx_2}{dx_1} = -\frac{u_{x_1}}{u_{x_2}}$, 从而 $MRS = \frac{u_{x_1}}{u_{x_2}}$ (以绝对值衡量斜率)。

Original (English)

Regarding the sign of σ , note that, in the movement from a to b , the slope of the indifference curve is always negative but increases towards 0. So, in absolute terms, the slope (MRS) declines (is getting smaller). Likewise, the ratio of x_2 to x_1 gets smaller, moving from a to b . Therefore, in moving from a to b both the ratio and MRS decline. Conversely, in the movement from b to a both increase. Consequently, as defined, the elasticity of substitution is non-negative, provided the level surface of the function is convex, as depicted.

翻译 (Chinese)

关于 σ 的符号, 注意在从 a 到 b 的移动中, 无差异曲线的斜率始终为负, 但朝 0 的方向增加。因此, 以绝对值来衡量, 斜率 (MRS) 下降 (变得更小)。同样地, 从 a 移动到 b , x_2 与 x_1 的比值也变小。因此, 在从 a 到 b 的移动中, 比值和 MRS 都下降。反之, 在从 b 到 a 的移动中, 两者都增加。因此, 按照定义, 只要函数的水平曲面是凸的 (如图所示), 替代弹性就是非负的。

Original (English)

Example 7.6: The constant elasticity of substitution function is:

$$u(x_1, x_2) = [\alpha x_1^\rho + (1 - \alpha)x_2^\rho]^{\frac{1}{\rho}}.$$

Therefore,

$$\begin{aligned} u_{x_1} &= \frac{1}{\rho} [\alpha x_1^\rho + (1 - \alpha)x_2^\rho]^{\frac{1}{\rho}-1} \alpha \rho x_1^{\rho-1}, \\ u_{x_2} &= \frac{1}{\rho} [\alpha x_1^\rho + (1 - \alpha)x_2^\rho]^{\frac{1}{\rho}-1} (1 - \alpha) \rho x_2^{\rho-1}. \end{aligned}$$

Therefore, MRS is:

$$\frac{u_{x_1}}{u_{x_2}} = \frac{\alpha x_1^{\rho-1}}{(1 - \alpha)x_2^{\rho-1}} = \frac{\alpha}{1 - \alpha} \left(\frac{x_1}{x_2} \right)^{\rho-1}.$$

Taking the $\ln$ of both sides,

$$\ln \frac{u_{x_1}}{u_{x_2}} = \ln \frac{\alpha}{1 - \alpha} + (\rho - 1) \ln \frac{x_1}{x_2}.$$

Therefore,

$$\ln \frac{x_2}{x_1} = -\frac{1}{\rho - 1} \ln \frac{\alpha}{1 - \alpha} + \frac{1}{\rho - 1} \ln \frac{u_{x_1}}{u_{x_2}}$$

and

$$\sigma = \frac{d \ln \left(\frac{x_2}{x_1} \right)}{d \ln \left(\frac{u_{x_1}}{u_{x_2}} \right)} = \frac{1}{1 - \rho}.$$

翻译 (Chinese)

例 7.6: 常替代弹性函数为:

$$u(x_1, x_2) = [\alpha x_1^\rho + (1 - \alpha)x_2^\rho]^{\frac{1}{\rho}}.$$

因此,

$$\begin{aligned} u_{x_1} &= \frac{1}{\rho} [\alpha x_1^\rho + (1 - \alpha)x_2^\rho]^{\frac{1}{\rho}-1} \alpha \rho x_1^{\rho-1}, \\ u_{x_2} &= \frac{1}{\rho} [\alpha x_1^\rho + (1 - \alpha)x_2^\rho]^{\frac{1}{\rho}-1} (1 - \alpha) \rho x_2^{\rho-1}. \end{aligned}$$

因此, MRS 为:

$$\frac{u_{x_1}}{u_{x_2}} = \frac{\alpha x_1^{\rho-1}}{(1-\alpha)x_2^{\rho-1}} = \frac{\alpha}{1-\alpha} \left(\frac{x_1}{x_2} \right)^{\rho-1}.$$

对两边取 $\ln$,

$$\ln \frac{u_{x_1}}{u_{x_2}} = \ln \frac{\alpha}{1-\alpha} + (\rho-1) \ln \frac{x_1}{x_2}.$$

因此,

$$\ln \frac{x_2}{x_1} = -\frac{1}{\rho-1} \ln \frac{\alpha}{1-\alpha} + \frac{1}{\rho-1} \ln \frac{u_{x_1}}{u_{x_2}}$$

且

$$\sigma = \frac{d \ln \left(\frac{x_2}{x_1} \right)}{d \ln \left(\frac{u_{x_1}}{u_{x_2}} \right)} = \frac{1}{1-\rho}.$$

Original (English)

The constant elasticity function can exhibit substantial variation in shape, depending on the value of ρ . Figure 7.5 illustrates three possible values for ρ . The figure suggests some properties of this function as ρ varies. When $\rho = 1$, the function is linear. As ρ decreases to 0, the function approaches the Cobb–Douglas function, $x_1^\alpha x_2^{1-\alpha}$, while as ρ goes to $-\infty$, the function approaches the Leontief function, $\min\{x_1, x_2\}$ (and as ρ tends to ∞ , the function tends to $\max\{x_1, x_2\}$).

翻译 (Chinese)

常弹性函数可以根据 ρ 的值展现出形状上的显著变化。图 7.5 展示了 ρ 的三种可能取值。该图提示了当 ρ 变化时此函数的一些性质。当 $\rho = 1$ 时, 函数是线性的。当 ρ 减小到 0 时, 函数趋近于柯布–道格拉斯函数 $x_1^\alpha x_2^{1-\alpha}$, 而当 ρ 趋于 $-\infty$ 时, 函数趋近于里昂惕夫函数 $\min\{x_1, x_2\}$ (当 ρ 趋于 ∞ 时, 函数趋于 $\max\{x_1, x_2\}$)。

Original (English)

Theorem 7.2: Given $u(x_1, x_2) = [\alpha x_1^\rho + (1-\alpha)x_2^\rho]^{\frac{1}{\rho}}$, then:

- $[\alpha x_1^\rho + (1-\alpha)x_2^\rho]^{\frac{1}{\rho}} \rightarrow x_1^\alpha x_2^{1-\alpha}$ as $\rho \rightarrow 0$,
- $[\alpha x_1^\rho + (1-\alpha)x_2^\rho]^{\frac{1}{\rho}} \rightarrow \min\{x_1, x_2\}$ as $\rho \rightarrow -\infty$.

翻译 (Chinese)

定理 7.2: 给定 $u(x_1, x_2) = [\alpha x_1^\rho + (1-\alpha)x_2^\rho]^{\frac{1}{\rho}}$, 则:

- $[\alpha x_1^\rho + (1-\alpha)x_2^\rho]^{\frac{1}{\rho}} \rightarrow x_1^\alpha x_2^{1-\alpha}$, 当 $\rho \rightarrow 0$ 时,
- $[\alpha x_1^\rho + (1-\alpha)x_2^\rho]^{\frac{1}{\rho}} \rightarrow \min\{x_1, x_2\}$, 当 $\rho \rightarrow -\infty$ 时。

Original (English)

Theorem 7.4: Suppose that F_i has continuous partial derivatives for all i and $|J(x^*)| \neq 0$ at (y^*, x^*) . Then there exist continuous functions $f^1, \dots, f^i, \dots, f^n$, such that $y_j = f^j(x)$, $j = 1, 2, \dots, n$, for x in a neighborhood of x^* . Furthermore, the effect on y_j of a change in x_i , $\frac{\partial y_j}{\partial x_i}$, is given by

$$\frac{\partial y_j}{\partial x_i} = \frac{|J^{ij}|}{|J|}.$$

翻译 (Chinese)

定理 7.4: 假设 F_i 对所有 i 均有连续偏导数, 且在 (y^*, x^*) 处 $|J(x^*)| \neq 0$. 则存在连续函数 $f^1, \dots, f^i, \dots, f^n$, 使得对于 x^* 的某个邻域内的 x , 有 $y_j = f^j(x)$, $j = 1, 2, \dots, n$. 此外, x_i 的变化对 y_j 的影响 $\frac{\partial y_j}{\partial x_i}$ 由下式给出:

$$\frac{\partial y_j}{\partial x_i} = \frac{|J^{ij}|}{|J|}.$$

Original (English)

To illustrate, return to the production problem discussed in Section 7.6.1 with a different specification of the production and constraint technology. Let demand, Q , be exogenously given and determining the required output. Suppose that the technology relating inputs x and y to output is $Q = x^\alpha y^\beta$. In addition, a technology constraint requires that $y = kx$. We wish to consider the impact of changes in the required production level Q on x and y . The equations relating the variables may be written:

$$F_1(x, y; Q, \alpha, \beta, k) = Q - x^\alpha y^\beta = 0, \quad F_2(x, y; Q, \alpha, \beta, k) = y - kx = 0.$$

These equations are depicted in Figure 7.11, where the impact of varying Q is illustrated.

$$J = \begin{pmatrix} -\alpha x^{\alpha-1} y^\beta & -\beta x^\alpha y^{\beta-1} \\ -k & 1 \end{pmatrix}, \quad \begin{pmatrix} -\frac{\partial F_1}{\partial Q} \\ -\frac{\partial F_2}{\partial Q} \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

Thus, $|J| = -\frac{Q}{x}(\alpha + \beta)$, and

$$\frac{\partial x}{\partial Q} = \frac{1}{|J|} \begin{vmatrix} 1 & -\beta x^\alpha y^{\beta-1} \\ 0 & 1 \end{vmatrix} = \frac{1}{|J|} = -\frac{x}{Q(\alpha + \beta)}, \quad \frac{\partial y}{\partial Q} = \frac{1}{|J|} \begin{vmatrix} -\alpha x^{\alpha-1} y^\beta & 1 \\ -k & 0 \end{vmatrix} = \frac{k}{|J|} = -\frac{kx}{Q(\alpha + \beta)} = -\frac{y}{Q(\alpha + \beta)}.$$

翻译 (Chinese)

为加以说明, 回到第 7.6.1 节讨论的生产问题, 但采用不同的生产技术和约束技术设定。设需求 Q 是外生给定的, 并决定了必需的产出。假设将投入 x 和 y 与产出联系起来的技术

为 $Q = x^\alpha y^\beta$ 。此外，一个技术约束要求 $y = kx$ 。我们希望考虑所需生产水平 Q 的变化对 x 和 y 的影响。将变量联系起来的方程可以写为：

$$F_1(x, y; Q, \alpha, \beta, k) = Q - x^\alpha y^\beta = 0, \quad F_2(x, y; Q, \alpha, \beta, k) = y - kx = 0.$$

这些方程在图 7.11 中描绘，其中展示了 Q 变化的影响。

$$J = \begin{pmatrix} -\alpha x^{\alpha-1} y^\beta & -\beta x^\alpha y^{\beta-1} \\ -k & 1 \end{pmatrix}, \quad \begin{pmatrix} -\frac{\partial F_1}{\partial Q} \\ -\frac{\partial F_2}{\partial Q} \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

因此， $|J| = -\frac{Q}{x}(\alpha + \beta)$ ，且

$$\frac{\partial x}{\partial Q} = \frac{1}{|J|} \begin{vmatrix} 1 & -\beta x^\alpha y^{\beta-1} \\ 0 & 1 \end{vmatrix} = \frac{1}{|J|} = -\frac{x}{Q(\alpha + \beta)}, \quad \frac{\partial y}{\partial Q} = \frac{1}{|J|} \begin{vmatrix} -\alpha x^{\alpha-1} y^\beta & 1 \\ -k & 0 \end{vmatrix} = \frac{k}{|J|} = -\frac{y}{Q(\alpha + \beta)}.$$

Original (English)

Example 7.13: Consider a macroeconomic model with two equations:

$$y = c(y - T(y), r) + I(y, r) + G, \quad M = L(y, r).$$

The first equation is called the IS curve and the second the LM curve in macroeconomics. In this widely studied model, a common task is to determine the impact of changes in policy parameters (government expenditure G and the money supply M) on income y and the interest rate r . These are called policy multipliers and are computed next. For example, $\frac{\partial y}{\partial G}$ is the policy multiplier relating government expenditure to output and measures the impact on output of an increase in government expenditure.

Write c_1 for the partial derivative of c with respect to its first argument, $c_r = \frac{\partial c}{\partial r}$, $I_r = \frac{\partial I}{\partial r}$, $L_r = \frac{\partial L}{\partial r}$ and $L_y = \frac{\partial L}{\partial y}$. Then, differentiating the first equation:

$$dy = c_1(1 - T')dy + c_r dr + I_y dy + I_r dr$$

gives the variation in r resulting from a variation in y when restricted to stay on this curve. Rearranging:

$$(1 - c_1(1 - T') + I_y)dy = (c_r + I_r)dr, \quad \frac{dr}{dy} = \frac{(1 - c_1(1 - T') + I_y)}{(c_r + I_r)}.$$

A common economic assumption is that $0 < (1 - c_1(1 - T') + I_y) < 1$ along with $c_r < 0$ and $I_r < 0$, so that $\frac{dr}{dy} < 0$. Similar calculations for the LM curve yield $\frac{dr}{dy} = -\frac{L_y}{L_r}$, which is positive under the assumptions $L_y > 0$ and $L_r < 0$.

翻译 (Chinese)

例 7.13: 考虑一个包含两个方程的宏观经济模型:

$$y = c(y - T(y), r) + I(y, r) + G, \quad M = L(y, r).$$

在宏观经济学中, 第一个方程称为 IS 曲线, 第二个称为 LM 曲线。在这个被广泛研究的模型中, 一个常见的任务是确定政策参数 (政府支出 G 和货币供给 M) 的变化对收入 y 和利率 r 的影响。这些称为政策乘数, 接下来加以计算。例如, $\frac{\partial y}{\partial G}$ 是将政府支出与产出联系起来的政策乘数, 衡量了政府支出增加对产出的影响。

记 c_1 为 c 对其第一个自变量的偏导数, $c_r = \frac{\partial c}{\partial r}$, $I_r = \frac{\partial I}{\partial r}$, $L_r = \frac{\partial L}{\partial r}$ 和 $L_y = \frac{\partial L}{\partial y}$ 。然后, 对第一个方程求微分:

$$dy = c_1(1 - T')dy + c_r dr + I_y dy + I_r dr$$

给出了当被限制停留在该曲线上时, y 的变化所引起的 r 的变化。重新整理:

$$(1 - c_1(1 - T') + I_y)dy = (c_r + I_r)dr, \quad \frac{dr}{dy} = \frac{(1 - c_1(1 - T') + I_y)}{(c_r + I_r)}.$$

一个常见的经济假设是 $0 < (1 - c_1(1 - T') + I_y) < 1$, 同时 $c_r < 0$ 和 $I_r < 0$, 因此 $\frac{dr}{dy} < 0$ 。对 LM 曲线的类似计算得 $\frac{dr}{dy} = -\frac{L_y}{L_r}$, 在假设 $L_y > 0$ 和 $L_r < 0$ 下该值为正。

Original (English)

In this model, the exogenous variables are government expenditure (G) and the money supply (M). The endogenous variables are income (y) and the rate of interest (r). (As a numerical example, let $T(y) = \frac{1}{2}y$, $c(y - T(y), r) = \frac{y - \frac{1}{2}y}{r}$, $I(y, r) = \frac{y}{r}$ and $L(y, r) = \frac{1}{y^2}r^{-1}$.) These equations may be rearranged in the general form given previously:

$$F_1(y, r, G, M) = y - c(y - T(y), r) - I(y, r) - G = 0, \quad F_2(y, r, G, M) = L(y, r) - M = 0.$$

Provided the Jacobian has full rank, we may view y and r as functions of G and M , $y = f^1(G, M)$, $r = f^2(G, M)$, and compute the corresponding partial derivatives according to the rule above. With this rearrangement, the Jacobian is obtained:

$$J = \begin{pmatrix} \frac{\partial F_1}{\partial y} & \frac{\partial F_1}{\partial r} \\ \frac{\partial F_2}{\partial y} & \frac{\partial F_2}{\partial r} \end{pmatrix} = \begin{pmatrix} 1 - c_1(1 - T_y) - I_y & -(c_r + I_r) \\ L_y & L_r \end{pmatrix}.$$

The Jacobian has determinant $|J| = [1 - c_1(1 - T_y) - I_y]L_r + L_y(c_r + I_r)$.

Observe that

$$-\frac{\partial F_1}{\partial G} = +1, \quad -\frac{\partial F_1}{\partial M} = 0; \quad -\frac{\partial F_2}{\partial G} = 0, \quad -\frac{\partial F_2}{\partial M} = +1.$$

Thus,

$$J^{yG} = \begin{pmatrix} 1 & -(c_r + I_r) \\ 0 & L_r \end{pmatrix}, \quad J^{yM} = \begin{pmatrix} 0 & -(c_r + I_r) \\ 1 & L_r \end{pmatrix},$$

$$J^{rG} = \begin{pmatrix} 1 - c_1(1 - T_y) - I_y & 1 \\ L_y & 0 \end{pmatrix}, \quad J^{rM} = \begin{pmatrix} 1 - c_1(1 - T_y) - I_y & 0 \\ L_y & 1 \end{pmatrix}.$$

Applying the theorem:

$$\frac{\partial y}{\partial G} = \frac{|J^{yG}|}{|J|} = \frac{L_r}{|J|}, \quad \frac{\partial y}{\partial M} = \frac{|J^{yM}|}{|J|} = \frac{c_r + I_r}{|J|}. \quad (7.11)$$

Similarly,

$$\frac{\partial r}{\partial G} = \frac{|J^{rG}|}{|J|} = -\frac{L_y}{|J|}, \quad \frac{\partial r}{\partial M} = \frac{|J^{rM}|}{|J|} = -\frac{[1 - c_1(1 - T_y) - I_y]}{|J|}. \quad (7.12)$$

翻译 (Chinese)

在此模型中, 外生变量是政府支出 (G) 和货币供给 (M)。内生变量是收入 (y) 和利率 (r)。(作为一个数值例子, 设 $T(y) = \frac{1}{2}y$, $c(y - T(y), r) = \frac{y - \frac{1}{2}y}{r}$, $I(y, r) = \frac{y}{r}$ 且 $L(y, r) = \frac{1}{y^2}r^{-1}$ 。) 这些方程可以按先前给出的一般形式重新整理:

$$F_1(y, r, G, M) = y - c(y - T(y), r) - I(y, r) - G = 0, \quad F_2(y, r, G, M) = L(y, r) - M = 0.$$

只要雅可比是满秩的, 我们就可以将 y 和 r 视为 G 和 M 的函数, $y = f^1(G, M)$, $r = f^2(G, M)$, 并根据上述规则计算相应的偏导数。通过这种重新整理, 得到雅可比:

$$J = \begin{pmatrix} \frac{\partial F_1}{\partial y} & \frac{\partial F_1}{\partial r} \\ \frac{\partial F_2}{\partial y} & \frac{\partial F_2}{\partial r} \end{pmatrix} = \begin{pmatrix} 1 - c_1(1 - T_y) - I_y & -(c_r + I_r) \\ L_y & L_r \end{pmatrix}.$$

雅可比行列式为 $|J| = [1 - c_1(1 - T_y) - I_y]L_r + L_y(c_r + I_r)$ 。

注意到

$$-\frac{\partial F_1}{\partial G} = +1, \quad -\frac{\partial F_1}{\partial M} = 0; \quad -\frac{\partial F_2}{\partial G} = 0, \quad -\frac{\partial F_2}{\partial M} = +1.$$

因此,

$$J^{yG} = \begin{pmatrix} 1 & -(c_r + I_r) \\ 0 & L_r \end{pmatrix}, \quad J^{yM} = \begin{pmatrix} 0 & -(c_r + I_r) \\ 1 & L_r \end{pmatrix},$$

$$J^{rG} = \begin{pmatrix} 1 - c_1(1 - T_y) - I_y & 1 \\ L_y & 0 \end{pmatrix}, \quad J^{rM} = \begin{pmatrix} 1 - c_1(1 - T_y) - I_y & 0 \\ L_y & 1 \end{pmatrix}.$$

应用定理:

$$\frac{\partial y}{\partial G} = \frac{|J^{yG}|}{|J|} = \frac{L_r}{|J|}, \quad \frac{\partial y}{\partial M} = \frac{|J^{yM}|}{|J|} = \frac{c_r + I_r}{|J|}. \quad (7.11)$$

类似地,

$$\frac{\partial r}{\partial G} = \frac{|J^{rG}|}{|J|} = -\frac{L_y}{|J|}, \quad \frac{\partial r}{\partial M} = \frac{|J^{rM}|}{|J|} = -\frac{[1 - c_1(1 - T_y) - I_y]}{|J|}. \quad (7.12)$$

Original (English)

Remark 7.7: It is worth noting that, for a slightly different perspective, these results may be obtained directly by totally differentiating the original equation system.

$$dy = c_1(1 - T_y)dy + c_r dr + I_y dy + I_r dr + dG, \quad L_y dy + L_r dr = dM,$$

$$[1 - c_1(1 - T_y) - I_y]dy - (c_r + I_r)dr = dG, \quad L_y dy + L_r dr = dM,$$

$$\begin{pmatrix} [1 - c_1(1 - T_y) - I_y] & -(c_r + I_r) \\ L_y & L_r \end{pmatrix} \begin{pmatrix} dy \\ dr \end{pmatrix} = \begin{pmatrix} dG \\ dM \end{pmatrix}.$$

Note that $A = J$ and that $|A| = [1 - c_1(1 - T_y) - I_y]L_r + L_y(c_r + I_r)$. Using Cramer's rule:

$$dy = \frac{1}{|A|} \begin{vmatrix} dG & -(c_r + I_r) \\ dM & L_r \end{vmatrix} = \frac{L_r dG + (c_r + I_r)dM}{|A|} = \frac{L_r}{|A|}dG + \frac{(c_r + I_r)}{|A|}dM,$$

and

$$dr = \frac{1}{|A|} \begin{vmatrix} [1 - c_1(1 - T_y) - I_y] & dG \\ L_y & dM \end{vmatrix} = \frac{[1 - c_1(1 - T_y) - I_y]dM - L_y dG}{|A|} = -\frac{L_y}{|A|}dG + \frac{[1 - c_1(1 - T_y) - I_y]}{|A|}dM.$$

This is the same result as appears in Equations 7.11 and 7.12.

翻译 (Chinese)

注 7.7: 值得注意的是, 从略为不同的角度来看, 这些结果可以通过直接对原始方程组进行全微分得到。

$$dy = c_1(1 - T_y)dy + c_r dr + I_y dy + I_r dr + dG, \quad L_y dy + L_r dr = dM,$$

$$[1 - c_1(1 - T_y) - I_y]dy - (c_r + I_r)dr = dG, \quad L_y dy + L_r dr = dM,$$

$$\begin{pmatrix} [1 - c_1(1 - T_y) - I_y] & -(c_r + I_r) \\ L_y & L_r \end{pmatrix} \begin{pmatrix} dy \\ dr \end{pmatrix} = \begin{pmatrix} dG \\ dM \end{pmatrix}.$$

注意 $A = J$ 且 $|A| = [1 - c_1(1 - T_y) - I_y]L_r + L_y(c_r + I_r)$ 。使用克拉默法则:

$$dy = \frac{1}{|A|} \begin{vmatrix} dG & -(c_r + I_r) \\ dM & L_r \end{vmatrix} = \frac{L_r dG + (c_r + I_r)dM}{|A|} = \frac{L_r}{|A|}dG + \frac{(c_r + I_r)}{|A|}dM,$$

和

$$dr = \frac{1}{|A|} \begin{vmatrix} [1 - c_1(1 - T_y) - I_y] & dG \\ L_y & dM \end{vmatrix} = \frac{[1 - c_1(1 - T_y) - I_y]dM - L_y dG}{|A|} = -\frac{L_y}{|A|}dG + \frac{[1 - c_1(1 - T_y) - I_y]}{|A|}dM$$

这与方程 (7.11) 和 (7.12) 中的结果相同。

Original (English)

The following discussion provides a graphical perspective on the calculations. A highway starts from point $(x, y) = (0, 0)$ and follows the path $y = x + x^2 - x^3$ (given the x coordinate, there is a unique y coordinate determined by this equation). The direction x is east and the direction y is north. The point (x, y) on this highway that is a distance r from the starting point satisfies $x^2 + y^2 = r^2$. (We assume that $\frac{2}{3} > r$.) This point is uniquely defined by these two equations.

Suppose that we move slightly further along the highway by increasing the distance r ; is the direction more northerly or more easterly?

From the starting point at radius r , consider an increase in the radius to r' with $r' - r > 0$ small. This changes the coordinates (x, y) slightly to (x', y') . The eastward movement is $x' - x$ and the northward movement is $y' - y$. Then

$$\frac{x' - x}{r' - r} \approx \frac{\partial x}{\partial r}, \quad \frac{y' - y}{r' - r} \approx \frac{\partial y}{\partial r}.$$

Thus, the northward movement is greater than the eastward movement if:

$$\frac{y' - y}{x' - x} = \frac{\frac{y' - y}{r' - r}}{\frac{x' - x}{r' - r}} \approx \frac{\frac{\partial y}{\partial r}}{\frac{\partial x}{\partial r}} > 1.$$

Note that, along the highway,

$$\frac{dy}{dx} = 1 + 2x - 3x^2 = 1 + (2 - 3x)x = 1 + 3\left(\frac{2}{3} - x\right)x.$$

So, $\frac{dy}{dx} > 1$ if $\frac{2}{3} - x > 0$ or $(2 - 3x) > 0$. At any point (x, y) on the circle $x^2 + y^2 = r^2$, and since $y^2 \geq 0$, $x^2 \leq r^2$, and since the solution is positive, $x < r$, so that $2 - 3x > 2 - 3r$ or $\frac{2}{3} - x > \frac{2}{3} - r$. Since $\frac{2}{3} > r$, $\frac{2}{3} - x \geq 0$. Therefore, at the point on the circle where the curve $y = x + x^2 - x^3$ cuts, the slope of the curve is greater than 1: at the solution, $\frac{dy}{dx} > 1$, and the northward movement is greater than the eastward movement.

翻译 (Chinese)

以下讨论为这些计算提供了一个图形视角。一条公路从点 $(x, y) = (0, 0)$ 出发，沿路径

$y = x + x^2 - x^3$ 延伸 (给定 x 坐标, 由该方程确定了唯一的 y 坐标)。方向 x 为东, 方向 y 为北。该公路上距离出发点为 r 的点 (x, y) 满足 $x^2 + y^2 = r^2$ 。(我们假设 $\frac{2}{3} > r$ 。) 该点由这两个方程唯一确定。

假设我们通过增加距离 r 而沿公路稍进一步; 方向是更偏北还是更偏东?

从半径为 r 的出发点, 考虑半径增加到 r' , 其中 $r' - r > 0$ 且很小。这使坐标 (x, y) 略微变化到 (x', y') 。向东移动为 $x' - x$, 向北移动为 $y' - y$ 。则

$$\frac{x' - x}{r' - r} \approx \frac{\partial x}{\partial r}, \quad \frac{y' - y}{r' - r} \approx \frac{\partial y}{\partial r}.$$

因此, 如果:

$$\frac{y' - y}{x' - x} = \frac{\frac{y' - y}{r' - r}}{\frac{x' - x}{r' - r}} \approx \frac{\frac{\partial y}{\partial r}}{\frac{\partial x}{\partial r}} > 1,$$

则向北移动大于向东移动。

注意, 沿公路,

$$\frac{dy}{dx} = 1 + 2x - 3x^2 = 1 + (2 - 3x)x = 1 + 3\left(\frac{2}{3} - x\right)x.$$

因此, 如果 $\frac{2}{3} - x > 0$ 或 $(2 - 3x) > 0$, 则 $\frac{dy}{dx} > 1$ 。在圆 $x^2 + y^2 = r^2$ 上的任意点 (x, y) 处, 且由于 $y^2 \geq 0$, 有 $x^2 \leq r^2$, 又因为解是正的, $x < r$, 因此 $2 - 3x > 2 - 3r$ 或 $\frac{2}{3} - x > \frac{2}{3} - r$ 。由于 $\frac{2}{3} > r$, 得 $\frac{2}{3} - x \geq 0$ 。因此, 在曲线 $y = x + x^2 - x^3$ 与圆相交的点处, 曲线的斜率大于 1: 在解处, $\frac{dy}{dx} > 1$, 向北移动大于向东移动。

Original (English)

The following example reconsiders this discussion by calculating the partial derivatives of y and x with respect to r , and comparing them.

Example 7.14: Recall:

$$x^2 + y^2 = r^2, \quad y - x - x^2 + x^3 = 0.$$

$$2x dx + 2y dy = 2r dr, \quad dy - dx - 2x dx + 3x^2 dx = 0.$$

$$\begin{pmatrix} 2x & 2y \\ 1 + 2x - 3x^2 & -1 \end{pmatrix} \begin{pmatrix} dx \\ dy \end{pmatrix} = \begin{pmatrix} 2r dr \\ 0 \end{pmatrix},$$

or, since only r is varied,

$$\begin{pmatrix} 2x & 2y \\ 1 + 2x - 3x^2 & -1 \end{pmatrix} \begin{pmatrix} \frac{\partial x}{\partial r} \\ \frac{\partial y}{\partial r} \end{pmatrix} = \begin{pmatrix} 2r \\ 0 \end{pmatrix}.$$

Using Cramer's rule, the coefficient determinant is $\Delta = -2x - 2y(1 + 2x - 3x^2)$, while $\Delta_1 = -2r$ and $\Delta_2 = -(1 + 2x - 3x^2)2r$. Then

$$\frac{\partial x}{\partial r} = \frac{\Delta_1}{\Delta}, \quad \frac{\partial y}{\partial r} = \frac{\Delta_2}{\Delta}$$

and

$$\frac{\frac{\partial y}{\partial r}}{\frac{\partial x}{\partial r}} = \frac{\Delta_2}{\Delta_1} = (1 + 2x - 3x^2) = (1 + (2 - 3x)x).$$

翻译 (Chinese)

下面的例子通过计算 y 和 x 关于 r 的偏导数并加以比较, 重新审视了这一讨论。

例 7.14: 回想:

$$x^2 + y^2 = r^2, \quad y - x - x^2 + x^3 = 0.$$

$$2x dx + 2y dy = 2r dr, \quad dy - dx - 2x dx + 3x^2 dx = 0.$$

$$\begin{pmatrix} 2x & 2y \\ 1 + 2x - 3x^2 & -1 \end{pmatrix} \begin{pmatrix} dx \\ dy \end{pmatrix} = \begin{pmatrix} 2r dr \\ 0 \end{pmatrix},$$

或者, 由于只有 r 在变化,

$$\begin{pmatrix} 2x & 2y \\ 1 + 2x - 3x^2 & -1 \end{pmatrix} \begin{pmatrix} \frac{\partial x}{\partial r} \\ \frac{\partial y}{\partial r} \end{pmatrix} = \begin{pmatrix} 2r \\ 0 \end{pmatrix}.$$

使用克拉默法则, 系数行列式为 $\Delta = -2x - 2y(1 + 2x - 3x^2)$, 而 $\Delta_1 = -2r$, $\Delta_2 = -(1 + 2x - 3x^2)2r$ 。则

$$\frac{\partial x}{\partial r} = \frac{\Delta_1}{\Delta}, \quad \frac{\partial y}{\partial r} = \frac{\Delta_2}{\Delta}$$

且

$$\frac{\frac{\partial y}{\partial r}}{\frac{\partial x}{\partial r}} = \frac{\Delta_2}{\Delta_1} = (1 + 2x - 3x^2) = (1 + (2 - 3x)x).$$

Exercises for Chapter 7 / 第 7 章习题

Original (English)

Exercise 7.1 The following simple macroeconomic model has four equations relating government policy variables to aggregate income and investment. Aggregate income equals consumption plus investment and government expenditure: $y = c + i + g$. Consumption is positively related to the real rate of interest: $c = \alpha + \beta y$, $\alpha, \beta > 0$. Investment is negatively related to the rate of interest, r : $i = \gamma - \delta r$, $\gamma, \delta > 0$. The money supply, m_s , equals the money demand, m_d , and this in turn

is positively related to the level of income ($\kappa > 0$) and negatively related to the rate of interest ($\lambda > 0$): $m_s = m_d = \kappa y - \lambda r$. Find the impact on y of (a) a small change in g , Δg , and (b) a small change in m , Δm .

Exercise 7.2 Consider a market defined by the following supply and demand curves: $q_d = a + bP_d + c\tilde{Y}$, $q_s = d + eP_s$, where $a, c, d, e > 0$, and $b < 0$, $a > d$. (a) (i) Compute the equilibrium price P and quantity q . (ii) Sign the comparative statics partial derivatives $\frac{\partial P}{\partial \tilde{Y}}$, $\frac{\partial q}{\partial \tilde{Y}}$. (b) Suppose a tax $t > 0$ is levied on producers: (i) Compute the new equilibrium price and quantity. (ii) Sign the derivatives $\frac{\partial q}{\partial t}$, $\frac{\partial P_d}{\partial t}$, $\frac{\partial P_s}{\partial t}$.

Exercise 7.3 Consider the following macroeconomic model of national income: $C = \alpha + \beta Y$, $\alpha > 0$, $0 < \beta < 1$; $Y = C + I + G$; $I = \gamma Y$, $0 < \gamma < 1$. What are the exogenous and endogenous variables? Write this system in matrix form and solve for the reduced form. Using the reduced form expressions, find the following comparative statics derivatives: $\frac{\partial C}{\partial G}$, $\frac{\partial Y}{\partial G}$, $\frac{\partial I}{\partial G}$.

Exercise 7.4 Consider the following macroeconomic model: $y = c(y, r) + i(r) + G$, $m = l(r, y)$. (a) Identify the exogenous and endogenous variables. (b) (i) What is the effect of an increase in government expenditure G , on output y and the interest rate r ? (ii) What is the effect of an increase in the money supply M , on output and the interest rate? (c) Consider what happens if we replace the above money market equation with the following: $m = \kappa y$. Repeat part (b) with this assumption. (d) Represent the two different systems in (y, r) space and show the above effect diagrammatically.

Exercise 7.5 Consider the following macroeconomic model: $S(Y, r) + T(Y) + IM = I(r) + G + x(e)$, $L(Y, r) = M$, where Y is national income, S is savings, IM is imports, x is exports, I is investment, G is government expenditure, M is money supply and e is the exchange rate. Using this model, determine how the equilibrium values of Y and r change with a change in the exchange rate e and with a change in the exogenous level of imports. Make the usual assumptions: $S_Y > 0$, $S_r > 0$, $T_Y > 0$, $I_r < 0$, $x_e < 0$, $L_Y > 0$, $L_r < 0$.

Exercise 7.6 Consider a production process using input x to produce output y . The relationship between inputs and outputs must satisfy two conditions: $x^2 + y^2 = \alpha$, $0 < \alpha < 1$; $y = x(1 - x)$, $0 < \beta$. This pair of equations has a unique non-negative solution ($x, y \geq 0$). Plot these functions in (x, y) space. At this solution, calculate $\frac{dx}{d\alpha}$, $\frac{dy}{d\alpha}$, $\frac{dx}{d\beta}$ and $\frac{dy}{d\beta}$. In relation to your graph, interpret the signs of the derivatives.

Exercise 7.7 Repeat the previous question, with input and output required to satisfy the following two conditions: $y - \ln(x + 1) = 0$, $xy = \theta$.

Exercise 7.8 Consider the following IS-LM model with an import sector: $y = c(y) + I(i) + G$, $m = L(y, i)$, $x(\rho) = z(y, \rho)$. (a) Calculate the slopes of the IS and LM curves. (b) Find $\frac{dy}{dG}$, $\frac{di}{dG}$

and $\frac{d\rho}{dG}$.

Exercise 7.9 An individual has utility function $u(x, y) = x^\alpha y^\beta$, where $0 < \alpha < 1$, $0 < \beta < 1$ and $0 < \alpha + \beta < 1$. The prices of x and y , respectively, are p and q , and the individual has income I . The marginal utility of x is $u_x(x, y) = \alpha x^{\alpha-1} y^\beta = \alpha \frac{u(x, y)}{x}$. Similarly, $u_y(x, y) = \beta x^\alpha y^{\beta-1} = \beta \frac{u(x, y)}{y}$. The utility maximizing levels of x and y are determined by the tangency condition $MRS = \frac{u_x}{u_y} = \frac{p}{q}$ and the budget constraint $px + qy = I$. (a) Write the pair of equations in the form $F_1(x, y; \alpha, \beta, p, q, I) = 0$, $F_2(x, y; \alpha, \beta, p, q, I) = 0$ and calculate $\frac{\partial x}{\partial I}$, $\frac{\partial x}{\partial p}$, $\frac{\partial x}{\partial q}$, $\frac{\partial x}{\partial \alpha}$, $\frac{\partial x}{\partial \beta}$. (b) Solve (1) and (2) directly for x and y , then take the partial derivatives of x with respect to the parameters, to confirm your answer in (a).

Exercise 7.10 A production possibilities (transformation) curve in two goods, x and y , is given by $T(x, y) = k$, where $T(x, y) = x^2 + y$. The social utility function is given by $u(x, y) = \alpha \ln(x) + \beta \ln(y)$. The efficient choices for x and y are given at the point where $MRS = MRT$ or $\frac{u_x}{u_y} = \frac{T_x}{T_y}$. Thus, there are two equations determining the solution. (a) Rearrange these to get two equations in the form $F_1(x, y; \alpha, \beta, k) = 0$, $F_2(x, y; \alpha, \beta, k) = 0$. (b) Use the implicit function theorem to determine $\frac{\partial x}{\partial k}$, $\frac{\partial y}{\partial k}$, $\frac{\partial x}{\partial \alpha}$, $\frac{\partial y}{\partial \alpha}$.

Exercise 7.13 A firm is required to produce output Q using inputs x and y . The production technology is $x^\alpha y^\beta$. In addition, the firm faces the technology constraint $y = ke^x$, $k > 0$. Find $\frac{\partial y}{\partial Q}$, $\frac{\partial x}{\partial Q}$, $\frac{\partial y}{\partial k}$, $\frac{\partial x}{\partial k}$. Also, try finding $\frac{\partial y}{\partial \alpha}$, $\frac{\partial x}{\partial \alpha}$.

Exercise 7.17 An individual has income or wealth, y , to allocate between present, c , and future, f , consumption. Suppose that the individual has utility function $u(c, f) = c \cdot f$. Income not consumed in the current period is invested at interest rate r to yield the amount available for future consumption: $f = (y - c)(1 + r)$. Since y is given, the choice of c determines f . Find the impact of a change in income, y , on consumption c .

翻译 (Chinese)

习题 7.1 以下简单的宏观经济模型有四个方程，将政府政策变量与总收入及投资联系起来。总收入等于消费加投资加政府支出： $y = c + i + g$ 。消费与实际利率正相关： $c = \alpha + \beta y$, $\alpha, \beta > 0$ 。投资与利率 r 负相关： $i = \gamma - \delta r$, $\gamma, \delta > 0$ 。货币供给 m_s 等于货币需求 m_d ，而后者与收入水平正相关 ($\kappa > 0$)，与利率负相关 ($\lambda > 0$)： $m_s = m_d = \kappa y - \lambda r$ 。求 (a) g 的微小变化 Δg 对 y 的影响，以及 (b) m 的微小变化 Δm 对 y 的影响。

习题 7.2 考虑由以下供需曲线定义的市场： $q_d = a + bP_d + c\tilde{Y}$, $q_s = d + eP_s$ ，其中 $a, c, d, e > 0$, $b < 0$, $a > d$ 。(a) (i) 计算均衡价格 P 和数量 q 。(ii) 确定比较静态偏导数 $\frac{\partial P}{\partial \tilde{Y}}$, $\frac{\partial q}{\partial \tilde{Y}}$ 的符号。(b) 假设对生产者征收税额 $t > 0$ ：(i) 计算新的均衡价格和数量。(ii) 确定导数 $\frac{\partial q}{\partial t}$, $\frac{\partial P_d}{\partial t}$, $\frac{\partial P_s}{\partial t}$ 的符号。

习题 7.3 考虑以下国民收入宏观经济模型: $C = \alpha + \beta Y$, $\alpha > 0$, $0 < \beta < 1$; $Y = C + I + G$; $I = \gamma Y$, $0 < \gamma < 1$ 。外生变量和内生变量分别是什么? 将此系统写成矩阵形式并求解简化形式。利用简化形式的表达式, 求以下比较静态导数: $\frac{\partial C}{\partial G}$, $\frac{\partial Y}{\partial G}$, $\frac{\partial I}{\partial G}$ 。

习题 7.9 一个体有柯布-道格拉斯效用函数 $u(x, y) = x^\alpha y^\beta$, 其中 $0 < \alpha < 1$, $0 < \beta < 1$ 且 $0 < \alpha + \beta < 1$ 。商品 x 和 y 的价格分别为 p 和 q , 个体有收入 I 。效用最大化的 x 和 y 水平由切点条件 $MRS = \frac{u_x}{u_y} = \frac{p}{q}$ 和预算约束 $px + qy = I$ 确定。(a) 将方程对写成 $F_1(x, y; \alpha, \beta, p, q, I) = 0$, $F_2(x, y; \alpha, \beta, p, q, I) = 0$ 的形式, 计算 $\frac{\partial x}{\partial I}$, $\frac{\partial x}{\partial p}$, $\frac{\partial x}{\partial q}$, $\frac{\partial x}{\partial \alpha}$, $\frac{\partial x}{\partial \beta}$ 。(b) 直接求解 x 和 y , 然后对参数求 x 的偏导数以验证 (a) 中的答案。

习题 7.17 一个体有收入或财富 y , 需在当前消费 c 和未来消费 f 之间分配。假设该个体的效用函数为 $u(c, f) = c \cdot f$ 。当期末消费的收入按利率 r 投资, 产生可用于未来消费的金额: $f = (y - c)(1 + r)$ 。由于 y 是给定的, c 的选择决定了 f 。求收入 y 的变化对消费 c 的影响。

6.12 Taylor Series / 泰勒级数

6.12.1 Introduction / 引言

Original (English)

Problems such as maximization of a function f or finding the root of an equation $f(x) = 0$ arise frequently in economics. However, often the function may be difficult to work with, motivating the use of approximations to the function f that are easier to study and provide insight regarding f . One relatively simple class of approximating functions are the polynomials. A function h is a polynomial in x if it has the form:

$$h(x) = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n,$$

where each a_i is a constant. So, h is the sum of terms where each term is a constant multiplied by x to the power of some integer, such as a_3x^3 , where a_3 is the constant and x is raised to the power of 3. The function h is called an n th order polynomial because the highest power is n (assuming $a_n \neq 0$). For example, $h(x) = 2 + x + 5x^2 + 8x^3$ is a third-order polynomial, as is $g(x) = 2x^3$. Under suitable circumstances, a large class of functions can be approximated by a polynomial, so that given the function f , one may find a polynomial such as h approximating f . This makes it possible to discover properties of f by study of the simpler function h and that fact is a primary motivation for studying these approximations. Taylor's theorem concerns the approximation of functions with polynomials and provides the theoretical basis for the polynomial approximations.

翻译 (Chinese)

诸如函数 f 的最大化或寻找方程 $f(x) = 0$ 的根等问题在经济学中经常出现。然而，函数往往难以处理，这促使人们使用更容易研究并能提供关于 f 的洞见的对 f 的近似。一类相对简单的近似函数是多项式。函数 h 是 x 的多项式，如果它具有如下形式：

$$h(x) = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n,$$

其中每个 a_i 是常数。因此， h 是各项之和，其中每一项都是一个常数乘以 x 的某个整数次幂，例如 a_3x^3 ，其中 a_3 是常数， x 被升到 3 次幂。函数 h 被称为 n 阶多项式，因为最高次幂为 n （假设 $a_n \neq 0$ ）。例如， $h(x) = 2 + x + 5x^2 + 8x^3$ 是一个三阶多项式， $g(x) = 2x^3$ 也是。在适当的条件下，很大一类函数可以用多项式来近似，因此给定函数 f ，我们可以找到一个多项式如 h 来近似 f 。这使得通过研究更简单的函数 h 来发现 f 的性质成为可能，这一事实是研究这些近似的主要动机。泰勒定理涉及用多项式对函数进行近似，并为多项式近似提供了理论基础。

Original (English)

Section 8.1.1 motivates the approach by illustrating how a quadratic function may be chosen to approximate a continuous function. Following this, in Section 8.1.2 the main theorem is stated. This asserts that any continuous differentiable function may be approximated by a polynomial function — the Taylor series expansion. Because Taylor series expansions involve approximation, there is a gap between the approximation and the actual function. However, the quality of the approximation improves as the length of the Taylor series expansion increases — the higher the order of the polynomial, the better the approximation. The improvement from lengthening the approximation is considered with an example in Section 8.1.3. Sections 8.2 to 8.4 provide a number of applications. Section 8.2 shows how Taylor series expansions may be used to address a variety of problems — such as relating concavity to the derivatives of a function, finding the roots of an equation, and generating numerical optimization routines to find maxima and minima of a function. In Section 8.3, expected utility is introduced. In this section, the theory is used to provide an approximate measure of utility loss from risk and a monetary measure of the cost incurred from bearing risk. Section 8.4 uses Taylor series to find the approximately optimal solution to a portfolio selection problem. Finally, Section 8.5 describes the multivariate version of Taylor's theorem and Section 8.6 provides a proof of the theorem.

翻译 (Chinese)

第 8.1.1 节通过说明如何选择一个二次函数来近似一个连续函数来激发该方法。此后，第 8.1.2 节陈述了主要定理。该定理断言任何连续可微函数都可以用多项式函数——泰勒级数

展开——来近似。由于泰勒级数展开涉及近似，在近似与实际函数之间存在差距。然而，随着泰勒级数展开长度的增加，近似的质量会提高——多项式的阶数越高，近似就越好。第 8.1.3 节通过一个例子考虑了延长近似所带来的改进。第 8.2 至 8.4 节提供了若干应用。第 8.2 节展示了泰勒级数展开如何用于处理各种问题——如将凹性与函数的导数联系起来、寻找方程的根、以及生成数值最优化程序以寻找函数的最大值和最小值。在第 8.3 节中，引入了期望效用理论。该节利用该理论提供了风险所致效用损失的近似度量以及承担风险所产生成本的货币度量。第 8.4 节使用泰勒级数来寻找投资组合选择问题的近似最优解。最后，第 8.5 节描述了泰勒定理的多元版本，第 8.6 节给出了该定理的证明。

6.12.2 Taylor Series Approximations: An Example / 泰勒级数近似：一个例子

Original (English)

To illustrate the way in which Taylor series may be used to approximate a function, consider the function:

$$f(x) = \ln(x+1) e^{-\frac{x}{2}}.$$

Although simple to describe mathematically, the function has reasonably complex behavior as x varies. Starting from $x = 0$, the function increases initially and then declines, “flattening out” to 0 as x becomes large. Often, functions such as this are difficult to study. For example, the function has a unique maximum, x^* , defined by $f'(x^*) = 0$, but this expression is very difficult to solve. One way to address this difficulty is to approximate f by another function, g , which is “close” to f but simpler to manipulate.

Consider the function, $g(x)$, which is given by the following formula:

$$g(x) = 0.1524 + 0.44285x - 0.17489x^2.$$

Observe that the function g “tracks” the function $f(x)$ to some degree — see Figure 8.1. Both functions, f and g , have the same value at $x = 1$ and are close when x is close to 1. So, g is a good approximation to f when x is close to 1 and g has a simple form.

翻译 (Chinese)

为说明泰勒级数可用于近似函数的方式，考虑函数：

$$f(x) = \ln(x+1) e^{-\frac{x}{2}}.$$

虽然在数学上描述简单, 但该函数随着 x 的变化具有相当复杂的行为。从 $x = 0$ 开始, 函数先增加然后下降, 随着 x 变大而“平坦化”趋于 0。通常, 这样的函数难以研究。例如, 该函数有一个唯一的最大值 x^* , 由 $f'(x^*) = 0$ 定义, 但这个表达式非常难以求解。解决这一困难的一种方法是用另一个函数 g 来近似 f , g “接近” f 但更易于操作。

考虑由以下公式给出的函数 $g(x)$:

$$g(x) = 0.1524 + 0.44285x - 0.17489x^2.$$

注意到函数 g 在某种程度上“跟踪”了函数 $f(x)$ ——见图 8.1。两个函数 f 和 g 在 $x = 1$ 处有相同的值, 且当 x 接近 1 时接近。因此, 当 x 接近 1 时, g 是 f 的一个良好近似, 且 g 具有简单的形式。

Original (English)

In fact, the function g is a second-order Taylor series expansion of f , defined according to the formula:

$$g(x) = f(1) + f'(1)(x - 1) + \frac{1}{2}f''(1)(x - 1)^2. \quad (8.1)$$

The logic for this is discussed later. The following calculations confirm that this formula in Equation 8.1 yields the expression for g given above. Observe that:

$$f'(x) = \frac{e^{-\frac{1}{2}x}}{(x+1)} - \frac{1}{2}\ln(x+1)e^{-\frac{1}{2}x}, \quad f'(1) = \frac{1}{2}e^{-\frac{1}{2}} - \frac{1}{2}\ln(2)e^{-\frac{1}{2}},$$

and

$$f''(x) = -\frac{x+2}{(x+1)^2}e^{-\frac{1}{2}x} + \frac{1}{4}\ln(x+1)e^{-\frac{1}{2}x}, \quad f''(1) = -\frac{3}{4}e^{-\frac{1}{2}} + \frac{1}{4}\ln(2)e^{-\frac{1}{2}}.$$

With these calculations in place, g may be obtained:

$$\begin{aligned} g(x) &= f(1) + f'(1)(x - 1) + \frac{1}{2}f''(1)(x - 1)^2 \\ &= 0.1524 + 0.44285x - 0.17489x^2, \end{aligned}$$

using the fact that $\ln(2) = 0.69314$ and $e^{-\frac{1}{2}} = 0.60653$. The function g is called a second-order Taylor series expansion of f around 1 since we used $f(1)$, $f'(1)$ and $f''(1)$ to obtain g .

翻译 (Chinese)

事实上, 函数 g 是 f 的一个二阶泰勒级数展开, 根据以下公式定义:

$$g(x) = f(1) + f'(1)(x - 1) + \frac{1}{2}f''(1)(x - 1)^2. \quad (8.1)$$

其逻辑将在后面讨论。下面的计算证实了方程 (8.1) 中的公式确实产生了上述 g 的表达式。注意到：

$$f'(x) = \frac{e^{-\frac{1}{2}x}}{(x+1)} - \frac{1}{2} \ln(x+1)e^{-\frac{1}{2}x}, \quad f'(1) = \frac{1}{2}e^{-\frac{1}{2}} - \frac{1}{2} \ln(2)e^{-\frac{1}{2}},$$

和

$$f''(x) = -\frac{x+2}{(x+1)^2}e^{-\frac{1}{2}x} + \frac{1}{4} \ln(x+1)e^{-\frac{1}{2}x}, \quad f''(1) = -\frac{3}{4}e^{-\frac{1}{2}} + \frac{1}{4} \ln(2)e^{-\frac{1}{2}}.$$

有了这些计算， g 可被获得：

$$\begin{aligned} g(x) &= f(1) + f'(1)(x-1) + \frac{1}{2}f''(1)(x-1)^2 \\ &= 0.1524 + 0.44285x - 0.17489x^2, \end{aligned}$$

利用了 $\ln(2) = 0.69314$ 和 $e^{-\frac{1}{2}} = 0.60653$ 。函数 g 被称为 f 在 1 附近的二阶泰勒级数展开，因为我们使用 $f(1)$, $f'(1)$ 和 $f''(1)$ 来获得 g 。

Original (English)

The ability to approximate a function (such as f), by a polynomial function, such as the quadratic function g , turns out to have many uses. One application is optimization. If the maximum of f is difficult to determine but g is close to f , then finding the maximum of g may be a good approximation to maximizing f . Note that $f'(x)$ may be written $f'(x) = \left[\frac{1}{(x+1)} - \frac{1}{2} \ln(x+1) \right] e^{-\frac{1}{2}x}$. Since for any x , $e^{-\frac{1}{2}x} \neq 0$, $f'(x) = 0$ implies $\frac{1}{(x+1)} - \frac{1}{2} \ln(x+1) = 0$: x^* satisfies $\frac{1}{(x^*+1)} - \frac{1}{2} \ln(x^*+1) = 0$. The solution for x^* is $x^* = 1.34575$. The function f has a unique maximum at x^* .

The function g has a maximum at $\hat{x} = 1.266$. Assume that, since g approximates f , the maximum of g might be close to the maximum of f , so if we now expand f around $\hat{x}$, we obtain a “new” function, $\hat{g}$; this should be a better approximation of f close to the maximum:

$$\hat{g}(x) = f(1.266) + f'(1.266)(x-1.266) + \frac{1}{2}f''(1.266)(x-1.266)^2 = 0.22901 + 0.307258x - 0.114577x^2.$$

Observe that $\hat{g}$ is maximized at $x = 1.341$, which is very close to x^* . These calculations suggest a simple procedure for locating the maximum of a function using a simple algorithm, and illustrate one important use of Taylor series expansions.

翻译 (Chinese)

用多项式函数（如二次函数 g ）来近似一个函数（如 f ）的能力被证明有许多用途。一个应用是最优化。如果 f 的最大值难以确定但 g 接近 f ，那么寻找 g 的最大值可能是最大化 f 的一个良好近似。注意 $f'(x)$ 可以写为 $f'(x) = \left[\frac{1}{(x+1)} - \frac{1}{2} \ln(x+1) \right] e^{-\frac{1}{2}x}$ 。由于对任意 x ,

$e^{-\frac{1}{2}x} \neq 0$, $f'(x) = 0$ 意味着 $\frac{1}{(x+1)} - \frac{1}{2} \ln(x+1) = 0$: x^* 满足 $\frac{1}{(x^*+1)} - \frac{1}{2} \ln(x^*+1) = 0$. x^* 的解为 $x^* = 1.34575$. 函数 f 在 x^* 处有唯一的最大值。

函数 g 在 $\hat{x} = 1.266$ 处有最大值。假设由于 g 近似 f , g 的最大值可能接近 f 的最大值, 因此如果我们现在在 $\hat{x}$ 附近展开 f , 就得到一个“新的”函数 $\hat{g}$; 这应该在最大值附近是 f 的一个更好的近似:

$$\hat{g}(x) = f(1.266) + f'(1.266)(x-1.266) + \frac{1}{2}f''(1.266)(x-1.266)^2 = 0.22901 + 0.307258x - 0.114577x^2.$$

注意到 $\hat{g}$ 在 $x = 1.341$ 处被最大化, 这与 x^* 非常接近。这些计算提示了使用一个简单算法来定位函数最大值的简单程序, 并说明了泰勒级数展开的一个重要用途。

6.12.3 Taylor Series Expansions / 泰勒级数展开

Original (English)

Taylor's theorem asserts that a continuous function with continuous derivatives may be approximated by a polynomial function. For notational convenience write $f^{(k)}(x) = \frac{d^k f}{dx^k}$ for the k th derivative of the function f . (As usual the lower derivatives are often written $f'(x) = \frac{df}{dx}$, $f''(x) = \frac{d^2 f}{dx^2}$ and $f'''(x) = \frac{d^3 f}{dx^3}$ instead of $f^{(1)}(x)$, $f^{(2)}(x)$ and $f^{(3)}(x)$.)

Theorem 8.1: Let f be a continuous function on an interval $[a, b]$ with continuous derivatives $f', f'', \dots, f^n$. Then given $x, x_0 \in (a, b)$:

$$f(x) = f(x_0) + f'(x_0)(x-x_0) + \frac{1}{2!}f''(x_0)(x-x_0)^2 + \frac{1}{3!}f'''(x_0)(x-x_0)^3 + \cdots + \frac{1}{n!}f^n(p)(x-x_0)^n \quad (8.2)$$

for some p between x and x_0 , and

$$f(x) \approx f(x_0) + f'(x_0)(x-x_0) + \frac{1}{2!}f''(x_0)(x-x_0)^2 + \frac{1}{3!}f'''(x_0)(x-x_0)^3 + \cdots + \frac{1}{n!}f^n(x_0)(x-x_0)^n, \quad (8.3)$$

when x is close to x_0 .

翻译 (Chinese)

泰勒定理断言, 一个具有连续导数的连续函数可以用一个多项式函数来近似。为了记号的方便, 将函数 f 的第 k 阶导数写为 $f^{(k)}(x) = \frac{d^k f}{dx^k}$. (与通常一样, 较低阶导数常写作 $f'(x) = \frac{df}{dx}$, $f''(x) = \frac{d^2 f}{dx^2}$ 和 $f'''(x) = \frac{d^3 f}{dx^3}$, 而不是 $f^{(1)}(x)$, $f^{(2)}(x)$ 和 $f^{(3)}(x)$.)

定理 8.1: 设 f 是区间 $[a, b]$ 上具有连续导数 $f', f'', \dots, f^n$ 的连续函数。则给定 $x, x_0 \in (a, b)$:

$$f(x) = f(x_0) + f'(x_0)(x-x_0) + \frac{1}{2!}f''(x_0)(x-x_0)^2 + \frac{1}{3!}f'''(x_0)(x-x_0)^3 + \cdots + \frac{1}{n!}f^n(p)(x-x_0)^n \quad (8.2)$$

其中 p 是 x 与 x_0 之间的某个点, 且

$$f(x) \approx f(x_0) + f'(x_0)(x - x_0) + \frac{1}{2!}f''(x_0)(x - x_0)^2 + \frac{1}{3!}f'''(x_0)(x - x_0)^3 + \cdots + \frac{1}{n!}f^n(x_0)(x - x_0)^n, \quad (8.3)$$

当 x 接近 x_0 时。

Original (English)

In Theorem 8.1, the expression on the right side is called a Taylor series expansion of f of order n around x_0 . Note that in one of the expansions, Equation 8.2, the last term is $\frac{1}{n!}f^n(p)(x - x_0)^n$ — the point p is chosen to give equality of the expansion with the function (and depends on both x and x_0). The second expansion, Equation 8.3, has last term $\frac{1}{n!}f^n(x_0)(x - x_0)^n$, and the expansion is approximately equal to $f(x)$. The term $n!$ is “ n factorial”, defined $n! = n(n-1)(n-2) \cdots 3 \cdot 2 \cdot 1$, so, for example, $3! = 3 \cdot 2 \cdot 1 = 6$. By convention, $0! = 1$. For example, a third-order expansion is:

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{1}{2!}f''(x_0)(x - x_0)^2 + \frac{1}{3!}f'''(p)(x - x_0)^3.$$

翻译 (Chinese)

在定理 8.1 中, 右边的表达式称为 f 在 x_0 附近的 n 阶泰勒级数展开。注意在其中一种展开 (8.2) 中, 最后一项是 $\frac{1}{n!}f^n(p)(x - x_0)^n$ ——点 p 被选择以使展开与函数相等 (且依赖于 x 和 x_0 两者)。第二种展开 (8.3) 的最后一项为 $\frac{1}{n!}f^n(x_0)(x - x_0)^n$, 且该展开近似等于 $f(x)$ 。项 $n!$ 是 “ n 阶乘”, 定义为 $n! = n(n-1)(n-2) \cdots 3 \cdot 2 \cdot 1$, 因此例如 $3! = 3 \cdot 2 \cdot 1 = 6$ 。按惯例, $0! = 1$ 。例如, 一个三阶展开为:

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{1}{2!}f''(x_0)(x - x_0)^2 + \frac{1}{3!}f'''(p)(x - x_0)^3.$$

Original (English)

To illustrate the formula, consider the quadratic function $f(x) = x^2$. The first and second derivatives are $f'(x) = 2x$ and $f''(x) = 2$ for all x , so that for any x_0 and any number p , $f''(x_0) = 2 = f''(p)$. An exact first-order expansion has the form $f(x) = f(x_0) + f'(p)(x - x_0)$ for some p , so that for the function $f(x) = x^2$:

$$f(x) = f(x_0) + f'(p) \cdot (x - x_0), \quad x^2 = x_0^2 + 2p \cdot (x - x_0).$$

This equality requires that $p = \frac{x_0+x}{2}$. To see this, rearrange the equation:

$$x^2 - x_0^2 = 2p \cdot (x - x_0), \quad (x - x_0)(x + x_0) = 2p \cdot (x - x_0), \quad (x_0 + x) = 2p, \quad \frac{(x_0 + x)}{2} = p.$$

Or, with approximation, $f(x) \approx f(x_0) + f'(x_0)(x - x_0) = [f(x_0) - f'(x_0)x_0] + f'(x_0) \cdot x$, a linear equation with intercept $[f(x_0) - f'(x_0)x_0]$ and slope $f'(x_0)$.

翻译 (Chinese)

为说明该公式, 考虑二次函数 $f(x) = x^2$ 。一阶和二阶导数对所有 x 分别为 $f'(x) = 2x$ 和 $f''(x) = 2$, 因此对任意 x_0 和任意数 p , $f''(x_0) = 2 = f''(p)$ 。一个精确的一阶展开具有形式 $f(x) = f(x_0) + f'(p)(x - x_0)$, 其中 p 是某个值, 因此对于函数 $f(x) = x^2$:

$$f(x) = f(x_0) + f'(p) \cdot (x - x_0), \quad x^2 = x_0^2 + 2p \cdot (x - x_0).$$

这个等式要求 $p = \frac{x_0+x}{2}$ 。要看到这一点, 重新整理方程:

$$x^2 - x_0^2 = 2p \cdot (x - x_0), \quad (x - x_0)(x + x_0) = 2p \cdot (x - x_0), \quad (x_0 + x) = 2p, \quad \frac{(x_0 + x)}{2} = p.$$

或者, 对于近似, $f(x) \approx f(x_0) + f'(x_0)(x - x_0) = [f(x_0) - f'(x_0)x_0] + f'(x_0) \cdot x$, 这是一个截距为 $[f(x_0) - f'(x_0)x_0]$ 、斜率为 $f'(x_0)$ 的线性方程。

Original (English)

A second-order Taylor series expansion of x^2 gives:

$$f(x) = x_0^2 + 2x_0(x - x_0) + \frac{1}{2} \cdot 2 \cdot (x - x_0)^2 = x_0^2 + 2x_0x - 2x_0^2 + x^2 - 2xx_0 + x_0^2 = x^2.$$

The expansion is exact because $f''(x) = 2$ for all values of x : the second derivative is a constant. Put differently, if $f'''(x) = 0$, so the third derivative is 0, then the second derivative must be constant and so a second-order expansion is exact if the third derivative is 0.

翻译 (Chinese)

x^2 的一个二阶泰勒级数展开给出:

$$f(x) = x_0^2 + 2x_0(x - x_0) + \frac{1}{2} \cdot 2 \cdot (x - x_0)^2 = x^2.$$

该展开是精确的, 因为对所有 x 的值, $f''(x) = 2$: 二阶导数是常数。换个说法, 如果 $f'''(x) = 0$, 即三阶导数为 0, 则二阶导数必为常数, 因此如果三阶导数为 0, 二阶展开是精确的。

6.12.4 Approximations and Accuracy / 近似与精度

Original (English)

As the length of the Taylor series expansion increases, the accuracy of the approximation increases, but the complexity of the approximation also increases with higher-order polynomials. So, there is a trade-off between increasing accuracy and increasing complexity. To see this, consider $f(x) = x^3$ and expansions around $x_0 = 1$. The first-order Taylor series expansion is $TSE_1(x) = 1 + 3(x - 1) = f(1) + f'(1)(x - 1)$; and the second-order Taylor series expansion is $TSE_2(x) = 1 + 3(x - 1) + 3(x - 1)^2 = f(1) + f'(1)(x - 1) + \frac{1}{2}f''(1)(x - 1)^2$. The error associated with the first-order Taylor series expansion is $E_1(x) = |f(x) - TSE_1(x)|$, and with the second-order expansion, $E_2(x) = |f(x) - TSE_2(x)|$.

翻译 (Chinese)

随着泰勒级数展开长度的增加,近似的精度会提高,但近似的复杂度也随着高阶多项式而增加。因此,在精度提高和复杂度增加之间存在一个权衡。要看到这一点,考虑 $f(x) = x^3$ 和在 $x_0 = 1$ 附近的展开。一阶泰勒级数展开为 $TSE_1(x) = 1 + 3(x - 1) = f(1) + f'(1)(x - 1)$; 二阶泰勒级数展开为 $TSE_2(x) = 1 + 3(x - 1) + 3(x - 1)^2 = f(1) + f'(1)(x - 1) + \frac{1}{2}f''(1)(x - 1)^2$ 。与一阶泰勒级数展开相关的误差为 $E_1(x) = |f(x) - TSE_1(x)|$, 与二阶展开相关的误差为 $E_2(x) = |f(x) - TSE_2(x)|$ 。

6.12.5 Applications of Taylor Series Expansions / 泰勒级数展开的应用

Original (English)

The following sections consider some applications of Taylor series expansions. The first application is a proof that concavity of a function implies that the second derivative is non-positive. Following this, numerical optimization routines based on Taylor series are developed. Then some applications on decision-making with risk are described.

翻译 (Chinese)

以下各节考虑泰勒级数展开的一些应用。第一个应用是证明函数的凹性意味着二阶导数非正。此后,开发了基于泰勒级数的数值最优化程序。然后描述了关于风险决策的一些应用。

Concavity and the Second Derivative / 凹性与二阶导数

Original (English)

The following theorem shows that concavity implies a non-positive second derivative; a similar argument may be used to show that convex functions have non-negative second derivatives.

Theorem 8.2: If f is concave and $f''(x)$ continuous then for any x , $f''(x) \leq 0$.

Proof: Pick x and let $\Delta x = \tilde{x} - x$, $\tilde{x}$ close to x . Note $\frac{1}{2}(x + \Delta x) + \frac{1}{2}(x - \Delta x) = x$ and f concave gives:

$$f(x) = f\left(\frac{1}{2}(x + \Delta x) + \frac{1}{2}(x - \Delta x)\right) \geq \frac{1}{2}f(x + \Delta x) + \frac{1}{2}f(x - \Delta x).$$

Therefore $2f(x) \geq f(x + \Delta x) + f(x - \Delta x)$. Now, Taylor's theorem gives:

$$f(y) = f(y_0) + f'(y_0)(y - y_0) + \frac{1}{2!}f''(\tilde{y})(y - y_0)^2,$$

where $\tilde{y}$ is between y and y_0 . First, take $y = x + \Delta x$, $y_0 = x$, so $y - y_0 = \Delta x$ and then

$$f(x + \Delta x) = f(x) + f'(x)\Delta x + \frac{1}{2}f''(\bar{x})(\Delta x)^2,$$

$\bar{x}$ between x and $x + \Delta x$. Next, take $y = x - \Delta x$, $y_0 = x$, so $y - y_0 = -\Delta x$ and then

$$f(x - \Delta x) = f(x) - f'(x)\Delta x + \frac{1}{2}f''(\hat{x})(\Delta x)^2,$$

$\hat{x}$ between x and $x - \Delta x$. Adding these together:

$$f(x + \Delta x) + f(x - \Delta x) = 2f(x) + \frac{1}{2}[f''(\bar{x}) + f''(\hat{x})](\Delta x)^2.$$

Recall from concavity that $2f(x) \geq f(x + \Delta x) + f(x - \Delta x)$, so

$$0 \geq f(x + \Delta x) + f(x - \Delta x) - 2f(x) = \frac{1}{2}[f''(\bar{x}) + f''(\hat{x})](\Delta x)^2.$$

Since $(\Delta x)^2 > 0$, this implies that $0 \geq \frac{1}{2}[f''(\bar{x}) + f''(\hat{x})]$. Since f'' is continuous, as $\Delta x \rightarrow 0$, $\bar{x}, \hat{x} \rightarrow x$, and so $f''(\bar{x}), f''(\hat{x}) \rightarrow f''(x)$. Therefore, $0 \geq f''(x)$.

翻译 (Chinese)

以下定理表明凹性意味着二阶导数非正；类似的论证可以用来证明凸函数具有非负的二阶导数。

定理 8.2: 如果 f 是凹的且 $f''(x)$ 连续, 则对任意 x , $f''(x) \leq 0$ 。

证明: 选 x 并令 $\Delta x = \tilde{x} - x$, $\tilde{x}$ 接近 x 。注意到 $\frac{1}{2}(x + \Delta x) + \frac{1}{2}(x - \Delta x) = x$, f 的凹性给出:

$$f(x) = f\left(\frac{1}{2}(x + \Delta x) + \frac{1}{2}(x - \Delta x)\right) \geq \frac{1}{2}f(x + \Delta x) + \frac{1}{2}f(x - \Delta x).$$

因此 $2f(x) \geq f(x + \Delta x) + f(x - \Delta x)$ 。现在, 泰勒定理给出:

$$f(y) = f(y_0) + f'(y_0)(y - y_0) + \frac{1}{2!}f''(\tilde{y})(y - y_0)^2,$$

其中 $\bar{y}$ 在 y 与 y_0 之间。首先, 取 $y = x + \Delta x$, $y_0 = x$, 因此 $y - y_0 = \Delta x$, 则

$$f(x + \Delta x) = f(x) + f'(x)\Delta x + \frac{1}{2}f''(\bar{x})(\Delta x)^2,$$

$\bar{x}$ 在 x 与 $x + \Delta x$ 之间。接着, 取 $y = x - \Delta x$, $y_0 = x$, 因此 $y - y_0 = -\Delta x$, 则

$$f(x - \Delta x) = f(x) - f'(x)\Delta x + \frac{1}{2}f''(\hat{x})(\Delta x)^2,$$

$\hat{x}$ 在 x 与 $x - \Delta x$ 之间。将这两者相加:

$$f(x + \Delta x) + f(x - \Delta x) = 2f(x) + \frac{1}{2}[f''(\bar{x}) + f''(\hat{x})](\Delta x)^2.$$

回想由凹性知 $2f(x) \geq f(x + \Delta x) + f(x - \Delta x)$, 因此

$$0 \geq f(x + \Delta x) + f(x - \Delta x) - 2f(x) = \frac{1}{2}[f''(\bar{x}) + f''(\hat{x})](\Delta x)^2.$$

由于 $(\Delta x)^2 > 0$, 这意味着 $0 \geq \frac{1}{2}[f''(\bar{x}) + f''(\hat{x})]$ 。由于 f'' 连续, 当 $\Delta x \rightarrow 0$ 时, $\bar{x}, \hat{x} \rightarrow x$, 从而 $f''(\bar{x}), f''(\hat{x}) \rightarrow f''(x)$ 。因此, $0 \geq f''(x)$ 。

Roots of a Function / 函数的根

Original (English)

Given a function f , a common problem is to find a root of f , a value for x such that $f(x^*) = 0$. Finding a root involves solving the equation $f(x) = 0$. In practice this can be difficult. However, using Taylor series there is a simple procedure that may be used to search for the roots of a function. Consider a first-order Taylor series expansion of f around some x_0 :

$$f(x) \approx f(x_0) + f'(x_0)(x - x_0).$$

Given x_0 , suppose x is chosen to set $f(x_0) + f'(x_0)(x - x_0) = 0$, with solution x_1 . Then $f(x_1) \approx f(x_0) + f'(x_0)(x_1 - x_0) = 0$, so that x_1 is “approximately” a root of f . If this process is repeated with x_1 replacing x_0 , a revised estimate, x_2 , for a root is obtained: $f(x_2) \approx f(x_1) + f'(x_1)(x_2 - x_1) = 0$. And so on. This process is easy to implement in practice. Assuming $f'(x_0) \neq 0$, the expression can be arranged to give an algorithm:

$$x_1 - x_0 = -\frac{f(x_0)}{f'(x_0)}, \quad x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}, \quad x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}.$$

This is the well-known Newton–Raphson method for finding roots.

翻译 (Chinese)

给定函数 f ，一个常见的问题是寻找 f 的根，即使得 $f(x^*) = 0$ 的 x 的值。寻找根涉及求解方程 $f(x) = 0$ 。在实践中这可能很困难。然而，利用泰勒级数，有一个简单的程序可用于搜索函数的根。考虑 f 在某个 x_0 附近的一阶泰勒级数展开：

$$f(x) \approx f(x_0) + f'(x_0)(x - x_0).$$

给定 x_0 ，假设选择 x 使得 $f(x_0) + f'(x_0)(x - x_0) = 0$ ，其解为 x_1 。则 $f(x_1) \approx f(x_0) + f'(x_0)(x_1 - x_0) = 0$ ，因此 x_1 “近似地”是 f 的一个根。如果用 x_1 替换 x_0 重复这一过程，则得到根的一个修正估计 x_2 ： $f(x_2) \approx f(x_1) + f'(x_1)(x_2 - x_1) = 0$ 。以此类推。这一过程在实践中易于实现。假设 $f'(x_0) \neq 0$ ，该表达式可以整理以给出一个算法：

$$x_1 - x_0 = -\frac{f(x_0)}{f'(x_0)}, \quad x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}, \quad x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}.$$

这就是著名的求解根的牛顿-拉弗森方法。

Numerical Optimization / 数值最优化

Original (English)

Suppose we wish to maximize some differentiable function $f(x)$. A maximum has the property that $f'(x) = 0$. How do we solve $f'(x) = 0$? One approach uses Taylor series expansions,

$$f(x) \approx f(x_0) + f'(x_0)(x - x_0) + \frac{1}{2}f''(x_0)(x - x_0)^2.$$

When x_0 is “close” to x , a value of x that maximizes the right-hand side (RHS) will also maximize the LHS. Differentiate with respect to x and set to 0. Since the RHS is a quadratic function of x , the derivative is linear in x . This yields

$$f'(x_0) + f''(x_0)(x - x_0) = 0.$$

Solving for x (call the solution x_1):

$$x_1 = x_0 - \frac{f'(x_0)}{f''(x_0)},$$

assuming that $f''(x_0) \neq 0$. Presumably, x_1 will be closer to a maximum of $f(x)$ than x_0 . We can continue the iteration:

$$x_n = x_{n-1} - \frac{f'(x_{n-1})}{f''(x_{n-1})},$$

assuming $f''(x_{n-1}) \neq 0$. If $\{x_n\}$ converges it will solve the problem $f'(x) = 0$.

翻译 (Chinese)

假设我们想最大化某个可微函数 $f(x)$ 。最大值具有性质 $f'(x) = 0$ 。如何求解 $f'(x) = 0$? 一种方法使用泰勒级数展开,

$$f(x) \approx f(x_0) + f'(x_0)(x - x_0) + \frac{1}{2}f''(x_0)(x - x_0)^2.$$

当 x_0 “接近” x 时, 最大化右边 (RHS) 的 x 的值也将最大化左边 (LHS)。对 x 求导并设为零。由于 RHS 是 x 的二次函数, 导数关于 x 是线性的。这给出

$$f'(x_0) + f''(x_0)(x - x_0) = 0.$$

解出 x (记其解为 x_1):

$$x_1 = x_0 - \frac{f'(x_0)}{f''(x_0)},$$

假设 $f''(x_0) \neq 0$ 。大致可以认为, x_1 会比 x_0 更接近 $f(x)$ 的最大值。我们可以继续迭代:

$$x_n = x_{n-1} - \frac{f'(x_{n-1})}{f''(x_{n-1})},$$

假设 $f''(x_{n-1}) \neq 0$ 。如果 $\{x_n\}$ 收敛, 它将解决 $f'(x) = 0$ 的问题。

Original (English)

The numerical optimization procedure generalizes directly to multivariate optimization. The second-order Taylor series approximation in the multivariate case is:

$$f(x) \approx f(x_0) + \nabla f(x_0) \cdot (x - x_0) + \frac{1}{2!}(x - x_0)'H(x_0)(x - x_0),$$

where $\nabla f(x_0)$ is the vector of partial derivatives of f evaluated at x_0 , and where $H(x_0)$ is the Hessian matrix of second-order partial derivatives, also evaluated at x_0 . Maximizing the right side by choice of the vector x gives:

$$\nabla f(x_0) + H(x_0)(x - x_0) = 0.$$

Assuming the Hessian matrix is non-singular, the solution x_1 satisfies

$$x_1 - x_0 = -H(x_0)^{-1}\nabla f(x_0),$$

and the resulting iteration is:

$$x_{i+1} - x_i = -H(x_i)^{-1}\nabla f(x_i).$$

翻译 (Chinese)

数值最优化程序可以直接推广到多元最优化。多元情形中的二阶泰勒级数近似为：

$$f(x) \approx f(x_0) + \nabla f(x_0) \cdot (x - x_0) + \frac{1}{2!}(x - x_0)'H(x_0)(x - x_0),$$

其中 $\nabla f(x_0)$ 是 f 在 x_0 处计算的偏导数向量， $H(x_0)$ 是二阶偏导数的海森矩阵，同样在 x_0 处计算。通过选择向量 x 最大化右边得到：

$$\nabla f(x_0) + H(x_0)(x - x_0) = 0.$$

假设海森矩阵非奇异，解 x_1 满足

$$x_1 - x_0 = -H(x_0)^{-1}\nabla f(x_0),$$

由此产生的迭代为：

$$x_{i+1} - x_i = -H(x_i)^{-1}\nabla f(x_i).$$

6.12.6 Expected Utility Theory and Behavior Towards Risk / 期望效用理论与风险态度

Original (English)

The presence of risk (variability) lowers the value of an asset: other things equal, comparing two assets, the one that has greater risk will have a lower value. In this section, Taylor series expansions are used to determine approximate welfare and monetary measures of the negative impact of risk. Before providing these measures, some introduction to utility and risk is necessary.

翻译 (Chinese)

风险（变异性）的存在降低了资产的价值：在其他条件相同的情况下，比较两种资产，风险较大的资产价值较低。在本节中，泰勒级数展开被用来确定风险负面影响的近似福利度和货币度量。在提供这些度量之前，有必要对效用和风险作一些介绍。

Expected Utility Theory / 期望效用理论

Original (English)

The usual consumer theory model has $u(x_1, x_2, \dots, x_n)$ = utility from consumption of the $(x_1, \dots, x_n)$ bundle of goods. In this model, the individual receives and consumes x_1 of good 1, x_2 of good 2 and so on. Suppose that one receives some random return instead of a sure consumption bundle. Then

this formulation is inappropriate: when a person receives a random return X , we cannot attach a utility to the return in the same way. Let X be a random return with values x_i and associated probabilities p_i . According to expected utility theory an individual has a utility function, u , defined on income. To calculate the utility associated with the random income or return stream X , calculate the utility of income in each state i , $u(x_i)$, multiply by the probability p_i to get $p_i u(x_i)$ and sum these terms, $\sum p_i u(x_i)$. This is called expected utility and denoted $E\{u(X)\}$. Now we can compare two random returns X and Y in expected utility terms: the individual prefers Y to X if $\sum q_j u(y_j) > \sum p_i u(x_i)$.

翻译 (Chinese)

通常的消费者理论模型有 $u(x_1, x_2, \dots, x_n)$ = 消费商品组合 $(x_1, \dots, x_n)$ 的效用。在该模型中，个体收到并消费 x_1 单位商品 1、 x_2 单位商品 2，等等。假设某人收到的是某种随机收益而不是确定的消费组合。那么这种表述就不合适了：当一个人收到随机收益 X 时，我们不能以同样的方式将效用附加到该收益上。设 X 是一个随机收益，取值为 x_i ，相应的概率为 p_i 。根据期望效用理论，个体有一个定义在收入上的效用函数 u 。要计算与随机收入或收益流 X 相关的效用，先计算每个状态 i 中的收入效用 $u(x_i)$ ，乘以概率 p_i 得 $p_i u(x_i)$ ，然后对这些项求和， $\sum p_i u(x_i)$ 。这称为期望效用，记为 $E\{u(X)\}$ 。现在我们可以用期望效用来比较两个随机收益 X 和 Y ：如果 $\sum q_j u(y_j) > \sum p_i u(x_i)$ ，则个体偏好 Y 胜于 X 。

Risk-Averse Preferences and Welfare Loss / 风险厌恶偏好与福利损失

Original (English)

In general, $u(\cdot)$ is assumed to be concave: this is said to represent risk-averse preferences. An individual is said to be risk averse if u is concave. Loosely speaking, risk-averse individuals prefer low variance for a given mean return.

Suppose x_1, x_2 are close to μ , so that we can use a Taylor series approximation:

$$\begin{aligned} u(x_1) &= u(\mu) + u'(\mu)(x_1 - \mu) + \frac{1}{2}u''(\mu)(x_1 - \mu)^2 + \frac{1}{3!}u'''(\xi_1)(x_1 - \mu)^3, \\ u(x_2) &= u(\mu) + u'(\mu)(x_2 - \mu) + \frac{1}{2}u''(\mu)(x_2 - \mu)^2 + \frac{1}{3!}u'''(\xi_2)(x_2 - \mu)^3. \end{aligned}$$

If $(x_i - \mu)$ is “small” then the last term is relatively small and can be ignored. Then,

$$\begin{aligned}
 E\{u(X)\} &\approx [p_1 u(\mu) + p_2 u(\mu)] + [p_1 u'(\mu)(x_1 - \mu) + p_2 u'(\mu)(x_2 - \mu)] \\
 &\quad + \frac{1}{2}[p_1 u''(\mu)(x_1 - \mu)^2 + p_2 u''(\mu)(x_2 - \mu)^2] \\
 &= [p_1 + p_2]u(\mu) + [p_1(x_1 - \mu) + p_2(x_2 - \mu)]u'(\mu) + \frac{1}{2}[p_1(x_1 - \mu)^2 + p_2(x_2 - \mu)^2]u''(\mu) \\
 &= u(\mu) + \frac{1}{2}u''(\mu)\sigma^2.
 \end{aligned} \tag{8.4}$$

If we write the last expression as $E\{u(X)\} - u(\mu) = \frac{1}{2}u''(\mu)\sigma^2$, we see that when u is concave, then u'' is negative — so that expected utility is less than $u(\mu)$. Note also that the difference depends on the variance of the random return. Thus, the welfare loss due to small risk is $\frac{1}{2}u''(\mu)\sigma^2$. So, $\frac{1}{2}u''(\mu)\sigma^2$ is the loss in utility or the disutility associated with the risk.

翻译 (Chinese)

一般而言, $u(\cdot)$ 被假设为凹的: 这被称为代表风险厌恶偏好。如果 u 是凹的, 则称个体是风险厌恶的。粗略地说, 风险厌恶的个体在给定平均收益下偏好低方差。

假设 x_1, x_2 接近 μ , 因此我们可以使用泰勒级数近似:

$$\begin{aligned}
 u(x_1) &= u(\mu) + u'(\mu)(x_1 - \mu) + \frac{1}{2}u''(\mu)(x_1 - \mu)^2 + \frac{1}{3!}u'''(\xi_1)(x_1 - \mu)^3, \\
 u(x_2) &= u(\mu) + u'(\mu)(x_2 - \mu) + \frac{1}{2}u''(\mu)(x_2 - \mu)^2 + \frac{1}{3!}u'''(\xi_2)(x_2 - \mu)^3.
 \end{aligned}$$

如果 $(x_i - \mu)$ 是“小的”, 则最后一项相对较小, 可以忽略。那么,

$$\begin{aligned}
 E\{u(X)\} &\approx [p_1 u(\mu) + p_2 u(\mu)] + [p_1 u'(\mu)(x_1 - \mu) + p_2 u'(\mu)(x_2 - \mu)] \\
 &\quad + \frac{1}{2}[p_1 u''(\mu)(x_1 - \mu)^2 + p_2 u''(\mu)(x_2 - \mu)^2] \\
 &= [p_1 + p_2]u(\mu) + [p_1(x_1 - \mu) + p_2(x_2 - \mu)]u'(\mu) + \frac{1}{2}[p_1(x_1 - \mu)^2 + p_2(x_2 - \mu)^2]u''(\mu) \\
 &= u(\mu) + \frac{1}{2}u''(\mu)\sigma^2.
 \end{aligned} \tag{8.4}$$

如果我们把最后的表达式写为 $E\{u(X)\} - u(\mu) = \frac{1}{2}u''(\mu)\sigma^2$, 我们看到当 u 是凹的时候, u'' 为负——因此期望效用小于 $u(\mu)$ 。还注意到, 该差异依赖于随机收益的方差。因此, 微小风险所导致的福利损失为 $\frac{1}{2}u''(\mu)\sigma^2$ 。所以, $\frac{1}{2}u''(\mu)\sigma^2$ 是与风险相关的效用损失或负效用。

Risk Aversion and the Cost of Small Risk / 风险厌恶与小风险的成本

Original (English)

Let income be represented by a random variable Y , where Y takes on the value y_i with probability p_i . Thus $\mu_y = \sum p_i y_i$ and $\sigma_y^2 = \sum p_i (y_i - \mu_y)^2$. Instead of asking how risk affects the level of utility, one can focus on the monetary cost of bearing risk. How much would a person pay to avoid facing the risk and receive a sure income μ_y ? If the individual receives μ_y and makes a payment of c , the amount received is $y = \mu_y - c$, and the individual is indifferent between the sure income y and facing the random return Y . The amount, c , is given as the solution of the equation: $u(\mu_y - c) = \sum p_i u(y_i)$. In general, solving this explicitly for c is very difficult. However, we can use Taylor series expansions to find an approximate solution. Expanding each side:

$$u(\mu_y - c) = u(\mu_y) - u'(\mu_y)c + [u'(\mu_y) - u'(\xi)]c \approx u(\mu_y) - u'(\mu_y)c, \text{ when } c \text{ is small.}$$

An expansion of $u(y_i)$ gives: $u(y_i) \approx u(\mu_y) + u'(\mu_y)(y_i - \mu_y) + \frac{1}{2}u''(\mu_y)(y_i - \mu_y)^2$. Multiplying by p_i and summing over i gives:

$$\sum p_i u(y_i) \approx u(\mu_y) + \frac{1}{2}u''(\mu_y)\sigma_y^2.$$

Combining these two approximations:

$$u(\mu_y) - u'(\mu_y)c \approx u(\mu_y) + \frac{1}{2}u''(\mu_y)\sigma_y^2.$$

Solving for c (approximately) gives:

$$c \approx \frac{1}{2} \left[-\frac{u''(\mu_y)}{u'(\mu_y)} \right] \sigma_y^2.$$

Thus the “cost” of the risk is proportional to σ_y^2 . The proportionality factor $-\frac{u''(\mu_y)}{u'(\mu_y)}$ is called the “degree of absolute risk aversion”.

翻译 (Chinese)

设收入由随机变量 Y 表示, 其中 Y 以概率 p_i 取值 y_i 。因此 $\mu_y = \sum p_i y_i$ 且 $\sigma_y^2 = \sum p_i (y_i - \mu_y)^2$ 。我们可以不关注风险如何影响效用水平, 而是关注承担风险的货币成本。一个人愿意支付多少钱来避免面对风险而获得确定收入 μ_y ? 如果个体收到 μ_y 并支付 c , 则收到的金额为 $y = \mu_y - c$, 且个体在确定收入 y 和面对随机收益 Y 之间无差异。金额 c 由方程 $u(\mu_y - c) = \sum p_i u(y_i)$ 的解给出。一般来说, 将其显式解出 c 是非常困难的。然而, 我们可以使用泰勒级数展开来找到一个近似解。展开两边:

$$u(\mu_y - c) = u(\mu_y) - u'(\mu_y)c + [u'(\mu_y) - u'(\xi)]c \approx u(\mu_y) - u'(\mu_y)c, \text{ 当 } c \text{ 很小时。}$$

对 $u(y_i)$ 展开得: $u(y_i) \approx u(\mu_y) + u'(\mu_y)(y_i - \mu_y) + \frac{1}{2}u''(\mu_y)(y_i - \mu_y)^2$ 。乘以 p_i 并对 i 求和得:

$$\sum p_i u(y_i) \approx u(\mu_y) + \frac{1}{2}u''(\mu_y)\sigma_y^2.$$

结合这两个近似:

$$u(\mu_y) - u'(\mu_y)c \approx u(\mu_y) + \frac{1}{2}u''(\mu_y)\sigma_y^2.$$

解出 c (近似地) 得:

$$c \approx \frac{1}{2} \left[-\frac{u''(\mu_y)}{u'(\mu_y)} \right] \sigma_y^2.$$

因此, 风险的“成本”与 σ_y^2 成正比。比例因子 $-\frac{u''(\mu_y)}{u'(\mu_y)}$ 被称为“绝对风险厌恶程度”。

6.12.7 Diversification and Portfolio Selection / 分散化与投资组合选择

Original (English)

Diversification of risk involves combining different random returns that are independent or have low correlation. In that way, the probability of all returns being low (or high) at the same time is small, so that extreme variation in return is less likely. This is briefly discussed and related to the cost of risk. Then approximately optimal portfolio selection using Taylor series expansions is considered.

翻译 (Chinese)

风险分散化涉及组合相互独立或具有低相关性的不同随机收益。通过这种方式, 所有收益同时处于低位 (或高位) 的概率很小, 因此收益的极端变动可能性更小。对此进行简要讨论并将其与风险成本联系起来。然后考虑利用泰勒级数展开进行近似最优投资组合选择。

Diversification / 分散化

Original (English)

Suppose that we have n projects. Assume that the return on each project is independent of the return on other projects. Project j has return $Y(j)$, a random variable with mean $\mu(j)$ and variance $\sigma^2(j)$. Let $\mu_j = \mu$, $\sigma^2(j) = \sigma^2$ — so we assume that each of the random variables has the same mean and variance. If an individual undertakes project j , the cost of risk is $c = \frac{1}{2} \left[-\frac{u''(\mu)}{u'(\mu)} \right] \sigma^2$.

Suppose instead the individual forms a portfolio containing $\frac{1}{n}$ of each project. Denote this portfolio $Z = \frac{1}{n} \sum_{j=1}^n Y(j)$. The mean is $E(Z) = \frac{1}{n} \sum_j \mu_j = \mu$ and $\text{Var}(Z) = \frac{1}{n} \sigma^2$, assuming the Y are independent. Using our earlier results, the cost of risk for Z is $c = \frac{1}{2} \left[-\frac{u''(\mu)}{u'(\mu)} \right] \frac{\sigma^2}{n}$. We observe that as the number (n) of projects increases, the cost of risk goes to 0. The portfolio Z has the same mean return as each of the $Y(j)$ assets but the variance of Z is $\frac{1}{n}$ the variance of each of the $Y(j)$.

翻译 (Chinese)

假设我们有 n 个项目。假设每个项目的收益与其他项目的收益相互独立。项目 j 有收益 $Y(j)$ ，是一个具有均值 $\mu(j)$ 和方差 $\sigma^2(j)$ 的随机变量。设 $\mu_j = \mu$, $\sigma^2(j) = \sigma^2$ ——因此我们假设每个随机变量具有相同的均值和方差。如果一个个体承担项目 j ，风险成本为 $c = \frac{1}{2} \left[-\frac{u''(\mu)}{u'(\mu)} \right] \sigma^2$ 。

假设改为该个体构建一个包含每个项目的 $\frac{1}{n}$ 的投资组合。记该组合为 $Z = \frac{1}{n} \sum_{j=1}^n Y(j)$ 。均值为 $E(Z) = \frac{1}{n} \sum_j \mu_j = \mu$ ，且假设 Y 相互独立， $\text{Var}(Z) = \frac{1}{n} \sigma^2$ 。利用我们之前的结果， Z 的风险成本为 $c = \frac{1}{2} \left[-\frac{u''(\mu)}{u'(\mu)} \right] \frac{\sigma^2}{n}$ 。我们观察到，随着项目数量 (n) 的增加，风险成本趋于 0。投资组合 Z 具有与每个 $Y(j)$ 资产相同的平均收益，但 Z 的方差是每个 $Y(j)$ 方差的 $\frac{1}{n}$ 。

Portfolio Selection / 投资组合选择**Original (English)**

This section studies the optimal division of a portfolio into a risky and a safe asset. The purchase of θ units of a risky asset yields an end-of-year gross return z per unit of risky asset purchased. Let $p_i = \text{Prob}(Z = z_i)$. The asset price is P (price at beginning of year), the individual's wealth at the beginning of the year is W . Thus, if the individual purchases θ units of the asset and if at the end of the year the risky asset return turns out to be z_i , then the individual's end-of-year wealth is $W - \theta P + \theta z_i = W + \theta(z_i - P)$. Therefore, the (end-of-year) expected utility is given by: $E\{u(W + \theta(Z - P))\} = \sum p_i u(W + \theta(z_i - P))$.

The individual's problem is to choose θ to solve: $\max_{\theta} \sum p_i u(W + \theta(z_i - P))$. Differentiating with respect to θ gives

$$\sum p_i u'(W + \theta(z_i - P))(z_i - P) = 0.$$

Solving this gives the optimal solution for θ . However, this may be difficult to solve, so that finding an approximate solution via a Taylor series approximation is useful. If $z_i - P$ is small,

$$u(W + \theta(z_i - P)) \approx u(W) + u'(W)\theta(z_i - P).$$

Then:

$$\sum p_i \{u(W) + u'(W)\theta(z_i - P)\}(z_i - P) \approx 0,$$

or

$$u'(W) \sum p_i (z_i - P) + u''(W)\theta \sum p_i (z_i - P)^2 \approx 0.$$

Focus on the terms: $\sum p_i(z_i - P) = \mu - P$, and $\sum p_i(z_i - P)^2 = \sigma^2 + (\mu - P)^2$. Therefore, the first-order condition is approximately:

$$u'(W)(\mu - P) + u''(W)\theta[\sigma^2 + (\mu - P)^2] \approx 0.$$

Solving for θ :

$$\theta \approx -\frac{u'(W)}{u''(W)} \frac{(\mu - P)}{\sigma^2 + (\mu - P)^2}.$$

翻译 (Chinese)

本节研究将投资组合最优划分为风险资产和安全资产。购买 θ 单位风险资产，每单位购买的风险资产产生年终总收益 z 。设 $p_i = \text{Prob}(Z = z_i)$ 。资产价格为 P （年初价格），个体年初的财富为 W 。因此，如果个体购买了 θ 单位该资产，且如果年终风险资产收益为 z_i ，则个体的年终财富为 $W - \theta P + \theta z_i = W + \theta(z_i - P)$ 。因此，（年终）期望效用由下式给出： $E\{u(W + \theta(Z - P))\} = \sum p_i u(W + \theta(z_i - P))$ 。

个体的问题是通过选择 θ 来求解： $\max_{\theta} \sum p_i u(W + \theta(z_i - P))$ 。对 θ 求导得

$$\sum p_i u'(W + \theta(z_i - P))(z_i - P) = 0.$$

解此给出 θ 的最优解。然而，这可能难以求解，因此通过泰勒级数近似找到一个近似解是有用的。如果 $z_i - P$ 很小，

$$u(W + \theta(z_i - P)) \approx u(W) + u'(W)\theta(z_i - P).$$

则：

$$\sum p_i \{u(W) + u'(W)\theta(z_i - P)\}(z_i - P) \approx 0,$$

或

$$u'(W) \sum p_i(z_i - P) + u''(W)\theta \sum p_i(z_i - P)^2 \approx 0.$$

关注这些项： $\sum p_i(z_i - P) = \mu - P$ ，且 $\sum p_i(z_i - P)^2 = \sigma^2 + (\mu - P)^2$ 。因此，一阶条件近似为：

$$u'(W)(\mu - P) + u''(W)\theta[\sigma^2 + (\mu - P)^2] \approx 0.$$

解出 θ ：

$$\theta \approx -\frac{u'(W)}{u''(W)} \frac{(\mu - P)}{\sigma^2 + (\mu - P)^2}.$$

6.12.8 The Multivariate Version of Taylor's Theorem / 泰勒定理的多元版本

Original (English)

Continuous functions of many variables can also be approximated by Taylor series expansions. Let $f(x) : \mathbb{R}^n \rightarrow \mathbb{R}$, be a function of n variables. Recall that the partial derivative of f with respect to x_i is denoted f_{x_i} and the second partial $f_{x_i x_j} = \frac{\partial^2 f}{\partial x_i \partial x_j}$. Write:

$$\nabla f = \begin{pmatrix} f_{x_1} \\ f_{x_2} \\ \vdots \\ f_{x_n} \end{pmatrix}, \quad H = \begin{pmatrix} f_{x_1 x_1} & f_{x_1 x_2} & \cdots & f_{x_1 x_n} \\ f_{x_2 x_1} & f_{x_2 x_2} & \cdots & f_{x_2 x_n} \\ \vdots & \vdots & \ddots & \vdots \\ f_{x_n x_1} & f_{x_n x_2} & \cdots & f_{x_n x_n} \end{pmatrix}, \quad x - x_0 = \begin{pmatrix} x_1 - x_{01} \\ x_2 - x_{02} \\ \vdots \\ x_n - x_{0n} \end{pmatrix}.$$

Also, write $\nabla f(x)$ and $H(x)$ to denote the matrix of derivatives evaluated at x . A Taylor series expansion of f to the second order around x_0 is

$$f(x) = f(x_0) + \nabla f(x_0) \cdot (x - x_0) + \frac{1}{2!}(x - x_0)' H(p)(x - x_0).$$

翻译 (Chinese)

多元连续函数也可以用泰勒级数展开来近似。设 $f(x) : \mathbb{R}^n \rightarrow \mathbb{R}$ 是一个 n 元函数。回想 f 关于 x_i 的偏导数记为 f_{x_i} ，二阶偏导数为 $f_{x_i x_j} = \frac{\partial^2 f}{\partial x_i \partial x_j}$ 。记：

$$\nabla f = \begin{pmatrix} f_{x_1} \\ f_{x_2} \\ \vdots \\ f_{x_n} \end{pmatrix}, \quad H = \begin{pmatrix} f_{x_1 x_1} & f_{x_1 x_2} & \cdots & f_{x_1 x_n} \\ f_{x_2 x_1} & f_{x_2 x_2} & \cdots & f_{x_2 x_n} \\ \vdots & \vdots & \ddots & \vdots \\ f_{x_n x_1} & f_{x_n x_2} & \cdots & f_{x_n x_n} \end{pmatrix}, \quad x - x_0 = \begin{pmatrix} x_1 - x_{01} \\ x_2 - x_{02} \\ \vdots \\ x_n - x_{0n} \end{pmatrix}.$$

同时，记 $\nabla f(x)$ 和 $H(x)$ 表示在 x 处计算的导数矩阵。 f 在 x_0 附近的二阶泰勒级数展开为

$$f(x) = f(x_0) + \nabla f(x_0) \cdot (x - x_0) + \frac{1}{2!}(x - x_0)' H(p)(x - x_0).$$

Original (English)

Theorem 8.3: Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$, be continuous with continuous first and second partial derivatives. Then given x, x_0 :

$$f(x) = f(x_0) + \nabla f(x_0) \cdot (x - x_0) + \frac{1}{2!}(x - x_0)' H(p)(x - x_0),$$

where p is on a line between x and x_0 : for some $\lambda \in (0, 1)$, $p = \lambda x_0 + (1 - \lambda)x$.

Example 8.10: Let $f(x, y) = x^2y^3$. So, $f_x(x, y) = 2xy^3$, $f_y(x, y) = 3x^2y^2$, $f_{xx} = 2y^3$, $f_{yy} = 6x^2y$ and $f_{xy} = 6xy^2$. Then a first-order Taylor series expansion of f around $x = 1, y = 1$ is:

$$f(x, y) \approx f(1, 1) + f_x(1, 1)(x - 1) + f_y(1, 1)(y - 1) = 1 + 2(x - 1) + 3(y - 1) = -4 + 3y + 2x.$$

The second-order Taylor series expansion has the form:

$$\begin{aligned} f(x, y) &\approx f(1, 1) + f_x(1, 1)(x - 1) + f_y(1, 1)(y - 1) \\ &\quad + \frac{1}{2}[f_{xx}(1, 1)(x - 1)^2 + 2f_{xy}(1, 1)(x - 1)(y - 1) + f_{yy}(1, 1)(y - 1)^2] \\ &= 1 + 2(x - 1) + 3(y - 1) + (x - 1)^2 + 6(x - 1)(y - 1) + 3(y - 1)^2 \\ &= 6 - 9y - 6x + 3y^2 + x^2 + 6xy. \end{aligned}$$

翻译 (Chinese)

定理 8.3: 设 $f: \mathbb{R}^n \rightarrow \mathbb{R}$ 是连续的, 且具有连续的一阶和二阶偏导数。则给定 x, x_0 :

$$f(x) = f(x_0) + \nabla f(x_0) \cdot (x - x_0) + \frac{1}{2!}(x - x_0)' H(p)(x - x_0),$$

其中 p 在 x 与 x_0 之间的连线上: 对某个 $\lambda \in (0, 1)$, $p = \lambda x_0 + (1 - \lambda)x$ 。

例 8.10: 设 $f(x, y) = x^2y^3$ 。则 $f_x(x, y) = 2xy^3$, $f_y(x, y) = 3x^2y^2$, $f_{xx} = 2y^3$, $f_{yy} = 6x^2y$ 且 $f_{xy} = 6xy^2$ 。 f 在 $x = 1, y = 1$ 附近的一阶泰勒级数展开为:

$$f(x, y) \approx f(1, 1) + f_x(1, 1)(x - 1) + f_y(1, 1)(y - 1) = 1 + 2(x - 1) + 3(y - 1) = -4 + 3y + 2x.$$

二阶泰勒级数展开具有如下形式:

$$\begin{aligned} f(x, y) &\approx f(1, 1) + f_x(1, 1)(x - 1) + f_y(1, 1)(y - 1) \\ &\quad + \frac{1}{2}[f_{xx}(1, 1)(x - 1)^2 + 2f_{xy}(1, 1)(x - 1)(y - 1) + f_{yy}(1, 1)(y - 1)^2] \\ &= 1 + 2(x - 1) + 3(y - 1) + (x - 1)^2 + 6(x - 1)(y - 1) + 3(y - 1)^2 \\ &= 6 - 9y - 6x + 3y^2 + x^2 + 6xy. \end{aligned}$$

6.12.9 Proof of Taylor's Theorem / 泰勒定理的证明

Original (English)

The proof of Taylor's theorem makes use of Rolle's theorem:

Theorem 8.4: Let f be a continuous differentiable function on the interval $[a, b]$. Suppose that $f(a) = f(b) = 0$. Then, there is a point $c \in (a, b)$ with $f'(c) = 0$.

Proof of Taylor's Theorem (Theorem 8.5): Consider the second-order expansion — the generalization to the n th-order case is straightforward. Let

$$g(z) = f(x) - f(z) - f'(z)(x - z) - \frac{K}{2}(x - z)^2.$$

Observe that $g(x) = 0$ and $g(x_0) = f(x) - f(x_0) - f'(x_0)(x - x_0) - \frac{K}{2}(x - x_0)^2$. Choose K to make this 0, so that $g(x_0) = 0$. Then $g(x) = g(x_0)$, so that by Rolle's theorem, there is a point between x_0 and x , say ξ with $g'(\xi) = 0$. Differentiating $g(z)$:

$$g'(z) = -f'(z) - f''(z)(x - z) + f'(z) + \frac{2K}{2}(x - z) = -f''(z)(x - z) + K(x - z).$$

At ξ , $g'(\xi) = -f''(\xi)(x - \xi) + K(x - \xi) = 0$. Since $\xi \neq x$, $f''(\xi) = K$.

For the n th-order case:

$$g(z) = f(x) - f(z) - f'(z)(x - z) - \frac{1}{2!}f''(z)(x - z)^2 - \cdots - \frac{1}{(n-1)!}f^{(n-1)}(z)(x - z)^{n-1} - \frac{K}{n!}(x - z)^n.$$

Again, $g(x) = 0$ and choose K so that $g(x_0) = 0$. Rolle's theorem implies that there is some ξ between x_0 and x with $g'(\xi) = 0$. In the derivative of g , all but two terms cancel:

$$g'(z) = -\frac{1}{(n-1)!}f^{(n)}(z)(x - z)^{n-1} + \frac{K}{(n-1)!}(x - z)^{n-1}.$$

At ξ , $g'(\xi) = -\frac{1}{(n-1)!}f^{(n)}(\xi)(x - \xi)^{n-1} + \frac{K}{(n-1)!}(x - \xi)^{n-1} = 0$. Since $x \neq \xi$, $f^{(n)}(\xi) = K$.

For the multivariate version, the proof is similar: let $g(t) = f(z + t\bar{z})$ and apply a second-order Taylor series expansion around 0 to this function of one variable.

翻译 (Chinese)

泰勒定理的证明利用了罗尔定理:

定理 8.4: 设 f 是区间 $[a, b]$ 上的连续可微函数。假设 $f(a) = f(b) = 0$ 。则存在一点 $c \in (a, b)$ 使得 $f'(c) = 0$ 。

泰勒定理的证明 (定理 8.5): 考虑二阶展开——推广到 n 阶情形是直接的。令

$$g(z) = f(x) - f(z) - f'(z)(x - z) - \frac{K}{2}(x - z)^2.$$

注意到 $g(x) = 0$ 且 $g(x_0) = f(x) - f(x_0) - f'(x_0)(x - x_0) - \frac{K}{2}(x - x_0)^2$ 。选择 K 使之为 0, 因此 $g(x_0) = 0$ 。则 $g(x) = g(x_0)$, 因此由罗尔定理, 存在 x_0 与 x 之间的一点, 记作 ξ , 使得 $g'(\xi) = 0$ 。对 $g(z)$ 求导:

$$g'(z) = -f'(z) - f''(z)(x - z) + f'(z) + \frac{2K}{2}(x - z) = -f''(z)(x - z) + K(x - z).$$

在 ξ 处, $g'(\xi) = -f''(\xi)(x - \xi) + K(x - \xi) = 0$ 。由于 $\xi \neq x$, 故 $f''(\xi) = K$ 。

对于 n 阶情形:

$$g(z) = f(x) - f(z) - f'(z)(x - z) - \frac{1}{2!}f''(z)(x - z)^2 - \cdots - \frac{1}{(n-1)!}f^{(n-1)}(z)(x - z)^{n-1} - \frac{K}{n!}(x - z)^n.$$

同样, $g(x) = 0$ 并选择 K 使得 $g(x_0) = 0$ 。罗尔定理意味着存在某个 x_0 与 x 之间的 ξ 使得 $g'(\xi) = 0$ 。在 g 的导数中, 除两项外所有项都消去了:

$$g'(z) = -\frac{1}{(n-1)!}f^{(n)}(z)(x - z)^{n-1} + \frac{K}{(n-1)!}(x - z)^{n-1}.$$

在 ξ 处, $g'(\xi) = -\frac{1}{(n-1)!}f^{(n)}(\xi)(x - \xi)^{n-1} + \frac{K}{(n-1)!}(x - \xi)^{n-1} = 0$ 。由于 $x \neq \xi$, 故 $f^{(n)}(\xi) = K$ 。

对于多元版本, 证明是类似的: 令 $g(t) = f(z + t\bar{z})$ 并对这个单变量函数应用在 0 附近的二阶泰勒级数展开。

Exercises for Chapter 8 / 第 8 章习题

Original (English)

Exercise 8.1 Consider the following functions: $f(x) = e^x - 1$, $g(x) = \ln(x + 1)$, $h(x) = x^2$, $u(x) = x^{\frac{1}{2}}$, $v(x) = \frac{1}{x}$, $w(x) = \sin(x)$. In all cases take $x > 0$. (a) Plot each function and determine whether each function is concave, convex or neither. (b) For each function, give the second-order Taylor series expansion.

Exercise 8.2 Give the second-order Taylor series expansions for the following functions: $f(x) = x^{\frac{1}{3}}$, $f(x) = \ln(x)$, $f(x) = \sin(x)$, $f(x) = \cos(x)$, $f(x) = e^{-rx}$, $f(x) = \sqrt{x}$, $f(x) = a^x$.

Exercise 8.5 A monopolist is faced with the following demand curve and cost function: $P(Q) = 154 - 2e^Q$, $C(Q) = 4Q + 6Q^2$. (a) Confirm that the profit function is concave. (b) Find the profit-maximizing level of output using numerical optimization.

Exercise 8.11 An individual has utility defined on income according to $u(y) = \ln(y + 1)$. The individual has initial income $\bar{y}$ and accepts a bet on a coin toss: if heads, the individual receives a dollar, if tails, the individual pays a dollar. The coin is fair (50/50 chance of heads or tails), thus the average utility from the bet is $\frac{1}{2}u(\bar{y} - 1) + \frac{1}{2}u(\bar{y} + 1)$. If the individual had not accepted the bet, the utility obtained is $u(\bar{y})$ for sure. Compare the average utility of the bet with the sure utility from not betting. Should the individual have accepted the bet?

Exercise 8.12 An individual has the utility function $u(y) = \ln(y + 1)$. The individual faces a random income next period: current income is \$100 and next period income will either fall by \$5

to \$95 or increase by \$5 to \$105 with 50/50 probabilities. The risk premium, c , is defined as the amount of money less than \$100 that makes the individual indifferent between accepting that sum for sure and facing the risk: $u(100 - c) = \frac{1}{2}u(95) + \frac{1}{2}u(105)$. Find the approximate value of c using Taylor series expansions.

Exercise 8.14 A student has just received a present of a new computer. The student's yearly income is \$50 and the student has a utility function $u(y) = y^{\frac{1}{2}}$. Unfortunately, there is a 10% chance that the computer will be stolen and the price of a new computer is \$5, so in the event of theft, the replacement will leave the student with only \$45 income for the year. Computer insurance is available at a cost of \$1. Use the calculations from a Taylor series expansion to decide whether the student should buy insurance or not.

Exercise 8.19 According to expected utility theory, faced with uncertain outcomes x_1 with probability p and x_2 with probability $(1-p)$, an individual associates utility $E(u) = pu(x_1) + (1-p)u(x_2)$. The risk premium, c , is defined: $u(\mu - c) = E(u)$, where $\mu = px_1 + (1-p)x_2$. In the case where $u(x) = \sqrt{x}$, $x_1 = 90$, $x_2 = 110$ and $p = \frac{1}{2}$, find c . An approximation for c is given by $c \approx -\frac{1}{2} \frac{u''(\mu)}{u'(\mu)} \sigma^2$. Estimate c using this approximation and compare it with the exact solution.

翻译 (Chinese)

习题 8.1 考虑以下函数: $f(x) = e^x - 1$, $g(x) = \ln(x+1)$, $h(x) = x^2$, $u(x) = x^{\frac{1}{2}}$, $v(x) = \frac{1}{x}$, $w(x) = \sin(x)$ 。在所有情形中取 $x > 0$ 。(a) 画出每个函数并判断它是凹的、凸的还是两者皆非。(b) 对每个函数, 给出二阶泰勒级数展开。

习题 8.2 给出以下函数的二阶泰勒级数展开: $f(x) = x^{\frac{1}{3}}$, $f(x) = \ln(x)$, $f(x) = \sin(x)$, $f(x) = \cos(x)$, $f(x) = e^{-rx}$, $f(x) = \sqrt{x}$, $f(x) = a^x$ 。

习题 8.5 一个垄断者面临以下需求曲线和成本函数: $P(Q) = 154 - 2e^Q$, $C(Q) = 4Q + 6Q^2$ 。(a) 确认利润函数是凹的。(b) 使用数值最优化求利润最大化的产出水平。

习题 8.11 一个个体有定义在收入上的效用 $u(y) = \ln(y+1)$ 。该个体有初始收入 $\bar{y}$ 并接受一个抛硬币赌局: 如果是正面, 个体收到一美元; 如果是反面, 个体支付一美元。硬币是公平的 (50/50 正面或反面概率), 因此该赌局的平均效用为 $\frac{1}{2}u(\bar{y}-1) + \frac{1}{2}u(\bar{y}+1)$ 。如果个体没有接受该赌局, 则确定获得效用 $u(\bar{y})$ 。比较该赌局的平均效用与不赌博的确定效用。该个体是否应该接受这个赌局?

习题 8.12 一个个体有效用函数 $u(y) = \ln(y+1)$ 。该个体下一期面临随机收入: 当前收入为 \$100, 下期收入将以 50/50 概率下降到 \$95 或上升到 \$105。风险溢价 c 定义为使个体在接受确定金额和面对风险之间无差异的、低于 \$100 的金额: $u(100 - c) = \frac{1}{2}u(95) + \frac{1}{2}u(105)$ 。使用泰勒级数展开求 c 的近似值。

习题 8.19 根据期望效用理论，面对以概率 p 取值 x_1 、以概率 $(1-p)$ 取值 x_2 的不确定结果，个体关联效用为 $E(u) = pu(x_1) + (1-p)u(x_2)$ 。风险溢价 c 定义为： $u(\mu - c) = E(u)$ ，其中 $\mu = px_1 + (1-p)x_2$ 。在 $u(x) = \sqrt{x}$, $x_1 = 90$, $x_2 = 110$ 且 $p = \frac{1}{2}$ 的情形中，求 c 。 c 的一个近似由 $c \approx -\frac{1}{2} \frac{u''(\mu)}{u'(\mu)} \sigma^2$ 给出。用此近似估计 c 并与其精确解进行比较。

6.13 Vectors / 向量

6.13.1 Introduction / 引言

Original (English)

A vector is an array of numbers, $x = (x_1, x_2, \dots, x_n)$, and may be interpreted in a number of ways: as the values of a set of variables, as a coordinate position in $\mathbb{R}^n$, as a direction (move x_i units in direction i , for $i = 1, \dots, n$), and so on. Depending on context, different interpretations are appropriate. In what follows, (Sections 9.2 and 9.3), some of the key summarizing features of a vector are considered: the length of a vector, the distance between vectors and the angle between vectors. As mentioned, from a geometric perspective, vectors can be interpreted as having direction — in subsequent chapters on optimization, direction of improvement of a function is central to the characterization of an optimal choice. As it turns out, this direction is represented by a vector of partial derivatives of the function, the gradient vector. Vectors implicitly define hyperplanes and halfspaces, key concepts in economic theory — Section 9.4 discusses these concepts. Finally, Section 9.5 describes Farkas' lemma, an old and important result in optimization theory.

翻译 (Chinese)

一个向量是一个数字阵列, $x = (x_1, x_2, \dots, x_n)$, 并且可以用多种方式来解释: 作为一组变量的值, 作为 $\mathbb{R}^n$ 中的一个坐标位置, 作为一个方向 (沿方向 i 移动 x_i 个单位, $i = 1, \dots, n$), 等等。根据上下文的不同, 应使用不同的解释。在下文中 (第 9.2 和 9.3 节), 考虑了向量的一些关键总结特征: 向量的长度、向量之间的距离和向量之间的夹角。如前所述, 从几何角度来看, 向量可以被解释为具有方向——在后续关于最优化的章节中, 函数改进的方向对最优选择的刻画至关重要。事实证明, 这个方向由函数的偏导数向量——梯度向量——来表示。向量隐式地定义了超平面和半空间, 这是经济理论中的关键概念——第 9.4 节讨论了这些概念。最后, 第 9.5 节描述了法卡斯引理, 一个最优化理论中古老而重要的结果。

6.13.2 Vectors: Length and Distance / 向量：长度与距离

Original (English)

The vector $x = (x_1, x_2, \dots, x_n)$ is a point in $\mathbb{R}^n$, a list of n real numbers. A vector may also be viewed in terms of direction as: x_1 units along axis 1, x_2 units along axis 2, and so on, as illustrated in Figure 9.1. Expressed that way, the vector x describes a direction from 0 to a point.

What is the length of the vector x , labeled c in Figure 9.1? To answer this, from the vector x form a right-angled triangle where the hypotenuse has length c and the adjacent and opposite have lengths a and b , respectively. Using Pythagoras' theorem, $a^2 + b^2 = c^2$, so that $c = \sqrt{a^2 + b^2}$. Thus, the length of the vector x , denoted $\|x\|$, is the length of the line c given by $\sqrt{x_1^2 + x_2^2}$:

$$\|x\| = \sqrt{x_1^2 + x_2^2}.$$

This extends to n dimensions directly. The length of a vector $x = (x_1, x_2, \dots, x_n)$ is denoted $\|x\|$ and defined

$$\|x\| = \sqrt{x_1^2 + x_2^2 + \dots + x_n^2} = \sqrt{\sum_{i=1}^n x_i^2} = \sqrt{x \cdot x}.$$

From the definition, the length of x and $-x$ are the same: $\|x\| = \|-x\|$. The sum and difference of two vectors x and y define vectors $x + y$ and $x - y$, as depicted in Figure 9.2.

The distance between two vectors, x, y , is the length of the vector $x - y$:

$$\|x - y\| = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}.$$

翻译 (Chinese)

向量 $x = (x_1, x_2, \dots, x_n)$ 是 $\mathbb{R}^n$ 中的一个点，一个 n 个实数的列表。向量也可以从方向的角度来看：沿轴 1 为 x_1 个单位，沿轴 2 为 x_2 个单位，等等，如图 9.1 所示。以这种方式表达，向量 x 描述了从 0 到一个点的方向。

向量 x 的长度（图 9.1 中标为 c ）是多少？为回答这个问题，从向量 x 构造一个直角三角形，其中斜边长度为 c ，邻边和对边长度分别为 a 和 b 。利用毕达哥拉斯定理， $a^2 + b^2 = c^2$ ，因此 $c = \sqrt{a^2 + b^2}$ 。于是，向量 x 的长度，记为 $\|x\|$ ，是由 $\sqrt{x_1^2 + x_2^2}$ 给出的线段 c 的长度：

$$\|x\| = \sqrt{x_1^2 + x_2^2}.$$

这直接推广到 n 维。向量 $x = (x_1, x_2, \dots, x_n)$ 的长度记为 $\|x\|$ ，并定义为

$$\|x\| = \sqrt{x_1^2 + x_2^2 + \dots + x_n^2} = \sqrt{\sum_{i=1}^n x_i^2} = \sqrt{x \cdot x}.$$

由定义, x 和 $-x$ 的长度相同: $\|x\| = \|-x\|$ 。两个向量 x 和 y 的和与差定义了向量 $x+y$ 和 $x-y$, 如图 9.2 所示。

两个向量 x, y 之间的距离是向量 $x-y$ 的长度:

$$\|x-y\| = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}.$$

Original (English)

Given a vector x , its length defines a circle or sphere with x in the boundary, so a circle with center 0 has radius $r = \|x\|$. Thus, the smaller the length of a vector, the smaller the circle on whose boundary it lies, as illustrated in Figure 9.3.

Given a vector z , equidistant points from z all lie on a circle centered on z of some fixed radius. The point w , closest to y and on the horizontal axis, has the property that $y-w$ is a vector perpendicular (orthogonal) to the horizontal axis. Let $e_1 = (1, 0)$. Then ce_1 is the vector e_1 scaled by the factor c . Any point w on the horizontal axis may be written as ce_1 , for some value of c . Consider the problem of choosing c to minimize the distance from y to the horizontal axis. This requires minimizing $\|y - ce_1\|$, or equivalently, minimizing:

$$\|y - ce_1\|^2 = (y - ce_1) \cdot (y - ce_1) = y \cdot y - 2cy \cdot e_1 + c^2 e_1 \cdot e_1.$$

Choosing c to minimize this expression gives: $-2y \cdot e_1 + 2ce_1 \cdot e_1 = 0$, so that

$$c^* = \frac{y \cdot e_1}{e_1 \cdot e_1}.$$

So, $w = c^*e_1$ is the point on the horizontal axis closest to y . Observe that

$$(y - w) \cdot e_1 = (y - c^*e_1) \cdot e_1 = y \cdot e_1 - c^*e_1 \cdot e_1 = y \cdot e_1 - y \cdot e_1 = 0.$$

The explanation is simple. The choice of w yields a vector $y-w$ with 0 in the first coordinate. Every vector on the horizontal axis has 0 in the second coordinate, so $(y-w) \cdot x = 0$ for any x on the horizontal axis.

翻译 (Chinese)

给定一个向量 x , 其长度定义了一个圆周或球面, x 在其边界上, 因此一个中心在 0 的圆具有半径 $r = \|x\|$ 。于是, 向量的长度越小, 其所在圆周的边界就越小, 如图 9.3 所示。

给定一个向量 z , 到 z 等距离的点都位于以 z 为中心、具有某个固定半径的圆上。点 w 是最接近 y 且在水平轴上的点, 它具有性质 $y-w$ 是一个与水平轴垂直 (正交) 的向量。令

$e_1 = (1, 0)$ 。则 ce_1 是向量 e_1 按因子 c 缩放的结果。水平轴上的任意点 w 可以写为 ce_1 ，其中 c 为某个值。考虑选择 c 以最小化从 y 到水平轴的距离的问题。这需要最小化 $\|y - ce_1\|$ ，或等价地，最小化：

$$\|y - ce_1\|^2 = (y - ce_1) \cdot (y - ce_1) = y \cdot y - 2cy \cdot e_1 + c^2 e_1 \cdot e_1.$$

选择 c 来最小化此表达式给出： $-2y \cdot e_1 + 2ce_1 \cdot e_1 = 0$ ，因此

$$c^* = \frac{y \cdot e_1}{e_1 \cdot e_1}.$$

因此， $w = c^*e_1$ 是水平轴上最接近 y 的点。注意到

$$(y - w) \cdot e_1 = (y - c^*e_1) \cdot e_1 = y \cdot e_1 - c^*e_1 \cdot e_1 = y \cdot e_1 - y \cdot e_1 = 0.$$

解释很简单。 w 的选择产生了一个在第一坐标中为 0 的向量 $y - w$ 。水平轴上的每个向量在第二坐标中为 0，因此对水平轴上的任意 x ， $(y - w) \cdot x = 0$ 。

6.13.3 Vectors: Direction and Angles / 向量：方向与角度

Original (English)

Since vectors indicate direction, we may compare the relative directions of different vectors. One particularly important comparison occurs when the vectors are orthogonal or “perpendicular”.

Definition 9.1: Two vectors a and b are said to be orthogonal if their inner product is 0, $a \cdot b = 0$.

In two dimensions, $a \cdot b = a_1b_1 + a_2b_2$, so that if the vectors are orthogonal, then $a_1b_1 + a_2b_2 = 0$, and rearranging, $\frac{a_2}{a_1} = -\frac{b_1}{b_2}$, provided a_1 and b_2 are both not equal to 0. From the definition, if a and b are orthogonal, then so are the vectors ca and db for any constants c and d , since $ca \cdot db = (cd)a \cdot b = 0$.

Consider two vectors, v and u . Two cases are depicted — where the angle is acute and where the angle is obtuse. In either case, drop a perpendicular line from the point v onto a point lying on the vector u , cu . Orthogonality requires that this line, $v - cu$, be orthogonal to u : $(v - cu) \cdot u = 0$. This implies that $v \cdot u - cu \cdot u = 0$, so that

$$c = \frac{v \cdot u}{u \cdot u} = \frac{v \cdot u}{\|u\|^2}.$$

The point cu is the closest point to v among the set of points $\{w \mid w = \lambda u, \text{ for some number } \lambda\}$.

The angle between vectors may be defined using the previous calculations. Using the cosine formula, considering Figure 9.6:

$$\cos(\theta) = \frac{\|cu\|}{\|v\|} = \frac{v \cdot u}{\|u\|\|v\|}.$$

In the case where the angle is acute, $c > 0$, $v \cdot u > 0$; when the angle is obtuse, $c < 0$, $v \cdot u < 0$. Notice that when $v \cdot u = 0$, $\cos(\theta) = 0$, corresponding to a 90° angle.

翻译 (Chinese)

由于向量指明方向，我们可以比较不同向量的相对方向。当向量正交或“垂直”时，这种比较尤为重要。

定义 9.1: 如果两个向量 a 和 b 的内积为 0，即 $a \cdot b = 0$ ，则称它们正交。

在二维中， $a \cdot b = a_1b_1 + a_2b_2$ ，因此如果向量正交，则 $a_1b_1 + a_2b_2 = 0$ ，重新整理得 $\frac{a_2}{a_1} = -\frac{b_1}{b_2}$ ，前提是 a_1 和 b_2 都不等于 0。由定义，如果 a 和 b 正交，则对于任意常数 c 和 d ，向量 ca 和 db 也正交，因为 $ca \cdot db = (cd)a \cdot b = 0$ 。

考虑两个向量 v 和 u 。两种情形被描绘出来——角度为锐角时和角度为钝角时。在任意一种情形中，从点 v 向位于向量 u 上的点 cu 作垂线。正交性要求这条线 $v - cu$ 与 u 正交： $(v - cu) \cdot u = 0$ 。这意味着 $v \cdot u - cu \cdot u = 0$ ，因此

$$c = \frac{v \cdot u}{u \cdot u} = \frac{v \cdot u}{\|u\|^2}.$$

点 cu 是点集 $\{w \mid w = \lambda u, \text{ 对某个数 } \lambda\}$ 中最接近 v 的点。

向量之间的角度可以用先前的计算来定义。利用余弦公式，考虑图 9.6:

$$\cos(\theta) = \frac{\|cu\|}{\|v\|} = \frac{v \cdot u}{\|u\|\|v\|}.$$

在角度为锐角的情形中， $c > 0$, $v \cdot u > 0$; 当角度为钝角时， $c < 0$, $v \cdot u < 0$ 。注意当 $v \cdot u = 0$ 时， $\cos(\theta) = 0$ ，对应于 90° 角。

6.13.4 Hyperplanes and Direction of Increase / 超平面与增加方向

Original (English)

Let $\lambda = (\lambda_1, \dots, \lambda_n) \neq 0$ be a vector of constants, c a constant and consider the equation:

$$\lambda \cdot x = \sum_{i=1}^n \lambda_i x_i = c.$$

The set of all x satisfying this equation defines a hyperplane:

$$H(\lambda, c) = \{x \mid \lambda \cdot x = c\}.$$

If y and z are on this hyperplane, then $\lambda \cdot y = \lambda \cdot z = c$, so that $\lambda \cdot (y - z) = 0$. Thus, λ and $(y - z)$ are orthogonal, as depicted in Figure 9.9. Movement along the hyperplane defined by λ and c is orthogonal to λ .

A hyperplane $H(\lambda, c) = \{x \mid \lambda \cdot x = c\}$ separates $\mathbb{R}^n$ into three regions:

$$H = \{x \mid \lambda \cdot x = c\}, \quad H^+ = \{x \mid \lambda \cdot x > c\}, \quad H^- = \{x \mid \lambda \cdot x < c\}.$$

On which sides of H do H^+ and H^- lie? Consider the point y and move from y along the direction λ a fraction $\alpha > 0$ to $y + \alpha\lambda$. Observe that

$$\lambda \cdot (y + \alpha\lambda) = \lambda \cdot y + \alpha\lambda \cdot \lambda = c + \alpha\|\lambda\|^2 > c.$$

Therefore, λ points into the region H^+ : starting at y and moving from y to $w = y + \alpha\lambda$, w is in H^+ . Equivalently, let $f(x) = \lambda \cdot x$, then λ points in the x direction on which $f(x)$ increases.

Furthermore, notice that the direction λ gives the greatest increase among all directions of equal length.

翻译 (Chinese)

设 $\lambda = (\lambda_1, \dots, \lambda_n) \neq 0$ 是一个常数向量, c 是一个常数, 考虑方程:

$$\lambda \cdot x = \sum_{i=1}^n \lambda_i x_i = c.$$

所有满足此方程的 x 的集合定义了一个超平面:

$$H(\lambda, c) = \{x \mid \lambda \cdot x = c\}.$$

如果 y 和 z 在此超平面上, 则 $\lambda \cdot y = \lambda \cdot z = c$, 因此 $\lambda \cdot (y - z) = 0$ 。于是, λ 与 $(y - z)$ 正交, 如图 9.9 所示。沿由 λ 和 c 定义的超平面的移动与 λ 正交。

一个超平面 $H(\lambda, c) = \{x \mid \lambda \cdot x = c\}$ 将 $\mathbb{R}^n$ 分成三个区域:

$$H = \{x \mid \lambda \cdot x = c\}, \quad H^+ = \{x \mid \lambda \cdot x > c\}, \quad H^- = \{x \mid \lambda \cdot x < c\}.$$

H^+ 和 H^- 位于 H 的哪一侧? 考虑点 y , 从 y 沿方向 λ 移动一个分数 $\alpha > 0$ 到 $y + \alpha\lambda$ 。注意到

$$\lambda \cdot (y + \alpha\lambda) = \lambda \cdot y + \alpha\lambda \cdot \lambda = c + \alpha\|\lambda\|^2 > c.$$

因此, λ 指向区域 H^+ : 从 y 出发, 从 y 移动到 $w = y + \alpha\lambda$, w 位于 H^+ 中。等价地, 设 $f(x) = \lambda \cdot x$, 则 λ 指向 $f(x)$ 在 x 方向上的增加方向。

此外，注意在所有等长方向中，方向 λ 给出了最大的增加。

Original (English)

If λ is normal (perpendicular) to H , then for any $\beta \neq 0$, $\beta\lambda$ is normal to the hyperplane. Dividing λ by $\|\lambda\|$ gives the unit normal, η , to the hyperplane H where $\eta = \frac{\lambda}{\|\lambda\|}$. The normal is orthogonal to H and has length 1, since $\|\eta\| = \frac{\|\lambda\|}{\|\lambda\|} = 1$.

翻译 (Chinese)

如果 λ 是 H 的法向量（垂直），则对任意 $\beta \neq 0$ ， $\beta\lambda$ 也是超平面的法向量。将 λ 除以 $\|\lambda\|$ 得到超平面 H 的单位法向量 $\eta = \frac{\lambda}{\|\lambda\|}$ 。该法向量与 H 正交且长度为 1，因为 $\|\eta\| = \frac{\|\lambda\|}{\|\lambda\|} = 1$ 。

6.13.5 Farkas' Lemma / 法卡斯引理

Original (English)

These ideas may be used to develop a well-known theorem in linear algebra and linear inequalities called Farkas' lemma. This result is particularly useful in optimization theory. Let A be an $m \times n$ matrix and b an $n \times 1$ vector.

Theorem 9.1: Either (1) $\{Ax = b, x \geq 0\}$, or (2) $\{A'y \geq 0 \text{ and } y'b > 0\}$ has a solution, but not both.

Proof: Suppose that (1) has no solution: $b \notin K \equiv \{Ax \mid x \geq 0\}$. Let $p \in \{Ax \mid x \geq 0\}$ solve $\min_{z \in K} \|b - z\|$. Since K is convex, p is unique. Let $w \geq 0$ satisfy $p = Aw$. The hyperplane perpendicular to $b - p$ separates b and K : $z \in K$, $(b - p) \cdot (z - p) \leq 0$. Equivalently:

$$(b - p) \cdot (Ax - Aw) \leq 0, \quad x \geq 0.$$

Put $y = b - p$, so $y \cdot (Ax - Aw) \leq 0$ for all $x \geq 0$. Let $x = w + e_i$, e_i a vector with 1 in the i th position and zero elsewhere. Then $A(x - w) = Ae_i = a_i$ the i th column of A . Thus, $y \cdot a_i \leq 0$, $\forall i$, or $y'A \leq 0$. Taking $x = 0$ in $y \cdot (Ax - Aw) \leq 0$ and recalling $Aw = p$ gives $-y \cdot p \leq 0$. Substituting $b - y$ for p , $-y \cdot (b - y) \leq 0$ or $-y \cdot b + y \cdot y \leq 0$ or $y \cdot y \leq y \cdot b$ and $y \cdot y > 0$ since $y \neq 0$, so $y \cdot b > 0$. Thus, if (1) does not hold, (2) must hold. Finally, (1) and (2) cannot both be satisfied: if $x \geq 0$, $Ax = b$ and if $y'A \leq 0$ then $y'b = y'Ax \leq 0$. This completes the proof.

翻译 (Chinese)

这些思想可以用来推导线性代数和线性不等式中的一个著名定理——法卡斯引理。该结果在最优化理论中特别有用。设 A 是一个 $m \times n$ 矩阵， b 是一个 $n \times 1$ 向量。

定理 9.1: (1) $\{Ax = b, x \geq 0\}$ 或 (2) $\{A'y \geq 0 \text{ 且 } y'b > 0\}$ 有解, 但两者不能同时成立。

证明: 假设 (1) 无解: $b \notin K \equiv \{Ax \mid x \geq 0\}$ 。设 $p \in \{Ax \mid x \geq 0\}$ 求解 $\min_{z \in K} \|b - z\|$ 。由于 K 是凸的, p 是唯一的。设 $w \geq 0$ 满足 $p = Aw$ 。垂直于 $b - p$ 的超平面将 b 与 K 分离: 对 $z \in K$, 有 $(b - p) \cdot (z - p) \leq 0$ 。等价地:

$$(b - p) \cdot (Ax - Aw) \leq 0, \quad x \geq 0.$$

令 $y = b - p$, 则对所有 $x \geq 0$ 有 $y \cdot (Ax - Aw) \leq 0$ 。取 $x = w + e_i$, 其中 e_i 是一个在第 i 个位置为 1、其余为零的向量。则 $A(x - w) = Ae_i = a_i$ 即 A 的第 i 列。因此, $y \cdot a_i \leq 0, \forall i$, 或 $y'A \leq 0$ 。在 $y \cdot (Ax - Aw) \leq 0$ 中取 $x = 0$ 并回想 $Aw = p$ 得 $-y \cdot p \leq 0$ 。将 $b - y$ 代入 p , 得 $-y \cdot (b - y) \leq 0$ 或 $-y \cdot b + y \cdot y \leq 0$ 或 $y \cdot y \leq y \cdot b$ 。由于 $y \neq 0, y \cdot y > 0$, 因此 $y \cdot b > 0$ 。因此, 如果 (1) 不成立, 则 (2) 必成立。最后, (1) 和 (2) 不能同时满足: 如果 $x \geq 0, Ax = b$ 且 $y'A \leq 0$, 则 $y'b = y'Ax \leq 0$ 。这就完成了证明。

Original (English)

Remark 9.1: Farkas' lemma has a simple geometric interpretation. In Figure 9.13, b is in the cone generated by the vectors a_1 and a_2 : $b = \lambda_1 a_1 + \lambda_2 a_2$ with $\lambda_i \geq 0$. The shaded region is defined as the set of y satisfying $a_1 \cdot y \leq 0$ and $a_2 \cdot y \leq 0$ or $A'y \leq 0$, where A is the matrix with columns a_1 and a_2 . As the figure illustrates, the region $\{y \mid A'y \leq 0\}$ is a subset of the region $\{y \mid b \cdot y \leq 0\}$. Thus, $Ax = b$ has a non-negative solution and $A'y \leq 0$ implies $b \cdot y \leq 0$.

Farkas' lemma may be expressed in other ways. One corollary is:

Lemma 2: Either $\{Ax \leq b, x \geq 0\}$ has a solution, or $\{A'y \geq 0, b \cdot y < 0, y \geq 0\}$ has a solution.

翻译 (Chinese)

注 9.1: 法卡斯引理有一个简单的几何解释。在图 9.13 中, b 位于由向量 a_1 和 a_2 生成的锥中: $b = \lambda_1 a_1 + \lambda_2 a_2$ 且 $\lambda_i \geq 0$ 。阴影区域定义为满足 $a_1 \cdot y \leq 0$ 和 $a_2 \cdot y \leq 0$ (或 $A'y \leq 0$) 的 y 的集合, 其中 A 是以 a_1 和 a_2 为列的矩阵。如该图所示, 区域 $\{y \mid A'y \leq 0\}$ 是区域 $\{y \mid b \cdot y \leq 0\}$ 的一个子集。因此, $Ax = b$ 有一个非负解且 $A'y \leq 0$ 意味着 $b \cdot y \leq 0$ 。

法卡斯引理还可以用其他方式表达。一个推论是:

引理 2: $\{Ax \leq b, x \geq 0\}$ 有解, 或者 $\{A'y \geq 0, b \cdot y < 0, y \geq 0\}$ 有解。

Exercises for Chapter 9 / 第 9 章习题

Original (English)

Exercise 9.1 Find the distance between the vectors $x = (5, 3)$ and $y = (4, 6)$.

Exercise 9.2 Let $x = (1, 3)$, $y = (2, 2)$ and $z = (1, -1)$. Find the angle between x and y , and that between x and z .

Exercise 9.3 Let $x = (4, 2)$ and $y = (3, 3)$. Find c such that $(y - cx) \cdot (y - cx) \leq (y - v) \cdot (y - v)$ for all $v = \tilde{c}x$, for some $\tilde{c}$. Confirm that $(y - cx) \cdot x = 0$.

Exercise 9.4 (a) Let $\lambda = (1, 2)$. Find the hyperplane $\lambda \cdot x = 2$. Noting that $(2, 0)$ is on this hyperplane, plot $(2, 0) + \lambda$. Identify the region $\{x \mid \lambda \cdot x \geq 2\}$. (b) Let $\lambda = (-1, 1)$. Find the hyperplane $\lambda \cdot x = 2$. Noting that $(-1, 1)$ is on this hyperplane, plot $(-1, 1) + \lambda$. Identify the region $\{x \mid \lambda \cdot x \geq 2\}$.

Exercise 9.5 Consider the matrix $R(\theta) = \begin{pmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{pmatrix}$, where θ is an angle measured in degrees. The transformation $R(\theta)x$ moves the vector x through an angle of θ degrees. (a) Letting $x = (1, 1)$, find $x' = R(45)x$. Confirm that $\cos(\theta) = \frac{1}{\sqrt{2}} = \frac{\|cx'\|}{\|x'\|}$. (b) Letting $y = (1, 3)$, find $y' = R(30)y$. Confirm that $\cos(\theta) = \frac{\sqrt{3}}{2} = \frac{\|cy'\|}{\|y'\|}$.

Exercise 9.7 Consider the following observations on two random variables, x and y : $x = (9, 7, 3, 5)'$, $y = (12, 8, 2, 6)'$. The correlation coefficient is defined: $r = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2 \sum(y_i - \bar{y})^2}}$, where $\bar{x}$ and $\bar{y}$ are the respective means. Find the correlation coefficient and relate it to the angle between the deviations around x and y .

翻译 (Chinese)

习题 9.1 求向量 $x = (5, 3)$ 和 $y = (4, 6)$ 之间的距离。

习题 9.2 设 $x = (1, 3)$, $y = (2, 2)$ 和 $z = (1, -1)$ 。求 x 与 y 之间的夹角, 以及 x 与 z 之间的夹角。

习题 9.3 设 $x = (4, 2)$ 和 $y = (3, 3)$ 。求 c 使得对所有 $v = \tilde{c}x$ (对某个 $\tilde{c}$), 有 $(y - cx) \cdot (y - cx) \leq (y - v) \cdot (y - v)$ 。验证 $(y - cx) \cdot x = 0$ 。

习题 9.4 (a) 设 $\lambda = (1, 2)$ 。求超平面 $\lambda \cdot x = 2$ 。注意 $(2, 0)$ 在此超平面上, 画出 $(2, 0) + \lambda$ 。识别区域 $\{x \mid \lambda \cdot x \geq 2\}$ 。(b) 设 $\lambda = (-1, 1)$ 。求超平面 $\lambda \cdot x = 2$ 。注意 $(-1, 1)$ 在此超平面上, 画出 $(-1, 1) + \lambda$ 。识别区域 $\{x \mid \lambda \cdot x \geq 2\}$ 。

习题 9.5 考虑矩阵 $R(\theta) = \begin{pmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{pmatrix}$, 其中 θ 是以度为单位的角。变换 $R(\theta)x$ 将向量 x 旋转 θ 度。(a) 令 $x = (1, 1)$, 求 $x' = R(45)x$ 。验证 $\cos(\theta) = \frac{1}{\sqrt{2}} = \frac{\|cx'\|}{\|x'\|}$ 。(b) 令 $y = (1, 3)$, 求 $y' = R(30)y$ 。验证 $\cos(\theta) = \frac{\sqrt{3}}{2} = \frac{\|cy'\|}{\|y'\|}$ 。

习题 9.7 考虑两个随机变量 x 和 y 的以下观测值: $x = (9, 7, 3, 5)'$, $y = (12, 8, 2, 6)'$ 。相关系数定义为: $r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$, 其中 $\bar{x}$ 和 $\bar{y}$ 是各自的均值。求相关系数并将其与 x 和 y 的离差向量之间的夹角联系起来。

6.14 Quadratic Forms / 二次型

6.14.1 10.1 Introduction / 引言

Original (English)

A quadratic form is given by the expression $x'Ax$ where x is an $n \times 1$ vector and A an $n \times n$ matrix. Quadratic forms arise in various branches of economics, and particularly in optimization theory to identify a candidate solution as a maximum or minimum. For example, in the study of quadratic functions, $ax^2 + bx + c = (ax + b)x + c$, this function is either “u-shaped” or the opposite, depending on whether a is positive or negative. So, the turning point either locates a minimum or a maximum depending on the sign of a . If, for example, $a > 0$, then large values of x correspond with large values of the function and the turning point corresponds to a minimum value of the function.

翻译 (Chinese)

二次型由表达式 $x'Ax$ 给出, 其中 x 是 $n \times 1$ 向量, A 是 $n \times n$ 矩阵。二次型出现在经济学的各个分支中, 特别是在最优化理论中用于判定候选解是最大值还是最小值。例如, 在研究二次函数 $ax^2 + bx + c = (ax + b)x + c$ 时, 该函数呈现 “U 形” 或倒 U 形, 取决于 a 为正还是为负。因此, 转折点确定的是最小值还是最大值取决于 a 的符号。例如, 若 $a > 0$, 则 x 的较大取值对应函数的较大取值, 而转折点对应函数的最小值。

Original (English)

Essentially the same situation occurs when the function is a quadratic in many variables: $x'Ax + bx + c$. In this case, the matrix A plays an analogous role to a in the scalar case: whether a “turning point” is a maximum or minimum depends on properties of A . Letting

$$f(x) = x'Ax + b'x + c = \sum_{i=1}^n \sum_{j=1}^n a_{ij}x_i x_j + \sum_{k=1}^n b_k x_k + c,$$

in the context of optimization, positive and negative definiteness are the central concepts. Section 10.2 introduces quadratic forms and shows how, in the two-variable case, properties of the A matrix determine the shape of the function f . In Section 10.3, positive and negative definiteness

are defined and the properties of positive and negative matrices are characterized in terms of determinant conditions.

翻译 (Chinese)

当函数是多变量二次函数 $x'Ax + bx + c$ 时，本质上同样的情况也会出现。此时，矩阵 A 扮演着与标量情形中 a 相类似的角色：“转折点”是最大值还是最小值取决于 A 的性质。令

$$f(x) = x'Ax + b'x + c = \sum_{i=1}^n \sum_{j=1}^n a_{ij}x_i x_j + \sum_{k=1}^n b_k x_k + c,$$

在最优化语境中，正定性和负定性是核心概念。第 10.2 节介绍二次型，并展示在两变量情形中，矩阵 A 的性质如何决定函数 f 的形状。在第 10.3 节中，定义了正定性和负定性，并用行列式条件刻画了正定和负定矩阵的性质。

Original (English)

When the function $f(x)$ is arbitrary (not necessarily quadratic), the criteria developed here extend directly by using a second-order quadratic approximation of the function f to identify local maxima or local minima. Finally, Section 10.4 considers the case where there are linear restrictions on x . A matrix A is positive definite if for all $x \neq 0$, $x'Ax > 0$ and negative definite when $x'Ax < 0$ for all non-zero x . So, for example, positive definiteness requires that for all non-zero x , $x'Ax > 0$. Suppose instead, that this requirement applies only to those non-zero x satisfying an additional condition, $Bx = 0$: under what circumstances is $x'Ax > 0$ for all non-zero x satisfying $Bx = 0$? This issue is considered in Section 10.4. The criteria developed there appear in the study of constrained optimization problems.

翻译 (Chinese)

当函数 $f(x)$ 为任意函数（未必是二次函数）时，这里建立的判据可直接推广：通过使用函数 f 的二阶二次逼近来识别局部极大值或局部极小值。最后，第 10.4 节考虑 x 上存在线性约束的情形。矩阵 A 是正定的，若对所有 $x \neq 0$ ，有 $x'Ax > 0$ ；是负定的，若对所有非零 x ，有 $x'Ax < 0$ 。例如，正定性要求对所有非零 x 有 $x'Ax > 0$ 。假设这一要求仅适用于那些满足附加条件 $Bx = 0$ 的非零 x ：在什么条件下，对所有满足 $Bx = 0$ 的非零 x 有 $x'Ax > 0$ ？第 10.4 节考虑了这一问题。那里建立的判据出现在约束最优化问题的研究中。

6.14.2 10.2 Quadratic Forms / 二次型

Original (English)

A quadratic form in n variables is given as:

$$x'Ax = (x_1, x_2, \dots, x_n) \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \sum_{ij} a_{ij} x_i x_j,$$

where a_{ij} is the element in the i th row and j th column of A . In the case of two variables:

$$x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \quad A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$$

and

$$x'Ax = \begin{pmatrix} x_1 & x_2 \end{pmatrix} \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = a_{11}x_1^2 + (a_{12} + a_{21})x_1x_2 + a_{22}x_2^2,$$

which is a quadratic form in the variables x_1 and x_2 . An important question concerns the sign of $x'Ax$. Under what conditions is the quadratic form always positive or always negative, regardless of the value of x ? The question arises in a variety of problems, and in particular in the area of optimization.

翻译 (Chinese)

n 个变量的二次型由下式给出:

$$x'Ax = (x_1, x_2, \dots, x_n) \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \sum_{ij} a_{ij} x_i x_j,$$

其中 a_{ij} 是 A 的第 i 行第 j 列元素。在两变量情形中:

$$x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \quad A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$$

且

$$x'Ax = \begin{pmatrix} x_1 & x_2 \end{pmatrix} \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = a_{11}x_1^2 + (a_{12} + a_{21})x_1x_2 + a_{22}x_2^2,$$

这是关于变量 x_1 和 x_2 的二次型。一个重要的问题涉及 $x'Ax$ 的符号: 在什么条件下, 二次型无论 x 取何值总是正的或总是负的? 该问题出现在各种问题中, 尤其是在最优化领域。

Original (English)

EXAMPLE 10.1: For example, if $a_{11} = 3$, $a_{22} = 4$ and $a_{12} + a_{21} = 3$, then the associated quadratic form is $q(x_1, x_2) = 3x_1^2 + 4x_2^2 + 3x_1x_2$. Let $f(x)$ be the quadratic function plotted in the figure:

$$z = f(x) = x_1 + x_2 + q(x_1, x_2) = x_1 + x_2 + 3x_1^2 + 4x_2^2 + 3x_1x_2.$$

This function is smallest when $(\bar{x}_1, \bar{x}_2) = (-\frac{5}{22}, -\frac{2}{11})$. That is: $f(\bar{x}_1, \bar{x}_2) \leq f(x_1, x_2)$ for all (x_1, x_2) . Notice that the “bowl” shape of the function is crucial: around $(\bar{x}_1, \bar{x}_2)$ the function curves up. This shape property is determined by the quadratic form $q(x_1, x_2)$, which has the property that $q(x_1, x_2) \geq 0$ for all (x_1, x_2) , and illustrates the relationship between quadratic forms and optimization.

翻译 (Chinese)

例 10.1: 例如, 若 $a_{11} = 3$, $a_{22} = 4$ 且 $a_{12} + a_{21} = 3$, 则相应的二次型为 $q(x_1, x_2) = 3x_1^2 + 4x_2^2 + 3x_1x_2$ 。令 $f(x)$ 为图中绘制的二次函数:

$$z = f(x) = x_1 + x_2 + q(x_1, x_2) = x_1 + x_2 + 3x_1^2 + 4x_2^2 + 3x_1x_2.$$

该函数在 $(\bar{x}_1, \bar{x}_2) = (-\frac{5}{22}, -\frac{2}{11})$ 处取得最小值。即: 对所有 (x_1, x_2) , 有 $f(\bar{x}_1, \bar{x}_2) \leq f(x_1, x_2)$ 。注意函数的“碗状”形状至关重要: 在 $(\bar{x}_1, \bar{x}_2)$ 附近, 函数向上弯曲。这一形状性质由二次型 $q(x_1, x_2)$ 决定, 该二次型具有性质 $q(x_1, x_2) \geq 0$ 对所有 (x_1, x_2) 成立, 这说明了二次型与最优化之间的关系。

Original (English)

In general, the sign of the quadratic form $x'Ax$ depends on both A and the vector x , as Example 10.2 illustrates.

EXAMPLE 10.2: Let $A = \begin{pmatrix} a & 0 \\ 0 & -\hat{a} \end{pmatrix}$ where $a > 0$ and $\hat{a} > 0$. Then

$$x'Ax = (x_1, x_2) \begin{pmatrix} a & 0 \\ 0 & -\hat{a} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = ax_1^2 - \hat{a}x_2^2.$$

Then

$$\begin{aligned} x'Ax > 0 & \quad \text{if} \quad ax_1^2 - \hat{a}x_2^2 > 0 \quad \text{or} \quad \frac{x_1^2}{x_2^2} > \frac{\hat{a}}{a} \\ x'Ax < 0 & \quad \text{if} \quad ax_1^2 - \hat{a}x_2^2 < 0 \quad \text{or} \quad \frac{x_1^2}{x_2^2} < \frac{\hat{a}}{a}. \end{aligned}$$

So, the sign of $x'Ax$ depends on the value of x and the parameters of A . If, instead,

$$A = \begin{pmatrix} a & 0 \\ 0 & \hat{a} \end{pmatrix}$$

then $x'Ax = ax_1^2 + \hat{a}x_2^2$, which is never negative since a and $\hat{a}$ are positive and $x_i^2 \geq 0$.

翻译 (Chinese)

一般而言, 二次型 $x'Ax$ 的符号既取决于 A 也取决于向量 x , 正如例 10.2 所示。

例 10.2: 令 $A = \begin{pmatrix} a & 0 \\ 0 & -\hat{a} \end{pmatrix}$, 其中 $a > 0$ 且 $\hat{a} > 0$ 。则

$$x'Ax = (x_1, x_2) \begin{pmatrix} a & 0 \\ 0 & -\hat{a} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = ax_1^2 - \hat{a}x_2^2.$$

那么

$$\begin{aligned} x'Ax > 0 & \quad \text{若} \quad ax_1^2 - \hat{a}x_2^2 > 0 \quad \text{或} \quad \frac{x_1^2}{x_2^2} > \frac{\hat{a}}{a} \\ x'Ax < 0 & \quad \text{若} \quad ax_1^2 - \hat{a}x_2^2 < 0 \quad \text{或} \quad \frac{x_1^2}{x_2^2} < \frac{\hat{a}}{a}. \end{aligned}$$

因此, $x'Ax$ 的符号取决于 x 的取值和 A 的参数。若采用

$$A = \begin{pmatrix} a & 0 \\ 0 & \hat{a} \end{pmatrix},$$

则 $x'Ax = ax_1^2 + \hat{a}x_2^2$, 由于 a 和 $\hat{a}$ 为正且 $x_i^2 \geq 0$, 该式永远不会为负。

Original (English)

Thus, as Example 10.2 illustrates, under suitable restrictions on A , the sign of a quadratic form, $x'Ax = ax_1^2 + \hat{a}x_2^2$, may be completely determined by the parameters of the A matrix.

Consider again the two-variable quadratic form:

$$\begin{aligned} \begin{pmatrix} x_1 & x_2 \end{pmatrix} \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} &= a_{11}x_1^2 + (a_{12} + a_{21})x_1x_2 + a_{22}x_2^2 \\ &= a_{11}x_1^2 + 2\tilde{a}x_1x_2 + a_{22}x_2^2, \quad \tilde{a} = \frac{1}{2}(a_{12} + a_{21}) \\ &= \begin{pmatrix} x_1 & x_2 \end{pmatrix} \begin{pmatrix} a_{11} & \tilde{a} \\ \tilde{a} & a_{22} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}. \end{aligned} \tag{10.1}$$

If $a_{11} = 0$, then $x = (1, 0)$ gives $x'Ax = 0$, so, for a positive or negative definite matrix, $a_{ii} \neq 0$ for all i . The expression $a_{11}x_1^2 + 2\tilde{a}x_1x_2 + a_{22}x_2^2$ may be rearranged (assume $a_{11} \neq 0$ or that $a_{22} \neq 0$ and with a symmetric calculation):

$$a_{11}x_1^2 + 2\tilde{a}x_1x_2 + a_{22}x_2^2 = a_{11} \left(x_1 + \frac{\tilde{a}}{a_{11}}x_2 \right)^2 + \left(a_{22} - \frac{\tilde{a}^2}{a_{11}} \right) x_2^2$$

$$= a_{11} \left(x_1 + \frac{\tilde{a}}{a_{11}} x_2 \right)^2 + \frac{a_{11}a_{22} - \tilde{a}^2}{a_{11}} x_2^2.$$

If $a_{11} > 0$ and $a_{11}a_{22} - \tilde{a}^2 > 0$, then this expression is positive if $(x_1, x_2) \neq (0, 0)$. Conversely, if $a_{11} < 0$ and $a_{11}a_{22} - \tilde{a}^2 > 0$, then the expression is negative for all $(x_1, x_2) \neq (0, 0)$. In either of the other two cases [$a_{11} > 0, a_{11}a_{22} - \tilde{a}^2 < 0$] and [$a_{11} < 0, a_{11}a_{22} - \tilde{a}^2 < 0$], the sign of $x'Ax$ can be positive or negative depending on the values of (x_1, x_2) . Consequently, the quadratic form is strictly positive for all $(x_1, x_2) \neq (0, 0)$ if and only if $a_{11} > 0$ and $a_{11}a_{22} - \tilde{a}^2 > 0$; and strictly negative for all $(x_1, x_2) \neq (0, 0)$ if and only if $a_{11} < 0$ and $a_{11}a_{22} - \tilde{a}^2 > 0$.

翻译 (Chinese)

因此, 如例 10.2 所示, 在对 A 施加适当限制下, 二次型 $x'Ax = ax_1^2 + \hat{a}x_2^2$ 的符号可以完全由矩阵 A 的参数决定。

再次考虑两变量二次型:

$$\begin{aligned} \begin{pmatrix} x_1 & x_2 \end{pmatrix} \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} &= a_{11}x_1^2 + (a_{12} + a_{21})x_1x_2 + a_{22}x_2^2 \\ &= a_{11}x_1^2 + 2\tilde{a}x_1x_2 + a_{22}x_2^2, \quad \tilde{a} = \frac{1}{2}(a_{12} + a_{21}) \\ &= \begin{pmatrix} x_1 & x_2 \end{pmatrix} \begin{pmatrix} a_{11} & \tilde{a} \\ \tilde{a} & a_{22} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}. \end{aligned} \quad (10.1)$$

若 $a_{11} = 0$, 则 $x = (1, 0)$ 给出 $x'Ax = 0$, 因此对于正定或负定矩阵, 对所有 i 有 $a_{ii} \neq 0$ 。表达式 $a_{11}x_1^2 + 2\tilde{a}x_1x_2 + a_{22}x_2^2$ 可重新排列 (假设 $a_{11} \neq 0$ 或 $a_{22} \neq 0$, 并进行对称计算):

$$\begin{aligned} a_{11}x_1^2 + 2\tilde{a}x_1x_2 + a_{22}x_2^2 &= a_{11} \left(x_1 + \frac{\tilde{a}}{a_{11}} x_2 \right)^2 + \left(a_{22} - \frac{\tilde{a}^2}{a_{11}} \right) x_2^2 \\ &= a_{11} \left(x_1 + \frac{\tilde{a}}{a_{11}} x_2 \right)^2 + \frac{a_{11}a_{22} - \tilde{a}^2}{a_{11}} x_2^2. \end{aligned}$$

若 $a_{11} > 0$ 且 $a_{11}a_{22} - \tilde{a}^2 > 0$, 则当 $(x_1, x_2) \neq (0, 0)$ 时该表达式为正。反之, 若 $a_{11} < 0$ 且 $a_{11}a_{22} - \tilde{a}^2 > 0$, 则对所有 $(x_1, x_2) \neq (0, 0)$ 该表达式为负。在其余两种情形 [$a_{11} > 0, a_{11}a_{22} - \tilde{a}^2 < 0$] 和 [$a_{11} < 0, a_{11}a_{22} - \tilde{a}^2 < 0$] 中, $x'Ax$ 的符号可正可负, 取决于 (x_1, x_2) 的取值。因此, 二次型对所有 $(x_1, x_2) \neq (0, 0)$ 严格为正当且仅当 $a_{11} > 0$ 且 $a_{11}a_{22} - \tilde{a}^2 > 0$; 对所有 $(x_1, x_2) \neq (0, 0)$ 严格为负当且仅当 $a_{11} < 0$ 且 $a_{11}a_{22} - \tilde{a}^2 > 0$ 。

Original (English)

EXAMPLE 10.3: This example illustrates how different values of the a_{ij} parameters in the A

matrix affect the sign of the quadratic form, Equation 10.1:

$$A_1 = \begin{pmatrix} 4 & 3 \\ 1 & 2 \end{pmatrix}, \quad x' A_1 x = 4x^2 + 4xy + 2y^2$$

$$A_2 = \begin{pmatrix} 4 & 3 \\ 5 & 2 \end{pmatrix}, \quad x' A_2 x = 4x^2 + 8xy + 2y^2.$$

Considering A_1 , $a_{11} = 4 > 0$, $a_{11}a_{22} - \tilde{a}^2 = 4 \cdot 2 - 2^2 = 4 > 0$. Similarly, for A_2 , $a_{11} = 4 > 0$, $a_{11}a_{22} - \tilde{a}^2 = 4 \cdot 2 - 4^2 = -8 < 0$. The pair of conditions $a_{11} > 0$ and $a_{11}a_{22} - \tilde{a}^2 > 0$ are satisfied by the matrix A_1 , but not by A_2 .

In the graphs, the quadratic form based on A_1 is always non-negative and strictly positive for $(x_1, x_2) \neq (0, 0)$, whereas the quadratic form based on A_2 is negative for some values of (x_1, x_2) . For example, at $(x_1, x_2) = (-1, 1)$, $4x^2 + 8xy + 2y^2 = 4(-1)^2 + 8(-1)(1) + 2(1)^2 = -2$.

翻译 (Chinese)

例 10.3: 本例说明矩阵 A 中 a_{ij} 参数的不同取值如何影响方程 (10.1) 中二次型的符号:

$$A_1 = \begin{pmatrix} 4 & 3 \\ 1 & 2 \end{pmatrix}, \quad x' A_1 x = 4x^2 + 4xy + 2y^2$$

$$A_2 = \begin{pmatrix} 4 & 3 \\ 5 & 2 \end{pmatrix}, \quad x' A_2 x = 4x^2 + 8xy + 2y^2.$$

对于 A_1 , $a_{11} = 4 > 0$, $a_{11}a_{22} - \tilde{a}^2 = 4 \cdot 2 - 2^2 = 4 > 0$ 。类似地, 对于 A_2 , $a_{11} = 4 > 0$, $a_{11}a_{22} - \tilde{a}^2 = 4 \cdot 2 - 4^2 = -8 < 0$ 。条件对 $a_{11} > 0$ 和 $a_{11}a_{22} - \tilde{a}^2 > 0$ 被矩阵 A_1 满足, 但不被 A_2 满足。

在图形中, 基于 A_1 的二次型总是非负的, 且对 $(x_1, x_2) \neq (0, 0)$ 严格为正; 而基于 A_2 的二次型对某些 (x_1, x_2) 取值为负。例如, 在 $(x_1, x_2) = (-1, 1)$ 处, $4x^2 + 8xy + 2y^2 = 4(-1)^2 + 8(-1)(1) + 2(1)^2 = -2$ 。

Original (English)

The next section describes a test to determine if a matrix is positive or negative definite.

翻译 (Chinese)

下一节将介绍判断矩阵是正定还是负定的检验方法。

6.14.3 10.3 Positive and Negative Definite Matrices / 正定和负定矩阵

Original (English)

For the following discussion let A be an $n \times n$ square matrix. Any vector x gives a quadratic form, $x'Ax$. Depending on the properties of the matrix, A , it may be that the expression $x'Ax$ is always positive or always negative (whenever $x \neq 0$). Definition 10.1 formalizes this concept and the discussion provides criteria for checking when a matrix is positive or negative definite.

Definition 10.1: Definite and semi-definite matrices:

- A is positive definite if $x'Ax > 0$, $x \neq 0$
- A is positive semi-definite if $x'Ax \geq 0$
- A is negative definite if $x'Ax < 0$, $x \neq 0$
- A is negative semi-definite if $x'Ax \leq 0$.

Observe that if A is positive (semi-)definite then $-A$ is negative (semi-)definite. If A is positive definite then A is positive semi-definite since $x'Ax > 0 \Rightarrow x'Ax \geq 0$. The converse is not true:

$$A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix},$$

so $x'Ax = (x_1 + x_2)^2 \geq 0$, but setting $x_1 = -x_2 \neq 0$ gives $x'Ax = 0$.

翻译 (Chinese)

在下面的讨论中，设 A 为 $n \times n$ 方阵。任意向量 x 给出一个二次型 $x'Ax$ 。根据矩阵 A 的性质，表达式 $x'Ax$ 可能总是正的或总是负的（只要 $x \neq 0$ ）。定义 10.1 将此概念形式化，随后的讨论给出了检验矩阵是否为正定或负定的判据。

定义 10.1：定矩阵与半定矩阵：

- A 是正定的，若 $x'Ax > 0$, $x \neq 0$
- A 是半正定的，若 $x'Ax \geq 0$
- A 是负定的，若 $x'Ax < 0$, $x \neq 0$
- A 是半负定的，若 $x'Ax \leq 0$.

注意，若 A 是正（半）定的，则 $-A$ 是负（半）定的。若 A 是正定的，则 A 是半正定的，因为 $x'Ax > 0 \Rightarrow x'Ax \geq 0$ 。反之不成立：

$$A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix},$$

故 $x'Ax = (x_1 + x_2)^2 \geq 0$, 但令 $x_1 = -x_2 \neq 0$ 得到 $x'Ax = 0$ 。

Original (English)

How can one determine if a matrix is positive or negative definite? Sometimes the nature of the problem provides an answer. For example, suppose that B is an $n \times k$ matrix and that $A = BB'$. Then $x'Ax = x'BB'x = z'z = \sum_{i=1}^n z_i^2 \geq 0$, where $z = B'x$. So, whenever A can be expressed as $A = BB'$, $x'Ax \geq 0$ for all x . Furthermore, if the rank of B is k , then for any $x \neq 0$, $z = B'x \neq 0$ so that $x'Ax = \sum_{i=1}^n z_i^2 > 0$ whenever $x \neq 0$.

In general, the matrix A will have no obvious structure implying the sign of the corresponding quadratic form and motivates the need for a systematic procedure to investigate definiteness of the matrix. The systematic procedure is based on the properties of “minors” derived from the matrix A .

翻译 (Chinese)

如何判断一个矩阵是正定还是负定的? 有时问题的性质本身就提供了答案。例如, 假设 B 是 $n \times k$ 矩阵且 $A = BB'$ 。则 $x'Ax = x'BB'x = z'z = \sum_{i=1}^n z_i^2 \geq 0$, 其中 $z = B'x$ 。因此, 只要 A 可表示为 $A = BB'$, 则对所有 x 有 $x'Ax \geq 0$ 。此外, 若 B 的秩为 k , 则对任意 $x \neq 0$, $z = B'x \neq 0$, 从而只要 $x \neq 0$ 就有 $x'Ax = \sum_{i=1}^n z_i^2 > 0$ 。

一般而言, 矩阵 A 可能没有明显的结构来暗示相应二次型的符号, 这就需要一个系统的程序来研究矩阵的定性。该系统程序基于由矩阵 A 导出的“子式”的性质。

Original (English)

Definition 10.2: Given a square matrix A , the determinant of any $k \times k$ matrix ($0 < k \leq n$) obtained by deleting any $n - k$ rows and $n - k$ columns is called a minor (or a k th-order minor to identify the dimensions of the matrix on which the determinant is derived).

- The (i, j) minor of a square matrix A , denoted m_{ij} , is the determinant of the matrix obtained from A by deleting the i th row and j th column.
- A minor obtained from a square matrix A as the determinant of a $k \times k$ matrix given by deleting $n - k$ matching rows and columns is called a (k th-order) principal minor.
- A principal minor is called a leading principal minor if it is the determinant of a submatrix obtained from A by deleting the last $(n - k)$ rows and columns, $0 < k \leq n$.

Definition 10.3: The trace, $\text{tr}(A)$, of a square matrix, A , is the sum of the diagonal elements of the matrix: $\text{tr}(A) = \sum_{i=1}^n a_{ii}$.

翻译 (Chinese)

定义 10.2: 给定方阵 A , 通过删除任意 $n-k$ 行和 $n-k$ 列得到的任意 $k \times k$ 矩阵 ($0 < k \leq n$) 的行列式称为子式 (或 k 阶子式, 以标明导出该行列式的矩阵的维数)。

- 方阵 A 的 (i, j) 子式, 记作 m_{ij} , 是从 A 中删除第 i 行和第 j 列得到的矩阵的行列式。
- 从方阵 A 中通过删除 $n-k$ 个匹配的行和列得到的 $k \times k$ 矩阵的行列式称为 (k 阶) 主子式。
- 若主子式是通过删除 A 的后 $(n-k)$ 行和列得到的子矩阵的行列式, 则称为顺序主子式, $0 < k \leq n$ 。

定义 10.3: 方阵 A 的迹, $\text{tr}(A)$, 是矩阵对角元素之和: $\text{tr}(A) = \sum_{i=1}^n a_{ii}$ 。

Original (English)

PROPOSITION 10.1: The following observations are useful:

1. If A is positive definite then $a_{ii} > 0, \forall i$; and if A is positive semi-definite then $a_{ii} \geq 0, \forall i$.
2. If A is positive definite $r(A) = n$.
3. If A is positive definite then $\text{tr}(A) > 0$.
4. The determinant of a symmetric positive definite matrix is positive.

PROOF: Let $x = (0, 0, \dots, 1, \dots, 0)$, a vector with 1 in the i th position and 0 elsewhere. Then $x'Ax = a_{ii}$. If A is positive definite then $a_{ii} > 0, \forall i$; and if A is positive semi-definite then $a_{ii} \geq 0, \forall i$. To show the second result, note that if A is positive definite, then for all $x \neq 0, x'Ax \neq 0$. Therefore, for all $x \neq 0, Ax \neq 0$, which implies linear independence of the columns of A , so that A has full rank, n . The positive determinant property follows from Theorem 10.1.

翻译 (Chinese)

命题 10.1: 以下观察结论是有用的:

1. 若 A 是正定的, 则 $a_{ii} > 0, \forall i$; 且若 A 是半正定的, 则 $a_{ii} \geq 0, \forall i$ 。
2. 若 A 是正定的, 则 $r(A) = n$ 。
3. 若 A 是正定的, 则 $\text{tr}(A) > 0$ 。
4. 对称正定矩阵的行列式为正。

证明: 令 $x = (0, 0, \dots, 1, \dots, 0)$, 即第 i 个位置为 1、其余位置为 0 的向量。则 $x'Ax = a_{ii}$ 。若 A 是正定的, 则 $a_{ii} > 0, \forall i$; 若 A 是半正定的, 则 $a_{ii} \geq 0, \forall i$ 。为证明第二个结果, 注

意到若 A 是正定的, 则对所有 $x \neq 0$, 有 $x'Ax \neq 0$ 。因此, 对所有 $x \neq 0$, 有 $Ax \neq 0$, 这意味着 A 的列线性无关, 故 A 具有满秩 n 。正行列式性质由定理 10.1 得出。

Original (English)

EXAMPLE 10.4: Consider the matrix A :

$$A = \begin{pmatrix} 1 & 0 \\ -4 & 1 \end{pmatrix}.$$

This matrix has positive trace, determinant and leading principal minors, but is not positive definite. (With $x = (2, 1)$, $x'Ax = -3$.) Notice that the matrix A is not symmetric.

翻译 (Chinese)

例 10.4: 考虑矩阵 A :

$$A = \begin{pmatrix} 1 & 0 \\ -4 & 1 \end{pmatrix}.$$

该矩阵具有正迹、正行列式和正顺序主子式, 但不是正定的。(取 $x = (2, 1)$, $x'Ax = -3$ 。)注意该矩阵 A 不是对称的。

10.3.1 Symmetric Matrices / 对称矩阵

Original (English)

Recall that a matrix A is symmetric if $A = A'$ (or $a_{ij} = a_{ji}$ for all i, j). The following considerations show that there is no loss of generality in considering only symmetric matrices when checking for positive definiteness. Consider any non-symmetric matrix B . Noting that $x'Bx = x'B'x$, if B is positive definite then so is $B + B'$ since $2x'Bx = x'Bx + x'B'x = x'(B + B')x$, so that $x'Bx > 0$ if and only if $x'(B + B')x > 0$. A matrix B is positive definite (semi-definite) if and only if the matrix $B + B'$ is positive definite (semi-definite). Note that $B + B'$ is symmetric. Therefore, to determine if B is positive definite (positive semi-definite) etc., one need only confirm that $(B + B')$ is positive definite (positive semi-definite). In discussing (semi-)definite matrices we need only consider symmetric matrices.

翻译 (Chinese)

回顾矩阵 A 是对称的, 若 $A = A'$ (或 $a_{ij} = a_{ji}$ 对所有 i, j)。以下讨论表明, 在检验正定性时只考虑对称矩阵并不失一般性。考虑任意非对称矩阵 B 。注意到 $x'Bx = x'B'x$, 若 B 是正定的, 则 $B + B'$ 也是正定的, 因为 $2x'Bx = x'Bx + x'B'x = x'(B + B')x$, 故 $x'Bx > 0$

当且仅当 $x'(B+B')x > 0$ 。矩阵 B 是正定（半正定）的当且仅当矩阵 $B+B'$ 是正定（半正定）的。注意 $B+B'$ 是对称的。因此，要判断 B 是否为正定（半正定）等，只需确认 $(B+B')$ 是正定（半正定）的即可。在讨论（半）定矩阵时，我们只需考虑对称矩阵。

10.3.2 Criteria for Positive and Negative Definiteness / 正定性和负定性的判据

Original (English)

In general, it is difficult to check positive or negative definiteness by inspecting a matrix. An alternative criterion is based on considering a sequence of determinants. Let A be a symmetric $n \times n$ matrix and let D_i be an $i \times i$ matrix — the submatrix of A obtained by deleting the last $n-i$ rows and the last $n-i$ columns. Thus $|D_i|$ is the i th leading principal minor. In words, for positive definiteness, the leading principal minors $\{|D_i|\}_{i=1}^n$ must all be positive; for negative definiteness they must alternate in sign, starting negative:

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1j} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2j} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{j1} & a_{j2} & \cdots & a_{jj} & \cdots & a_{jn} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nj} & \cdots & a_{nn} \end{pmatrix}, \quad D_i = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1i} \\ a_{21} & a_{22} & \cdots & a_{2i} \\ \vdots & \vdots & \ddots & \vdots \\ a_{i1} & a_{i2} & \cdots & a_{ii} \end{pmatrix}.$$

With this notation:

Theorem 10.1: Let A be a symmetric matrix. The matrix A is positive definite if and only if $|D_i| > 0$, $i = 1, \dots, n$. That is:

$$|D_1| > 0, |D_2| > 0, |D_3| > 0, \dots, |D_n| > 0.$$

The matrix A is negative definite if and only if $(-1)^i |D_i| > 0$, $i = 1, \dots, n$. That is:

$$|D_1| < 0, |D_2| > 0, |D_3| < 0, \dots, (-1)^n |D_n| > 0.$$

Thus, with positive definiteness, $|D_i|$ is positive for all i , whereas with negative definiteness, $|D_i|$ alternates in sign, beginning negative.

翻译 (Chinese)

一般而言，仅凭观察矩阵来检验正定性或负定性是困难的。另一种判据基于考虑一系列行列式。设 A 为对称 $n \times n$ 矩阵， D_i 为 $i \times i$ 矩阵——通过删除 A 的后 $n-i$ 行和后 $n-i$ 列

得到的子矩阵。因此 $|D_i|$ 是第 i 个顺序主子式。简言之, 对于正定性, 顺序主子式 $\{|D_i|\}_{i=1}^n$ 必须全部为正; 对于负定性, 它们必须符号交替, 且从负开始:

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1j} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2j} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{j1} & a_{j2} & \cdots & a_{jj} & \cdots & a_{jn} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nj} & \cdots & a_{nn} \end{pmatrix}, \quad D_i = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1i} \\ a_{21} & a_{22} & \cdots & a_{2i} \\ \vdots & \vdots & \ddots & \vdots \\ a_{i1} & a_{i2} & \cdots & a_{ii} \end{pmatrix}.$$

在此记号下:

定理 10.1: 设 A 为对称矩阵。矩阵 A 是正定的当且仅当 $|D_i| > 0$, $i = 1, \dots, n$ 。即:

$$|D_1| > 0, |D_2| > 0, |D_3| > 0, \dots, |D_n| > 0.$$

矩阵 A 是负定的当且仅当 $(-1)^i |D_i| > 0$, $i = 1, \dots, n$ 。即:

$$|D_1| < 0, |D_2| > 0, |D_3| < 0, \dots, (-1)^n |D_n| > 0.$$

因此, 对于正定性, $|D_i|$ 对所有 i 为正; 而对于负定性, $|D_i|$ 符号交替, 且从负开始。

Original (English)

In the 2×2 case, because the matrix is symmetric, $a_{12} = a_{21}$ and $a_{11} \neq 0$, the quadratic form is:

$$x'Ax = a_{11}x_1^2 + 2a_{12}x_1x_2 + a_{22}x_2^2 = a_{11} \left(x_1 + \frac{a_{12}}{a_{11}}x_2 \right)^2 + \left(a_{22} - \frac{a_{12}^2}{a_{11}} \right) x_2^2.$$

From this expression, it is clear that if $a_{11} > 0$ and $a_{11}a_{22} - a_{12}^2 > 0$, then provided $x_1 \neq 0$ or $x_2 \neq 0$, $x'Ax > 0$. Note that a_{11} is the determinant of the matrix D_1 , and $a_{11}a_{22} - a_{12}^2$ is the determinant of the matrix D_2 . From these calculations, note that to check positive or negative definiteness, there are at most n determinants to evaluate.

翻译 (Chinese)

在 2×2 情形中, 由于矩阵是对称的, $a_{12} = a_{21}$ 且 $a_{11} \neq 0$, 二次型为:

$$x'Ax = a_{11}x_1^2 + 2a_{12}x_1x_2 + a_{22}x_2^2 = a_{11} \left(x_1 + \frac{a_{12}}{a_{11}}x_2 \right)^2 + \left(a_{22} - \frac{a_{12}^2}{a_{11}} \right) x_2^2.$$

从该表达式可以清楚地看出, 若 $a_{11} > 0$ 且 $a_{11}a_{22} - a_{12}^2 > 0$, 则只要 $x_1 \neq 0$ 或 $x_2 \neq 0$, 就有 $x'Ax > 0$ 。注意 a_{11} 是矩阵 D_1 的行列式, $a_{11}a_{22} - a_{12}^2$ 是矩阵 D_2 的行列式。从这些计算可知, 检验正定性或负定性最多需要计算 n 个行列式。

10.3.3 Positive and Negative Semi-Definiteness / 半正定性和半负定性

Original (English)

The next theorem gives similar conditions for positive and negative semi-definiteness and uses the following notation. Let $\tilde{D}_i$ be any submatrix of A containing i rows and columns obtained by deleting any $n - i$ rows and the corresponding columns (if the k th row is deleted, then the k th column is also deleted). For example, if A is an $n \times n$ matrix:

$$\tilde{D}_i = \begin{pmatrix} a_{kk} & a_{kj} & \cdots & a_{kl} & \cdots & a_{kr} \\ a_{jk} & a_{jj} & \cdots & a_{jl} & \cdots & a_{jr} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{lk} & a_{lj} & \cdots & a_{ll} & \cdots & a_{lr} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{rk} & a_{rj} & \cdots & a_{rl} & \cdots & a_{rr} \end{pmatrix}.$$

The determinant of any $\tilde{D}_i$ is an i th-order principal minor. For example, if A is a 3×3 matrix:

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix},$$

then there are three $\tilde{D}_i$ matrices:

$$\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}, \begin{pmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{pmatrix}, \begin{pmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{pmatrix},$$

and hence three principal minors of order 2.

Theorem 10.2: Let A be a symmetric matrix. The matrix A is positive semi-definite if and only if:

$$\text{all } |\tilde{D}_1| \geq 0, \text{ all } |\tilde{D}_2| \geq 0, \dots \text{ all } |\tilde{D}_i| \geq 0, \dots \text{ all } |\tilde{D}_n| \geq 0.$$

The matrix A is negative semi-definite if and only if:

$$\text{all } |\tilde{D}_1| \leq 0, \text{ all } |\tilde{D}_2| \geq 0, \dots \text{ all } (-1)^i |\tilde{D}_i| \geq 0, \dots \text{ all } (-1)^n |\tilde{D}_n| \geq 0.$$

Observe that the statements for positive and negative semi-definiteness are not obtained by replacing strict inequalities by non-strict inequalities. To check for positive semi-definiteness, it is necessary to check that all 1st-order principal minors are positive, all 2nd-order principal minors are positive, all 3rd-order principal minors are positive, and so on. There are $\binom{n}{1} = \frac{n!}{(n-1)!1!}$ 1st-order principal minors, $\frac{n!}{(n-2)!2!}$ 2nd-order principal minors, $\frac{n!}{(n-3)!3!}$ 3rd-order principal minors, and so on. There are $\sum_{i=1}^n \frac{n!}{(n-i)!i!} = 2^n - 1$ such minors in total.

翻译 (Chinese)

下一个定理给出了半正定性和半负定性的类似条件，并使用了以下记号。设 $\tilde{D}_i$ 是 A 的任意包含 i 行和 i 列的子矩阵，通过删除任意 $n-i$ 行及相应的列（若第 k 行被删除，则第 k 列也被删除）得到。例如，若 A 是 $n \times n$ 矩阵：

$$\tilde{D}_i = \begin{pmatrix} a_{kk} & a_{kj} & \cdots & a_{kl} & \cdots & a_{kr} \\ a_{jk} & a_{jj} & \cdots & a_{jl} & \cdots & a_{jr} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{lk} & a_{lj} & \cdots & a_{ll} & \cdots & a_{lr} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{rk} & a_{rj} & \cdots & a_{rl} & \cdots & a_{rr} \end{pmatrix}.$$

任意 $\tilde{D}_i$ 的行列式是一个 i 阶主子式。例如，若 A 是 3×3 矩阵：

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix},$$

则有三个 $\tilde{D}_i$ 矩阵：

$$\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}, \begin{pmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{pmatrix}, \begin{pmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{pmatrix},$$

因此有三个 2 阶主子式。

定理 10.2: 设 A 为对称矩阵。矩阵 A 是半正定的当且仅当：

$$\text{所有 } |\tilde{D}_1| \geq 0, \text{ 所有 } |\tilde{D}_2| \geq 0, \dots \text{ 所有 } |\tilde{D}_i| \geq 0, \dots \text{ 所有 } |\tilde{D}_n| \geq 0.$$

矩阵 A 是半负定的当且仅当：

$$\text{所有 } |\tilde{D}_1| \leq 0, \text{ 所有 } |\tilde{D}_2| \geq 0, \dots \text{ 所有 } (-1)^i |\tilde{D}_i| \geq 0, \dots \text{ 所有 } (-1)^n |\tilde{D}_n| \geq 0.$$

注意，半正定性和半负定性的表述并非简单地将严格不等式替换为非严格不等式而得到。要检验半正定性，必须检验所有一阶主子式非负、所有二阶主子式非负、所有三阶主子式非负，等等。一阶主子式有 $\binom{n}{1} = \frac{n!}{(n-1)!1!}$ 个，二阶主子式有 $\frac{n!}{(n-2)!2!}$ 个，三阶主子式有 $\frac{n!}{(n-3)!3!}$ 个，等等。总共有 $\sum_{i=1}^n \frac{n!}{(n-i)!i!} = 2^n - 1$ 个这样的子式。

6.14.4 10.4 Definiteness with Equality Constraints / 带等式约束的定性

Original (English)

In constrained optimization problems, checking for maxima or minima leads to considering definiteness of a quadratic form subject to linear constraints: for example, that $x'Ax > 0$ for all x satisfying $Bx = 0$, for some $m \times n$ matrix B , where

$$B = \begin{pmatrix} b_{11} & b_{12} & \cdots & b_{1n} \\ b_{21} & b_{22} & \cdots & b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ b_{m1} & b_{m2} & \cdots & b_{mn} \end{pmatrix}.$$

First, consider the simplest case where there is one constraint so $Bx = 0$ may be written $b'x = 0$ where $b = (b_1, b_2, \dots, b_n)$. Form the bordered matrix:

$$M_{kk} = \begin{pmatrix} 0 & b_1 & b_2 & \cdots & b_k \\ b_1 & a_{11} & a_{12} & \cdots & a_{1k} \\ b_2 & a_{21} & a_{22} & \cdots & a_{2k} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ b_k & a_{k1} & a_{k2} & \cdots & a_{kk} \end{pmatrix},$$

or, writing b^k for the first k elements of b and A_{kk} for the matrix obtained from A by deleting the last $k+1, k+2, \dots, n$ rows and columns, M_{kk} may be written

$$M_{kk} = \begin{pmatrix} 0 & b^k \\ b^k & A_{kk} \end{pmatrix}.$$

Theorem 10.3: Let A be a symmetric matrix and $b_1 \neq 0$. Then,

1. $x'Ax > 0$ for all $x \neq 0$ and $b'x = 0$ if and only if

$$|M_{kk}| < 0, \quad k \geq 2$$

2. $x'Ax < 0$ for all $x \neq 0$ and $b'x = 0$ if and only if

$$(-1)^k |M_{kk}| > 0, \quad k \geq 2$$

翻译 (Chinese)

在约束最优化问题中, 检验最大值或最小值需要考虑在线性约束下二次型的定性: 例如, 对某个 $m \times n$ 矩阵 B , 对所有满足 $Bx = 0$ 的 x 有 $x'Ax > 0$, 其中

$$B = \begin{pmatrix} b_{11} & b_{12} & \cdots & b_{1n} \\ b_{21} & b_{22} & \cdots & b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ b_{m1} & b_{m2} & \cdots & b_{mn} \end{pmatrix}.$$

首先考虑只有一个约束的最简单情形, 此时 $Bx = 0$ 可写为 $b'x$, 其中 $b = (b_1, b_2, \dots, b_n)$ 。构造加边矩阵:

$$M_{kk} = \begin{pmatrix} 0 & b_1 & b_2 & \cdots & b_k \\ b_1 & a_{11} & a_{12} & \cdots & a_{1k} \\ b_2 & a_{21} & a_{22} & \cdots & a_{2k} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ b_k & a_{k1} & a_{k2} & \cdots & a_{kk} \end{pmatrix},$$

或者, 用 b^k 表示 b 的前 k 个元素, 用 A_{kk} 表示从 A 中删除后 $k+1, k+2, \dots, n$ 行和列得到的矩阵, M_{kk} 可写为

$$M_{kk} = \begin{pmatrix} 0 & b^{k'} \\ b^k & A_{kk} \end{pmatrix}.$$

定理 10.3: 设 A 为对称矩阵且 $b_1 \neq 0$ 。则,

1. 对所有 $x \neq 0$ 且 $b'x = 0$ 有 $x'Ax > 0$ 当且仅当

$$|M_{kk}| < 0, \quad k \geq 2$$

2. 对所有 $x \neq 0$ 且 $b'x = 0$ 有 $x'Ax < 0$ 当且仅当

$$(-1)^k |M_{kk}| > 0, \quad k \geq 2$$

Original (English)

In the case where $m > 1$, similar computations apply. Then,

$$M_{kk} = \begin{pmatrix} 0 & \cdots & 0 & b_{11} & b_{12} & \cdots & b_{1k} \\ \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & b_{m1} & b_{m2} & \cdots & b_{mk} \\ b_{11} & \cdots & b_{m1} & a_{11} & a_{12} & \cdots & a_{1k} \\ b_{12} & \cdots & b_{m2} & a_{21} & a_{22} & \cdots & a_{2k} \\ \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ b_{1k} & \cdots & b_{mk} & a_{k1} & a_{k2} & \cdots & a_{kk} \end{pmatrix}.$$

The matrix M_{kk} has $m+k$ rows and columns. Writing A_{kk} and B_{mk} for the component matrices,

$$M_{kk} = \begin{pmatrix} 0 & B'_{mk} \\ B_{mk} & A_{kk} \end{pmatrix}.$$

Theorem 10.4: Let A be a symmetric matrix and let $|B_{mm}| \neq 0$. Then,

1. $x'Ax > 0$ for all $x \neq 0$ with $Bx = 0$ if and only if:

$$(-1)^m |M_{kk}| > 0, \quad k = m+1, \dots, n.$$

2. $x'Ax < 0$ for all $x \neq 0$ with $Bx = 0$ if and only if:

$$(-1)^k |M_{kk}| > 0, \quad k = m+1, \dots, n.$$

翻译 (Chinese)

在 $m > 1$ 的情形中, 类似的计算适用。此时,

$$M_{kk} = \begin{pmatrix} 0 & \cdots & 0 & b_{11} & b_{12} & \cdots & b_{1k} \\ \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & b_{m1} & b_{m2} & \cdots & b_{mk} \\ b_{11} & \cdots & b_{m1} & a_{11} & a_{12} & \cdots & a_{1k} \\ b_{12} & \cdots & b_{m2} & a_{21} & a_{22} & \cdots & a_{2k} \\ \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ b_{1k} & \cdots & b_{mk} & a_{k1} & a_{k2} & \cdots & a_{kk} \end{pmatrix}.$$

矩阵 M_{kk} 有 $m+k$ 行和列。用 A_{kk} 和 B_{mk} 表示分量矩阵,

$$M_{kk} = \begin{pmatrix} 0 & B'_{mk} \\ B_{mk} & A_{kk} \end{pmatrix}.$$

定理 10.4: 设 A 为对称矩阵且 $|B_{mm}| \neq 0$ 。则,

1. 对所有 $x \neq 0$ 且 $Bx = 0$ 有 $x'Ax > 0$ 当且仅当:

$$(-1)^m |M_{kk}| > 0, \quad k = m+1, \dots, n.$$

2. 对所有 $x \neq 0$ 且 $Bx = 0$ 有 $x'Ax < 0$ 当且仅当:

$$(-1)^k |M_{kk}| > 0, \quad k = m+1, \dots, n.$$

Original (English)

REMARK 10.1: This characterization may be expressed in different ways. Let I_p be an identity matrix of dimension p . Then:

$$\begin{pmatrix} 0 & I_k \\ I_m & 0 \end{pmatrix} \begin{pmatrix} 0 & B'_{mk} \\ B_{mk} & A_{kk} \end{pmatrix} \begin{pmatrix} 0 & I_m \\ I_k & 0 \end{pmatrix} = \begin{pmatrix} A_{kk} & B'_{mk} \\ B_{mk} & 0 \end{pmatrix}.$$

From this, noting that

$$\begin{vmatrix} 0 & I_m \\ I_k & 0 \end{vmatrix} = \begin{vmatrix} 0 & I_k \\ I_m & 0 \end{vmatrix},$$

$$|M_{kk}| = \begin{vmatrix} 0 & I_k \\ I_m & 0 \end{vmatrix}^2 \begin{vmatrix} A_{kk} & B'_{mk} \\ B_{mk} & 0 \end{vmatrix} = \begin{vmatrix} A_{kk} & B'_{mk} \\ B_{mk} & 0 \end{vmatrix}.$$

Therefore, the determinantal conditions given previously can also be expressed in terms of matrices in the form:

$$\begin{pmatrix} A_{kk} & B'_{mk} \\ B_{mk} & 0 \end{pmatrix}.$$

REMARK 10.2: If there is just one constraint, $m = 1$, then constrained positiveness requires $|M_{kk}| < 0$, $k = m+1, \dots, n$. More generally, with $m > 1$, m odd gives $|M_{kk}| < 0$, $k = m+1, \dots, n$, while with m even the inequality is reversed so that $|M_{kk}| > 0$, $k = m+1, \dots, n$. In contrast, for constrained negatives $|M_{kk}|$ alternates in sign starting positive at $k = m+1$ if m is odd and starting negative if m is even.

翻译 (Chinese)

注记 10.1: 这种刻画可以用不同方式表达。令 I_p 为 p 维单位矩阵。则：

$$\begin{pmatrix} 0 & I_k \\ I_m & 0 \end{pmatrix} \begin{pmatrix} 0 & B'_{mk} \\ B_{mk} & A_{kk} \end{pmatrix} \begin{pmatrix} 0 & I_m \\ I_k & 0 \end{pmatrix} = \begin{pmatrix} A_{kk} & B'_{mk} \\ B_{mk} & 0 \end{pmatrix}.$$

由此，注意到

$$\begin{vmatrix} 0 & I_m \\ I_k & 0 \end{vmatrix} = \begin{vmatrix} 0 & I_k \\ I_m & 0 \end{vmatrix},$$

$$|M_{kk}| = \begin{vmatrix} 0 & I_k \\ I_m & 0 \end{vmatrix}^2 \begin{vmatrix} A_{kk} & B'_{mk} \\ B_{mk} & 0 \end{vmatrix} = \begin{vmatrix} A_{kk} & B'_{mk} \\ B_{mk} & 0 \end{vmatrix}.$$

因此，前述行列式条件也可以用如下形式的矩阵表示：

$$\begin{pmatrix} A_{kk} & B'_{mk} \\ B_{mk} & 0 \end{pmatrix}.$$

注记 10.2: 若只有一个约束, $m = 1$, 则约束正定性要求 $|M_{kk}| < 0$, $k = m + 1, \dots, n$ 。更一般地, 当 $m > 1$ 时, m 为奇数给出 $|M_{kk}| < 0$, $k = m + 1, \dots, n$; 而 m 为偶数时不等式反转, 即 $|M_{kk}| > 0$, $k = m + 1, \dots, n$ 。与之相对, 对于约束负定性, $|M_{kk}|$ 的符号交替, 若 m 为奇数则从 $k = m + 1$ 处为正开始, 若 m 为偶数则从负开始。

Exercises for Chapter 10 / 第 10 章习题

Original (English)

Exercise 10.1 Consider the following symmetric matrices:

$$A = \begin{pmatrix} 3 & -1 & 1 \\ -1 & 3 & -2 \\ 1 & -2 & 2 \end{pmatrix}, \quad B = \begin{pmatrix} 3 & 2 & 1 \\ 2 & 12 & 4 \\ 1 & 4 & 2 \end{pmatrix}.$$

Confirm that both A and B are positive definite.

Exercise 10.2 Consider the following matrices:

$$A = \begin{pmatrix} -3 & 2 & 5 \\ 2 & -6 & 1 \\ 5 & 1 & -13 \end{pmatrix}, \quad B = \begin{pmatrix} -5 & 2 & 1 \\ 2 & -1 & -1 \\ 1 & -1 & -7 \end{pmatrix}.$$

Confirm that both A and B are negative definite.

Exercise 10.3 Consider the matrix:

$$A = \begin{pmatrix} 4 & -\frac{1}{3} \\ -\frac{1}{3} & \frac{2}{9} \end{pmatrix}.$$

Show that A is positive definite for all $x = (x_1, x_2)$ satisfying $(\frac{1}{2}, \frac{2}{3}) \cdot (x_1, x_2) = 0$.

Exercise 10.4 Consider the matrix:

$$A = \begin{pmatrix} 4 & -\frac{4}{3} \\ -\frac{4}{3} & \frac{32}{9} \end{pmatrix}.$$

Show that A is positive definite for all $x = (x_1, x_2)$ satisfying $(2, \frac{8}{3}) \cdot (x_1, x_2) = 0$.

Exercise 10.5 Consider the matrix:

$$Q(\alpha, \beta, \gamma) = \begin{pmatrix} \alpha(\beta - 1) & \alpha(\gamma - 1) \\ \alpha(\beta - 1) & \gamma(\beta - 1) \end{pmatrix}.$$

Let $x'Q(\alpha, \beta, \gamma)x$ be a quadratic form. Also, let $b = (w, r, p)$ be a vector of strictly positive entries. Suppose that $\alpha = \beta = \gamma = \frac{1}{4}$. Show that $x'Q(\alpha, \beta, \gamma)x < 0$ for all x satisfying $b'x = 0$.

翻译 (Chinese)

习题 10.1 考虑以下对称矩阵:

$$A = \begin{pmatrix} 3 & -1 & 1 \\ -1 & 3 & -2 \\ 1 & -2 & 2 \end{pmatrix}, \quad B = \begin{pmatrix} 3 & 2 & 1 \\ 2 & 12 & 4 \\ 1 & 4 & 2 \end{pmatrix}.$$

验证 A 和 B 都是正定的。

习题 10.2 考虑以下矩阵:

$$A = \begin{pmatrix} -3 & 2 & 5 \\ 2 & -6 & 1 \\ 5 & 1 & -13 \end{pmatrix}, \quad B = \begin{pmatrix} -5 & 2 & 1 \\ 2 & -1 & -1 \\ 1 & -1 & -7 \end{pmatrix}.$$

验证 A 和 B 都是负定的。

习题 10.3 考虑矩阵:

$$A = \begin{pmatrix} 4 & -\frac{1}{3} \\ -\frac{1}{3} & \frac{2}{9} \end{pmatrix}.$$

证明 A 对所有满足 $(\frac{1}{2}, \frac{2}{3}) \cdot (x_1, x_2) = 0$ 的 $x = (x_1, x_2)$ 是正定的。

习题 10.4 考虑矩阵:

$$A = \begin{pmatrix} 4 & -\frac{4}{3} \\ -\frac{4}{3} & \frac{32}{9} \end{pmatrix}.$$

证明 A 对所有满足 $(2, \frac{8}{3}) \cdot (x_1, x_2) = 0$ 的 $x = (x_1, x_2)$ 是正定的。

习题 10.5 考虑矩阵:

$$Q(\alpha, \beta, \gamma) = \begin{pmatrix} \alpha(\beta - 1) & \alpha(\gamma - 1) \\ \alpha(\beta - 1) & \gamma(\beta - 1) \end{pmatrix}.$$

令 $x'Q(\alpha, \beta, \gamma)x$ 为二次型。同时, 令 $b = (w, r, p)$ 为严格正分量的向量。假设 $\alpha = \beta = \gamma = \frac{1}{4}$ 。证明对所有满足 $b'x = 0$ 的 x 有 $x'Q(\alpha, \beta, \gamma)x < 0$ 。

6.15 Multivariate Optimization / 多变量最优化

6.15.1 11.1 Introduction / 引言

Original (English)

This chapter is concerned with multivariate optimization, finding values for $(x_1, x_2, \dots, x_n)$ to make the function $f(x_1, x_2, \dots, x_n)$ as large or small as possible. In Section 11.2, the two-variable case is described; Section 11.3 considers the n -variable case. The extension from two to n variables is natural. Section 11.2 provides sufficient conditions for a local optimum and provides motivation for those conditions. Also, in this section, the envelope theorem is described. Section 11.3 provides sufficient conditions for both local and global optima. These conditions are developed in terms of concavity of the objective function and the properties of Hessian matrices in Section 11.4.

翻译 (Chinese)

本章讨论多变量最优化，即寻找 $(x_1, x_2, \dots, x_n)$ 的值以使函数 $f(x_1, x_2, \dots, x_n)$ 尽可能大或尽可能小。第 11.2 节描述了两变量情形；第 11.3 节考虑了 n 变量情形。从两个变量推广到 n 个变量是自然的。第 11.2 节给出了局部最优的充分条件，并提供了这些条件的动机。此外，该节还描述了包络定理。第 11.3 节给出了局部和全局最优的充分条件。这些条件在第 11.4 节中通过目标函数的凹性和 Hessian 矩阵的性质得到了展开。

6.15.2 11.2 The Two-Variable Case / 两变量情形

Original (English)

Consider the problem of choosing (x, y) to maximize $z = f(x, y)$, a function of two variables. The first requirement is that no unilateral variations in x or y improve the value of f , the first-order conditions:

$$f_x(x, y) = \frac{\partial f(x, y)}{\partial x} = 0 \quad (11.1)$$

$$f_y(x, y) = \frac{\partial f(x, y)}{\partial y} = 0. \quad (6.1)$$

The total differential is $dz = f_x dx + f_y dy$. This gives the change in the value of the function due to the variation (dx, dy) in x and y from the value (x, y) . Note that dz depends on (x, y) since $dz = f_x(x, y)dx + f_y(x, y)dy$. When the choice of x and y maximizes f , there can be no changes dx and dy that raise the value of f , as required by the partial derivative conditions above — that f_x and f_y are both 0. Consider the function plotted in Figure 11.1, $f(x, y) = 1 - x^2 - y^2$. There, at $(0, 0)$, $f_x = f_y = 0$, and it is clear from the figure that this point locates the highest point of the function. Confirming that a point is the maximizing value requires a comparison similar to the univariate test $f''(x) < 0$. The next theorem summarizes the key result for local optimality of a point (x^*, y^*) .

翻译 (Chinese)

考虑选择 (x, y) 以最大化两变量函数 $z = f(x, y)$ 的问题。第一个要求是 x 或 y 的任何单方面变动都不能改善 f 的值，即一阶条件：

$$f_x(x, y) = \frac{\partial f(x, y)}{\partial x} = 0 \quad (11.1)$$

$$f_y(x, y) = \frac{\partial f(x, y)}{\partial y} = 0. \quad (6.2)$$

全微分为 $dz = f_x dx + f_y dy$ 。这给出了由于 x 和 y 从 (x, y) 处的变动 (dx, dy) 所导致的函数值的变化。注意 dz 依赖于 (x, y) ，因为 $dz = f_x(x, y)dx + f_y(x, y)dy$ 。当 x 和 y 的选择使 f 最大化时，不可能存在能提高 f 的值的变动 dx 和 dy ，这是由上述偏导数条件——即 f_x 和 f_y 均为 0——所要求的。考虑图 11.1 中绘制的函数 $f(x, y) = 1 - x^2 - y^2$ 。在 $(0, 0)$ 处， $f_x = f_y = 0$ ，从图中可以清楚地看出该点确定了函数的最高点。确认一个点是最大值点需要类似于单变量检验 $f''(x) < 0$ 的比较。下一个定理总结了关于点 (x^*, y^*) 局部最优性的关键结果。

Original (English)

Theorem 11.1: Suppose that (x^*, y^*) satisfy

$$(1a) \quad f_x(x^*, y^*) = f_y(x^*, y^*) = 0$$

$$(1b) \quad f_{xx}(x^*, y^*) < 0, f_{yy}(x^*, y^*) < 0 \text{ and } f_{xx}(x^*, y^*)f_{yy}(x^*, y^*) - f_{xy}^2(x^*, y^*) > 0,$$

then (x^*, y^*) is a local maximum.

$$(2a) \quad f_x(x^*, y^*) = f_y(x^*, y^*) = 0$$

$$(2b) \quad f_{xx}(x^*, y^*) > 0, f_{yy}(x^*, y^*) > 0 \text{ and } f_{xx}(x^*, y^*)f_{yy}(x^*, y^*) - f_{xy}^2(x^*, y^*) > 0,$$

then (x^*, y^*) is a local minimum.

The following examples illustrate the use of the theorem.

翻译 (Chinese)

定理 11.1: 假设 (x^*, y^*) 满足

$$(1a) \quad f_x(x^*, y^*) = f_y(x^*, y^*) = 0$$

$$(1b) \quad f_{xx}(x^*, y^*) < 0, f_{yy}(x^*, y^*) < 0 \text{ 且 } f_{xx}(x^*, y^*)f_{yy}(x^*, y^*) - f_{xy}^2(x^*, y^*) > 0,$$

则 (x^*, y^*) 是局部极大值。

$$(2a) \quad f_x(x^*, y^*) = f_y(x^*, y^*) = 0$$

(2b) $f_{xx}(x^*, y^*) > 0$, $f_{yy}(x^*, y^*) > 0$ 且 $f_{xx}(x^*, y^*)f_{yy}(x^*, y^*) - f_{xy}^2(x^*, y^*) > 0$,

则 (x^*, y^*) 是局部极小值。

下面的例子说明了该定理的应用。

Original (English)

EXAMPLE 11.1: Suppose that a firm has a product that is produced at unit cost:

$$\min_{x,y} c(x, y) = e^x e^y - \ln x - \ln y.$$

What are the unit cost minimizing levels of x and y ? The first-order conditions are:

$$c_x = e^x e^y - \frac{1}{x} = 0, \quad c_y = e^x e^y - \frac{1}{y} = 0.$$

Thus, $\frac{1}{x} = \frac{1}{y}$ and $x = y$. From this, $e^{2x} = \frac{1}{x}$, so that solving $xe^{2x} = 1$ gives the solution ($x = 0.4263$).

The second-order conditions are:

$$c_{xx} = e^x e^y + \frac{1}{x^2}, \quad c_{yy} = e^x e^y + \frac{1}{y^2}, \quad c_{xy} = e^x e^y.$$

So, $c_{xx} > 0$ and

$$\begin{aligned} c_{xx}c_{yy} - c_{xy}^2 &= \left(e^x e^y + \frac{1}{x^2}\right) \left(e^x e^y + \frac{1}{y^2}\right) - [e^x e^y]^2 \\ &= e^x e^y \frac{1}{y^2} + e^x e^y \frac{1}{x^2} + \frac{1}{x^2} \frac{1}{y^2} > 0. \end{aligned}$$

翻译 (Chinese)

例 11.1: 假设某企业的产品单位成本为:

$$\min_{x,y} c(x, y) = e^x e^y - \ln x - \ln y.$$

使单位成本最小化的 x 和 y 水平是多少? 一阶条件为:

$$c_x = e^x e^y - \frac{1}{x} = 0, \quad c_y = e^x e^y - \frac{1}{y} = 0.$$

因此, $\frac{1}{x} = \frac{1}{y}$ 且 $x = y$ 。由此, $e^{2x} = \frac{1}{x}$, 故解 $xe^{2x} = 1$ 得到解 ($x = 0.4263$)。二阶条件为:

$$c_{xx} = e^x e^y + \frac{1}{x^2}, \quad c_{yy} = e^x e^y + \frac{1}{y^2}, \quad c_{xy} = e^x e^y.$$

因此, $c_{xx} > 0$ 且

$$\begin{aligned} c_{xx}c_{yy} - c_{xy}^2 &= \left(e^x e^y + \frac{1}{x^2}\right) \left(e^x e^y + \frac{1}{y^2}\right) - [e^x e^y]^2 \\ &= e^x e^y \frac{1}{y^2} + e^x e^y \frac{1}{x^2} + \frac{1}{x^2} \frac{1}{y^2} > 0. \end{aligned}$$

Original (English)

EXAMPLE 11.2: Suppose that a firm has profit function

$$f(x, y) = \sqrt{x} + \ln(y + 1) - x - \frac{1}{2}y.$$

What values of x and y maximize profit?

In this example, the computations are simplified because the function may be written $f(x, y) = f_1(x) + f_2(y)$: $f(x, y) = [\sqrt{x} - x] + [\ln(y + 1) - \frac{1}{2}y]$, and the cross-partial derivatives vanish.

The first-order conditions are:

$$f_x = \frac{1}{2\sqrt{x}} - 1 = 0, \quad f_y = \frac{1}{y+1} - \frac{1}{2} = 0.$$

So, $\frac{1}{\sqrt{x}} = 2$ and $2 = y + 1$ giving $x = \frac{1}{4}$, $y = 1$. The second derivatives are:

$$f_{xx} = -\frac{1}{4}x^{-\frac{3}{2}}, \quad f_{yy} = -(y+1)^{-2}, \quad f_{xy} = 0.$$

So, f_{xx} and f_{yy} are both negative in a neighborhood of $(x^*, y^*) = (\frac{1}{4}, 1)$, and $f(x, y) < f(\frac{1}{4}, 1)$ for $(x, y) \neq (\frac{1}{4}, 1)$.

REMARK 11.1: In Example 11.2, the function $f(x, y)$ has cross-partial derivatives that are 0, so that $f_{xy}(x, y) = 0$ for all (x, y) . Later, in Equation 11.2, with $f_x(x^*, y^*) = f_y(x^*, y^*) = 0$ and $f_{xy}(x^*, y^*) = f_{yx}(x^*, y^*) = 0$, the Taylor series expansion reduces to:

$$f(x, y) = f(x^*, y^*) + [f_{xx}(\bar{x}, \bar{y})(\Delta x)^2 + f_{yy}(\bar{x}, \bar{y})(\Delta y)^2].$$

So, if $f_{xx}(\bar{x}, \bar{y}) < 0$ and $f_{yy}(\bar{x}, \bar{y}) < 0$, since $(\Delta x)^2$ and $(\Delta y)^2$ are both positive, $f(x, y) < f(x^*, y^*)$. If $f_{xx}(x^*, y^*) < 0$ and $f_{yy}(x^*, y^*) < 0$, then this will also hold for (x, y) close to (x^*, y^*) , so that for all (x, y) close to (x^*, y^*) , $f(x, y) < f(x^*, y^*)$.

翻译 (Chinese)

例 11.2: 假设某企业的利润函数为

$$f(x, y) = \sqrt{x} + \ln(y + 1) - x - \frac{1}{2}y.$$

什么样的 x 和 y 值能使利润最大化?

在本例中, 计算得以简化, 因为函数可写为 $f(x, y) = f_1(x) + f_2(y)$: $f(x, y) = [\sqrt{x} - x] + [\ln(y + 1) - \frac{1}{2}y]$, 且交叉偏导数为零。

一阶条件为:

$$f_x = \frac{1}{2\sqrt{x}} - 1 = 0, \quad f_y = \frac{1}{y+1} - \frac{1}{2} = 0.$$

因此, $\frac{1}{\sqrt{x}} = 2$ 和 $2 = y + 1$, 得到 $x = \frac{1}{4}$, $y = 1$ 。二阶导数为:

$$f_{xx} = -\frac{1}{4}x^{-\frac{3}{2}}, \quad f_{yy} = -(y+1)^{-2}, \quad f_{xy} = 0.$$

因此, 在 $(x^*, y^*) = (\frac{1}{4}, 1)$ 的邻域内, f_{xx} 和 f_{yy} 均为负, 对 $(x, y) \neq (\frac{1}{4}, 1)$ 有 $f(x, y) < f(\frac{1}{4}, 1)$ 。

注记 11.1: 在例 11.2 中, 函数 $f(x, y)$ 的交叉偏导数为 0, 故对所有 (x, y) 有 $f_{xy}(x, y) = 0$ 。稍后在方程 11.2 中, 当 $f_x(x^*, y^*) = f_y(x^*, y^*) = 0$ 且 $f_{xy}(x^*, y^*) = 0$ 时, 泰勒级数展开简化为:

$$f(x, y) = f(x^*, y^*) + [f_{xx}(\bar{x}, \bar{y})(\Delta x)^2 + f_{yy}(\bar{x}, \bar{y})(\Delta y)^2].$$

因此, 若 $f_{xx}(\bar{x}, \bar{y}) < 0$ 且 $f_{yy}(\bar{x}, \bar{y}) < 0$, 由于 $(\Delta x)^2$ 和 $(\Delta y)^2$ 均为正, 有 $f(x, y) < f(x^*, y^*)$ 。若在 (x^*, y^*) 处 $f_{xx}(x^*, y^*) < 0$ 且 $f_{yy}(x^*, y^*) < 0$, 则在 (x, y) 接近 (x^*, y^*) 时这也成立, 从而对所有接近 (x^*, y^*) 的 (x, y) 有 $f(x, y) < f(x^*, y^*)$ 。

Original (English)

EXAMPLE 11.3: A problem in physics leads to the following objective. Minimize $f(x_1, x_2)$, where f is defined:

$$f(x_1, x_2) = \frac{1}{2}(\mu_1 x_1^2 + \mu_3(x_2 - x_1)^2 + \mu_2 x_2^2) - b x_2, \quad \mu_i > 0, i = 1, 2, 3.$$

Differentiating with respect to x_1 and x_2 gives:

$$f_{x_1} = \mu_1 x_1 - \mu_3(x_2 - x_1) = 0, \quad f_{x_2} = \mu_3(x_2 - x_1) + \mu_2 x_2 - b = 0.$$

The second derivatives are: $f_{x_1 x_1} = \mu_1 + \mu_3$, $f_{x_1 x_2} = -\mu_3$, $f_{x_2 x_2} = \mu_2 + \mu_3$. This gives the matrix H as

$$H = \begin{pmatrix} f_{x_1 x_1} & f_{x_1 x_2} \\ f_{x_1 x_2} & f_{x_2 x_2} \end{pmatrix} = \begin{pmatrix} (\mu_1 + \mu_3) & -\mu_3 \\ -\mu_3 & (\mu_2 + \mu_3) \end{pmatrix}.$$

To solve the first-order conditions, observe that they can be rearranged as:

$$\begin{pmatrix} (\mu_1 + \mu_3) & -\mu_3 \\ -\mu_3 & (\mu_2 + \mu_3) \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ b \end{pmatrix}.$$

Cramer's rule may be used to solve for x_1 and x_2 . The determinant of H is

$$|H| = (\mu_1 + \mu_3)(\mu_2 + \mu_3) - \mu_3^2 = \mu_1 \mu_2 + \mu_1 \mu_3 + \mu_2 \mu_3.$$

So,

$$x_1 = \frac{1}{|H|} \begin{vmatrix} 0 & -\mu_3 \\ b & (\mu_2 + \mu_3) \end{vmatrix}, \quad x_2 = \frac{1}{|H|} \begin{vmatrix} (\mu_1 + \mu_3) & 0 \\ -\mu_3 & b \end{vmatrix}.$$

Since

$$x_1 = \frac{b\mu_3}{\mu_1\mu_2 + \mu_1\mu_3 + \mu_2\mu_3}, \quad x_2 = \frac{b(\mu_1 + \mu_3)}{\mu_1\mu_2 + \mu_1\mu_3 + \mu_2\mu_3}.$$

$$f_{x_1x_1} = \mu_1 + \mu_3 > 0, \quad f_{x_1x_1}f_{x_2x_2} - f_{x_1x_2}^2 = \mu_1\mu_2 + \mu_1\mu_3 + \mu_2\mu_3 > 0,$$

the second-order condition for a minimum is satisfied.

翻译 (Chinese)

例 11.3: 物理学中的一个问题导出以下目标函数。最小化 $f(x_1, x_2)$, 其中 f 定义为:

$$f(x_1, x_2) = \frac{1}{2}(\mu_1x_1^2 + \mu_3(x_2 - x_1)^2 + \mu_2x_2^2) - bx_2, \quad \mu_i > 0, i = 1, 2, 3.$$

对 x_1 和 x_2 求导得:

$$f_{x_1} = \mu_1x_1 - \mu_3(x_2 - x_1) = 0, \quad f_{x_2} = \mu_3(x_2 - x_1) + \mu_2x_2 - b = 0.$$

二阶导数为: $f_{x_1x_1} = \mu_1 + \mu_3$, $f_{x_1x_2} = -\mu_3$, $f_{x_2x_2} = \mu_2 + \mu_3$ 。由此得到矩阵 H :

$$H = \begin{pmatrix} (\mu_1 + \mu_3) & -\mu_3 \\ -\mu_3 & (\mu_2 + \mu_3) \end{pmatrix}.$$

为求解一阶条件, 注意到它们可重新排列为:

$$\begin{pmatrix} (\mu_1 + \mu_3) & -\mu_3 \\ -\mu_3 & (\mu_2 + \mu_3) \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ b \end{pmatrix}.$$

可用 Cramer 法则求解 x_1 和 x_2 。 H 的行列式为

$$|H| = (\mu_1 + \mu_3)(\mu_2 + \mu_3) - \mu_3^2 = \mu_1\mu_2 + \mu_1\mu_3 + \mu_2\mu_3.$$

故

$$x_1 = \frac{b\mu_3}{\mu_1\mu_2 + \mu_1\mu_3 + \mu_2\mu_3}, \quad x_2 = \frac{b(\mu_1 + \mu_3)}{\mu_1\mu_2 + \mu_1\mu_3 + \mu_2\mu_3}.$$

由于

$$f_{x_1x_1} = \mu_1 + \mu_3 > 0, \quad f_{x_1x_1}f_{x_2x_2} - f_{x_1x_2}^2 = \mu_1\mu_2 + \mu_1\mu_3 + \mu_2\mu_3 > 0,$$

极小值的二阶条件得到满足。

11.2.1 Motivation for the Second-Order Conditions / 二阶条件的动机

Original (English)

Recall that in the univariate case, a Taylor series expansion gave:

$$f(x) = f(x^*) + f'(x^*)(x - x^*) + \frac{1}{2!}f''(p)(x - x^*)^2$$

for some p between x and x^* . So, with $f'(x^*) = 0$, $f(x) = f(x^*) + \frac{1}{2}f''(p)(x - x^*)^2$, so that if $f''(p) < 0$ for all p close to x^* , then $f(x) < f(x^*)$ and x^* is a local maximum. Essentially the same ideas apply to the multivariate case. In the two-variable case, a Taylor series expansion of f around (x^*, y^*) , $(x, y) = (x^* + \Delta x, y^* + \Delta y)$ is given by:

$$f(x, y) = f(x^*, y^*) + [f_x(x^*, y^*)\Delta x + f_y(x^*, y^*)\Delta y] \quad (11.2)$$

$$+ [f_{xx}(\bar{x}, \bar{y})(\Delta x)^2 + 2f_{xy}(\bar{x}, \bar{y})\Delta x\Delta y + f_{yy}(\bar{x}, \bar{y})(\Delta y)^2] \quad (11.3)$$

$$\approx f(x^*, y^*) + [f_x(x^*, y^*)\Delta x + f_y(x^*, y^*)\Delta y] \\ + [f_{xx}(x^*, y^*)(\Delta x)^2 + 2f_{xy}(x^*, y^*)\Delta x\Delta y + f_{yy}(x^*, y^*)(\Delta y)^2],$$

where $(\bar{x}, \bar{y})$ is on the line segment joining the points (x, y) and (x^*, y^*) :

$$(\bar{x}, \bar{y}) \in \{(\tilde{x}, \tilde{y}) \mid (\tilde{x}, \tilde{y}) = \theta(x, y) + (1 - \theta)(x^*, y^*), \theta \in [0, 1]\},$$

and where $\Delta x = x - x^*$ and $\Delta y = y - y^*$. The expression may be written in matrix terms:

$$f(x, y) = f(x^*, y^*) + [f_x(x^*, y^*)\Delta x + f_y(x^*, y^*)\Delta y] \quad (11.4)$$

$$+ \frac{1}{2}(\Delta x \ \Delta y)H(\bar{x}, \bar{y}) \begin{pmatrix} \Delta x \\ \Delta y \end{pmatrix} \quad (11.5)$$

$$\approx f(x^*, y^*) + [f_x(x^*, y^*)\Delta x + f_y(x^*, y^*)\Delta y] \\ + \frac{1}{2}(\Delta x \ \Delta y)H(x^*, y^*) \begin{pmatrix} \Delta x \\ \Delta y \end{pmatrix},$$

where

$$H(x, y) = \begin{pmatrix} f_{xx}(x, y) & f_{xy}(x, y) \\ f_{yx}(x, y) & f_{yy}(x, y) \end{pmatrix},$$

is called the Hessian matrix. In the case where $f(x, y) = 1 - x^2 - y^2$,

$$H(x, y) = \begin{pmatrix} -2 & 0 \\ 0 & -2 \end{pmatrix},$$

so, the expression in Equation 11.4 reduces to

$$f(x, y) = f(x^*, y^*) + [f_x(x^*, y^*)\Delta x + f_y(x^*, y^*)\Delta y] - 2[(\Delta x)^2 + (\Delta y)^2],$$

and at (x^*, y^*) , if $f_x(x^*, y^*) = f_y(x^*, y^*) = 0$ the expression reduces to $f(x, y) = -2[(\Delta x)^2 + (\Delta y)^2]$. Since this is less than or equal to 0 for all values of Δx and Δy , the value of f at (x^*, y^*) is at least as large as at any other (x, y) value. In this case, and in general, the properties of the Hessian matrix are crucial in determining whether a point satisfying the first-order conditions (Equations 11.1) determines a local maximum (or minimum).

翻译 (Chinese)

回顾在单变量情形中，泰勒级数展开给出：

$$f(x) = f(x^*) + f'(x^*)(x - x^*) + \frac{1}{2!}f''(p)(x - x^*)^2$$

对某个介于 x 和 x^* 之间的 p 。因此，当 $f'(x^*) = 0$ 时， $f(x) = f(x^*) + \frac{1}{2}f''(p)(x - x^*)^2$ ，故若对所有接近 x^* 的 p 有 $f''(p) < 0$ ，则 $f(x) < f(x^*)$ 且 x^* 为局部极大值。本质上相同的思想适用于多变量情形。在两变量情形中， f 在 (x^*, y^*) 附近的泰勒级数展开， $(x, y) = (x^* + \Delta x, y^* + \Delta y)$ ，由下式给出：

$$f(x, y) = f(x^*, y^*) + [f_x(x^*, y^*)\Delta x + f_y(x^*, y^*)\Delta y] \quad (11.2)$$

$$+ [f_{xx}(\bar{x}, \bar{y})(\Delta x)^2 + 2f_{xy}(\bar{x}, \bar{y})\Delta x\Delta y + f_{yy}(\bar{x}, \bar{y})(\Delta y)^2] \quad (11.3)$$

其中 $(\bar{x}, \bar{y})$ 在连接点 (x, y) 和 (x^*, y^*) 的线段上：

$$(\bar{x}, \bar{y}) \in \{(\tilde{x}, \tilde{y}) \mid (\tilde{x}, \tilde{y}) = \theta(x, y) + (1 - \theta)(x^*, y^*), \theta \in [0, 1]\},$$

且 $\Delta x = x - x^*$ ， $\Delta y = y - y^*$ 。该表达式可用矩阵形式写出：

$$f(x, y) = f(x^*, y^*) + [f_x(x^*, y^*)\Delta x + f_y(x^*, y^*)\Delta y] + \frac{1}{2}(\Delta x \ \Delta y)H(\bar{x}, \bar{y}) \begin{pmatrix} \Delta x \\ \Delta y \end{pmatrix},$$

其中

$$H(x, y) = \begin{pmatrix} f_{xx}(x, y) & f_{xy}(x, y) \\ f_{yx}(x, y) & f_{yy}(x, y) \end{pmatrix},$$

称为 Hessian 矩阵。在 $f(x, y) = 1 - x^2 - y^2$ 的情形中，

$$H(x, y) = \begin{pmatrix} -2 & 0 \\ 0 & -2 \end{pmatrix},$$

故上式简化为 $f(x, y) = -2[(\Delta x)^2 + (\Delta y)^2]$ 。由于这对所有 Δx 和 Δy 的取值都小于等于 0， f 在 (x^*, y^*) 处的值至少与在任何其他 (x, y) 处的值一样大。在这一情形以及一般情形

中, Hessian 矩阵的性质对于确定满足一阶条件 (方程 11.1) 的点是否为局部极大值 (或极小值) 至关重要。

Original (English)

Note that the entries (partial derivatives) in the Hessian matrix depend on x and y : writing $H(x, y)$ emphasizes that the entries are evaluated at (x, y) . If the first-order conditions hold at (x^*, y^*) , then $f_x(x^*, y^*) = f_y(x^*, y^*) = 0$. In this case, Equations 11.2 and 11.3 become:

$$f(x, y) = f(x^*, y^*) + [f_{xx}(\bar{x}, \bar{y})(\Delta x)^2 + 2f_{xy}(\bar{x}, \bar{y})\Delta x\Delta y + f_{yy}(\bar{x}, \bar{y})(\Delta y)^2] \quad (11.6)$$

$$\approx f(x^*, y^*) + [f_{xx}(x^*, y^*)(\Delta x)^2 + 2f_{xy}(x^*, y^*)\Delta x\Delta y + f_{yy}(x^*, y^*)(\Delta y)^2], \quad (11.7)$$

where $(\bar{x}, \bar{y})$ is on the line segment joining (x, y) and (x^*, y^*) . In terms of the Hessian matrix:

$$f(x, y) = f(x^*, y^*) + (\Delta x \ \Delta y)H(\bar{x}, \bar{y}) \begin{pmatrix} \Delta x \\ \Delta y \end{pmatrix} \quad (11.8)$$

$$\approx f(x^*, y^*) + (\Delta x \ \Delta y)H(x^*, y^*) \begin{pmatrix} \Delta x \\ \Delta y \end{pmatrix}. \quad (11.9)$$

The sign of this expression depends on properties of the matrix:

$$H(x, y) = \begin{pmatrix} f_{xx}(x, y) & f_{xy}(x, y) \\ f_{xy}(x, y) & f_{yy}(x, y) \end{pmatrix}.$$

The key property is called negative definiteness.

Definition 11.1: A square symmetric matrix $A = \begin{pmatrix} a & b \\ b & c \end{pmatrix}$ is said to be negative definite if for all $z = (z_1 \ z_2) \neq (0, 0)$, $z'Az < 0$. The matrix is said to be positive definite if $z'Az > 0$ for all such z .

Recall the criteria for negative and positive definiteness:

PROPOSITION 11.1: In terms of the elements of the matrix, positive definiteness and negative definiteness are characterized as follows:

1. A is negative definite if and only if $a < 0$ and $ac - b^2 > 0$, and
2. A is positive definite if and only if $a > 0$ and $ac - b^2 > 0$.

REMARK 11.2: This implies that if A is negative (positive) definite and $\tilde{A}$ is (sufficiently) close to A , then $\tilde{A}$ is negative (positive) definite. For example, if $a < 0$ and $ac - b^2 > 0$ and $\tilde{a}, \tilde{b}, \tilde{c}$ are close to a, b, c , respectively, then $\tilde{a} < 0$ and $\tilde{a}\tilde{c} - \tilde{b}^2 > 0$, so that $\tilde{A}$ is negative definite.

翻译 (Chinese)

注意 Hessian 矩阵中的元素（偏导数）依赖于 x 和 y ：记作 $H(x, y)$ 以强调这些元素是在 (x, y) 处取值的。若一阶条件在 (x^*, y^*) 处成立，则 $f_x(x^*, y^*) = f_y(x^*, y^*) = 0$ 。此时，方程 11.2 和 11.3 变为：

$$f(x, y) = f(x^*, y^*) + [f_{xx}(\bar{x}, \bar{y})(\Delta x)^2 + 2f_{xy}(\bar{x}, \bar{y})\Delta x\Delta y + f_{yy}(\bar{x}, \bar{y})(\Delta y)^2] \quad (11.6)$$

$$\approx f(x^*, y^*) + [f_{xx}(x^*, y^*)(\Delta x)^2 + 2f_{xy}(x^*, y^*)\Delta x\Delta y + f_{yy}(x^*, y^*)(\Delta y)^2], \quad (11.7)$$

其中 $(\bar{x}, \bar{y})$ 在连接 (x, y) 和 (x^*, y^*) 的线段上。用 Hessian 矩阵表示：

$$f(x, y) = f(x^*, y^*) + (\Delta x \ \Delta y)H(\bar{x}, \bar{y}) \begin{pmatrix} \Delta x \\ \Delta y \end{pmatrix}.$$

该表达式的符号取决于矩阵的性质：

$$H(x, y) = \begin{pmatrix} f_{xx}(x, y) & f_{xy}(x, y) \\ f_{xy}(x, y) & f_{yy}(x, y) \end{pmatrix}.$$

关键的性质称为负定性。

定义 11.1：称对称方阵 $A = \begin{pmatrix} a & b \\ b & c \end{pmatrix}$ 为负定的，若对所有 $z = (z_1 \ z_2) \neq (0, 0)$ ，有 $z'Az < 0$ 。称该矩阵为正定的，若对所有这样的 z 有 $z'Az > 0$ 。

回顾负定性和正定性的判据：

命题 11.1：用矩阵元素表示，正定性和负定性有如下刻画：

1. A 是负定的当且仅当 $a < 0$ 且 $ac - b^2 > 0$,
2. A 是正定的当且仅当 $a > 0$ 且 $ac - b^2 > 0$ 。

注记 11.2：这意味着若 A 是负（正）定的，且 $\tilde{A}$ （充分）接近 A ，则 $\tilde{A}$ 是负（正）定的。例如，若 $a < 0$ 且 $ac - b^2 > 0$ ，且 $\tilde{a}, \tilde{b}, \tilde{c}$ 分别接近 a, b, c ，则 $\tilde{a} < 0$ 且 $\tilde{a}\tilde{c} - \tilde{b}^2 > 0$ ，故 $\tilde{A}$ 是负定的。

Original (English)

Consequently, $H(x, y)$ is negative definite at (x, y) if:

$$f_{xx}(x, y) < 0 \quad \text{and} \quad f_{xx}(x, y)f_{yy}(x, y) - f_{xy}^2(x, y) > 0.$$

Likewise, $H(x, y)$ is positive definite at (x, y) if:

$$f_{xx}(x, y) > 0 \quad \text{and} \quad f_{xx}(x, y)f_{yy}(x, y) - f_{xy}^2(x, y) > 0.$$

REMARK 11.3: As noted earlier, if $f_{xx}(x^*, y^*) < 0$, then for $[f_{xx}(x^*, y^*)f_{yy}(x^*, y^*) - f_{xy}^2(x^*, y^*)] > 0$ to hold, it must be that $f_{yy}(x^*, y^*) < 0$, since $f_{xy}^2(x^*, y^*) = (f_{xy}(x^*, y^*))^2 \geq 0$. So, condition 1(b) in Theorem 11.1 is equivalent to $f_{xx}(x^*, y^*) < 0$ and $f_{xx}(x^*, y^*)f_{yy}(x^*, y^*) - f_{xy}^2(x^*, y^*) > 0$. An analogous equivalence applies in the local minimum case.

Given the previous discussion, the main theorem, Theorem 11.1, is straightforward to prove.

PROOF (Theorem 11.1): Suppose that at (x^*, y^*) , $f_x(x^*, y^*) = f_y(x^*, y^*) = 0$, and suppose also that condition 1(b) is satisfied, so that $H(x^*, y^*)$ is negative definite. Given a small number ε , let $-\varepsilon < \Delta x < \varepsilon$ and $-\varepsilon < \Delta y < \varepsilon$. Since $(\bar{x}, \bar{y})$ lies between (x^*, y^*) and $(x^* + \Delta x, y^* + \Delta y)$ in Equation 11.8, $(\bar{x}, \bar{y})$ is within ε of (x^*, y^*) . In view of the preceding observation, it must be that $H(\bar{x}, \bar{y})$ is negative definite for small ε (since $H(\bar{x}, \bar{y})$ is close to $H(x^*, y^*)$). Therefore,

$$f(x, y) = f(x^*, y^*) + (\Delta x \ \Delta y)H(\bar{x}, \bar{y}) \begin{pmatrix} \Delta x \\ \Delta y \end{pmatrix} \quad \text{and} \quad (\Delta x \ \Delta y)H(\bar{x}, \bar{y}) \begin{pmatrix} \Delta x \\ \Delta y \end{pmatrix} < 0,$$

so it must be that $f(x, y) < f(x^*, y^*)$. Similar reasoning applies to the case of a local minimum.

As mentioned in Remark 11.3, the conditions $f_{xx}(x, y) < 0$ and $f_{xx}(x, y)f_{yy}(x, y) - f_{xy}^2(x, y) > 0$ together imply that $f_{yy}(x, y) < 0$ so that the pair of conditions $[f_{xx}(x, y) < 0$ and $f_{xx}(x, y)f_{yy}(x, y) - f_{xy}^2(x, y) > 0]$ are equivalent to $[f_{xx}(x, y) < 0, f_{yy}(x, y) < 0$ and $f_{xx}(x, y)f_{yy}(x, y) - f_{xy}^2(x, y) > 0]$. However, as a practical matter, it is often useful to check the simpler conditions $f_{xx}(x, y) < 0$ and $f_{yy}(x, y) < 0$ first: if either fails, then the condition $f_{xx}(x, y)f_{yy}(x, y) - f_{xy}^2(x, y) > 0$ cannot be satisfied.

翻译 (Chinese)

因此, $H(x, y)$ 在 (x, y) 处为负定, 若:

$$f_{xx}(x, y) < 0 \quad \text{且} \quad f_{xx}(x, y)f_{yy}(x, y) - f_{xy}^2(x, y) > 0.$$

类似地, $H(x, y)$ 在 (x, y) 处为正定, 若:

$$f_{xx}(x, y) > 0 \quad \text{且} \quad f_{xx}(x, y)f_{yy}(x, y) - f_{xy}^2(x, y) > 0.$$

注记 11.3: 如前所述, 若 $f_{xx}(x^*, y^*) < 0$, 则要使 $[f_{xx}(x^*, y^*)f_{yy}(x^*, y^*) - f_{xy}^2(x^*, y^*)] > 0$ 成立, 必须有 $f_{yy}(x^*, y^*) < 0$, 因为 $f_{xy}^2(x^*, y^*) \geq 0$ 。因此定理 11.1 中的条件 1(b) 等价于 $f_{xx}(x^*, y^*) < 0$ 且 $f_{xx}(x^*, y^*)f_{yy}(x^*, y^*) - f_{xy}^2(x^*, y^*) > 0$ 。类似的等价性适用于局部极小值情形。

基于上述讨论, 主定理 (定理 11.1) 的证明是直接的。

证明 (定理 11.1): 假设在 (x^*, y^*) 处, $f_x(x^*, y^*) = f_y(x^*, y^*) = 0$, 并假设条件 1(b) 成立, 故 $H(x^*, y^*)$ 是负定的。给定一个小数 ε , 令 $-\varepsilon < \Delta x < \varepsilon$ 且 $-\varepsilon < \Delta y < \varepsilon$ 。由于方程 11.8 中的 $(\bar{x}, \bar{y})$ 介于 (x^*, y^*) 和 $(x^* + \Delta x, y^* + \Delta y)$ 之间, $(\bar{x}, \bar{y})$ 在 (x^*, y^*) 的 ε 范围内。鉴于前面的观察, 对于小的 ε , 必有 $H(\bar{x}, \bar{y})$ 为负定 (因为 $H(\bar{x}, \bar{y})$ 接近 $H(x^*, y^*)$)。因此,

$$f(x, y) = f(x^*, y^*) + (\Delta x \ \Delta y)H(\bar{x}, \bar{y}) \begin{pmatrix} \Delta x \\ \Delta y \end{pmatrix} \quad \text{且} \quad (\Delta x \ \Delta y)H(\bar{x}, \bar{y}) \begin{pmatrix} \Delta x \\ \Delta y \end{pmatrix} < 0,$$

故必有 $f(x, y) < f(x^*, y^*)$ 。类似的推理适用于局部极小值情形。

如注记 11.3 所述, $f_{xx}(x, y) < 0$ 和 $f_{xx}(x, y)f_{yy}(x, y) - f_{xy}^2(x, y) > 0$ 合在一起意味着 $f_{yy}(x, y) < 0$ 。然而, 在实际操作中, 通常先检验较简单的条件 $f_{xx}(x, y) < 0$ 和 $f_{yy}(x, y) < 0$ 是有用的: 若其中任何一个不成立, 则条件 $f_{xx}(x, y)f_{yy}(x, y) - f_{xy}^2(x, y) > 0$ 不可能满足。

Original (English)

REMARK 11.4: In fact, since the Taylor series expansion $f(x, y) = f(x^*, y^*) + (\Delta x \ \Delta y)H(\bar{x}, \bar{y}) \begin{pmatrix} \Delta x \\ \Delta y \end{pmatrix}$ holds generally (given any (x, y) , and (x^*, y^*) with $f_x(x^*, y^*) = f_y(x^*, y^*) = 0$, there is $(\bar{x}, \bar{y})$ such that this equation is satisfied). So, for example, if $H(x, y)$ were negative definite for all (x, y) , then $f(x, y) < f(x^*, y^*)$ for all $(x, y) \neq (x^*, y^*)$ and the first-order conditions give a global maximum.

With a view to generalization to the many-variable case, the second-order conditions can be expressed directly in terms of submatrices of the Hessian matrix H . Considering H , write H_1 for the matrix consisting of just the top left entry $H_1 = (f_{xx})$, and H_2 for the matrix H . Because the determinant of a matrix with one row and column is just the single entry of that matrix (if $A = [a]$, $|A| = a$),

$$|H_1| = f_{xx}$$

and

$$|H_2| = f_{xx}f_{yy} - f_{xy}^2.$$

Consequently:

REMARK 11.5: The second-order conditions for a maximum are

$$|H_1| < 0 \quad \text{and} \quad |H_2| > 0.$$

The second-order conditions for a minimum are

$$|H_1| > 0 \quad \text{and} \quad |H_2| > 0.$$

In the case of two variables, the benefit from expressing the second-order conditions in terms of determinants derived from the Hessian matrix is not significant. However, with three or more variables, the matrix presentation is more succinct. (See Section 11.3.)

翻译 (Chinese)

注记 11.4: 实际上, 由于泰勒级数展开 $f(x, y) = f(x^*, y^*) + (\Delta x \ \Delta y)H(\bar{x}, \bar{y}) \begin{pmatrix} \Delta x \\ \Delta y \end{pmatrix}$ 普遍成立 (给定任意 (x, y) 和满足 $f_x(x^*, y^*) = f_y(x^*, y^*) = 0$ 的 (x^*, y^*) , 存在 $(\bar{x}, \bar{y})$ 使该方程成立)。因此, 例如, 若 $H(x, y)$ 对所有 (x, y) 是负定的, 则对所有 $(x, y) \neq (x^*, y^*)$ 有 $f(x, y) < f(x^*, y^*)$, 且一阶条件给出全局极大值。

为向多变量情形推广, 二阶条件可直接用 Hessian 矩阵 H 的子矩阵表示。考虑 H , 记 H_1 为仅包含左上角元素的矩阵 $H_1 = (f_{xx})$, 记 H_2 为矩阵 H 。由于单行单列矩阵的行列式恰为该矩阵的单个元素 (若 $A = [a]$, $|A| = a$),

$$|H_1| = f_{xx}, \quad |H_2| = f_{xx}f_{yy} - f_{xy}^2.$$

因此:

注记 11.5: 极大值的二阶条件为

$$|H_1| < 0 \quad \text{且} \quad |H_2| > 0.$$

极小值的二阶条件为

$$|H_1| > 0 \quad \text{且} \quad |H_2| > 0.$$

在二变量情形中, 用由 Hessian 矩阵导出的行列式来表示二阶条件的优势并不显著。然而, 当有三个或更多变量时, 矩阵表示更为简洁。(见第 11.3 节。)

11.2.2 Failure of the Second-Order Conditions / 二阶条件失效的情形

Original (English)

The second-order conditions confirm that at a candidate optimum, in a neighborhood of that point the function “bends down” in all directions in the case of maximization, and “bends up” in all directions in the case of minimization. There are a variety of ways in which this condition may fail. The following examples illustrate two common cases. Consider the function

$$f(x, y) = x^2 - y^2.$$

The function is depicted in Figure 11.2 and is sometimes called a saddle function in view of its shape.

As the figure indicates, the point $(0, 0)$ satisfies the first-order conditions but from there variations in x raise f while variations in y reduce f . Observe that $f_x = 2x$, $f_y = -2y$, $f_{xx} = 2$, $f_{yy} = -2$ and $f_{xy} = 0$, so the Hessian matrix is:

$$H = \begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix}.$$

Thus, $|H_1| = 2$ and $|H_2| = -2$ so that neither the conditions for a local maximum or a local minimum are satisfied.

As a second example, consider the function:

$$f(x, y) = x^2y.$$

This function is plotted in Figure 11.3. Again, the point $(0, 0)$ satisfies the first-order conditions since $f_x = 2xy$, $f_y = x^2$, but the function twists at $(0, 0)$. The Hessian matrix is:

$$H = \begin{pmatrix} 2y & 2x \\ 2x & 0 \end{pmatrix}.$$

So $|H_1| = 2y$ and $|H_2| = -4x^2$. Both of these are 0 at $(x, y) = (0, 0)$.

翻译 (Chinese)

二阶条件确认了在候选最优点处，极大化时函数在该点邻域内沿所有方向“向下弯曲”，极小化时沿所有方向“向上弯曲”。这一条件可能以多种方式失效。下面的例子说明了两种常见情形。考虑函数

$$f(x, y) = x^2 - y^2.$$

该函数如图 11.2 所示，因其形状有时被称为鞍形函数。

如图所示，点 $(0, 0)$ 满足一阶条件，但从该点出发， x 的变动提高 f 而 y 的变动降低 f 。注意到 $f_x = 2x$ ， $f_y = -2y$ ， $f_{xx} = 2$ ， $f_{yy} = -2$ 且 $f_{xy} = 0$ ，故 Hessian 矩阵为：

$$H = \begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix}.$$

因此， $|H_1| = 2$ ， $|H_2| = -2$ ，既不符合局部极大值的条件也不符合局部极小值的条件。

作为第二个例子，考虑函数：

$$f(x, y) = x^2y.$$

该函数如图 11.3 所示。同样，点 $(0,0)$ 满足一阶条件，因为 $f_x = 2xy$ ， $f_y = x^2$ ，但函数在 $(0,0)$ 处发生扭转。Hessian 矩阵为：

$$H = \begin{pmatrix} 2y & 2x \\ 2x & 0 \end{pmatrix}.$$

故 $|H_1| = 2y$ ， $|H_2| = -4x^2$ ，两者在 $(x,y) = (0,0)$ 处均为 0。

11.2.3 The Envelope Theorem / 包络定理

Original (English)

Consider the problem of maximizing a function of x when the function depends on some exogenous parameter θ : $f(x, \theta)$. For example: $f(x, \theta) = x^{\frac{1}{2}} - \theta x$. The usual procedure is to set the first derivative to zero: $f_x(x, \theta) = 0$. This then gives the solution $x = x(\theta)$ as a function of θ , chosen to satisfy $f_x(x(\theta), \theta) = 0$. Note that if θ varies, so does $x(\theta)$ to maintain the equality of the partial derivative with 0. As θ varies, so will the optimal choice of x to maintain the identity $f_x(x(\theta), \theta) \equiv 0$. In the example, $f(x, \theta) = x^{\frac{1}{2}} - \theta x$ gives $f_x(x, \theta) = \frac{1}{2}x^{-\frac{1}{2}} - \theta = 0$. Rearranging, $\frac{1}{2} = \theta x^{\frac{1}{2}}$ or $x(\theta) = \frac{1}{4\theta^2}$. If the optimal choice of x given θ , $x(\theta)$, is substituted into the function f , then $f(x(\theta), \theta)$ is the value of the function being maximized, at the solution. Note that $f(x(\theta), \theta)$ depends only on θ ; the value of the optimized function may be written $f^*(\theta) \equiv f(x(\theta), \theta)$. In the example:

$$f^*(\theta) = f(x(\theta), \theta) = \left(\frac{1}{4\theta^2}\right)^{\frac{1}{2}} - \theta \left(\frac{1}{4\theta^2}\right) = \frac{1}{2\theta} - \frac{1}{4\theta} = \frac{1}{4\theta}.$$

Now consider the following question. If θ varies, how much will the value of the maximized function vary? That is, what is the value of $\frac{df^*(\theta)}{d\theta}$? Differentiating:

$$\frac{df^*(\theta)}{d\theta} = f_x(x(\theta), \theta) \frac{dx(\theta)}{d\theta} + f_\theta(x(\theta), \theta).$$

Note now that from the first-order condition $f_x(x(\theta), \theta) = 0$ at each value of θ , so that for each θ ,

$$\frac{df^*(\theta)}{d\theta} = f_\theta(x(\theta), \theta).$$

In words, the change in the maximized value of the function, $f^*(\theta)$, as the parameter (θ) changes, is equal to the derivative of the (unoptimized) function, $f(x, \theta)$, with respect to the parameter, with the derivative evaluated at the solution $(x(\theta))$: $f_\theta(x(\theta), \theta)$. In the example, $f^*(\theta) = \frac{1}{4\theta}$, so that

$$\frac{df^*(\theta)}{d\theta} = -\frac{1}{4\theta^2} = -x(\theta).$$

If we differentiate $f(x, \theta) = x^{\frac{1}{2}} - \theta x$ with respect to θ and evaluate at the solution $x(\theta)$, we get $-x(\theta)$.

翻译 (Chinese)

考虑在函数依赖于某个外生参数 θ 时最大化 x 的函数的问题: $f(x, \theta)$ 。例如: $f(x, \theta) = x^{\frac{1}{2}} - \theta x$ 。通常的做法是令一阶导数为零: $f_x(x, \theta) = 0$ 。这给出解 $x = x(\theta)$, 作为 θ 的函数, 被选定以满足 $f_x(x(\theta), \theta) = 0$ 。注意若 θ 变化, $x(\theta)$ 也随之变化以维持偏导数等于零的等式。在例子中, $f(x, \theta) = x^{\frac{1}{2}} - \theta x$ 给出 $f_x(x, \theta) = \frac{1}{2}x^{-\frac{1}{2}} - \theta = 0$ 。重新排列得 $\frac{1}{2} = \theta x^{\frac{1}{2}}$ 或 $x(\theta) = \frac{1}{4\theta^2}$ 。若将给定 θ 下 x 的最优选择 $x(\theta)$ 代入函数 f , 则 $f(x(\theta), \theta)$ 是在该解处被最大化函数的取值。记最优函数值为 $f^*(\theta) \equiv f(x(\theta), \theta)$ 。在例子中:

$$f^*(\theta) = f(x(\theta), \theta) = \frac{1}{4\theta}.$$

现在考虑以下问题。若 θ 变化, 被最大化函数的值将变化多少? 即, $\frac{df^*(\theta)}{d\theta}$ 的值是多少? 求导:

$$\frac{df^*(\theta)}{d\theta} = f_x(x(\theta), \theta) \frac{dx(\theta)}{d\theta} + f_\theta(x(\theta), \theta).$$

现在注意到由一阶条件 $f_x(x(\theta), \theta) = 0$ 在每个 θ 值处成立, 所以对每个 θ ,

$$\frac{df^*(\theta)}{d\theta} = f_\theta(x(\theta), \theta).$$

换言之, 被最大化函数的值 $f^*(\theta)$ 随参数 θ 变化而产生的变化, 等于 (未最优化) 函数 $f(x, \theta)$ 对参数的偏导数在解 $x(\theta)$ 处的取值: $f_\theta(x(\theta), \theta)$ 。在例子中, $f^*(\theta) = \frac{1}{4\theta}$, 故

$$\frac{df^*(\theta)}{d\theta} = -\frac{1}{4\theta^2} = -x(\theta).$$

若我们对 $f(x, \theta) = x^{\frac{1}{2}} - \theta x$ 关于 θ 求偏导并在解 $x(\theta)$ 处取值, 得到 $-x(\theta)$ 。

Original (English)

This observation extends to the general case. Let $f(x_1, x_2, \dots, x_n, \alpha_1, \alpha_2, \dots, \alpha_m)$. For ease of notation write $x = (x_1, x_2, \dots, x_n)$ and $\alpha = (\alpha_1, \alpha_2, \dots, \alpha_m)$. The first-order conditions are $f_{x_i}(x, \alpha) = 0$, $i = 1, 2, \dots, n$. Solving these gives $x(\alpha) = (x_1(\alpha), x_2(\alpha), \dots, x_n(\alpha))$. Then the optimized value of the function is $f^*(\alpha) = f(x(\alpha), \alpha)$. To calculate the change in f^* due to a change in one α , say α_i , take the partial derivative of $f^*(\alpha)$ with respect to α_i . This gives

$$\frac{\partial f^*(\alpha)}{\partial \alpha_i} = \sum_{j=1}^n \frac{\partial f(x(\alpha), \alpha)}{\partial x_j} \frac{\partial x_j(\alpha)}{\partial \alpha_i} + \frac{\partial f(x(\alpha), \alpha)}{\partial \alpha_i}.$$

Note now that from the first-order conditions, the partial derivative of $f(x, \alpha)$ evaluated at $x = x(\alpha)$ is equal to 0 for each $j = 1, 2, \dots, n$: $\frac{\partial f(x(\alpha), \alpha)}{\partial x_j} = 0, \forall j$. Thus,

$$\frac{\partial f^*(\alpha)}{\partial \alpha_i} = \frac{\partial f(x(\alpha), \alpha)}{\partial \alpha_i}.$$

翻译 (Chinese)

这一观察推广到一般情形。设 $f(x_1, x_2, \dots, x_n, \alpha_1, \alpha_2, \dots, \alpha_m)$ 。为记号简便, 记 $x = (x_1, x_2, \dots, x_n)$, $\alpha = (\alpha_1, \alpha_2, \dots, \alpha_m)$ 。一阶条件为 $f_{x_i}(x, \alpha) = 0, i = 1, 2, \dots, n$ 。解之得到 $x(\alpha) = (x_1(\alpha), x_2(\alpha), \dots, x_n(\alpha))$ 。则最优函数值为 $f^*(\alpha) = f(x(\alpha), \alpha)$ 。为计算 f^* 因某一 α_i 的变化而产生的变化, 求 $f^*(\alpha)$ 对 α_i 的偏导数:

$$\frac{\partial f^*(\alpha)}{\partial \alpha_i} = \sum_{j=1}^n \frac{\partial f(x(\alpha), \alpha)}{\partial x_j} \frac{\partial x_j(\alpha)}{\partial \alpha_i} + \frac{\partial f(x(\alpha), \alpha)}{\partial \alpha_i}.$$

现在注意到由一阶条件, $f(x, \alpha)$ 的偏导数在 $x = x(\alpha)$ 处的取值对每个 $j = 1, 2, \dots, n$ 等于 0: $\frac{\partial f(x(\alpha), \alpha)}{\partial x_j} = 0, \forall j$ 。因此,

$$\frac{\partial f^*(\alpha)}{\partial \alpha_i} = \frac{\partial f(x(\alpha), \alpha)}{\partial \alpha_i}.$$

6.15.3 11.3 Optimization: n Variables / n 变量最优化

Original (English)

Consider the problem of maximizing or minimizing a function f : $\max f(x_1, \dots, x_n)$ or $\min f(x_1, \dots, x_n)$. For convenience, write $f(x)$, where $x = (x_1, \dots, x_n)$. The first derivatives of f are $f_i = \frac{\partial f}{\partial x_i}$, $i = 1, \dots, n$. Set $\nabla f = (f_1, f_2, \dots, f_n)$. If it is the case that at a point x , the partial derivative of f with respect to x_i is not equal to 0, say $f_i(x) = \frac{\partial f(x)}{\partial x_i} > 0$, this means that at the point x the value of f is increased if x_i is increased from x_i . This implies that for x to be a local maximum of the function f , it must be the case that every partial derivative is 0. A necessary condition for x to be a local maximum is that partial derivatives at x are 0:

$$\frac{\partial f(x)}{\partial x_i} = 0, \quad i = 1, \dots, n.$$

Under what conditions can one be sure that x is a local maximum? In other words, what are necessary and sufficient conditions for x to be a local maximum? Let $H(p)$ be the Hessian matrix

evaluated at p :

$$H = \begin{pmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \frac{\partial^2 f}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{pmatrix} = \begin{pmatrix} f_{11} & f_{12} & \cdots & f_{1n} \\ f_{21} & f_{22} & \cdots & f_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ f_{n1} & f_{n2} & \cdots & f_{nn} \end{pmatrix}.$$

In general, H will depend on x . This dependence is made explicit by writing $H(x)$. Determining if H is positive or negative definite is possible by checking a set of determinantal conditions. Define H_j to be the $j \times j$ matrix obtained from H by deleting the last $n - j$ rows and columns:

$$H_j = \begin{pmatrix} f_{11} & f_{12} & \cdots & f_{1j} \\ f_{21} & f_{22} & \cdots & f_{2j} \\ \vdots & \vdots & \ddots & \vdots \\ f_{j1} & f_{j2} & \cdots & f_{jj} \end{pmatrix}.$$

Theorem 11.2: Suppose that x^* satisfies:

$$\frac{\partial f(x^*)}{\partial x_i} = 0, \quad i = 1, \dots, n.$$

If

$$|H_1| < 0, |H_2| > 0, |H_3| < 0, \dots, \text{ or } (-1)^j |H_j| > 0, \quad j = 1, \dots, n,$$

then x^* is a local maximum. If

$$|H_1| > 0, |H_2| > 0, |H_3| > 0, \dots, \text{ or } |H_j| > 0, \quad j = 1, \dots, n,$$

then x^* is a local minimum.

翻译 (Chinese)

考虑最大化或最小化函数 f 的问题: $\max f(x_1, \dots, x_n)$ 或 $\min f(x_1, \dots, x_n)$ 。为方便起见, 记 $f(x)$, 其中 $x = (x_1, \dots, x_n)$ 。 f 的一阶导数为 $f_i = \frac{\partial f}{\partial x_i}, i = 1, \dots, n$ 。记 $\nabla f = (f_1, f_2, \dots, f_n)$ 。若在某点 x 处, f 对 x_i 的偏导数不等于 0, 譬如 $f_i(x) > 0$, 这意味着在点 x 处增加 x_i 会增大 f 的值。这意味着若 x 是函数 f 的局部极大值点, 则每个偏导数必须为 0。 x 成为局部极大值的必要条件是在 x 处偏导数为 0:

$$\frac{\partial f(x)}{\partial x_i} = 0, \quad i = 1, \dots, n.$$

在什么条件下可以确信 x 是局部极大值？换言之， x 成为局部极大值的充分必要条件是什么？令 $H(p)$ 为在 p 处取值的 Hessian 矩阵：

$$H = \begin{pmatrix} f_{11} & f_{12} & \cdots & f_{1n} \\ f_{21} & f_{22} & \cdots & f_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ f_{n1} & f_{n2} & \cdots & f_{nn} \end{pmatrix}.$$

一般而言， H 依赖于 x 。这一依赖性通过记作 $H(x)$ 来显式表达。通过检验一组行列式条件可以判断 H 是正定还是负定。定义 H_j 为从 H 中删除后 $n-j$ 行和列得到的 $j \times j$ 矩阵。 H_j 由 H 的前 j 行和前 j 列组成。

定理 11.2： 假设 x^* 满足：

$$\frac{\partial f(x^*)}{\partial x_i} = 0, \quad i = 1, \dots, n.$$

若

$$|H_1| < 0, |H_2| > 0, |H_3| < 0, \dots, (-1)^j |H_j| > 0, j = 1, \dots, n,$$

则 x^* 是局部极大值。若

$$|H_1| > 0, |H_2| > 0, |H_3| > 0, \dots, |H_j| > 0, j = 1, \dots, n,$$

则 x^* 是局部极小值。

11.3.1 Motivation for the Hessian Conditions / Hessian 条件的动机

Original (English)

To motivate the previous discussion, consider a Taylor series expansion of f . The matrix version of Taylor's theorem gives:

$$f(\tilde{x}) = f(x) + \nabla f(x) \cdot (\tilde{x} - x) + (\tilde{x} - x)' H(p) (\tilde{x} - x),$$

where p is between $\tilde{x}$ and x . If x satisfies: $f_i(x) = 0, \forall i$, then

$$f(\tilde{x}) = f(x) + (\tilde{x} - x)' H(p) (\tilde{x} - x). \quad (11.14)$$

From this expression, whether $f(\tilde{x})$ is larger or smaller than $f(x)$ depends entirely on $H(p)$. If, for all vectors, $z \neq 0$, $z' H(p) z < 0$, then $H(p)$ is negative definite. From Equation 11.14 this implies $f(\tilde{x}) < f(x)$. If, for all vectors, $z \neq 0$, $z' H(p) z > 0$ the matrix $H(p)$ is positive definite

and Equation 11.14 implies $f(\tilde{x}) > f(x)$. So the sufficiency conditions depend on properties of the Hessian matrix H , and the associated quadratic form, characterized in terms of positive and negative definiteness.

Theorem 11.3: The matrix $H(x)$ is negative definite if:

$$|H_1(x)| < 0, |H_2(x)| > 0, |H_3(x)| < 0, \dots, (-1)^j |H_j(x)| > 0, j = 1, \dots, n.$$

The matrix $H(x)$ is positive definite if:

$$|H_1(x)| > 0, |H_2(x)| > 0, |H_3(x)| > 0, \dots, |H_j(x)| > 0, j = 1, \dots, n.$$

Note that if $H(x)$ satisfies the conditions for positive or negative definiteness, then provided p is close to x , $H(p)$ will also satisfy those conditions. If $\tilde{x}$ is close to x in Equation 11.14, then so is p . So, for example, if at x , $f_i(x) = 0$, for all i , and $H(x)$ is negative definite, then so is $H(p)$ and $f(\tilde{x}) < f(x)$.

翻译 (Chinese)

为给出前述讨论的动机, 考虑 f 的泰勒级数展开。泰勒定理的矩阵形式给出:

$$f(\tilde{x}) = f(x) + \nabla f(x) \cdot (\tilde{x} - x) + (\tilde{x} - x)' H(p) (\tilde{x} - x),$$

其中 p 介于 $\tilde{x}$ 和 x 之间。若 x 满足: $f_i(x) = 0, \forall i$, 则

$$f(\tilde{x}) = f(x) + (\tilde{x} - x)' H(p) (\tilde{x} - x). \quad (11.14)$$

从该表达式可知, $f(\tilde{x})$ 是大于还是小于 $f(x)$ 完全取决于 $H(p)$ 。若对所有向量 $z \neq 0$, $z' H(p) z < 0$, 则 $H(p)$ 是负定的。由方程 11.14, 这意味着 $f(\tilde{x}) < f(x)$ 。若对所有向量 $z \neq 0$, $z' H(p) z > 0$, 矩阵 $H(p)$ 是正定的, 则方程 11.14 意味着 $f(\tilde{x}) > f(x)$ 。因此充分性条件取决于 Hessian 矩阵 H 的性质, 以及相应的二次型, 这些性质由正定性和负定性来刻画。

定理 11.3: 矩阵 $H(x)$ 是负定的, 若:

$$|H_1(x)| < 0, |H_2(x)| > 0, |H_3(x)| < 0, \dots, (-1)^j |H_j(x)| > 0, j = 1, \dots, n.$$

矩阵 $H(x)$ 是正定的, 若:

$$|H_1(x)| > 0, |H_2(x)| > 0, |H_3(x)| > 0, \dots, |H_j(x)| > 0, j = 1, \dots, n.$$

注意若 $H(x)$ 满足正定性或负定性条件, 则只要 p 接近 x , $H(p)$ 也将满足那些条件。若方程 11.14 中 $\tilde{x}$ 接近 x , 则 p 也接近 x 。因此, 例如, 若在 x 处对所有 i 有 $f_i(x) = 0$, 且 $H(x)$ 是负定的, 则 $H(p)$ 也是负定的, 从而 $f(\tilde{x}) < f(x)$ 。

11.3.2 Definiteness and Second-Order Conditions / 定性与二阶条件

Original (English)

Recall that the Hessian matrix is the matrix of second partial derivatives of the function. Because the order of differentiation does not matter, $f_{x_i x_j} = f_{x_j x_i}$, the Hessian matrix is symmetric and the expression $z'H(p)z$ is a quadratic form. The main theorem giving second-order conditions in terms of the Hessian matrix is:

Theorem 11.4: Let x^* satisfy $f_i(x^*) = 0, i = 1, \dots, n$. Then

1. x^* is a local maximum if $H(\tilde{x})$ is negative semi-definite, $\tilde{x}$ close to x^* .
2. x^* is a unique local maximum if $H(x^*)$ is negative definite.
3. x^* is a global maximum if $H(x)$ is negative semi-definite $\forall x$.
4. x^* is a unique global maximum if $H(x)$ is negative definite $\forall x$.
5. x^* is a local minimum if $H(x)$ is positive semi-definite.
6. x^* is a unique local minimum if $H(x^*)$ is positive definite, $\tilde{x}$ close to x^* .
7. x^* is a global minimum if $H(x)$ is positive semi-definite $\forall x$.
8. x^* is a unique global minimum if $H(x)$ is positive definite $\forall x$.

PROOF: These results are easy to verify by considering the Taylor series expansion:

$$f(\tilde{x}) = f(x^*) + \nabla f(x^*) \cdot (\tilde{x} - x^*) + (\tilde{x} - x^*)' H(p) (\tilde{x} - x^*)$$

and if at x^* , $f_i(x^*) = 0$ so $\nabla f(x^*) = 0$, then the expansion gives:

$$f(\tilde{x}) = f(x^*) + (\tilde{x} - x^*)' H(p) (\tilde{x} - x^*).$$

For example, if $H(\tilde{x})$ is negative semi-definite, $\tilde{x}$ close to x^* , then this will also be satisfied by p in the Taylor series expansion, since p is between $\tilde{x}$ and x^* : $H(\tilde{x})$ is negative semi-definite when $\tilde{x}$ is close to x^* . In this case, $(\tilde{x} - x^*)' H(p) (\tilde{x} - x^*) \leq 0$, so that $f(\tilde{x}) \leq f(x^*)$. This confirms that x^* is a local maximum. A similar discussion applies to the other results.

翻译 (Chinese)

回顾 Hessian 矩阵是函数的二阶偏导数矩阵。由于求导次序无关紧要, $f_{x_i x_j} = f_{x_j x_i}$, 故 Hessian 矩阵是对称的, 且表达式 $z'H(p)z$ 是二次型。用 Hessian 矩阵给出二阶条件的主定理为:

定理 11.4: 设 x^* 满足 $f_i(x^*) = 0, i = 1, \dots, n$ 。则

1. x^* 是局部极大值, 若 $H(\tilde{x})$ 是半负定的, $\tilde{x}$ 接近 x^* 。

2. x^* 是唯一局部极大值, 若 $H(x^*)$ 是负定的。
3. x^* 是全局极大值, 若 $H(x)$ 对所有 x 是半负定的。
4. x^* 是唯一全局极大值, 若 $H(x)$ 对所有 x 是负定的。
5. x^* 是局部极小值, 若 $H(x)$ 是半正定的。
6. x^* 是唯一局部极小值, 若 $H(x^*)$ 是正定的, $\tilde{x}$ 接近 x^* 。
7. x^* 是全局极小值, 若 $H(x)$ 对所有 x 是半正定的。
8. x^* 是唯一全局极小值, 若 $H(x)$ 对所有 x 是正定的。

证明: 通过考察泰勒级数展开可容易验证这些结果:

$$f(\tilde{x}) = f(x^*) + \nabla f(x^*) \cdot (\tilde{x} - x^*) + (\tilde{x} - x^*)' H(p) (\tilde{x} - x^*)$$

若在 x^* 处 $f_i(x^*) = 0$, 故 $\nabla f(x^*) = 0$, 则展开给出:

$$f(\tilde{x}) = f(x^*) + (\tilde{x} - x^*)' H(p) (\tilde{x} - x^*).$$

例如, 若 $H(\tilde{x})$ 是半负定的, $\tilde{x}$ 接近 x^* , 则泰勒级数展开中的 p 也将满足该性质, 因为 p 介于 $\tilde{x}$ 和 x^* 之间。此时, $(\tilde{x} - x^*)' H(p) (\tilde{x} - x^*) \leq 0$, 故 $f(\tilde{x}) \leq f(x^*)$ 。这确认了 x^* 是局部极大值。类似的讨论适用于其他结果。

Original (English)

REMARK 11.6: When f is additively separable, the second order conditions are simpler. The function f is additively separable if $f(x_1, \dots, x_n) = \sum_{i=1}^n f^i(x_i) = f^1(x_1) + \dots + f^n(x_n)$. The Hessian has the form:

$$H = \begin{pmatrix} \frac{\partial^2 f^1}{\partial x_1^2} & 0 & \cdots & 0 \\ 0 & \frac{\partial^2 f^2}{\partial x_2^2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \frac{\partial^2 f^n}{\partial x_n^2} \end{pmatrix}.$$

Then f is negative definite if and only if $\frac{\partial^2 f^i}{\partial x_i^2} < 0, \forall i$.

翻译 (Chinese)

注记 11.6: 当 f 是可加分离的时, 二阶条件更简单。函数 f 是可加分离的, 若 $f(x_1, \dots, x_n) = \sum_{i=1}^n f^i(x_i)$ 。Hessian 矩阵具有如下形式:

$$H = \begin{pmatrix} \frac{\partial^2 f^1}{\partial x_1^2} & 0 & \cdots & 0 \\ 0 & \frac{\partial^2 f^2}{\partial x_2^2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \frac{\partial^2 f^n}{\partial x_n^2} \end{pmatrix}.$$

则 f 是负定的当且仅当 $\frac{\partial^2 f}{\partial x_i^2} < 0, \forall i$ 。

6.15.4 11.4 Concavity, Convexity and the Hessian Matrix / 凹性、凸性与 Hessian 矩阵

Original (English)

The next theorem relates concavity and convexity to the “definiteness” of the Hessian.

Theorem 11.5:

1. If f is strictly concave, then f is concave; and concavity of f is equivalent to negative semi-definiteness of the Hessian matrix $H(x)$, for all x .
2. If the Hessian matrix $H(x)$ is negative definite for all x , then f is strictly concave.
3. If f is strictly convex then f is convex; and convexity of f implies that the Hessian matrix $H(x)$ is positive semi-definite for all x .
4. If the Hessian matrix $H(x)$ is positive definite for all x , then f is strictly convex.

REMARK 11.7: Note that strict concavity of a function f does not imply that $H(x)$ is negative definite for all x ; and strict convexity does not imply that $H(x)$ is positive definite for all x . For example, consider $f(x) = -(x-1)^4$, a strictly concave function. Differentiating gives $f''(x) = -12(x-1)^2$. Note that $f''(1) = 0$ so that f'' is only negative semi-definite despite the fact that the function f is strictly concave.

翻译 (Chinese)

下一个定理将凹性和凸性与 Hessian 矩阵的“定性”联系起来。

定理 11.5:

1. 若 f 是严格凹的，则 f 是凹的；且 f 的凹性等价于 Hessian 矩阵 $H(x)$ 对所有 x 是半负定的。
2. 若 Hessian 矩阵 $H(x)$ 对所有 x 是负定的，则 f 是严格凹的。
3. 若 f 是严格凸的，则 f 是凸的；且 f 的凸性意味着 Hessian 矩阵 $H(x)$ 对所有 x 是半正定的。
4. 若 Hessian 矩阵 $H(x)$ 对所有 x 是正定的，则 f 是严格凸的。

注记 11.7: 注意函数的严格凹性并不意味着 $H(x)$ 对所有 x 是负定的；严格凸性也不意味着 $H(x)$ 对所有 x 是正定的。例如，考虑 $f(x) = -(x-1)^4$ ，这是一个严格凹函数。求导得 $f''(x) = -12(x-1)^2$ 。注意 $f''(1) = 0$ ，因此 f'' 仅是半负定的，尽管函数 f 是严格凹的。

Exercises for Chapter 11 / 第 11 章习题

Original (English)

Exercise 11.1 The total weekly revenue (in dollars) that a company obtains from selling goods x and y is:

$$R(x, y) = -\frac{1}{4}x^2 - \frac{3}{8}y^2 - \frac{1}{4}xy + 300x + 240y.$$

The total weekly cost attributed to production is

$$C(x, y) = 180x + 140y + 5000.$$

Find the profit-maximizing levels of x and y and check the second-order conditions.

Exercise 11.2 A television relay station will serve three towns, A, B and C, located in the (x, y) plane at coordinates $A = (30, 20)$, $B = (-20, 10)$ and $C = (10, -10)$. Plot the location of the three towns in the (x, y) plane and find the location of a transmitter, chosen to minimize the sum of the squares of the distances of the transmitter from each of the towns. For two points (x_1, y_1) and (x_2, y_2) , the distance between them is $[(x_1 - x_2)^2 + (y_1 - y_2)^2]^{\frac{1}{2}}$.

Exercise 11.3 Consider the following duopoly model. There are two firms supplying a market where demand is given by $p(Q) = a - bQ$. Firm i produces q_i units of output and so the total level of production is $Q = q_1 + q_2$. Both firms face the same constant marginal cost, so the cost of producing q_i for firm i is cq_i . Thus the profit functions of firms 1 and 2 respectively, are given by:

$$\begin{aligned}\pi_1(q_1, q_2) &= p(q_1 + q_2)q_1 - cq_1 = (a - b[q_1 + q_2])q_1 - cq_1 \\ \pi_2(q_1, q_2) &= p(q_1 + q_2)q_2 - cq_2 = (a - b[q_1 + q_2])q_2 - cq_2.\end{aligned}$$

(a) Suppose that each firm takes the output of the other firm as fixed, and calculates its own best choice of quantity. In this case, firm i solves $\frac{\partial \pi_i}{\partial q_i} = 0$, taking as given the value of q_j . Calculate for the linear demand case the equations $\frac{\partial \pi_1}{\partial q_1} = 0$, $\frac{\partial \pi_2}{\partial q_2} = 0$. Solve for q_1 and q_2 and denote the solutions q_1^d and q_2^d , where d denotes duopoly. Compare the value of $Q^d = q_1^d + q_2^d$ with the monopoly level of output. (b) Suppose now that we have n firms instead of two. This case is called oligopoly. Let the output of the i th firm be q_i^o , where o denotes oligopoly, and see if you can solve for $Q^o = q_1^o + q_2^o + \cdots + q_n^o$. (c) Suppose now that a government tax per unit of output is imposed on each firm, with tax t_i per unit on firm i . Then the profit functions become:

$$\begin{aligned}\pi_1(q_1, q_2; t_1) &= p(q_1 + q_2)q_1 - cq_1 - t_1q_1 = (a - b[q_1 + q_2])q_1 - cq_1 - t_1q_1 \\ \pi_2(q_1, q_2; t_2) &= p(q_1 + q_2)q_2 - cq_2 - t_2q_2 = (a - b[q_1 + q_2])q_2 - cq_2 - t_2q_2.\end{aligned}$$

Repeat the exercise in (a), and solve for q_1 and q_2 . Maximize T with respect to t_1 and t_2 , and check that the second-order conditions for a maximum are satisfied.

Exercise 11.4 Consider a firm that sells its output, y , at a fixed price p . In the production process, the firm creates pollution. To control pollution, the firm can engage in pollution abatement, a . The technology is as follows: the cost of producing y units of output when operating at abatement level a is given by $c(y, a)$, and the level of pollution emission is $e(y, a)$. We assume that for the cost function, $c_y > 0$, $c_{yy} > 0$, $c_a > 0$ and $c_{aa} > 0$. Similarly, for the emission process, e , we assume that $e_y > 0$, $e_{yy} > 0$, $e_a < 0$ and $e_{aa} > 0$. In addition, we assume that $c_{ay}(= c_{ya})$ and $e_{ay}(= e_{ya})$ are both positive. Finally, the government imposes a (unit) tax, τ , on the level of pollution emission. The profit function for the firm is then given by:

$$\pi(y, a) = py - c(y, a) - \tau e(y, a).$$

(a) Find the first-order conditions for profit maximization with respect to choice of (y, a) . (b) Determine the second-order conditions. (c) In the case where $c(y, a) = y^2a + ay$ and $e(y, a) = y - a^2 + ay$, α, β, γ parameters, find the profit-maximizing levels of y and a . Check the second-order conditions. From the solution to the first-order conditions calculate $\partial y / \partial \tau$ and $\partial a / \partial \tau$.

Exercise 11.5 A firm's costs of plant operation are a function of two variables, x and y . The cost function is given by:

$$c(x, y) = 3x^2 - 2xy + y^2 + 4x + 3y.$$

Give the solution to the first-order conditions and confirm that the solution minimizes the cost function.

Exercise 11.6 A box with edge lengths x , y and z has volume $V = xyz$. The surface area of a box with these dimensions (open at the top) is: $S = xy + 2yz + 2xz$. Suppose you have to choose a box size with volume 32; what choice of x , y and z is optimal to minimize surface area? Note that, in this question, there are only two choice variables once the volume is fixed. The variable z can be substituted out: $z = 32/xy$. Find the optimal choices of the variables to minimize surface area S , and check that the second-order conditions for a minimum are satisfied.

Exercise 11.7 Consider a firm that sells its output, y , at a fixed price p . In the production process, the firm creates pollution. To control pollution, the firm can engage in pollution abatement, a . The technology is as follows: the cost of producing y units of output when operating at abatement level a is given by $c(y, a)$, and the level of pollution emission is $e(y, a)$. Finally, the government imposes a (unit) tax, τ , on the level of pollution emission. The profit function for the firm is then given by:

$$\pi(y, a) = py - c(y, a) - \tau e(y, a).$$

Give the first- and second-order conditions for a maximum. Find the profit-maximizing levels of output and abatement when the cost function is given by $c(y, a) = y^2 + \frac{1}{2}a^2$, and $e(y, a) = y(10 - a)$.

Check that the second-order conditions are satisfied. We assume that the tax rate is less than 1 ($0 < \tau < 1$), and $p > 10$.

Exercise 11.8 Find the maximizing values of (x, y) for the following function, and confirm that the second-order conditions are satisfied:

$$f(x, y) = \sqrt{x} + \ln y - x^2 - 3y.$$

Exercise 11.9 Find the maximizing values of (x, y) for the following function, and confirm that the second-order conditions are satisfied:

$$f(x, y) = a \ln x + b \ln y - \frac{1}{2}c(x^2 + y^2).$$

Exercise 11.10 Find the maximizing values of (x, y) for the following function, and confirm that the second-order conditions are satisfied:

$$f(x, y) = \ln(ax + by) - \frac{1}{2}c(x^2 + y^2).$$

翻译 (Chinese)

习题 11.1 某公司销售商品 x 和 y 的每周总收入（美元）为：

$$R(x, y) = -\frac{1}{4}x^2 - \frac{3}{8}y^2 - \frac{1}{4}xy + 300x + 240y.$$

生产相关的每周总成本为

$$C(x, y) = 180x + 140y + 5000.$$

求利润最大化的 x 和 y 水平，并检验二阶条件。

习题 11.2 一个电视中继站将为三个城镇 A、B 和 C 服务，它们位于 (x, y) 平面上的坐标分别为 $A = (30, 20)$ ， $B = (-20, 10)$ 和 $C = (10, -10)$ 。在 (x, y) 平面上绘制三个城镇的位置，并求出发射机的位置，使其最小化发射机到每个城镇距离的平方和。对于两点 (x_1, y_1) 和 (x_2, y_2) ，它们之间的距离为 $[(x_1 - x_2)^2 + (y_1 - y_2)^2]^{\frac{1}{2}}$ 。

习题 11.3 考虑以下双寡头模型。两家企业供应一个市场，需求由 $p(Q) = a - bQ$ 给出。企业 i 生产 q_i 单位产出，因此总生产水平为 $Q = q_1 + q_2$ 。两家企业面临相同的固定边际成本，故企业 i 生产 q_i 的成本为 cq_i 。因此企业 1 和企业 2 的利润函数分别为：

$$\pi_1(q_1, q_2) = (a - b[q_1 + q_2])q_1 - cq_1, \quad \pi_2(q_1, q_2) = (a - b[q_1 + q_2])q_2 - cq_2.$$

(a) 假设每家企业将另一家企业的产出视为给定，并计算自身的最优产量选择。求解 q_1 和 q_2 并将解记为 q_1^d 和 q_2^d 。比较 $Q^d = q_1^d + q_2^d$ 与垄断产出水平。(b) 假设有 n 家企业而非两

家。求解 $Q^o = q_1^o + q_2^o + \cdots + q_n^o$ 。(c) 假设对每家企业征收单位产出税。求解 q_1 和 q_2 , 最大化 T 关于 t_1 和 t_2 , 并检验极大值的二阶条件。

习题 11.4 一家企业以固定价格 p 销售其产出 y 。控制污染的减排水平为 a 。生产成本为 $c(y, a)$, 污染排放水平为 $e(y, a)$ 。假设 $c_y > 0$, $c_{yy} > 0$, $c_a > 0$, $c_{aa} > 0$; $e_y > 0$, $e_{yy} > 0$, $e_a < 0$, $e_{aa} > 0$ 。利润函数为 $\pi(y, a) = py - c(y, a) - \tau e(y, a)$ 。(a) 求利润最大化的一阶条件。(b) 确定二阶条件。(c) 在 $c(y, a) = y^2 a + ay$ 和 $e(y, a) = y - a^2 + ay$ 的情形下, 求利润最大化的 y 和 a 并对 τ 求偏导。

习题 11.5 某企业工厂运营成本是两变量 x 和 y 的函数: $c(x, y) = 3x^2 - 2xy + y^2 + 4x + 3y$ 。给出一阶条件的解并确认该解最小化成本函数。

习题 11.6 一个棱长分别为 x 、 y 和 z 的盒子体积为 $V = xyz$, 无盖表面积为 $S = xy + 2yz + 2xz$ 。假设需选择体积为 32 的盒子尺寸, 求最优的 x, y, z 以最小化表面积 S , 并检验极小值的二阶条件。

习题 11.7 类似习题 11.4, 当 $c(y, a) = y^2 + \frac{1}{2}a^2$ 和 $e(y, a) = y(10 - a)$ 时, 求利润最大化的产出和减排水平, 并检验二阶条件。

习题 11.8 求函数 $f(x, y) = \sqrt{x} + \ln y - x^2 - 3y$ 的最大化值, 并确认二阶条件。

习题 11.9 求函数 $f(x, y) = a \ln x + b \ln y - \frac{1}{2}c(x^2 + y^2)$ 的最大化值, 并确认二阶条件。

习题 11.10 求函数 $f(x, y) = \ln(ax + by) - \frac{1}{2}c(x^2 + y^2)$ 的最大化值, 并确认二阶条件。

Original (English)

Exercise 11.11 Consider the problem of maximizing a function, $f(x, y; a)$, by choice of x and y , where a is some parameter. Suppose that the solution values for x and y are $x(a)$ and $y(a)$, respectively. The optimized value of the function is then $f^*(a) = f(x(a), y(a); a)$. (a) Explain why $\frac{df^*(a)}{da} = \frac{\partial f(x(a), y(a); a)}{\partial a}$. (b) Illustrate this in the one-variable case, where $f(x; a) = a \ln(x) - \frac{1}{a}x$.

Exercise 11.12 Consider the following problem. Minimize $f(x, y) = (x - a)^2 + (y - b)^2 - \frac{1}{2}cxy$. (a) Find the minimizing values for x and y . (Assume that $c < 1$.) (b) Confirm that the second-order conditions are satisfied. Let $f^*(a, b, c)$ be the value of the function at the solution. (c) Find $\frac{\partial f^*(a, b, c)}{\partial a}$ and $\frac{\partial f^*(a, b, c)}{\partial c}$.

Exercise 11.13 A firm maximizes profit, where profit is equal to revenue minus cost. Output (Q) depends on two inputs, K and L , related according to the production function $Q = f(K, L)$. Output sells at price p and the costs of the inputs K and L are r and w , respectively. So the profit maximization problem is:

$$\max_{K, L} \pi(K, L) = pf(K, L) - rK - wL.$$

Give the first- and second-order conditions for profit maximization. In the particular case where f is $f(K, L) = K^\alpha L^\beta$ with $\alpha > 0$, $\beta > 0$ and $0 < \alpha + \beta < 1$, verify that the second-order conditions are satisfied and solve for the optimal choice of (K, L) .

Exercise 11.14 A standard econometric problem is determining the ordinary least squares estimator (OLS). In the underlying model, the i th observation on one variable y is denoted y_i and the corresponding observation on another variable x is denoted x_i . It is postulated that y and x are linearly related with some error, $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$. If a and b are chosen as the intercept and slope, respectively, then the estimated error associated with the i th observation is $e_i = y_i - a - bx_i$. The OLS estimator is obtained by minimizing the sum of squares of the estimated errors, $\sum_{i=1}^n e_i^2$, by choice of a and b . Thus, the OLS estimator is obtained by solving the problem:

$$\min_{a,b} S = \sum_{i=1}^n e_i^2 = \min_{a,b} \sum_{i=1}^n (y_i - a - bx_i)^2.$$

Find the OLS estimator and confirm the second-order conditions for a minimum.

Exercise 11.15 A monopolist can sell a product in three different countries. The demand in each country is:

$$Q_1 = 50 - \frac{1}{2}P_1, \quad Q_2 = 50 - \frac{1}{3}P_2, \quad Q_3 = 75 - \frac{1}{4}P_3.$$

The cost function for the monopolist is $C(Q) = 30 + Q^2$, where $Q = Q_1 + Q_2 + Q_3$. Calculate the amount of the product that the profit-maximizing monopolist should sell in each country and check the second-order conditions for a maximum.

Exercise 11.16 For the following functions, calculate the Hessians and determine if they are positive or negative definite:

- (a) $f(x, y) = -2x^2 - 7y^2 + xy + 12x + 37y$
- (b) $f(x, y) = 10x^2 + 5y^2 + 3x - 12y + 9xy$
- (c) $f(x, y, z) = 3x^2 + 12y^2 + 5z^2 - 13x + 12y + 2xz - 3yz$
- (d) $f(x, y, z) = 2x + 3y - 5xz + 2xy - 6x^2 - 8y^2 - 11z^2$.

Exercise 11.17 For the following function, find all extreme points and check if they are maxima or minima:

$$f(x, y) = 3y^3 + 2x^2 + 4xy + 8y^2 + 2y.$$

Exercise 11.18 Given the three-input Cobb–Douglas production function, $Y = AL^\alpha K^\beta M^\gamma$, where L is the amount of labour, K the amount of capital and M the amount of other materials used, prove that the second-order conditions for a profit maximization hold if and only if $\alpha + \beta + \gamma < 1$ ($\alpha, \beta, \gamma > 0$). The good sells for a price of a dollar per unit. The costs of labour, capital and

materials are, respectively, w , r and p . Find the Hessian matrix and confirm that it is negative definite.

Exercise 11.19 The geometric average of three (positive) numbers x , y and z is given by $(xyz)^{\frac{1}{3}}$. Suppose that $x + y + z = k$. Show that the geometric average is maximized when $x = y = z$.

Exercise 11.20 A firm uses two inputs x , y to produce Q according to $Q = A(x^\alpha + y^\beta)$, with $0 < \alpha, \beta < 1$. The production cost is given by $c(x, y)$, so that profit is $\pi(x, y) = pA(x^\alpha + y^\beta) - c(x, y)$, where p is the price of Q . Give the first- and second-order conditions for a maximum. In the case where $c(x, y) = ax + by + dxy$, $a, b, d > 0$, find the maximizing values of x, y and show that the conditions for a maximum are satisfied.

Exercise 11.21 For the function $f(x, y) = 10x^2 + 10y^2 + 3x - 12y + 10xy$, show that the Hessian is positive definite.

Exercise 11.22 For the function $f(x, y, z) = 2x + 3y - 5xz + 2xy - 6x^2 - 9y^2 - 12z^2$, show that the Hessian is negative definite.

翻译 (Chinese)

习题 11.11 考虑通过选择 x 和 y 最大化函数 $f(x, y; a)$ 的问题, 其中 a 是某个参数。设 $x(a)$ 和 $y(a)$ 分别为 x 和 y 的解值。最优函数值为 $f^*(a) = f(x(a), y(a); a)$ 。(a) 解释为什么 $\frac{df^*(a)}{da} = \frac{\partial f(x(a), y(a); a)}{\partial a}$ 。(b) 在单变量情形中说明此结果, 其中 $f(x; a) = a \ln(x) - \frac{1}{a}x$ 。

习题 11.12 最小化 $f(x, y) = (x - a)^2 + (y - b)^2 - \frac{1}{2}cxy$ 。(a) 求 x 和 y 的最小化值。(假设 $c < 1$ 。)(b) 确认二阶条件成立。令 $f^*(a, b, c)$ 为解处的函数值。(c) 求 $\frac{\partial f^*(a, b, c)}{\partial a}$ 和 $\frac{\partial f^*(a, b, c)}{\partial c}$ 。

习题 11.13 企业最大化利润, 产出 $Q = f(K, L)$, 价格为 p , 投入成本为 r 和 w 。在 $f(K, L) = K^\alpha L^\beta$, $\alpha, \beta > 0$, $0 < \alpha + \beta < 1$ 的情形下, 验证二阶条件并求解 (K, L) 的最优选择。

习题 11.14 普通最小二乘 (OLS) 估计量通过最小化 $\sum_{i=1}^n (y_i - a - bx_i)^2$ 得到。求 OLS 估计量并确认极小值的二阶条件。

习题 11.15 垄断者在三个不同国家销售产品。需求为 $Q_1 = 50 - \frac{1}{2}P_1$, $Q_2 = 50 - \frac{1}{3}P_2$, $Q_3 = 75 - \frac{1}{4}P_3$, 成本函数为 $C(Q) = 30 + Q^2$ 。计算利润最大化垄断者在每个国家销售的产品数量并检验二阶条件。

习题 11.16 对以下函数, 计算 Hessian 矩阵并判断它们是正定还是负定: (a) $f(x, y) = -2x^2 - 7y^2 + xy + 12x + 37y$ (b) $f(x, y) = 10x^2 + 5y^2 + 3x - 12y + 9xy$ (c) $f(x, y, z) = 3x^2 + 12y^2 + 5z^2 - 13x + 12y + 2xz - 3yz$ (d) $f(x, y, z) = 2x + 3y - 5xz + 2xy - 6x^2 - 8y^2 - 11z^2$ 。

习题 11.17 求函数 $f(x, y) = 3y^3 + 2x^2 + 4xy + 8y^2 + 2y$ 的所有极值点, 并判断它们是极

大值还是极小值。

习题 11.18 给定三投入 Cobb–Douglas 生产函数 $Y = AL^\alpha K^\beta M^\gamma$ ，证明利润最大化的二阶条件成立当且仅当 $\alpha + \beta + \gamma < 1$ 。求 Hessian 矩阵并确认它是负定的。

习题 11.19 三个正数 x, y, z 的几何平均为 $(xyz)^{\frac{1}{3}}$ 。假设 $x + y + z = k$ 。证明当 $x = y = z$ 时几何平均最大化。

习题 11.20 企业使用两种投入 x, y 生产 $Q = A(x^\alpha + y^\beta)$ 。在 $c(x, y) = ax + by + dxy$ 的情形下，求出 x, y 的最大化值，并证明极大值条件成立。

习题 11.21 对函数 $f(x, y) = 10x^2 + 10y^2 + 3x - 12y + 10xy$ ，证明 Hessian 矩阵是正定的。

习题 11.22 对函数 $f(x, y, z) = 2x + 3y - 5xz + 2xy - 6x^2 - 9y^2 - 12z^2$ ，证明 Hessian 矩阵是负定的。

6.16 Equality-Constrained Optimization / 等式约束最优化

6.16.1 12.1 Introduction / 引言

Original (English)

Optimization of a function $f(x_1, \dots, x_n)$ involves choosing the x_i variables to make the function f as large or as small as possible. With constrained optimization, permissible choices for the x_i are restricted or constrained. In what follows, such constraints are represented by a function, g , whereby the x_i variables must satisfy a condition of the form $g(x_1, \dots, x_n) = c$. For example, in the standard utility maximization problem, the task is to maximize utility $u(x_1, \dots, x_n)$ subject to a budget constraint $p_1x_1 + \dots + p_nx_n = c$, where p_i is the price of good i and c is the income available.

Constraints limit the choices that are feasible. In the case of two goods, the budget constraint is $p_1x_1 + p_2x_2 = c$ so that the feasible pairs lie on a line. In this case, there is one constraint with two choice variables. The choice of one variable determines the choice of the other: given x_1 , x_2 is determined as $x_2 = \frac{c}{p_2} - \frac{p_1}{p_2}x_1$. Suppose that in addition to the budget constraint there is a second constraint – that the two goods are complements that must be consumed in a set ratio: $\frac{x_2}{x_1} = r$ or $x_2 = rx_1$. Given this, from $p_1x_1 + p_2x_2 = c$ one has $p_1x_1 + p_2rx_1 = c$ or $(p_1 + p_2r)x_1 = c$ or $x_1 = \frac{c}{(p_1 + p_2r)}$ and $x_2 = \frac{cr}{(p_1 + p_2r)}$. Thus, there is a unique pair satisfying the two constraints. This observation is generally true; if the number of constraints equals the number of choice variables,

there are generally at most a finite number of solutions to the set of constraints. For this reason, the set of constraints is generally less than the number of choice variables.

The next section provides a motivating example showing how the Lagrangian technique relates to the familiar first-order condition (marginal rate of substitution equals the price ratio) for an optimum in the consumer choice problem. Section 12.2 considers the case of two variables and one constraint. The case of n variables and one constraint is discussed in Section 12.4. It turns out that the Lagrange multiplier appearing in the definition of the Lagrangian has a significant economic interpretation: it provides a measure of the value of the constraint. Specifically, it measures the marginal change in the value of the objective resulting from a small change in the constraint. This is examined in Section 12.5. In the case of unconstrained optimization, identification of sufficient conditions for a global optimum is simplified when the objective has a specific shape, particularly concavity or convexity. Similarly, with constrained optimization, under suitable conditions, discussed in Section 12.6, the local conditions for an optimum are sufficient for the solution to be a global optimum.

翻译 (Chinese)

函数 $f(x_1, \dots, x_n)$ 的最优化涉及选择 x_i 变量以使函数 f 尽可能大或尽可能小。在约束最优化中, x_i 的容许选择受到限制或约束。在下文中, 这种约束由一个函数 g 表示, 其中 x_i 变量必须满足形如 $g(x_1, \dots, x_n) = c$ 的条件。例如, 在标准的效用最大化问题中, 任务是在预算约束 $p_1x_1 + \dots + p_nx_n = c$ 下最大化效用 $u(x_1, \dots, x_n)$, 其中 p_i 为商品 i 的价格, c 为可支配收入。

约束限制了可行的选择。在两种商品的情形中, 预算约束为 $p_1x_1 + p_2x_2 = c$, 因此可行的商品对位于一条直线上。此时, 有两个选择变量和一个约束。一个变量的选择决定了另一个变量的选择: 给定 x_1 , x_2 由 $x_2 = \frac{c}{p_2} - \frac{p_1}{p_2}x_1$ 确定。若约束个数等于选择变量的个数, 通常至多存在有限个解。因此, 约束的个数通常小于选择变量的个数。

接下来一节给出一个引导性例子, 展示拉格朗日方法与消费者选择问题中熟悉的一阶条件(边际替代率等于价格比)之间的关系。第 12.2 节讨论了二变量单约束的情形。第 12.4 节讨论了 n 变量单约束的情形。出现在拉格朗日函数定义中的拉格朗日乘数具有重要的经济解释: 它提供了约束价值的一个度量。具体而言, 它度量了约束的微小变化所导致的目标函数值的边际变化。这在第 12.5 节中进行了考察。在无约束最优化中, 当目标函数具有特定形状(特别是凹性或凸性)时, 全局最优的充分条件的识别得以简化。类似地, 在约束最优化中, 在第 12.6 节讨论的适当条件下, 局部最优条件是全局最优的充分条件。

12.1.1 A Motivating Example / 引导性例子

Original (English)

The optimization techniques described next are based on the use of Lagrangian functions. The discussion here shows how that formulation may be related to optimization ideas introduced earlier. Consider the problem of maximizing $u(y, l)$ where y is income and l is leisure. Work is time not used for leisure, $24 - l$, and income is given by the wage w times the amount of time spent working, so income and leisure are connected according to the equation or constraint $y = w(24 - l)$. Assume that the marginal benefit of both income and leisure is positive: $u_y > 0$ and $u_l > 0$. Maximizing $u(y, l)$ subject to the constraint $y = w(24 - l)$, is an equality-constrained optimization problem.

The condition $y = w(24 - l)$ implies that there is only one variable to be chosen – once l is chosen, the value of y is determined. Using this fact, y can be eliminated from the problem – the constraint can be substituted into the objective to give $\hat{u}(l) = u(w(24 - l), l)$ and the function $\hat{u}$ can be maximized with respect to l as an unconstrained problem. Differentiating and setting to 0:

$$\hat{u}' = 0 \Rightarrow -u_y w + u_l = 0 \Rightarrow \frac{u_l}{u_y} = w.$$

For example, if $u(y, l) = \ln y + \ln l$, then $\hat{u}(l) = \ln(w(24 - l)) + \ln(l)$. Differentiating $\hat{u}(l)$ and setting the derivative to 0:

$$\hat{u}'(l) = 0 = \frac{-w}{w(24 - l)} + \frac{1}{l} \Rightarrow -l + 24 - l = 0 = 24 - 2l.$$

This has solution $l = \frac{24}{2} = 12$, $y = w(24 - 12) = 12w$. The second-order condition for a maximum is easy to check: $\hat{u}''(l) = -\frac{1}{(24-l)^2} - \frac{1}{l^2} < 0$. Thus, using the constraint to solve for one variable in terms of the other leads to an unconstrained problem in one variable that can be solved as the unconstrained maximization of a function of one variable.

An alternative approach to constrained optimization is to define a function called a Lagrangian. The Lagrangian is denoted $\mathcal{L}$. It is the original function plus the constraint multiplied by a “Lagrange multiplier”. The arguments of the function $\mathcal{L}$ are the choice variables and a Lagrange multiplier, λ . In the present example, the Lagrangian is defined:

$$\mathcal{L}(y, l, \lambda) = U(y, l) + \lambda(y - w(24 - l)).$$

This is now viewed as the objective and maximized with respect to all variables, including the Lagrangian multiplier.

$$\mathcal{L}_y = u_y + \lambda = 0 \tag{12.1}$$

$$\mathcal{L}_l = u_l + \lambda w = 0 \tag{12.2}$$

$$\mathcal{L}_\lambda = y - w(24 - l) = 0. \tag{12.3}$$

The first two conditions give $\frac{u_l}{u_y} = w$, and this condition imposes the tangency of the indifference curve to the income-leisure constraint. The third equation gives $y - w(24 - l) = 0$. Solving these determines the optimal values for l and y .

To illustrate, consider the case where $u(l, y) = \ln l + \ln y$, so that $u_y = \frac{1}{y}$, $u_l = \frac{1}{l}$, and so $w = \frac{u_l}{u_y} = \frac{1/l}{1/y} = \frac{y}{l}$, i.e. $wl = y$. Since $y = w(24 - l) = wl$, this gives $wl = 24w - lw$ or $24w = 2lw$. Therefore $l = 12$ and $y = wl = 12w$. This is the same solution as was obtained by substituting the constraint into the objective function. From Equation 12.2, $u_l + \lambda w = 0$. This equation determines λ : $\lambda = -\frac{u_l}{w} = -\frac{1}{wl} = -\frac{1}{12w}$.

翻译 (Chinese)

接下来描述的最优化技术基于拉格朗日函数的使用。考虑在约束 $y = w(24 - l)$ 下最大化 $u(y, l)$ 的问题（收入与闲暇）。将约束代入目标函数可化为无约束问题： $\hat{u}(l) = u(w(24 - l), l)$ ，求导并令其为零： $\hat{u}' = 0 \Rightarrow -u_y w + u_l = 0 \Rightarrow \frac{u_l}{u_y} = w$ 。

例如，若 $u(y, l) = \ln y + \ln l$ ，则 $\hat{u}(l) = \ln(w(24 - l)) + \ln(l)$ ，令导数为零得 $l = 12$, $y = 12w$ 。另一种方法是定义拉格朗日函数：

$$\mathcal{L}(y, l, \lambda) = U(y, l) + \lambda(y - w(24 - l)).$$

将其视为目标函数，对所有变量（包括拉格朗日乘数）求最大化：

$$\mathcal{L}_y = u_y + \lambda = 0 \quad (6.3)$$

$$\mathcal{L}_l = u_l + \lambda w = 0 \quad (6.4)$$

$$\mathcal{L}_\lambda = y - w(24 - l) = 0. \quad (6.5)$$

前两个条件给出 $\frac{u_l}{u_y} = w$ ，即无差异曲线与收入-闲暇约束的切点条件。解之得 $l = 12$, $y = 12w$ ，与代入法结果相同。 $\lambda = -\frac{u_l}{w} = -\frac{1}{12w}$ 。

6.16.2 12.2 The Two-Variable Case / 两变量情形

Original (English)

The following discussion considers in general terms the case where there are two choice variables and one equality constraint. Let the objective function be $f(x, y)$ and let the constraint be $c = g(x, y)$. Throughout, assume that $(x, y) \in Z$ where Z is an open set in $\mathbb{R}^2$. Thus, if (x^*, y^*) are in Z , all points (x, y) sufficiently close to (x^*, y^*) are in Z . Furthermore, f and g are assumed to be

continuously differentiable. The task is to maximize or minimize $f(x, y)$ subject to (x, y) , satisfying the constraint $c = g(x, y)$.

Definition 12.1: A point $(\bar{x}, \bar{y})$ is a constrained local maximum if $g(\bar{x}, \bar{y}) = 0$ and $f(\bar{x}, \bar{y}) \geq f(x, y)$ for all (x, y) in a neighborhood of $(\bar{x}, \bar{y})$ and satisfying $g(x, y) = 0$. A constrained local minimum is similarly defined.

To identify necessary and sufficient conditions for $(\bar{x}, \bar{y})$ to be a constrained local maximum or minimum it is useful to formulate the Lagrangian, $\mathcal{L}$:

$$\mathcal{L}(x, y, \lambda) = f(x, y) + \lambda[c - g(x, y)]. \quad (12.4)$$

In addition, define the bordered Hessian, $\bar{H}$:

$$\bar{H} = \begin{pmatrix} 0 & g_x & g_y \\ g_x & f_{xx} - \lambda g_{xx} & f_{xy} - \lambda g_{xy} \\ g_y & f_{xy} - \lambda g_{xy} & f_{yy} - \lambda g_{yy} \end{pmatrix}. \quad (12.5)$$

With this notation:

Theorem 12.1: Consider the problem of maximizing or minimizing the function $f(x, y)$ subject to a constraint $g(x, y) = c$. Let $\mathcal{L}(x, y, \lambda) = f(x, y, \lambda) + \lambda[c - g(x, y)]$. Suppose $(\bar{x}, \bar{y}, \bar{\lambda})$ satisfies:

$$\mathcal{L}_x = f_x(\bar{x}, \bar{y}) - \bar{\lambda}g_x(\bar{x}, \bar{y}) = 0 \quad (12.6)$$

$$\mathcal{L}_y = f_y(\bar{x}, \bar{y}) - \bar{\lambda}g_y(\bar{x}, \bar{y}) = 0 \quad (12.7)$$

$$\mathcal{L}_\lambda = c - g(\bar{x}, \bar{y}) = 0. \quad (12.8)$$

Then,

- if $|\bar{H}| > 0$, $(\bar{x}, \bar{y})$ is a constrained local maximum;
- if $|\bar{H}| < 0$, $(\bar{x}, \bar{y})$ is a constrained local minimum.

Assuming the derivatives are non-zero, conditions 12.6 and 12.7 imply that

$$\frac{f_x(\bar{x}, \bar{y})}{f_y(\bar{x}, \bar{y})} = \frac{g_x(\bar{x}, \bar{y})}{g_y(\bar{x}, \bar{y})}.$$

This, along with the constraint, $c - g(\bar{x}, \bar{y}) = 0$, yields two equations in the two unknowns, $(\bar{x}, \bar{y})$. These may be solved for $(\bar{x}, \bar{y})$ and $\bar{\lambda}$ solved residually from Equations 12.6 or 12.7.

翻译 (Chinese)

以下讨论一般情况下两选择变量和一个等式约束的情形。设目标函数为 $f(x, y)$ ，约束为

$c = g(x, y)$ 。假设 f 和 g 是连续可微的。任务是在满足约束 $c = g(x, y)$ 的条件下最大化或最小化 $f(x, y)$ 。

定义 12.1: 点 $(\bar{x}, \bar{y})$ 是约束局部极大值, 若 $g(\bar{x}, \bar{y}) = 0$ 且对邻域内所有满足 $g(x, y) = 0$ 的 (x, y) 有 $f(\bar{x}, \bar{y}) \geq f(x, y)$ 。约束局部极小值类似定义。

为找到必要条件与充分条件, 构造拉格朗日函数 $\mathcal{L}$:

$$\mathcal{L}(x, y, \lambda) = f(x, y) + \lambda[c - g(x, y)]. \quad (12.4)$$

并定义加边 Hessian 矩阵 $\bar{H}$:

$$\bar{H} = \begin{pmatrix} 0 & g_x & g_y \\ g_x & f_{xx} - \lambda g_{xx} & f_{xy} - \lambda g_{xy} \\ g_y & f_{xy} - \lambda g_{xy} & f_{yy} - \lambda g_{yy} \end{pmatrix}. \quad (12.5)$$

定理 12.1: 考虑在约束 $g(x, y) = c$ 下最大化或最小化 $f(x, y)$ 的问题。设 $(\bar{x}, \bar{y}, \bar{\lambda})$ 满足一阶条件 $\mathcal{L}_x = 0, \mathcal{L}_y = 0, \mathcal{L}_\lambda = 0$ 。则:

- 若 $|\bar{H}| > 0$, $(\bar{x}, \bar{y})$ 是约束局部极大值;
- 若 $|\bar{H}| < 0$, $(\bar{x}, \bar{y})$ 是约束局部极小值。

在导数非零的假设下, 条件 (12.6) 和 (12.7) 意味着 $\frac{f_x}{f_y} = \frac{g_x}{g_y}$ 。这与约束 $c - g(x, y) = 0$ 一起给出关于 $(\bar{x}, \bar{y})$ 的两个方程。

Original (English)

REMARK 12.2: Theorem 12.1 gives sufficient conditions for a local maximum. But a local (or global) maximum may fail to satisfy these conditions. Suppose that $f(x, y) = xy$ and $g(x, y) = (x + y - 1)^2$. Consider the problem of maximizing f subject to the constraint $g(x, y) = 0$. The constraint is satisfied if and only if $x + y - 1 = 0$ or $x + y = 1$ or $y = 1 - x$. Substituting $y = 1 - x$ into the objective and maximizing $\hat{f}(x) = x(1 - x)$ gives $\hat{f}'(x) = 1 - 2x$ and $\hat{f}''(x) = -2$ so that the local and global maxima are found at $x = y = \frac{1}{2}$. But observe that $g_x(x, y) = g_y(x, y) = (x + y - 1)$ so that at the solution $(\bar{x}, \bar{y}) = (\frac{1}{2}, \frac{1}{2})$, $g_x(\bar{x}, \bar{y}) = g_y(\bar{x}, \bar{y}) = 0$. However, $(f_x, f_y) = (y, x)$ so that $(f_x(\bar{x}, \bar{y}), f_y(\bar{x}, \bar{y})) = (\frac{1}{2}, \frac{1}{2})$. So, for example, in Theorem 12.1, the condition $f_x(\bar{x}, \bar{y}) = \lambda g_x(\bar{x}, \bar{y})$ fails, although $(\bar{x}, \bar{y})$ is the global optimum. Because $(g_x(\bar{x}, \bar{y}), g_y(\bar{x}, \bar{y})) = (0, 0)$, it is impossible to write $f_x(\bar{x}, \bar{y}) = \lambda g_x(\bar{x}, \bar{y})$.

The issue here is that at the solution, $(g_x(\bar{x}, \bar{y}), g_y(\bar{x}, \bar{y})) = (0, 0)$. Theorem 12.2 confirms this observation.

Theorem 12.2: Let f and g be continuously differentiable functions on an open set Z in $\mathbb{R}^2$. Suppose that $(\bar{x}, \bar{y})$ is a constrained local maximum or minimum of f , subject to the constraint $g(x, y) = c$. Suppose that $(g_x(\bar{x}, \bar{y}), g_y(\bar{x}, \bar{y})) \neq (0, 0)$. Then there is a λ such that

$$\nabla f(\bar{x}, \bar{y}) = \lambda \nabla g(\bar{x}, \bar{y}).$$

Therefore, when $(g_x(\bar{x}, \bar{y}), g_y(\bar{x}, \bar{y})) \neq (0, 0)$, the conditions $f_x = \lambda g_x$, $f_y = \lambda g_y$ and $g = 0$ give three equations in three unknowns, $(\bar{x}, \bar{y}, \lambda)$, which may be solved to identify a candidate solution to the optimization problem.

翻译 (Chinese)

注记 12.2: 定理 12.1 给出了局部极大值的充分条件。但局部（或全局）极大值可能不满足这些条件。设 $f(x, y) = xy$ 且 $g(x, y) = (x + y - 1)^2$ 。在约束 $g(x, y) = 0$ （即 $x + y = 1$ ）下最大化 f ，代入得 $\hat{f}(x) = x(1 - x)$ ，解之得 $x = y = \frac{1}{2}$ 。但在该解处， $g_x = g_y = 0$ ，故 $f_x = \lambda g_x$ 的条件失效。

定理 12.2: 设 f 和 g 是 $\mathbb{R}^2$ 中开集 Z 上的连续可微函数。若 $(\bar{x}, \bar{y})$ 是约束局部极值且 $(g_x(\bar{x}, \bar{y}), g_y(\bar{x}, \bar{y})) \neq (0, 0)$ ，则存在 λ 使得 $\nabla f(\bar{x}, \bar{y}) = \lambda \nabla g(\bar{x}, \bar{y})$ 。

6.16.3 12.3 Motivation for First- and Second-Order Conditions / 一阶与二阶条件的动机

12.3.1 The First-Order Condition / 一阶条件

Original (English)

Assume that we can write y as a function of x : $y = h(x)$. Then, by substituting for y the problem reduces to $\max_x f(x, h(x))$. The original problem with two choice variables and a constraint has been reduced to a one-variable unconstrained problem – maximizing the function $\hat{f}(x) \stackrel{\text{def}}{=} f(x, h(x))$. The following calculations show that the solution from $\max_x f(x, h(x))$ is the same as that obtained from the Lagrangian method. Maximizing $f(x, h(x))$ with respect to x , the first-order condition is

$$0 = \frac{d\hat{f}}{dx} = f_x + f_y h', \quad \text{where } h' = \frac{dy}{dx} = -\frac{g_x}{g_y}.$$

Thus,

$$0 = f_x + f_y \left(-\frac{g_x}{g_y} \right), \quad \text{so that } \frac{f_x}{g_x} = \frac{f_y}{g_y}.$$

This condition and $c = g(x, y)$ give the same conditions as obtained from the Lagrangian first-order conditions.

翻译 (Chinese)

假设可将 y 写成 x 的函数 $y = h(x)$ 。代入后问题简化为 $\max_x f(x, h(x))$ 。计算表明由 $\max_x f(x, h(x))$ 得到的解与拉格朗日方法得到的解相同。最大化 $f(x, h(x))$ 的一阶条件为 $0 = f_x + f_y h'$, 其中 $h' = -\frac{g_x}{g_y}$ 。故 $\frac{f_x}{g_x} = \frac{f_y}{g_y}$ 。此条件与 $c = g(x, y)$ 给出了与拉格朗日一阶条件相同的条件。

12.3.2 The Second-Order Condition / 二阶条件**Original (English)**

Maximizing the function $\hat{f}(x) = f(x, h(x))$ gives a first-order condition for an optimum. The second-order condition, $\hat{f}''(x) < 0$ for a maximum or $\hat{f}''(x) > 0$ for a minimum, may also be related to the bordered Hessian condition. Evaluating the second derivative:

$$\frac{d^2 \hat{f}}{dx^2} = f_{xx} + 2f_{xy}h' + f_{yy}(h')^2 + f_y h''.$$

After substantial algebraic manipulation (using $h' = -g_x/g_y$ and $\lambda = f_y/g_y$), one obtains:

$$\frac{d^2 \hat{f}}{dx^2} \cdot (g_y^2) = -\{(f_{xx} - \lambda g_{xx})g_y^2 - 2(f_{xy} - \lambda g_{xy})g_x g_y + (f_{yy} - \lambda g_{yy})g_x^2\}.$$

Now, consider the determinant $|\bar{H}|$:

$$|\bar{H}| = -\{(f_{xx} - \lambda g_{xx})g_y^2 + 2(f_{xy} - \lambda g_{xy})g_x g_y + (f_{yy} - \lambda g_{yy})g_x^2\} = -\frac{d^2 \hat{f}}{dx^2} \cdot (g_y^2).$$

Therefore, $|\bar{H}|$ has the opposite sign to $\frac{d^2 \hat{f}}{dx^2}$. For a maximum, the condition $\frac{d^2 \hat{f}}{dx^2} < 0$ is sufficient and this is equivalent to $|\bar{H}| > 0$. For a minimum, $|\bar{H}| < 0$.

Finally, since $\mathcal{L}_{xx} = f_{xx} - \lambda g_{xx}$, etc., the determinant may be written succinctly:

$$|\bar{H}| = \begin{vmatrix} 0 & g_x & g_y \\ g_x & \mathcal{L}_{xx} & \mathcal{L}_{xy} \\ g_y & \mathcal{L}_{yx} & \mathcal{L}_{yy} \end{vmatrix}.$$

翻译 (Chinese)

最大化 $\hat{f}(x) = f(x, h(x))$ 的二阶条件 $\hat{f}''(x) < 0$ (极大值) 或 $\hat{f}''(x) > 0$ (极小值) 可与加边 Hessian 条件建立联系。计算二阶导数并经过代数运算得到:

$$\frac{d^2 \hat{f}}{dx^2} \cdot (g_y^2) = -\{(f_{xx} - \lambda g_{xx})g_y^2 - 2(f_{xy} - \lambda g_{xy})g_x g_y + (f_{yy} - \lambda g_{yy})g_x^2\}.$$

而行列式 $|\bar{H}|$ 恰好等于 $-\frac{d^2\hat{f}}{dx^2} \cdot (g_y^2)$ 。因此 $|\bar{H}|$ 与 $\frac{d^2\hat{f}}{dx^2}$ 反号。对极大值, $\frac{d^2\hat{f}}{dx^2} < 0$ 等价于 $|\bar{H}| > 0$; 对极小值, $|\bar{H}| < 0$ 。用 $\mathcal{L}$ 表示, 可简写为:

$$|\bar{H}| = \begin{vmatrix} 0 & g_x & g_y \\ g_x & \mathcal{L}_{xx} & \mathcal{L}_{xy} \\ g_y & \mathcal{L}_{yx} & \mathcal{L}_{yy} \end{vmatrix}.$$

12.3.3 A Geometric Perspective / 几何视角

Original (English)

With just one constraint, g , and one objective, f , constrained optimization involves locating a point where the objective is tangent to the constraint and then confirming that this is an optimum. The perpendicular to a curve $f(x, y) = c$ is proportional to the vector of partial derivatives, $(f_x(x, y), f_y(x, y))$. Furthermore, this vector points in the direction of greatest local increase in the function and therefore has a direct connection to the optimization problem. The slope of the level surface is $-\frac{f_x}{f_y}$. The line perpendicular to the tangent to the level surface is proportional to $(f_x(x^*, y^*), f_y(x^*, y^*))$.

In the economic context of utility maximization where $u(x, y) = x^\alpha y^\beta$, the gradient $\nabla u(x, y) = (\frac{\alpha u}{x}, \frac{\beta u}{y})$ is perpendicular to the level surface (indifference curve). The constraint has the form $g(x, y) = px + qy = c$, so $\nabla g(x, y) = (p, q)$ is constant. At the optimum, $\nabla u(z^*) = \lambda \nabla g(z^*)$, where the gradient of utility is proportional to the price vector – this is the first-order condition.

翻译 (Chinese)

在只有一个约束 g 和一个目标函数 f 的情形中, 约束最优化涉及找到一个点, 在该点处目标函数与约束相切, 然后确认这是最优解。曲线 $f(x, y) = c$ 的垂线方向与偏导数向量 (f_x, f_y) 成比例。该向量指向函数局部增加最快的方向。在效用最大化的经济语境中, 无差异曲线的梯度 $\nabla u(x, y)$ 垂直于无差异曲线。预算约束为 $g(x, y) = px + qy = c$, 故 $\nabla g = (p, q)$ 为常数。在最优点处, $\nabla u = \lambda \nabla g$, 即效用梯度与价格向量成比例——这是一阶条件。

6.16.4 12.4 The n -Variable Case / n 变量情形

Original (English)

This section considers the problem of maximizing or minimizing a function $f(x_1, x_2, \dots, x_n)$ subject to the constraint $g(x_1, x_2, \dots, x_n) = c$. The procedure is similar to the two-variable case: obtain

first-order conditions relating the partial derivatives of the objective to those of the constraint and then confirm that the solution to these equations actually is a local maximum or local minimum. The generalization of the Lagrangian to n variables is straightforward. Let

$$\mathcal{L} = f(x_1, x_2, \dots, x_n) + \lambda(c - g(x_1, \dots, x_n)).$$

The first-order conditions are:

$$\frac{\partial \mathcal{L}}{\partial x_i} = \mathcal{L}_i = f_i(x) - \lambda g_i(x) = 0, \quad \frac{\partial \mathcal{L}}{\partial \lambda} = \mathcal{L}_\lambda = c - g(x) = 0.$$

Let $(\bar{x}, \bar{\lambda})$ be a solution to these $n + 1$ equations.

翻译 (Chinese)

本节考虑在约束 $g(x_1, \dots, x_n) = c$ 下最大化或最小化 $f(x_1, \dots, x_n)$ 的问题。过程与两变量情形类似：首先得到联系目标函数偏导数和约束偏导数的一阶条件，然后确认这些方程的解确实是局部极大值或极小值。拉格朗日函数向 n 个变量的推广是直接的：

$$\mathcal{L} = f(x_1, \dots, x_n) + \lambda(c - g(x_1, \dots, x_n)).$$

一阶条件为 $\mathcal{L}_i = f_i(x) - \lambda g_i(x) = 0$ 和 $\mathcal{L}_\lambda = c - g(x) = 0$ 。设 $(\bar{x}, \bar{\lambda})$ 为这 $n + 1$ 个方程的解。

12.4.1 Quadratic Forms and Bordered Matrices / 二次型与加边矩阵

Original (English)

A matrix A determines a quadratic form according to the expression $x'Ax$. The matrix A is said to be negative or positive definite if $x'Ax < 0$ or $x'Ax > 0$ for all non-zero vectors x . In constrained optimization, there is a comparable criterion based on bordered matrices.

Theorem 12.3: Let A be an $n \times n$ symmetric matrix and let b be an $n \times 1$ vector with $b \neq 0$.

Also, let $C = C_n = \begin{pmatrix} 0 & b' \\ b & A \end{pmatrix}$ and

$$C_j = \begin{pmatrix} 0 & b_1 & b_2 & \cdots & b_j \\ b_1 & a_{11} & a_{12} & \cdots & a_{1j} \\ b_2 & a_{21} & a_{22} & \cdots & a_{2j} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ b_j & a_{j1} & a_{j2} & \cdots & a_{jj} \end{pmatrix}.$$

Then

- $\xi' A \xi < 0$, for all $\xi \neq 0$ with $b' \xi = 0$, if and only if $(-1)^j |C_j| > 0$, for all $j = 2, 3, \dots, n$.
- $\xi' A \xi > 0$ for all $\xi \neq 0$ with $b' \xi = 0$, if and only if $|C_j| < 0$, for all $j = 2, 3, \dots, n$.

翻译 (Chinese)

矩阵 A 通过表达式 $x' A x$ 确定二次型。在约束最优化中，存在基于加边矩阵的类似判据。

定理 12.3: 设 A 为 $n \times n$ 对称矩阵， b 为 $n \times 1$ 向量且 $b \neq 0$ 。令 C_j 为加边矩阵：

$$C_j = \begin{pmatrix} 0 & b_1 & \cdots & b_j \\ b_1 & a_{11} & \cdots & a_{1j} \\ \vdots & \vdots & \ddots & \vdots \\ b_j & a_{j1} & \cdots & a_{jj} \end{pmatrix}.$$

则对所有 $\xi \neq 0$ 且 $b' \xi = 0$ 有 $\xi' A \xi < 0$ 当且仅当 $(-1)^j |C_j| > 0$, $j = 2, \dots, n$ 。类似地， $\xi' A \xi > 0$ 当且仅当 $|C_j| < 0$, $j = 2, \dots, n$ 。

12.4.2 Bordered Hessians and Optimization / 加边 Hessian 与最优化

Original (English)

In the present context of n -variable optimization, $(x_1, \dots, x_n)$ are chosen to maximize the objective $f(x_1, \dots, x_n)$, subject to the constraint $g(x_1, \dots, x_n) = c$. Define the bordered Hessian matrices:

$$\bar{H} = \begin{pmatrix} 0 & g_1 & g_2 & \cdots & g_n \\ g_1 & \mathcal{L}_{11} & \mathcal{L}_{12} & \cdots & \mathcal{L}_{1n} \\ g_2 & \mathcal{L}_{21} & \mathcal{L}_{22} & \cdots & \mathcal{L}_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ g_n & \mathcal{L}_{n1} & \mathcal{L}_{n2} & \cdots & \mathcal{L}_{nn} \end{pmatrix}, \quad \bar{H}_j = \begin{pmatrix} 0 & g_1 & g_2 & \cdots & g_j \\ g_1 & \mathcal{L}_{11} & \mathcal{L}_{12} & \cdots & \mathcal{L}_{1j} \\ g_2 & \mathcal{L}_{21} & \mathcal{L}_{22} & \cdots & \mathcal{L}_{2j} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ g_j & \mathcal{L}_{j1} & \mathcal{L}_{j2} & \cdots & \mathcal{L}_{jj} \end{pmatrix}.$$

Theorem 12.4: Let $\mathcal{L}(x_1, \dots, x_n, \lambda) = f(x_1, \dots, x_n) + \lambda(c - g(x_1, \dots, x_n))$ and suppose that at $(\bar{x}, \bar{\lambda})$ the following are satisfied:

$$\frac{\partial \mathcal{L}}{\partial x_i} = \mathcal{L}_i = 0, \quad i = 1, \dots, n, \quad \frac{\partial \mathcal{L}}{\partial \lambda} = \mathcal{L}_\lambda = 0.$$

If, in addition, the bordered Hessian $\bar{H}$ satisfies (1) at $\bar{x}$, then $\bar{x}$ is a local maximum; and if $\bar{H}$ satisfies (2) at $\bar{x}$, then $\bar{x}$ is a local minimum, where

1. $|\bar{H}_2| > 0, |\bar{H}_3| < 0, |\bar{H}_4| > 0, \dots, (-1)^n |\bar{H}_n| > 0$
2. $|\bar{H}_2| < 0, |\bar{H}_3| < 0, \dots, |\bar{H}_n| < 0$.

翻译 (Chinese)

在 n 变量最优化的语境中, 定义加边 Hessian 矩阵 $\bar{H}$ 和 $\bar{H}_j$ (如上所示)。

定理 12.4: 设 $(\bar{x}, \bar{\lambda})$ 满足一阶条件 $\mathcal{L}_i = 0, i = 1, \dots, n$ 和 $\mathcal{L}_\lambda = 0$ 。若加边 Hessian $\bar{H}$ 在 $\bar{x}$ 处满足:

1. $|\bar{H}_2| > 0, |\bar{H}_3| < 0, |\bar{H}_4| > 0, \dots, (-1)^n |\bar{H}_n| > 0$, 则 $\bar{x}$ 是局部极大值;
2. $|\bar{H}_2| < 0, |\bar{H}_3| < 0, \dots, |\bar{H}_n| < 0$, 则 $\bar{x}$ 是局部极小值。

Original (English)

Theorem 12.5: Let f and g be continuously differentiable functions on an open set Z in $\mathbb{R}^n$. Suppose that $(\bar{x})$ is a constrained local maximum or minimum of f , subject to the constraint $g(x) = c$. Suppose that $\nabla g(\bar{x}) \neq 0$. Then there is a λ such that

$$\nabla f(\bar{x}) = \lambda \nabla g(\bar{x}).$$

The sign of the Lagrange multiplier: Sometimes it is useful to know the sign of the Lagrangian multiplier at the solution. However, with equality-constrained optimization, the multiplier may be positive or negative, depending on the way the constraint is expressed. When ∇f and ∇g “point” in the same direction, then for some $\lambda > 0$, $\nabla f = \lambda \nabla g$. When ∇f and ∇g “point” in opposite directions, then for some $\lambda < 0$, $\nabla f = \lambda \nabla g$. The sign of the Lagrange multiplier is indeterminate in equality-constrained optimization.

翻译 (Chinese)

定理 12.5: 设 f 和 g 是 $\mathbb{R}^n$ 中开集 Z 上的连续可微函数。若 $\bar{x}$ 是约束局部极值且 $\nabla g(\bar{x}) \neq 0$, 则存在 λ 使得 $\nabla f(\bar{x}) = \lambda \nabla g(\bar{x})$ 。

拉格朗日乘数的符号: 在等式约束最优化中, 乘数可正可负, 取决于约束的表达式。当 ∇f 和 ∇g “指向” 相同方向时, $\lambda > 0$; 指向相反时, $\lambda < 0$ 。因此等式约束最优化中拉格朗日乘数的符号是不确定的。

6.16.5 12.5 Interpretation of the Lagrange Multiplier / 拉格朗日乘数的解释

Original (English)

The Lagrangian is $\mathcal{L} = f(x_1, \dots, x_n) + \lambda(c - g(x_1, \dots, x_n))$. This has first-order conditions:

$$f_i(x_1, \dots, x_n) - \lambda g_i(x_1, \dots, x_n) = 0, \quad c - g(x_1, \dots, x_n) = 0.$$

Let $(\bar{x}, \bar{\lambda})$ be a solution. Note that $\bar{x}$ and $\bar{\lambda}$ depend on c : as c varies they will also. Thus $\bar{x}$ and $\bar{\lambda}$ may be written as functions of c : $\bar{x}(c), \bar{\lambda}(c)$. The value of the objective at c is: $\mathcal{L}^*(c) = f(\bar{x}_1(c), \dots, \bar{x}_n(c)) + \bar{\lambda}(c)[c - g(\bar{x}_1(c), \dots, \bar{x}_n(c))]$. This function may be differentiated with respect to c to give:

$$\frac{d\mathcal{L}^*(c)}{dc} = \sum_{i=1}^n [f_i - \bar{\lambda}(c)g_i] \frac{d\bar{x}_i}{dc} + \frac{d\bar{\lambda}}{dc} [c - g(\bar{x}(c))] + \bar{\lambda}(c) = \bar{\lambda}(c).$$

This follows using the first-order conditions. Thus, $\frac{d\mathcal{L}^*}{dc} = \lambda$, the marginal impact on the value of the objective due to changing the constraint “a little”, is equal to the Lagrange multiplier.

EXAMPLE 12.5 (Utility maximization): Consider the standard utility maximization problem:

$$\max_{x,y} u(x, y) = \ln x + \ln y, \quad \text{subject to} \quad px + qy = I.$$

The Lagrangian is: $\mathcal{L} = \ln x + \ln y + \lambda(I - px - qy)$. First-order conditions give $x = \frac{I}{2p}$, $y = \frac{I}{2q}$, $\lambda = \frac{2}{I}$. The optimized utility is $u(x(I), y(I)) = \ln \frac{I^2}{4pq}$. Differentiating: $\frac{du}{dI} = \frac{2}{I} = \lambda$. Thus the Lagrange multiplier measures the marginal utility of income.

翻译 (Chinese)

拉格朗日函数为 $\mathcal{L} = f(x) + \lambda(c - g(x))$ 。设 $(\bar{x}, \bar{\lambda})$ 为解。 $\bar{x}$ 和 $\bar{\lambda}$ 依赖于 c 。求导得：

$$\frac{d\mathcal{L}^*(c)}{dc} = \sum_{i=1}^n [f_i - \bar{\lambda}g_i] \frac{d\bar{x}_i}{dc} + \frac{d\bar{\lambda}}{dc} [c - g] + \bar{\lambda} = \bar{\lambda}.$$

因此 $\frac{d\mathcal{L}^*}{dc} = \lambda$ ：目标函数值因约束“微小”变化而产生的边际影响等于拉格朗日乘数。

例 12.5 (效用最大化)： $\max \ln x + \ln y$, s.t. $px + qy = I$ 。解之得 $x = \frac{I}{2p}$, $y = \frac{I}{2q}$, $\lambda = \frac{2}{I}$ 。最优效用为 $u = \ln \frac{I^2}{4pq}$ ，故 $\frac{du}{dI} = \frac{2}{I} = \lambda$ 。因此拉格朗日乘数度量了收入的边际效用。

6.16.6 12.6 Optimization: Concavity and Convexity / 最优化：凹性与凸性

Original (English)

In a constrained optimization problem, the shape of the objective function and the constraints play an important role in characterizing global optima – just as in the unconstrained case. For this discussion, assume that the domain of the function is convex.

Definition 12.2: A set X is convex if $x, y \in X \Rightarrow \theta x + (1 - \theta)y \in X, \forall \theta \in [0, 1]$.

Definition 12.3: Concavity and quasiconcavity.

1. A function f is concave if $f(\theta x + (1 - \theta)y) \geq \theta f(x) + (1 - \theta)f(y)$, $\forall x, y \in X$, $\forall \theta \in [0, 1]$.
2. A function f is strictly concave if $f(\theta x + (1 - \theta)y) > \theta f(x) + (1 - \theta)f(y)$, $x \neq y$, $\forall \theta \in (0, 1)$.
3. A function f is quasiconcave if $f(\theta x + (1 - \theta)y) \geq \min\{f(x), f(y)\}$, $\forall x, y \in X$, $\forall \theta \in [0, 1]$.
4. A function f is strictly quasiconcave if $f(\theta x + (1 - \theta)y) > \min\{f(x), f(y)\}$, $x \neq y$, $\forall \theta \in (0, 1)$.
5. A function f is explicitly quasiconcave if it is quasiconcave and $f(x) > f(y) \Rightarrow f(\theta x + (1 - \theta)y) > f(y)$, $\forall \theta \in (0, 1)$.

PROPOSITION 12.1: If f is (strictly) concave then f is (strictly) quasiconcave. **PROPOSITION 12.2:** If f is strictly quasiconcave then f is explicitly quasiconcave.

As discussed earlier, if f is concave then the Hessian matrix $H(x)$ is negative semi-definite for all x . In fact, f is concave if and only if $H(x)$ is negative semi-definite for all x .

Theorem 12.6: Let $\mathcal{L}(x_1, \dots, x_n, \lambda) = f(x_1, \dots, x_n) + \lambda(c - g(x_1, \dots, x_n))$, where

- (a) f is explicitly quasiconcave
- (b) $\{x \mid g(x) = c\}$ is convex.

If $(\bar{x}, \bar{\lambda})$ satisfy,

1. $\mathcal{L}_i(\bar{x}, \bar{\lambda}) = 0$, $i = 1, \dots, n$, $\mathcal{L}_\lambda(\bar{x}, \bar{\lambda}) = 0$
2. $|\bar{H}_2(\bar{x})| > 0$, $|\bar{H}_3(\bar{x})| < 0$, $\dots$, $(-1)^n |\bar{H}_n(\bar{x})| > 0$,

then $\bar{x}$ is a global maximum of the constrained program.

翻译 (Chinese)

在约束最优化问题中，目标函数和约束的形状对刻画全局最优起着重要作用——正如无约束情形一样。

定义 12.2: 集合 X 是凸的，若 $x, y \in X \Rightarrow \theta x + (1 - \theta)y \in X$, $\forall \theta \in [0, 1]$ 。

定义 12.3: 凹性与拟凹性。

1. f 是凹的，若 $f(\theta x + (1 - \theta)y) \geq \theta f(x) + (1 - \theta)f(y)$ 。
2. f 是严格凹的，若 $f(\theta x + (1 - \theta)y) > \theta f(x) + (1 - \theta)f(y)$, $x \neq y$ 。
3. f 是拟凹的，若 $f(\theta x + (1 - \theta)y) \geq \min\{f(x), f(y)\}$ 。
4. f 是严格拟凹的，若 $f(\theta x + (1 - \theta)y) > \min\{f(x), f(y)\}$, $x \neq y$ 。
5. f 是显式拟凹的，若它是拟凹的且 $f(x) > f(y) \Rightarrow f(\theta x + (1 - \theta)y) > f(y)$ 。

命题 12.1: 若 f (严格) 凹，则 f (严格) 拟凹。**命题 12.2:** 若 f 严格拟凹，则 f 显式拟凹。

定理 12.6: 若 f 是显式拟凹的, $\{x \mid g(x) = c\}$ 是凸的, 且 $(\bar{x}, \bar{\lambda})$ 满足一阶和二阶条件 (加边 Hessian 行列式符号交替), 则 $\bar{x}$ 是约束规划的全局极大值。

12.6.1 Quasiconcavity and Bordered Hessians / 拟凹性与加边 Hessian

Original (English)

Just as there is a close connection between concavity and the Hessian matrix, there is a similar connection for quasiconcavity, except that the Hessian matrix is replaced by a bordered Hessian matrix. Augment the Hessian matrix with a border of first derivatives as follows. Let

$$Q = \begin{pmatrix} 0 & f_1 & f_2 & \cdots & f_n \\ f_1 & f_{11} & f_{12} & \cdots & f_{1n} \\ f_2 & f_{21} & f_{22} & \cdots & f_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ f_n & f_{n1} & f_{n2} & \cdots & f_{nn} \end{pmatrix}, \quad Q_j = \begin{pmatrix} 0 & f_1 & f_2 & \cdots & f_j \\ f_1 & f_{11} & f_{12} & \cdots & f_{1j} \\ f_2 & f_{21} & f_{22} & \cdots & f_{2j} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ f_j & f_{j1} & f_{j2} & \cdots & f_{jj} \end{pmatrix}.$$

Theorem 12.7:

1. If f is quasiconcave then $|Q_2(x)| \leq 0, |Q_3(x)| \geq 0, \dots, (-1)^n |Q_n(x)| \geq 0, \forall x$.
2. If $|Q_1(x)| < 0, |Q_2(x)| > 0, \dots, (-1)^n |Q_n(x)| > 0, \forall x$, then f is quasiconcave.
3. If f is quasiconvex then $|Q_2(x)| \leq 0, |Q_3(x)| \leq 0, \dots, |Q_n(x)| \leq 0, \forall x$.
4. If $|Q_1(x)| < 0, |Q_2(x)| < 0, \dots, |Q_n(x)| < 0, \forall x$, then f is quasiconvex.

Theorem 12.8: If the bordered Hessian satisfies the conditions for f to be quasiconcave, then the bordered Hessian satisfies the sufficient conditions for a local maximum.

翻译 (Chinese)

正如凹性与 Hessian 矩阵之间存在密切联系, 拟凹性也有类似联系, 只是 Hessian 矩阵被替换为加边 Hessian 矩阵。用一阶导数作为边框对 Hessian 矩阵进行增广, 得到矩阵 Q 和 Q_j 。

定理 12.7: 给出了拟凹性/拟凸性与加边 Hessian 行列式符号条件之间的关系。拟凹性要求 $|Q_2| \leq 0, |Q_3| \geq 0, \dots, (-1)^n |Q_n| \geq 0$; 拟凸性要求 $|Q_j| \leq 0, \forall j$ 。

定理 12.8: 若加边 Hessian 满足拟凹性条件, 则该加边 Hessian 满足局部极大值的充分条件。

6.16.7 12.7 Optimization with Many Constraints / 多约束最优化

Original (English)

Let $f(x_1, \dots, x_n)$ be the objective function and $g^k(x_1, \dots, x_n) = c_k$, $k = 1, \dots, m$ a collection of m constraints. Assume that $m < n$. Define the Lagrangian:

$$\mathcal{L} = f(x_1, \dots, x_n) + \lambda_1[c_1 - g^1(x_1, \dots, x_n)] + \dots + \lambda_m[c_m - g^m(x_1, \dots, x_n)].$$

Define the matrix Dg as the $m \times n$ Jacobian of the constraints, and define the bordered matrix C_r (combining the constraint Jacobian with the Hessian of $\mathcal{L}$).

Theorem 12.9 (Quadratic forms with multiple constraints): Suppose that A is symmetric and B_m is non-singular.

1. Then $x'Ax > 0$ for every x with $Bx = 0$, if and only if: $(-1)^m|C_r| < 0$, $r = m+1, \dots, n$.
2. Then $x'Ax < 0$ for every x with $Bx = 0$, if and only if: $(-1)^r|C_r| > 0$, $r = m+1, \dots, n$.

Theorem 12.10: Suppose that $\bar{x}$ is feasible (satisfies all the constraints). If $(\bar{x}, \bar{\lambda}_1, \dots, \bar{\lambda}_m) = (\bar{x}, \bar{\lambda})$ satisfies:

1. $\mathcal{L}_i(\bar{x}, \bar{\lambda}) = 0$, $i = 1, \dots, n$, $\mathcal{L}_{\lambda_k}(\bar{x}, \bar{\lambda}) = 0$, $k = 1, \dots, m$
2. $(-1)^r|C_r| > 0$, $r = m+1, \dots, n$,

then $\bar{x}$ is a constrained local maximum. If condition 2 is replaced by $(-1)^m|C_r| > 0$, $r = m+1, \dots, n$, then $\bar{x}$ is a constrained local minimum.

翻译 (Chinese)

设目标函数为 f , 约束为 $g^k(x) = c_k$, $k = 1, \dots, m$ ($m < n$). 定义拉格朗日函数 $\mathcal{L}$ 包含 m 个拉格朗日乘数。定义约束的雅可比矩阵 Dg 和加边矩阵 C_r 。

定理 12.9: 给出了多约束下二次型正定/负定的行列式条件。

定理 12.10: 若 $(\bar{x}, \bar{\lambda})$ 满足一阶条件且 $(-1)^r|C_r| > 0$, $r = m+1, \dots, n$, 则 $\bar{x}$ 是约束局部极大值。若代之以 $(-1)^m|C_r| > 0$, 则为约束局部极小值。例如, 当 $m = 1$ 时, 这与定理 12.4 的条件一致。

Exercises for Chapter 12 / 第 12 章习题

Original (English)

Exercise 12.1 A firm has the production function $Y = 30L^{\frac{1}{2}}K^{\frac{2}{3}}$. (a) What is the cost-minimizing level of L and K if the firm must produce 1000 units? Check the second-order conditions. (b) By how much will cost increase if the firm must increase output by one unit? (c) Given this production function, can we find a profit-maximizing level of output? Why or why not?

Exercise 12.2 Assume that a monopolist producing one good can sell to two distinct markets and price discriminate. The demand function in each market is

$$D_1 = 300 - \frac{1}{3}P_1, \quad D_2 = 300 - \frac{1}{2}P_2.$$

The total cost to the monopolist is $C = 2D_1^2 + 3D_1D_2 + D_2^2$. Also, owing to a government restriction, the total output of the firm must be 100 units. What is the profit-maximizing output in each market? Check the second-order conditions.

Exercise 12.3 Consider a firm that produces beer and pretzels. It must sell a bag of pretzels for every four bottles of beer. If its profit function is

$$\pi = 150B - 4B^2 + 150P - 3P^2 + 10BP + 100$$

($P = \#$ of bags of pretzels, $B = \#$ beers), what is the profit-maximizing level of output?

Exercise 12.4 The geometric average of two positive numbers, x and y , is $(xy)^{\frac{1}{2}}$. Consider the task of minimizing the geometric average, subject to the condition $x + y = 1$. Show, using the method of Lagrange multipliers, that the optimal choice of (x, y) is $(\frac{1}{2}, \frac{1}{2})$ and confirm that this gives the constrained minimum.

Exercise 12.5 A swimming pool is 25 meters times 100 meters. In coordinates, a position (x, y) gives length and distance traveled from $(0, 0)$. Thus $(15, 20)$ denotes the position of a swimmer who swam 20 meters along the x axis and 15 meters along the y axis from $(0, 0)$. There are four corners $(0, 0)$, $(0, 25)$, $(100, 0)$ and $(100, 25)$. The diagonal line connecting corners $(0, 25)$ and $(100, 0)$ is given by $4y + x = 100$. The distance from corner $(0, 0)$ to coordinate (x, y) is $(x^2 + y^2)^{\frac{1}{2}}$. A swimmer wishes to swim from the $(0, 0)$ corner, and cross the diagonal line in the shortest distance possible. What coordinate position should the person swim to? Use the method of Lagrange multipliers to determine the answer.

Exercise 12.6 A firm wishes to produce a level of output Q at minimum cost. Two inputs, capital (K) and labour (L), are used to produce Q with the production function given by $Q = K^\alpha + L^\beta$, where $0 < \alpha, \beta < 1$. The wage rate is ω and the cost of capital (per unit) is r , so that total cost given (K, L) is $C = rK + \omega L$. (a) Set up the Lagrangian to solve the problem of finding the cost-minimizing levels of K and L required to produce Q . (b) Solve for K , L and the Lagrange multiplier, λ . (c) Check that the second-order conditions for a minimum are satisfied. (d) Observe that in your solution to part (b), both K and L depend on Q . To emphasize this dependency, write

$K(Q)$ and $L(Q)$. Thus the cost of producing the output level Q is $C(Q) = rK(Q) + \omega L(Q)$. Find $dC(Q)/dQ$ and compare your answer with λ .

Exercise 12.7 Dr. Who lives at coordinates $(x, y) = (3, 12)$. A road in the (x, y) plane follows the path: $k = x^2 - 6x + y$, where $0 < k < 3$. On the bus journey home, each traveller must give the driver “drop-off” coordinates. Unfortunately for Dr. Who, the bus does not travel by his house. (a) What drop-off coordinates should Dr. Who give the driver so that Dr. Who is as close to his house as possible? (Hint: minimize $d = (x_1 - x_2)^2 + (y_1 - y_2)^2$, since minimizing $\sqrt{d}$ is equivalent to minimizing d .) (b) Confirm that the second-order conditions are satisfied. (c) The solution depends on k . Let $d(k)$ be the distance at the solution from the drop-off point to Dr. Who’s house. Calculate $\frac{d[d(k)]}{dk}$ and relate it to the Lagrange multiplier in (a).

Exercise 12.8 A firm sets both price and quantity in a market. It is known that the demand function for this market is $P = k/Q$. By law, the firm is required to choose a price-quantity combination lying on the demand curve. The cost of setting (P, Q) combinations is $C(P, Q) = P + Q^2$. (a) Minimize $C(P, Q)$ subject to the demand constraint. (b) Confirm that the second-order conditions are satisfied. (c) The optimal choices of P and Q depend on (α, k) : $P(\alpha, k)$, $Q(\alpha, k)$, so that the cost at the solution is $C^*(\alpha, k) = C(P(\alpha, k), Q(\alpha, k))$. Find $\partial C^*(\alpha, k)/\partial k$ and compare this with the Lagrange multiplier obtained in (a).

Exercise 12.9 Maximize xy subject to the constraint $x^2 + y = k$, where k is some positive constant. Confirm that the second-order conditions for a constrained optimum are satisfied. Interpret the Lagrange multiplier.

Exercise 12.10 Minimize $(x - 1)^2 + y^2$ subject to the constraint $x^2 + k = y$, where k is some positive constant. Confirm that the second-order conditions for a constrained optimum are satisfied. Interpret the Lagrange multiplier.

Exercise 12.11 Consider the following constrained optimization problem: maximize $f(x, y) = y - bx^2$ subject to the constraint: $y = k + a \ln x$. Here k , a and b are positive constants. (a) Plot the problem. (b) Find the solution values for x and y . (c) Confirm that the second-order conditions are satisfied. (d) Let the solution be $x(a, b, k)$ and $y(a, b, k)$. Then the value of the objective at the solution is $f^*(a, b, k) = f(x(a, b, k), y(a, b, k))$. Find $\frac{\partial f^*(a, b, k)}{\partial a}$. Find $\frac{\partial f^*(a, b, k)}{\partial k}$ and interpret this in terms of the Lagrange multiplier.

Exercise 12.12 An individual faces the problem of minimizing expenditure subject to reaching some fixed level of utility $\bar{u}$. There are two goods, x and y , and the utility function is given by $u(x, y) = x^\alpha y^\beta$, $0 < 2\alpha < 1$, $\beta > 0$. The cost of x (per unit) is p and the cost of y (per unit) is q , so the cost of the bundle (x, y) is $px + qy$. (a) Set up the Lagrangian for this problem. Find the optimal values of x and y and the value of the Lagrange multiplier. Check the second-order conditions.

(b) The solution depends on p , q and $\bar{u}$ so write $x(p, q, \bar{u})$, $y(p, q, \bar{u})$, $\lambda(p, q, \bar{u})$. The expenditure required to achieve utility level $\bar{u}$ given prices p and q is therefore $e(p, q, \bar{u}) = px(p, q, \bar{u}) + qy(p, q, \bar{u})$. Show that

$$\frac{\partial e(p, q, \bar{u})}{\partial p} = x(p, q, \bar{u}) \quad \text{and} \quad \frac{\partial e(p, q, \bar{u})}{\partial q} = y(p, q, \bar{u}).$$

翻译 (Chinese)

习题 12.1 某企业的生产函数为 $Y = 30L^{\frac{1}{2}}K^{\frac{2}{3}}$ 。(a) 若企业必须生产 1000 单位, 求成本最小化的 L 和 K 水平, 并检验二阶条件。(b) 若企业必须增加一单位产出, 成本将增加多少?(c) 给定此生产函数, 能否找到利润最大化的产出水平? 为什么?

习题 12.2 假设一家生产单一商品的垄断者可以向两个不同的市场销售并实行价格歧视。每个市场的需求函数为 $D_1 = 300 - \frac{1}{3}P_1$, $D_2 = 300 - \frac{1}{2}P_2$ 。总成本为 $C = 2D_1^2 + 3D_1D_2 + D_2^2$ 。由于政府限制, 企业的总产出必须为 100 单位。求每个市场的利润最大化产出并检验二阶条件。

习题 12.3 一家企业生产啤酒和椒盐脆饼干, 必须每四瓶啤酒搭配一袋脆饼干销售。利润函数为 $\pi = 150B - 4B^2 + 150P - 3P^2 + 10BP + 100$ 。求利润最大化的产出水平。

习题 12.4 两个正数 x 和 y 的几何平均为 $(xy)^{\frac{1}{2}}$ 。在条件 $x + y = 1$ 下最小化几何平均。使用拉格朗日乘数法证明最优选择为 $(\frac{1}{2}, \frac{1}{2})$, 并确认这给出了约束极小值。

习题 12.5 游泳池尺寸为 25 米乘以 100 米。对角线连接 $(0, 25)$ 和 $(100, 0)$, 方程为 $4y + x = 100$ 。游泳者希望从 $(0, 0)$ 角游到对角线的最短距离。使用拉格朗日乘数法确定坐标位置。

习题 12.6 企业用资本 K 和劳动 L 生产 $Q = K^\alpha + L^\beta$ ($0 < \alpha, \beta < 1$)。成本为 $C = rK + \omega L$ 。(a) 建立拉格朗日函数求解成本最小化的 K 和 L 。(b) 求解 K 、 L 和拉格朗日乘数 λ 。(c) 检验极小值的二阶条件。(d) 求 $dC(Q)/dQ$ 并与 λ 比较。

习题 12.7 Dr. Who 的家在 $(3, 12)$ 。道路沿 $k = x^2 - 6x + y$ 延伸。求到家的距离最短的落客坐标。使用拉格朗日乘数法。检验二阶条件。计算 $\frac{d[d(k)]}{dk}$ 并与拉格朗日乘数建立联系。

习题 12.8 市场需求为 $P = k/Q$ 。设定价格-数量组合的成本为 $C(P, Q) = P + Q^2$ 。(a) 在需求约束下最小化 $C(P, Q)$ 。(b) 确认二阶条件。(c) 求 $\partial C^*(\alpha, k)/\partial k$ 并与拉格朗日乘数比较。

习题 12.9 在约束 $x^2 + y = k$ ($k > 0$) 下最大化 xy 。确认约束最优的二阶条件成立。解释拉格朗日乘数。

习题 12.10 在约束 $x^2 + k = y$ ($k > 0$) 下最小化 $(x - 1)^2 + y^2$ 。确认二阶条件。解释拉格朗日乘数。

习题 12.11 在约束 $y = k + a \ln x$ 下最大化 $f(x, y) = y - bx^2$ ($k, a, b > 0$)。 (a) 绘制问题图形。 (b) 求 x 和 y 的解。 (c) 确认二阶条件。 (d) 求 $\frac{\partial f^*(a, b, k)}{\partial a}$ 和 $\frac{\partial f^*(a, b, k)}{\partial k}$, 并用拉格朗日乘数解释。

习题 12.12 在达到固定效用水平 $\bar{u}$ 的约束下最小化支出。效用函数 $u(x, y) = x^\alpha y^\beta$, 成本为 $px + qy$ 。 (a) 建立拉格朗日函数, 求 x 、 y 和乘数的最优值, 检验二阶条件。 (b) 证明 $\frac{\partial e(p, q, \bar{u})}{\partial p} = x(p, q, \bar{u})$ 和 $\frac{\partial e(p, q, \bar{u})}{\partial q} = y(p, q, \bar{u})$ 。

第十三章 Inequality-Constrained Optimization / 不等式约束最优化

13.1 Introduction / 引言

Original (English)

This chapter considers the problem of maximizing or minimizing a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ subject to inequality constraints such as $g(x) \leq c$ or $g(x) \geq c$. This type of problem arises frequently in economics. For example, maximizing output $y = f(x_1, x_2)$ subject to expenditure constraint, $g(x_1, x_2) = w_1x_1 + w_2x_2 \leq c$, is a typical problem.

翻译 (Chinese)

本章研究在不等式约束（如 $g(x) \leq c$ 或 $g(x) \geq c$ ）下最大化或最小化函数 $f : \mathbb{R}^n \rightarrow \mathbb{R}$ 的问题。这类问题在经济学中经常出现。例如，在支出约束 $g(x_1, x_2) = w_1x_1 + w_2x_2 \leq c$ 下最大化产出 $y = f(x_1, x_2)$ ，就是一个典型问题。

Original (English)

In Section 13.2, the constrained optimization is formally defined and the main result (Theorem 13.1) is given. All the results in this chapter are derived from this theorem and the constraint qualification. Section 13.2.1 discusses the constraint qualification, a (mild) condition that the derivatives of the constraints must satisfy to guarantee that the "standard" first-order conditions are necessary for a local optimum. In Section 13.2.2, the complementary slackness condition is discussed: on non-binding constraints we can take the Lagrange multipliers to be 0. Section 13.2.3 discusses the logical equivalence of the maximization and minimization programs. In Section 13.2.4, conditions are given under which a local optimum is also a global optimum. In Section 13.2.5, problems with non-negativity constraints are considered. The necessary conditions for a local optimum may be expressed in terms of the Kuhn–Tucker conditions, and this is done in

Theorems 13.5 and 13.6, which are derived directly from the main result, Theorem 13.1. The remaining sections deal with a few miscellaneous topics.

翻译 (Chinese)

第 13.2 节正式定义了约束最优化问题，并给出了主要结果（定理 13.1）。本章中的所有结果都从该定理和约束资格条件推导而来。第 13.2.1 节讨论了约束资格条件——这是约束的导数必须满足的一个（温和的）条件，用以保证标准一阶条件是局部最优解的必要条件。第 13.2.2 节讨论了互补松弛条件：对于非紧约束，我们可以令拉格朗日乘子为零。第 13.2.3 节讨论了最大化与最小化问题的逻辑等价性。第 13.2.4 节给出了局部最优解也是全局最优解的条件。第 13.2.5 节考虑了带有非负性约束的问题。局部最优解的必要条件可以用 Kuhn–Tucker 条件来表达，这在定理 13.5 和 13.6 中完成，这两个定理直接由定理 13.1 推导而来。其余各节讨论了一些其他主题。

13.1.1 A motivating example / 一个引入性例子

Original (English)

To develop some of the issues, consider the following optimization example.

EXAMPLE 13.1:

$$\max_{x_1, x_2} x_1 x_2$$

subject to:

$$x_1^2 + x_2^2 \leq 1, \quad x_2 \geq \alpha x_1.$$

Thus, the objective is $f(x_1, x_2) = x_1 x_2$, and these constraints may be written $g_1(x_1, x_2) = x_1^2 + x_2^2 - 1 \leq 0$ and $g_2(x_1, x_2) = -\alpha x_1 + x_2 \leq 0$ (or more precisely $g_2(x_1, x_2) = -\alpha x_1 + x_2$). The figure depicts the problem for two different values of the parameter α .

翻译 (Chinese)

为阐明其中涉及的问题，考虑以下最优化例子。

例 13.1:

$$\max_{x_1, x_2} x_1 x_2$$

约束条件为:

$$x_1^2 + x_2^2 \leq 1, \quad x_2 \geq \alpha x_1.$$

因此, 目标函数为 $f(x_1, x_2) = x_1x_2$, 约束可写为 $g_1(x_1, x_2) = x_1^2 + x_2^2 - 1 \leq 0$ 和 $g_2(x_1, x_2) = -\alpha x_1 + x_2 \leq 0$ (或更精确地, $g_2(x_1, x_2) = -\alpha x_1 + x_2$)。图形描绘了参数 α 取两个不同值时的情况。

Original (English)

At parameter value $\tilde{\alpha}$, the constraints are $g_1(x_1, x_2) \leq 0$ and $\tilde{g}_2(x_1, x_2) \leq 0$ and the solution, x^a , satisfies $g_1(x^a) = 0$, $\tilde{g}_2(x^a) < 0$. Constraint g_1 is active or binding at x^a but constraint $\tilde{g}_2$ is not active at x^a . At parameter value $\hat{\alpha}$, the constraints are $g_1(x_1, x_2) \leq 0$ and $\hat{g}_2(x_1, x_2) \leq 0$, and, at the solution, x^b satisfies $g_1(x^b) = 0$, $\hat{g}_2(x^b) = 0$. Both constraints g_1 and $\hat{g}_2$ are active at x^b . Depending on circumstance, a constraint may or may not be binding, so that the equations defining the solution must reflect this possibility.

In the case where $\alpha = \tilde{\alpha}$, the slope of the objective is equal to the slope of the active constraint. In the case where $\alpha = \hat{\alpha}$, the slope of the objective lies between the slopes of the constraints. The point x^b is located at the solution to the two constraints: $x^b = (0.9285, 0.3714)$. The slope of constraint 1 is 0.4 and the slope of constraint 2 is $-\frac{0.9285}{0.3714} = -2.5$. The slope of the objective is -0.2908 . Thus, since the slope of the objective lies between the slopes of the two constraints, the slope of the objective may be written as a weighted average of the slopes of the constraints.

翻译 (Chinese)

在参数值 $\tilde{\alpha}$ 下, 约束为 $g_1(x_1, x_2) \leq 0$ 和 $\tilde{g}_2(x_1, x_2) \leq 0$, 解 x^a 满足 $g_1(x^a) = 0$, $\tilde{g}_2(x^a) < 0$ 。约束 g_1 在 x^a 处是积极的 (紧的), 但约束 $\tilde{g}_2$ 在 x^a 处不是积极的。在参数值 $\hat{\alpha}$ 下, 约束为 $g_1(x_1, x_2) \leq 0$ 和 $\hat{g}_2(x_1, x_2) \leq 0$, 解 x^b 满足 $g_1(x^b) = 0$, $\hat{g}_2(x^b) = 0$ 。两个约束 g_1 和 $\hat{g}_2$ 在 x^b 处都是积极的。根据具体情况, 约束可能是紧的也可能不是, 因此定义解的方程必须反映这种可能性。

在 $\alpha = \tilde{\alpha}$ 的情形中, 目标函数的斜率等于积极约束的斜率。在 $\alpha = \hat{\alpha}$ 的情形中, 目标函数的斜率介于两个约束的斜率之间。点 x^b 位于两个约束的解处: $x^b = (0.9285, 0.3714)$ 。约束 1 的斜率为 0.4, 约束 2 的斜率为 $-\frac{0.9285}{0.3714} = -2.5$ 。目标函数的斜率为 -0.2908 。因此, 由于目标函数的斜率介于两个约束的斜率之间, 目标函数的斜率可以写成两个约束斜率的加权平均值。

Original (English)

REMARK 13.1: Rather than express conditions in terms of slopes, it is standard practice to use the normal to a curve to represent the slope. Thus, with $f(x_1, x_2) = x_1x_2$ evaluated at x^b ,

$$\nabla f \stackrel{\text{def}}{=} (f_{x_1}, f_{x_2}) = (x_2, x_1) = (0.3714, 0.9285)$$

is the normal to the curve $f(x_1, x_2)$ at (x_1, x_2) . As a vector, this has slope $\frac{x_1}{x_2}$, which is the negative of the reciprocal of the slope of the objective. Similarly, the normal of the linear constraint g_2 is $\nabla g_2 = (\frac{\partial g_2}{\partial x_1}, \frac{\partial g_2}{\partial x_2}) = (-0.4, 1)$ and the normal of the quadratic constraint g_1 is $\nabla g_1 = (\frac{\partial g_1}{\partial x_1}, \frac{\partial g_1}{\partial x_2}) = (2x_1, 2x_2) = (1.857, 0.7428)$.

翻译 (Chinese)

注 13.1: 与其用斜率来表示条件, 标准做法是使用曲线的法向量。因此, 对于 $f(x_1, x_2) = x_1x_2$ 在 x^b 处的取值:

$$\nabla f \stackrel{\text{def}}{=} (f_{x_1}, f_{x_2}) = (x_2, x_1) = (0.3714, 0.9285)$$

是曲线 $f(x_1, x_2)$ 在 (x_1, x_2) 处的法向量。其斜率为 $\frac{x_1}{x_2}$, 这是目标函数斜率的负倒数。类似地, 线性约束 g_2 的法向量为 $\nabla g_2 = (\frac{\partial g_2}{\partial x_1}, \frac{\partial g_2}{\partial x_2}) = (-0.4, 1)$, 二次约束 g_1 的法向量为 $\nabla g_1 = (\frac{\partial g_1}{\partial x_1}, \frac{\partial g_1}{\partial x_2}) = (2x_1, 2x_2) = (1.857, 0.7428)$ 。

Original (English)

The normal of the objective may be expressed as a positive linear combination of the constraint normals:

$$\begin{aligned} \nabla f &= \lambda_1 \nabla g_1 + \lambda_2 \nabla g_2 \\ \begin{pmatrix} 0.3714 \\ 0.9285 \end{pmatrix} &= 0.3448 \begin{pmatrix} 1.8570 \\ 0.7428 \end{pmatrix} + 0.6724 \begin{pmatrix} -0.4 \\ 1 \end{pmatrix}. \end{aligned} \quad (13.1)$$

The observations in Example 13.1 apply for the general case.

翻译 (Chinese)

目标函数的法向量可以表示为约束法向量的正线性组合:

$$\begin{aligned} \nabla f &= \lambda_1 \nabla g_1 + \lambda_2 \nabla g_2 \\ \begin{pmatrix} 0.3714 \\ 0.9285 \end{pmatrix} &= 0.3448 \begin{pmatrix} 1.8570 \\ 0.7428 \end{pmatrix} + 0.6724 \begin{pmatrix} -0.4 \\ 1 \end{pmatrix}. \end{aligned} \quad (13.1)$$

例 13.1 中的观察适用于一般情形。

13.1.2 Overview / 概述

Original (English)

Given a function $h(x)$, the normal to the function at x is the gradient $\nabla h = (h_{x_1}, \dots, h_{x_n})$. The key characterizing condition for an optimum is that the gradient of the objective may be written as a weighted sum of the gradients of the constraints: $\nabla f = \sum_{j=1}^k \lambda_j \nabla g^j$, with $\lambda_j \geq 0$ for all j , as in Theorem 13.1.

Provided a constraint qualification is satisfied, this condition along with complementary slackness conditions determines a system of equations that may be solved to provide necessary conditions for a local optimum. All results here are derived from these conditions. Sufficiency conditions are discussed in Section 13.2.4: concavity of the objective and convexity of the feasible set imply that a solution to the necessary conditions yields a global maximum for the constrained problem. Finally, the necessary conditions can be specialized to include non-negativity conditions in Section 13.2.5.

翻译 (Chinese)

给定函数 $h(x)$ ，函数在 x 处的法向量即为梯度 $\nabla h = (h_{x_1}, \dots, h_{x_n})$ 。最优解的核心刻画条件是：目标函数的梯度可以写成各约束梯度的加权和： $\nabla f = \sum_{j=1}^k \lambda_j \nabla g^j$ ，其中 $\lambda_j \geq 0$ ，如定理 13.1 所述。

只要满足约束资格条件，这一条件连同互补松弛条件就确定了一个方程组，可以求解以提供局部最优解的必要条件。本文中的所有结果都从这些条件推导而来。充分条件在第 13.2.4 节讨论：目标函数的凹性和可行集的凸性意味着必要条件的一个解就是约束问题的全局最大值。最后，必要条件可以在第 13.2.5 节中特殊化以包含非负性条件。

13.2 Optimization / 最优化

Original (English)

Since a constraint of the form $g(x) \leq c$ may be written as $g(x) - c \leq 0$ and constraints of the form $g(x) \geq c$ may be written as $g(x) - c \geq 0$ or as $-g(x) \leq -c$, we can express all such constraints in the form $g(x) \leq 0$ (or $g(x) \geq 0$). Equality constraints may be dealt with in the same way since $g(x) = c$ is equivalent to $g(x) \leq c$ and $g(x) \geq c$, at the cost of having two constraints instead of one. Non-negativity constraints may be expressed by setting $g(x) = -x$, the condition $g(x) \leq 0$ implies $x \geq 0$.

An inequality-constrained maximization problem is:

$$\max f(x) \quad \text{subject to} \quad g^j(x) \leq 0, \quad j = 1, \dots, r. \quad (13.2)$$

The function $f(x)$ is the objective and g^j are the constraints.

翻译 (Chinese)

由于 $g(x) \leq c$ 形式的约束可写为 $g(x) - c \leq 0$, 而 $g(x) \geq c$ 形式的约束可写为 $g(x) - c \geq 0$ 或 $-g(x) \leq -c$, 我们可以将所有此类约束表达为 $g(x) \leq 0$ (或 $g(x) \geq 0$) 的形式。等式约束也可以用同样方式处理, 因为 $g(x) = c$ 等价于 $g(x) \leq c$ 且 $g(x) \geq c$, 代价是需要两个约束而非一个。非负性约束可以通过设 $g(x) = -x$ 来表达, 条件 $g(x) \leq 0$ 蕴含 $x \geq 0$ 。

不等式约束最大化问题为:

$$\max f(x) \quad \text{subject to} \quad g^j(x) \leq 0, j = 1, \dots, r. \quad (13.2)$$

函数 $f(x)$ 是目标函数, g^j 是约束集。

Original (English)

Definition 13.1: A point x is feasible if it satisfies all constraints: $g^j(x) \leq 0, \forall j$. A constraint g^j is active at feasible x if $g^j(x) = 0$, and otherwise inactive. Let $\mathcal{A}(x)$ be the set of active constraints at x : $j \in \mathcal{A}(x)$ iff $g^j(x) = 0$.

Definition 13.2: A point x is a constrained local maximum of f if there is a neighborhood $N(x)$ such that $f(x) \geq f(x')$ for all $x' \in N(x) \cap \{x \mid g^l(x) \leq 0, l = 1, \dots, r\}$.

翻译 (Chinese)

定义 13.1: 如果点 x 满足所有约束 $g^j(x) \leq 0 (\forall j)$, 则称其为可行点。如果在可行点 x 处有 $g^j(x) = 0$, 则称约束 g^j 是积极的, 否则是非积极的。令 $\mathcal{A}(x)$ 为 x 处积极约束的集合: $j \in \mathcal{A}(x)$ 当且仅当 $g^j(x) = 0$ 。

定义 13.2: 如果存在邻域 $N(x)$ 使得对所有的 $x' \in N(x) \cap \{x \mid g^l(x) \leq 0, l = 1, \dots, r\}$ 有 $f(x) \geq f(x')$, 则 x 是 f 的约束局部最大值。

Original (English)

The following discussion identifies necessary conditions using Farkas' lemma (Remark 13.2), applied to the equation system:

$$\begin{pmatrix} \frac{\partial g^1(x)}{\partial x_1} & \dots & \frac{\partial g^r(x)}{\partial x_1} \\ \vdots & \ddots & \vdots \\ \frac{\partial g^1(x)}{\partial x_n} & \dots & \frac{\partial g^r(x)}{\partial x_n} \end{pmatrix} \begin{pmatrix} \lambda_1 \\ \vdots \\ \lambda_r \end{pmatrix} = \begin{pmatrix} \frac{\partial f(x)}{\partial x_1} \\ \vdots \\ \frac{\partial f(x)}{\partial x_n} \end{pmatrix} \quad (13.3)$$

Each row: $\frac{\partial f(x)}{\partial x_i} = \sum_{j=1}^r \lambda_j \frac{\partial g^j(x)}{\partial x_i}$.

Writing $\nabla g(x)$ for the $n \times r$ matrix and $\lambda = (\lambda_1, \dots, \lambda_r)$:

$$\nabla g(x) \cdot \lambda = \nabla f(x). \quad (13.5)$$

Subject to a qualification, if the equation system does not have a non-negative solution, then x is not a local optimum.

翻译 (Chinese)

以下讨论利用 Farkas 引理（注 13.2）来确定必要条件，将其应用于方程组：

$$\begin{pmatrix} \frac{\partial g^1(x)}{\partial x_1} & \dots & \frac{\partial g^r(x)}{\partial x_1} \\ \vdots & \ddots & \vdots \\ \frac{\partial g^1(x)}{\partial x_n} & \dots & \frac{\partial g^r(x)}{\partial x_n} \end{pmatrix} \begin{pmatrix} \lambda_1 \\ \vdots \\ \lambda_r \end{pmatrix} = \begin{pmatrix} \frac{\partial f(x)}{\partial x_1} \\ \vdots \\ \frac{\partial f(x)}{\partial x_n} \end{pmatrix} \quad (13.3)$$

每一行： $\frac{\partial f(x)}{\partial x_i} = \sum_{j=1}^r \lambda_j \frac{\partial g^j(x)}{\partial x_i}$ 。

将 $n \times r$ 矩阵记为 $\nabla g(x)$, $\lambda = (\lambda_1, \dots, \lambda_r)$:

$$\nabla g(x) \cdot \lambda = \nabla f(x). \quad (13.5)$$

在满足资格条件的前提下，如果该方程组没有非负解，则 x 不是局部最优解。

Original (English)

Theorem 13.1: Given program 13.2, suppose that at x there is some z with $\nabla g^j(x) \cdot z < 0$ for each active constraint j . If x is a constrained local maximum then the system must have a non-negative solution $\lambda = (\lambda_1, \dots, \lambda_r)$, $\lambda_j \geq 0$, $j = 1, \dots, r$.

REMARK 13.2 (Farkas' lemma): If A is an $n \times r$ matrix then exactly one holds:

1. $Ax = b$ has a solution $x \geq 0$, or
2. There exists y with $A'y \geq 0$ and $b'y > 0$.

REMARK 13.3: The program 13.2 includes the possibility of imposing non-negativity constraints. If $g^k(x) = -x_k$, then $g^k(x) \leq 0$ is equivalent to $x_k \geq 0$. In Section 13.2.5, a special case is considered where some constraints are non-negativity conditions. There, the Lagrangian together with complementary slackness conditions define the Kuhn–Tucker conditions.

REMARK 13.4: In Equation 13.3, there are n equations in $n + r$ variables (x, λ) . Complementary slackness asserts $\lambda_j g^j = 0$ for each j , giving an additional r equations. Provided the constraint

qualification is satisfied, a necessary condition is that the system of $n+r$ equations in $n+r$ unknowns has a solution with $\lambda_j \geq 0$.

REMARK 13.5: At the solution, ∇f points in the direction of increase of f , and ∇g points out from the feasible region. For x to be an optimum, ∇f and ∇g must point in the same direction, so $\nabla f = \lambda \nabla g$ with $\lambda > 0$.

翻译 (Chinese)

定理 13.1: 给定规划 13.2, 假设在 x 处存在某个 z 使得对每个积极约束 j 有 $\nabla g^j(x) \cdot z < 0$ 。如果 x 是约束局部最大值, 则该系统必须有一个非负解 $\lambda = (\lambda_1, \dots, \lambda_r)$, $\lambda_j \geq 0$, $j = 1, \dots, r$ 。

注 13.2 (Farkas 引理): 若 A 为 $n \times r$ 矩阵, 则以下条件恰有一个成立:

1. $Ax = b$ 有解 $x \geq 0$, 或
2. 存在 y 使得 $A'y \geq 0$ 且 $b'y > 0$ 。

注 13.3: 规划 13.2 包含了施加非负性约束的可能性。若 $g^k(x) = -x_k$, 则 $g^k(x) \leq 0$ 等价于 $x_k \geq 0$ 。第 13.2.5 节考虑了部分约束为非负性条件的特殊情况, 其中拉格朗日函数与互补松弛条件共同定义了 Kuhn-Tucker 条件。

注 13.4: 方程 13.3 中有 n 个方程, $n+r$ 个变量 (x, λ) 。互补松弛条件断言对每个 j 有 $\lambda_j g^j = 0$, 提供了额外的 r 个方程。在满足约束资格条件的前提下, $n+r$ 个方程与 $n+r$ 个未知数的系统有解 ($\lambda_j \geq 0$) 是必要条件。

注 13.5: 在解处, ∇f 指向 f 增加的方向, 而 ∇g 指向可行区域之外。要使 x 成为最优解, ∇f 和 ∇g 必须指向同一方向, 因此 $\nabla f = \lambda \nabla g$ ($\lambda > 0$)。

13.2.1 The constraint qualification / 约束资格条件

Original (English)

In Theorem 13.1, the constraint qualification requires that there exists some z with $\nabla g^j(x) \cdot z < 0$ for each active constraint j . An alternative constraint qualification is given in terms of the linear independence of the gradients of the active constraints. Let $\mathcal{A}(x)\nabla g(x)$ be the matrix obtained by deleting the columns corresponding to inactive constraints. Suppose $\mathcal{A}(x)\nabla g(x)$ has rank equal to the number of active constraints – the gradients of the active constraints are linearly independent (so the number of active constraints must be no greater than n). Then there exists some z with $\nabla g^j(x) \cdot z < 0$ for each active constraint j .

翻译 (Chinese)

在定理 13.1 中, 约束资格条件要求存在某个 z 使得对每个积极约束 j 有 $\nabla g^j(x) \cdot z < 0$ 。另一种约束资格条件以积极约束梯度的线性独立性来表达。令 $\mathcal{A}(x)\nabla g(x)$ 为删除非积极约束对应列后得到的矩阵。假设 $\mathcal{A}(x)\nabla g(x)$ 的秩等于积极约束的个数——积极约束的梯度线性无关 (因此积极约束的个数必不大于 n)。则存在某个 z 使得对每个积极约束 j 有 $\nabla g^j(x) \cdot z < 0$ 。

Original (English)

When the constraint qualification fails, the representation in Equation 13.5 may be impossible to satisfy. Example 13.6 illustrates: two circle constraints $g_1(x) = (x_1 - 3)^2 + (x_2 - 3)^2 \leq 8$ and $g_2(x) = (x_1 - 7)^2 + (x_2 - 7)^2 \leq 8$ have only one point in common $x = (5, 5)$. At this point $\nabla g_1(5, 5) = (4, 4)$ and $\nabla g_2(5, 5) = (-4, -4)$, which are linearly dependent. There is no point z with $\nabla g_i(5, 5) \cdot z < 0$ for $i = 1, 2$. Unless $\nabla f(5, 5) = (\theta, \theta)$ for some θ , it is impossible to write $\nabla f = \lambda_1 \nabla g_1 + \lambda_2 \nabla g_2$.

翻译 (Chinese)

当约束资格条件不满足时, 方程 13.5 的表示可能无法做到。例 13.6 说明了这一点: 两个圆约束 $g_1(x) = (x_1 - 3)^2 + (x_2 - 3)^2 \leq 8$ 和 $g_2(x) = (x_1 - 7)^2 + (x_2 - 7)^2 \leq 8$ 只有一个公共点 $x = (5, 5)$ 。在该点处 $\nabla g_1(5, 5) = (4, 4)$, $\nabla g_2(5, 5) = (-4, -4)$, 两者线性相关。不存在点 z 使得对 $i = 1, 2$ 有 $\nabla g_i(5, 5) \cdot z < 0$ 。除非 $\nabla f(5, 5) = (\theta, \theta)$, 否则不可能写出 $\nabla f = \lambda_1 \nabla g_1 + \lambda_2 \nabla g_2$ 。

13.2.2 Complementary slackness / 互补松弛**Original (English)**

Consider the modified program obtained by deleting constraints not binding at x :

$$\max f(x) \quad \text{subject to} \quad g^j(x) \leq 0, \quad j \in \mathcal{A}(x). \quad (13.7)$$

For this program, x is a constrained local maximum. Applying Theorem 13.1, there exists $\tilde{\lambda}_j \geq 0$, $j \in \mathcal{A}(x)$ with $\nabla f(x) = \sum_{j \in \mathcal{A}(x)} \tilde{\lambda}_j \nabla g^j(x)$. For $j \notin \mathcal{A}(x)$, let $\tilde{\lambda}_j = 0$. The condition $\tilde{\lambda}_j g^j = 0$ for all j is the complementary slackness condition.

Theorem 13.2: Given program 13.2, suppose that at x the constraint qualification is satisfied. If x is a constrained local maximum, then there exists $\lambda_j \geq 0$, $j = 1, \dots, r$, with:

$$\frac{\partial f(x)}{\partial x_i} = \sum_{j=1}^r \lambda_j \frac{\partial g^j(x)}{\partial x_i}, \quad i = 1, \dots, n, \quad (13.8)$$

and $\lambda_j = 0$ if $g^j(x) < 0$ (equivalently $\lambda_j g^j(x) = 0, j = 1, \dots, r$).

翻译 (Chinese)

考虑通过删除在 x 处非紧的约束而得到的修正规划:

$$\max f(x) \quad \text{subject to} \quad g^j(x) \leq 0, j \in \mathcal{A}(x). \quad (13.7)$$

对于该规划, x 是约束局部最大值。应用定理 13.1, 存在 $\tilde{\lambda}_j \geq 0$ ($j \in \mathcal{A}(x)$) 使得 $\nabla f(x) = \sum_{j \in \mathcal{A}(x)} \tilde{\lambda}_j \nabla g^j(x)$ 。对于 $j \notin \mathcal{A}(x)$, 令 $\tilde{\lambda}_j = 0$ 。对所有 j 成立的条件 $\tilde{\lambda}_j g^j = 0$ 即为互补松弛条件。

定理 13.2: 给定规划 13.2, 假设在 x 处约束资格条件成立。若 x 是约束局部最大值, 则存在 $\lambda_j \geq 0$ ($j = 1, \dots, r$) 使得:

$$\frac{\partial f(x)}{\partial x_i} = \sum_{j=1}^r \lambda_j \frac{\partial g^j(x)}{\partial x_i}, \quad i = 1, \dots, n, \quad (13.8)$$

且若 $g^j(x) < 0$ 则 $\lambda_j = 0$ (等价地, $\lambda_j g^j(x) = 0, j = 1, \dots, r$)。

13.2.3 Minimization and maximization / 最小化与最大化

Original (English)

Minimizing $f(x)$ subject to $g^j(x) \geq 0$ is equivalent to maximizing $\hat{f}(x) = -f(x)$ subject to $\hat{g}^j(x) = -g^j(x) \leq 0$. Applying Theorem 13.2 gives:

Theorem 13.3: If at x there is some z with $\nabla g^l(x) \cdot z > 0$ for each active constraint l , and x is a local minimum, then there exists $\lambda_j \geq 0$ such that $\nabla f(x) = \sum_{j=1}^r \lambda_j \nabla g^j(x)$ and $\lambda_j g^j(x) = 0$ for all j .

翻译 (Chinese)

在 $g^j(x) \geq 0$ 约束下最小化 $f(x)$ 等价于在 $\hat{g}^j(x) = -g^j(x) \leq 0$ 约束下最大化 $\hat{f}(x) = -f(x)$ 。应用定理 13.2 即得:

定理 13.3: 若在 x 处存在某个 z 使得对每个积极约束 l 有 $\nabla g^l(x) \cdot z > 0$, 且 x 是局部最小值, 则存在 $\lambda_j \geq 0$ 使得 $\nabla f(x) = \sum_{j=1}^r \lambda_j \nabla g^j(x)$ 且对所有 j 有 $\lambda_j g^j(x) = 0$ 。

13.2.4 Global and local optima / 全局最优与局部最优

Original (English)

Theorem 13.4: Suppose f is concave and the feasible region convex. Then, given the constraint qualification, any x satisfying Equation 13.3 solves the optimization problem 13.2.

PROOF: If f is concave, $\nabla f(x) \cdot (x' - x) \geq f(x') - f(x)$. If there is feasible x' with $f(x') > f(x)$, then $\nabla f(x) \cdot (x' - x) > 0$. Because the feasible set is convex, for every binding constraint j , $\nabla g^j(x) \cdot (x' - x) \leq 0$. Put $z = x' - x$. Then $0 < \nabla f(x) \cdot z$ and $\nabla g^j(x) \cdot z \leq 0$ for each binding j . From the necessary condition $\nabla f = \sum \lambda_j \nabla g^j$ with $\lambda_j = 0$ for non-binding constraints: $0 < \nabla f \cdot z = \sum \lambda_j \nabla g^j \cdot z \leq 0$, a contradiction. An analogous result holds for minimization with f convex.

翻译 (Chinese)

定理 13.4: 假设 f 是凹函数且可行区域为凸集。那么, 在满足约束资格条件的前提下, 任何满足方程 13.3 的 x 即为最优化问题 13.2 的解。

证明: 若 f 是凹函数, $\nabla f(x) \cdot (x' - x) \geq f(x') - f(x)$ 。若存在可行点 x' 使得 $f(x') > f(x)$, 则 $\nabla f(x) \cdot (x' - x) > 0$ 。因可行集为凸集, 对每个紧约束 j 有 $\nabla g^j(x) \cdot (x' - x) \leq 0$ 。令 $z = x' - x$ 。则 $0 < \nabla f(x) \cdot z$ 且对每个紧约束 j 有 $\nabla g^j(x) \cdot z \leq 0$ 。由必要条件 $\nabla f = \sum \lambda_j \nabla g^j$ (非紧约束的 $\lambda_j = 0$): $0 < \nabla f \cdot z = \sum \lambda_j \nabla g^j \cdot z \leq 0$, 矛盾。对于 f 为凸函数的最小化问题有类似结果成立。

13.2.5 Non-negativity constraints / 非负性约束

Original (English)

Consider:

$$\max f(x) \quad \text{s.t.} \quad g^j(x) \leq 0, \quad j = 1, \dots, r; \quad x_i \geq 0, \quad i = 1, \dots, n. \quad (13.11)$$

With μ_i denoting the multiplier on constraint $x_i \geq 0$ (i.e., $-x_i \leq 0$):

$$\frac{\partial f(x)}{\partial x_i} = \sum_{j=1}^r \lambda_j \frac{\partial g^j(x)}{\partial x_i} - \mu_i, \quad i = 1, \dots, n. \quad (13.14)$$

Complementary slackness gives $\lambda_j g^j(x) = 0$ and $\mu_i x_i = 0$. With $\mu_i \geq 0$:

$$\frac{\partial f(x)}{\partial x_i} - \sum_{j=1}^r \lambda_j \frac{\partial g^j(x)}{\partial x_i} \leq 0, \quad x_i \left[\frac{\partial f(x)}{\partial x_i} - \sum_{j=1}^r \lambda_j \frac{\partial g^j(x)}{\partial x_i} \right] = 0, \quad x_i \geq 0. \quad (13.15)$$

Theorem 13.5 (Kuhn–Tucker conditions for maximization): Let $L = f(x) - \sum_{j=1}^r \lambda_j g^j(x)$. Provided the constraint qualification is satisfied, a local maximum satisfies:

$$\frac{\partial L}{\partial x_i} \leq 0; \quad x_i \frac{\partial L}{\partial x_i} = 0; \quad x_i \geq 0 \quad (i = 1, \dots, n); \quad \frac{\partial L}{\partial \lambda_j} \geq 0; \quad \lambda_j \frac{\partial L}{\partial \lambda_j} = 0; \quad \lambda_j \geq 0 \quad (j = 1, \dots, r). \quad (13.18)$$

Theorem 13.6 (Kuhn–Tucker conditions for minimization): For $\min f(x)$ s.t. $g^j(x) \geq 0, x_i \geq 0$, with $L = f(x) - \sum \lambda_j g^j(x)$, a local minimum satisfies:

$$\frac{\partial L}{\partial x_i} \geq 0; \quad x_i \frac{\partial L}{\partial x_i} = 0; \quad x_i \geq 0; \quad \frac{\partial L}{\partial \lambda_j} \leq 0; \quad \lambda_j \frac{\partial L}{\partial \lambda_j} = 0; \quad \lambda_j \geq 0. \quad (13.22)$$

翻译 (Chinese)

考虑:

$$\max f(x) \quad \text{s.t.} \quad g^j(x) \leq 0, \quad j = 1, \dots, r; \quad x_i \geq 0, \quad i = 1, \dots, n. \quad (13.11)$$

令 μ_i 表示约束 $x_i \geq 0$ (即 $-x_i \leq 0$) 的乘子:

$$\frac{\partial f(x)}{\partial x_i} = \sum_{j=1}^r \lambda_j \frac{\partial g^j(x)}{\partial x_i} - \mu_i, \quad i = 1, \dots, n. \quad (13.14)$$

互补松弛条件给出 $\lambda_j g^j(x) = 0$ 和 $\mu_i x_i = 0$ 。由 $\mu_i \geq 0$:

$$\frac{\partial f(x)}{\partial x_i} - \sum_{j=1}^r \lambda_j \frac{\partial g^j(x)}{\partial x_i} \leq 0, \quad x_i \left[\frac{\partial f(x)}{\partial x_i} - \sum_{j=1}^r \lambda_j \frac{\partial g^j(x)}{\partial x_i} \right] = 0, \quad x_i \geq 0. \quad (13.15)$$

定理 13.5 (最大化的 Kuhn–Tucker 条件): 令 $L = f(x) - \sum_{j=1}^r \lambda_j g^j(x)$ 。在满足约束资格条件的前提下, 局部最大值满足:

$$\frac{\partial L}{\partial x_i} \leq 0; \quad x_i \frac{\partial L}{\partial x_i} = 0; \quad x_i \geq 0 \quad (i = 1, \dots, n); \quad \frac{\partial L}{\partial \lambda_j} \geq 0; \quad \lambda_j \frac{\partial L}{\partial \lambda_j} = 0; \quad \lambda_j \geq 0 \quad (j = 1, \dots, r). \quad (13.18)$$

定理 13.6 (最小化的 Kuhn–Tucker 条件): 对于 $\min f(x)$ s.t. $g^j(x) \geq 0, x_i \geq 0$, 令 $L = f(x) - \sum \lambda_j g^j(x)$, 局部最小值满足:

$$\frac{\partial L}{\partial x_i} \geq 0; \quad x_i \frac{\partial L}{\partial x_i} = 0; \quad x_i \geq 0; \quad \frac{\partial L}{\partial \lambda_j} \leq 0; \quad \lambda_j \frac{\partial L}{\partial \lambda_j} = 0; \quad \lambda_j \geq 0. \quad (13.22)$$

13.2.6 Necessary conditions and non-convexities / 必要条件与非凸性

Original (English)

The necessary conditions for a local optimum are not sufficient. Example 13.11 revisits Example 13.3: maximize $f(x, y) = \beta \ln x + y$ subject to $g_1(x, y) = y^2 + (x - 1)^2 \leq 2$ and $g_2(x, y) = xy \leq k = 1.12$. The necessary conditions are satisfied at two points: the true optimum and a second point that is not a local maximum. At the non-optimal point, the constraint set is not convex, so Theorem 13.4 does not apply.

翻译 (Chinese)

局部最优解的必要条件并非充分的。例 13.11 重新审视了例 13.3：在约束 $g_1(x, y) = y^2 + (x - 1)^2 \leq 2$ 和 $g_2(x, y) = xy \leq k = 1.12$ 下最大化 $f(x, y) = \beta \ln x + y$ 。必要条件在两个点处得到满足：真正的最大值点，以及另一个不是局部最大值的点。在非最优解点处，约束集不是凸的，因此定理 13.4 不适用。

13.2.7 The Lagrange multiplier / 拉格朗日乘子

Original (English)

Consider a maximization program with objective f and a single constraint g . At a point x on the boundary, ∇f should point out from the feasible region. If ∇g also points out, at a tangency $\nabla f = \lambda \nabla g$ for some positive λ . For ∇g to point out from the feasible set, the feasible set must be defined $\{x \mid g(x) \leq 0\}$. Conversely, if minimizing, ∇f should point into the feasible region at the candidate solution x on the boundary. To write $\nabla f = \lambda \nabla g$ with $\lambda > 0$, ∇g must point into the feasible region, so we define the feasible set as $\{x \mid g(x) \geq 0\}$.

翻译 (Chinese)

考虑具有目标函数 f 和单个约束 g 的最大化问题。在边界上的点 x 处， ∇f 应指向可行区域之外。若 ∇g 也指向外部，则在切点处 $\nabla f = \lambda \nabla g$ ($\lambda > 0$)。要使 ∇g 指向可行集之外，可行集必须定义为 $\{x \mid g(x) \leq 0\}$ 。反过来，若是最小化问题， ∇f 应指向边界上候选解 x 的可行区域内部。要写出 $\nabla f = \lambda \nabla g$ ($\lambda > 0$)， ∇g 必须指向可行区域内部，因此我们将可行集定义为 $\{x \mid g(x) \geq 0\}$ 。

13.2.8 Equality and inequality constraints / 等式与不等式约束

Original (English)

Consider minimizing $f(x)$ subject to $h^l(x) = 0, l = 1, \dots, m; g^j(x) \leq 0, j = 1, \dots, r$.

Theorem 13.7: Suppose f, h^l, g^j are C^1 . Let x be a local minimum. Then there exist numbers $\gamma, \mu_1, \dots, \mu_m, \lambda_1, \dots, \lambda_r$, not all 0, such that:

$$\gamma \nabla f(x) + \sum_{l=1}^m \mu_l \nabla h^l(x) + \sum_{j=1}^r \lambda_j \nabla g^j(x) = 0. \quad (13.27)$$

In addition: $\gamma \geq 0$ and $\lambda_j \geq 0$; for each j with $g^j(x) < 0, \lambda_j = 0$; if the gradients of all h^l and active g^j are linearly independent, then γ can be chosen equal to 1.

Theorem 13.8: Suppose the conditions of Theorem 13.7 hold, the constraint qualification is satisfied, f is convex and the constraint set convex. Then x minimizes f subject to the constraints.

翻译 (Chinese)

考虑在约束 $h^l(x) = 0$ ($l = 1, \dots, m$); $g^j(x) \leq 0$ ($j = 1, \dots, r$) 下最小化 $f(x)$ 。

定理 13.7: 假设 f, h^l, g^j 均为 C^1 。令 x 为局部最小值。则存在不全为零的数 $\gamma, \mu_1, \dots, \mu_m, \lambda_1, \dots, \lambda_r$ 使得:

$$\gamma \nabla f(x) + \sum_{l=1}^m \mu_l \nabla h^l(x) + \sum_{j=1}^r \lambda_j \nabla g^j(x) = 0. \quad (13.27)$$

此外: $\gamma \geq 0$ 且 $\lambda_j \geq 0$; 对于满足 $g^j(x) < 0$ 的每个 $j, \lambda_j = 0$; 若所有 h^l 和积极 g^j 的梯度线性无关, 则 γ 可以取为 1。

定理 13.8: 假设定理 13.7 的条件成立, 约束资格条件满足, f 是凸函数且约束集为凸集。则 x 在约束下最小化 f 。

Exercises for Chapter 13 / 第 13 章习题

Original (English)

Exercise 13.1: Let $f(x, y) = y - x^\alpha - \beta x, \alpha, \beta > 1, g(x, y) = y^2 - 1 - \ln(x + 1)$. Solve $\max f(x, y)$ subject to $g(x, y) \leq 0$ and $x, y \geq 0$, using the Kuhn–Tucker conditions.

Exercise 13.2: Let $f(x, y) = x^\alpha y, g_1(x, y) = ax + by - 2a$ ($a = b > 0$), $g_2(x, y) = y + cx - 1$ ($c = \frac{1}{10}, \beta = 2$). Solve $\max f(x, y)$ subject to $g_i(x, y) \leq 0$ for $i = 1, 2$ and $x, y \geq 0$, using Kuhn–Tucker conditions.

Exercise 13.3: Let $f(x, y) = x^2y$, $g_1(x, y) = ax + by - 2a$ ($a = b > 0$), $g_2(x, y) = y + cx - 1$ ($c = \frac{1}{10}$, $\beta = 2$). Solve using Kuhn–Tucker conditions.

Exercise 13.4: Let $f(x, y) = 14 - x^2 - y^2 - 2x + 2y$, $g(x, y) = x + y - 4$. Solve $\max f(x, y)$ subject to $g(x, y) \leq 0$ and $x, y \geq 0$, using Kuhn–Tucker conditions.

Exercise 13.5: Let $f(x, y) = xy$, $g_1(x, y) = 1 - y + (x - 1)^3$, $g_2(x, y) = 2y - 2 + 4(x - 1)^3$. Consider $\max f(x, y)$ subject to $g_i(x, y) \leq 0$, $i = 1, 2$. Confirm the solution occurs at $(x, y) = (1, 1)$, but at this point there is no non-negative solution (λ_1, λ_2) to $\nabla f = \lambda_1 \nabla g_1 + \lambda_2 \nabla g_2$. Explain why the Kuhn–Tucker conditions fail.

翻译 (Chinese)

习题 13.1: 令 $f(x, y) = y - x^\alpha - \beta x$, $\alpha, \beta > 1$, $g(x, y) = y^2 - 1 - \ln(x + 1)$ 。在 $g(x, y) \leq 0$ 且 $x, y \geq 0$ 下求解 $\max f(x, y)$, 使用 Kuhn–Tucker 条件。

习题 13.2: 令 $f(x, y) = x^\alpha y$, $g_1(x, y) = ax + by - 2a$ ($a = b > 0$), $g_2(x, y) = y + cx - 1$ ($c = \frac{1}{10}$, $\beta = 2$)。在 $g_i(x, y) \leq 0$ ($i = 1, 2$) 且 $x, y \geq 0$ 下求解 $\max f(x, y)$, 使用 Kuhn–Tucker 条件。

习题 13.3: 令 $f(x, y) = x^2y$, 约束条件同习题 13.2。使用 Kuhn–Tucker 条件求解。

习题 13.4: 令 $f(x, y) = 14 - x^2 - y^2 - 2x + 2y$, $g(x, y) = x + y - 4$ 。在 $g(x, y) \leq 0$ 且 $x, y \geq 0$ 下求解 $\max f(x, y)$, 使用 Kuhn–Tucker 条件。

习题 13.5: 令 $f(x, y) = xy$, $g_1(x, y) = 1 - y + (x - 1)^3$, $g_2(x, y) = 2y - 2 + 4(x - 1)^3$ 。考虑在 $g_i(x, y) \leq 0$ ($i = 1, 2$) 下最大化 $f(x, y)$ 。验证解出现在 $(x, y) = (1, 1)$, 但在该点处 $\nabla f = \lambda_1 \nabla g_1 + \lambda_2 \nabla g_2$ 没有非负解 (λ_1, λ_2) 。解释为什么 Kuhn–Tucker 条件失效。

第十四章 Integration / 积分

14.1 Introduction / 引言

Original (English)

The integral of a function f defines a new function, which may have the interpretation of an area or as the "reverse" of differentiation. For example, the computation of consumer surplus and dead-weight loss involve the calculation of area. Given a demand function $P(Q)$ and market price p , the consumer surplus is defined as the area under the demand curve above price and is used as a measure of the welfare generated by the existence of that market. The technique of integration provides a means of determining this area. In a second type of problem, it is useful to identify a function as the derivative of another function. For example, the rate of return on an asset – the change in the value of the asset per unit of time – may be viewed as the derivative of the value of the asset. Thus, knowing $v'(t)$, it is necessary to determine $v(t)$ from this derivative. These problems are solved by integration.

In what follows, the integral is formally defined. Section 14.4 considers welfare measures of producer and consumer surplus. Section 14.5 considers cost and profit for a firm. Taxation and dead-weight loss are examined in Section 14.6. Sections 14.7, 14.8, 14.9 and 14.10 discuss present-value calculations, Leibnitz's rule, measures of inequality and Ramsey pricing. Section 14.11 explores various measures of welfare.

翻译 (Chinese)

函数 f 的积分定义了一个新函数，它可以理解为面积，也可以理解为微分的“逆运算”。例如，消费者剩余和无谓损失的计算涉及面积计算。给定需求函数 $P(Q)$ 和市场价格 p ，消费者剩余定义为需求曲线之下、价格之上的面积，用于衡量市场存在所产生的福利。积分技术提供了确定该面积的方法。在第二类问题中，将某个函数识别为另一个函数的导数是很有用的。例如，资产的回报率——资产价值每单位时间的变化——可以视为资产价值的导数。因此，已知 $v'(t)$ ，需要由此导数确定 $v(t)$ 。这些问题都可以通过积分解决。

接下来, 将正式定义函数的积分。第 14.4 节考虑生产者和消费者剩余的福利度量。第 14.5 节考虑企业的成本与利润计算。税收与无谓损失在第 14.6 节讨论。第 14.7、14.8、14.9 和 14.10 节分别讨论现值计算、莱布尼茨法则、不平等度量与拉姆齐定价。第 14.11 节深入探讨各种福利度量。

14.2 The integral / 积分

Original (English)

Let f be a continuous function defined on $[a, b]$. The integral is denoted $\int_a^b f(x)dx$. Partition $[a, b]$ into subintervals $[x_1, x_2], [x_2, x_3], \dots, [x_{k-1}, x_k]$ with $a = x_1$ and $b = x_k$. The norm or mesh of the partition is $|P| = \max_i(x_{i+1} - x_i)$. For each $[x_i, x_{i+1}]$, let $s_i \in [x_i, x_{i+1}]$ and consider:

$$R(P, s) = \sum_{i=1}^{k-1} f(s_i)(x_{i+1} - x_i). \quad (14.1)$$

Definition 14.1: f is (Riemann) integrable on $[a, b]$ if there is a number I such that for every $\epsilon > 0$ there is a $\delta > 0$ such that if $|P| < \delta$, then $|R(P, s) - I| < \epsilon$ for any $s = (s_1, \dots, s_k)$. In this case, $I = \int_a^b f(x)dx$.

From a different perspective, integration is the "reverse" of differentiation. With a fixed, define $F(b) = \int_a^b f(x)dx$. Then $F(b + \epsilon) - F(b) = \int_b^{b+\epsilon} f(x)dx$. If ϵ is small, with f continuous, f is approximately constant on $[b, b + \epsilon]$ and equal to $f(b)$. Then $F(b + \epsilon) - F(b) \approx f(b)\epsilon$. As $\epsilon \rightarrow 0$, $F'(b) = f(b)$. From this perspective, integration recovers a function F whose derivative is f .

翻译 (Chinese)

令 f 为定义在 $[a, b]$ 上的连续函数。积分记为 $\int_a^b f(x)dx$ 。将 $[a, b]$ 划分为子区间 $[x_1, x_2], [x_2, x_3], \dots, [x_{k-1}, x_k]$ 其中 $a = x_1$, $b = x_k$ 。划分的范数或网格为 $|P| = \max_i(x_{i+1} - x_i)$ 。对每个 $[x_i, x_{i+1}]$, 令 $s_i \in [x_i, x_{i+1}]$, 考虑:

$$R(P, s) = \sum_{i=1}^{k-1} f(s_i)(x_{i+1} - x_i). \quad (14.1)$$

定义 14.1: 如果存在数 I , 使得对任意 $\epsilon > 0$, 存在 $\delta > 0$, 当 $|P| < \delta$ 时, 对任何 $s = (s_1, \dots, s_k)$ 均有 $|R(P, s) - I| < \epsilon$, 则称 f 在 $[a, b]$ 上 (黎曼) 可积。此时 $I = \int_a^b f(x)dx$ 。

从另一个角度看, 积分是微分的"逆运算"。固定 a , 定义 $F(b) = \int_a^b f(x)dx$ 。则 $F(b + \epsilon) - F(b) = \int_b^{b+\epsilon} f(x)dx$ 。若 ϵ 很小, 由 f 的连续性, f 在 $[b, b + \epsilon]$ 上近似为常数且等于 $f(b)$ 。

则 $F(b + \epsilon) - F(b) \approx f(b)\epsilon$ 。当 $\epsilon \rightarrow 0$ 时, $F'(b) = f(b)$ 。从这个角度看, 积分恢复了一个导数为 f 的函数 F 。

14.3 Common integrals and rules of integration / 常见积分与积分法则

Original (English)

It is common to write $\int f(x)dx$ to denote a function $F(x)$ with $F'(x) = f(x)$. F is not unique since adding a constant leaves the derivative unchanged. The definite integral satisfies $F(b) - F(a) = \int_a^b f(x)dx$.

Standard indefinite integrals:

$$\begin{aligned}\int x^n dx &= \frac{1}{n+1}x^{n+1} & \int \frac{1}{x} dx &= \ln x, \quad x > 0 \\ \int e^{ax} dx &= \frac{1}{a}e^{ax} & \int \sin(ax) dx &= -\frac{1}{a}\cos(ax) \\ \int \cos(ax) dx &= \frac{1}{a}\sin(ax)\end{aligned}$$

翻译 (Chinese)

通常用 $\int f(x)dx$ 表示满足 $F'(x) = f(x)$ 的函数 $F(x)$ 。 F 不唯一, 因为加上常数后导数不变。定积分满足 $F(b) - F(a) = \int_a^b f(x)dx$ 。

标准不定积分:

$$\begin{aligned}\int x^n dx &= \frac{1}{n+1}x^{n+1} & \int \frac{1}{x} dx &= \ln x, \quad x > 0 \\ \int e^{ax} dx &= \frac{1}{a}e^{ax} & \int \sin(ax) dx &= -\frac{1}{a}\cos(ax) \\ \int \cos(ax) dx &= \frac{1}{a}\sin(ax)\end{aligned}$$

14.4 Measuring welfare: surplus / 度量福利: 剩余

14.4.1 Consumer surplus / 消费者剩余

Original (English)

Suppose six consumers each require one unit of the good with willingness to pay: $wp_1 = 8, wp_2 = 6, wp_3 = 5, wp_4 = 4, wp_5 = 3, wp_6 = 1$. Each pays $p = 3$. Total consumer surplus is:

$$CS = (8 - 3) + (6 - 3) + (5 - 3) + (4 - 3) + (3 - 3) = 11.$$

If $D(p)$ is demand and $p(q)$ inverse demand:

$$CS = \int_3^8 D(p)dp = \int_0^4 [p(q) - 3]dq.$$

翻译 (Chinese)

假设有六个消费者，每人需要一单位商品，支付意愿分别为： $wp_1 = 8$, $wp_2 = 6$, $wp_3 = 5$, $wp_4 = 4$, $wp_5 = 3$, $wp_6 = 1$ 。每人支付 $p = 3$ 。总消费者剩余为：

$$CS = (8 - 3) + (6 - 3) + (5 - 3) + (4 - 3) + (3 - 3) = 11.$$

若 $D(p)$ 为需求函数， $p(q)$ 为反需求函数：

$$CS = \int_3^8 D(p)dp = \int_0^4 [p(q) - 3]dq.$$

14.4.2 Producer surplus / 生产者剩余

Original (English)

Firm supply is given by marginal cost. Let c_i denote the extra cost of the i th unit. Producer surplus is the sum of price less cost for each additional unit. With $S(p)$ the supply function, total surplus is:

$$\int_{\frac{1}{2}}^8 [D(p) - S(p)]dp.$$

In inverse form, $p_d(q)$ and $p_s(q)$:

$$\text{Total surplus} = \int_0^{\bar{q}} [p_d(q) - p_s(q)]dq,$$

where $\bar{q}$ satisfies $p_d(\bar{q}) = p_s(\bar{q})$.

翻译 (Chinese)

企业供给由边际成本给出。令 c_i 表示第 i 单位的额外成本。生产者剩余是每单位价格减去成本之和。以 $S(p)$ 表示供给函数，总剩余为：

$$\int_{\frac{1}{2}}^8 [D(p) - S(p)]dp.$$

以反函数形式 $p_d(q)$ 和 $p_s(q)$ 表示：

$$\text{总剩余} = \int_0^{\bar{q}} [p_d(q) - p_s(q)]dq,$$

其中 $\bar{q}$ 满足 $p_d(\bar{q}) = p_s(\bar{q})$ 。

14.4.3 Consumer surplus and welfare / 消费者剩余与福利

Original (English)

Suppose utility is $u(x, y) = x^{\frac{1}{2}} + y - px$. Maximizing by choice of x gives $\frac{1}{2}x^{-\frac{1}{2}} - p = 0$, or $x(p) = \frac{1}{4p^2}$, and inverse demand $p(x) = \frac{1}{2\sqrt{x}}$. Without good x , welfare is $u_0 = y$. With x available at price p , welfare is $u = y + \frac{1}{4p}$. Thus $u - u_0 = \frac{1}{4p}$. Consumer surplus is:

$$CS = \int_p^\infty x(t)dt = \frac{1}{4} \int_p^\infty t^{-2}dt = \frac{1}{4p}.$$

Thus $u - u_0 = CS$. With quasilinear utility, there is an exact connection between welfare measured by utility variation and consumer surplus.

翻译 (Chinese)

假设效用函数为 $u(x, y) = x^{\frac{1}{2}} + y - px$ 。通过选择 x 最大化效用给出 $\frac{1}{2}x^{-\frac{1}{2}} - p = 0$, 即 $x(p) = \frac{1}{4p^2}$, 反需求函数为 $p(x) = \frac{1}{2\sqrt{x}}$ 。没有商品 x 时, 福利为 $u_0 = y$ 。当 x 以价格 p 可获得时, 福利为 $u = y + \frac{1}{4p}$ 。因此 $u - u_0 = \frac{1}{4p}$ 。消费者剩余为:

$$CS = \int_p^\infty x(t)dt = \frac{1}{4} \int_p^\infty t^{-2}dt = \frac{1}{4p}.$$

因此 $u - u_0 = CS$ 。在拟线性效用下, 效用变化度量的福利与消费者剩余之间存在精确联系。

14.5 Cost, supply and profit / 成本、供给与利润

Original (English)

For a price-taking firm, profit at price p is $\pi(q) = pq - c(q)$, where $c(q)$ is total cost. Fixed cost $F = c(0)$. Variable cost $c_v(q) = c(q) - c(0)$. Maximizing profit gives $p = c'(q) = mc(q)$ - price equals marginal cost. The area under the marginal cost curve gives variable cost:

$$\int_0^q c'(q) dq = c(q) - c(0) = c_v(q).$$

Producer surplus equals profit plus fixed cost: $\pi(q) + F = pq - c_v(q)$. Average cost $AC(q) = \frac{c(q)}{q} = \frac{c_v(q)}{q} + \frac{F}{q}$. Average cost is minimized where $MC = AC$.

翻译 (Chinese)

对于价格接受企业, 在价格 p 下的利润为 $\pi(q) = pq - c(q)$, 其中 $c(q)$ 为总成本。固定成

本 $F = c(0)$ 。可变成本 $c_v(q) = c(q) - c(0)$ 。最大化利润给出 $p = c'(q) = mc(q)$ ——价格等于边际成本。边际成本曲线下的面积给出可变成本：

$$\int_0^q c'(q) dq = c(q) - c(0) = c_v(q).$$

生产者剩余等于利润加固定成本： $\pi(q) + F = pq - c_v(q)$ 。平均成本 $AC(q) = \frac{c(q)}{q} = \frac{c_v(q)}{q} + \frac{F}{q}$ 。平均成本在 $MC = AC$ 处最小化。

14.6 Taxation and consumer surplus / 税收与消费者剩余

Original (English)

Taxation imposes a burden and changes consumer behavior, leading to dead-weight loss. With demand $P(Q)$ and ad-valorem tax at rate t , $P_t = P_0(1+t)$. Dead-weight loss is:

$$D(t) = \int_{Q_t}^{Q_0} P(Q) dQ - P_0(Q_0 - Q_t). \quad (14.2)$$

EXAMPLE 14.1: Let $P(Q) = \frac{1}{Q}$ with exogenous P_0 . Then $Q_0 = \frac{1}{P_0}$, $Q_t = \frac{1}{P_0(1+t)}$. Dead-weight loss: $D(t) = \ln(1+t) - \frac{t}{1+t}$. We have $D'(t) = \frac{t}{(1+t)^2} > 0$ and $D''(t) = \frac{1-t}{(1+t)^3}$. Dead-weight loss is increasing at an increasing rate.

翻译 (Chinese)

税收施加负担并改变消费者行为，导致无谓损失。对于需求函数 $P(Q)$ 和税率为 t 的从价税， $P_t = P_0(1+t)$ 。无谓损失为：

$$D(t) = \int_{Q_t}^{Q_0} P(Q) dQ - P_0(Q_0 - Q_t). \quad (14.2)$$

例 14.1：令 $P(Q) = \frac{1}{Q}$ ， P_0 为外生给定。则 $Q_0 = \frac{1}{P_0}$ ， $Q_t = \frac{1}{P_0(1+t)}$ 。无谓损失： $D(t) = \ln(1+t) - \frac{t}{1+t}$ 。我们有 $D'(t) = \frac{t}{(1+t)^2} > 0$ 且 $D''(t) = \frac{1-t}{(1+t)^3}$ 。无谓损失以递增速率增加。

14.7 Present value / 现值

Original (English)

A firm has revenue flow $R(t)$ and cost $c(T)$ with discount rate r . Present value (profit) from operating to period T is:

$$\Pi(T) = \int_0^T R(t)e^{-rt}dt - c(T).$$

For how long should the firm operate? With $R(t) = kt$ and $c(T) = ce^{rT}$:

$$\Pi(T) = k \int_0^T te^{-rt}dt - ce^{rT}.$$

Using integration by parts, $\int_0^T te^{-rt}dt = \frac{1}{r^2} - \frac{1}{r^2}e^{-rT} - \frac{1}{r}Te^{-rT}$. The first-order condition gives $kTe^{-rT} - cre^{rT} = 0$.

翻译 (Chinese)

企业具有收入流 $R(t)$ 和成本 $c(T)$ ，贴现率为 r 。运营到时期 T 的现值（利润）为：

$$\Pi(T) = \int_0^T R(t)e^{-rt}dt - c(T).$$

企业应运营多长时间？对于 $R(t) = kt$ 和 $c(T) = ce^{rT}$ ：

$$\Pi(T) = k \int_0^T te^{-rt}dt - ce^{rT}.$$

使用分部积分法， $\int_0^T te^{-rt}dt = \frac{1}{r^2} - \frac{1}{r^2}e^{-rT} - \frac{1}{r}Te^{-rT}$ 。一阶条件给出 $kTe^{-rT} - cre^{rT} = 0$ 。

14.8 Leibnitz's rule / 莱布尼茨法则

Original (English)

Let $f(x, k)$ be continuous and $g(k) = \int_{a(k)}^{b(k)} f(x, k)dx$. Leibnitz's rule:

$$\frac{dg}{dk} = \int_{a(k)}^{b(k)} \frac{\partial f}{\partial k}dx + f(b(k), k)\frac{db(k)}{dk} - f(a(k), k)\frac{da(k)}{dk}.$$

Applied to the present-value calculation with $a(T) = 0$, $b(T) = T$:

$$\frac{d}{dT} \int_0^T R(t)e^{-rt}dt = \int_0^T \frac{\partial}{\partial T}(R(t)e^{-rt})dt + R(T)e^{-rT}\frac{dT}{dT} - 0 = R(T)e^{-rT}.$$

翻译 (Chinese)

令 $f(x, k)$ 连续， $g(k) = \int_{a(k)}^{b(k)} f(x, k)dx$ 。莱布尼茨法则：

$$\frac{dg}{dk} = \int_{a(k)}^{b(k)} \frac{\partial f}{\partial k}dx + f(b(k), k)\frac{db(k)}{dk} - f(a(k), k)\frac{da(k)}{dk}.$$

应用于现值计算, $a(T) = 0$, $b(T) = T$:

$$\frac{d}{dT} \int_0^T R(t)e^{-rt} dt = \int_0^T \frac{\partial}{\partial T} (R(t)e^{-rt}) dt + R(T)e^{-rT} \frac{dT}{dT} - 0 = R(T)e^{-rT}.$$

14.9 Inequality measures / 不平等度量

Original (English)

The Lorenz curve identifies the share of national income belonging to different percentiles of the population, ordered by relative income. Let F be a distribution function of income with density $f(y) > 0$. Given $p = F(y)$, $y(p)$ is the income level such that $100p\%$ have income $\leq y(p)$. The Lorenz curve:

$$L(p) = \frac{\int_0^{y(p)} xf(x)dx}{\int_0^\infty xf(x)dx} = p \frac{\bar{\mu}_p}{\bar{\mu}},$$

where $\bar{\mu}_p$ is the mean income of the poorest fraction p and $\bar{\mu}$ the population mean.

The Gini coefficient:

$$G = \frac{\frac{1}{2} - \int_0^1 L(p)dp}{\frac{1}{2}} = 1 - 2 \int_0^1 L(p)dp = \frac{A}{A+B}.$$

$L(p)$ is convex with $L'(p) = \frac{y(p)}{\bar{\mu}}$ and $L''(p) = \frac{1}{\bar{\mu}f(y(p))} > 0$.

翻译 (Chinese)

洛伦兹曲线表示按相对收入排序后, 不同人口百分位所拥有的国民收入份额。令 F 为收入分布函数, 密度为 $f(y) > 0$ 。给定 $p = F(y)$, 则 $y(p)$ 是使得 $100p\%$ 人口收入 $\leq y(p)$ 的收入水平。洛伦兹曲线:

$$L(p) = \frac{\int_0^{y(p)} xf(x)dx}{\int_0^\infty xf(x)dx} = p \frac{\bar{\mu}_p}{\bar{\mu}},$$

其中 $\bar{\mu}_p$ 是最穷的 p 比例人口的平均收入, $\bar{\mu}$ 为总体平均收入。

基尼系数:

$$G = \frac{\frac{1}{2} - \int_0^1 L(p)dp}{\frac{1}{2}} = 1 - 2 \int_0^1 L(p)dp = \frac{A}{A+B}.$$

$L(p)$ 是凸的, $L'(p) = \frac{y(p)}{\bar{\mu}}$, $L''(p) = \frac{1}{\bar{\mu}f(y(p))} > 0$ 。

14.10 Ramsey pricing / 拉姆齐定价

Original (English)

Efficient pricing sets price equal to marginal cost, but with fixed costs this is not viable. A firm supplies two markets with inverse demands p_x and p_y , variable cost $c(x, y)$, and fixed cost F . Surplus:

$$S(x, y) = \int_0^x p_x(\tilde{x})d\tilde{x} + \int_0^y p_y(\tilde{y})d\tilde{y} - c(x, y).$$

Maximizing surplus subject to revenue covering cost gives the Ramsey pricing rule: the markup of price over marginal cost in each market is proportional to the reciprocal of the elasticity in that market:

$$\frac{p_x - c_x}{p_x} = \frac{1}{k\epsilon_x}, \quad \frac{p_y - c_y}{p_y} = \frac{1}{k\epsilon_y}.$$

翻译 (Chinese)

有效定价要求价格等于边际成本，但在存在固定成本时这不可行。一家企业供应两个市场，反需求函数为 p_x 和 p_y ，可变成本为 $c(x, y)$ ，固定成本为 F 。剩余为：

$$S(x, y) = \int_0^x p_x(\tilde{x})d\tilde{x} + \int_0^y p_y(\tilde{y})d\tilde{y} - c(x, y).$$

在收入覆盖成本的约束下最大化剩余，得到拉姆齐定价规则：每个市场中价格超出边际成本的加成与该市场需求弹性的倒数成比例：

$$\frac{p_x - c_x}{p_x} = \frac{1}{k\epsilon_x}, \quad \frac{p_y - c_y}{p_y} = \frac{1}{k\epsilon_y}.$$

14.11 Welfare measures / 福利度量

14.11.1 Consumer surplus / 消费者剩余

Original (English)

Maximizing $u(x_1, \dots, x_n)$ subject to $\sum p_k x_k = m$ yields Marshallian demand $x_i(p_i, p_{-i}, m)$. When p_i increases to p'_i , consumer surplus change is:

$$\Delta CS = \int_{p_i}^{p'_i} x_i(\tilde{p}_i, p_j, m)d\tilde{p}_i. \quad (14.8)$$

EXAMPLE 14.6: With $u(x_i, x_j) = a \ln x_i + b \ln x_j$, demand is $x_i = \frac{a}{a+b} \frac{m}{p_i}$. For a price rise from p_i to p'_i :

$$\Delta CS = \frac{a}{a+b} m [\ln(p'_i) - \ln(p_i)].$$

翻译 (Chinese)

在 $\sum p_k x_k = m$ 约束下最大化 $u(x_1, \dots, x_n)$ 给出马歇尔需求 $x_i(p_i, p_{-i}, m)$ 。当 p_i 增加到 p'_i 时, 消费者剩余变化为:

$$\Delta CS = \int_{p_i}^{p'_i} x_i(\tilde{p}_i, p_{-i}, m) d\tilde{p}_i. \quad (14.8)$$

例 14.6: 对于 $u(x_i, x_j) = a \ln x_i + b \ln x_j$, 需求为 $x_i = \frac{a}{a+b} \frac{m}{p_i}$ 。价格从 p_i 上升到 p'_i :

$$\Delta CS = \frac{a}{a+b} m [\ln(p'_i) - \ln(p_i)].$$

14.11.2 Compensating variation / 补偿变化**Original (English)**

When p_i increases to p'_i , compensating variation (CV) is the additional income required to maintain the original utility level u_0 . The expenditure function $e(p_i, p_{-i}, u_0)$ gives the minimum expenditure to achieve utility u_0 at prices (p_i, p_{-i}) . Then:

$$CV = e(p'_i, p_{-i}, u_0) - e(p_i, p_{-i}, u_0) = \int_{p_i}^{p'_i} x_i^h(\tilde{p}_i, p_{-i}, u_0) d\tilde{p}_i. \quad (14.14)$$

The derivative of the expenditure function with respect to price gives Hicksian demand:

$$\frac{\partial e(p_i, p_j, u_0)}{\partial p_i} = x_i^h(p_i, p_j, u_0). \quad (14.11)$$

翻译 (Chinese)

当 p_i 增加到 p'_i 时, 补偿变化 (CV) 是维持原有效用水平 u_0 所需的额外收入。支出函数 $e(p_i, p_{-i}, u_0)$ 给出在价格 (p_i, p_{-i}) 下达到效用 u_0 的最小支出。则:

$$CV = e(p'_i, p_{-i}, u_0) - e(p_i, p_{-i}, u_0) = \int_{p_i}^{p'_i} x_i^h(\tilde{p}_i, p_{-i}, u_0) d\tilde{p}_i. \quad (14.14)$$

支出函数对价格的导数给出希克斯需求:

$$\frac{\partial e(p_i, p_j, u_0)}{\partial p_i} = x_i^h(p_i, p_j, u_0). \quad (14.11)$$

14.11.3 Equivalent variation / 等价变化

Original (English)

Equivalent variation (EV) asks what income reduction would have the same impact on utility as the price increase. With original utility u_0 and new utility u_1 after the price increase:

$$EV = e(p'_i, p_{-i}, u_1) - e(p_i, p_{-i}, u_1) = \int_{p_i}^{p'_i} x_i^h(\tilde{p}_i, p_{-i}, u_1) d\tilde{p}_i. \quad (14.20)$$

翻译 (Chinese)

等价变化 (EV) 考察什么样的收入减少会对效用产生与价格上涨相同的影响。设原有效用为 u_0 ，价格上涨后的新效用为 u_1 ：

$$EV = e(p'_i, p_{-i}, u_1) - e(p_i, p_{-i}, u_1) = \int_{p_i}^{p'_i} x_i^h(\tilde{p}_i, p_{-i}, u_1) d\tilde{p}_i. \quad (14.20)$$

14.11.4 Comparison of welfare measures / 福利度量的比较

Original (English)

The Slutsky equation connects Hicksian and Marshallian demand:

$$\frac{\partial x_i^h}{\partial p_i} = \frac{\partial x_i}{\partial p_i} + \frac{\partial x_i}{\partial m} x_i^h. \quad (14.23)$$

For a normal good ($\frac{\partial x_i}{\partial m} > 0$), $\frac{\partial x_i^h}{\partial p_i} < 0$:

$$\frac{\partial x_i}{\partial p_i} < \frac{\partial x_i^h}{\partial p_i} < 0. \quad (14.25)$$

Thus for a price increase: $EV < CS < CV$. In the quasilinear case with no income effects, all three measures coincide.

翻译 (Chinese)

斯卢茨基方程将希克斯需求与马歇尔需求联系起来：

$$\frac{\partial x_i^h}{\partial p_i} = \frac{\partial x_i}{\partial p_i} + \frac{\partial x_i}{\partial m} x_i^h. \quad (14.23)$$

对于正常品 ($\frac{\partial x_i}{\partial m} > 0$), $\frac{\partial x_i^h}{\partial p_i} < 0$ ：

$$\frac{\partial x_i}{\partial p_i} < \frac{\partial x_i^h}{\partial p_i} < 0. \quad (14.25)$$

因此对于价格上涨： $EV < CS < CV$ 。在拟线性情形（无收入效应）下，三种度量一致。

Exercises for Chapter 14 / 第 14 章习题

Original (English)

Exercise 14.1: With demand $Q = \left(\frac{p}{\alpha}\right)^{1/(\beta-1)}$, $0 < \beta < 1$, and constant marginal cost c , calculate consumer surplus under perfect competition.

Exercise 14.2: With demand $Q = \left(\frac{p}{\alpha}\right)^{\frac{1}{\beta-1}}$, $0 < \beta < 1$, calculate consumer surplus with a per-unit tax at rate t (competitive markets, price equals marginal cost c).

Exercise 14.3: Compare consumer surplus in Exercises 14.1 and 14.2.

Exercise 14.4: Demand is $q = a + bp$, $b < 0$. If the supplier can perfectly price discriminate, find total revenue $R^D(q)$.

Exercise 14.5: Given $P_d(Q)$ and $P_s(Q)$, with equilibrium Q_0 satisfying $P_d(Q_0) = P_s(Q_0)$. A value-added tax gives Q_t defined by $P_d(Q_t) = (1+t)P_s(Q_t)$. With $P_d(Q) = \frac{a}{Q}$ and $P_s(Q) = c + dQ$, $a = \frac{2c^2}{d}$, solve for Q_0 , Q_t and dead-weight loss.

Exercise 14.6: Repeat Exercise 14.5 with $P_d(Q) = a - bQ^2$ and $P_s(Q) = c + dQ^2$, $a > c$.

Exercise 14.7: With $u(x, y) = \alpha \ln(x) + \beta \ln(y)$, to achieve u_0 minimize $px + qy$ s.t. $u_0 = u(x, y)$. If the price of x increases from p to $p' > p$, how much more money is needed to maintain utility u_0 ?

Exercise 14.8: With $u(x, y) = \alpha \ln x + \beta \ln y$, prices p, q , income I , achieving utility level u . If x 's price rises from p to p' , how much more income is needed to be as well off as before?

翻译 (Chinese)

习题 14.1: 需求为 $Q = \left(\frac{p}{\alpha}\right)^{1/(\beta-1)}$, $0 < \beta < 1$, 边际成本为常数 c 。计算完全竞争下的消费者剩余。

习题 14.2: 需求为 $Q = \left(\frac{p}{\alpha}\right)^{\frac{1}{\beta-1}}$, $0 < \beta < 1$ 。计算征收每单位税率 t 时的消费者剩余 (竞争性市场, 价格等于边际成本 c)。

习题 14.3: 比较习题 14.1 和 14.2 中的消费者剩余。

习题 14.4: 需求为 $q = a + bp$, $b < 0$ 。若供应商能够完全价格歧视, 求总收入 $R^D(q)$ 。

习题 14.5: 给定 $P_d(Q)$ 和 $P_s(Q)$, 均衡量 Q_0 满足 $P_d(Q_0) = P_s(Q_0)$ 。增值税导致新数量 Q_t 由 $P_d(Q_t) = (1+t)P_s(Q_t)$ 定义。对于 $P_d(Q) = \frac{a}{Q}$ 和 $P_s(Q) = c + dQ$, $a = \frac{2c^2}{d}$, 求解 Q_0 、 Q_t 和无谓损失。

习题 14.6: 用 $P_d(Q) = a - bQ^2$ 和 $P_s(Q) = c + dQ^2$ ($a > c$) 重复习题 14.5。

习题 14.7: 效用为 $u(x, y) = \alpha \ln(x) + \beta \ln(y)$ 。为达到 u_0 ，在 $u_0 = u(x, y)$ 约束下最小化 $px + qy$ 。若 x 的价格从 p 上升到 $p' > p$ ，需要多少额外货币来维持效用 u_0 ？

习题 14.8: 效用为 $u(x, y) = \alpha \ln x + \beta \ln y$ ，价格为 p, q ，收入为 I ，当前达到效用水平 u 。若商品 x 的价格从 p 上升到 p' ，需要多少额外收入才能维持与之前相同的福利水平（即达到效用水平 u ）？

第十五章 Eigenvalues and Eigenvectors / 特征值与特征向量

15.1 Introduction / 引言

Original (English)

Eigenvalues and eigenvectors are scalars and vectors associated with any square matrix and identify key properties of the matrix. They have widespread use in economics. Consider a dynamic system $y_{t+1} = Ay_t$, where the evolution is governed by powers of A : $y_{t+j} = A^j y_t$. This task is greatly simplified by using eigenvectors and eigenvalues. Markov processes evolve according to $x_{t+1} = x_t T$, and a steady state $x^* = x^* T$ is associated with a specific eigenvalue of T . In least squares estimation, the distribution of estimated errors involves a matrix M whose eigenvalues play a useful role. In certain optimization problems, the optimum location relates to a symmetric matrix having $x'Ax > 0$ for all x . The study of each of these problems is facilitated through eigenvalues and eigenvectors.

Section 15.2 defines eigenvalues and eigenvectors. Section 15.3 describes dynamic models that illustrate their use. The Perron–Frobenius theorem is summarized in Section 15.4. Section 15.5 describes applications to webpage ranking and Leslie matrices. Section 15.6 discusses symmetric matrices. Section 15.7 treats complex eigenvalues. Section 15.8 considers diagonalization. Section 15.9 describes important properties of eigenvalues.

翻译 (Chinese)

特征值和特征向量是与任意方阵相关联的标量和向量，能够识别矩阵的关键性质。它们在经济中有广泛应用。考虑动态系统 $y_{t+1} = Ay_t$ ，其演化由 A 的幂控制： $y_{t+j} = A^j y_t$ 。利用特征向量和特征值可以大大简化这一任务。马尔可夫过程按照 $x_{t+1} = x_t T$ 演化，稳态 $x^* = x^* T$ 与矩阵 T 的某个特定特征值相关联。在最小二乘估计中，估计误差的分布涉及矩阵 M ，其特征值发挥着有用的作用。在某些最优化问题中，最优解的位置与满足 $x'Ax > 0$ 的对称矩阵密切相关。这些问题的研究都通过特征值和特征向量得到简化。

第 15.2 节定义了特征值和特征向量。第 15.3 节描述了说明其应用的动态模型。Perron–Frobenius 定理在第 15.4 节总结。第 15.5 节描述了网页排名和 Leslie 矩阵的应用。第 15.6 节讨论了对称矩阵。第 15.7 节处理复特征值。第 15.8 节考虑对角化。第 15.9 节描述了特征值的重要性质。

15.2 Definitions and basic properties / 定义与基本性质

Original (English)

Consider a square $n \times n$ matrix A . Given a vector x , this is transformed by A to $y = Ax$. In the special case where y is proportional to x , $y = \lambda x$, x is called an eigenvector of A .

EXAMPLE 15.1: Let

$$A = \begin{pmatrix} 4 & 6 \\ 2 & 8 \end{pmatrix}, x_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, x_2 = \begin{pmatrix} -3 \\ 1 \end{pmatrix}, \lambda_1 = 10, \lambda_2 = 2.$$

Notice $Ax_1 = 10x_1$ and $Ax_2 = 2x_2$. Applying A to x_1 scales x_1 up by 10; applying A to x_2 scales x_2 up by 2. For both vectors, the direction is unchanged.

Definition 15.1: Given a square matrix A , a non-zero vector x is an eigenvector of A if there is some λ such that $Ax = \lambda x$. The scalar λ is the eigenvalue associated with x (and may be 0).

Note that $Ax = \lambda x \implies A(cx) = \lambda(cx)$, so if x is an eigenvector, so is cx for any $c \neq 0$. Eigenvectors are conventionally normalized.

Since $Ax = \lambda x$ implies $(A - \lambda I)x = 0$, for $x \neq 0$, $(A - \lambda I)$ must be singular, so $|A - \lambda I| = 0$. This determinantal equation defines an n th-order polynomial in λ called the characteristic equation.

Definition 15.2: The characteristic polynomial of A is $p(\lambda) = |A - \lambda I|$.

For a 2×2 matrix:

$$|A - \lambda I| = \lambda^2 - \text{tr}(A)\lambda + \det(A). \quad (15.1)$$

The roots are $\lambda_1, \lambda_2 = \frac{\text{tr}(A) \pm \sqrt{\text{tr}(A)^2 - 4\det(A)}}{2}$.

翻译 (Chinese)

考虑 $n \times n$ 方阵 A 。给定向量 x ，经 A 变换为 $y = Ax$ 。在 y 与 x 成比例的特殊情形中， $y = \lambda x$ ，则称 x 为矩阵 A 的特征向量。

例 15.1：令

$$A = \begin{pmatrix} 4 & 6 \\ 2 & 8 \end{pmatrix}, x_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, x_2 = \begin{pmatrix} -3 \\ 1 \end{pmatrix}, \lambda_1 = 10, \lambda_2 = 2.$$

注意 $Ax_1 = 10x_1$ 和 $Ax_2 = 2x_2$ 。将 A 作用于 x_1 将其缩放 10 倍；作用于 x_2 将其缩放 2 倍。对两个向量，方向均未改变。

定义 15.1: 给定方阵 A ，若存在某个数 λ 使得 $Ax = \lambda x$ ，则称非零向量 x 为 A 的特征向量。标量 λ 称为与 x 相关联的特征值（可以为 0）。

注意 $Ax = \lambda x \implies A(cx) = \lambda(cx)$ ，因此若 x 是特征向量，则对任意 $c \neq 0$ ， cx 也是特征向量。通常通过标准化来处理这一非唯一性。

由于 $Ax = \lambda x$ 蕴含 $(A - \lambda I)x = 0$ ，对于 $x \neq 0$ ， $(A - \lambda I)$ 必为奇异矩阵，故 $|A - \lambda I| = 0$ 。这个行列式方程定义了关于 λ 的 n 次多项式，称为特征方程。

定义 15.2: A 的特征多项式为 $p(\lambda) = |A - \lambda I|$ 。

对于 2×2 矩阵：

$$|A - \lambda I| = \lambda^2 - \operatorname{tr}(A)\lambda + \det(A). \quad (15.1)$$

根为 $\lambda_1, \lambda_2 = \frac{\operatorname{tr}(A) \pm \sqrt{\operatorname{tr}(A)^2 - 4\det(A)}}{2}$ 。

Original (English)

REMARK 15.1: Left eigenvectors satisfy $yA = \lambda y$. The set of left eigenvalues is the same as the set of right eigenvalues (since $|A' - \lambda I| = |A - \lambda I|$), but the eigenvectors generally differ. Left and right eigenvectors satisfy orthogonality conditions.

For an $n \times n$ matrix A , the characteristic polynomial has n roots, which may be real or complex and may be repeated. From the fundamental theorem of algebra:

Theorem 15.1: Let $p(z) = a_n z^n + \cdots + a_0$ with $a_n \neq 0$. Then p has n roots $z_1, \dots, z_n$.

Even if a matrix has only real entries, eigenvalues and eigenvectors may be complex.

Theorem 15.2: Let A have distinct eigenvalues $\lambda_1, \dots, \lambda_n$ with eigenvectors $x_1, \dots, x_n$. Then $x_1, \dots, x_n$ are linearly independent.

REMARK 15.5 (Cayley–Hamilton theorem): A matrix satisfies its own characteristic equation: $p(A) = a_n A^n + \cdots + a_1 A + a_0 I = 0$.

If the eigenvalues are distinct, eigenvectors are independent. Diagonalization is possible: $AX = X\Lambda$, where Λ is diagonal with eigenvalues on the diagonal and X is the matrix of eigenvectors. Then $X^{-1}AX = \Lambda$ and $A = X\Lambda X^{-1}$.

Theorem 15.3: A is diagonalizable iff it has n linearly independent eigenvectors. A sufficient condition is that all eigenvalues be distinct.

翻译 (Chinese)

注 15.1: 左特征向量满足 $yA = \lambda y$ 。左特征值的集合与右特征值的集合相同 (因 $|A' - \lambda I| = |A - \lambda I|$)，但特征向量通常不同。左、右特征向量满足正交性条件。

对于 $n \times n$ 矩阵 A ，特征多项式有 n 个根，可以是实的或复的，可以有重根。由代数学基本定理：

定理 15.1: 令 $p(z) = a_n z^n + \cdots + a_0$, $a_n \neq 0$ 。则 p 有 n 个根 $z_1, \dots, z_n$ 。

即使矩阵只有实元素，特征值和特征向量也可能是复的。

定理 15.2: 令 A 有互异的特征值 $\lambda_1, \dots, \lambda_n$ 及相应的特征向量 $x_1, \dots, x_n$ 。则 $x_1, \dots, x_n$ 线性无关。

注 15.5 (Cayley–Hamilton 定理): 矩阵满足其自身的特征方程： $p(A) = a_n A^n + \cdots + a_1 A + a_0 I = 0$ 。

若特征值互异，特征向量线性无关。对角化是可能的： $AX = X\Lambda$ ，其中 Λ 是对角线上为特征值的对角矩阵， X 是特征向量矩阵。则 $X^{-1}AX = \Lambda$ 且 $A = X\Lambda X^{-1}$ 。

定理 15.3: A 可对角化当且仅当它有 n 个线性无关的特征向量。特征值全部互异是充分条件。

15.3 An application: dynamic models / 应用：动态模型

Original (English)

Consider $y_t = Ay_{t-1}$, so $y_t = A^t y_0$. With distinct eigenvalues, $A = X\Lambda X^{-1}$, so $A^k = X\Lambda^k X^{-1}$. Then:

$$y_t = A^t y_0 = X\Lambda^t X^{-1} y_0 = \sum_{j=1}^n (w^j y_0) x_j \lambda_j^t. \quad (15.2)$$

If λ_1 is the largest eigenvalue in absolute value, then for large t , $y_t \approx \lambda_1^t (w^1 y_0) x_1$ - y_t is approximately proportional to the eigenvector with the largest eigenvalue.

Definition 15.3: The spectral radius of A , $\rho(A) = \max\{|\lambda| : \lambda \text{ is an eigenvalue of } A\}$.

翻译 (Chinese)

考虑 $y_t = Ay_{t-1}$ ，故 $y_t = A^t y_0$ 。对于互异特征值， $A = X\Lambda X^{-1}$ ，因此 $A^k = X\Lambda^k X^{-1}$ 。则：

$$y_t = A^t y_0 = X\Lambda^t X^{-1} y_0 = \sum_{j=1}^n (w^j y_0) x_j \lambda_j^t. \quad (15.2)$$

若 λ_1 是绝对值最大的特征值，则对大的 t ， $y_t \approx \lambda_1^t (w^1 y_0) x_1$ —— y_t 近似正比于具有最大特征值的特征向量。

定义 15.3: A 的谱半径， $\rho(A) = \max\{|\lambda| : \lambda \text{ 是 } A \text{ 的特征值}\}$ 。

15.3.1 Supply and demand dynamics / 供需动态

Original (English)

Consider excess demand $e = d + Ap$. With equilibrium p^* satisfying $d = -Ap^*$ and price adjustment $p_{t+1} - p_t = e_t = A(p_t - p^*)$. Writing $v_t = p_t - p^*$: $v_{t+1} = (I + A)v_t = Bv_t$. With distinct eigenvalues, $B = Q\Lambda Q^{-1}$, so $v_{t+1} = Q\Lambda^t Q^{-1}v_0$. The price path moves quickly to be proportional to the eigenvector corresponding to the dominant eigenvalue.

翻译 (Chinese)

考虑超额需求 $e = d + Ap$ 。均衡价格 p^* 满足 $d = -Ap^*$ ，价格调整 $p_{t+1} - p_t = e_t = A(p_t - p^*)$ 。记 $v_t = p_t - p^*$: $v_{t+1} = (I + A)v_t = Bv_t$ 。对于互异特征值， $B = Q\Lambda Q^{-1}$ ，因此 $v_{t+1} = Q\Lambda^t Q^{-1}v_0$ 。价格路径迅速趋向与主导特征值相应的特征向量成正比。

15.3.2 A model of population growth / 人口增长模型

Original (English)

Consider a population where mature individuals produce one offspring per period and never die. Tracking mature and young: $m_{t+1} = m_t + y_t$, $y_{t+1} = m_t$. In matrix form:

$$\begin{pmatrix} m_{t+1} \\ y_{t+1} \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} m_t \\ y_t \end{pmatrix}.$$

The total population follows the Fibonacci equation: $x_t = x_{t-1} + x_{t-2}$, $t \geq 2$, with $x_0 = 0$, $x_1 = 1$. In matrix form:

$$\begin{pmatrix} x_{t+1} \\ z_{t+1} \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x_t \\ z_t \end{pmatrix}, \quad z_t = x_{t-1}.$$

The eigenvalues are $\lambda_1, \lambda_2 = \frac{1 \pm \sqrt{5}}{2}$ with eigenvectors $q_1 = (\lambda_1, 1)$, $q_2 = (\lambda_2, 1)$. The long-run solution:

$$x_{t+1} = \frac{1}{\sqrt{5}} \left[\left(\frac{1 + \sqrt{5}}{2} \right)^{t+1} - \left(\frac{1 - \sqrt{5}}{2} \right)^{t+1} \right].$$

翻译 (Chinese)

考虑一个种群，成年个体每期产生一个后代且永不死亡。跟踪成年和幼年数量： $m_{t+1} = m_t + y_t$, $y_{t+1} = m_t$ 。矩阵形式：

$$\begin{pmatrix} m_{t+1} \\ y_{t+1} \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} m_t \\ y_t \end{pmatrix}.$$

总人口遵循斐波那契方程： $x_t = x_{t-1} + x_{t-2}$, $t \geq 2$, $x_0 = 0$, $x_1 = 1$ 。矩阵形式：

$$\begin{pmatrix} x_{t+1} \\ z_{t+1} \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x_t \\ z_t \end{pmatrix}, \quad z_t = x_{t-1}.$$

特征值为 $\lambda_1, \lambda_2 = \frac{1 \pm \sqrt{5}}{2}$ ，特征向量为 $q_1 = (\lambda_1, 1)$, $q_2 = (\lambda_2, 1)$ 。长期解：

$$x_{t+1} = \frac{1}{\sqrt{5}} \left[\left(\frac{1 + \sqrt{5}}{2} \right)^{t+1} - \left(\frac{1 - \sqrt{5}}{2} \right)^{t+1} \right].$$

15.4 The Perron–Frobenius theorem / Perron–Frobenius 定理

Original (English)

The Perron–Frobenius theorem gives properties of eigenvalues and eigenvectors of square matrices with non-negative or positive entries. A positive matrix (or vector) has all entries > 0 ; a non-negative matrix (or vector) has all entries ≥ 0 .

Theorem 15.4: Let A be a positive square matrix ($A \gg 0$). Then:

1. There is a real positive eigenvalue λ^* with $\lambda^* = \rho(A)$. This eigenvalue is simple.
2. All other eigenvalues λ satisfy $|\lambda| < \lambda^*$.
3. λ^* has a real positive eigenvector x , $x_i > 0$ for all i .
4. x is the only real non-negative eigenvector.
5. λ^* satisfies: $\min_i \sum_j a_{ij} \leq \lambda^* \leq \max_i \sum_j a_{ij}$.

Theorem 15.5: Let A be a non-negative irreducible square matrix. Then similar results hold: there is a real positive $\lambda^* = \rho(A)$ with positive eigenvector, unique up to scalar multiple. For any other eigenvalue λ , $|\lambda| \leq \lambda^*$.

翻译 (Chinese)

Perron–Frobenius 定理给出了具有非负或正元素的方阵的特征值和特征向量的性质。正矩阵（或向量）的所有元素 > 0 ；非负矩阵（或向量）的所有元素 ≥ 0 。

定理 15.4: 令 A 为正矩阵 ($A \gg 0$)。则:

1. 存在正实特征值 λ^* , $\lambda^* = \rho(A)$, 该特征值是简单的。
2. 所有其他特征值 λ 满足 $|\lambda| < \lambda^*$ 。
3. λ^* 有正实特征向量 x , 对所有的 i 有 $x_i > 0$ 。
4. x 是唯一的实非负特征向量。
5. λ^* 满足: $\min_i \sum_j a_{ij} \leq \lambda^* \leq \max_i \sum_j a_{ij}$ 。

定理 15.5: 令 A 为非负不可约方阵。则类似结果成立: 存在正实特征值 $\lambda^* = \rho(A)$ 及正特征向量（在标量倍数意义下唯一）。对任何其他特征值 λ , $|\lambda| \leq \lambda^*$ 。

15.5 Some applications / 一些应用

15.5.1 Webpage ranking / 网页排名

Original (English)

The Internet can be viewed as a collection of interconnected nodes (webpages). The adjacency matrix $A = \{\ell_{ij}\}$ has $\ell_{ij} = 1$ if page i links to page j . Let $h(i)$ be the number of outlinks from i . Define transition probabilities $t_{ij} = \frac{1}{h(i)}$ if $\ell_{ij} = 1$ and $h(i) > 0$, else 0. The page rank r_i satisfies:

$$r_i = \sum_{\{j|\ell_{ji}=1\}} \frac{r_j}{h(j)} = \sum_j t_{ji} r_j.$$

Writing $r = (r_1, \dots, r_n)$, we seek $r = rT$, or $T'r = r$. By the Perron–Frobenius theorem, there is a unique eigenvalue–eigenvector pair (λ, r) with $\lambda > 0$ and $r \geq 0$ satisfying $T'r = r$. Since $T\mathbf{1} = \mathbf{1}$, $\lambda = 1$ and the ranking vector is the eigenvector associated with eigenvalue 1.

Under iteration, T^k converges to a matrix with all rows equal to the (normalized) eigenvector associated with the dominant eigenvalue.

翻译 (Chinese)

互联网可以视为相互连接的节点（网页）的集合。邻接矩阵 $A = \{\ell_{ij}\}$ 中，若网页 i 链接到网页 j ，则 $\ell_{ij} = 1$ 。令 $h(i)$ 为从 i 出发的链接数。定义转移概率 $t_{ij} = \frac{1}{h(i)}$ （若 $\ell_{ij} = 1$ 且

$h(i) > 0$), 否则为 0。网页排名 r_i 满足:

$$r_i = \sum_{\{j|\ell_{ji}=1\}} \frac{r_j}{h(j)} = \sum_j t_{ji} r_j.$$

记 $r = (r_1, \dots, r_n)$, 我们寻求 $r = rT$, 即 $T'r = r$ 。由 Perron–Frobenius 定理, 存在唯一的特征值–特征向量对 (λ, r) ($\lambda > 0$, $r \geq 0$) 满足 $T'r = r$ 。因 $T\mathbf{1} = \mathbf{1}$, 故 $\lambda = 1$, 排名向量即为特征值 1 对应的特征向量。

在迭代下, T^k 收敛到一个所有行都等于主导特征值对应 (标准化) 特征向量的矩阵。

15.5.2 Leslie matrices / Leslie 矩阵

Original (English)

Consider a species with maximum lifespan n . An individual aged j gives birth to f_j offspring and survives with probability s_j . The population evolves as $x_{t+1} = Lx_t$, where:

$$L = \begin{pmatrix} f_1 & f_2 & \cdots & f_{n-1} & f_n \\ s_1 & 0 & \cdots & 0 & 0 \\ 0 & s_2 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & s_{n-1} & 0 \end{pmatrix}.$$

A long-run steady state satisfies $x = Lx$, so x is an eigenvector of L . If there is a positive eigenvalue λ_1 with norm larger than any other, the long-run population is proportional to $x_1 \lambda_1^t z_{01}$.

翻译 (Chinese)

考虑一个最大寿命为 n 期的物种。年龄为 j 的个体生育 f_j 个后代, 并以概率 s_j 存活到下一期。种群按 $x_{t+1} = Lx_t$ 演化, 其中:

$$L = \begin{pmatrix} f_1 & f_2 & \cdots & f_{n-1} & f_n \\ s_1 & 0 & \cdots & 0 & 0 \\ 0 & s_2 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & s_{n-1} & 0 \end{pmatrix}.$$

长期稳态满足 $x = Lx$, 因此 x 是 L 的特征向量。若存在范数大于其他所有特征值的正特征值 λ_1 , 则长期种群正比于 $x_1 \lambda_1^t z_{01}$ 。

15.6 Eigenvalues and eigenvectors of symmetric matrices / 对称矩阵的特征值与特征向量

Original (English)

For symmetric matrices, eigenvalues and eigenvectors have additional properties.

Proposition 15.1: For an $n \times n$ symmetric matrix:

1. The eigenvalues are all real.
2. If eigenvalues are distinct, eigenvectors are orthogonal.
3. n linearly independent orthogonal eigenvectors can always be found.

Definition 15.4: A square matrix C is orthogonal if $C'C = I$.

Proposition 15.2: If C is orthogonal, it is non-singular.

Proposition 15.4: If A is symmetric, there is an orthogonal matrix C such that $C'AC$ is diagonal with eigenvalues on the diagonal.

Proposition 15.8: If A is positive definite, then all eigenvalues are real and $\lambda_i > 0$ for all i . If positive semi-definite, $\lambda_i \geq 0$.

翻译 (Chinese)

对于对称矩阵，特征值和特征向量具有额外的性质。

命题 15.1: 对于 $n \times n$ 对称矩阵：

1. 所有特征值都是实数。
2. 若特征值互异，则特征向量正交。
3. 总能找到 n 个线性无关的正交特征向量。

定义 15.4: 若方阵 C 满足 $C'C = I$ ，则称其为正交矩阵。

命题 15.2: 若 C 正交，则它非奇异。

命题 15.4: 若 A 对称，则存在正交矩阵 C 使得 $C'AC$ 为对角矩阵，对角线上为特征值。

命题 15.8: 若 A 正定，则所有特征值均为实数且 $\lambda_i > 0$ 。若半正定，则 $\lambda_i \geq 0$ 。

15.7 Real and complex eigenvalues and eigenvectors / 实与复特征值和特征向量

Original (English)

Even if all entries of a matrix are real, eigenvalues and eigenvectors may be complex. Some basic complex algebra is necessary. A complex number $z = a + ib$ has conjugate $\bar{z} = a - bi$, norm $|z| = \sqrt{a^2 + b^2}$.

For complex vectors, the inner product is defined as $x \cdot y = \sum x_i \bar{y}_i$, which is not symmetric: $x \cdot y = \overline{y \cdot x}$.

Definition 15.6: A complex-valued square matrix A with $A^* = A$ is Hermitian (where A^* is the conjugate transpose).

Definition 15.7: A is normal if $A^*A = AA^*$. Hermitian matrices are normal.

Definition 15.8: A is unitary if $A^*A = AA^* = I$, so $A^{-1} = A^*$.

翻译 (Chinese)

即使矩阵的所有元素都是实数，特征值和特征向量也可能是复的。一些基本的复代数知识是必要的。复数 $z = a + ib$ 的共轭为 $\bar{z} = a - bi$ ，范数为 $|z| = \sqrt{a^2 + b^2}$ 。

对于复向量，内积定义为 $x \cdot y = \sum x_i \bar{y}_i$ ，它不对称： $x \cdot y = \overline{y \cdot x}$ 。

定义 15.6: 满足 $A^* = A$ 的复值方阵 A 称为 Hermitian 矩阵（其中 A^* 是共轭转置）。

定义 15.7: 若 $A^*A = AA^*$ ，则称 A 为正规矩阵。Hermitian 矩阵都是正规矩阵。

定义 15.8: 若 $A^*A = AA^* = I$ ，则称 A 为酉矩阵，此时 $A^{-1} = A^*$ 。

15.8 Diagonalization / 对角化

Original (English)

Definition 15.9: A square matrix A is diagonalizable if there is a non-singular matrix Q and a diagonal matrix D such that $A = QDQ^{-1}$.

Theorem 15.6: A is diagonalizable iff it has n linearly independent eigenvectors.

Definition 15.10: A is unitarily diagonalizable if there is a unitary matrix P such that $P^{-1}AP = P^*AP = \Lambda$ (diagonal).

Theorem 15.7: A is unitarily diagonalizable iff it is normal.

翻译 (Chinese)

定义 15.9: 若存在非奇异矩阵 Q 和对角矩阵 D 使得 $A = QDQ^{-1}$, 则方阵 A 可对角化。

定理 15.6: A 可对角化当且仅当它有 n 个线性无关的特征向量。

定义 15.10: 若存在酉矩阵 P 使得 $P^{-1}AP = P^*AP = \Lambda$ (对角矩阵), 则称 A 可酉对角化。

定理 15.7: A 可酉对角化当且仅当它是正规矩阵。

15.9 Properties of eigenvalues and eigenvectors / 特征值与特征向量的性质

Original (English)

Theorem 15.8: Let $\lambda_1, \dots, \lambda_n$ be the roots of A (possibly complex). Then:

- $\text{tr}(A) = \sum_{i=1}^n \lambda_i$.
- $\det(A) = \prod_{i=1}^n \lambda_i$.
- $\text{tr}(A^k) = \sum_i \lambda_i^k$.

Theorem 15.9: If A is Hermitian, all eigenvalues are real.

Theorem 15.10: For any square matrix A :

- (i) The roots of A' are conjugates of roots of A .
- (ii) If A is non-singular, roots of A^{-1} are λ_i^{-1} .
- (iii) If q is real and λ an eigenvalue of A , then $q\lambda$ is an eigenvalue of qA .
- (iv) If q is a positive integer, λ^q is an eigenvalue of A^q .

Proposition 15.9: $r(A)$ = number of non-zero roots of A .

翻译 (Chinese)

定理 15.8: 令 $\lambda_1, \dots, \lambda_n$ 为 A 的 (可能为复的) 根。则:

- $\text{tr}(A) = \sum_{i=1}^n \lambda_i$.
- $\det(A) = \prod_{i=1}^n \lambda_i$.

- $\text{tr}(A^k) = \sum_i \lambda_i^k$ 。

定理 15.9: 若 A 是 Hermitian 矩阵, 则所有特征值均为实数。

定理 15.10: 对任意方阵 A :

- (i) A' 的根是 A 的根的共轭。
- (ii) 若 A 非奇异, 则 A^{-1} 的根为 λ_i^{-1} 。
- (iii) 若 q 为实数, λ 为 A 的特征值, 则 $q\lambda$ 是 qA 的特征值。
- (iv) 若 q 为正整数, λ^q 是 A^q 的特征值。

命题 15.9: $r(A) = A$ 的非零根的个数。

15.9.1 Largest eigenvalues and eigenvalue multiplicity / 最大特征值与特征值重数

Original (English)

Given eigenvectors $v_1, \dots, v_n$ and eigenvalues $\lambda_1, \dots, \lambda_n$, let $z = \sum \alpha_i v_i$. Then $A^t z = \sum \alpha_i \lambda_i^t v_i$. If $|\lambda_1| > |\lambda_i|$ for $i \neq 1$, then for large t , $A^t z \approx \lambda_1^t \alpha_1 v_1$.

More generally, let $\rho = \max_i |\lambda_i|$ and $\hat{\lambda}_i = \lambda_i / \rho$. Assume $\lambda_1 = \rho$. Let $\mathcal{I} = \{i \mid |\lambda_i| = \rho\}$. Then for $j \notin \mathcal{I}$, $\hat{\lambda}_j^t \rightarrow 0$ as $t \rightarrow \infty$, and $A^t z \approx \rho^t \sum_{j \in \mathcal{I}} \alpha_j v_j$.

Definition 15.11: If root λ_j occurs exactly k times, it has algebraic multiplicity k . If $k = 1$, it is a simple root.

Definition 15.12: The geometric multiplicity of an eigenvalue is the number of linearly independent eigenvectors with that eigenvalue.

If n_i is the algebraic multiplicity and m_i the geometric multiplicity of λ_i , then $m_i \leq n_i$.

翻译 (Chinese)

给定特征向量 $v_1, \dots, v_n$ 和特征值 $\lambda_1, \dots, \lambda_n$, 令 $z = \sum \alpha_i v_i$ 。则 $A^t z = \sum \alpha_i \lambda_i^t v_i$ 。若对 $i \neq 1$ 有 $|\lambda_1| > |\lambda_i|$, 则对大的 t , $A^t z \approx \lambda_1^t \alpha_1 v_1$ 。

更一般地, 令 $\rho = \max_i |\lambda_i|$ 和 $\hat{\lambda}_i = \lambda_i / \rho$ 。设 $\lambda_1 = \rho$ 。令 $\mathcal{I} = \{i \mid |\lambda_i| = \rho\}$ 。则对 $j \notin \mathcal{I}$, 当 $t \rightarrow \infty$ 时 $\hat{\lambda}_j^t \rightarrow 0$, 且 $A^t z \approx \rho^t \sum_{j \in \mathcal{I}} \alpha_j v_j$ 。

定义 15.11: 若根 λ_j 恰好出现 k 次, 则其代数重数为 k 。若 $k = 1$, 则为单根。

定义 15.12: 特征值的几何重数是具有该特征值的线性无关特征向量的个数。

若 n_i 为 λ_i 的代数重数, m_i 为其几何重数, 则 $m_i \leq n_i$ 。

Exercises for Chapter 15 / 第 15 章习题

Original (English)

Exercise 15.1: Let $A = \begin{pmatrix} 1 & 2 \\ 4 & 3 \end{pmatrix}$. Give the characteristic equation and find eigenvalues and eigenvectors.

Exercise 15.2: Let $B = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 4 & 2 \\ 4 & 0 & 2 \end{pmatrix}$. Give the characteristic equation and find eigenvalues and eigenvectors.

Exercise 15.3: Let $A = \begin{pmatrix} -\frac{6}{4} & \frac{6}{4} \\ -\frac{12}{4} & \frac{11}{4} \end{pmatrix}$. Let λ_1, λ_2 be eigenvalues with eigenvectors x_1, x_2 . Use $A = X\Lambda X^{-1}$ to show $A^k \rightarrow 0$ as $k \rightarrow \infty$.

Exercise 15.4: Let A be $n \times n$. Suppose the sum of absolute values of entries in any column is ≤ 1 . Show all eigenvalues satisfy $|\lambda| \leq 1$.

Exercise 15.5: Show A is non-singular iff all eigenvalues are non-zero.

Exercise 15.6: For $A = \begin{pmatrix} 2 & 5 \\ 4 & 3 \end{pmatrix}$, $p(\lambda) = \lambda^2 - 5\lambda - 14 = 0$. Verify the Cayley–Hamilton theorem: $A^2 - 5A - 14I = 0$.

Exercise 15.7: Using the Cayley–Hamilton theorem, show how to compute A^k in terms of A and I for a 2×2 matrix.

Exercise 15.8: Show how the Cayley–Hamilton theorem may be used to find the inverse of A .

Exercise 15.9: For $B = \begin{pmatrix} 1 & 5 \\ 2 & 3 \end{pmatrix}$, find eigenvalues and eigenvectors.

Exercise 15.10: Find eigenvalues of $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$.

Exercise 15.11: For complex matrix $A = \begin{pmatrix} 2+2i & 4+i \\ 3+2i & 5+2i \end{pmatrix}$, calculate the determinant and inverse.

翻译 (Chinese)

习题 15.1: 令 $A = \begin{pmatrix} 1 & 2 \\ 4 & 3 \end{pmatrix}$ 。给出特征方程并求特征值和特征向量。

习题 15.2: 令 $B = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 4 & 2 \\ 4 & 0 & 2 \end{pmatrix}$ 。给出特征方程并求特征值和特征向量。

习题 15.3: 令 $A = \begin{pmatrix} -\frac{6}{4} & \frac{6}{4} \\ -\frac{12}{4} & \frac{11}{4} \end{pmatrix}$ 。令 λ_1, λ_2 为特征值, x_1, x_2 为特征向量。利用 $A = X\Lambda X^{-1}$ 证明当 $k \rightarrow \infty$ 时 $A^k \rightarrow 0$ 。

习题 15.4: 令 A 为 $n \times n$ 矩阵。假设任意列中元素的绝对值之和 ≤ 1 。证明所有特征值满足 $|\lambda| \leq 1$ 。

习题 15.5: 证明 A 非奇异当且仅当所有特征值非零。

习题 15.6: 对于 $A = \begin{pmatrix} 2 & 5 \\ 4 & 3 \end{pmatrix}$, $p(\lambda) = \lambda^2 - 5\lambda - 14 = 0$ 。验证 Cayley-Hamilton 定理: $A^2 - 5A - 14 = 0$ 。

习题 15.7: 利用 Cayley-Hamilton 定理, 说明如何对 2×2 矩阵用 A 和 I 来计算 A^k 。

习题 15.8: 说明 Cayley-Hamilton 定理如何用于求 A 的逆。

习题 15.9: 对于 $B = \begin{pmatrix} 1 & 5 \\ 2 & 3 \end{pmatrix}$, 求特征值和特征向量。

习题 15.10: 求 $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ 的特征值。

习题 15.11: 对于复矩阵 $A = \begin{pmatrix} 2+2i & 4+i \\ 3+2i & 5+2i \end{pmatrix}$, 计算行列式和逆矩阵。

第一部分

Dynamic Models, Probability & Distributions / 动态模型、概率与分布

第十六章 Differential Equations / 微分方程

16.1 Introduction / 引言

Original (English)

In this chapter, dynamic models are considered where some variable varies continuously as a function of some other variable (such as time). If y is a function of x , then assuming differentiability, the derivative is written $\frac{dy}{dx}$ or $y'(x)$. Knowing y , it is easy to determine the derivative $y'(x)$. However, in a variety of problems, the situation is reversed — the derivative, $y'(x)$ is known, but not the actual function $y(x)$. And in this class of problem, the task is to determine $y(x)$ from knowledge of $y'(x)$, called solving the differential equation. Section 16.1.1 provides a few motivating examples to illustrate how such dynamics may arise. Generally, it is difficult to determine $y(x)$ from knowledge of x . For example, given $y(x) = \ln(1+x^2) + xe^x$, differentiation gives $y'(x) = \frac{2x}{1+x^2} + e^x + xe^x = \frac{2x}{1+x^2} + (1+x)e^x$. But if given $y'(x) = \frac{2x}{1+x^2} + (1+x)e^x$, it is more difficult to see that $y(x) = \ln(1+x^2) + xe^x$.

翻译 (Chinese)

本章考虑动态模型，其中某个变量作为另一个变量（如时间）的函数连续变化。若 y 是 x 的函数，则在可微的假设下，导数记为 $\frac{dy}{dx}$ 或 $y'(x)$ 。已知 y ，容易确定导数 $y'(x)$ 。然而，在许多问题中，情况是相反的——导数 $y'(x)$ 已知，而实际函数 $y(x)$ 未知。在这类问题中，任务是从 $y'(x)$ 的知识确定 $y(x)$ ，这称为求解微分方程。第 16.1.1 节提供了一些启发性例子，说明这种动力学如何产生。一般来说，从对 x 的知识确定 $y(x)$ 是困难的。例如，给定 $y(x) = \ln(1+x^2) + xe^x$ ，求导得 $y'(x) = \frac{2x}{1+x^2} + e^x + xe^x = \frac{2x}{1+x^2} + (1+x)e^x$ 。但若给定 $y'(x) = \frac{2x}{1+x^2} + (1+x)e^x$ ，则更难看出 $y(x) = \ln(1+x^2) + xe^x$ 。

Original (English)

The general approach to this problem is to categorize differential equations into classes of problems of varying difficulty. In a first-order differential equation, the dynamic relation is expressed by a function f with $y' = f(y, x)$. If this can be written in the form $a_1(x)y'(x) + a_0(x)y = b(x)$ it is called a first-order linear differential equation. Similarly, a second-order differential equation has the form $y''(x) = f(y', y, x)$, where y'' is the second derivative of the (unknown) function y . Again, the task is to find the function $y(x)$ that satisfies this equation. Depending on the function f , this problem may be extremely difficult to solve. In Section 16.2, first-order linear differential equations are considered. These arise in a number of economic models — such as growth models. Turning to non-linear differential equations, these are typically difficult to solve but there are some specific functional forms for which a solution may easily be found. Two non-linear first-order differential equations (the logistic and Bernoulli equations) are introduced in Section 16.3. Given a first-order differential equation, $y'(x) = f(y, x)$, a fundamental question concerns the existence of a solution: is there a unique function y that satisfies this equation? Section 16.4 provides sufficient conditions for the existence of a unique solution. Second-order linear differential equations are considered in Section 16.5. Finally, Section 16.6 introduces linear systems of differential equations and uses that framework to study price dynamics in a multimarket system.

翻译 (Chinese)

解决此问题的一般方法是将微分方程按难度分类。在一阶微分方程中，动态关系由一个函数 f 以 $y' = f(y, x)$ 表示。如果可以写成 $a_1(x)y'(x) + a_0(x)y = b(x)$ 的形式，则称为一阶线性微分方程。类似地，二阶微分方程具有形式 $y''(x) = f(y', y, x)$ ，其中 y'' 是（未知）函数 y 的二阶导数。同样，任务是找到满足该方程的函数 $y(x)$ 。取决于函数 f ，此问题可能极难求解。第 16.2 节讨论一阶线性微分方程，这些方程出现在许多经济模型中，如增长模型。转向非线性微分方程，这些通常难以求解，但有一些特定的函数形式可以容易地找到解。第 16.3 节介绍了两个非线性一阶微分方程（逻辑斯蒂方程和伯努利方程）。给定一阶微分方程 $y'(x) = f(y, x)$ ，一个基本问题是解的存在性：是否存在唯一的函数 y 满足该方程？第 16.4 节提供了唯一解存在的充分条件。第 16.5 节讨论二阶线性微分方程。最后，第 16.6 节介绍了线性微分方程组，并用该框架研究多市场体系中的价格动态。

16.1.1 Preliminary discussion / 初步讨论

Original (English)

A differential equation is a rule governing the change in some variable. For example, suppose there is a stock of wealth invested and growing at the rate r . Thus, if the current stock is $y(t)$, and the value a short time later is $y(t + \Delta t)$ then the percentage increase is given by $\frac{y(t+\Delta t)-y(t)}{y(t)}$. The

increase per unit of time over the time period t to $t + \Delta t$ is given by $\frac{1}{\Delta t} \frac{y(t+\Delta t) - y(t)}{y(t)} = \frac{1}{y(t)} \frac{y(t+\Delta t) - y(t)}{\Delta t}$. If $y(t)$ varies smoothly with t , then this is approximately $\frac{1}{y(t)} \frac{dy}{dt}$, as Δt goes to 0. With a constant growth rate of r , if $y(t)$ is the stock of wealth at time t , the percentage change in the stock is given by the equation $\frac{1}{y(t)} \frac{dy}{dt} = r$, so that

$$\frac{dy}{dt} - ry(t) = 0.$$

This is a differential equation where the change in y over time is governed by the current value of y and the parameter r . A solution to the differential equation is a function $y(t)$ satisfying the expression $\frac{dy}{dt} - ry(t) = 0$ for each t . This differential equation is easily solved, recognizing that $\frac{d \ln y(t)}{dt} = \frac{1}{y(t)} \frac{dy}{dt}$, so that

$$\ln y(t) - \ln y(t_0) = \int_{t_0}^t \frac{d \ln y(t)}{dt} dt = \int_{t_0}^t \frac{1}{y(t)} \frac{dy(t)}{dt} dt = \int_{t_0}^t r dt = r(t - t_0).$$

Thus, $\ln y(t) = \ln y(t_0) + r(t - t_0)$, so that $y(t) = e^{\ln y(t_0) + r(t - t_0)} = y(t_0)e^{r(t - t_0)}$.

翻译 (Chinese)

微分方程是控制某个变量变化规律的方程。例如，假设有一笔财富以速率 r 投资并增长。因此，若当前存量为 $y(t)$ ，短时间后的价值为 $y(t + \Delta t)$ ，则百分比增长由 $\frac{y(t+\Delta t) - y(t)}{y(t)}$ 给出。在时间区间 t 到 $t + \Delta t$ 上每单位时间的增长由 $\frac{1}{\Delta t} \frac{y(t+\Delta t) - y(t)}{y(t)} = \frac{1}{y(t)} \frac{y(t+\Delta t) - y(t)}{\Delta t}$ 给出。若 $y(t)$ 随 t 平滑变化，则当 Δt 趋于 0 时，这近似为 $\frac{1}{y(t)} \frac{dy}{dt}$ 。在增长率为常数 r 的情况下，若 $y(t)$ 是时刻 t 的财富存量，则存量的百分比变化由方程 $\frac{1}{y(t)} \frac{dy}{dt} = r$ 给出，从而

$$\frac{dy}{dt} - ry(t) = 0.$$

这是一个微分方程，其中 y 随时间的变化由 y 的当前值和参数 r 控制。微分方程的解是一个函数 $y(t)$ ，对每个 t 满足表达式 $\frac{dy}{dt} - ry(t) = 0$ 。此微分方程容易求解，注意到 $\frac{d \ln y(t)}{dt} = \frac{1}{y(t)} \frac{dy}{dt}$ ，从而

$$\ln y(t) - \ln y(t_0) = \int_{t_0}^t \frac{d \ln y(t)}{dt} dt = \int_{t_0}^t \frac{1}{y(t)} \frac{dy(t)}{dt} dt = \int_{t_0}^t r dt = r(t - t_0).$$

因此， $\ln y(t) = \ln y(t_0) + r(t - t_0)$ ，从而 $y(t) = e^{\ln y(t_0) + r(t - t_0)} = y(t_0)e^{r(t - t_0)}$ 。

Original (English)

More generally, given $y(x)$, the function $y(x)$ is an unknown function of the variable x (or t when it is convenient to emphasize time), but the derivative $y'(x)$ is specified in terms of some function f :

$$\frac{dy}{dx} = f(y, x).$$

Such an equation is called an ordinary differential equation of order 1. Writing $y^{(n)}$ for the n th derivative of y with respect to x , an equation of the form

$$y^{(n)} = f(y^{(n-1)}, y^{(n-2)}, \dots, y^{(1)}, y, x)$$

is called an n th-order ordinary differential equation. A solution is a function $y(x)$ satisfying the equation.

The next example illustrates a simple application (a first-order linear differential equation), determining the flow of payments required to amortize a loan in some fixed period of time.

翻译 (Chinese)

更一般地, 给定 $y(x)$, 函数 $y(x)$ 是变量 x (或为强调时间而用 t) 的未知函数, 但导数 $y'(x)$ 由某个函数 f 确定:

$$\frac{dy}{dx} = f(y, x).$$

这样的方程称为一阶常微分方程。记 $y^{(n)}$ 为 y 关于 x 的 n 阶导数, 则形如

$$y^{(n)} = f(y^{(n-1)}, y^{(n-2)}, \dots, y^{(1)}, y, x)$$

的方程称为 n 阶常微分方程。解是满足该方程的函数 $y(x)$ 。

下一个例子说明了一个简单的应用 (一阶线性微分方程), 确定在某个固定期限内摊销贷款所需的支付流。

16.2 First-order linear differential equations / 一阶线性微分方程

Original (English)

A first-order linear differential equation has the form:

$$a_1(x) \frac{dy}{dx} + a_0(x)y = b(x).$$

This expression defines the movement of y , $\frac{dy}{dx}$, in terms of its current value, y , and the value of a variable x . A solution to the differential equation is a function $y(x)$ satisfying this equation on some interval in the domain of x .

翻译 (Chinese)

一阶线性微分方程具有如下形式:

$$a_1(x) \frac{dy}{dx} + a_0(x)y = b(x).$$

该表达式以 y 的当前值和变量 x 的值定义了 y 的运动 $\frac{dy}{dx}$ 。微分方程的解是一个在 x 的定义域上某个区间内满足该方程的函数 $y(x)$ 。

16.2.1 Constant coefficients / 常系数

Original (English)

When the coefficients a_1 , a_0 and b do not depend on x and $a_1(x) \neq 0$, $a_1 \frac{dy}{dx} + a_0 y = b$ may be rewritten $\frac{dy}{dx} + \frac{a_0}{a_1} y = \frac{b}{a_1}$, giving the simple first-order differential equation:

$$\frac{dy}{dx} + \alpha y = \beta. \quad (16.1)$$

A particular solution, y , is any function satisfying Equation 16.1. If y_a and y_b satisfy Equation 16.1, then $y_c = y_a - y_b$ satisfies the homogeneous Equation:

$$\frac{dy}{dx} + \alpha y = 0. \quad (16.2)$$

A solution to the homogeneous equation is called a complementary solution. Therefore, any two particular solutions differ by a complementary solution. So, given any particular solution, y_p , to Equation 16.1, every particular solution, y , to Equation 16.1 is equal to y_p plus a complementary solution, y_c : $y = y_p + y_c$. The family of particular solutions is called the general solution.

Therefore, every solution to Equation 16.1 may be found by finding a particular solution and adding a complementary solution. Using this observation, the general solution may be obtained by finding a particular solution and then all complementary solutions.

For a particular solution, with constant coefficients, this is easy to find: set $\frac{dy}{dx} = 0$ and pick the constant y to solve $\alpha y = \beta$: $y_p(x) = \frac{\beta}{\alpha}$. Next, consider the homogeneous case where $\beta = 0$, $\frac{dy}{dx} + \alpha y = 0$, and rearrange to give $\frac{1}{y} \frac{dy}{dx} = -\alpha$. Integrating,

$$\int_{x_0}^x \frac{1}{y} \frac{dy}{dx} dx = \int_{x_0}^x \frac{1}{y(\tilde{x})} y'(\tilde{x}) d\tilde{x} = \ln y(x) - \ln y(x_0) = - \int_{x_0}^x \alpha d\tilde{x} = -\alpha(x - x_0).$$

Therefore, $\ln y(x) = [\ln y(x_0) + \alpha x_0] - \alpha x$. So, the complementary solution is:

$$y_c(x) = [e^{\ln y(x_0) + \alpha x_0}] e^{-\alpha x} = c_0 e^{-\alpha x}.$$

Note that c_0 is an indeterminate constant of integration, so this expression is a family of functions. Combining the particular and complementary solutions gives:

$$y(x) = y_p(x) + y_c(x) = \frac{\beta}{\alpha} + c_0 e^{-\alpha x}. \quad (16.3/16.4)$$

If the value of y is given at x_0 , $y_0 = y(x_0)$, then $y_0 = \frac{\beta}{\alpha} + c_0 e^{-\alpha x_0}$ or $y_0 - \frac{\beta}{\alpha} = c_0 e^{-\alpha x_0}$ or $c_0 = \left[y_0 - \frac{\beta}{\alpha} \right] e^{\alpha x_0}$. Therefore,

$$y(x) = y_p(x) + y_c(x) = \frac{\beta}{\alpha} + \left[y_0 - \frac{\beta}{\alpha} \right] e^{-\alpha(x-x_0)}. \quad (16.5/16.6)$$

翻译 (Chinese)

当系数 a_1 、 a_0 和 b 不依赖于 x 且 $a_1(x) \neq 0$ 时, $a_1 \frac{dy}{dx} + a_0 y = b$ 可重写为 $\frac{dy}{dx} + \frac{a_0}{a_1} y = \frac{b}{a_1}$, 得到简单的一阶微分方程:

$$\frac{dy}{dx} + \alpha y = \beta. \quad (16.1)$$

特解 y 是满足方程 16.1 的任意函数。若 y_a 和 y_b 满足方程 16.1, 则 $y_c = y_a - y_b$ 满足齐次方程:

$$\frac{dy}{dx} + \alpha y = 0. \quad (16.2)$$

齐次方程的解称为互补解。因此, 任意两个特解之间相差一个互补解。所以, 给定方程 16.1 的任意特解 y_p , 方程 16.1 的每个特解 y 等于 y_p 加上某个互补解 y_c : $y = y_p + y_c$ 。所有特解的族称为通解。

因此, 方程 16.1 的每个解都可以通过找到一个特解并加上一个互补解来求得。利用这一观察, 通解可以通过找到一个特解然后找到所有互补解来获得。

对于常系数情况, 特解很容易找到: 令 $\frac{dy}{dx} = 0$ 并选择常数 y 使得 $\alpha y = \beta$: $y_p(x) = \frac{\beta}{\alpha}$ 。接下来, 考虑齐次情况, 即 $\beta = 0$, $\frac{dy}{dx} + \alpha y = 0$, 重排得 $\frac{1}{y} \frac{dy}{dx} = -\alpha$ 。积分得,

$$\int_{x_0}^x \frac{1}{y} \frac{dy}{dx} dx = \int_{x_0}^x \frac{1}{y(\tilde{x})} y'(\tilde{x}) d\tilde{x} = \ln y(x) - \ln y(x_0) = - \int_{x_0}^x \alpha d\tilde{x} = -\alpha(x - x_0).$$

因此, $\ln y(x) = [\ln y(x_0) + \alpha x_0] - \alpha x$ 。故互补解为:

$$y_c(x) = [e^{\ln y(x_0) + \alpha x_0}] e^{-\alpha x} = c_0 e^{-\alpha x}.$$

注意 c_0 是一个未定的积分常数, 因此该表达式是一个函数族。将特解和互补解组合起来得到:

$$y(x) = y_p(x) + y_c(x) = \frac{\beta}{\alpha} + c_0 e^{-\alpha x}. \quad (16.3/16.4)$$

若已知 y 在 x_0 处的值 $y_0 = y(x_0)$, 则 $y_0 = \frac{\beta}{\alpha} + c_0 e^{-\alpha x_0}$ 或 $y_0 - \frac{\beta}{\alpha} = c_0 e^{-\alpha x_0}$, 即 $c_0 = [y_0 - \frac{\beta}{\alpha}] e^{\alpha x_0}$ 。因此,

$$y(x) = y_p(x) + y_c(x) = \frac{\beta}{\alpha} + \left[y_0 - \frac{\beta}{\alpha} \right] e^{-\alpha(x-x_0)}. \quad (16.5/16.6)$$

Original (English)

REMARK 16.1: What happens to the solution function $y(x)$ as x becomes very large? From the solution 16.4, it is clear that the answer depends on α . If $\alpha > 0$, then $e^{-\alpha x}$ goes to 0 as x becomes large, and if $\alpha < 0$, then $e^{-\alpha x}$ goes to infinity (or minus infinity, depending on c_0) as x becomes large. Call the first case “stable” and the second “unstable”. Considering the complementary equation, $\frac{dy}{dx} + \alpha y = 0$, a trial solution for y is $y(x) = e^{rx}$, so that the equation becomes $0 = re^{rx} + \alpha e^{rx} = (r + \alpha)e^{rx}$ giving $(r + \alpha) = 0$. This is called the characteristic polynomial of the homogeneous equation and has root $r = -\alpha$. Therefore, the dynamic system is stable if the characteristic equation has a negative root.

EXAMPLE 16.2: The Harrod–Domar model of economic growth postulates that output and capital are related according the formula $K = \nu Y$, where K is the aggregate amount of capital, Y national output and ν a coefficient determining the amount of capital required per unit of output produced. Savings, S , is related to output according to the formula, $S = sY$, so that savings is proportional to Y . Investment I equals saving S , $I = S$. Capital changes over time owing to two effects: investment raises capital and depreciation reduces capital, where depreciation is assumed to be at rate δ . Therefore, over an increment of time Δt , the change in capital is

$$\Delta K = (I - \delta K)\Delta t = (sY - \delta \nu Y)\Delta t,$$

using $I = sY$ and $K = \nu Y$. Since $K = \nu Y$, $\Delta K = \nu \Delta Y$ and so,

$$\nu \Delta Y = (I - \delta K)\Delta t, \quad \nu \Delta Y = \left(\frac{s}{\nu} - \delta \right) Y \Delta t, \quad \frac{\Delta Y}{\Delta t} = \frac{s - \delta \nu}{\nu} Y.$$

In continuous time, this is written: $\frac{dY}{dt} = (\frac{s}{\nu} - \delta)Y$, or $\frac{dY}{dt} - (\frac{s}{\nu} - \delta)Y = 0$. From the earlier discussion, this has solution:

$$Y(t) = Y(0) \cdot e^{(\frac{s}{\nu} - \delta)t}.$$

EXAMPLE 16.3: The linear demand and supply model is given by the equations:

$$Q_d = \alpha - \beta p, \quad Q_s = \gamma + \delta p,$$

where $\alpha \gg 0$ and $\beta, \delta > 0$. Define a dynamic model by setting $\frac{dp}{dt} = \kappa(Q_d - Q_s)$, where $\kappa > 0$ is a constant. Thus, $\frac{dp}{dt} = \kappa[\alpha - \gamma - (\beta + \delta)p]$ or

$$\frac{dp}{dt} + \kappa(\beta + \delta)p = \kappa(\alpha - \gamma).$$

Solving,

$$p(t) = c_0 e^{-\kappa(\beta+\delta)t} + \frac{\alpha - \gamma}{\beta + \delta}.$$

At $t = 0$, $p(0) = c_0 + \frac{\alpha - \gamma}{\beta + \delta}$, so that $c_0 = p(0) - \frac{\alpha - \gamma}{\beta + \delta}$, and substituting this gives,

$$p(t) = \left(p(0) - \frac{\alpha - \gamma}{\beta + \delta} \right) e^{-\kappa(\beta+\delta)t} + \frac{\alpha - \gamma}{\beta + \delta}.$$

EXAMPLE 16.4: According to Newton's second law, $F = ma$, force equals mass times acceleration. If s denotes speed or velocity, $\frac{ds}{dt} = a(t)$, acceleration is equal to change in speed. Thus $F = m \frac{ds}{dt}$. Consider an object falling from the sky. In a vacuum, gravity creates force proportional to mass: mg where g is a gravity parameter (g is roughly 9.8 m/s^2). So, for an object falling in a vacuum, $mg = m \frac{ds}{dt}$. However, in the air, an object with mass encounters resistance proportional to speed, tending to offset gravity. Suppose the proportionality factor is k , so that the acceleration formula becomes $m \frac{ds}{dt} = mg - ks(t)$. Or, $\frac{ds}{dt} + \frac{k}{m}s(t) = g$. Terminal velocity $\frac{ds}{dt} = 0$ occurs when speed satisfies $mg - ks(t) = 0$.

Consider the differential equation describing a falling object: $\frac{ds}{dt} + \frac{k}{m}s(t) = g$. Suppose that at time $t_0 = 0$, the object begins falling so that, initially, the speed is 0: $s_0 = s(t_0) = 0$. According to the formula:

$$s(t) = \left(s_0 - \frac{gm}{k} \right) e^{-kt} + \frac{gm}{k} = \frac{gm}{k}(1 - e^{-kt}).$$

Thus, the speed converges to $s(\infty) = \frac{gm}{k}$, so that the speed at time t as a fraction of terminal velocity is $f = \frac{s(t)}{s(\infty)} = (1 - e^{-kt})$. Therefore, $e^{-kt} = (1 - f)$ and $-kt = \ln(1 - f)$ so that $t = -\frac{1}{k} \ln(1 - f)$. For a skydiver in an outstretched position, terminal velocity is in the approximate range of 120–130 miles per hour.

翻译 (Chinese)

注 16.1: 当 x 变得非常大时, 解函数 $y(x)$ 会怎样? 从解 16.4 可以清楚地看出答案取决于 α 。若 $\alpha > 0$, 则当 x 变大时 $e^{-\alpha x}$ 趋于 0; 若 $\alpha < 0$, 则当 x 变大时 $e^{-\alpha x}$ 趋于无穷 (或负无穷, 取决于 c_0)。称第一种情况为“稳定”, 第二种为“不稳定”。考虑互补方程 $\frac{dy}{dx} + \alpha y = 0$, 对 y 的试探解为 $y(x) = e^{rx}$, 使得方程变为 $0 = re^{rx} + \alpha e^{rx} = (r + \alpha)e^{rx}$, 得到 $(r + \alpha) = 0$ 。这称为齐次方程的特征多项式, 具有根 $r = -\alpha$ 。因此, 若特征方程有负根, 则动态系统是稳定的。

例 16.2: 哈罗德-多马经济增长模型假设产出与资本按公式 $K = \nu Y$ 相关, 其中 K 是资本总量, Y 是国民产出, ν 是生产每单位产出所需资本量的系数。储蓄 S 按公式 $S = sY$ 与产出相关, 因此储蓄与 Y 成比例。投资 I 等于储蓄 S , $I = S$ 。资本随时间变化源于两种效应: 投资增加资本, 折旧减少资本, 其中折旧假设以速率 δ 发生。因此, 在时间增量

Δt 内, 资本的变化为 $\Delta K = (I - \delta K)\Delta t = (sY - \delta\nu Y)\Delta t$, 利用了 $I = sY$ 和 $K = \nu Y$ 。由于 $K = \nu Y$, $\Delta K = \nu\Delta Y$, 故 $\nu\Delta Y = (s - \delta\nu)Y\Delta t$, $\frac{\Delta Y}{\Delta t} = \frac{s - \delta\nu}{\nu}Y$ 。在连续时间中, 这写为: $\frac{dY}{dt} = (\frac{s}{\nu} - \delta)Y$, 或 $\frac{dY}{dt} - (\frac{s}{\nu} - \delta)Y = 0$ 。根据前面的讨论, 其解为: $Y(t) = Y(0) \cdot e^{(\frac{s}{\nu} - \delta)t}$ 。

例 16.3: 线性需求与供给模型由下列方程给出: $Q_d = \alpha - \beta p$, $Q_s = \gamma + \delta p$, 其中 $\alpha \gg 0$ 且 $\beta, \delta > 0$ 。通过设定 $\frac{dp}{dt} = \kappa(Q_d - Q_s)$ 来定义动态模型, 其中 $\kappa > 0$ 是常数。因此, $\frac{dp}{dt} = \kappa[\alpha - \gamma - (\beta + \delta)p]$ 或 $\frac{dp}{dt} + \kappa(\beta + \delta)p = \kappa(\alpha - \gamma)$ 。求解得 $p(t) = c_0 e^{-\kappa(\beta + \delta)t} + \frac{\alpha - \gamma}{\beta + \delta}$ 。在 $t = 0$ 处, $p(0) = c_0 + \frac{\alpha - \gamma}{\beta + \delta}$, 故 $c_0 = p(0) - \frac{\alpha - \gamma}{\beta + \delta}$, 代入得 $p(t) = (p(0) - \frac{\alpha - \gamma}{\beta + \delta})e^{-\kappa(\beta + \delta)t} + \frac{\alpha - \gamma}{\beta + \delta}$ 。

例 16.4: 根据牛顿第二定律, $F = ma$, 力等于质量乘加速度。若 s 表示速度, 则 $\frac{ds}{dt} = a(t)$, 加速度等于速度的变化。因此 $F = m\frac{ds}{dt}$ 。考虑一个从空中落下的物体。在真空中, 重力产生与质量成比例的力: mg , 其中 g 是以加速度衡量的重力参数 (g 约为 9.8 m/s^2)。因此, 对于在真空中下落的物体, $mg = m\frac{ds}{dt}$ 。然而, 在空气中, 有质量的物体遇到与速度成比例的阻力, 倾向于抵消重力。假设比例因子为 k , 则加速度公式变为 $m\frac{ds}{dt} = mg - ks(t)$ 。或 $\frac{ds}{dt} + \frac{k}{m}s(t) = g$ 。终极速度 $\frac{ds}{dt} = 0$ 发生在速度满足 $mg - ks(t) = 0$ 时。

考虑描述下落物体的微分方程: $\frac{ds}{dt} + \frac{k}{m}s(t) = g$ 。假设在时刻 $t_0 = 0$ 物体开始下落, 因此初始速度为 0: $s_0 = s(t_0) = 0$ 。根据公式: $s(t) = (s_0 - \frac{gm}{k})e^{-kt} + \frac{gm}{k} = \frac{gm}{k}(1 - e^{-kt})$ 。因此, 速度收敛到 $s(\infty) = \frac{gm}{k}$, 故时刻 t 的速度占终极速度的比例为 $f = \frac{s(t)}{s(\infty)} = (1 - e^{-kt})$ 。因此, $e^{-kt} = (1 - f)$ 且 $-kt = \ln(1 - f)$, 故 $t = -\frac{1}{k}\ln(1 - f)$ 。对于四肢伸展的跳伞者, 终极速度大约在每小时 120–130 英里的范围内。

16.2.2 Variable coefficients / 变系数

Original (English)

A general first-order (linear) differential equation has the form:

$$a_1(x)\frac{dy}{dx} + a_0(x)y = b(x), \quad (16.7)$$

where $a_1(x) \neq 0, \forall x$. Equation 16.7 is called the inhomogeneous equation. The homogeneous equation is obtained when $b(x) = 0, \forall x$:

$$a_1(x)\frac{dy}{dx} + a_0(x)y = 0. \quad (16.8)$$

To solve Equation 16.7 write it in the form:

$$\frac{dy}{dx} + \frac{a_0(x)}{a_1(x)}y = \frac{b(x)}{a_1(x)},$$

or, with $\alpha(x) = \frac{a_0(x)}{a_1(x)}$, and $\beta(x) = \frac{b(x)}{a_1(x)}$,

$$\frac{dy}{dx} + \alpha(x)y = \beta(x). \quad (16.9)$$

The corresponding homogeneous equation is:

$$\frac{dy}{dx} + \alpha(x)y = 0.$$

As noted earlier, a solution to Equation 16.9, y_p , is called a particular solution. The family of all particular solutions is called the general solution. If y_c is a solution of the homogeneous equation (the complementary solution) and y_p is a particular solution, then $y_p + y_c$ is also a particular solution. Note that if y_c is a solution to the homogeneous equation, then so is ky_c for any constant k (since $\frac{dy_c}{dx} + \alpha(x)y_c = 0$ implies $k\frac{dy_c}{dx} + \alpha(x)ky_c = 0$). Therefore, if there is a solution, there will be an infinite number of solutions, requiring some initial condition to provide a unique solution.

Considering the homogeneous component, $\frac{dy}{dx} + \alpha(x)y = 0$, rewrite as:

$$\frac{1}{y} \frac{dy}{dx} = -\alpha(x).$$

Integrating, $\ln y(x) - \ln y(x_0) = -\int_{x_0}^x \alpha(\tilde{x})d\tilde{x}$, so that

$$y(x) = y(x_0)e^{-\int_{x_0}^x \alpha(\tilde{x})d\tilde{x}} \quad (16.10)$$

is a solution of the homogeneous equation. We can check this by differentiating:

$$\frac{dy(x)}{dx} = y(x_0)[- \alpha(x)]e^{-\int_{x_0}^x \alpha(\tilde{x})d\tilde{x}} = [-\alpha(x)]y(x).$$

Solving the general (inhomogeneous) case, given in Equation 16.9, to obtain a particular solution makes use of integrating factors, described next.

翻译 (Chinese)

一般的一阶（线性）微分方程具有形式：

$$a_1(x)\frac{dy}{dx} + a_0(x)y = b(x), \quad (16.7)$$

其中对所有 x 有 $a_1(x) \neq 0$ 。方程 16.7 称为非齐次方程。当对所有 x 有 $b(x) = 0$ 时得到齐次方程：

$$a_1(x)\frac{dy}{dx} + a_0(x)y = 0. \quad (16.8)$$

为求解方程 16.7，将其写成： $\frac{dy}{dx} + \frac{a_0(x)}{a_1(x)}y = \frac{b(x)}{a_1(x)}$ ，或记 $\alpha(x) = \frac{a_0(x)}{a_1(x)}$ ， $\beta(x) = \frac{b(x)}{a_1(x)}$ ，

$$\frac{dy}{dx} + \alpha(x)y = \beta(x). \quad (16.9)$$

相应的齐次方程为: $\frac{dy}{dx} + \alpha(x)y = 0$ 。

如前所述, 方程 16.9 的一个解 y_p 称为特解。所有特解的族称为通解。若 y_c 是齐次方程的解 (互补解), y_p 是特解, 则 $y_p + y_c$ 也是特解。注意, 若 y_c 是齐次方程的解, 则对任意常数 k , ky_c 也是齐次方程的解 (因为 $\frac{dy_c}{dx} + \alpha(x)y_c = 0$ 意味着 $k\frac{dy_c}{dx} + \alpha(x)ky_c = 0$)。因此, 若存在一个解, 则将有无穷多个解, 需要某个初始条件来提供唯一解。

考虑齐次部分 $\frac{dy}{dx} + \alpha(x)y = 0$, 重写为 $\frac{1}{y}\frac{dy}{dx} = -\alpha(x)$ 。积分得 $\ln y(x) - \ln y(x_0) = -\int_{x_0}^x \alpha(\tilde{x})d\tilde{x}$, 故

$$y(x) = y(x_0)e^{-\int_{x_0}^x \alpha(\tilde{x})d\tilde{x}} \quad (16.10)$$

是齐次方程的解。我们可通过求导来验证:

$$\frac{dy(x)}{dx} = y(x_0)[- \alpha(x)]e^{-\int_{x_0}^x \alpha(\tilde{x})d\tilde{x}} = [-\alpha(x)]y(x).$$

求解方程 16.9 给出的一般 (非齐次) 情况以获得特解, 需要用到积分因子, 下面描述。

Integrating factors / 积分因子

Original (English)

Returning to the general equation, Equation 16.9,

$$\frac{dy(x)}{dx} + \alpha(x)y(x) = \beta(x),$$

observe that the left side is similar in form to $q'(x)y(x) + q(x)y'(x)$, which is the derivative of $q(x)y(x)$. If Equation 16.9 is expressed in this way, recovery of the function $y(x)$ is simplified. To carry out this procedure, multiply the equation by some function $q(x)$ (called an integrating factor) to give:

$$q(x)\frac{dy(x)}{dx} + q(x)\alpha(x)y(x) = q(x)\beta(x).$$

If $q'(x) = q(x)\alpha(x)$, then the expression is

$$q(x)\frac{dy(x)}{dx} + q'(x)y(x) = \frac{d}{dx}\{q(x)y(x)\} = q(x)\beta(x). \quad (16.11)$$

In this case, integrating both sides of Equation 16.11 gives:

$$\int_{x_0}^x \frac{d}{d\tilde{x}}\{q(\tilde{x})y(\tilde{x})\}d\tilde{x} = q(x)y(x) - q(x_0)y(x_0) = \int_{x_0}^x q(\tilde{x})\beta(\tilde{x})d\tilde{x}, \quad (16.12)$$

or

$$q(x)y(x) = q(x_0)y(x_0) + \int_{x_0}^x q(\tilde{x})\beta(\tilde{x})d\tilde{x}, \quad (16.13)$$

so that a solution for $y(x)$ is given:

$$y(x) = \frac{1}{q(x)} \left[q(x_0)y(x_0) + \int_{x_0}^x q(\tilde{x})\beta(\tilde{x})d\tilde{x} \right]. \quad (16.14)$$

What is q ? Since $q'(x) = q(x)\alpha(x)$, $\frac{q'(x)}{q(x)} = \alpha(x)$ so that $\ln q(x)|_{x_0}^x = \int_{x_0}^x \alpha(\tilde{x})d\tilde{x}$, or $\ln \frac{q(x)}{q(x_0)} = \int_{x_0}^x \alpha(\tilde{x})d\tilde{x}$, so that

$$\frac{q(x)}{q(x_0)} = e^{\int_{x_0}^x \alpha(\tilde{x})d\tilde{x}} \quad \text{or} \quad q(x) = q(x_0)e^{\int_{x_0}^x \alpha(\tilde{x})d\tilde{x}}.$$

Therefore,

$$y(x) = \frac{y(x_0) + \int_{x_0}^x e^{\int_{x_0}^{\tilde{x}} \alpha(\tilde{x})d\tilde{x}} \beta(\tilde{x})d\tilde{x}}{e^{\int_{x_0}^x \alpha(\tilde{x})d\tilde{x}}}. \quad (16.15)$$

Notice that when $\beta(x) = 0$ for all x , this expression reduces to Equation 16.10, the solution to the homogeneous equation.

翻译 (Chinese)

回到一般方程 16.9, $\frac{dy(x)}{dx} + \alpha(x)y(x) = \beta(x)$, 注意到左边在形式上类似于 $q'(x)y(x) + q(x)y'(x)$, 后者是 $q(x)y(x)$ 的导数。若方程 16.9 以这种方式表示, 则函数 $y(x)$ 的恢复得以简化。为执行此过程, 将方程乘以某个函数 $q(x)$ (称为积分因子), 得到:

$$q(x)\frac{dy(x)}{dx} + q(x)\alpha(x)y(x) = q(x)\beta(x).$$

若 $q'(x) = q(x)\alpha(x)$, 则该表达式为

$$q(x)\frac{dy(x)}{dx} + q'(x)y(x) = \frac{d}{dx}\{q(x)y(x)\} = q(x)\beta(x). \quad (16.11)$$

在此情形下, 对方程 16.11 两边积分得:

$$\int_{x_0}^x \frac{d}{d\tilde{x}}\{q(\tilde{x})y(\tilde{x})\}d\tilde{x} = q(x)y(x) - q(x_0)y(x_0) = \int_{x_0}^x q(\tilde{x})\beta(\tilde{x})d\tilde{x}, \quad (16.12)$$

或

$$q(x)y(x) = q(x_0)y(x_0) + \int_{x_0}^x q(\tilde{x})\beta(\tilde{x})d\tilde{x}, \quad (16.13)$$

从而 $y(x)$ 的解由下式给出:

$$y(x) = \frac{1}{q(x)} \left[q(x_0)y(x_0) + \int_{x_0}^x q(\tilde{x})\beta(\tilde{x})d\tilde{x} \right]. \quad (16.14)$$

q 是什么? 由于 $q'(x) = q(x)\alpha(x)$, $\frac{q'(x)}{q(x)} = \alpha(x)$, 故 $\ln q(x)|_{x_0}^x = \int_{x_0}^x \alpha(\tilde{x})d\tilde{x}$, 即 $\ln \frac{q(x)}{q(x_0)} = \int_{x_0}^x \alpha(\tilde{x})d\tilde{x}$, 因此

$$\frac{q(x)}{q(x_0)} = e^{\int_{x_0}^x \alpha(\tilde{x})d\tilde{x}} \quad \text{或} \quad q(x) = q(x_0)e^{\int_{x_0}^x \alpha(\tilde{x})d\tilde{x}}.$$

因此,

$$y(x) = \frac{y(x_0) + \int_{x_0}^x e^{\int_{x_0}^{\tilde{x}} \alpha(\tilde{x}) d\tilde{x}} \beta(\tilde{x}) d\tilde{x}}{e^{\int_{x_0}^x \alpha(\tilde{x}) d\tilde{x}}}. \quad (16.15)$$

注意, 当对所有 x 有 $\beta(x) = 0$ 时, 该表达式退化为方程 16.10, 即齐次方程的解。

Original (English)

EXAMPLE 16.5: Consider the linear demand and supply model of Example 16.3 with dynamic adjustment modified to $\frac{dp}{dt} = \kappa(t)(Q_d - Q_s)$, where κ now depends on time and with $Q_d = \alpha - \beta p$ and $Q_s = \gamma + \delta p$. Dynamic adjustment is now given by:

$$\frac{dp}{dt} + \kappa(t)(\beta + \delta)p = \kappa(t)(\alpha - \gamma).$$

From Equation 16.15,

$$p(t) = \frac{p(t_0) + \int_{t_0}^t e^{\int_{t_0}^{\tilde{t}} \kappa(\tilde{t})(\beta + \delta) d\tilde{t}} \kappa(\tilde{t})(\alpha - \gamma) d\tilde{t}}{e^{\int_{t_0}^t \kappa(\tilde{t})(\beta + \delta) d\tilde{t}}}.$$

To simplify this expression, let $t_0 = 0$ and set $\kappa(t) = \theta \cdot t$. Then

$$\int_{t_0}^t \kappa(\tilde{t})(\beta + \delta) d\tilde{t} = (\beta + \delta) \frac{1}{2} \theta t^2 = kt^2,$$

where $k = (\beta + \delta) \frac{\theta}{2}$. Next,

$$\int_{t_0}^t e^{\int_{t_0}^{\tilde{t}} \kappa(\tilde{t})(\beta + \delta) d\tilde{t}} \kappa(\tilde{t})(\alpha - \gamma) d\tilde{t} = (\alpha - \gamma) \int_0^t e^{k\tilde{t}^2} \theta \tilde{t} d\tilde{t} = (\alpha - \gamma) \frac{1}{2k} [e^{kt^2} - 1] = \frac{(\alpha - \gamma)}{(\beta + \delta)} [e^{kt^2} - 1].$$

Thus,

$$p(t) = \frac{p(0) + \frac{(\alpha - \gamma)}{(\beta + \delta)} [e^{kt^2} - 1]}{e^{kt^2}} = p(0)e^{-kt^2} + \frac{(\alpha - \gamma)}{(\beta + \delta)} [1 - e^{-kt^2}].$$

Writing $p_e = \frac{(\alpha - \gamma)}{(\beta + \delta)}$, the market clearing price $p(t)$ is a weighted average of the initial price $p(0)$ and the market clearing price p_e : $p(t) = p(0)e^{-kt^2} + p_e[1 - e^{-kt^2}]$.

EXAMPLE 16.6:

$$\frac{dy}{dx} + axy = x.$$

Here, $\alpha(x) = ax$ so that $\int_{x_0}^x \alpha(\tilde{x}) d\tilde{x} = \frac{a}{2}[x^2 - x_0^2]$. Thus,

$$e^{\int_{x_0}^x \alpha(\tilde{x}) d\tilde{x}} = e^{\frac{a}{2}x^2} e^{-\frac{a}{2}x_0^2} = k(x_0)e^{\frac{a}{2}x^2}$$

so that $q(x) = q(x_0)k(x_0)e^{\frac{a}{2}x^2} = c_0e^{\frac{a}{2}x^2}$. Multiplying both sides of $\frac{dy}{dx} + axy = x$ by $c_0e^{\frac{a}{2}x^2}$ gives

$$\left[\frac{dy}{dx} + axy \right] e^{\frac{a}{2}x^2} = xe^{\frac{a}{2}x^2}$$

or $\frac{d}{dx} \left[y e^{\frac{a}{2}x^2} \right] = x e^{\frac{a}{2}x^2}$, so that

$$\int_{x_0}^x \frac{d}{d\tilde{x}} \left[y e^{\frac{a}{2}\tilde{x}^2} \right] d\tilde{x} = \int_{x_0}^x \tilde{x} e^{\frac{a}{2}\tilde{x}^2} d\tilde{x}$$

or

$$y(x) e^{\frac{a}{2}x^2} - y(x_0) e^{\frac{a}{2}x_0^2} = \int_{x_0}^x \tilde{x} e^{\frac{a}{2}\tilde{x}^2} d\tilde{x} = \frac{1}{a} \left[e^{\frac{1}{2}ax^2} - e^{\frac{1}{2}ax_0^2} \right]$$

or $y(x) e^{\frac{a}{2}x^2} = c_0 + \frac{1}{a} e^{\frac{1}{2}ax^2}$. Therefore,

$$y(x) = c_0 e^{-\frac{a}{2}x^2} + \frac{1}{a}.$$

翻译 (Chinese)

例 16.5: 考虑例 16.3 的线性供需模型, 其中动态调整修改为 $\frac{dp}{dt} = \kappa(t)(Q_d - Q_s)$, 其中 κ 现在依赖于时间, 且 $Q_d = \alpha - \beta p$, $Q_s = \gamma + \delta p$. 动态调整现在由 $\frac{dp}{dt} + \kappa(t)(\beta + \delta)p = \kappa(t)(\alpha - \gamma)$ 给出。由方程 16.15, 为简化此表达式, 令 $t_0 = 0$ 并设 $\kappa(t) = \theta \cdot t$. 则 $\int_{t_0}^t \kappa(\tilde{t})(\beta + \delta) d\tilde{t} = (\beta + \delta) \frac{1}{2} \theta t^2 = k t^2$, 其中 $k = (\beta + \delta) \frac{\theta}{2}$. 接下来, $\int_{t_0}^t e^{\int_{t_0}^{\tilde{t}} \kappa(\tilde{t})(\beta + \delta) d\tilde{t}} \kappa(\tilde{t})(\alpha - \gamma) d\tilde{t} = (\alpha - \gamma) \int_0^t e^{k\tilde{t}^2} \theta \tilde{t} d\tilde{t} = \frac{(\alpha - \gamma)}{(\beta + \delta)} [e^{k t^2} - 1]$. 因此, $p(t) = p(0) e^{-k t^2} + \frac{(\alpha - \gamma)}{(\beta + \delta)} [1 - e^{-k t^2}]$. 记 $p_e = \frac{(\alpha - \gamma)}{(\beta + \delta)}$, 市场出清价格 $p(t)$ 是初始价格 $p(0)$ 和市场出清价格 p_e 的加权平均: $p(t) = p(0) e^{-k t^2} + p_e [1 - e^{-k t^2}]$.

例 16.6: $\frac{dy}{dx} + axy = x$. 此处 $\alpha(x) = ax$, 故 $\int_{x_0}^x \alpha(\tilde{x}) d\tilde{x} = \frac{a}{2} [x^2 - x_0^2]$. 因此, $e^{\int_{x_0}^x \alpha(\tilde{x}) d\tilde{x}} = e^{\frac{a}{2}x^2} e^{-\frac{a}{2}x_0^2}$, 故 $q(x) = c_0 e^{\frac{a}{2}x^2}$. 将 $\frac{dy}{dx} + axy = x$ 两边乘以 $c_0 e^{\frac{a}{2}x^2}$ 得到 $[\frac{dy}{dx} + axy] e^{\frac{a}{2}x^2} = x e^{\frac{a}{2}x^2}$, 或 $\frac{d}{dx} [y e^{\frac{a}{2}x^2}] = x e^{\frac{a}{2}x^2}$, 从而 $y(x) e^{\frac{a}{2}x^2} - y(x_0) e^{\frac{a}{2}x_0^2} = \frac{1}{a} [e^{\frac{1}{2}ax^2} - e^{\frac{1}{2}ax_0^2}]$. 因此, $y(x) = c_0 e^{-\frac{a}{2}x^2} + \frac{1}{a}$.

16.3 Some non-linear first-order differential equations / 一些非线性一阶微分方程

Original (English)

This section considers two well-known non-linear first-order differential equations that have exact solutions: the logistic and Bernoulli differential equations.

翻译 (Chinese)

本节考虑两个具有精确解的著名非线性一阶微分方程: 逻辑斯蒂 (Logistic) 微分方程和伯努利 (Bernoulli) 微分方程。

16.3.1 The logistic model / 逻辑斯蒂模型

Original (English)

The linear model $\frac{dy}{dx} + ay = 0$ has solution $y(x) = y_0 e^{-ax}$ so that as x becomes large $y(x)$ tends to 0 or to ∞ , depending on whether a is positive or negative. Models of population growth where the habitat is finite cannot be modeled by this equation over long periods of time. The logistic model provides a growth model that asymptotes to a finite limit. Although the differential equation is non-linear, it is easily solved.

Consider

$$\frac{\dot{y}}{y} = \alpha \left(1 - \frac{y}{k}\right) = \alpha \left(\frac{k-y}{k}\right) = \beta[k-y], \quad \beta = \frac{\alpha}{k}, \quad \alpha > 0, \quad k > 0.$$

From this expression, when $k - y > 0$ or $y < k$, $\dot{y} > 0$ and the value of y grows. Similarly, when $k - y < 0$ the value of y decreases. At $y = k$, $\dot{y} = 0$, and k is the steady state value for y . The equation may be rearranged as,

$$\frac{1}{y} \frac{k}{k-y} \dot{y} = \alpha.$$

Observe that $\frac{1}{y} + \frac{1}{k-y} = \frac{(k-y)+y}{y(k-y)} = \frac{k}{y(k-y)}$. Consequently,

$$\frac{1}{y} \frac{k}{k-y} \dot{y} = \frac{1}{y} \dot{y} + \frac{1}{k-y} \dot{y} = \alpha,$$

and so

$$\int_{x_0}^x \frac{1}{y} \frac{k}{k-y} \dot{y} dx = \int_{x_0}^x \frac{1}{y} \dot{y} dx + \int_{x_0}^x \frac{1}{k-y} \dot{y} dx = \alpha[x - x_0],$$

or

$$\ln y(x) - \ln y(x_0) - [\ln(k - y(x)) - \ln(k - y(x_0))] = \alpha[x - x_0].$$

This yields $\ln \frac{k-y(x)}{y(x)} = c_0 - \alpha x$, or $\frac{k-y(x)}{y(x)} = c e^{-\alpha x}$. Therefore,

$$y(x) = \frac{k}{1 + \tilde{c} e^{-\alpha x}}, \quad \tilde{c} = \frac{1}{c}.$$

The value of $\tilde{c}$ is determined by the initial condition $y(x_0) = y_0$: $y_0 = \frac{k}{1 + \tilde{c} e^{-\alpha x_0}}$ or $\tilde{c} = \left(\frac{k}{y_0} - 1\right) e^{\alpha x_0}$. The function $y(x) = \frac{k}{1 + \tilde{c} e^{-\alpha x}}$ is called the logistic function (Figure 16.1).

翻译 (Chinese)

线性模型 $\frac{dy}{dx} + ay = 0$ 具有解 $y(x) = y_0 e^{-ax}$, 因此当 x 变大时, $y(x)$ 趋于 0 或趋于 ∞ , 取决于 a 是正还是负。对于栖息地有限的人口增长模型, 该方程不能在长时间内建模。逻辑斯蒂模型提供了一个渐近于有限极限的增长模型。尽管该微分方程是非线性的, 但它很容易求解。

考虑

$$\frac{\dot{y}}{y} = \alpha \left(1 - \frac{y}{k}\right) = \alpha \left(\frac{k-y}{k}\right) = \beta[k-y], \quad \beta = \frac{\alpha}{k}, \quad \alpha > 0, \quad k > 0.$$

由该表达式, 当 $k-y > 0$ 或 $y < k$ 时, $\dot{y} > 0$, y 的值增长。类似地, 当 $k-y < 0$ 时, y 的值减小。在 $y = k$ 处, $\dot{y} = 0$, k 是 y 的稳态值。该方程可重排为 $\frac{1}{y} \frac{k}{k-y} \dot{y} = \alpha$ 。注意到 $\frac{1}{y} + \frac{1}{k-y} = \frac{k}{y(k-y)}$ 。因此, $\frac{1}{y} \dot{y} + \frac{1}{k-y} \dot{y} = \alpha$, 积分得 $\ln y(x) - \ln y(x_0) - [\ln(k-y(x)) - \ln(k-y(x_0))] = \alpha[x - x_0]$ 。这给出 $\ln \frac{k-y(x)}{y(x)} = c_0 - \alpha x$, 或 $\frac{k-y(x)}{y(x)} = ce^{-\alpha x}$ 。因此,

$$y(x) = \frac{k}{1 + \tilde{c}e^{-\alpha x}}, \quad \tilde{c} = \frac{1}{c}.$$

$\tilde{c}$ 的值由初始条件 $y(x_0) = y_0$ 确定: $y_0 = \frac{k}{1 + \tilde{c}e^{-\alpha x_0}}$ 或 $\tilde{c} = (\frac{k}{y_0} - 1)e^{\alpha x_0}$ 。函数 $y(x) = \frac{k}{1 + \tilde{c}e^{-\alpha x}}$ 称为逻辑斯蒂函数 (图 16.1)。

16.3.2 The Bernoulli equation / 伯努利方程

Original (English)

One particular non-linear differential equation that has appeared in economic applications is the Bernoulli equation. This is a differential equation of the form:

$$\frac{dy}{dx} + a(x)y = \beta(x)y^\gamma, \quad \gamma \neq 0, 1.$$

Dividing by y^γ ,

$$\frac{1}{y^\gamma} \frac{dy}{dx} + a(x)y^{1-\gamma} = \beta(x).$$

Define $w = y^{1-\gamma}$, so that $w' = (1-\gamma)y^{-\gamma}y'$ or $\frac{dw}{dx} = (1-\gamma)\frac{1}{y^\gamma}\frac{dy}{dx}$. With this change of variables:

$$\frac{1}{1-\gamma} \frac{dw}{dx} + a(x)w = \beta(x).$$

This is a first-order linear differential equation, which may be solved for $w(x)$ as described previously. The function $y(x)$ is then obtained as $y(x) = w(x)^{\frac{1}{1-\gamma}}$.

EXAMPLE 16.7: The Solow growth model provides a differential equation governing the long-run dynamics of per capita capital. A constant returns to scale production function $Y = f(K, L)$ is assumed, so that $y = \frac{Y}{L} = \frac{1}{L}f(K, L) = f(\frac{K}{L}, \frac{L}{L}) = f(k)$, where $k = \frac{K}{L}$. For example, $f(K, L) = K^\alpha L^{1-\alpha}$ gives $f(k) = k^\alpha$, with $0 < \alpha < 1$. Savings, S , is a fixed proportion, s , of income, Y , and investment I is equal to saving: $I = S = sY$. Capital growth per unit of time is equal to investment less depreciation: $I - \delta K$: $\Delta K = (sY - \delta K)\Delta t$ or, in continuous time, $\frac{dK}{dt} = (sY - \delta K)$. Therefore, $\frac{1}{K} \frac{dK}{dt} = s \frac{Y}{K} - \delta$. Since $\frac{Y}{K} = \frac{Y/L}{K/L} = \frac{y}{k}$,

$$\frac{1}{K} \frac{dK}{dt} = s \frac{y}{k} - \delta.$$

To convert to per capita terms, observe that

$$\frac{dk}{dt} = \frac{L \frac{dK}{dt} - K \frac{dL}{dt}}{L^2} = \frac{1}{L} \frac{dK}{dt} - \frac{K}{L} \frac{1}{L} \frac{dL}{dt}.$$

So, with $\frac{1}{k} = \frac{L}{K}$,

$$\frac{1}{k} \frac{dk}{dt} = \frac{1}{K} \frac{dK}{dt} - \frac{1}{L} \frac{dL}{dt}.$$

Let $n = \frac{1}{L} \frac{dL}{dt}$ be the continuous time growth rate of the labor supply, so that

$$\frac{1}{k} \frac{dk}{dt} = \frac{1}{K} \frac{dK}{dt} - n.$$

Collecting terms,

$$\frac{1}{K} \frac{dK}{dt} = \frac{1}{k} \frac{dk}{dt} + n = s \frac{y}{k} - \delta.$$

Therefore,

$$\frac{dk}{dt} = sy - (\delta + n)k = sf(k) - (\delta + n)k. \quad (16.16)$$

From this expression, there are two points where k is constant: at 0, and at $\hat{k}$, defined by $sf(\hat{k}) - (\delta + n)\hat{k} = 0$.

With $f(k) = k^\alpha$, the differential equation becomes

$$\frac{dk}{dt} = sk^\alpha - (\delta + n)k$$

or $\frac{dk}{dt} + (\delta + n)k = sk^\alpha$, which is a simple Bernoulli equation. Rearrange to get:

$$\frac{1}{k^\alpha} \frac{dk}{dt} + (\delta + n)k^{1-\alpha} = s$$

and define $w = k^{1-\alpha}$, so that

$$\frac{dw}{dt} = (1 - \alpha)k^{-\alpha} \frac{dk}{dt} = (1 - \alpha) \frac{1}{k^\alpha} \frac{dk}{dt} \quad \text{or} \quad \frac{1}{1 - \alpha} \frac{dw}{dt} = \frac{1}{k^\alpha} \frac{dk}{dt}.$$

With this change of variable, the differential equation is:

$$\frac{1}{1 - \alpha} \frac{dw}{dt} + (\delta + n)w = s \quad \text{or} \quad \frac{dw}{dt} + (1 - \alpha)(\delta + n)w = (1 - \alpha)s.$$

The solution to a differential equation of the form $\frac{dw}{dt} + aw = b$ has the form:

$$w(t) = w(0)e^{-at} + \frac{b}{a} = w(0)e^{-[(1-\alpha)(\delta+n)]t} + \frac{s}{\delta + n}.$$

Since $k = w^{\frac{1}{1-\alpha}}$,

$$k(t) = \left(k(0)^{1-\alpha} e^{-[(1-\alpha)(\delta+n)]t} + \frac{s}{\delta + n} \right)^{\frac{1}{1-\alpha}}.$$

At $t \rightarrow \infty$, $k(t) \rightarrow \left(\frac{s}{\delta+n}\right)^{\frac{1}{1-\alpha}}$. From Equation 16.16, $\frac{dk}{dt} = 0$ when $sk^\alpha - (\delta+n)k = 0$ or $k = \left(\frac{s}{\delta+n}\right)^{\frac{1}{1-\alpha}}$, so the solution gives the same steady state.

In equilibrium, $sk^\alpha - (n+\delta)k = 0$, so the amount of output per capita saved, sk^α , equals $(n+\delta)k$. Total output per capita is k^α and so consumption per capita, c , is $c = k^\alpha - sk^\alpha = k^\alpha - (n+\delta)k$. This is maximized when $\alpha k^{\alpha-1} - (n+\delta) = 0$, or

$$k = \left(\frac{\alpha}{n+\delta}\right)^{\frac{1}{1-\alpha}}.$$

The solution is called the golden rule level of capital, the level that maximizes per capita consumption. Comparing this with the long-run steady state level of $k = \left(\frac{s}{n+\delta}\right)^{\frac{1}{1-\alpha}}$; these are equal when $s = \alpha$. The savings rate that maximizes the long-run per capita consumption is called the golden rule savings rate.

翻译 (Chinese)

在经济应用中出现的一个特殊非线性微分方程是伯努利方程。这是一个形如 $\frac{dy}{dx} + a(x)y = \beta(x)y^\gamma$ 的微分方程，其中 $\gamma \neq 0, 1$ 。除以 y^γ 得 $\frac{1}{y^\gamma} \frac{dy}{dx} + a(x)y^{1-\gamma} = \beta(x)$ 。定义 $w = y^{1-\gamma}$ ，则 $w' = (1-\gamma)y^{-\gamma}y'$ 或 $\frac{dw}{dx} = (1-\gamma)\frac{1}{y^\gamma} \frac{dy}{dx}$ 。经过变量替换： $\frac{1}{1-\gamma} \frac{dw}{dx} + a(x)w = \beta(x)$ 。这是一阶线性微分方程，可按前面描述的方法求解 $w(x)$ 。然后函数 $y(x)$ 通过 $y(x) = w(x)^{\frac{1}{1-\gamma}}$ 得到。

例 16.7: 索洛增长模型提供了控制人均资本长期动态的微分方程。假设规模报酬不变的生产函数 $Y = f(K, L)$ ，因此 $y = \frac{Y}{L} = f(k)$ ，其中 $k = \frac{K}{L}$ 。例如， $f(K, L) = K^\alpha L^{1-\alpha}$ 给出 $f(k) = k^\alpha$ ，其中 $0 < \alpha < 1$ 。储蓄 S 是收入 Y 的固定比例 s ，投资 I 等于储蓄： $I = S = sY$ 。每单位时间的资本增长等于投资减去折旧： $\Delta K = (sY - \delta K)\Delta t$ ，或在连续时间中， $\frac{dK}{dt} = (sY - \delta K)$ 。因此， $\frac{1}{K} \frac{dK}{dt} = s\frac{Y}{K} - \delta$ 。由于 $\frac{Y}{K} = \frac{Y/L}{K/L} = \frac{y}{k}$ ， $\frac{1}{K} \frac{dK}{dt} = s\frac{y}{k} - \delta$ 。

为转化为人均形式，注意到 $\frac{dk}{dt} = \frac{1}{L} \frac{dK}{dt} - \frac{K}{L} \frac{1}{L} \frac{dL}{dt}$ 。因此， $\frac{1}{k} \frac{dk}{dt} = \frac{1}{K} \frac{dK}{dt} - n$ ，其中 $n = \frac{1}{L} \frac{dL}{dt}$ 是劳动力供给的连续时间增长率。整理得 $\frac{dk}{dt} = sy - (\delta+n)k = sf(k) - (\delta+n)k$ (方程 16.16)。从该表达式可知， k 有两个常数点：在 0 处，以及在由 $sf(\hat{k}) - (\delta+n)\hat{k} = 0$ 定义的 $\hat{k}$ 处。

当 $f(k) = k^\alpha$ 时，微分方程变为 $\frac{dk}{dt} + (\delta+n)k = sk^\alpha$ ，这是一个简单的伯努利方程。定义 $w = k^{1-\alpha}$ ，经过变量替换得 $\frac{dw}{dt} + (1-\alpha)(\delta+n)w = (1-\alpha)s$ 。解的形式为 $w(t) = w(0)e^{-[(1-\alpha)(\delta+n)]t} + \frac{s}{\delta+n}$ 。由于 $k = w^{\frac{1}{1-\alpha}}$ ， $k(t) = (k(0)^{1-\alpha}e^{-[(1-\alpha)(\delta+n)]t} + \frac{s}{\delta+n})^{\frac{1}{1-\alpha}}$ 。当 $t \rightarrow \infty$ 时， $k(t) \rightarrow \left(\frac{s}{\delta+n}\right)^{\frac{1}{1-\alpha}}$ 。

在均衡中， $sk^\alpha - (n+\delta)k = 0$ 。人均消费 $c = k^\alpha - sk^\alpha = k^\alpha - (n+\delta)k$ 。这在 $\alpha k^{\alpha-1} - (n+\delta) = 0$ 时最大化，即 $k = \left(\frac{\alpha}{n+\delta}\right)^{\frac{1}{1-\alpha}}$ 。该解称为资本黄金律水平，即最大化人均消费的水平。与长期稳态水平 $k = \left(\frac{s}{n+\delta}\right)^{\frac{1}{1-\alpha}}$ 比较，当 $s = \alpha$ 时二者相等。最大化长期人均消费的储蓄率称为黄金律储蓄率。

16.4 First-order differential equations: existence of solutions / 一阶微分方程：解的存在性

Original (English)

Consider the (differential) equation

$$\frac{dy}{dx} = f(x, y),$$

associating the change in y to the current values of x and y . When $f(x, y) = \beta(x) - \alpha(x)y$, this gives the first-order differential equation $\frac{dy}{dx} + \alpha(x)y = \beta(x)$. A solution to the equation on the interval (a, b) is a function $y(x)$ satisfying $\frac{dy(x)}{dx} = f(x, y(x))$ for $x \in (a, b)$. Two natural questions arise. Does the differential equation have a solution? And is the solution unique?

Definition 16.1: Let Q be a region in $\mathbb{R}^2$. Say that the function $f(x, y)$ satisfies a uniform Lipschitz condition in Q if there is some constant, K , such that for any pairs $(x, y_1), (x, y_2) \in Q$:

$$|f(x, y_1) - f(x, y_2)| \leq K|y_1 - y_2|.$$

With this definition, a key theorem on the existence of a solution to a differential equation is the following.

Theorem 16.1: Given the differential equation:

$$\frac{dy}{dx} = f(x, y),$$

with initial condition $y(x_0) = y_0$, where f is a continuous function on $Q = [x_0 - a, x_0 + a] \times [y_0 - b, y_0 + b]$. Then for some $a > 0$, there is a function y defined on $[x_0 - a, x_0 + a]$ with $y(x_0) = y_0$ and $y'(x) = f(x, y(x))$, $x \in [x_0 - a, x_0 + a]$. If f satisfies a uniform Lipschitz condition on Q , the solution is unique.

In the case of a linear differential equation, $f(x, y) = \beta(x) - \alpha(x)y$, with $\alpha(x)$ continuous,

$$|f(x, y_1) - f(x, y_2)| = |\alpha(x)y_1 - \alpha(x)y_2| \leq |\alpha(x)||y_1 - y_2| \leq K|y_1 - y_2|,$$

where $K = \max_{x \in [x_0 - a, x_0 + a]} |\alpha(x)|$, and so the uniform Lipschitz condition is satisfied.

翻译 (Chinese)

考虑（微分）方程 $\frac{dy}{dx} = f(x, y)$ ，它将 y 的变化与 x 和 y 的当前值联系起来。当 $f(x, y) = \beta(x) - \alpha(x)y$ 时，这给出了一阶微分方程 $\frac{dy}{dx} + \alpha(x)y = \beta(x)$ 。在区间 (a, b) 上该方程的解是一个函数 $y(x)$ ，对 $x \in (a, b)$ 满足 $\frac{dy(x)}{dx} = f(x, y(x))$ 。自然地出现两个问题：微分方程是否有解？解是否唯一？

定义 16.1: 设 Q 是 $\mathbb{R}^2$ 中的一个区域。称函数 $f(x, y)$ 在 Q 上满足一致 Lipschitz 条件, 如果存在某个常数 K , 使得对任意 $(x, y_1), (x, y_2) \in Q$:

$$|f(x, y_1) - f(x, y_2)| \leq K|y_1 - y_2|.$$

定理 16.1: 给定微分方程 $\frac{dy}{dx} = f(x, y)$ 及初始条件 $y(x_0) = y_0$, 其中 f 是 $Q = [x_0 - a, x_0 + a] \times [y_0 - b, y_0 + b]$ 上的连续函数。则对某个 $a > 0$, 存在定义在 $[x_0 - a, x_0 + a]$ 上的函数 y , 满足 $y(x_0) = y_0$ 且 $y'(x) = f(x, y(x))$, $x \in [x_0 - a, x_0 + a]$ 。若 f 在 Q 上满足一致 Lipschitz 条件, 则解是唯一的。

在线性微分方程的情形下, $f(x, y) = \beta(x) - \alpha(x)y$, 其中 $\alpha(x)$ 连续,

$$|f(x, y_1) - f(x, y_2)| = |\alpha(x)y_1 - \alpha(x)y_2| \leq |\alpha(x)||y_1 - y_2| \leq K|y_1 - y_2|,$$

其中 $K = \max_{x \in [x_0 - a, x_0 + a]} |\alpha(x)|$, 故一致 Lipschitz 条件得到满足。

16.5 Second- and higher-order differential equations / 二阶及高阶微分方程

Original (English)

An equation is an n th-order linear differential equation if it has the form:

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1(x) \frac{dy}{dx} + a_0(x)y = b(x).$$

A second-order differential equation has the form:

$$a_2(x) \frac{d^2 y}{dx^2} + a_1(x) \frac{dy}{dx} + a_0(x)y = b(x).$$

In general, this is difficult to solve for the function y . The remainder of the discussion focuses on the second-order differential equation with constant coefficients.

翻译 (Chinese)

若方程具有以下形式, 则称为 n 阶线性微分方程:

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1(x) \frac{dy}{dx} + a_0(x)y = b(x).$$

二阶微分方程具有形式:

$$a_2(x) \frac{d^2 y}{dx^2} + a_1(x) \frac{dy}{dx} + a_0(x)y = b(x).$$

一般来说, 这很难求解函数 y 。接下来的讨论侧重于常系数的二阶微分方程。

16.5.1 Constant coefficients / 常系数

Original (English)

When the coefficients are constant, the second-order linear differential equation has the form:

$$a_2 \frac{d^2 y}{dx^2} + a_1 \frac{dy}{dx} + a_0 y = b.$$

This equation has a particular solution $\bar{y} = \frac{b}{a_0}$. Consider the homogeneous equation,

$$a_2 \frac{d^2 y}{dx^2} + a_1 \frac{dy}{dx} + a_0 y = 0.$$

Observe that if y_1 and y_2 are solutions to the homogeneous equation, then so is $y = c_1 y_1 + c_2 y_2$ for any c_1 and c_2 .

A candidate solution $y(x) = ke^{rx}$ has the property that $y'(x) = rke^{rx} = ry(x)$ and $y''(x) = r^2 ke^{rx} = r^2 y(x)$, so that if this function satisfies the homogeneous equation then

$$a_2 r^2 y(x) + a_1 r y(x) + a_0 y(x) = (a_2 r^2 + a_1 r + a_0) y(x) = 0.$$

Since $y(x) = ke^{rx} \neq 0$, this requires $a_2 r^2 + a_1 r + a_0 = 0$. This quadratic expression has two roots:

$$r_1, r_2 = \frac{-a_1 \pm \sqrt{a_1^2 - 4a_2 a_0}}{2a_2}.$$

The solution to the differential equation depends on the roots, which may be real and distinct, real and repeated or complex.

$$y(x) = \begin{cases} c_1 e^{r_1 x} + c_2 e^{r_2 x} + \frac{b}{a_0}, & \text{roots real and distinct} \\ c_1 e^{rx} + c_2 x e^{rx} + \frac{b}{a_0}, & \text{roots real and equal} \\ c_1 e^{\alpha x} \cos(\beta x) + c_2 e^{\alpha x} \sin(\beta x) + \frac{b}{a_0}, & \text{roots complex.} \end{cases}$$

The first two cases are easy to confirm, the third involves some computation. These issues are developed in Section 16.5.2.

REMARK 16.2: Considering stability, in the first case it is necessary for r_1 and r_2 to be negative, and in the second case for r to be negative. In the complex case, the roots are repeated and α is the real part of both roots, so in that case, it is necessary for α to be negative. In sum, a sufficient condition for stability of the solution is that the real parts of the roots be negative.

翻译 (Chinese)

当系数为常数时，二阶线性微分方程具有形式：

$$a_2 \frac{d^2 y}{dx^2} + a_1 \frac{dy}{dx} + a_0 y = b.$$

该方程有一个特解 $\bar{y} = \frac{b}{a_0}$ 。考虑齐次方程 $a_2 \frac{d^2 y}{dx^2} + a_1 \frac{dy}{dx} + a_0 y = 0$ 。注意到若 y_1 和 y_2 是齐次方程的解，则对任意 c_1 和 c_2 ， $y = c_1 y_1 + c_2 y_2$ 也是。

试探解 $y(x) = ke^{rx}$ 具有性质 $y'(x) = ry(x)$ 和 $y''(x) = r^2 y(x)$ ，因此若该函数满足齐次方程，则 $(a_2 r^2 + a_1 r + a_0)y(x) = 0$ 。由于 $y(x) = ke^{rx} \neq 0$ ，这要求 $a_2 r^2 + a_1 r + a_0 = 0$ 。此二次方程有两个根： $r_1, r_2 = \frac{-a_1 \pm \sqrt{a_1^2 - 4a_2 a_0}}{2a_2}$ 。

微分方程的解取决于根的类型：实且不同、实且相同、或复数。通解包括以上三种情形。

注 16.2：关于稳定性，在第一种情况下需要 r_1 和 r_2 为负，在第二种情况下 r 为负。在复数情况下， α 是两根的实部，因此需要 α 为负。总之，解稳定的充分条件是特征根的实部为负。

16.5.2 Derivation of solution / 解的推导

Original (English)

Regarding these roots, there are three distinct regions of the parameter space:

1. Roots real and distinct: $a_1^2 - 4a_2 a_0 > 0$. The general solution of the homogeneous equation has the form: $y(x) = c_1 e^{r_1 x} + c_2 e^{r_2 x}$.
2. Roots real and equal: $a_1^2 - 4a_2 a_0 = 0$, so that $r_1 = r_2 = r$. The general solution of the homogeneous equation has the form: $y(x) = c_1 e^{rx} + c_2 x e^{rx}$.
3. Roots complex: $a_1^2 - 4a_2 a_0 < 0$, so that $r_1 = \alpha + i\beta$ and $r_2 = \alpha - i\beta$, where $\alpha = -\frac{a_1}{2a_2}$ and $\beta = \sqrt{4a_2 a_0 - a_1^2}/2a_2$. The general solution has the form: $y(x) = c_1 e^{r_1 x} + c_2 e^{r_2 x}$.

According to Euler's formula, for any real number x , $e^{ix} = \cos(x) + i \sin(x)$ (Figure 16.2). Thus,

$$e^{r_1 x} = e^{\alpha x + i\beta x} = e^{\alpha x} e^{i\beta x} = e^{\alpha x} [\cos(\beta x) + i \sin(\beta x)]$$

and

$$e^{r_2 x} = e^{\alpha x - i\beta x} = e^{\alpha x} e^{i(-\beta x)} = e^{\alpha x} [\cos(-\beta x) + i \sin(-\beta x)] = e^{\alpha x} [\cos(\beta x) - i \sin(\beta x)].$$

Therefore,

$$y(x) = c_1 e^{r_1 x} + c_2 e^{r_2 x} = e^{\alpha x} [(c_1 + c_2) \cos(\beta x) + i(c_1 - c_2) \sin(\beta x)].$$

Since y is a real-valued function, solutions of the homogeneous equation have the form:

$$y(x) = c_1 e^{\alpha x} \cos(\beta x) + c_2 e^{\alpha x} \sin(\beta x) = e^{\alpha x} [c_1 \cos(\beta x) + c_2 \sin(\beta x)],$$

where c_1 and c_2 are real numbers. This may be rearranged using a right-angled triangle:

$$y(x) = h e^{\alpha x} \cos(\beta x - \theta),$$

where $h = \sqrt{c_1^2 + c_2^2}$, $\cos(\theta) = c_1/h$, $\sin(\theta) = c_2/h$.

EXAMPLE 16.8: According to Hooke's law, the force acting on an object is proportional to its displacement from equilibrium: $F = -ky$. According to Newton's second law, $F = ma = m \frac{d^2 y}{dt^2}$. Hence

$$m \frac{d^2 y}{dt^2} + ky = 0.$$

Rearranging, $\frac{d^2 y}{dt^2} + \frac{k}{m}y = 0$. The associated roots are $r_1, r_2 = \pm i\sqrt{\frac{k}{m}}$. Thus, the roots are complex and the solution is:

$$y(t) = h \cos\left(\sqrt{\frac{k}{m}} \cdot t - \theta\right) = y_0 \cos\left(\sqrt{\frac{k}{m}} \cdot (t - t_0)\right).$$

This example may be generalized when the spring is damped: $m \frac{d^2 y}{dt^2} + q \frac{dy}{dt} + ky = 0$ (with real roots if and only if $q^2 - 4mk > 0$).

Theorem 16.2: The n th-order linear differential equation:

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1(x) \frac{dy}{dx} + a_0(x)y = b(x)$$

with initial conditions $y(x_0) = y_0$, $y'(x_0) = y_1$, $y''(x_0) = y_2$, ..., $y^{(n-1)}(x_0) = y_{n-1}$ has a unique solution.

翻译 (Chinese)

对于这些根, 参数空间有三个不同的区域: (1) 根为实且不同: $a_1^2 - 4a_2a_0 > 0$, 齐次方程通解为 $y(x) = c_1 e^{r_1 x} + c_2 e^{r_2 x}$; (2) 根为实且相同: $a_1^2 - 4a_2a_0 = 0$, 通解为 $y(x) = c_1 e^{rx} + c_2 x e^{rx}$; (3) 根为复数: $a_1^2 - 4a_2a_0 < 0$, $r_1, r_2 = \alpha \pm i\beta$, 其中 $\alpha = -\frac{a_1}{2a_2}$, $\beta = \sqrt{4a_2a_0 - a_1^2}/2a_2$.

根据欧拉公式, 对任意实数 x , $e^{ix} = \cos(x) + i\sin(x)$ (图 16.2)。因此, 通过代数推导, 齐次方程的解具有形式: $y(x) = c_1 e^{\alpha x} \cos(\beta x) + c_2 e^{\alpha x} \sin(\beta x)$, 这可以重排为 $y(x) = h e^{\alpha x} \cos(\beta x - \theta)$ 。

例 16.8: 根据胡克定律, 作用在物体上的力与其偏离均衡的位移成比例: $F = -ky$ 。根据牛顿第二定律, $F = m \frac{d^2 y}{dt^2}$ 。因此 $m \frac{d^2 y}{dt^2} + ky = 0$, 根为 $r_1, r_2 = \pm i\sqrt{\frac{k}{m}}$ 。解为 $y(t) = y_0 \cos(\sqrt{\frac{k}{m}} \cdot (t - t_0))$ 。此例可推广到阻尼弹簧情形: $m \frac{d^2 y}{dt^2} + q \frac{dy}{dt} + ky = 0$ (当且仅当 $q^2 - 4mk > 0$ 时有实根)。

定理 16.2: n 阶线性微分方程在给定 n 个初始条件下具有唯一解。

16.6 Systems of differential equations / 微分方程组

Original (English)

Let A be an $n \times n$ matrix and γ an $n \times 1$ vector of constants. A system of first-order linear differential equations is given by:

$$\frac{dy}{dx}(x) + Ay(x) = \gamma. \quad (16.17)$$

To simplify the discussion, assume that A has n linearly independent eigenvectors v_i and corresponding eigenvalues λ_i , then for each i , $Av_i = \lambda_i v_i$. Let Λ be the diagonal matrix with λ_i in the i th row and column and let P be the matrix with v_i in the i th column. Then this collection of n equations may be written as $AP = P\Lambda$. Therefore, $P^{-1}AP = P^{-1}P\Lambda = \Lambda$. Considering Equation 16.17, define $z = P^{-1}y$, so that $y = Pz$ and substitute into Equation 16.17 to obtain:

$$P \frac{dz}{dx}(x) + APz(x) = \gamma.$$

Multiplying by P^{-1} :

$$\frac{dz}{dx}(x) + P^{-1}APz(x) = P^{-1}\gamma$$

or

$$\frac{dz}{dx}(x) + \Lambda z(x) = P^{-1}\gamma = \tilde{\gamma}.$$

Because Λ is diagonal, this system has the form:

$$\frac{dz_i}{dx}(x) + \lambda_i z_i(x) = \tilde{\gamma}_i, \quad i = 1, \dots, n, \quad (16.18)$$

a collection of n first-order linear differential equations with solutions $z_i(x) = c_i e^{-\lambda_i x} + \tilde{\gamma}_i$. Given initial values $y_i(x_0)$, values $z_i(x_0)$ are determined according to $z(0) = P^{-1}y(0)$, and this determines c_i and $\tilde{\gamma}_i$.

Since $y = Pz$,

$$y_i(x) = \sum_{j=1}^n P_{ij} z_j = \sum_{j=1}^n P_{ij} [c_j e^{-\lambda_j x} + \tilde{\gamma}_j] = \sum_{j=1}^n \pi_{ij} e^{-\lambda_j x} + \tilde{\gamma}_j, \quad (16.19)$$

where $\pi_{ij} = P_{ij} c_j$ and $\tilde{\gamma}_i = \sum_{j=1}^n P_{ij} \tilde{\gamma}_j$.

The homogeneous equation $y' + Ay = 0$ solves to give $y_c(x) = \sum_{j=1}^n v_j c_j e^{-\lambda_j x}$ as the complementary solution of the homogeneous equation and a particular solution is given as the solution to $Ay = \gamma$ or $y_p = A^{-1}\gamma$. A general solution of the homogeneous equation is then:

$$y(x) = \sum_{j=1}^n v_j c_j e^{-\lambda_j x} + A^{-1}\gamma.$$

EXAMPLE 16.9: Consider the two-good market given by:

$$\text{Market 1 : } D_1 = S_1 \quad \text{or} \quad a_0 + a_1 p_1 + a_2 p_2 = b_0 + b_1 p_1 + b_2 p_2$$

$$\text{Market 2 : } D_2 = S_2 \quad \text{or} \quad \alpha_0 + \alpha_1 p_1 + \alpha_2 p_2 = \beta_0 + \beta_1 p_1 + \beta_2 p_2.$$

Suppose that price dynamics are linear, with price adjusting to eliminate excess demand over time:

$\frac{dp_i}{dt} = \kappa(D_i(t) - S_i(t))$. Thus:

$$\begin{aligned} \frac{dp_1}{dt} &= \kappa(a_0 - b_0) + \kappa(a_1 - b_1)p_1 + \kappa(a_2 - b_2)p_2 \\ \frac{dp_2}{dt} &= \kappa(\alpha_0 - \beta_0) + \kappa(\alpha_1 - \beta_1)p_1 + \kappa(\alpha_2 - \beta_2)p_2. \end{aligned}$$

The system may be written: $\frac{dp}{dt} = Ap + \gamma$ or $\frac{dp}{dt} - Ap = \gamma$. With specific parameter values, the matrix A has eigenvalues $\lambda_1 = -\frac{1}{4}$, $\lambda_2 = -\frac{1}{2}$, with corresponding eigenvectors $v_1 = (1, 2)$ and $v_2 = (3, 1)$. Because the roots are negative between 0 and -1 , the system converges to $p^e = -A^{-1}\gamma$.

The price dynamics may be written:

$$p(t) = V\Lambda(t)V^{-1}(p(0) - p^e) + p^e.$$

where $\Lambda(t)$ is the diagonal matrix with $e^{\lambda_i t}$ as the i th diagonal term.

EXAMPLE 16.10: Let

$$\begin{pmatrix} y_1' \\ y_2' \end{pmatrix} = \begin{pmatrix} \frac{1}{5} & 0 \\ \frac{2}{5} & \frac{4}{5} \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} + \begin{pmatrix} 2 \\ 4 \end{pmatrix}.$$

The coefficient matrix A has eigenvalues $\lambda_1 = \frac{1}{5}$ and $\lambda_2 = \frac{4}{5}$. The corresponding eigenvectors are $v_1 = (-\frac{3}{2}, 1)$ and $v_2 = (0, 1)$. The general solution is:

$$\begin{pmatrix} y_1(x) \\ y_2(x) \end{pmatrix} = \begin{pmatrix} -\frac{3}{2}c_1 e^{-\frac{1}{5}x} + 10 \\ c_1 e^{-\frac{1}{5}x} + c_2 e^{-\frac{4}{5}x} \end{pmatrix}.$$

翻译 (Chinese)

设 A 为 $n \times n$ 矩阵, γ 为 $n \times 1$ 常数向量。一阶线性微分方程组为 $\frac{dy}{dx}(x) + Ay(x) = \gamma$ (方程 16.17)。假设 A 有 n 个线性无关的特征向量 v_i 及对应特征值 λ_i , 则 $Av_i = \lambda_i v_i$ 。令 Λ 为对角矩阵, P 为特征向量矩阵, 则 $AP = P\Lambda$, $P^{-1}AP = \Lambda$ 。定义 $z = P^{-1}y$, 得 $\frac{dz}{dx} + \Lambda z = P^{-1}\gamma$ 。由于 Λ 是对角的, 系统分解为 n 个一阶线性微分方程: $\frac{dz_i}{dx} + \lambda_i z_i = \tilde{\gamma}_i$ (方程 16.18), 解为 $z_i(x) = c_i e^{-\lambda_i x} + \tilde{\gamma}_i$ 。通解为 $y(x) = \sum_{j=1}^n v_j c_j e^{-\lambda_j x} + A^{-1}\gamma$ 。

例 16.9: 考虑两商品市场, 价格动态为 $\frac{dp_i}{dt} = \kappa(D_i(t) - S_i(t))$ 。系统可写为 $\frac{dp}{dt} - Ap = \gamma$ 。对于特定的参数值, 矩阵 A 的特征值为 $\lambda_1 = -\frac{1}{4}$, $\lambda_2 = -\frac{1}{2}$, 对应的特征向量为 $v_1 = (1, 2)$ 和

$v_2 = (3, 1)$ 。由于根为负，系统收敛到 $p^e = -A^{-1}\gamma$ 。价格动态可写为 $p(t) = V\Lambda(t)V^{-1}(p(0) - p^e) + p^e$ 。

例 16.10: 对给定的系数矩阵 A ，特征值为 $\lambda_1 = \frac{1}{5}$ 和 $\lambda_2 = \frac{4}{5}$ ，对应特征向量为 $v_1 = (-\frac{3}{2}, 1)$ 和 $v_2 = (0, 1)$ 。通解如上所示。

16.7 Stability / 稳定性

Original (English)

The long-run behavior of a dynamic system depends on the roots of the characteristic equation. A sufficient condition for stability of the dynamic system is that the roots of the characteristic polynomial have negative real parts. Consider the polynomial:

$$p(\lambda) = a_0\lambda^n + a_1\lambda^{n-1} + a_2\lambda^{n-2} + \cdots + a_{n-1}\lambda + a_n,$$

where $a_0 > 0$.

Definition 16.2: The polynomial $p(\lambda)$ is called stable if all roots of p have negative real part.

The Routh–Hurwitz theorem provides a means to determine the stability of a polynomial. Define the $n \times n$ (Hurwitz) matrix, H , as in Equation 16.20.

Theorem 16.3: The polynomial $p(\lambda)$ is stable if and only if all the leading principal minors of H are positive. More formally, letting H_j denote the matrix obtained from H by deleting the last $n - j$ rows and columns for $j = 1, \dots, n$, the polynomial is stable if and only if $|H_j| > 0$ for all j .

In the case of a second-order polynomial,

$$H = \begin{pmatrix} a_1 & 0 \\ a_0 & a_2 \end{pmatrix}, \quad H_1 = \{a_1\}, \quad H_2 = H.$$

so that $|H_1| = a_1$ and $|H_2| = a_1a_2$. These are positive if and only if a_1 and a_2 are positive. And, since $a_0 > 0$, stability requires that all coefficients in the characteristic polynomial $p(\lambda) = a_0\lambda^2 + a_1\lambda + a_2$ are positive.

Consider Example 16.9 with coefficient matrix A . The corresponding characteristic equation is $|A - \lambda I| = \lambda^2 + \frac{3}{4}\lambda + \frac{1}{8}$, so that all coefficients are positive. The matrix has eigenvalues $\lambda_1 = -\frac{1}{4}$, $\lambda_2 = -\frac{1}{2}$, leading to stable dynamics of the system.

翻译 (Chinese)

动态系统的长期行为取决于特征方程的根。动态系统稳定的充分条件是特征多项式的根具有负实部。考虑多项式 $p(\lambda) = a_0\lambda^n + a_1\lambda^{n-1} + \cdots + a_{n-1}\lambda + a_n$ ，其中 $a_0 > 0$ 。

定义 16.2: 若 p 的所有根具有负实部，则称多项式 $p(\lambda)$ 是稳定的。

Routh–Hurwitz 定理提供了判断多项式稳定性的方法。定义 $n \times n$ (Hurwitz) 矩阵 H (方程 16.20)。

定理 16.3: $p(\lambda)$ 是稳定的当且仅当 H 的所有顺序主子式为正。更形式化地，令 H_j 表示从 H 中删除最后 $n - j$ 行和列得到的矩阵 ($j = 1, \dots, n$)，则多项式是稳定的当且仅当对所有 j 有 $|H_j| > 0$ 。

对于二阶多项式， $H = \begin{pmatrix} a_1 & 0 \\ a_0 & a_2 \end{pmatrix}$ ，故 $|H_1| = a_1$ ， $|H_2| = a_1a_2$ 。这些为正当且仅当 a_1 和 a_2 为正。由于 $a_0 > 0$ ，稳定性要求特征多项式 $p(\lambda) = a_0\lambda^2 + a_1\lambda + a_2$ 的所有系数为正。例 16.9 中特征方程为 $\lambda^2 + \frac{3}{4}\lambda + \frac{1}{8}$ ，所有系数为正，特征值为 $\lambda_1 = -\frac{1}{4}$ ， $\lambda_2 = -\frac{1}{2}$ ，导致系统稳定的动态。

Exercises for Chapter 16 / 第 16 章习题

Original (English)

Exercise 16.1 Consider a loan with outstanding value $P(t)$ at time t . The growth of the debt at time t , over a small instant of time, Δt , is $P(t)r\Delta t$, where r is the continuous time rate of interest. Suppose that repayments are divided into two periods: an early period with no interest — from time $t = 0$ to τ and then on a continuous basis at the rate d or a repayment of $d\Delta t$ over the period of time Δt . Find the value of d that amortizes the loan over the length of time $T > \tau$.

Exercise 16.2 Suppose that $\frac{dy}{dt} - \frac{3}{4}y(t) = 5$ and y satisfies the initial condition, $y(0) = 0$. Find $y(t)$.

Exercise 16.3 The elasticity of demand is $\eta(p) = \frac{p}{Q} \frac{dQ(p)}{dp} = f(p)$. Find the demand function when the elasticity of demand is: (a) $f(p) = c$, (b) $f(p) = pc$, (c) $f(p) = cp$. The parameter c may be positive or negative depending on context.

Exercise 16.4 Suppose that $\frac{dy}{dx} + \alpha(x)y(x) = 0$. Find $y(x)$ by direct integration.

Exercise 16.5 Suppose that $\frac{dy}{dx} + \frac{1}{2}y(x) = x$ and y satisfies the initial condition, $y(0) = 2$. Find $y(x)$.

Exercise 16.6 Suppose that $\frac{dy}{dt} - \alpha y(t) = t$, where α is a constant and y satisfies the initial condition, $y(t) = y(t_0)$ at $t = t_0$. Find $y(t)$.

Exercise 16.7 The theory of population growth from Malthus suggested that as the population grows, competition for resources will work to inhibit that population growth. One model that captures this feature is the logistic growth model: $\frac{dP}{dt} = aP(1 - \frac{P}{P})$. Find the growth rate of the population as a function of t .

Exercise 16.8 Consider a production function $y = f(k) = k^{\frac{1}{2}}$. Suppose that k evolves over time according to the formula: $\frac{dk}{dt} = sk^{\frac{1}{2}} - (\delta + n)k = \frac{1}{20}k^{\frac{1}{2}} - (\frac{1}{10} + \frac{1}{30})k$, where $s = \frac{1}{20}$, $\delta = \frac{1}{10}$ and $n = \frac{1}{30}$ are the savings, depreciation and population growth rates. Find $k(t)$.

Exercise 16.9 Suppose that $\frac{d^2y}{dt^2} - \frac{3}{4}\frac{dy(t)}{dt} + \frac{1}{8}y(t) = 5$ and y satisfies the initial conditions, $y(0) = 0$, $y(1) = 1$. Find $y(t)$.

Exercise 16.10 Consider the system:

$$\begin{pmatrix} \frac{dy_1(t)}{dt} \\ \frac{dy_2(t)}{dt} \end{pmatrix} + \begin{pmatrix} \frac{7}{4} & -\frac{3}{4} \\ \frac{5}{2} & -1 \end{pmatrix} \begin{pmatrix} y_1(t) \\ y_2(t) \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}.$$

(a) Find the eigenvalues and eigenvectors of the coefficient matrix A , showing that $v_1 = (3, 5)$ and $v_2 = (1, 2)$ are eigenvectors of A . (b) Let P be the matrix of eigenvectors; define $z = P^{-1}y$. Express the system in terms of z . (c) Solve for z .

翻译 (Chinese)

习题 16.1 考虑一笔在时刻 t 未偿价值为 $P(t)$ 的贷款。债务在时刻 t 、在微小时间段 Δt 内的增长为 $P(t)r\Delta t$ ，其中 r 为连续时间利率。假设还款分为两个阶段：早期无利息阶段（从 $t = 0$ 到 τ ），然后以速率 d 连续还款。求在时间长度 $T > \tau$ 内摊销贷款所需的 d 值。

习题 16.2 假设 $\frac{dy}{dt} - \frac{3}{4}y(t) = 5$ 且 y 满足初始条件 $y(0) = 0$ 。求 $y(t)$ 。

习题 16.3 需求弹性为 $\eta(p) = \frac{p}{Q} \frac{dQ(p)}{dp} = f(p)$ 。当需求弹性为 (a) $f(p) = c$, (b) $f(p) = pc$, (c) $f(p) = cp$ 时，求需求函数。

习题 16.4 假设 $\frac{dy}{dx} + \alpha(x)y(x) = 0$ 。通过直接积分求 $y(x)$ 。

习题 16.5 假设 $\frac{dy}{dx} + \frac{1}{2}y(x) = x$ 且 y 满足初始条件 $y(0) = 2$ 。求 $y(x)$ 。

习题 16.6 假设 $\frac{dy}{dt} - \alpha y(t) = t$ ，其中 α 为常数且 y 满足初始条件 $y(t) = y(t_0)$ ($t = t_0$)。求 $y(t)$ 。

习题 16.7 马尔萨斯人口增长理论认为，随着人口增长，资源竞争将抑制人口增长。逻辑斯蒂增长模型 $\frac{dP}{dt} = aP(1 - \frac{P}{P})$ 捕捉了这一特征。求人口增长率作为 t 的函数。

习题 16.8 考虑生产函数 $y = f(k) = k^{\frac{1}{2}}$ 。假设 k 按公式 $\frac{dk}{dt} = sk^{\frac{1}{2}} - (\delta + n)k$ 随时间演化，其中 $s = \frac{1}{20}$, $\delta = \frac{1}{10}$, $n = \frac{1}{30}$ 分别为储蓄率、折旧率和人口增长率。求 $k(t)$ 。

习题 16.9 假设 $\frac{d^2y}{dt^2} - \frac{3}{4}\frac{dy(t)}{dt} + \frac{1}{8}y(t) = 5$ 且 y 满足初始条件 $y(0) = 0$, $y(1) = 1$ 。求 $y(t)$ 。

习题 16.10 对于给定的微分方程组, (a) 求系数矩阵 A 的特征值和特征向量, 证明 $v_1 = (3, 5)$ 和 $v_2 = (1, 2)$ 是 A 的特征向量; (b) 用特征向量矩阵 P 定义 $z = P^{-1}y$ 并以 z 表示系统; (c) 求解 z 。

第十七章 Linear Difference Equations / 线性差分方程

17.1 Introduction / 引言

Original (English)

In the study of many problems, the change in a variable is most naturally modeled in discrete units of time. For many bank loans, interest is computed on a monthly basis so that the level of debt is updated on a monthly basis. Government data are routinely released on a quarterly or annual basis, firms report profit each quarter. Thus, unemployment data or profit data are seen as sequences $u_t, u_{t-1}, \dots$ or $\pi_t, \pi_{t-1}, \dots$. When the data evolve according to a systematic pattern, difference equations may be used to study long-run behavior.

Section 17.2 provides some introductory examples to motivate the use of difference equations. Difference equations are categorized by the order of the difference equation. In a first-order difference equation the current value of the variable of interest depends on its value in the previous period; in an n th-order difference equation, the current value depends on the value in the previous n periods. The entire discussion here considers only first- and second-order difference equations, developed in Sections 17.3 and 17.4, respectively. Section 17.3.3 describes the well-known cobweb model of cycles in the demand and price of a single good and illustrates the process of solving a first-order difference equation. A generalization of the first-order difference equation is considered in Section 17.3.4. In Section 17.4.5 some examples of second-order difference equations are considered, in particular the multiplier-accelerator model from macroeconomic theory. Finally, Section 17.5 considers vector difference equations.

翻译 (Chinese)

在许多问题的研究中, 变量的变化最自然地建模为离散时间单位。对于许多银行贷款, 利息按月计算, 因此债务水平按月更新。政府数据通常按季度或年度发布, 公司每季度报告

利润。因此，失业数据或利润数据被视为序列 $u_t, u_{t-1}, \dots$ 或 $\pi_t, \pi_{t-1}, \dots$ 。当数据按照系统模式演变时，差分方程可用于研究长期行为。

第 17.2 节提供了一些介绍性例子来说明差分方程的应用。差分方程按阶数分类。在一阶差分方程中，关注变量的当前值取决于其前一期的值；在 n 阶差分方程中，当前值取决于前 n 期的值。此处的整个讨论仅考虑一阶和二阶差分方程，分别在第 17.3 节和第 17.4 节中展开。第 17.3.3 节描述了著名的蛛网模型，该模型描述了单一商品需求和价格的周期，并说明了求解一阶差分方程的过程。第 17.3.4 节考虑了一阶差分方程的推广。第 17.4.5 节考虑了二阶差分方程的一些例子，特别是宏观经济理论中的乘数-加速数模型。最后，第 17.5 节考虑向量差分方程。

17.2 Motivating examples / 启发性例子

Original (English)

Suppose that you obtain a loan from a bank, to be repaid over time. At the end of each period, interest is computed on the outstanding balance and a payment is made against the loan. Suppose that at the beginning of period t the outstanding balance is y_t . At the end of the period, the outstanding balance has grown to $(1+r)y_t$ when the interest rate is r . Then a payment, m , is made to the bank. So, at the beginning of period $t+1$, the outstanding balance is $y_{t+1} = (1+r)y_t - m$. This may be rewritten as

$$y_{t+1} - (1+r)y_t = -m. \quad (17.1)$$

What sequence $\{y_t\}$ might satisfy this rule? Observe that if y_t were constant over time, $y_t = \bar{y}$, then $\bar{y} - (1+r)\bar{y} = -m$ or $\bar{y} = \frac{m}{r}$. In this case, the outstanding debt does not change and the repayment just covers interest on the principal. From the lender's perspective, the loan is an asset, generating m dollars per period in perpetuity.

Consider an asset that pays m dollars at the end of each year. In present-value terms, the present value of the cash flow is:

$$PV = \frac{1}{(1+r)}m + \frac{1}{(1+r)^2}m + \frac{1}{(1+r)^3}m + \cdots = m \frac{1}{(1+r)} \frac{1+r}{r} = \frac{m}{r}. \quad (17.2)$$

The loan of $\bar{y}$ with repayment only of interest generates a yearly cash flow of m dollars. The value $\bar{y}$ satisfies $r\bar{y} = m$; m is the repayment each period that exactly covers interest accrued on the outstanding amount $\bar{y}$, so the principal $\bar{y}$ is unchanged from period to period and equals PV .

Notice that if there were some sequence $\{\tilde{y}_t\}$ satisfying $\tilde{y}_{t+1} - (1+r)\tilde{y}_t = 0$, then $(\bar{y} + \tilde{y}_{t+1}) - (1+r)(\bar{y} + \tilde{y}_t) = -m$, so that the sequence $\{y_t\} = \{\bar{y} + \tilde{y}_t\}$ also satisfies Equation 17.1. Considering

$\tilde{y}_{t+1} - (1+r)\tilde{y}_t = 0$, this expression asserts that $\tilde{y}_{t+1} = (1+r)\tilde{y}_t$, so that $\tilde{y}_{t+1} = A(1+r)^t$ for $t \geq 1$, with A arbitrary. Thus, $y_t = A(1+r)^t + \frac{m}{r}$. Suppose that some terminal date T is set such that the loan is fully repaid: $0 = y_T = A(1+r)^T + \frac{m}{r}$. Therefore, $A = -\frac{m}{r} \frac{1}{(1+r)^T}$, and

$$y_t = \frac{m}{r} \left[1 - \frac{(1+r)^t}{(1+r)^T} \right]. \quad (17.3)$$

If the repayment per period is m and the loan is repaid in T periods, the amount borrowed must have been: $y_0 = \frac{m}{r} [1 - \frac{1}{(1+r)^T}]$.

As a final motivating example, consider an employment model where workers transition in and out of employment. Let $p(e | e)$ be the probability of being employed next period given currently employed, and $p(u | e)$ the probability of being unemployed next period given currently employed. $p(e | u)$ and $p(u | u)$ denote the corresponding probabilities for an individual currently unemployed. From the definitions, $p(e | e) + p(u | e) = 1$ and $p(e | u) + p(u | u) = 1$. If l is the labor force, then $l = e_t + u_t$. Of those currently employed, $e_t p(e | e)$ remain employed and $e_t p(u | e)$ become unemployed. Of those currently unemployed, $u_t p(e | u)$ become employed and $u_t p(u | u)$ remain unemployed. Thus, unemployment next period is

$$u_{t+1} = e_t p(u | e) + u_t p(u | u) = (l - u_t) p(u | e) + u_t p(u | u) = l p(u | e) + u_t [p(u | u) - p(u | e)] = k + \lambda u_t,$$

where $k = l p(u | e)$ and $\lambda = [p(u | u) - p(u | e)]$. In the steady state, $u_{t+1} = u_t = \bar{u}$, and so $\bar{u} = \frac{k}{1-\lambda}$. Also, $u_{t+1} - \bar{u} = \lambda(u_t - \bar{u})$, so that $u_{t+j} = \lambda^j(u_t - \bar{u})$.

翻译 (Chinese)

假设你从银行获得一笔贷款，需随时间偿还。在每个时期末，对未偿余额计算利息，并进行还款。假设在时期 t 开始时未偿余额为 y_t 。在时期末，当利率为 r 时，未偿余额增长为 $(1+r)y_t$ 。然后向银行支付 m 。因此，在时期 $t+1$ 开始时，未偿余额为 $y_{t+1} = (1+r)y_t - m$ 。这可以重写为 $y_{t+1} - (1+r)y_t = -m$ （方程 17.1）。

什么样的序列 $\{y_t\}$ 满足此规则？若 y_t 随时间保持不变， $y_t = \bar{y}$ ，则 $\bar{y} - (1+r)\bar{y} = -m$ 或 $\bar{y} = \frac{m}{r}$ 。在此情况下，未偿债务不发生变化，还款仅覆盖本金利息。从贷款人角度看，贷款是一种资产，每期产生 m 美元的永久现金流。每期支付 m 的现金流现值为 $PV = \frac{m}{r}$ （方程 17.2）。

若存在某个序列 $\{\tilde{y}_t\}$ 满足 $\tilde{y}_{t+1} - (1+r)\tilde{y}_t = 0$ ，则 $\tilde{y}_{t+1} = A(1+r)^t$ 。因此 $y_t = A(1+r)^t + \frac{m}{r}$ 。假设设定某个终端日期 T 使贷款完全偿清： $0 = y_T$ ，则 $y_t = \frac{m}{r} [1 - \frac{(1+r)^t}{(1+r)^T}]$ 。若每期还款为 m 且在 T 期内偿清，则借款额为 $y_0 = \frac{m}{r} [1 - \frac{1}{(1+r)^T}]$ 。

在就业模型中，设 $p(e | e)$ 为当前就业者下一期保持就业的概率， $p(u | e)$ 为当前就业者下一期失业的概率。类似定义 $p(e | u)$ 和 $p(u | u)$ 。劳动力总量 $l = e_t + u_t$ 。则下一期失业

为 $u_{t+1} = k + \lambda u_t$, 其中 $k = lp(u | e)$, $\lambda = [p(u | u) - p(u | e)]$ 。稳态中 $\bar{u} = \frac{k}{1-\lambda}$, 且 $u_{t+j} = \lambda^j(u_t - \bar{u})$ 。

17.3 First-order linear difference equations / 一阶线性差分方程

Original (English)

A first-order linear difference equation describes the movement of some variable y_t over time according to an equation of the form:

$$a_1 y_t + a_0 y_{t-1} = f(t).$$

Assume $a_1 \neq 0$. When $a_1 \neq 0$, the equation can be rearranged:

$$y_t + a y_{t-1} = \theta(t), \quad (17.4)$$

where $a = \frac{a_0}{a_1}$ and $\theta(t) = \frac{1}{a_1} f(t)$. This is the general form of the first-order linear difference equation.

A solution to this dynamic equation is a sequence $\{y_t\}$ satisfying the equation for each t .

Definition 17.1: A function y_t^p is called a particular solution of $y_t + a y_{t-1} = \theta(t)$ if $y_t^p + a y_{t-1}^p = \theta(t)$, $\forall t$. The collection of all particular solutions is called the general solution.

In general, there are many particular solutions. A unique particular solution is selected by imposing an initial condition.

Definition 17.2: A function y_t^c is called a complementary solution if $y_t^c + a y_{t-1}^c = 0$, $\forall t$.

If y_t and x_t are two particular solutions then $z_t = y_t - x_t$ satisfies $z_t + a z_{t-1} = 0$. Consequently, given any particular solution x_t , any other particular solution y_t may be written $y_t = x_t + z_t$, for some complementary solution z_t .

REMARK 17.1: Multiplying any complementary solution y_t^c by a constant k gives a complementary solution $k y_t^c$. And, if x and y are complementary solutions, then for any numbers α and β , $\alpha x + \beta y$ is a complementary solution. There are many complementary and (hence) many particular solutions.

翻译 (Chinese)

一阶线性差分方程描述某个变量 y_t 随时间按形如 $a_1 y_t + a_0 y_{t-1} = f(t)$ 的方程运动。假设

$a_1 \neq 0$, 方程可重排为 $y_t + ay_{t-1} = \theta(t)$ (方程 17.4), 其中 $a = \frac{a_0}{a_1}$, $\theta(t) = \frac{1}{a_1}f(t)$ 。这是一阶线性差分方程的一般形式。该动态方程的解是一个序列 $\{y_t\}$, 对每个 t 满足方程。

定义 17.1: 若对所有 t 有 $y_t^p + ay_{t-1}^p = \theta(t)$, 则称函数 y_t^p 为 $y_t + ay_{t-1} = \theta(t)$ 的特解。所有特解的集合称为通解。一般来说, 存在许多特解, 通过施加初始条件选择唯一的特解。

定义 17.2: 若对所有 t 有 $y_t^c + ay_{t-1}^c = 0$, 则称函数 y_t^c 为互补解。若 y_t 和 x_t 是两个特解, 则 $z_t = y_t - x_t$ 满足 $z_t + az_{t-1} = 0$ 。

注 17.1: 将任何互补解乘以常数 k 仍得到互补解。若 x 和 y 是互补解, 则对任意数 α 和 β , $\alpha x + \beta y$ 也是互补解。

17.3.1 Solving first-order linear difference equations / 求解一阶线性差分方程

Original (English)

These observations suggest a general method of solution for the first-order linear difference equation:

1. Find a complementary solution, y_t^c , to the homogeneous equation: $y_t + ay_{t-1} = 0$.
2. Find a particular solution, y_t^p , to the inhomogeneous equation: $y_t + ay_{t-1} = \theta(t)$.
3. Combine the complementary and particular solutions to give the general solution: $y_t = y_t^c + y_t^p$.
4. Use the initial condition to select a particular solution.

To simplify the discussion, the case where $\theta(t)$ is constant and the case where it is variable are discussed separately.

翻译 (Chinese)

这些观察提示了一阶线性差分方程的一般求解方法: (1) 求齐次方程 $y_t + ay_{t-1} = 0$ 的互补解 y_t^c ; (2) 求非齐次方程 $y_t + ay_{t-1} = \theta(t)$ 的特解 y_t^p ; (3) 组合互补解和特解得到通解: $y_t = y_t^c + y_t^p$; (4) 利用初始条件选择特解。为简化讨论, 分别讨论 $\theta(t)$ 为常数和可变的情况。

17.3.2 The constant forcing function / 常值强制函数

Original (English)

The function $\theta(t)$ is sometimes called the forcing function of the difference equation. When this is constant, $\theta(t) = c$ for all t , the difference equation is:

$$y_t + ay_{t-1} = c.$$

The homogeneous equation is $y_t + ay_{t-1} = 0$. Finding a complementary solution is straightforward: $y_t = -ay_{t-1} = (-a)^2 y_{t-2} = (-a)^3 y_{t-3} = \cdots$, so it is clear that $y_t = k(-a)^t$ satisfies this expression for any constant k . When $a = -1$, $k(-a)^t = k$.

The inhomogeneous equation $y_t + ay_{t-1} = c$ has a constant particular solution if $a \neq -1$: pick $y_t = \bar{y}$ where $\bar{y} + a\bar{y} = c$, so $y_t = \bar{y} = \frac{c}{1+a}$ is a solution. If $a = -1$, the equation is $y_t - y_{t-1} = c$, so that $y_t = ct + k$ for any number k .

Combining the complementary and particular solutions,

$$y_t = \begin{cases} k(-a)^t + \frac{c}{1+a}, & a \neq -1 \\ y_0 + ct, & a = -1. \end{cases}$$

Imposing an initial condition eliminates the arbitrary parameter. With initial condition y_0 ,

$$y_t = \begin{cases} \left(y_0 - \frac{c}{1+a}\right)(-a)^t + \frac{c}{1+a}, & a \neq -1 \\ y_0 + ct, & a = -1. \end{cases}$$

EXAMPLE 17.1: $y_t - \frac{1}{2}y_{t-1} = 4$, $y_0 = 1$. The particular solution is $y_t^p = \frac{4}{1-\frac{1}{2}} = 8$. The complementary solution is $y_t^c = A(\frac{1}{2})^t$. This gives $y_t = A(\frac{1}{2})^t + 8$, and using $y_0 = 1$ gives $1 = y_0 = A + 8$, $A = -7$. Therefore, $y_t = (-7)(\frac{1}{2})^t + 8$.

EXAMPLE 17.2: $y_t - y_{t-1} = 4$, $y_0 = 1$. The particular solution is $y_t^p = 4t$ and the complementary solution is $y_t^c = A$. This gives $y_t = A + 4t$, so $1 = y_0 = A$. Thus, $y_t = 1 + 4t$.

翻译 (Chinese)

函数 $\theta(t)$ 有时称为差分方程的强制函数。当其为常数时, 即对所有 t 有 $\theta(t) = c$, 差分方程为 $y_t + ay_{t-1} = c$ 。齐次方程为 $y_t + ay_{t-1} = 0$ 。求互补解很直接: $y_t = -ay_{t-1} = (-a)^2 y_{t-2} = \cdots$, 故 $y_t = k(-a)^t$ 对任意常数 k 满足该表达式。

非齐次方程 $y_t + ay_{t-1} = c$ 在 $a \neq -1$ 时有常数特解: $y_t = \bar{y} = \frac{c}{1+a}$ 。若 $a = -1$, 方程为 $y_t - y_{t-1} = c$, $y_t = ct + k$ 。组合互补解和特解, 并施加初始条件 y_0 , 得到上述形式。

例 17.1: $y_t - \frac{1}{2}y_{t-1} = 4$, $y_0 = 1$ 。特解为 $y_t^p = 8$ 。互补解为 $y_t^c = A(\frac{1}{2})^t$ 。 $y_t = A(\frac{1}{2})^t + 8$, 由 $y_0 = 1$ 得 $A = -7$ 。故 $y_t = (-7)(\frac{1}{2})^t + 8$ 。

例 17.2: $y_t - y_{t-1} = 4$, $y_0 = 1$ 。特解为 $y_t^p = 4t$, 互补解为 $y_t^c = A$ 。 $y_t = 1 + 4t$ 。

17.3.3 The cobweb model / 蛛网模型

Original (English)

This is a supply and demand model with the key feature that supply in the current period depends on the previous period's price. Thus, in any given period, price alone adjusts to clear the market:

$$D_t = a + bp_t, \quad a > 0, b < 0, \quad S_t = c + dp_{t-1}, \quad 0 < c < a, d > 0.$$

Market clearing each period gives: $D_t = a + bp_t = c + dp_{t-1} = S_t$.

In a stationary equilibrium: $p_t = p_{t-1} = p^e$, $a + bp^e = c + dp^e \Rightarrow p^e = \frac{a-c}{d-b}$.

Dynamic behavior of price / 价格的动态行为

With market clearing each period, $D_t = S_t$, rearranging gives $bp_t - dp_{t-1} = c - a$, so that:

$$p_t - \frac{d}{b}p_{t-1} = \frac{c-a}{b},$$

or $p_t + \alpha p_{t-1} = \beta$, where $\alpha = -\frac{d}{b}$ and $\beta = \frac{c-a}{b}$. Note that $\alpha = -\frac{d}{b} > 0$ since $d > 0, b < 0$. The general solution is:

$$p_t = k \left(\frac{d}{b} \right)^t + \frac{\beta}{1+\alpha} = k \left(\frac{d}{b} \right)^t + p^e.$$

With an initial price p_0 , $k = p_0 - p^e$ and this gives the particular solution:

$$p_t = (p_0 - p^e) \left(\frac{d}{b} \right)^t + p^e.$$

This is stable ($p_t \rightarrow p^e$) provided $\left| \frac{d}{b} \right| < 1$.

Stability / 稳定性

In terms of elasticities, at p^e , $\frac{\varepsilon_S(p^e)}{\varepsilon_D(p^e)} = \frac{d}{b}$. Since $d > 0$ and $b < 0$, $\frac{d}{b} < 0$. There are three cases:

1. $\frac{d}{b} < -1$ ($\left| \frac{d}{b} \right| > 1$): supply more responsive than demand, unstable dynamics.
2. $\frac{d}{b} = -1$ ($\left| \frac{d}{b} \right| = 1$): equal magnitude slopes, cycles occur.
3. $\frac{d}{b} > -1$ ($\left| \frac{d}{b} \right| < 1$): supply less responsive than demand, stable dynamics.

翻译 (Chinese)

这是一个供需模型，其关键特征是当期供给取决于前期价格。在市场出清下： $D_t = a + bp_t = c + dp_{t-1} = S_t$ 。在平稳均衡中： $p^e = \frac{a-c}{d-b}$ 。

每期市场出清给出 $bp_t - dp_{t-1} = c - a$, 故 $p_t + \alpha p_{t-1} = \beta$, 其中 $\alpha = -\frac{d}{b}$, $\beta = \frac{c-a}{b}$. 通解为 $p_t = k(\frac{d}{b})^t + p^e$. 施加初始价格 p_0 得 $p_t = (p_0 - p^e)(\frac{d}{b})^t + p^e$. 当 $|\frac{d}{b}| < 1$ 时稳定。

以弹性表示, 在 p^e 处, $\frac{\varepsilon_S(p^e)}{\varepsilon_D(p^e)} = \frac{d}{b}$. 存在三种情况: (1) $\frac{d}{b} < -1$: 供给比需求更敏感, 动态不稳定; (2) $\frac{d}{b} = -1$: 出现周期; (3) $\frac{d}{b} > -1$: 供给不如需求敏感, 动态稳定。

17.3.4 Equations with variable forcing function / 变系数强制函数方程

Original (English)

Considering again the equation $y_t + ay_{t-1} = \theta(t)$. The homogeneous part is the same regardless of $\theta(t)$, and the complementary solution is unchanged: $y_t = k(-a)^t$. Finding a particular solution is more complex. The usual practice is to postulate a functional form for the particular solution that is similar to the functional form of the forcing function θ . Three cases are considered:

Case 1: $\theta(t) = gd^t$

$$y_t = \begin{cases} \left(y_0 - \frac{gd}{a+d}\right)(-a)^t + \frac{gd^{t+1}}{a+d}, & a+d \neq 0 \\ y_0(-a)^t - \frac{g}{a}td^{t+1}, & a+d = 0. \end{cases} \quad (17.5)$$

Case 2: $\theta(t) = \gamma t + \delta$

$$y_t = \left(y_0 - \frac{\delta}{1+a} - \frac{\gamma a}{(1+a)^2}\right)(-a)^t + \frac{\gamma}{1+a}t + \frac{\delta}{1+a} + \frac{\gamma a}{(1+a)^2}, \quad a \neq -1. \quad (17.6)$$

Case 3: $\theta(t) = \gamma t^2 + \delta$

$$y_t = [y_0 - \eta_3](-a)^t + \eta_1 t^2 + \eta_2 t + \eta_3, \quad a \neq -1. \quad (17.7)$$

Detailed calculations deriving these expressions are given in Section 17.7.1.

EXAMPLE 17.3: $y_t - \frac{1}{2}y_{t-1} = 5(\frac{1}{4})^t$, $y_0 = 1$. This corresponds to case 1 with $a = -\frac{1}{2}$, $d = \frac{1}{4}$. So $y_t = 6(\frac{1}{2})^t - 20(\frac{1}{4})^{t+1}$.

EXAMPLE 17.4: $y_t - \frac{1}{2}y_{t-1} = 5(\frac{1}{2})^t$, $y_0 = 1$. $y_t = 6(\frac{1}{2})^t - 10t(\frac{1}{2})^{t+1}$.

翻译 (Chinese)

再次考虑方程 $y_t + ay_{t-1} = \theta(t)$. 齐次部分相同, 互补解不变: $y_t = k(-a)^t$. 求特解更复杂。通常做法是假设特解的函数形式与强制函数 θ 的函数形式相似。讨论了三种情况 (方程 17.5、17.6、17.7), 详细推导见第 17.7.1 节。

例 17.3: $y_t - \frac{1}{2}y_{t-1} = 5(\frac{1}{4})^t$, $y_0 = 1$ 。属于情形 1, $y_t = 6(\frac{1}{2})^t - 20(\frac{1}{4})^{t+1}$ 。

例 17.4: $y_t - \frac{1}{2}y_{t-1} = 5(\frac{1}{2})^t$, $y_0 = 1$ 。 $y_t = 6(\frac{1}{2})^t - 10t(\frac{1}{2})^{t+1}$ 。

17.4 Second-order linear difference equations / 二阶线性差分方程

Original (English)

The simplest form of second-order difference equation is:

$$y_t + a_1y_{t-1} + a_2y_{t-2} = c.$$

As in the case of first-order difference equations, the problem of finding a solution is dealt with as follows:

- Solve the inhomogeneous equation to obtain the particular solution.
- Solve the homogeneous equation to obtain the complementary solution.
- Obtain the general solution as the sum of the particular and complementary solutions.
- Use initial conditions to identify a unique solution.

翻译 (Chinese)

二阶差分方程的最简单形式为 $y_t + a_1y_{t-1} + a_2y_{t-2} = c$ 。与一阶情形相同, 通过求解非齐次方程得特解, 求解齐次方程得互补解, 两者之和为通解, 利用初始条件确定唯一解。

17.4.1 The inhomogeneous equation / 非齐次方程

Original (English)

The form of solution for the inhomogeneous equation depends on the values of a_1 and a_2 . The simplest candidate particular solution is a constant solution $y_t = y$ and if this satisfies the inhomogeneous equation: $y(1 + a_1 + a_2) = c$, y is determined as $y = \frac{c}{1+a_1+a_2}$.

However, $y(1 + a_1 + a_2) = c$ is insoluble if $(1 + a_1 + a_2) = 0$ (and $c \neq 0$). In this case, try a particular solution $y = \gamma t$, which yields $\gamma = -\frac{c}{a_1+2a_2}$, provided $a_1 + 2a_2 \neq 0$.

Finally, if both $(1 + a_1 + a_2)$ and $(a_1 + 2a_2)$ equal 0, try $y = \gamma t^2$, giving $\gamma = \frac{c}{2}$. So, depending on the parameters, a particular solution is given by:

$$y_t^p = \begin{cases} \frac{c}{1+a_1+a_2}, & a_1 + a_2 \neq -1 \\ -\frac{c}{a_1+2a_2} t, & a_1 + a_2 = -1, a_1 + 2a_2 \neq 0 \\ \frac{c}{2} t^2, & a_1 + a_2 = -1, a_1 + 2a_2 = 0. \end{cases} \quad (17.8)$$

翻译 (Chinese)

非齐次方程解的形式取决于 a_1 和 a_2 的值。最简单的候选特解是常数解 $y_t = y$ ，若满足非齐次方程，则 $y = \frac{c}{1+a_1+a_2}$ 。若 $1 + a_1 + a_2 = 0$ ，尝试 $y = \gamma t$ ，得 $\gamma = -\frac{c}{a_1+2a_2}$ 。若两者均为零，尝试 $y = \gamma t^2$ ，得 $\gamma = \frac{c}{2}$ 。特解如表 17.8 所示。

17.4.2 The homogeneous equation / 齐次方程

Original (English)

Now, consider the homogeneous equation, $y_t + a_1 y_{t-1} + a_2 y_{t-2} = 0$. As a trial complementary solution consider $y_t^c = k b^t$. This gives the equation: $k b^t + a_1 k b^{t-1} + a_2 k b^{t-2} = 0$. Canceling k gives the quadratic equation:

$$b^2 + a_1 b + a_2 = 0.$$

Use the usual formula to find the roots λ_1 and λ_2 :

$$\lambda_1, \lambda_2 = \frac{-a_1 \pm \sqrt{a_1^2 - 4a_2}}{2}.$$

So, $y_t = k \lambda_i^t$ satisfies the complementary equation for any value of k . More generally, any y_t of the form $y_t = k_1 \lambda_1^t + k_2 \lambda_2^t$ satisfies the complementary equation. The dynamic behavior depends critically on the roots, with three possibilities:

1. Both roots real and distinct ($a_1^2 - 4a_2 > 0$).
2. Both roots real but repeated ($\lambda_1 = \lambda_2$, when $a_1^2 - 4a_2 = 0$).
3. Both roots complex and distinct ($a_1^2 - 4a_2 < 0$).

Depending on which case applies, the complementary solution is:

$$y_t^c = \begin{cases} k_1 \lambda_1^t + k_2 \lambda_2^t, & \text{roots real and distinct} \\ k_1 \lambda^t + k_2 t \lambda^t, & \text{roots real and equal} \\ r^t [B_1 \cos(\omega t) + B_2 \sin(\omega t)], & \text{roots complex,} \end{cases} \quad (17.9)$$

where $r = |\lambda_1| = |\lambda_2| = \sqrt{a_2}$ and $\omega = \cos^{-1}(-\frac{a_1}{2\sqrt{a_2}})$ for complex roots. The derivation is given in Section 17.7.2.

翻译 (Chinese)

考虑齐次方程 $y_t + a_1 y_{t-1} + a_2 y_{t-2} = 0$ 。试探互补解 $y_t^c = kb^t$ ，代入得 $b^2 + a_1 b + a_2 = 0$ 。根为 $\lambda_1, \lambda_2 = \frac{-a_1 \pm \sqrt{a_1^2 - 4a_2}}{2}$ 。动态行为取决于三种情况：(1) 实且不同；(2) 实且相同；(3) 复数。互补解形式如方程 17.9 所示。

17.4.3 The general solution and stability / 通解与稳定性

Original (English)

Together, Equations 17.8 and 17.9 define the solution: $y_t = y_t^p + y_t^c$. Given initial conditions y_0 and y_1 , the constants k_1 and k_2 are uniquely determined.

For stability: from the particular solution, if $a_1 + a_2 = -1$, then the system diverges, so a necessary condition for stability is $a_1 + a_2 \neq -1$. For the complementary solution:

- Real distinct roots: stability requires $|\lambda_1| < 1$ and $|\lambda_2| < 1$, which translates to $-2 < a_1 < 2$, $1 + a_1 + a_2 > 0$ and $1 - a_1 + a_2 > 0$.
- Real equal roots: stability requires $|\lambda| < 1$ ($-2 < a_1 < 2$) and $a_1 + a_2 \neq -1$.
- Complex distinct roots: stability requires $|\lambda_1| = |\lambda_2| < 1$, i.e., $a_2 < 1$.

翻译 (Chinese)

方程 17.8 和 17.9 共同定义了通解 $y_t = y_t^p + y_t^c$ 。给定初始条件 y_0 和 y_1 ，常数 k_1 和 k_2 唯一确定。关于稳定性， $a_1 + a_2 = -1$ 是必要的稳定条件。稳定性区域由参数 a_1 和 a_2 决定 (图 17.3)。

17.4.4 The Samuelson multiplier-accelerator model / 萨缪尔森乘数-加速数模型

Original (English)

EXAMPLE 17.6: Aggregate output equals aggregate consumption plus investment plus government spending:

$$Y_t = C_t + I_t + G, \quad C_t = \alpha Y_{t-1}, \quad 0 < \alpha < 1, \quad I_t = \beta(C_t - C_{t-1}), \quad \beta > 0.$$

Substitution gives: $Y_t - \alpha Y_{t-1} - \beta \alpha (Y_{t-1} - Y_{t-2}) = G$, or

$$Y_t - \alpha(1 + \beta)Y_{t-1} + \alpha\beta Y_{t-2} = G.$$

Solving for the particular solution: $Y_t^p = \frac{G}{1 - \alpha(1 + \beta) + \alpha\beta} = \frac{G}{1 - \alpha}$. The roots are $\lambda_1, \lambda_2 = \frac{\alpha(1 + \beta) \pm \sqrt{[\alpha(1 + \beta)]^2 - 4\alpha\beta}}{2}$.

Three cases: (1) Real distinct roots when $[\alpha(1 + \beta)]^2 > 4\alpha\beta$, stable when $\alpha < 1$. (2) Real equal roots when $[\alpha(1 + \beta)]^2 = 4\alpha\beta$, stable when $|\alpha| < 2$. (3) Complex roots when $[\alpha(1 + \beta)]^2 < 4\alpha\beta$, stable when $0 < \alpha < 1$.

EXAMPLE 17.7: Adding a monetary sector with money supply $\bar{m}$ and demand $m_t = m_0 + m_1 Y_{t-1} + m_2 r_t$, and investment $I_t = \beta(C_t - C_{t-1}) + i_0 r_t$, gives a second-order difference equation. The characteristic equation is $\lambda^2 - \left[(1 + \beta) - \frac{i_0 m_1}{m_2}\right] \lambda + \beta\alpha = 0$. Stability conditions follow from the general case.

翻译 (Chinese)

例 17.6: 总产出等于消费加投资加政府支出: $Y_t = C_t + I_t + G$, $C_t = \alpha Y_{t-1}$, $I_t = \beta(C_t - C_{t-1})$ 。代入得 $Y_t - \alpha(1 + \beta)Y_{t-1} + \alpha\beta Y_{t-2} = G$ 。特解为 $Y_t^p = \frac{G}{1 - \alpha}$ 。根据参数有三种情况。

例 17.7: 在例 17.6 基础上添加货币部门, 货币供给为 $\bar{m}$, 货币需求为 $m_t = m_0 + m_1 Y_{t-1} + m_2 r_t$, 投资为 $I_t = \beta(C_t - C_{t-1}) + i_0 r_t$, 得到二阶差分方程。特征方程为 $\lambda^2 - \left[(1 + \beta) - \frac{i_0 m_1}{m_2}\right] \lambda + \beta\alpha = 0$ 。稳定性条件从一般情形推出。

17.5 Vector difference equations / 向量差分方程

Original (English)

Consider a dynamic system describing the movement of two variables x_t and y_t over time:

$$a_{11}x_t + a_{12}y_t + b_{11}x_{t-1} + b_{12}y_{t-1} = c_1, \quad a_{21}x_t + a_{22}y_t + b_{21}x_{t-1} + b_{22}y_{t-1} = c_2.$$

Let $z_t = \begin{pmatrix} x_t \\ y_t \end{pmatrix}$, $A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$, $B = \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix}$, $C = \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$, so that $Az_t + Bz_{t-1} = C$.

A complementary solution solves $Az_t^c + Bz_{t-1}^c = 0$; the particular solution solves $Az_t^p + Bz_{t-1}^p = C$; and the general solution is the sum: $z_t = z_t^c + z_t^p$.

17.5.1 The particular solution / 特解

To find the particular solution, set $z_t^p = \bar{z}$. Then $A\bar{z} + B\bar{z} = C$ or $(A + B)\bar{z} = C$. Assume $A + B$ is non-singular, so $z_t^p = (A + B)^{-1}C$.

17.5.2 The complementary solution / 互补解

Try a solution of the form $z_t^c = \begin{pmatrix} s \\ u \end{pmatrix} b^t$. Divide by b^{t-1} to get $(Ab + B) \begin{pmatrix} s \\ u \end{pmatrix} = 0$. If $(s, u) \neq 0$, then $|Ab + B| = 0$, giving a quadratic in b : $k_1 b^2 + k_2 b + k_3 = 0$. Denote the roots b_1, b_2 . The system $[Ab_i + B] \begin{pmatrix} s \\ u \end{pmatrix} = 0$ has solution (s_i, u_i) , determined up to a scalar multiple. This gives the complementary solution:

$$z_t^c = \begin{pmatrix} x_t^c \\ y_t^c \end{pmatrix} = \begin{pmatrix} \alpha s_1 b_1^t + \beta s_2 b_2^t \\ \alpha u_1 b_1^t + \beta u_2 b_2^t \end{pmatrix}.$$

If the roots are real and repeated, try $z_t^c = \begin{pmatrix} s_1 b_1^t + s_2 t b_1^t \\ u_1 b_1^t + u_2 t b_1^t \end{pmatrix}$.

EXAMPLE 17.8: Consider an oligopoly of two firms. The first-order conditions give $Q_{1t} + \frac{1}{2}Q_{2t-1} = \frac{a-c}{2b}$ and $Q_{2t} + \frac{1}{2}Q_{1t-1} = \frac{a-c}{2b}$. The system $Az_t + Bz_{t-1} = C$ has roots $b_1 = -\frac{1}{2}$, $b_2 = -\frac{1}{2}$ (repeated and real). The solution converges since $|\frac{1}{2}|^t \rightarrow 0$ and $t|\frac{1}{2}|^t \rightarrow 0$.

翻译 (Chinese)

考虑描述两个变量 x_t 和 y_t 随时间的运动的动态系统: $Az_t + Bz_{t-1} = C$ 。互补解满足 $Az_t^c + Bz_{t-1}^c = 0$; 特解满足 $Az_t^p + Bz_{t-1}^p = C$; 通解为二者之和。特解为 $z_t^p = (A + B)^{-1}C$ (假设 $A + B$ 非奇异)。互补解通过设 $z_t^c = \begin{pmatrix} s \\ u \end{pmatrix} b^t$ 并解 $|Ab + B| = 0$ 求得。

例 17.8: 考虑双寡头古诺模型。一阶条件给出方程组, 根为 $b_1 = b_2 = -\frac{1}{2}$ (重实根)。由于 $|\frac{1}{2}|^t \rightarrow 0$, 解是收敛的。

17.6 The n -variable case / n 变量情形

Original (English)

Consider a system of n equations governing the evolution of the n -vector x over time:

$$x_t - Ax_{t-1} = c. \quad (17.10)$$

Provided $(I - A)$ is invertible, $\bar{x} = (I - A)^{-1}c$ is a particular solution. With distinct eigenvectors, $A = U\Lambda U^{-1}$ where Λ is diagonal. Then

$$x_t = U\Lambda^t U^{-1}(x_0 - \bar{x}) + \bar{x}.$$

Provided all roots of A are less than 1, Λ^t converges to the 0 matrix and x_t converges to $\bar{x}$.

翻译 (Chinese)

考虑 n 个方程的系统控制 n 维向量 x 随时间的演化: $x_t - Ax_{t-1} = c$ (方程 17.10)。若 $(I - A)$ 可逆, $\bar{x} = (I - A)^{-1}c$ 是特解。当特征向量不同时, $A = U\Lambda U^{-1}$, 则 $x_t = U\Lambda^t U^{-1}(x_0 - \bar{x}) + \bar{x}$ 。若 A 的所有根小于 1, Λ^t 收敛到零矩阵, x_t 收敛到 $\bar{x}$ 。

17.7 Miscellaneous calculations / 补充计算

Original (English)

The following sections provide detailed computations underlying the results stated in Sections 17.3.4 and 17.4.2, covering solutions for various forcing functions, roots of the characteristic equation, and stability regions. The derivations confirm the formulas for $\theta(t) = gd^t$, $\theta(t) = \gamma t + \delta$, $\theta(t) = \gamma t^2 + \delta$, and verify the complementary solution forms for real distinct, real repeated, and complex roots (including de Moivre's theorem application). The stability regions are derived in three cases: real distinct roots (conditions $-2 < a_1 < 2$, $1 + a_1 + a_2 > 0$, $1 - a_1 + a_2 > 0$), real equal roots ($a_1 + a_2 \neq -1$, $-2 < a_1 < 2$), and complex distinct roots ($a_2 < 1$).

翻译 (Chinese)

以下各节提供了第 17.3.4 节和第 17.4.2 节所述结果的详细计算, 涵盖各种强制函数的解、特征方程的根以及稳定性区域。推导确认了 $\theta(t) = gd^t$ 、 $\theta(t) = \gamma t + \delta$ 、 $\theta(t) = \gamma t^2 + \delta$ 的公式, 并验证了实不同根、实重根和复数根 (包括棣莫弗定理的应用) 的互补解形式。稳定性区域按三种情况推导: 实不同根、实重根和复数根。

Exercises for Chapter 17 / 第 17 章习题

Original (English)

Exercise 17.1 Let y_0 be the initial value of y . Find the long-run dynamics of the following four systems:

$$(a) y_t = \frac{1}{2}y_{t-1} + 2, \quad (b) y_t = -\frac{1}{2}y_{t-1} + 2, \quad (c) y_t = 2y_{t-1} + 2, \quad (d) y_t = -2y_{t-1} + 2.$$

Exercise 17.2 Suppose that an amount P_0 is borrowed at time $t = 0$. The interest rate is r so after one period the debt grows to $P_0(1 + r)$. Each period a repayment d is made, so that at the

end of period 1, the outstanding debt is $P_1 = P_0(1+r) - d$. (a) Find the equation governing the evolution of P_t . (b) Given a period T , find the value of d that amortizes (pays off) the loan exactly at the end of period T .

Exercise 17.3 In a labor-market model, suppose that $p(u | e) = 0.05$ and $p(u | u) = 0.3$. If u_t is the unemployment rate, then $u_{t+1} = p(u | e)(1 - u_t) + p(u | u)u_t$. Find the long-run dynamic behavior of u_t .

Exercise 17.4 Consider the dynamic model $y_{t+1} = ay_t + c$. (a) Plot the equation $y_{t+1} = ay_t + c$ and the steady-state line $y_{t+1} = y_t$ for: (i) $a = \frac{1}{2}$, $c = 4$; (ii) $a = -\frac{1}{2}$, $c = 4$; (iii) $a = 2$, $c = -1$. (b) In each case, trace the dynamics from the initial value of $y_0 = 1$.

Exercise 17.5 Let y_0 be the initial value of y . Find the long-run dynamics of: (a) $y_t = \frac{1}{2}y_{t-1} + 3t$, (b) $y_t = -\frac{1}{2}y_{t-1} + 3t$, (c) $y_t = 2y_{t-1} + 3t$, (d) $y_t = -2y_{t-1} + 3t$.

Exercise 17.6 Let y_0 be the initial value of y . Find the long-run dynamics of: (a) $y_t = \frac{1}{2}y_{t-1} + 3^t$, (b) $y_t = -\frac{1}{2}y_{t-1} + 2^t$, (c) $y_t = 2y_{t-1} + (\frac{1}{2})^t$, (d) $y_t = -2y_{t-1} + 2^t$.

翻译 (Chinese)

习题 17.1 设 y_0 为 y 的初值。求四个系统 (a)–(d) 的长期动态。

习题 17.2 $t = 0$ 时借入 P_0 。利率为 r ，每期还款 d ， $P_1 = P_0(1+r) - d$ 。(a) 求 P_t 的演化方程；(b) 给定期限 T ，求恰好在该期限末偿清贷款所需的 d 值。

习题 17.3 劳动力市场模型中， $p(u | e) = 0.05$ ， $p(u | u) = 0.3$ 。若 u_t 为失业率， $u_{t+1} = p(u | e)(1 - u_t) + p(u | u)u_t$ 。求 u_t 的长期动态行为。

习题 17.4 动态模型 $y_{t+1} = ay_t + c$ 。(a) 对三种参数情形画出方程和稳态线；(b) 每种情形下从 $y_0 = 1$ 追踪动态。

习题 17.5 求四个系统 (a)–(d) 的长期动态。

习题 17.6 求四个系统 (a)–(d) 的长期动态。

第十八章 Probability and Distributions / 概率与分布

18.1 Introduction / 引言

Original (English)

A probability distribution gives the frequency of occurrence of some variable. For example, the income distribution is summarized by a function F , where $F(y)$ gives the fraction of the population with income at or below y . If a person is selected at random from the population, then the probability that the person will have income below y is $F(y)$. Similarly, one may consider the age distribution, represented by G , so that $G(a)$ is the fraction of people with age no larger than a , or a wage distribution where $H(w)$ gives the proportion of people earning a wage no larger than w . Thus, a distribution provides information regarding the frequency of a certain characteristic or property in a population. Considering the income distribution, suppose a person is selected at random from the population and income recorded. At the selection process, let Y be the as yet to be determined income. This unknown number is called a random variable. The probability that the person selected at random will have an income below y is denoted $P(Y \leq y)$ and this is equal to $F(y)$. In Section 18.2, random variables are introduced and briefly described. Intuitively, every distribution function describes the behavior of a random variable. Section 18.3 introduces the distribution function and its density.

In some cases, a random variable may only have a finite number of possible values. If the random variable has a finite or countable number of possible outcomes, say that the random variable is discrete; if there are a continuum of possible outcomes, say the random variable is a continuous variable. These categorizations lead to an analogous categorization of distributions into discrete distributions and continuous distributions. Section 18.4 considers a few important discrete distributions (the Bernoulli, binomial and Poisson distributions). Finally, Section 18.5 describes the primary distributions used in hypothesis testing in economics: the uniform, normal, t , chi-square

and F distributions.

翻译 (Chinese)

概率分布给出某个变量出现的频率。例如，收入分布由一个函数 F 概括，其中 $F(y)$ 给出收入在 y 或以下的人口比例。若从总体中随机选择一人，则该人收入低于 y 的概率为 $F(y)$ 。类似地，可以考虑年龄分布 G ， $G(a)$ 是年龄不超过 a 的人口比例；或工资分布 $H(w)$ ， $H(w)$ 给出工资不高于 w 的人口比例。因此，分布提供了关于总体中某种特征或属性频率的信息。在选择过程中，令 Y 表示待确定的收入，这个未知数称为随机变量。随机选择的人收入低于 y 的概率记为 $P(Y \leq y)$ ，这等于 $F(y)$ 。第 18.2 节引入并简要描述随机变量。直观上，每个分布函数描述了一个随机变量的行为。第 18.3 节介绍分布函数及其密度。

在某些情况下，随机变量只有有限个可能值。若随机变量有有限或可数个可能结果，称为离散随机变量；若有连续统的可能结果，称为连续随机变量。这种分类导致将分布类似地分为离散分布和连续分布。第 18.4 节考虑几个重要的离散分布（伯努利、二项和泊松分布）。第 18.5 节描述经济学假设检验中使用的主要分布：均匀、正态、 t 、卡方和 F 分布。

18.2 Random variables / 随机变量

Original (English)

Consider the experiment of tossing a six-sided die with each side equally likely. For the experiment, there are six possible outcomes $\{1, 2, 3, 4, 5, 6\}$ and each outcome is equally likely with probability $\frac{1}{6}$. We can formalize this by denoting the outcome of the die toss by X , where X can take on any one of the six possible values. Write $P(X = i)$ to denote the probability that X takes the value i . With the probability of each outcome assigned a number, we can determine the probability that the number drawn is less than or equal to 3 ($P(X \leq 3) = P(X = 1) + P(X = 2) + P(X = 3)$), that the probability is greater than 3, and so on. The notion of a random variable gives a formal way of examining random outcomes with associated probabilities.

Any experiment with a finite number of outcomes $\{x_1, x_2, \dots, x_n\}$ can be represented in this way: if outcome x_i has probability p_i , the associated random variable has corresponding probability $P(X = x_i) = p_i$. With this formalism, one can identify defining features of the random process. The mean is denoted μ and defined $\mu = \sum_{i=1}^n p_i x_i$, and the variance, σ^2 , is defined $\sigma^2 = \sum_{i=1}^n p_i (x_i - \mu)^2$. The mean provides one measure of “central tendency”. While the mean provides a measure of “central tendency”, variance measures dispersion. In the die-tossing case, $\mu = \sum_i p_i x_i = \frac{1}{6} \cdot 1 + \frac{1}{6} \cdot 2 + \dots + \frac{1}{6} \cdot 6 = 3\frac{1}{2}$, and $\sigma^2 = 17\frac{1}{2}$.

This formalism allows one to study arbitrary situations where outcomes are unknown. For a discrete random variable X that has outcome x_i with probability p_i , the mean $\mu = \sum x_i p_i$ is often called the expected value of X and written $E\{X\}$. Similarly, the variance $\sigma^2 = \sum (x_i - \mu)^2 p_i$ is written $E\{(X - \mu)^2\}$.

If the random variable is continuous, there is a continuum of possible outcomes. Some random events are most suitably modeled using continuous variables. For example, consider a device for filling cans of oil. Design specification requires that each can be filled to, say, 330 ml. In operation, few cans will be filled exactly; although the average fill may be 330 ml, a can selected at random may have, for example, 328.755 ml. However, most cans will be filled to a level close to 330 ml. Describing the dispersion around 330 ml is most easily modeled with a continuous random variable.

翻译 (Chinese)

考虑掷一个六面骰子的实验，每个面等可能。实验有六个可能结果 $\{1, 2, 3, 4, 5, 6\}$ ，每个结果的概率为 $\frac{1}{6}$ 。我们可以通过用 X 表示掷骰结果来形式化， X 可以取六个可能值中的任意一个。记 $P(X = i)$ 表示 X 取值 i 的概率。有了每个结果的概率，我们可以确定抽到的数小于等于 3 的概率，等等。随机变量的概念提供了一种形式化方式来考察带有相关概率的随机结果。

任何具有有限个结果 $\{x_1, x_2, \dots, x_n\}$ 的实验都可以这样表示。均值记为 μ ，定义为 $\mu = \sum_{i=1}^n p_i x_i$ ；方差 σ^2 定义为 $\sigma^2 = \sum_{i=1}^n p_i (x_i - \mu)^2$ 。均值提供了“中心趋势”的一种度量，方差度量离散程度。在掷骰情形中， $\mu = 3\frac{1}{2}$ ， $\sigma^2 = 17\frac{1}{2}$ 。

对于离散随机变量 X ，其取值为 x_i 的概率为 p_i ，均值 $\mu = \sum x_i p_i$ 通常称为 X 的期望值，记为 $E\{X\}$ 。类似地，方差 $\sigma^2 = \sum (x_i - \mu)^2 p_i$ 记为 $E\{(X - \mu)^2\}$ 。若随机变量是连续的，则存在连续统的可能结果。某些随机事件最适合用连续变量建模。

18.3 The distribution function and density / 分布函数与密度

Original (English)

A random variable takes on different values with specific probabilities. The function attaching probabilities to outcomes is called the distribution function (or the related function, the density). The distribution function, F , associated with the random variable X is defined

$$F(x) = P(X \leq x). \quad (18.1)$$

Thus, $F(x)$ is the probability that X will be no larger than x . Since $P(X \leq x) \leq P(X \leq x')$ whenever $x \leq x'$, $F(x) \leq F(x')$ for $x \leq x'$, so that $F(x)$ is non-decreasing in x . Given F , other probabilities may easily be computed. For example, $P(x_a < X \leq x_b) = P(X \leq x_b) - P(X \leq x_a) = F(x_b) - F(x_a)$. A random variable X distributed on an interval $[a, b]$ satisfies $P(X < a) = P(X > b) = 0$. At a , $F(a) = 0$ and at b , $F(b) = 1$. The smallest such interval $[a, b]$ is called the support of the distribution.

In the case of discrete random variables, the distribution function is defined in the same way: $F(x) = P(X \leq x)$. Since X takes values from a set $\{x_1, x_2, \dots, x_k, \dots\}$, with $x_i < x_{i+1}$ for all i , $F(x) = \sum_{x_i \leq x} P(X = x_i)$, so that the distribution is a step function increasing by the amount $P(X = x_i) = p_i$ at $x = x_i$.

In general, a distribution function may have features of continuous and discrete distributions.

For a continuous distribution F , for any x , $F(x) - F(x - \Delta x) \rightarrow 0$ as $\Delta x \rightarrow 0$. When F is continuous and differentiable, denote the derivative of F by $f(x) = F'(x)$; this is called the density of F . From the definition of a derivative,

$$f(x) = \lim_{\Delta x \rightarrow 0} \frac{F(x + \Delta x) - F(x)}{\Delta x}.$$

Thus, when $\Delta x > 0$ is small, $f(x)\Delta x \approx F(x + \Delta x) - F(x) = P(x < X \leq x + \Delta x)$. Therefore, $f(x)\Delta x$ is approximately the probability that the random variable X will fall in the interval $[x, x + \Delta x]$. From the definition of f ,

$$\int_a^b f(x)dx = F(b) - F(a) = P(a < X \leq b),$$

so the distribution function F can be obtained from the density f .

While the distribution function has values between 0 and 1, the density function can have arbitrarily large positive values. If F is a distribution function with $F(a) = 0$, $F(b) = 1$ and density f , then $\int_a^b f(x)dx = F(b) - F(a) = 1$, so that the area under f is 1.

翻译 (Chinese)

随机变量以特定的概率取不同的值。将概率赋予结果的函数称为分布函数（或相关函数，密度）。与随机变量 X 相关的分布函数 F 定义为 $F(x) = P(X \leq x)$ （方程 18.1）。因此， $F(x)$ 是 X 不大于 x 的概率。由于当 $x \leq x'$ 时有 $P(X \leq x) \leq P(X \leq x')$ ， $F(x)$ 在 x 上非递减。给定 F ，其他概率可以轻松计算。例如， $P(x_a < X \leq x_b) = F(x_b) - F(x_a)$ 。分布在区间 $[a, b]$ 上的随机变量 X 满足 $P(X < a) = P(X > b) = 0$ ， $F(a) = 0$ ， $F(b) = 1$ 。最小的此类区间 $[a, b]$ 称为分布的支撑。

在离散随机变量情形下, $F(x) = \sum_{x_i \leq x} P(X = x_i)$, 分布是一个阶梯函数。分布函数可能同时具有连续和离散特征。对于连续分布 F , 当 F 连续且可微时, 记 F 的导数为 $f(x) = F'(x)$, 称为 F 的密度。 $f(x)\Delta x$ 近似等于随机变量 X 落在区间 $[x, x + \Delta x]$ 内的概率。 $\int_a^b f(x)dx = F(b) - F(a) = P(a < X \leq b)$ 。若 F 有支撑 $[a, b]$, 则 $\int_a^b f(x)dx = 1$ 。

18.3.1 The conditional distribution / 条件分布

Original (English)

The probability that the random variable falls between a and b ($a < b$) is given by $F(b) - F(a)$. If b is the largest possible value, $F(b) = 1$, then the probability that the random variable is larger than a is $1 - F(a)$. The conditional probability that X lies between $\underline{a}$ and $\bar{a}$ given $X > a$ is:

$$P(X \leq \bar{a} \mid X > a) = \frac{F(\bar{a}) - F(a)}{1 - F(a)}.$$

The logic is simple: the relative likelihood of each outcome adds up to 1, with

$$1 = \frac{F(\bar{a}) - F(a)}{1 - F(a)} + \frac{1 - F(\bar{a})}{1 - F(a)}.$$

Given that X is above a , the conditional probability of X falling in $[a, \bar{a}]$ is $\frac{F(\bar{a}) - F(a)}{1 - F(a)}$.

翻译 (Chinese)

随机变量落在 a 和 b 之间的概率为 $F(b) - F(a)$ 。若 b 是最大可能值, 则 $F(b) = 1$, 随机变量大于 a 的概率为 $1 - F(a)$ 。给定 $X > a$ 时 X 在 $[a, \bar{a}]$ 内的条件概率为 $P(X \leq \bar{a} \mid X > a) = \frac{F(\bar{a}) - F(a)}{1 - F(a)}$ 。逻辑是每个结果的相对可能性加起来等于 1。

18.3.2 Joint distributions / 联合分布

Original (English)

Given two random variables X and Y , let the probability that X is in a set A and Y in a set B be $P(X \in A, Y \in B)$.

Definition 18.1: The joint distribution of two random variables X, Y is a function $F(x, y)$ giving the probability that X is no larger than x and Y is no larger than y :

$$F(x, y) = P(X \leq x \text{ and } Y \leq y). \quad (18.3)$$

Assuming sufficient differentiability, the density $f(x, y)$ associated with F is defined as $f(x, y) = \frac{\partial^2 F}{\partial x \partial y}$.

EXAMPLE 18.1: $F(x, y) = \frac{xy}{(2-xy)}$, $0 \leq x \leq 1$, $0 \leq y \leq 1$, with corresponding density $f(x, y) = \frac{1}{(2-xy)^2} + \frac{3yx}{(2-xy)^3} + \frac{2x^2y^2}{(2-xy)^3}$.

Independence / 独立性

Definition 18.2: The random variables X and Y are said to be independent if for any events A and B ,

$$P(X \in A, Y \in B) = P(X \in A)P(Y \in B). \quad (18.5)$$

An equivalent condition is $P(X \leq x, Y \leq y) = P(X \leq x)P(Y \leq y)$, $\forall x, y$. In terms of the distribution function, two random variables X and Y are independent if the joint distribution may be written $F(x, y) = F_X(x)F_Y(y)$ for all x, y . Equivalently, X and Y are independent if the density factors satisfy $f(x, y) = f_X(x)f_Y(y)$.

With n random variables, the condition is $F(x_1, \dots, x_n) = F_{X_1}(x_1) \cdots F_{X_n}(x_n)$.

Two random variables X and Y have a jointly normal distribution if

$$f(x, y) = \frac{1}{2\pi\sigma_x\sigma_y\sqrt{1-\rho^2}} \exp \left\{ -\frac{1}{2(1-\rho^2)} \left[\left(\frac{x-\mu_x}{\sigma_x} \right)^2 + \left(\frac{y-\mu_y}{\sigma_y} \right)^2 - 2\rho \frac{(x-\mu_x)(y-\mu_y)}{\sigma_x\sigma_y} \right] \right\},$$

where $\rho = \frac{\text{Cov}(X, Y)}{\sigma_x\sigma_y}$ is the correlation coefficient. In the special case where $\rho = 0$, $f(x, y) = f(x)f(y)$.

REMARK 18.1: For the normal distribution, zero correlation implies independence.

翻译 (Chinese)

给定两个随机变量 X 和 Y , $P(X \in A, Y \in B)$ 表示 X 在集合 A 中且 Y 在集合 B 中的概率。

定义 18.1: X, Y 的联合分布是函数 $F(x, y) = P(X \leq x \text{ and } Y \leq y)$ (方程 18.3)。在充分可微的假设下, 密度 $f(x, y) = \frac{\partial^2 F}{\partial x \partial y}$ 。

例 18.1: $F(x, y) = \frac{xy}{(2-xy)}$ (图 18.4)。

定义 18.2: 若对任意事件 A 和 B , $P(X \in A, Y \in B) = P(X \in A)P(Y \in B)$, 则称 X 和 Y 是独立的。等价地, $F(x, y) = F_X(x)F_Y(y)$ 或 $f(x, y) = f_X(x)f_Y(y)$ 。两个随机变量具有联合正态分布, 其中 ρ 为相关系数。**注 18.1:** 对于正态分布, 零相关意味着独立。

18.4 Discrete distributions / 离散分布

Original (English)

A discrete distribution is characterized by a set of possible outcomes $\{x_1, x_2, \dots\}$ and corresponding probabilities $\{p_1, p_2, \dots\}$. The expectation is $\mu = \sum_i p_i x_i$; the variance $\sigma^2 = \sum_i p_i (x_i - \mu)^2$. The following describes the Bernoulli, binomial and Poisson distributions.

翻译 (Chinese)

离散分布由可能结果的集合 $\{x_1, x_2, \dots\}$ 和相应的概率 $\{p_1, p_2, \dots\}$ 刻画。期望为 $\mu = \sum_i p_i x_i$ ；方差为 $\sigma^2 = \sum_i p_i (x_i - \mu)^2$ 。以下描述伯努利、二项和泊松分布。

18.4.1 The Bernoulli distribution / 伯努利分布

Original (English)

Consider a random variable X with two possible outcomes, 0 or 1. Suppose $P(X = 1) = p$, $P(X = 0) = 1 - p$. Then $E\{X\} = p \cdot 1 + (1 - p) \cdot 0 = p$; and the variance $\sigma^2 = p(1 - p)^2 + (1 - p)(0 - p)^2 = p(1 - p)$.

翻译 (Chinese)

考虑具有两个可能结果 0 或 1 的随机变量 X 。设 $P(X = 1) = p$, $P(X = 0) = 1 - p$ 。则 $E\{X\} = p$, 方差 $\sigma^2 = p(1 - p)$ 。

18.4.2 The binomial distribution / 二项分布

Original (English)

Suppose n coins are drawn randomly and each tossed. The probability of exactly k ones and $n - k$ zeros is $p^k(1 - p)^{n-k}$. The number of ways of having k ones in n positions is $\binom{n}{k} = \frac{n!}{(n-k)!k!}$. Thus, the probability of having k ones is:

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{(n-k)}, \quad k = 0, \dots, n. \quad (18.9)$$

According to the binomial theorem, $(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^k y^{n-k}$. With $x = p$, $y = 1 - p$, we have $1 = (p + 1 - p)^n = \sum_{k=0}^n \binom{n}{k} p^k (1 - p)^{n-k}$, so the probabilities add up to 1.

More generally, if items can be categorized into r distinct groups with k_i items of category i ($k = k_1 + \dots + k_r$), the number of distinct patterns is $\frac{n!}{(n-k)!k_1!k_2!\dots k_r!}$.

翻译 (Chinese)

假设随机抽取 n 枚硬币并掷出。恰好有 k 个 1 和 $n-k$ 个 0 的概率为 $p^k(1-p)^{n-k}$ 。 n 个位置中有 k 个 1 的方式数为 $\binom{n}{k} = \frac{n!}{(n-k)!k!}$ 。因此, 有 k 个 1 的概率为 $P(X = k) = \binom{n}{k}p^k(1-p)^{(n-k)}$ (方程 18.9)。根据二项式定理, 概率之和为 1。

更一般地, 若物品分为 r 个不同类别, 第 i 类有 k_i 个物品, 则不同模式的数量为 $\frac{n!}{(n-k)!k_1!k_2!\cdots k_r!}$ 。

18.4.3 The Poisson distribution / 泊松分布

Original (English)

Let X be a random variable denoting the number of occurrences in one unit of time. The Poisson distribution assigns probability:

$$P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}, \quad k = 0, 1, \dots \quad (18.10)$$

where $\lambda > 0$ is a given parameter. The expected value is $E\{X\} = \lambda$ and the variance is $\text{Var}(X) = \lambda$, so the mean and variance are equal.

When the unit of time is τ , $P(X = k) = \frac{e^{-\lambda\tau}(\lambda\tau)^k}{k!}$, $k = 0, 1, \dots$

18.4.4 The Poisson process / 泊松过程

For each $t \geq 0$, let X_t be a random variable denoting the number of occurrences in the time interval $[0, t]$, with distribution $P(X_t = k) = \frac{e^{-\lambda t}(\lambda t)^k}{k!}$. Furthermore, suppose X_t has independent increments, so that $X_t - X_s$ is independent of $X_q - X_p$ for $t > s \geq q > p$ with

$$P(X_t - X_s = k) = \frac{e^{-\lambda(t-s)}[\lambda(t-s)]^k}{k!}. \quad (18.12)$$

REMARK 18.2: The Poisson process is characterized by simple postulates: for a short length of time Δ , (a) $P(X_{t+\Delta} - X_t = 0) = 1 - \lambda\Delta + o(\Delta)$, (b) $P(X_{t+\Delta} - X_t = 1) = \lambda\Delta + o(\Delta)$, (c) $P(X_{t+\Delta} - X_t \geq 2) = o(\Delta)$; and independent increments. Then the family satisfies Equation 18.12.

翻译 (Chinese)

设 X 为表示单位时间内事件发生次数的随机变量。泊松分布指定概率为 $P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}$, $k = 0, 1, \dots$ (方程 18.10), 其中 $\lambda > 0$ 为给定参数。期望 $E\{X\} = \lambda$, 方差 $\text{Var}(X) = \lambda$, 故均值和方差相等。

当时间单位为 τ 时, $P(X = k) = \frac{e^{-\lambda\tau}(\lambda\tau)^k}{k!}$ 。

泊松过程：对每个 $t \geq 0$ ，设 X_t 表示时间区间 $[0, t]$ 内事件的发生次数。此外，设 X_t 具有独立增量。**注 18.2：**泊松过程由简单的公设刻画：对很短的时段 Δ ，满足概率条件并具有独立增量。

18.5 Continuous distributions / 连续分布

Original (English)

In this section, continuous distributions are considered, characterized by a cumulative distribution F and density $f = F'$. In this case, the mean is $E\{X\} = \mu = \int xf(x)dx$ and variance $E\{(X-\mu)^2\} = \sigma^2 = \int (x-\mu)^2 f(x)dx$. Similarly, given a function $h(x)$, $E\{h(X)\} = \int h(x)f(x)dx$.

The following discussion introduces the primary distributions used in hypothesis testing in economics: the uniform, normal, t , chi-square and F distributions.

翻译 (Chinese)

本节考虑连续分布，其特征为累积分布 F 和密度 $f = F'$ 。此时，均值 $E\{X\} = \mu = \int xf(x)dx$ ，方差 $E\{(X-\mu)^2\} = \sigma^2 = \int (x-\mu)^2 f(x)dx$ 。类似地， $E\{h(X)\} = \int h(x)f(x)dx$ 。以下介绍经济学假设检验中使用的主要分布：均匀、正态、 t 、卡方和 F 分布。

18.5.1 The uniform distribution / 均匀分布

Original (English)

Consider the task of selecting a number at random from the interval $[a, b]$. With each number “equally likely”, this random variable is said to be uniformly distributed on $[a, b]$ with distribution

$$F(x) = \frac{1}{b-a}(x-a), \quad x \in [a, b].$$

The corresponding density is $f(x) = \frac{1}{b-a}$. The mean is $\mu = \frac{1}{2}(b+a)$ and the variance is $\sigma^2 = \frac{1}{12}(b-a)^2$.

翻译 (Chinese)

考虑从区间 $[a, b]$ 中随机选取一个数。每个数“等可能”，该随机变量称为在 $[a, b]$ 上均匀分布，分布函数为 $F(x) = \frac{1}{b-a}(x-a)$ 。密度为 $f(x) = \frac{1}{b-a}$ 。均值 $\mu = \frac{1}{2}(b+a)$ ，方差 $\sigma^2 = \frac{1}{12}(b-a)^2$ 。

18.5.2 The exponential distribution / 指数分布

Original (English)

A common model of the time of occurrence of an event uses the exponential distribution. This has density:

$$f(x) = \lambda e^{-\lambda x}, \quad x \geq 0.$$

This has cumulative distribution $F(x) = 1 - e^{-\lambda x}$ with $x \geq 0$. The distribution has mean $\mu = \frac{1}{\lambda}$ and variance $\sigma^2 = \frac{1}{\lambda^2}$.

One interesting feature of the exponential distribution is that the conditional distribution and the unconditional distribution are the same: $P(X \leq \bar{a} \mid X > a) = 1 - e^{-\lambda s} = F(s)$, where $s = \bar{a} - a$. The distribution of the wait time is the same at time $x = a$, given no arrival, as it is at time $x = 0$.

翻译 (Chinese)

事件发生时间的常见模型使用指数分布。密度为 $f(x) = \lambda e^{-\lambda x}$, $x \geq 0$ 。累积分布为 $F(x) = 1 - e^{-\lambda x}$ 。均值 $\mu = \frac{1}{\lambda}$, 方差 $\sigma^2 = \frac{1}{\lambda^2}$ 。指数分布的一个有趣特征是条件分布与无条件分布相同: $P(X \leq \bar{a} \mid X > a) = 1 - e^{-\lambda s} = F(s)$ 。等待时间的分布在时刻 $x = a$ (给定未到达) 与在时刻 $x = 0$ 是相同的。

18.5.3 The normal distribution / 正态分布

Original (English)

The normal distribution is often used to model the spread of a characteristic in a population. The normal distribution has density:

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \quad -\infty < x < \infty, \quad (18.13)$$

with corresponding cumulative distribution:

$$F(x) = \int_{-\infty}^x f(\xi) d\xi = \int_{-\infty}^x \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(\xi-\mu)^2}{2\sigma^2}} d\xi. \quad (18.14)$$

In the special case where $\mu = 0$ and $\sigma^2 = 1$, the distribution is called the standard normal distribution, with density $\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}$ and cumulative distribution $\Phi(x) = \int_{-\infty}^x \phi(\xi) d\xi$.

Notice that $\Phi(-1.645) = 0.05$ and $\Phi(1.645) = 0.95$, so that a random variable drawn from a standard normal distribution will fall below -1.645 with probability 0.05 and above 1.645 with probability 0.05.

Theorem 18.1: If X is a normally distributed random variable with mean μ and variance σ^2 , then $Z = a + bX$ has a normal distribution with mean $a + b\mu$ and variance $b^2\sigma^2$.

Given a normal random variable X with mean μ and variance σ^2 , the transformation $\frac{X-\mu}{\sigma}$ has the standard normal distribution. Thus, for example, if a random variable is drawn from a normal distribution, there is a 95% probability that it will lie within 1.96 standard deviations of the mean.

Theorem 18.2: Suppose that $X_1, \dots, X_n$ are normally distributed random variables. Then $\bar{X} = \sum_{i=1}^n a_i X_i$ is normally distributed with mean $\sum_{i=1}^n a_i \mu_i$. If the random variables are independent then the variance is $\sum_{i=1}^n a_i^2 \sigma_i^2$.

翻译 (Chinese)

正态分布常用于对总体中某特征的分布进行建模。正态分布的密度为 $f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$ (方程 18.13), 相应的累积分布为 $F(x) = \int_{-\infty}^x f(\xi) d\xi$ (方程 18.14)。当 $\mu = 0$ 且 $\sigma^2 = 1$ 时, 称为标准正态分布, 密度为 $\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}$, 累积分布为 $\Phi(x)$ 。 $\Phi(-1.645) = 0.05$, $\Phi(1.645) = 0.95$ 。

定理 18.1: 若 X 为正态分布随机变量, 均值为 μ , 方差为 σ^2 , 则 $Z = a + bX$ 具有均值 $a + b\mu$ 和方差 $b^2\sigma^2$ 的正态分布。变换 $\frac{X-\mu}{\sigma}$ 具有标准正态分布。因此, 从正态分布中抽取的随机变量有 95% 的概率落在均值两侧 1.96 个标准差之内。

定理 18.2: 正态分布随机变量的线性组合服从正态分布。若各变量独立, 则方差为 $\sum a_i^2 \sigma_i^2$ 。

18.5.4 Moment-generating functions / 矩母函数

Original (English)

The expectation of a function $h(x)$ is $E\{h(X)\} = \int h(x)f(x)dx$. One special case occurs when $h(x) = e^{tx}$. The expectation is called the moment-generating function (assuming the integral exists):

$$m_X(t) = E\{e^{tX}\} = \int e^{tx} f(x) dx.$$

Theorem 18.3: Suppose that $m_X(t)$ is defined on an interval $(-\epsilon, \epsilon)$, $\epsilon > 0$. Then m_X uniquely determines the distribution.

One important property: if $X_1, \dots, X_n$ are independent and $Y = X_1 + \dots + X_n$, then $m_Y(t) = m_{X_1}(t) \cdots m_{X_n}(t)$. In the i.i.d. case, $m_Y(t) = m_X(t)^n$.

For the normal distribution, the moment-generating function is $m(t) = e^{\mu t + \frac{1}{2}t^2\sigma^2}$. Thus, the sum of n i.i.d. normal random variables has moment-generating function $m(t)^n = e^{n\mu t + \frac{1}{2}t^2 n\sigma^2}$, a normal distribution with mean $n\mu$ and variance $n\sigma^2$.

翻译 (Chinese)

函数 $h(x)$ 的期望为 $E\{h(X)\} = \int h(x)f(x)dx$ 。当 $h(x) = e^{tx}$ 时, 期望称为矩母函数: $m_X(t) = E\{e^{tX}\}$ 。

定理 18.3: 若 $m_X(t)$ 在区间 $(-\epsilon, \epsilon)$ 上定义 ($\epsilon > 0$), 则 m_X 唯一确定分布。一个重要性质是: 若 $X_1, \dots, X_n$ 独立且 $Y = X_1 + \dots + X_n$, 则 $m_Y(t) = m_{X_1}(t) \cdots m_{X_n}(t)$ 。对于正态分布, 矩母函数为 $m(t) = e^{ut + \frac{1}{2}t^2\sigma^2}$ 。 n 个独立同分布正态变量之和服从正态分布, 均值为 $n\mu$, 方差为 $n\sigma^2$ 。

18.5.5 The chi-square distribution / 卡方分布

Original (English)

A chi-square distribution with k degrees of freedom (k a positive integer) is defined by density:

$$f(x) = \frac{1}{2^{\frac{k}{2}}\Gamma(\frac{k}{2})} x^{\frac{k}{2}-1} e^{-\frac{1}{2}x}, \quad x > 0, \quad (18.15)$$

where $\Gamma(\cdot)$ is the gamma function $\Gamma(q) = \int_0^\infty \xi^{q-1} e^{-\xi} d\xi$. The chi-square distribution with n degrees of freedom has mean n and variance $2n$.

REMARK 18.4: If $q > 1$, $\Gamma(q) = (q-1)\Gamma(q-1)$. $\Gamma(1) = 1$, $\Gamma(\frac{1}{2}) = \sqrt{\pi}$. If x is an integer then $\Gamma(x+1) = x!$.

REMARK 18.5: The chi-square distribution is a special case of the gamma distribution with density $f(x) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}$, where $\alpha = \frac{k}{2}$ and $\beta = \frac{1}{2}$.

Theorem 18.4: (1) If X has a standard normal distribution, then X^2 has a chi-square distribution with one degree of freedom. (2) If $X_1, \dots, X_n$ are independent standard normal random variables, then $Z = X_1^2 + \dots + X_n^2$ has a chi-square distribution with n degrees of freedom.

翻译 (Chinese)

自由度为 k 的卡方分布由密度 $f(x) = \frac{1}{2^{\frac{k}{2}}\Gamma(\frac{k}{2})} x^{\frac{k}{2}-1} e^{-\frac{1}{2}x}$ 定义 (方程 18.15), 其中 $\Gamma(\cdot)$ 为伽玛函数。 n 自由度卡方分布均值为 n , 方差为 $2n$ 。

注 18.4: 若 $q > 1$, $\Gamma(q) = (q-1)\Gamma(q-1)$ 。 $\Gamma(1) = 1$, $\Gamma(\frac{1}{2}) = \sqrt{\pi}$ 。

注 18.5: 卡方分布是伽玛分布的特例。**定理 18.4:** (1) 若 X 为标准正态, 则 X^2 服从自由度 1 的卡方分布; (2) n 个独立标准正态变量的平方和服从自由度 n 的卡方分布。

18.5.6 The t distribution / t 分布

Original (English)

The random variable X is said to have a t distribution with k degrees of freedom if the density is:

$$f(x) = \frac{\Gamma(\frac{1}{2}(k+1))}{\Gamma(\frac{k}{2})} \frac{1}{\sqrt{k\pi}} \frac{1}{(1 + \frac{x^2}{k})^{\frac{k+1}{2}}}, \quad -\infty < x < \infty. \quad (18.17)$$

The t distribution with k degrees of freedom has mean 0 ($k > 1$) and variance $\frac{k}{k-2}$ ($k > 2$).

Theorem 18.5: Suppose that Y has a standard normal distribution and Z has a chi-square distribution with k degrees of freedom, with Y and Z independent. Then the random variable $X = \frac{Y}{\sqrt{Z/k}}$ has a t distribution with k degrees of freedom.

REMARK 18.6: Given a collection of i.i.d. standard normal random variables with mean 0 and variance 1, let $\bar{x} = \frac{1}{n} \sum_i x_i$ and $s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$. Then $\bar{x}$ and s^2 are independently distributed; $\bar{x}$ has a normal distribution with mean 0 and variance $\frac{1}{n}$; $(n-1)s^2$ has a chi-square distribution with $n-1$ degrees of freedom. Therefore $\frac{\bar{x}}{s/\sqrt{n}}$ has a t distribution with $n-1$ degrees of freedom.

翻译 (Chinese)

自由度为 k 的 t 分布密度如方程 18.17 所示。 k 自由度的 t 分布均值为 0 ($k > 1$)，方差为 $\frac{k}{k-2}$ ($k > 2$)。

定理 18.5: 若 Y 为标准正态， Z 为自由度 k 的卡方，且独立，则 $X = \frac{Y}{\sqrt{Z/k}}$ 服从自由度 k 的 t 分布。

注 18.6: 给定一组独立同分布标准正态随机变量， $\bar{x}$ 服从均值为 0、方差为 $\frac{1}{n}$ 的正态分布； $(n-1)s^2$ 服从自由度 $n-1$ 的卡方分布。因此 $\frac{\bar{x}}{s/\sqrt{n}}$ 服从自由度 $n-1$ 的 t 分布。

18.5.7 The F distribution / F 分布

Original (English)

The random variable X is said to have the F distribution with degrees of freedom m and n if the density is:

$$f(x) = \frac{\Gamma(\frac{m+n}{2})}{\Gamma(\frac{m}{2})\Gamma(\frac{n}{2})} \left(\frac{m}{n}\right)^{\frac{m}{2}} \frac{x^{\frac{m-2}{2}}}{(1 + \frac{m}{n}x)^{\frac{m+n}{2}}}, \quad x > 0. \quad (18.18)$$

Usually, m and n are positive integers. The F distribution with degrees of freedom m and n has mean $\frac{n}{n-2}$ ($n > 2$) and variance $\frac{2n^2(m+n-2)}{m(n-2)^2(n-4)}$ ($n > 4$).

Theorem 18.6: Suppose that Y and Z are independent chi-square distributions with m and n degrees of freedom, respectively. Then the random variable $X = \frac{Y/m}{Z/n}$ has an F distribution with degrees of freedom m and n .

翻译 (Chinese)

自由度为 m 和 n 的 F 分布密度如方程 18.18 所示。 F 分布均值为 $\frac{n}{n-2}$ ($n > 2$)，方差为 $\frac{2n^2(m+n-2)}{m(n-2)^2(n-4)}$ ($n > 4$)。

定理 18.6: 若 Y 和 Z 是独立的卡方分布，自由度分别为 m 和 n ，则随机变量 $X = \frac{Y/m}{Z/n}$ 服从自由度 m 和 n 的 F 分布。

Exercises for Chapter 18 / 第 18 章习题

Original (English)

Exercise 18.1 Suppose that an instructor has 100 questions available and must write an exam consisting of four questions. A question may be asked only once in the exam. How many different exams can the instructor set? (Two exams are different if there is at least one question on one exam that is not on the other.)

Exercise 18.2 Suppose that an instructor has 100 questions available and must write an exam consisting of four questions. Furthermore, each question must come from a distinct category: A, B, C, D. The instructor has 15 category A questions, 25 category B questions, 40 category C questions and 30 category D questions. (a) How many different exams can the instructor set? (b) Suppose instead that the exam consists of eight questions, and two must be drawn from each section. How many distinct exams are possible?

Exercise 18.3 Let $f(x) = a(1 - x^4)$ for $0 < x < 1$ and 0 otherwise be a probability density function. Find a .

Exercise 18.4 Let $f(x) = a(1 - x^n)$ for $0 < x < 1$ and 0 otherwise be a probability density function where n is a positive integer. Find a .

Exercise 18.5 Let X be a random variable with density $f(x) = 4x^3$ for $0 < x < 1$ and 0 otherwise. Find $\text{Prob}(\frac{1}{3} < X < \frac{2}{3})$.

Exercise 18.6 For a random variable X , the moments of the distribution are defined: $\mu_1 = E\{X\}$, $\mu_2 = E\{X^2\}$, ..., $\mu_k = E\{X^k\}$ with the mean $\mu = \mu_1$. Show that the variance σ^2 satisfies $\sigma^2 = \mu_2 - \mu_1^2$.

Exercise 18.7 The moment-generating function for a discrete random variable X is defined: $m(t) = \sum_x e^{tx} p(x)$, where $p(x) = \text{Prob}(X = x)$. Show that $m'(0) = \sum_{i=1}^n x_i p(x_i)$ and $m''(0) = \sum_{i=1}^n x_i^2 p(x_i)$.

Exercise 18.8 The Bernoulli distribution is: $f(x) = p^x(1-p)^{1-x}$ for $x = 0, 1$ and 0 otherwise. (a) Find the mean μ and variance σ^2 . (b) Find the moment-generating function of f . (c) Use the moment-generating function to find μ and σ^2 .

Exercise 18.9 The exponential distribution is: $f(x) = \lambda e^{-\lambda x}$ for $x > 0$ and 0 for $x \leq 0$. (a) Find the mean and variance. (b) Find the moment-generating function.

Exercise 18.10 For the exponential distribution, the hazard function is defined: $h(x) = \frac{f(x)}{1-F(x)}$. Show that, for the exponential function, the hazard function is constant.

Exercise 18.11 The Poisson distribution is: $f(x) = \frac{e^{-\lambda} \lambda^x}{x!}$ for $x = 0, 1, 2, \dots$ and 0 otherwise. Suppose that a coffee shop has an average of 24 customers per hour. (a) What is the probability that the shop will have six or fewer customers in any given hour? (b) Considering two successive hours, what is the probability that the shop will have six or fewer customers in the first hour and more than six customers in the second hour?

Exercise 18.12 Suppose X and Y are two independent random variables with Poisson distributions: $\frac{e^{-\lambda_x} \lambda_x^x}{x!}$ and $\frac{e^{-\lambda_y} \lambda_y^y}{y!}$. Then $Z = X + Y$ has a Poisson distribution with parameter $\lambda_z = \lambda_x + \lambda_y$. Suppose two shops A and B have an average number of customers per hour of 5 and 10, respectively. What is the probability that, together, the two stores will have fewer than 20 customers in any given hour?

Exercise 18.13 Use the fact that, for any real number λ , $e^\lambda = \sum_{x=0}^{\infty} \frac{\lambda^x}{x!}$ to show that the moment-generating function for the Poisson distribution is $m(t) = e^{-\lambda} e^{\lambda e^t}$. From this conclude that the Poisson distribution has mean λ .

Exercise 18.14 Suppose that a ball will fall onto the real number line. The distribution of the point of impact is given by a standard normal distribution. Suppose you can place an interval of length $k > 0$ on the line. How should the interval be located to maximize the probability that the ball will fall within the interval?

Exercise 18.15 Let X be a standard normal random variable. Define the random variable $Z = \sigma X^2$ where $\sigma > 0$. Find the distribution of Z .

Exercise 18.16 The random variable $T_n = \frac{X}{\sqrt{Z/n}}$, where X is standard normal and Z a chi-square variable with n degrees of freedom, has a t distribution with n degrees of freedom. Use the law of large numbers to argue that, as n becomes large, the distribution of a T variable converges to that of a standard normal distribution.

Exercise 18.17 The random variable $F_n = \frac{Y/m}{Z/n}$, where Y, Z are independent chi-square variables with m and n degrees of freedom, has an F distribution. Use the law of large numbers to argue that, as n becomes large, the distribution of an F variable converges to that of a chi-square (with m degrees of freedom) divided by its degrees of freedom.

翻译 (Chinese)

习题 18.1 教师有 100 道题目，需出一份包含四道题的试卷。每题只能用一次。可以出多少份不同的试卷？

习题 18.2 教师需从 A、B、C、D 四类中各选一题。A 类 15 题，B 类 25 题，C 类 40 题，D 类 30 题。(a) 可出多少份不同试卷？(b) 若试卷需八道题，每类各抽两题，有多少种不同试卷？

习题 18.3 设 $f(x) = a(1 - x^4)$ 对 $0 < x < 1$ ，否则为 0，是概率密度函数。求 a 。

习题 18.4 设 $f(x) = a(1 - x^n)$ 对 $0 < x < 1$ ，否则为 0，是概率密度函数， n 为正整数。求 a 。

习题 18.5 设 X 是密度为 $f(x) = 4x^3$ ($0 < x < 1$) 的随机变量。求 $\text{Prob}(\frac{1}{3} < X < \frac{2}{3})$ 。

习题 18.6 对随机变量 X ，矩定义为 $\mu_k = E\{X^k\}$ 。证明方差 $\sigma^2 = \mu_2 - \mu_1^2$ 。

习题 18.7 离散随机变量的矩母函数 $m(t) = \sum_x e^{tx} p(x)$ 。证明 $m'(0) = \sum x_i p(x_i)$ 且 $m''(0) = \sum x_i^2 p(x_i)$ 。

习题 18.8 伯努利分布： $f(x) = p^x(1-p)^{1-x}$ ， $x = 0, 1$ 。(a) 求均值和方差；(b) 求矩母函数；(c) 用矩母函数求均值和方差。

习题 18.9 指数分布： $f(x) = \lambda e^{-\lambda x}$ ， $x > 0$ 。(a) 求均值和方差；(b) 求矩母函数。

习题 18.10 风险函数 $h(x) = \frac{f(x)}{1-F(x)}$ 。证明对指数分布，风险函数为常数。

习题 18.11 泊松分布： $f(x) = \frac{e^{-\lambda} \lambda^x}{x!}$ 。咖啡店平均每小时 24 位顾客。(a) 任意一小时内顾客数不超过六的概率？(b) 连续两小时内第一小时不超六且第二小时超六的概率？

习题 18.12 若 X 和 Y 独立且服从泊松分布，则 $Z = X + Y$ 服从参数 $\lambda_z = \lambda_x + \lambda_y$ 的泊松分布。A 店平均每小时 5 位顾客，B 店 10 位。两店合计任意一小时内顾客少于 20 的概率？

习题 18.13 利用 $e^\lambda = \sum_{x=0}^{\infty} \frac{\lambda^x}{x!}$ 证明泊松分布的矩母函数为 $m(t) = e^{-\lambda} e^{\lambda e^t}$ ，并得出均值为 λ 。

习题 18.14 一个球落向实数轴。撞击点的分布是标准正态分布。你可以放置一个长度为 $k > 0$ 的区间。如何放置以最大化球落入该区间的概率？

习题 18.15 设 X 为标准正态随机变量。定义 $Z = \sigma X^2$ ($\sigma > 0$)。求 Z 的分布。

习题 18.16 利用大数定律论证，当 n 变大时， T 变量的分布收敛到标准正态分布。

习题 18.17 利用大数定律论证，当 n 变大时， F 变量的分布收敛到卡方分布 (m 自由度) 除以其自由度。